MATHEMATICS
ANNOTATED READING NOTES

PDE

Partial Differential Equations

英文精读札记与中文边注

Classical PDE and Sobolev Spaces

Reading companion to Lawrence C. Evans
Partial Differential Equations, Second Edition

基于 Evans 原著的学习整理

$$\partial_t u - \Delta u = 0$$

LUMENBOOK

2026-11-09

澄明讲义 / 数学阅读札记

How to Use These Notes

These notes follow Lawrence C. Evans's *Partial Differential Equations*, second edition. The English PDF supplies the chapter order and mathematical source. The supplied Chinese translation is a reference for terminology and reading support.

English text and margin notes

Read the main argument first. Each Chinese margin note addresses a specific mathematical issue: the role of a hypothesis, the structure of an operator, a counterexample, a limiting case, or the connection to a later method. Headings name the mathematical issue. Longer calculations and proofs belong in the main text and are marked as editorial supplements.

边注的数学任务

边注应回答可检验的问题：某条件在哪里使用？删去后有什么反例？方程的主部如何决定方法？不以句式分析、常用词翻译或正文复述占据边栏。

Expanded proofs

An expanded proof will state the assumptions, explain the strategy, and fill in omitted calculations. Each use of approximation, integration by parts, compactness, or passage to a limit must identify the applicable hypotheses. Source theorem numbers will be recorded explicitly; the notes' own numbering does not automatically coincide with the original book.

Current status

Prefaces and Chapters 1–12 are annotated. Chapter 2 emphasizes harmonic and heat-ball means; Chapter 3 develops characteristics, Hopf–Lax and entropy solutions. Chapter 4 treats representation methods, asymptotics and analytic Cauchy theory. Chapter 5 develops Sobolev theory; Chapter 6 applies it to weak elliptic solutions, regularity, maximum principles and eigenvalues. Appendices A–E remain outlines. Source comparison, editorial checking, and independent review are recorded separately in the project documentation.

The introduction precedes Part I, as in the source. The three parts contain Chapters 2–4, 5–7, and 8–12. Appendices A–E supply background material. The bilingual glossary is an added study aid. Detailed subsection headings will be added when their source pages are studied.

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Preface to the second edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xi–xii; PDF pages 12–13. Chinese reference: PDF page 9.

原著序言全文；中文边注为整理者补充，英文印次信息按所提供的英文 PDF 保留。

Let me thank everyone who over the past decade has provided me with suggestions and corrections for improving the first edition of this book.

I am extraordinarily grateful. Although I have not always followed these many pieces of advice and criticism, I have thought carefully about them all. So many people have helped me out that it is unfortunately no longer feasible to list all their names. I have also received great help from everyone at the AMS, especially Sergei Gelfand, Stephen Moye and Arlene O'Sean. The NSF has generously supported my research during the writing of both the original edition of the book and this revision. I will continue to maintain lists of errors on my homepage, accessible through the `math.berkeley.edu` website.

When you write a big book on a big subject, the temptation is to include everything.

A critic famously once imagined Tolstoy during the writing of *War and Peace*: "The book is long, but even if it were twice as long, if it were three times as long, there would always be scenes that have been omitted, and these Tolstoy, waking up in the middle of the night, must have regretted. There must have been a night when it occurred to him that he had not included a yacht race…" (G. Moore, *Avowals*).

This image notwithstanding, I have tried to pack into this second edition as many fascinating new topics in partial differential equations (PDE) as I could manage, most notably in the new Chapter 12 on nonlinear wave equations. There are new sections on Noether's Theorem and on local minimizers in the calculus of variations, on the Radon transform, on Turing instabilities for reaction-diffusion systems, etc. I have rewritten and expanded the previous discussions on blow-up of solutions, on group and phase velocities, and on several further subjects. I have also updated and greatly increased citations to books in the bibliography and have moved

爆破与延拓判据

有限时间爆破必须说明失控的量：可能是解的幅度、梯度，或某个 Sobolev 范数。局部存在与全局存在之间的关键，是控制延拓定理所要求的范数。

references to research articles to within the text. There are countless further minor modifications in notation and wording. Most importantly, I have added about 80 new exercises, most quite interesting and some rather elaborate. There are now over 200 in total.

And there is a yacht race among the problems for Chapter 10.

LCE
January, 2010
Berkeley

This third printing of the second edition corrects all known mistakes, as of February, 2022.

印次差异

英文此处为 2022 年；中译本对应写 2015 年，并附译者注说明参考了 2020 年勘误。此处按英文 PDF 保留，不能混作同一印次。

Preface to the first edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xiii–xv; PDF pages 14–16. Chinese reference: PDF pages 11–12.

原著序言全文；中文边注为整理者补充。保留作者的写作原则、致谢和署名。

I present in this book a wide-ranging survey of many important topics in the theory of partial differential equations (PDE), with particular emphasis on various modern approaches.

I have made a huge number of editorial decisions about what to keep and what to toss out, and can only claim that this selection seems to me about right. I of course include the usual formulas for solutions of the usual linear PDE, but also devote large amounts of exposition to energy methods within Sobolev spaces, to the calculus of variations, to conservation laws, etc.

My general working principles in the writing have been these:

a. PDE theory is (mostly) not restricted to two independent variables. Many texts describe PDE as if functions of the two variables (x, y) or (x, t) were all that matter. This emphasis seems to me misleading, as modern discoveries concerning many types of equations, both linear and nonlinear, have allowed for the rigorous treatment of these in any number of dimensions. I also find it unsatisfactory to "classify" partial differential equations: this is possible in two variables, but creates the false impression that there is some kind of general and useful classification scheme available in general.

高维中的主部

二阶线性算子的主部由矩阵 $A = (a^{ij})$ 描述，其二次型 $\xi^T A \xi$ 决定椭圆性等结构。二维判别式只能覆盖这一框架的特殊情形，不能据此统一处理任意阶数、方程组和非线性。

b. Many interesting equations are nonlinear.

My view is that overall we know too much about linear PDE and too little about nonlinear PDE. I have accordingly introduced nonlinear concepts early in the text and have tried hard to emphasize everywhere nonlinear analogues of the linear theory.

c. Understanding generalized solutions is fundamental. Many of the partial differential equations we study, especially nonlinear first-order equations, do not in general possess smooth solutions. It is therefore essential to devise some kind of proper notion of generalized or weak solution. This is an important but subtle undertaking, and much of the hardest material in this book concerns the uniqueness of appropriately defined weak solutions.

广义解仍需选择原则

守恒律的分布弱形式容许不连续解，但单凭积分恒等式一般无法保证唯一性，还需熵条件。Hamilton–Jacobi 方程则采用黏性解；解的定义必须配合方程结构。

d. PDE theory is not a branch of functional analysis. Whereas certain classes of equations can profitably be viewed as generating abstract operators between Banach spaces, the insistence on an overly abstract viewpoint, and consequent ignoring of deep calculus and measure theoretic estimates, is ultimately limiting.

估计与抽象框架的分工

把 PDE 写成算子方程后，仍须证明适用条件：例如椭圆问题的强制性来自能量估计，非线性极限常需紧性或单调性。抽象存在定理不会自动提供这些针对具体方程的估计。

e. Notation is a nightmare.

I have really tried to introduce consistent notation, which works for all the important classes of equations studied. This attempt is sometimes at variance with notational conventions within a given subarea.

f. Good theory is (almost) as useful as exact formulas. I incorporate this principle into the overall organization of the text, which is subdivided into three parts, roughly mimicking the historical development of PDE theory itself. Part I concerns the search for explicit formulas for solutions, and Part II the abandoning of this quest in favor of general theory asserting the existence and other properties of solutions for linear equations. Part III is the mostly modern endeavor of fashioning general theory for important classes of nonlinear PDE.

估计提供可检验的结论

若能建立 $\|u-\tilde{u}\|_Y \le C\|g-\tilde{g}\|_X$，就同时得到唯一性与定量稳定性。这类结论即使没有显式公式，也能说明数据误差如何传到解。

Let me also explicitly comment here that I intend the development within each section to be rigorous and complete (exceptions being the frankly heuristic treatment of asymptotics in §4.5 and an occasional reference to a research paper). This means that even locally within each chapter the topics do not necessarily progress logically from "easy" to "hard" concepts. There are many difficult proofs and computations early on, but as compensation many easier ideas later. The student should certainly omit on first reading some of the more arcane proofs.

渐近展开的证明责任

形式上把幂级数代入 PDE 只能给出系数关系。要证明渐近公式有效，还须在指定区域和范数中估计余项；没有余项控制的形式展开不能作为解存在或误差界的证明。

I wish next to emphasize that this is a textbook, and not a reference book. I have tried everywhere to present the essential ideas in the clearest possible settings, and therefore have almost never established sharp versions of any of the theorems. Research articles and advanced monographs, many of them listed in the Bibliography, provide such precision and generality. My goal has rather been to explain, as best I can, the many fundamental ideas of the subject within fairly simple contexts.

I have greatly profited from the comments and thoughtful suggestions of many of my colleagues, friends and students, in particular: S. Antman,

J. Bang, X. Chen, A. Chorin, M. Christ, J. Cima, P. Colella, J. Cooper, M. Crandall, B. Driver, M. Feldman, M. Fitzpatrick, R. Gariepy, J. Goldstein, D. Gomes, O. Hald, W. Han, W. Hrusa, T. Ilmanen, I. Ishii, I. Israel, R. Jerrard, C. Jones, B. Kawohl, S. Koike, J. Lewis, T.-P. Liu, H. Lopes, J. McLaughlin, K. Miller, J. Morford, J. Neu, M. Portilheiro, J. Ralston, F. Rezakhanlou, W. Schlag, D. Serre, P. Souganidis, J. Strain, W. Strauss, M. Struwe, R. Temam, B. Tvedt, J.-L. Vazquez, M. Weinstein, P. Wolfe, and Y. Zheng.

I especially thank Tai-Ping Liu for many years ago writing out for me the first draft of what is now Chapter 11.

I am extremely grateful for the suggestions and lists of mistakes from earlier drafts of this book sent to me by many readers, and I encourage others to send me their comments, at `evans@math.berkeley.edu`. I have come to realize that I must be more than slightly mad to try to write a book of this length and complexity, but I am not yet crazy enough to think that I have made no mistakes. I will therefore maintain a listing of errors which come to light, and will make this accessible through the `math.berkeley.edu` homepage.

Faye Yeager at UC Berkeley has done a really magnificent job typing and updating these notes, and Jaya Nagendra heroically typed an earlier version at the University of Maryland. My deepest thanks to both.

I have been supported by the NSF during much of the writing, most recently under grant DMS-9424342.

LCE
August, 1997
Berkeley

Chapter 1

Introduction

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 1; printed pp. 1–13; PDF pages 18–30.

Chinese reference: 中译本：引言，书内第 1–9 页；PDF 第 19–27 页。

Editorial convention. The source exposition and all five source problems are retained below. Chinese margin notes and passages headed "Editorial supplement" are added by the editor. Only the two equations numbered (1) and (2) in the source carry those source numbers here; supplementary displays are unnumbered.

This chapter surveys the principal theoretical issues concerning the solving of partial differential equations.

To follow the subsequent discussion, the reader should first of all turn to Appendix A and look over the notation presented there, particularly the multiindex notation for partial derivatives.

1.1 Partial Differential Equations

A *partial differential equation* (PDE) is an equation involving an unknown function of two or more variables and certain of its partial derivatives.

Using the notation explained in Appendix A, we can write out symbolically a typical PDE, as follows. Fix an integer $k \geq 1$ and let U denote an open subset of $\mathbb{R}^n$.

Definition (source, §1.1). An expression of the form

$$F(D^k u(x), D^{k-1}u(x), \dots, Du(x), u(x), x) = 0 \quad (x \in U) \tag{1}$$

is called a *kth-order partial differential equation*, where

$$F : \mathbb{R}^{n^k} \times \mathbb{R}^{n^{k-1}} \times \cdots \times \mathbb{R}^n \times \mathbb{R} \times U \to \mathbb{R}$$

is given and $u : U \to \mathbb{R}$ is the unknown.

We solve the PDE if we find all u verifying (1), possibly only among those functions satisfying certain auxiliary boundary conditions on some part Γ of ∂U. By finding the solutions we mean, ideally, obtaining simple, explicit solutions, or, failing that, deducing the existence and other properties of solutions.

Definitions (source, §1.1).

(i) The partial differential equation (1) is called *linear* if it has the form

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

for given functions a_α $(|\alpha| \le k), f$. This linear PDE is *homogeneous* if $f \equiv 0$.

(ii) The PDE (1) is *semilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iii) The PDE (1) is *quasilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(D^{k-1}u, \dots, Du, u, x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iv) The PDE (1) is *fully nonlinear* if it depends nonlinearly upon the highest order derivatives.

导数数组与对称性

原书把 $D^k u$ 写成含 n^k 个分量的数组。若 $u \in C^k$，混合偏导可交换，独立分量只有 $\binom{n+k-1}{k}$ 个。前者按求导次序储存，后者按多重指标计数；两种记号并不矛盾。

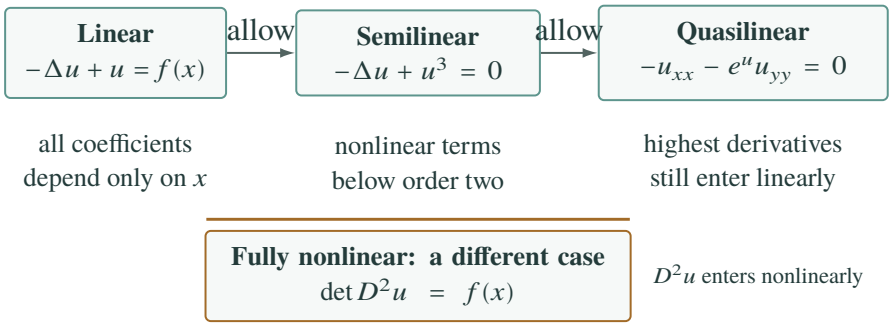


Diagram T1.1. Classify by the highest derivatives. The first three classes allow successively more dependence on lower-order variables. In the last example the highest derivatives themselves enter nonlinearly. All displayed examples are second order.

分类看最高阶导数

先找阶数 k，再看 $D^k u$ 如何出现。半线性：最高阶系数只依赖 x；拟线性：系数可依赖低阶导数。例如 $u_t + uu_x = 0$ 是一阶拟线性，而 $u_t + uu_x + u_{xxx} = 0$ 是三阶半线性。

A *system of partial differential equations* is, informally speaking, a collection of several PDE for several unknown functions.

Definition (source, §1.1). Let k be a positive integer. An expression of the form

$$\mathbf{F}(D^k\mathbf{u}(x), \dots, D\mathbf{u}(x), \mathbf{u}(x), x) = \mathbf{0} \quad (x \in U) \tag{2}$$

is called a *kth-order system of partial differential equations*, where

$$\mathbf{F} : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \cdots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times U \to \mathbb{R}^m$$

is given and

$$\mathbf{u} : U \to \mathbb{R}^m, \qquad \mathbf{u} = (u^1, \dots, u^m)$$

is the unknown.

Here we are supposing that the system comprises the same number m of scalar equations as unknowns $(u^1, \dots, u^m)$. This is the most common circumstance, although other systems may have fewer or more equations than unknowns. Systems are classified in the obvious way as being linear, semilinear, etc.

Notation. We write "PDE" as an abbreviation for both the singular "partial differential equation" and the plural "partial differential equations".

Editorial supplement: the multiindex convention

For $\alpha = (\alpha_1, \dots, \alpha_n) \in \{0, 1, 2, \dots\}^n$, set

$$|\alpha| = \sum_{i=1}^{n} \alpha_i, \qquad D^\alpha u = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} u.$$

The *principal part* of a linear operator of order k is $\sum_{|\alpha|=k} a_\alpha(x) D^\alpha$; the terms of smaller order are its lower-order terms. Here $D^0 u = u$. For example, when $n = 2$, $D^{(2,1)}u = u_{x_1 x_1 x_2}$ and $|(2,1)| = 3$.

The words linear, semilinear and quasilinear describe nested forms. When a single classification is requested, we report the most restrictive applicable form: linear first, then semilinear, then quasilinear. These definitions concern the displayed differential operator. An algebraic rewriting that changes its form need not preserve its classification.

等价零点不保结构

把 $Lu = 0$ 改写成 $(Lu)^3 = 0$，经典解集不变，但新表达式对最高阶导数非线性，且在线性化时可能退化。因此分类必须说明采用哪个算子表达式，不能只看方程的解集。

1.2 Examples

There is no general theory known concerning the solvability of all partial differential equations. Such a theory is extremely unlikely to exist, given the rich variety of physical, geometric, and probabilistic phenomena which can be modeled by PDE. Instead, research focuses on various particular partial differential equations that are important for applications within and outside of mathematics, with the hope that insight from the origins of these PDE can give clues as to their solutions.

Following is a list of many specific partial differential equations of interest in current research. This listing is intended merely to familiarize the reader with the names and forms of various famous PDE. To display most clearly the mathematical structure of these equations, we have mostly set relevant physical constants to unity. We will later discuss the origin and interpretation of many of these PDE.

Throughout $x \in U$, where U is an open subset of $\mathbb{R}^n$, and $t \geq 0$. Also $Du = D_x u = (u_{x_1}, \dots, u_{x_n})$ denotes the gradient of u with respect to the spatial variable $x = (x_1, \dots, x_n)$. The variable t always denotes time.

1.2.1 Single partial differential equations

a. Linear equations

1. Laplace's equation
$$\Delta u = \sum_{i=1}^{n} u_{x_i x_i} = 0.$$
2. Helmholtz's (or eigenvalue) equation
$$-\Delta u = \lambda u.$$
3. Linear transport equation
$$u_t + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$
4. Liouville's equation
$$u_t + \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

5. Heat (or diffusion) equation

$$u_t - \Delta u = 0.$$

6. Schrödinger's equation

$$\mathrm{i}u_t + \Delta u = 0.$$

7. Kolmogorov's equation

$$u_t - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

8. Fokker–Planck equation

$$u_t - \sum_{i,j=1}^{n} (a^{ij} u)_{x_i x_j} - \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

9. Wave equation

$$u_{tt} - \Delta u = 0.$$

10. Klein–Gordon equation

$$u_{tt} - \Delta u + m^2 u = 0.$$

11. Telegraph equation

$$u_{tt} + 2du_t - u_{xx} = 0.$$

12. General wave equation

$$u_{tt} - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

13. Airy's equation

$$u_t + u_{xxx} = 0.$$

14. Beam equation

$$u_{tt} + u_{xxxx} = 0.$$

b. Nonlinear equations

1. Eikonal equation

$$|Du| = 1.$$

2. Nonlinear Poisson equation

$$-\Delta u = f(u).$$

随流不变与质量守恒

输运方程沿 $\dot{X} = b(X,t)$ 保持 $u(X(t),t)$ 不变。Liouville 方程展开后多出 $(\operatorname{div} b)u$，描述密度随体积伸缩而变。只有 $\operatorname{div} b = 0$ 时两式一致。

算子与形式伴随

令 $L = a^{ij}\partial_{ij} - b^i\partial_i$（重复指标求和）。分部积分给出 $L^*\rho = \partial_{ij}(a^{ij}\rho) + \partial_i(b^i\rho)$。两式分别为 $u_t = Lu$ 与 $\rho_t = L^*\rho$，漂移项符号因此不同。

阶数与时间初值

热方程、波方程都是二阶 PDE，但前者时间一阶，后者时间二阶，通常分别需给 $u(0)$ 与 $(u(0), u_t(0))$。梁方程总阶数为四、时间阶数为二；总阶数不能直接决定初值个数。

3. p-Laplacian equation

$$\operatorname{div}(|Du|^{p-2}Du) = 0.$$

4. Minimal surface equation

$$\operatorname{div}\left(\frac{Du}{(1+|Du|^2)^{1/2}}\right) = 0.$$

5. Monge–Ampère equation

$$\det(D^2u) = f.$$

6. Hamilton–Jacobi equation

$$u_t + H(Du, x) = 0.$$

7. Scalar conservation law

$$u_t + \operatorname{div}\mathbf{F}(u) = 0.$$

8. Inviscid Burgers' equation

$$u_t + uu_x = 0.$$

9. Scalar reaction-diffusion equation

$$u_t - \Delta u = f(u).$$

10. Porous medium equation

$$u_t - \Delta(u^\gamma) = 0.$$

11. Nonlinear wave equation

$$u_{tt} - \Delta u + f(u) = 0.$$

12. Korteweg–deVries (KdV) equation

$$u_t + uu_x + u_{xxx} = 0.$$

13. Nonlinear Schrödinger equation

$$\mathrm{i}u_t + \Delta u = f(|u|^2)u.$$

退化与奇异

对 p-Laplacian，$p > 2$ 时主部在 $Du = 0$ 退化；$1 < p < 2$ 时其线性化系数在那里奇异。散度形式及相应弱形式仍有意义，不能把只在 $Du \neq 0$ 合法的展开直接用于临界点。

多孔介质的扩散系数

在 $u > 0$ 的光滑区，$\Delta(u^\gamma) = \gamma u^{\gamma-1}\Delta u + \gamma(\gamma-1)u^{\gamma-2}|Du|^2$。通常取 $\gamma > 1, u \geq 0$，主部在 $u = 0$ 退化；$0 < \gamma < 1$ 对应快扩散情形。

1.2.2 Systems of partial differential equations

a. Linear systems

1. Equilibrium equations of linear elasticity

$$\mu\Delta\mathbf{u} + (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

2. Evolution equations of linear elasticity

$$\mathbf{u}_{tt} - \mu\Delta\mathbf{u} - (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

3. Maxwell's equations

$$\begin{cases}\mathbf{E}_t = \operatorname{curl}\mathbf{B},\\ \mathbf{B}_t = -\operatorname{curl}\mathbf{E},\\ \operatorname{div}\mathbf{B} = \operatorname{div}\mathbf{E} = 0.\end{cases}$$

b. Nonlinear systems

1. System of conservation laws

$$\mathbf{u}_t + \operatorname{div}\mathbf{F}(\mathbf{u}) = \mathbf{0}.$$

2. Reaction-diffusion system

$$\mathbf{u}_t - \Delta\mathbf{u} = \mathbf{f}(\mathbf{u}).$$

3. Euler's equations for incompressible, inviscid flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

4. Navier–Stokes equations for incompressible, viscous flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} - \Delta\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

约束的传播

真空 Maxwell 系统中，$\partial_t \operatorname{div}\mathbf{E} = 0$、$\partial_t \operatorname{div}\mathbf{B} = 0$，因为散度作用于旋度为零。散度条件是初值必须满足且由演化保持的约束，不能视作任意额外数据。

See Zwillinger [Zw] for a much more extensive listing of interesting PDE.

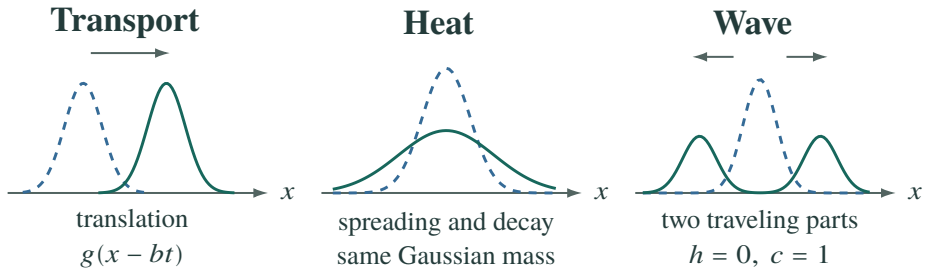


Diagram T1.2. Three evolution mechanisms. Dashed curves are initial profiles and solid curves are later profiles. Transport translates a shape; heat spreads a Gaussian while preserving its integral; the one-dimensional wave equation with zero initial velocity splits a profile into two half-amplitude translates. The panel axes are schematic and use separate scales.

Editorial supplement: two principal-part calculations

For $1 < p < \infty$ and a C^2 function at a point where $Du \neq 0$, the chain rule gives

$$\operatorname{div}(|Du|^{p-2}Du) = \sum_{i,j=1}^{n} A_{ij}(Du)u_{x_i x_j},$$

$$A(q) = |q|^{p-2}\left(I + (p-2)\frac{q \otimes q}{|q|^2}\right).$$

The eigenvalue in the direction of q is $(p-1)|q|^{p-2}$; in every orthogonal direction it is $|q|^{p-2}$. This proves ellipticity at $q \neq 0$ and explains its failure to be uniform as $|q| \to 0$ or $|q| \to \infty$ when $p \neq 2$. At $p = 2$ the equation is exactly Laplace's equation.

Similarly, the minimal surface equation expands as

$$\frac{1}{\sqrt{1 + |Du|^2}} \sum_{i,j=1}^{n} \left(\delta_{ij} - \frac{u_{x_i}u_{x_j}}{1 + |Du|^2}\right) u_{x_i x_j} = 0.$$

The matrix in parentheses has eigenvalues 1 transverse to Du and $(1 + |Du|^2)^{-1}$ parallel to Du. Thus it is positive definite for each finite gradient, and a bound on $|Du|$ supplies a uniform ellipticity bound. Both calculations exhibit quasilinear second-order equations.

压力是约束乘子

对光滑不可压流取散度，得到 $-\Delta p = \sum_{i,j}(\partial_i u^j)(\partial_j u^i)$。压力由速度及边界条件耦合确定；没有独立的压力演化方程。仅看速度的扩散项会漏掉这一非局部约束。

1.3 Strategies for Studying PDE

As explained in §1.1 our goal is the discovery of ways to solve partial differential equations of various sorts, but—as should now be clear in view of the many diverse examples set forth in §1.2—this is no easy task. And indeed the very question of what it means to "solve" a given PDE can be subtle, depending in large part on the particular structure of the problem at hand.

1.3.1 Well-posed problems, classical solutions

The informal notion of a well-posed problem captures many of the desirable features of what it means to solve a PDE. We say that a given problem for a partial differential equation is well-posed if

(i) the problem in fact has a solution;
(ii) this solution is unique; and
(iii) the solution depends continuously on the data given in the problem.

The last condition is particularly important for problems arising from physical applications: we would prefer that our (unique) solution changes only a little when the conditions specifying the problem change a little. (For many problems, on the other hand, uniqueness is not to be expected. In these cases the primary mathematical tasks are to classify and to characterize the solutions.)

Now clearly it would be desirable to "solve" PDE in such a way that (i)–(iii) hold. But notice that we still have not carefully defined what we mean by a "solution". Should we ask, for example, that a "solution" u must be real analytic or at least infinitely differentiable? This might be desirable, but perhaps we are asking too much. Maybe it would be wiser to require a solution of a PDE of order k to be at least k times continuously differentiable. Then at least all the derivatives which appear in the statement of the PDE will exist and be continuous, although maybe certain higher derivatives will not exist. Let us informally call a solution with this much smoothness a classical solution of the PDE: this is certainly the most obvious notion of solution.

So by solving a partial differential equation in the classical sense we mean if possible to write down a formula for a classical solution satisfying (i)–(iii) above, or at least to show such a solution exists, and to deduce various of its properties.

Monge–Ampère 的分支

当 $n \geq 2$，行列式非线性地依赖 Hessian。在 D^2u 正定的凸解分支上，线性化系数为正定的余子式矩阵。离开该分支，椭圆性没有保证；仅写 $\det D^2u = f$ 还不足以指定椭圆问题。

适定性依赖拓扑

连续依赖必须指定数据空间 X 和解空间 Y。常见目标是 $\|u-\tilde{u}\|_Y \leq C\|g-\tilde{g}\|_X$。同一方程更换范数、时间方向或边界条件后，适定性可能改变。

1.3.2 Weak solutions and regularity

But can we achieve this? The answer is that certain specific partial differential equations (e.g. Laplace's equation) can be solved in the classical sense, but many others, if not most others, cannot. Consider for instance the scalar conservation law

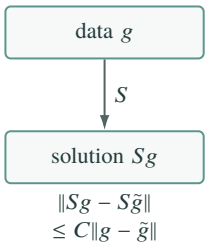
$$u_t + F(u)_x = 0.$$

We will see in §3.4 that this PDE governs various one-dimensional phenomena involving fluid dynamics, and in particular models the formation and propagation of shock waves. Now a shock wave is a curve of discontinuity of the solution u; and so if we wish to study conservation laws, and recover the underlying physics, we must surely allow for solutions u which are not continuously differentiable or even continuous. In general, as we shall see, the conservation law has no classical solutions but is well-posed if we allow for properly defined generalized or weak solutions.

This is all to say that we may be forced by the structure of the particular equation to abandon the search for smooth, classical solutions. We must instead, while still hoping to achieve the well-posedness conditions (i)–(iii), investigate a wider class of candidates for solutions. And in fact, even for those PDE which turn out to be classically solvable, it is often most expedient initially to search for some appropriate kind of weak solution.

The point is this: if from the outset we demand that our solutions be very regular, say k-times continuously differentiable, then we are usually going to have a really hard time finding them, as our proofs must then necessarily include possibly intricate demonstrations that the functions we are building are in fact smooth enough. A far more reasonable strategy is to consider as separate the existence and the smoothness (or regularity) problems. The idea is to define for a given PDE a reasonably wide notion of a weak solution, with the expectation that since we are not asking too much by way of smoothness of this weak solution, it may be easier to establish its existence, uniqueness, and continuous dependence on the given data. Thus, to repeat, it is often wise to aim at proving well-posedness in some appropriate class of weak or generalized solutions.

图 T1.M：连续依赖的方向



先说明数据与解各用什么范数，再检验解映射是否连续。存在、唯一和连续依赖是三个不同问题。

经典解可局部存在

“没有经典解”不能理解成任意初值、任意短时间都无解。光滑 Burgers 初值通常先产生局部经典解，再因特征线相交形成激波。失效的是一般情形下全局保持光滑的要求。

Now, as noted above, for various partial differential equations this is the best that can be done. For other equations we can hope that our weak solution may turn out after all to be smooth enough to qualify as a classical solution. This leads to the question of regularity of weak solutions. As we will see, it is often the case that the existence of weak solutions depends upon rather simple estimates plus ideas of functional analysis, whereas the regularity of the weak solutions, when true, usually rests upon many intricate calculus estimates.

弱形式与唯一性

分部积分把导数转移给光滑试验函数，降低对 u 的正则性要求。但守恒律的分布弱解通常不唯一；还需熵条件筛选物理解。“弱解存在”本身不等于问题已经适定。

Let me explicitly note here that once we are past Part I (Chapters 2–4), our efforts will be largely devoted to proving mathematically the existence of solutions to various sorts of partial differential equations, and not so much to deriving formulas for these solutions. This may seem wasted or misguided effort, but in fact mathematicians are like theologians: we regard existence as the prime attribute of what we study. But unlike theologians, we need not always rely upon faith alone.

1.3.3 Typical difficulties

Following are some vague but general principles, which may be useful to keep in mind:

(i) Nonlinear equations are more difficult than linear equations; and, indeed, the more the nonlinearity affects the higher derivatives, the more difficult the PDE is.
(ii) Higher-order PDE are more difficult than lower-order PDE.
(iii) Systems are harder than single equations.
(iv) Partial differential equations entailing many independent variables are harder than PDE entailing few independent variables.
(v) For most partial differential equations it is not possible to write out explicit formulas for solutions.

None of these assertions is without important exceptions.

Editorial supplement: estimates, instability, and shocks

The following three calculations make the preceding distinctions precise. They are editorial additions, not proofs stated in the source introduction.

1. A forward heat estimate. Let U be a bounded smooth domain, and let u, v be sufficiently smooth solutions of $u_t - \Delta u = 0$ with the same homogeneous Dirichlet boundary condition. For $w = u - v$, multiplication

by w and integration by parts give

$$\frac{1}{2}\frac{d}{dt}\int_U w^2\,dx + \int_U |Dw|^2\,dx = \int_{\partial U} w\,\partial_\nu w\,dS = 0.$$

Consequently, for every time within their common interval of existence,

$$\|u(t) - v(t)\|_{L^2(U)} \le \|u(0) - v(0)\|_{L^2(U)}.$$

This proves uniqueness in this smooth class and an L^2 continuous-dependence estimate. It does not prove existence. Extending the calculation to weak solutions requires a justification for the time derivative and the test function w; the formal identity alone is not that justification.

边界项决定估计

这里消失的是 $w\partial_\nu w$，依据为相同的齐次 Dirichlet 条件。若边值不同，就不能直接令边界项为零。同样的能量计算必须随边界条件调整，不能只从方程形式照抄结论。

2. Reversing time destroys this stability. On $U = (0, \pi)$ fix $T > 0$, with zero boundary values, and define

$$u_j(x,t) = e^{j^2(T-t)}\sin(jx), \qquad 0 \le t \le T.$$

Every u_j is a smooth forward heat solution. Its terminal and initial norms satisfy

$$\|u_j(T)\|_2 = \sqrt{\pi/2}, \qquad \|u_j(0)\|_2 = e^{j^2T}\sqrt{\pi/2}.$$

More decisively, $v_j = e^{-j^2T/2}u_j$ satisfies

$$\|v_j(T)\|_2 \longrightarrow 0, \qquad \|v_j(0)\|_2 \longrightarrow \infty.$$

Thus the map from terminal data to the initial solution cannot be continuous at zero in L^2, even when restricted to these smooth solvable data. No claim of nonexistence is needed for this instability argument.

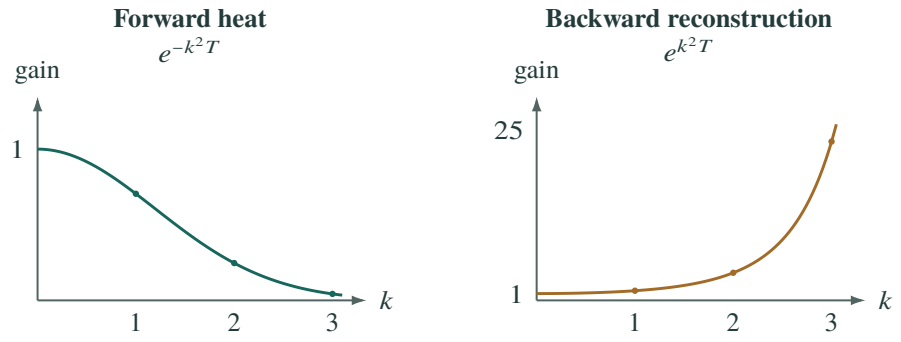


Diagram T1.3. High frequencies distinguish stability from solvability. For the Dirichlet mode $\sin(kx)$ on $(0, \pi)$, forward heat evolution multiplies amplitude by e^{-k^2T}; reconstruction multiplies by its reciprocal. Here $T = 0.35$ and vertical scales differ. Arbitrarily small terminal errors in high modes can therefore give large reconstructed initial errors.

3. Smooth data can create a Burgers singularity. For $u_t + uu_x = 0$ on the line, let $u(x,0) = g(x)$, where $g \in C^1(\mathbb{R})$ and both g and g' are bounded. Along a characteristic starting at ξ,

$$X(\xi,t) = \xi + tg(\xi), \qquad u(X(\xi,t),t) = g(\xi).$$

Before $\partial_\xi X = 1 + tg'(\xi)$ vanishes, inversion and the chain rule give

$$u_x(X(\xi,t),t) = \frac{g'(\xi)}{1+tg'(\xi)}.$$

If $g'(\xi_0) < 0$, this denominator vanishes at $t = -1/g'(\xi_0) > 0$, so the formula cannot continue as a C^1 solution through that event. If the negative minimum $m = \min_{\mathbb{R}} g'$ is attained, the first such time is $T_* = -1/m$. A bounded amplitude therefore does not prevent derivative blow-up.

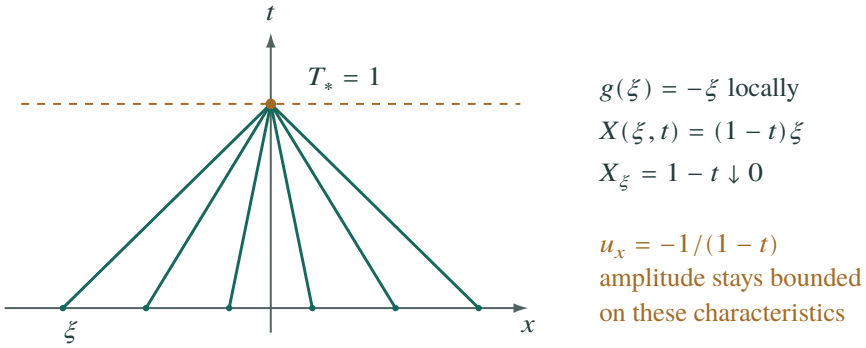


Diagram T1.4. Loss of invertibility precedes a shock. For the local initial profile $g(\xi) = -\xi$, the characteristics meet at $t = 1$. The characteristic formula is valid for $t < 1$; the meeting point depicts failure of a classical continuation, rather than several values of an admissible solution at one point.

To include a discontinuity, write $F(u) = u^2/2$. A locally bounded weak solution with initial data g must satisfy

$$\int_0^\infty \int_{\mathbb{R}} (u\varphi_t + F(u)\varphi_x)\,dx\,dt + \int_{\mathbb{R}} g(x)\varphi(x,0)\,dx = 0$$

for every smooth test function compactly supported in $\mathbb{R} \times [0,\infty)$. The initial term follows from time integration by parts; compact support removes the spatial boundary term.

For a piecewise constant state u_L to the left of $x = st$ and u_R to the right, the delta coefficient in $u_t + F(u)_x$ is $-s(u_R - u_L) + F(u_R) - F(u_L)$. It vanishes precisely when

$$s(u_R - u_L) = F(u_R) - F(u_L).$$

This is the Rankine–Hugoniot condition. For Burgers with $u_L \neq u_R$, it gives $s = (u_L + u_R)/2$. Yet it permits both compressive and expansive

爆破的量必须说明

Burgers 特征线上 $u = g(\xi)$ 始终有界，失控的是 u_x。这叫梯度爆破，不能误写成 $|u| \to \infty$。局部存在定理的延拓判据通常取决于某个特定范数，而不是笼统的“解变大”。

jumps. The entropy criterion selects a compressive shock, $u_L > u_R$, for which $u_R < s < u_L$; a jump with $u_L < u_R$ is not the entropy solution. This calculation explains why merely imposing the weak identity is insufficient. The general entropy theory belongs to the later chapters.

1.4 Overview

This textbook is divided into three major Parts.

Part I: Representation Formulas for Solutions

Here we identify those important partial differential equations for which in certain circumstances explicit or more-or-less explicit formulas can be had for solutions. The general progression of the exposition is from direct formulas for certain linear equations to far less concrete representation formulas, of a sort, for various nonlinear PDE.

Chapter 2 is a detailed study of four exactly solvable partial differential equations: the linear transport equation, Laplace's equation, the heat equation, and the wave equation. These PDE, which serve as archetypes for the more complicated equations introduced later, admit directly computable solutions, at least in the case that there is no domain whose boundary geometry complicates matters. The explicit formulas are augmented by various indirect, but easy and attractive, "energy"-type arguments, which serve as motivation for the developments in Chapters 6, 7 and thereafter.

Chapter 3 continues the theme of searching for explicit formulas, now for general first-order nonlinear PDE. The key insight is that such PDE can, locally at least, be transformed into systems of ordinary differential equations (ODE), the characteristic equations. We stipulate that once the problem becomes "only" the question of integrating a system of ODE, it is in principle solved, sometimes quite explicitly. The derivation of the characteristic equations given in the text is very simple and does not require any geometric insights. It is in truth so easy to derive the characteristic equations that no real purpose is had by dealing with the quasilinear case first.

We introduce also the Hopf–Lax formula for Hamilton–Jacobi equations (§3.3) and the Lax–Oleinik formula for scalar conservation laws (§3.4). (Some knowledge of measure theory is useful here but is not essential.) These sections provide an early acquaintance with the global theory of these important nonlinear PDE and so motivate the later Chapters 10 and 11.

Chapter 4 is a grab bag of techniques for explicitly (or kind of explicitly) solving various linear and nonlinear partial differential equations, and the reader should study only whatever seems interesting. The section on the Fourier transform is, however, essential. The Cauchy–Kovalevskaya Theorem appears at the very end. Although this is basically the only general existence theorem in the subject, and thus logically should perhaps be regarded as central, in practice these power series methods are not so prevalent.

局部存在不等于稳定

Cauchy–Kovalevskaya 定理要求解析系数、解析初值及非特征初始面，给出局部解析解。它不直接给出光滑或 Sobolev 数据下的连续依赖，也不保证全局解。

Part II: Theory for Linear Partial Differential Equations

Next we abandon the search for explicit formulas and instead rely on functional analysis and relatively easy "energy" estimates to prove the existence of weak solutions to various linear PDE. We investigate also the uniqueness and regularity of such solutions and deduce various other properties.

Chapter 5 is an introduction to Sobolev spaces, the proper setting for the study of many linear and nonlinear partial differential equations via energy methods. This is a hard chapter, the real worth of which is only later revealed, and requires some basic knowledge of Lebesgue measure theory. However, the requirements are not really so great, and the review in Appendix E should suffice. In my opinion there is no particular advantage in considering only the Sobolev spaces with exponent $p = 2$, and indeed insisting upon this obscures the two central inequalities, those of Gagliardo–Nirenberg–Sobolev (§5.6.1) and of Morrey (§5.6.2).

指数与维数

Sobolev 空间的指数不能随意固定。当 $1 \le p < n$，临界指数 $p^* = np/(n-p)$ 控制可积性增益；当 $p > n$，Morrey 估计给出 Hölder 连续性。维数在这里直接决定可得的正则性。

In Chapter 6 we vastly generalize our knowledge of Laplace's equation to other second-order elliptic equations. Here we work through a rather complete treatment of existence, uniqueness and regularity theory for solutions, including the maximum principle, and also a reasonable introduction to the study of eigenvalues, including a discussion of the principal eigenvalue for nonselfadjoint operators.

Chapter 7 expands the energy methods to a variety of linear partial differential equations characterizing evolutions in time. We broaden our earlier investigation of the heat equation to general second-order parabolic PDE and of the wave equation to general second-order hyperbolic PDE. We study as well linear first-order hyperbolic systems, with the aim of motivating the developments concerning nonlinear systems of conservation laws in Chapter 11. The concluding Section 7.4 presents the alternative functional analytic method of semigroups for building solutions.

(Missing from this long Part II on linear partial differential equations is any discussion of distribution theory or potential theory. These are

important topics, but for our purposes seem dispensable, even in a book of such length. These omissions do not slow us up much and make room for more nonlinear theory.)

Part III: Theory for Nonlinear Partial Differential Equations

This section parallels for nonlinear PDE the development in Part II but is far less unified in its approach, as the various types of nonlinearity must be treated in quite different ways.

Chapter 8 commences the general study of nonlinear partial differential equations with an extensive discussion of the calculus of variations. Here we set forth a careful derivation of the direct method for deducing the existence of minimizers and discuss also a variety of variational systems and constrained problems, as well as minimax methods. Variational theory is the most useful and accessible of the methods for nonlinear PDE, and so this chapter is fundamental.

Chapter 9 is, rather like Chapter 4 earlier, a gathering of assorted other techniques of use for nonlinear elliptic and parabolic partial differential equations. We encounter here monotonicity and fixed point methods and a variety of other devices, mostly involving the maximum principle. We study as well certain nice aspects of nonlinear semigroup theory, to complement the linear semigroup theory from Chapter 7.

不同的广义解机制

守恒律用积分恒等式表达守恒，再加熵不等式；Hamilton–Jacobi 方程的黏性解用接触试验函数表达微分不等式。两者的稳定性和唯一性机制不同，不能仅以“弱导数存在”统一定义。

Chapter 10 is an introduction to the modern theory of Hamilton–Jacobi PDE and in particular to the notion of “viscosity solutions”. We encounter also the connections with the optimal control of ODE, through dynamic programming.

Chapter 11 picks up from Chapter 3 the discussion of conservation laws, now systems of conservation laws. Unlike the general theoretical developments in Chapters 5–9, for which Sobolev spaces provide the proper abstract framework, we are forced to employ here direct linear algebra and calculus computations. We pay particular attention to the solution of Riemann’s problem and to entropy criteria.

Chapter 12, an introduction to nonlinear wave equations, is new with this edition. We provide long time and short time existence theorems for certain quasilinear wave equations and an in-depth examination of semilinear wave equations, especially for subcritical and critical power nonlinearities in three space dimensions. To complement these existence theorems, the final section identifies various criteria ensuring nonexistence of solutions.

Appendices A–E provide for the reader’s convenience some background material, with selected proofs, on inequalities, linear functional

analysis, measure theory, etc.

The Bibliography is an updated and extensive listing of interesting PDE books to consult for further information. Since this is a textbook and not a reference monograph, I have mostly not attempted to track down and document the original sources for the myriads of ideas and methods we will encounter. The mathematical literature for partial differential equations is truly vast, but the books cited in the Bibliography should at least provide a starting point for locating the primary sources. (Citations to selected research papers appear throughout the text.)

1.5 Problems

Source problems (Evans, Chapter 1, Problems 1–5)

The numbering in this list is the source problem numbering. Solutions in the next subsection are editorial additions.

1. Classify each of the partial differential equations in §1.2 as follows:
   (a) Is the PDE linear, semilinear, quasilinear or fully nonlinear?
   (b) What is the order of the PDE?
2. Let k be a positive integer. Show that a smooth function defined on $\mathbb{R}^n$ has in general
$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$$
distinct partial derivatives of order k.
(Hint: This is the number of ways of inserting $n-1$ dividers | within a row of k symbols $\circ$: for example, $\circ\circ \,||\, \circ\circ\circ \,|\, \circ \,|\, \circ\circ\circ \,||\, \circ\circ\circ\circ|$. Explain why each such pattern corresponds to precisely one of the partial derivatives of order k.)

The next exercises provide some practice with the multiindex notation introduced in Appendix A.

3. Prove the Multinomial Theorem:
$$(x_1 + \cdots + x_n)^k = \sum_{|\alpha|=k} \binom{|\alpha|}{\alpha} x^\alpha,$$
where $\binom{|\alpha|}{\alpha} := |\alpha|!/\alpha!$, $\alpha! = \alpha_1! \cdots \alpha_n!$, and $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. The sum is taken over all multiindices $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $|\alpha| = k$.
4. Prove Leibniz's formula:
$$D^\alpha(uv) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v,$$

where $u, v : \mathbb{R}^n \to \mathbb{R}$ are smooth, $\binom{\alpha}{\beta} := \alpha!/[\beta!(\alpha-\beta)!]$, and $\beta \le \alpha$ means $\beta_i \le \alpha_i$ $(i = 1, \dots, n)$.

5. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is smooth. Prove

$$f(x) = \sum_{|\alpha| \le k} \frac{1}{\alpha!} D^\alpha f(0) x^\alpha + O(|x|^{k+1}) \qquad \text{as } x \to 0$$

for each $k = 1, 2, \dots$. This is Taylor's formula in multiindex notation. (Hint: Fix $x \in \mathbb{R}^n$ and consider the function of one variable $g(t) := f(tx)$.)

Editorial supplement: solutions to Problems 1–5

Problem 1: classification. The following tables use the most restrictive applicable class, for generic nonlinearities. The order is the largest total derivative order in space and time. Coefficients and forcing terms are prescribed, not additional unknowns, unless explicitly stated otherwise.

Scalar linear equation (§1.2.1a)	Class	Order
1. Laplace	Linear	2
2. Helmholtz	Linear	2
3. Transport	Linear	1
4. Liouville	Linear	1
5. Heat	Linear	2
6. Schrödinger	Linear	2
7. Kolmogorov	Linear	2
8. Fokker–Planck	Linear	2
9. Wave	Linear	2
10. Klein–Gordon	Linear	2
11. Telegraph	Linear	2
12. General wave	Linear	2
13. Airy	Linear	3
14. Beam	Linear	4

Scalar nonlinear equation (§1.2.1b)	Class	Order
1. Eikonal	Fully nonlinear	1
2. Nonlinear Poisson	Semilinear	2
3. p-Laplacian	Quasilinear	2
4. Minimal surface	Quasilinear	2
5. Monge–Ampère	Fully nonlinear	2
6. Hamilton–Jacobi	Fully nonlinear	1
7. Scalar conservation law	Quasilinear	1
8. Inviscid Burgers	Quasilinear	1
9. Reaction-diffusion	Semilinear	2
10. Porous medium	Quasilinear	2
11. Nonlinear wave	Semilinear	2
12. KdV	Semilinear	3
13. Nonlinear Schrödinger	Semilinear	2

The p-Laplacian entry assumes $1 < p < \infty$, $p \neq 2$, using its classical expansion away from $Du = 0$ as in §1.2. The porous medium entry assumes $\gamma > 1$ and uses the expansion where $u > 0$. The Monge–Ampère entry assumes $n \geq 2$ and given f. If $H(q, x)$ is affine in q, Hamilton–Jacobi is quasilinear (in fact linear in this particular displayed form). If $f(u)$ is affine, the nonlinear Poisson, reaction-diffusion and nonlinear wave equations become linear. For the Schrödinger equations the unknown is complex-valued; one may equivalently use its real and imaginary parts as a real system.

先约定参数与分支

表中按一般非线性列出。$p = 2$ 时 p-Laplacian 变为线性；$\gamma = 1$ 时多孔介质式变为热方程；$n = 1$ 且 f 已知时 Monge–Ampère 式是 $u_{xx} = f$。忽略这些例外会把分类误当成方程名称的固定属性。

System (§1.2.2)	Class	Order
a1. Elastic equilibrium	Linear	2
a2. Elastic evolution	Linear	2
a3. Maxwell	Linear	1
b1. Conservation laws	Quasilinear	1
b2. Reaction-diffusion	Semilinear	2
b3. Incompressible Euler	Quasilinear	1
b4. Incompressible Navier–Stokes	Semilinear	2

For a smooth scalar flux, the chain rule gives

$$\operatorname{div} \mathbf{F}(u) = \sum_i (F^i)'(u) u_{x_i},$$

hence its quasilinear form. For a system with m components, $\mathbf{F} : \mathbb{R}^m \to \mathbb{R}^{m \times n}$ and $(\operatorname{div} \mathbf{F}(\mathbf{u}))^a = \sum_i \partial_{x_i} F^{ai}(\mathbf{u})$. Affine fluxes give linear equations or systems.

For the incompressible flow entries, the unknowns are $(\mathbf{u}, p)$. Navier–Stokes has constant coefficients on the second derivatives of $\mathbf{u}$; its nonlinear convection term has order one. This is the sense of the semilinear classification in the table. The divergence equation has order one, and there

are no second derivatives of p in the displayed system. It is a system of mixed differential orders, not an unconstrained uniformly parabolic system for all $n+1$ unknowns. Euler has first-order derivatives with coefficients depending on $\mathbf{u}$, hence is quasilinear.

Problem 2: counting independent derivative operators.

Expanded proof (editorial solution to source Problem 2).

For a C^k function, the order of differentiation does not matter. Thus a distinct derivative operator of order k is determined by a multiindex $\alpha \in \{0,1,2,\ldots\}^n$ satisfying $\alpha_1 + \cdots + \alpha_n = k$. Represent α_i by α_i circles in the ith compartment between $n-1$ dividers. Empty compartments are allowed: they encode $\alpha_i = 0$. This is a bijection.

Among the $k+n-1$ positions, choose the $n-1$ divider positions, giving $\binom{k+n-1}{n-1} = \binom{k+n-1}{k}$. For $n = 1$ there are no dividers and exactly one derivative, as the same formula predicts. This counts operators; special functions may have several derivatives with equal values. □

Problem 3: the multinomial theorem.

Expanded proof (editorial solution to source Problem 3).

Expand the product of k copies of $(x_1 + \cdots + x_n)$ by distributivity. A term x^α is obtained by selecting x_i from exactly α_i of the factors. First choose α_1 of the k factors, then α_2 of the remaining factors, and so on. The number of such choices is

$$\binom{k}{\alpha_1}\binom{k-\alpha_1}{\alpha_2}\cdots\binom{\alpha_n}{\alpha_n} = \frac{k!}{\alpha_1!\cdots\alpha_n!}.$$

Grouping all terms with the same multiindex proves the formula. The identity also holds for $k = 0$, with the single multiindex $\alpha = 0$. □

两类组合系数

多项式展开中 $\binom{k}{\alpha} = k!/\alpha!$，要求 $|\alpha| = k$。乘积求导中 $\binom{\alpha}{\beta} = \prod_i \binom{\alpha_i}{\beta_i}$，逐坐标分配导数。后者一般不等于 $\binom{|\alpha|}{|\beta|}$。

Problem 4: the multiindex product rule.

Expanded proof (editorial solution to source Problem 4).

Apply the one-variable product rule α_i times in coordinate x_i, for each $i = 1,\ldots,n$. If exactly β_i of these derivatives fall on u, the remaining $\alpha_i - \beta_i$ fall on v, and the coefficient is $\binom{\alpha_i}{\beta_i}$. Since mixed derivatives commute under the smoothness assumption, the resulting term is $D^\beta u\, D^{\alpha-\beta} v$, with coefficient

$$\prod_{i=1}^{n}\binom{\alpha_i}{\beta_i} = \frac{\alpha!}{\beta!(\alpha-\beta)!}.$$

Summing over all coordinatewise choices $0 \le \beta_i \le \alpha_i$ proves the identity. For $\alpha = 0$ it reduces to $uv = uv$. □

Problem 5: Taylor's formula with a uniform remainder.

Expanded proof (editorial solution to source Problem 5).

It suffices that $f \in C^{k+1}$ on a neighborhood of 0. Choose $r > 0$ so the closed ball $\overline{B}_r(0)$ lies inside this neighborhood. For $|x| < r$ set $g_x(t) = f(tx)$, $0 \le t \le 1$. Repeated differentiation and the multinomial theorem give, for $0 \le j \le k+1$,

$$g_x^{(j)}(t) = \left(\sum_{i=1}^{n} x_i \partial_{x_i}\right)^j f(tx) = \sum_{|\alpha|=j} \frac{j!}{\alpha!} x^\alpha D^\alpha f(tx).$$

The one-variable Taylor formula with integral remainder gives

$$f(x) = \sum_{j=0}^{k} \frac{g_x^{(j)}(0)}{j!} + R_k(x),$$
$$R_k(x) = \frac{1}{k!} \int_0^1 (1-t)^k g_x^{(k+1)}(t)\, dt.$$

Substituting the derivative formula converts the finite sum into

$$\sum_{|\alpha|\le k} \frac{D^\alpha f(0)}{\alpha!} x^\alpha.$$

Continuity and compactness give a finite common bound

$$M = \max_{\substack{|\alpha|=k+1 \\ y\in\overline{B}_r(0)}} |D^\alpha f(y)|.$$

Using $|x^\alpha| \le |x|^{k+1}$ and $\int_0^1 (1-t)^k\, dt = 1/(k+1)$, we obtain

$$|R_k(x)| \le M \left(\sum_{|\alpha|=k+1} \frac{1}{\alpha!}\right) |x|^{k+1}.$$

The constant depends only on n, k and the indicated local derivative bound, not on the direction of x. This proves the required $O(|x|^{k+1})$. □

1.6 References

Klainerman's article [Kl] is a nice modern overview of the field of partial differential equations.

Good general texts and monographs on PDE include Arnold [Ar2], Courant–Hilbert [C-H], Craig [Cr], DiBenedetto [DB1], Folland [F1], Friedman [Fr2], Garabedian [G], John [J2], Jost [Jo], Levy–Shearer [L-S] McOwen [MO], Mikhailov [M], Olver [O2] Petrovsky [Py], Rauch [R1], Renardy–Rogers [R-R], Showalter [Sh], Smirnov [Sm], Smoller [S], Strauss [St2], Taylor [Ta], Thoe–Zachmanoglou [T-Z], Vasy [Vy], Zauderer [Za], and many others. The prefaces to Arnold [Ar2] and to Bernstein [Bt] are particularly interesting reading. Zwillinger's handbook [Zw] on differential equations is a useful compendium of methods for PDE.

Editorial source note. The bracketed keys above are Evans's bibliography identifiers, retained as source identifiers rather than local citation numbers. Their full entries are in the supplied English source, printed p. 689 onward (PDF page 706 onward). This list records the supplied printing; it is not an updated literature survey. The supplied Chinese printing has a shorter list, omitting, for example, Craig [Cr], Jost [Jo], Levy–Shearer [L-S], Olver [O2], and Vasy [Vy]. The English printing controls the transcription here.

余项须对方向一致

逐条射线使用一元 Taylor 公式还不够：余项常数可能依赖方向。在固定闭球上统一控制所有 $k+1$ 阶导数，才得到真正的多元 $O(|x|^{k+1})$。仅假设 C^k 通常只能得到 $o(|x|^k)$。

Part I

Representation Formulas for Solutions

Chapter 2

Four Important Linear Partial Differential Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 2; printed pp. 17–87; PDF pages 34–104.

Chinese reference: 中译本：书内第 13 页起；球均值见第 18–19 页，热球均值见第 35–37 页。

Editorial study edition. The exposition and proofs are reconstructed in the order of Evans, Chapter 2; they are not a verbatim transcription. Source theorem numbers are identified separately from our chapter-wide numbering. Chinese mathematical notes, worked examples, figures, and exercise solutions are editorial additions.

Our main route is

$$\text{fundamental solutions} \longrightarrow \text{representation formulas} \longrightarrow \text{estimates and uniqueness.}$$

The central comparison is between harmonic means (§2.2.2) and the heat-ball mean (§2.3.2). The latter is a positive weighted average over past space-time, not a uniform average over an ordinary ball.

四个模型四种机制

传输沿特征线搬运；Laplace 方程描述静态平衡；热方程耗散并向未来扩散；波动方程保存能量并有限速传播。相同的“平均”符号背后可能是完全不同的测度。

Conventions and prerequisites. $\omega_n = |B_1|$ and $|\partial B_1| = n\omega_n$. The elliptic operator is $-\Delta$; the heat operator is $L = \partial_t - \Delta_x$; the wave speed is one unless stated otherwise. $C^{2,1}$ means two continuous spatial derivatives and one continuous time derivative. Basic multivariable calculus, Green's identities, polar integration, dominated convergence, convolution, and smooth cutoffs are prerequisites. Boundary correctors on general domains are explicitly conditional on their existence; ball and half-space kernels are constructed here.

2.1 Transport Equation

2.1.1 Characteristics and the initial-value problem (Evans §2.1.1)

For fixed $b \in \mathbb{R}^n$, consider $u_t + b \cdot Du = 0$, $u(x,0) = g(x)$. Through (x,t) trace the line

$$X(s) = x + (s-t)b, \qquad 0 \le s \le t.$$

For a C^1 solution, the chain rule gives

$$\frac{d}{ds} u(X(s),s) = Du(X(s),s) \cdot b + u_t(X(s),s) = 0.$$

Integrating from 0 to t proves necessity of

$$\boxed{u(x,t) = g(x-tb).}$$

Conversely, if $g \in C^1$, differentiation of this formula yields $u_t = -b \cdot Dg(x-tb)$ and $Du = Dg(x-tb)$, proving the PDE and initial condition. This establishes both existence and uniqueness in the classical class, rather than merely guessing a candidate.

回溯特征线

计算当前点 (x,t) 的值，要沿速度 b 回到初始平面，落点是 $x-tb$。正负号可用一维脉冲向右移动检验：$g(x-bt)$ 的峰位于 $x = bt$。

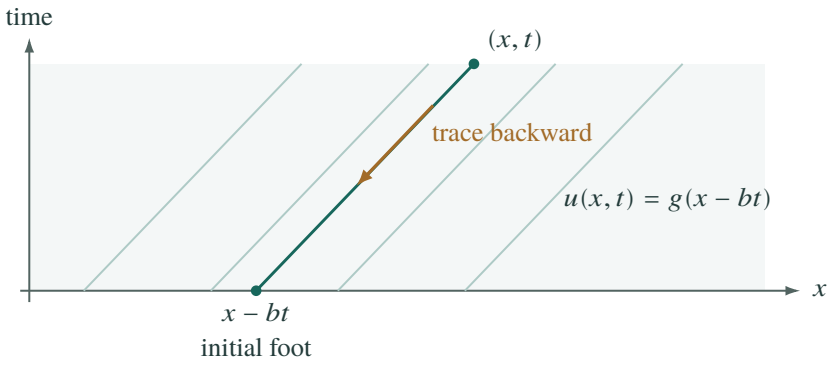


Diagram T2.1. Read a transport solution backward. The solution is constant on each characteristic when the source is zero. With $b > 0$, the forward line tilts right, while evaluation at a target requires its leftward initial foot $x - bt$. The orange arrow indicates evaluation backward in time, not a negative physical velocity.

Nonsmooth data and weak solutions. For $g \in L^1_{\mathrm{loc}}$, the same formula is locally integrable. For a compactly supported smooth test function on $\mathbb{R}^n \times [0,\infty)$, change variables $x = z + tb$:

$$\begin{aligned}\int_0^\infty \int u(\phi_t + b \cdot D\phi)\,dx\,dt &= \int_{\mathbb{R}^n} g(z) \int_0^\infty \frac{d}{dt}\phi(z+tb,t)\,dt\,dz \\ &= -\int_{\mathbb{R}^n} g(z)\phi(z,0)\,dz.\end{aligned}$$

Fubini is justified by compact test support and local integrability. This is the distributional formulation with the initial datum included. There is no need to assign a classical derivative at a moving jump. Translations preserve L^p norms: $\|u(\cdot,t)\|_p = \|g\|_p$ whenever $g \in L^p$.

2.1.2 Forcing and Duhamel's principle (Evans §2.1.2)

If $u_t + b \cdot Du = f$ and $u(x,0) = g(x)$, then

$$\frac{d}{ds}u(X(s),s) = f(X(s),s).$$

Integration along the same characteristic gives

$$\boxed{u(x,t) = g(x - tb) + \int_0^t f(x - (t-s)b, s)\, ds.}$$

For C^1 data differentiate under the integral, with the upper endpoint contributing $f(x,t)$; the remaining integrals cancel between u_t and $b \cdot Du$. This directly verifies the formula.

For the additional constant term cu, multiply along characteristics by e^{cs}:

$$u(x,t) = e^{-ct}g(x - tb) + \int_0^t e^{-c(t-s)}f(x - (t-s)b, s)\, ds.$$

The damping factor belongs to the elapsed time $t - s$, not to s alone.

源项沿轨迹累积

在时间 s 注入的源，之后还要平移 $(t-s)b$ 才到达当前点。Duhamel 的共同思想是把每个时刻的小冲量，按齐次方程传播后相加。

2.2 Laplace's Equation

2.2.1 Fundamental solution (Evans §2.2.1, Theorem 1)

A radial function $v(x) = F(r)$, $r = |x| > 0$, satisfies

$$\Delta v = F''(r) + \frac{n-1}{r}F'(r).$$

Thus harmonicity off zero gives $(r^{n-1}F')' = 0$. Integration yields a constant plus r^{2-n} if $n \geq 3$, or a constant plus $\log r$ if $n = 2$. Choose the constants so that the inward flux is one:

$$\Phi(x) = \begin{cases} [n(n-2)\omega_n]^{-1}|x|^{2-n}, & n \geq 3, \\ -(2\pi)^{-1}\log|x|, & n = 2. \end{cases} \qquad -\Delta\Phi = \delta_0.$$

符号由通量固定

我们求解 $-\Delta u = f$，故要求 $-\int_{\partial B_r}\partial_\nu\Phi = 1$。若改用 Δ 作基本算子，基本解及 Green 公式的符号也要一起改。

Expanded proof (the distributional singularity).

The displayed function is locally integrable and harmonic away from zero, and $\partial_r\Phi = -1/(n\omega_n r^{n-1})$. For $\phi \in C_c^\infty$, apply Green's identity on a large ball containing its support, with $\overline{B}_\varepsilon$ removed:

$$\int_{|x|>\varepsilon} \Phi(-\Delta\phi) = \int_{\partial B_\varepsilon} (\Phi\partial_r\phi - \phi\partial_r\Phi)\, dS.$$

The first term tends to zero: it is $O(\varepsilon)$ for $n \geq 3$, and $O(\varepsilon|\log\varepsilon|)$ for $n = 2$. The second is the normalized spherical mean of ϕ, tending to $\phi(0)$. Local integrability permits passage in the volume integral. Hence $-\Delta\Phi = \delta_0$ as a distribution. □

For $f \in C_c^2(\mathbb{R}^n)$ put $u = \Phi * f$. Write $u(x) = \int \Phi(y) f(x-y)\,dy$ and differentiate f, not the singular kernel. On each compact set of x, all these integrals have common bounded support in y and are dominated by an integrable multiple of $|\Phi|$. Thus $u \in C^2$ and

$$-\Delta u(x) = \int \Phi(y)\,(-\Delta_y f(x-y))\,dy = f(x)$$

by the distribution identity just proved. This avoids an illegal unqualified differentiation of $D^2\Phi$ at zero. For $n = 1$, $\Phi(x) = -|x|/2$ has the same normalization.

2.2.2 Mean-value formulas (Evans §2.2.2, Theorems 2–3)

Write $\omega_n = |B_1|$ and $\sigma_{n-1} = |\partial B_1| = n\omega_n$. For $\overline{B}_r(x) \subset U$ define the *normalized* means

$$S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u\,dS, \qquad A_u(x,r) = \frac{1}{\omega_n r^n} \int_{B_r(x)} u\,dy.$$

The sphere is a boundary of dimension $n-1$; the ball has dimension n. These are two different averaging measures.

平均不是积分本身

球面平均除以面积，球体平均除以体积。对常数 $u = 1$，两者都必须等于 1；漏掉 r^{n-1} 或 r^n 就会破坏这一检验。

Theorem 2.1 (Mean-value identities)
If $u \in C^2(U)$ and $\Delta u = 0$, then

$$u(x) = S_u(x,r) = A_u(x,r) \quad \text{whenever } \overline{B}_r(x) \subset U.$$

More generally, for any C^2 function,

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u(y)\,dy. \tag{M}$$

Expanded proof (all radius and surface factors).

1. Remove the moving surface before differentiating. Set $y = x + r\theta$ on the unit sphere. Its surface Jacobian is r^{n-1}, so

$$S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} u(x + r\theta)\,dS_\theta.$$

On a fixed compact range of radii, Du is continuous and bounded. Differentiation under this fixed finite integral is legitimate:

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n} \int_{\partial B_1} Du(x + r\theta)\cdot\theta\,dS_\theta.$$

Transforming back, $\theta = \nu(y)$ is the outward unit normal, giving

$$\partial_r S_u(x,r) = \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} Du\cdot\nu\,dS.$$

The divergence theorem on the smooth ball changes the flux into $\int_{B_r(x)} \Delta u$. This proves (M).

2. Identify the constant. If $\Delta u = 0$, the derivative in (M) vanishes for every admissible r. Hence the spherical mean is independent of radius. Continuity gives

$$|S_u(x,r) - u(x)| \le \sup_{|y-x|\le r} |u(y) - u(x)| \longrightarrow 0.$$

The constant is therefore exactly $u(x)$, not an unspecified constant.

3. Recover the ball average by adding spherical shells. Polar integration yields

$$\int_{B_r(x)} u = \int_0^r \int_{\partial B_\rho(x)} u \, dS \, d\rho = \int_0^r n\omega_n \rho^{n-1} u(x) \, d\rho = \omega_n r^n u(x).$$

Divide by the volume. This also proves, for general continuous u,

$$A_u(x,r) = \frac{n}{r^n} \int_0^r \rho^{n-1} S_u(x,\rho) \, d\rho. \tag{SB}$$

□

Can one recover spheres from balls? Yes. Differentiating $r^n A_u(x,r)$ in (SB) gives

$$S_u(x,r) = A_u(x,r) + \frac{r}{n} \partial_r A_u(x,r).$$

Consequently a continuous function with the ball-mean identity for every small radius also has the sphere-mean identity. A single radius at a single center is not sufficient.

Theorem 2.2 (Converse, including a continuous version)
If $u \in C^2(U)$ has either mean identity for every sufficiently small ball at each center, then $\Delta u = 0$. If $u \in C(U)$ has this property at every center and all locally admissible small radii, it is smooth and harmonic.

Expanded proof.

For C^2 functions Taylor-expand at x with a uniform $o(r^2)$ error. Symmetry of the sphere cancels linear and mixed terms; its normalized second moments are $\mathbb{E}(\theta_i\theta_j) = \delta_{ij}/n$. The ball has moments $\mathbb{E}(z_i z_j) = r^2\delta_{ij}/(n+2)$. Thus

$$S_u(x,r) = u(x) + \frac{r^2}{2n}\Delta u(x) + o(r^2),$$

$$A_u(x,r) = u(x) + \frac{r^2}{2(n+2)}\Delta u(x) + o(r^2). \tag{T}$$

图 T2.M：球体由球壳相加



shell volume
$n\omega_n \rho^{n-1}\, d\rho$

先对每个半径做球面平均，再乘球壳体积并积分。体积平均的径向权重不是均匀的。

方程如何变成平均

关键中介是通量：$\Delta u = 0$ 令每个球的总法向通量为零，通量又恰好是球面平均对半径的导数。故平均不随尺度变化。

An exact identity for arbitrarily small r forces $\Delta u(x) = 0$.

For a merely continuous u, first turn ball means into sphere means using (SB), if necessary. Take a radial smooth mollifier of mass one, supported in B_ε. In polar coordinates,

$$(\rho_\varepsilon * u)(x) = \int_0^\varepsilon \rho_\varepsilon(r) \int_{\partial B_r(x)} u \, dS \, dr = u(x).$$

The convolution is smooth on the smaller domain; hence u is smooth there and throughout U. Now apply the C^2 converse. This proves the smoothness assertion without differentiating an initially nonsmooth function. □

Subharmonic and superharmonic functions. If $\Delta v \geq 0$, (M) implies $S_v(x,r) \geq v(x)$ and then $A_v(x,r) \geq v(x)$. If $\Delta v \leq 0$, reverse both inequalities. For $-\Delta u = f$, integrating (M) and changing the order of integration gives, for $n \geq 3$,

$$u(x) = S_u(x,r) + \int_{B_r(x)} \frac{|y-x|^{2-n} - r^{2-n}}{n(n-2)\omega_n} f(y)\, dy. \tag{P}$$

Indeed $\int_{|y-x|}^r \rho^{1-n}\, d\rho = (|y-x|^{2-n} - r^{2-n})/(n-2)$. For $n = 2$ replace the kernel by $(2\pi)^{-1}\log(r/|y-x|)$. The kernel is nonnegative: a positive source raises $u(x)$ above its boundary average under the convention $-\Delta u = f$.

Concrete checks. In two dimensions, $u(y) = y_1^2 - y_2^2$ and $u(y) = y_1 y_2$ have $\Delta u = 0$, so both means reproduce $u(x)$ at every center. For $v(y) = |y|^2$,

$$S_v(0,r) = r^2, \qquad A_v(0,r) = \frac{n}{n+2} r^2, \qquad v(0) = 0.$$

For $n = 1$, the sphere consists of two endpoints: $S_u(x,r) = [u(x-r) + u(x+r)]/2$. Harmonic functions are affine, and the same identities remain valid with $\omega_1 = 2$.

2.2.3 Consequences of averaging (Evans §2.2.3)

Theorem 2.3 (Strong maximum principle; source Theorems 4–5)
If U is bounded and $u \in C^2(U) \cap C(\overline{U})$ is harmonic, its maximum and minimum are attained on ∂U. If U is connected and a maximum is attained inside, u is constant. Consequently the classical Dirichlet problem $-\Delta u = f$, $u = g$ has at most one solution in this class.

二阶误差读出 Laplace

普通函数的平均一般不等于中心值；首个对称性允许的误差就是 $r^2\Delta u$。调和性让这一误差及全部更高阶平均误差精确抵消。

别混同一元中值定理

这里是积分平均精确重现中心值，不是声称“存在某个中间点”。后面的热球公式也保持中心值，但必须改变平均的几何区域和权重。

Expanded proof.

Let $M = \max_{\overline{U}} u$ and suppose $u(x) = M$ at an interior point. The ball identity gives $\int_{B_r(x)} (M - u) = 0$. The integrand is nonnegative and continuous. If it were positive at one point, continuity would give a positive integral on a smaller ball, a contradiction. Thus $u = M$ throughout $B_r(x)$. The set $\{u = M\}$ is open and relatively closed in U. On a connected component containing an interior maximum it is the whole component, and continuity transfers this value to that component's boundary, which lies in ∂U. This proves the boundary maximum statement even if U has several components. Apply the argument to $-u$ for minima. The difference of two Dirichlet solutions is harmonic with zero boundary values, so its maximum and minimum are both zero. □

正权平均的等号条件

平均值达到所有被平均数的上界，意味着它们几乎处处都达到该上界；连续性再把“几乎处处”提升为处处。热球的强最大值原理使用同一机制。

Smoothness and derivative estimates (source Theorems 6–7). Radial convolution equals u by the mean identity, as proved in §2.2.2, so harmonic functions are C^∞. Fix a radial mollifier ρ with support in B_1. For $B_r(x) \Subset U$, use a convolution radius $r/2$ near x:

$$|D^\alpha u(x)| = \left| \int D^\alpha \rho_{r/2}(x - y) u(y)\, dy \right| \le C_{\alpha,n} r^{-n-|\alpha|} \|u\|_{L^1(B_r(x))}. \tag{D}$$

All derivatives land on the smooth kernel; its scaling explains the power of r. With a first derivative the same argument gives $\|Du\|_{L^\infty(B_{R-\delta})} \le C_n \delta^{-1} \|u\|_{L^\infty(B_R)}$.

> **Theorem 2.4** (Liouville and bounded Poisson representation; source 8–9)
> *A bounded harmonic function on $\mathbb{R}^n$ is constant. For $n \ge 3$ and $f \in C_c^2$, every bounded solution of $-\Delta u = f$ is $\Phi * f + C$.*

Expanded proof.

Apply (D) for $|\alpha| = 1$ on $B_R(x)$: $|Du(x)| \le C_n R^{-1} \|u\|_\infty \to 0$ as $R \to \infty$. Thus $Du = 0$ and connectedness makes u constant. For $n \ge 3$, $\Phi * f$ is continuous and tends to zero at infinity, hence bounded. Subtract it from u and apply Liouville. For $n = 2$, the logarithmic potential need not be bounded at infinity, so that final step cannot be asserted with the same hypotheses. □

> **Theorem 2.5** (Analyticity; source Theorem 10)
> *Harmonic functions are real analytic in the interior.*

Expanded proof (quantitative iteration rather than smoothness alone).

Fix $B_R(x_0) \Subset U$, with $M = \|u\|_{L^\infty(B_R)}$. Every derivative of u is harmonic, because derivatives commute with Δ. Iterate the first-derivative

estimate k times on concentric balls whose successive radii differ by $R/(2k)$. For any $|\alpha| = k$ this gives

$$\|D^\alpha u\|_{L^\infty(B_{R/2})} \le M(2C_n k/R)^k \le M(2eC_n/R)^k k!.$$

The last inequality follows from $\log(k!) = \sum_{j=1}^{k} \log j \ge \int_1^k \log s\, ds \ge k\log k - k$. For h small enough that the segment from x_0 to $x_0 + h$ stays in $B_{R/2}$, apply one-dimensional Taylor's theorem to $q(t) = u(x_0 + th)$. Its kth derivative is

$$q^{(k)}(t) = \sum_{|\alpha|=k} \frac{k!}{\alpha!} h^\alpha D^\alpha u(x_0 + th).$$

Dividing the remainder by $k!$ and using $\sum_{|\alpha|=k} |h^\alpha|/\alpha! = |h|_1^k/k!$ bounds it by $M(2eC_n|h|_1/R)^k$, tending to zero for sufficiently small h. Thus the multivariable Taylor series equals u near x_0. □

Theorem 2.6 (Harnack; source Theorem 11)
For connected $V \Subset U$ and nonnegative harmonic u, $\sup_V u \le C_{U,V,n} \inf_V u$.

Expanded proof.

Choose $\rho > 0$ with every $B_{2\rho}(z)$, $z \in V$, inside U. If $x, y \in V$ and $|x - y| \le \rho$, then $B_\rho(y) \subset B_{2\rho}(x)$ and positivity gives

$$u(x) = \frac{1}{|B_{2\rho}|} \int_{B_{2\rho}(x)} u \ge 2^{-n} \frac{1}{|B_\rho|} \int_{B_\rho(y)} u = 2^{-n} u(y).$$

Exchange x, y for the other inequality. Compactness gives a finite $\rho/4$ net of V. Connectedness permits a chain between any two net points with successive centers separated by at most ρ; one may obtain this by covering a path in V and deleting loops in the finite cover graph. There is a uniform finite bound on the chain length. Multiplying the local factors proves the global comparison. If one value is zero, the same inequalities force every value in V to be zero, so the statement includes that case. □

光滑不等于解析

C^∞ 只说明每阶导数存在；解析还需控制高阶导数的增长。这里的 $k!$ 型估计让 Taylor 余项趋零，不能省略这一环。

2.2.4 Green and Poisson kernels (Evans §2.2.4)

For sufficiently smooth u, v, the two Green identities are

$$\int_U Du \cdot Dv = -\int_U v\Delta u + \int_{\partial U} v\partial_\nu u,$$

$$\int_U (u\Delta v - v\Delta u) = \int_{\partial U} (u\partial_\nu v - v\partial_\nu u).$$

The second is the difference of the first identity in the two orders.

Boundary correction. For fixed $x \in U$, suppose a harmonic corrector h_x with $h_x(y) = \Phi(y - x)$ on ∂U has been constructed. Set $G(x, y) = \Phi(y - x) - h_x(y)$. Then

$$-\Delta_y G(x, y) = \delta_x, \qquad G(x, y) = 0 \quad (y \in \partial U).$$

General-domain existence of this corrector is not proved by the notation; it belongs to Dirichlet existence theory later in the book. The following representation is conditional on such a sufficiently regular Green function. Explicit examples below avoid that existence issue.

Green 函数是修正后的点源

基本解有正确的点源奇性，却通常不满足边界条件。减去调和修正项保留奇性、消掉边界值；“定义修正项”本身不等于证明它存在。

Theorem 2.7 (Representation and symmetry; source Theorems 12–13)
For $-\Delta u = f$, $u = g$ on the boundary,

$$u(x) = \int_U G(x, y) f(y)\, dy - \int_{\partial U} g(y)\, \partial_{\nu_y} G(x, y)\, dS_y. \qquad \text{(G)}$$

Furthermore $G(x, y) = G(y, x)$.

Expanded proof.

Apply the second Green identity on $U \setminus \overline{B}_\varepsilon(x)$ to u and $G(x, \cdot)$. Away from x, $\Delta G = 0$. On the outer boundary $G = 0$, leaving $\int g \partial_\nu G$. On the inner boundary the singular normal derivative has total flux one, with normal directed into the removed ball; the resulting term tends to $u(x)$. Terms involving $G \partial_\nu u$ tend to zero at rates $O(\varepsilon)$ or $O(\varepsilon |\log \varepsilon|)$. Thus $\int_U Gf = \int_{\partial U} g \partial_\nu G + u(x)$, giving (G).

For symmetry, apply the same second identity to $G(x, \cdot), G(z, \cdot)$ with small disjoint balls about x, z removed. Both are harmonic on the remaining domain and zero on the outer boundary. At the sphere around x, the term with the singular normal derivative tends to $G(z, x)$ and the opposite product tends to zero; the sphere around z gives $-G(x, z)$. Their sum is zero, proving symmetry. □

The upper half-space (source Theorem 14). For $x_n > 0$ let $x^* = (x', -x_n)$. Reflection gives $G(x, y) = \Phi(y - x) - \Phi(y - x^*)$. The singularity of the image lies outside the domain, and the two distances agree on $y_n = 0$. Its boundary normal derivative gives

$$P(x, z) = \frac{2x_n}{n\omega_n (|x' - z|^2 + x_n^2)^{n/2}}, \qquad z \in \mathbb{R}^{n-1}.$$

$$u(x) = \int_{\mathbb{R}^{n-1}} P(x, z) g(z)\, dz.$$

The kernel is harmonic in x, positive, and has mass one. To check the mass without imposing boundary decay, put $z = x' + x_n w$; the remaining integral is $[2/(n\omega_n)] \int (1 + |w|^2)^{-n/2}\, dw = 1$. Its value follows by

polar coordinates, or the Gaussian integral and Gamma substitution. For bounded continuous g, split the boundary integral into $|z - x_0'| < \delta$ and its complement. Uniform continuity near x_0' controls the first error; the latter kernel mass tends to zero as $x \to (x_0', 0)$. This proves boundary recovery and, by differentiation on compact interior sets, harmonicity. Uniqueness here is among bounded solutions: a bounded zero-boundary harmonic function extends oddly across the plane, is harmonic by the reflection argument in Problem 10, and is constant by Liouville, hence zero.

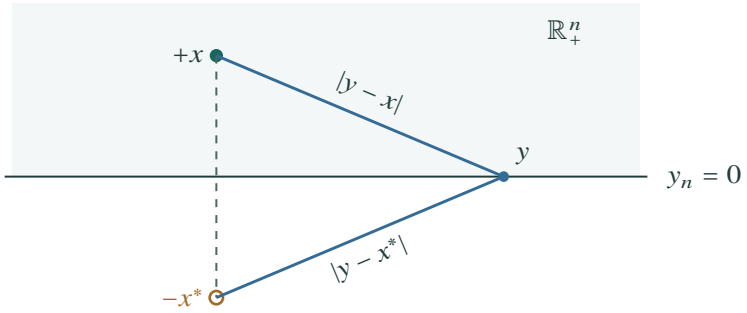


Diagram T2.2. A reflected point source enforces the boundary value. The solid source at x is inside the upper half-space; the image at x^* is outside and has the opposite sign. Every boundary point y is equidistant from the pair, so their potentials cancel there. The labels indicate source signs, not signed coordinate locations.

The ball (source Theorem 15). For $n \geq 3$, $0 < |x| < R$, put $x^* = R^2 x/|x|^2$. An explicit Green function is

$$G(x, y) = \Phi(y - x) - (R/|x|)^{n-2} \Phi(y - x^*).$$

On $|y| = R$, $|y - x^*| = (R/|x|)|y - x|$, so the boundary value is zero. At $x = 0$ the formula has the limit $\Phi(y) - \Phi(R)$. In two dimensions use instead

$$G(x, y) = \frac{1}{2\pi} \log \frac{|x|\,|y - x^*|}{R|y - x|},$$

with the corresponding center limit $(2\pi)^{-1} \log(R/|y|)$. Differentiating in the boundary normal gives, for every $n \geq 2$,

$$P_R(x, y) = \frac{R^2 - |x|^2}{n\omega_n R|x - y|^n}, \qquad u(x) = \int_{\partial B_R} P_R(x, y) g(y)\, dS_y. \tag{PB}$$

Expanded proof (why the boundary formula solves the problem).

The constructed Green function and (G) applied to $u = 1$ give $\int_{\partial B_R} P_R = 1$. This uses a known constant harmonic function, so does

not assume existence for arbitrary data. For $|y| = R$, direct differentiation yields

$$\Delta_x\big[(R^2 - |x|^2)|x-y|^{-n}\big] = 2n(R^2 - |y|^2)|x-y|^{-n-2} = 0.$$

Thus the integral is harmonic; all differentiations are dominated on compact subsets separated from the sphere. For $x \to y_0 \in \partial B_R$ split $g(y) - g(y_0)$ into a small spherical neighborhood and its complement. Positivity and mass one control the near part by the modulus of continuity. On the complement the denominator stays bounded below while $R^2 - |x|^2 \to 0$, so the far part tends to zero. This proves the boundary data. The maximum principle gives uniqueness. At $x = 0$, P_R is constant and (PB) reduces exactly to the spherical mean identity. □

2.2.5 Energy and Dirichlet's principle (Evans §2.2.5, Theorems 16–17)

Let U be bounded and smooth enough for the following integrations. For two $C^2(\overline{U})$ solutions of the same Dirichlet problem, their difference w has $\Delta w = 0$ and $w|_{\partial U} = 0$. Green's first identity gives $\int_U |Dw|^2 = -\int_U w\Delta w + \int_{\partial U} w\partial_\nu w = 0$. Thus w is constant on each component, and its boundary values make it zero.

For $-\Delta u = f$, $u = g$, define

$$J(v) = \int_U \left(\tfrac{1}{2}|Dv|^2 - fv\right) dx.$$

For $v = u + w$ with $w|_{\partial U} = 0$,

$$J(v) - J(u) = \int_U (Du\cdot Dw - fw) + \tfrac{1}{2}\int_U |Dw|^2 = \tfrac{1}{2}\int_U |Dw|^2 \geq 0.$$

Here the linear term vanishes by Green's identity and the PDE. Conversely, if a smooth u minimizes J for its boundary data, test $v = u + \varepsilon\phi$ with $\phi \in C_c^\infty(U)$ and both signs of ε. Differentiating the quadratic polynomial at zero gives $\int Du\cdot D\phi = \int f\phi$, hence $-\Delta u = f$. This proves the equivalence in the smooth class; existence of a minimizer in a Sobolev class uses the later functional-analytic theory.

中心平均与偏心平均

从球心看，边界各方向权重相同；从偏心点看，靠近该点的边界权重更大，变成 Poisson 核。球面均值是 Poisson 表示的球心特例。

2.3 Heat Equation

2.3.1 The Gaussian kernel and Duhamel (Evans §2.3.1)

Heat balance with unit diffusivity is $u_t = \Delta u$. The scaling $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$ suggests a mass-preserving kernel $t^{-n/2}F(x/\sqrt{t})$. Substitution gives $\Delta F + \frac{1}{2}z\cdot DF + \frac{n}{2}F = 0$. Taking $F(z) = Ce^{-|z|^2/4}$ satisfies this equation;

从第二章走向第五章

能量只含一阶导数，所以弱形式 $\int Du\cdot D\phi = \int f\phi$ 自然进入 H^1。第二章的显式公式与第五章的函数空间方法在这里接上。

the product of one-dimensional Gaussian integrals fixes $C = (4\pi)^{-n/2}$. Thus

$$K(x,t) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)} \quad (t>0), \qquad K = 0 \quad (t \le 0).$$

The value at the single point $(0,0)$ is interpreted distributionally. Direct differentiation verifies

$$K_t = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K, \qquad \Delta K = \left(-\frac{n}{2t} + \frac{|x|^2}{4t^2}\right) K.$$

Also $K > 0$, $\int K(x,t)\,dx = 1$, and $\int_{|x|>\delta} K(x,t)\,dx \to 0$ as $t \downarrow 0$. The last assertion follows from $x = \sqrt{t}\,z$ and the integrable Gaussian tail.

Theorem 2.8 (Source Theorem 1)
For bounded continuous g,

$$u(x,t) = \int_{\mathbb{R}^n} K(x-y,t) g(y)\,dy$$

is smooth for $t > 0$, solves the heat equation, and tends to $g(x_0)$ as $(x,t) \to (x_0,0)$.

Expanded proof.

On compact subsets of $t > 0$, every kernel derivative is a polynomial times a Gaussian with an integrable uniform majorant. Differentiate under the integral and use $K_t = \Delta K$. For initial convergence use mass one:

$$u(x,t) - g(x_0) = \int K(x-y,t)\,[g(y) - g(x_0)]\,dy.$$

Given $\varepsilon > 0$, continuity makes the bracket smaller than ε for $|y - x_0| < \delta$. For $|x - x_0| < \delta/2$, the far part is bounded by $2\|g\|_\infty$ times a Gaussian tail outside $B_{\delta/2}(x)$, which tends to zero. This proves the joint limit. Completing the square in a product of two Gaussians gives $K_t * K_s = K_{t+s}$, the semigroup law. □

For a time-dependent source with zero initial data, Duhamel gives

$$v(x,t) = \int_0^t (K_{t-s} * f(\cdot,s))(x)\,ds.$$

For example assume f is smooth on the time interval, with uniformly compact spatial support and bounded indicated derivatives. Change to $\tau = t - s$ if needed to keep differentiations away from the singular upper endpoint. Moving spatial derivatives to f gives

$$v_t = f(x,t) + \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds, \qquad \Delta v = \int_0^t K_{t-s} * \Delta f(\cdot,s)\,ds.$$

抛物缩放为何是平方

空间二阶导数缩放为 λ^2，时间一阶导数也必须缩放为 λ^2 才保持方程。热球的高度因此正比 r^2，不是 r。

瞬时传播不等于大量传播

$K(x,t) > 0$ 对任意 x、$t > 0$ 成立，所以非零非负初值立即在全空间产生正值；远处的影响可极小，但不是严格为零。

The endpoint term uses $K_\varepsilon * f(\cdot,t) \to f(\cdot,t)$. The bounds on $f, D^2 f$ provide domination, proving $Lv = f$ and $v(\cdot,0) = 0$. Adding $K_t * g$ solves the full initial-value problem (source Theorem 2). We use these explicit smoothness assumptions rather than claim that arbitrary continuous forcing gives classical second derivatives without further justification.

2.3.2 Heat-ball mean values (Evans §2.3.2, Theorem 3)

We now expand the connection with harmonic means. The region, the time orientation, and the weight are all essential.

1. The kernel determines the geometry

Let $K(z,\tau) = (4\pi\tau)^{-n/2} e^{-|z|^2/(4\tau)}$, $\tau > 0$. For a target (x,t) define

$$E_r(x,t) = \{(y,s) : s < t,\ K(x-y, t-s) > r^{-n}\}.$$

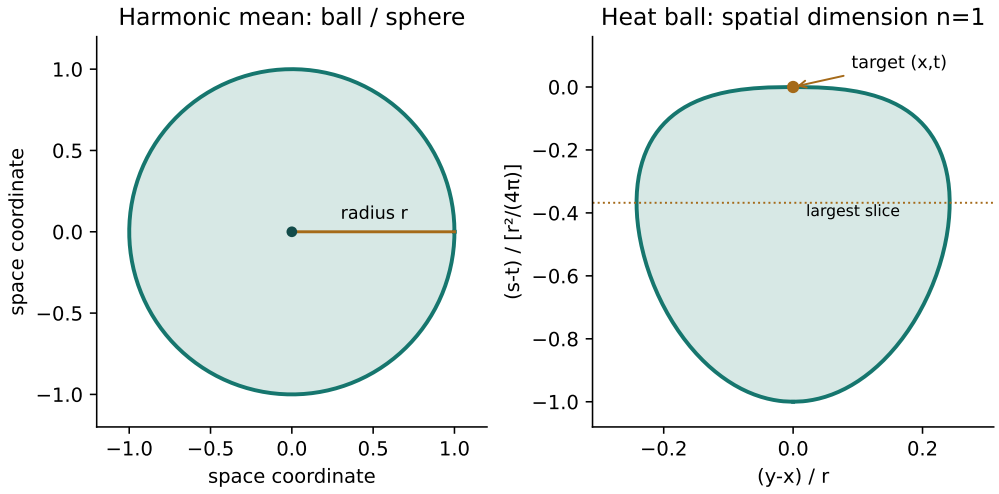
The open or closed choice on the ordinary boundary does not affect the integral. Put $\tau = t - s$ and $a = r^2/(4\pi)$. Taking logarithms gives

$$0 < \tau < a, \qquad |y - x| < R_r(\tau), \qquad R_r(\tau)^2 = 2n\tau \log(a/\tau). \quad \text{(HB)}$$

Each past slice is an ordinary spatial ball. Its radius vanishes at both tips and is largest at $\tau = a/e$:

$$\max R_r(\tau) = r\sqrt{\frac{n}{2\pi e}}.$$

Thus r is a kernel-level scale, not the largest spatial radius.



Editorial figure. Left: an elliptic ball and its boundary. Right: a one-dimensional spatial section of a heat ball, with time increasing toward the target. Spatial and temporal axes use different scales.

热球为何朝向过去

核 $K(x-y, t-s)$ 只在 $s < t$ 有意义。热球下端在 $t - r^2/(4\pi)$，上端是目标时刻 t；它不是以 (x,t) 为对称中心的时空球。

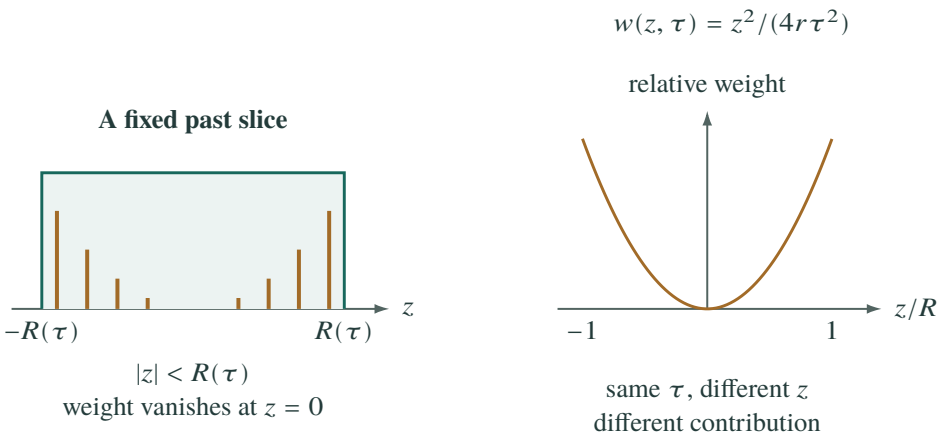


Diagram T2.3. A heat ball carries a nonuniform averaging measure. In one spatial dimension, fix a past time $\tau = t - s$. The blue band marks the allowed interval, while orange heights illustrate the factor $z^2/(4r\tau^2)$. The right panel shows its quadratic spatial dependence up to a fixed positive factor. This is a slice of the full space–time probability measure, not a separately normalized conditional density.

2. Statement and full proof

Theorem 2.9 (Watson–Evans mean-value formula)
If $u \in C^{2,1}$ solves $Lu = u_t - \Delta u = 0$ on a neighborhood of $\overline{E}_r(x,t)$, then

$$\boxed{u(x,t) = \frac{1}{4r^n} \iint_{E_r(x,t)} u(y,s) \frac{|x-y|^2}{(t-s)^2}\, dy\, ds.} \tag{H}$$

The nonnegative measure

$$d\mu_r(y,s) = \frac{\mathbf{1}_{E_r(x,t)}}{4r^n} \frac{|x-y|^2}{(t-s)^2}\, dy\, ds$$

has mass one, so this is $u(x,t) = \mathbb{E}_{\mu_r}[u(Y,S)]$.

Expanded proof (including the source's omitted normalization).

A. Compute the mass exactly. At fixed τ, $\int_{B_R} |z|^2\, dz = n\omega_n R^{n+2}/(n+2)$. With $b = n/2$ and $a = r^2/(4\pi)$, polar integration gives

$$\begin{aligned} I_r &:= \iint_{E_r(0,0)} \frac{|y|^2}{s^2}\, dy\, ds \\ &= \frac{n\omega_n}{n+2} (2n)^{b+1} \int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau. \end{aligned}$$

Set $q = \log(a/\tau)$, so $\tau = ae^{-q}$. Then

$$\int_0^a \tau^{b-1} [\log(a/\tau)]^{b+1}\, d\tau = a^b \int_0^\infty e^{-bq} q^{b+1}\, dq = a^b \frac{\Gamma(b+2)}{b^{b+2}}.$$

区域和权重缺一不可

均匀热球抽样使用 $dy\, ds/|E_r|$；这里使用 $|x-y|^2/[4r^n(t-s)^2]$。即使区域画对了，丢掉权重仍然不是均值定理。

Here $\Gamma(c) = \int_0^\infty z^{c-1}e^{-z}\,dz$; the last equality uses $z = bq$. Substitute $\omega_n = \pi^b/\Gamma(b+1)$, $a^b = r^n/(4\pi)^b$, and $\Gamma(b+2) = (b+1)\Gamma(b+1)$. Cancellation yields

$$I_r = 4r^n, \qquad \mu_r(E_r) = 1.$$

This also proves integrability of the seemingly singular top-tip weight for every $n \geq 1$.

B. Differentiate after fixing the domain. Translate the target to $(0,0)$; thus integration times s are negative. Scaling of K gives $E_r = \{(ry, r^2 s) : (y,s) \in E_1\}$. Define

$$\phi(r) = r^{-n} \iint_{E_r} u(y,s)\frac{|y|^2}{s^2}\,dy\,ds = \iint_{E_1} u(ry, r^2 s)\frac{|y|^2}{s^2}\,dy\,ds.$$

The Jacobian is r^{n+2} and the weight contributes r^{-2}, explaining the cancellation. First suppose u smooth. Differentiate on E_1, then scale back:

$$\phi'(r) = A + B,$$

$$A = r^{-n-1}\iint_{E_r}(y\cdot Du)\frac{|y|^2}{s^2}, \qquad B = r^{-n-1}\iint_{E_r} 2u_s\frac{|y|^2}{s}. \tag{AB}$$

Bounded derivatives on compact sets and the integrable weights justify this operation for radii in a compact admissible interval.

C. Find a boundary-vanishing function. Set

$$\psi_r = \log\left(r^n K(y,-s)\right) = n\log r - \frac{n}{2}\log(-4\pi s) + \frac{|y|^2}{4s}.$$

It is positive inside E_r, zero on its lateral boundary, and

$$D_y\psi_r = \frac{y}{2s}, \qquad (\psi_r)_s = -\frac{n}{2s} - \frac{|y|^2}{4s^2}. \tag{L}$$

Since $2|y|^2/s = 4y\cdot D\psi_r$, spatial integration by parts gives

$$B = -4r^{-n-1}\iint_{E_r}\psi_r(nu_s + y\cdot Du_s).$$

Integrate the last term in time and insert (L):

$$\begin{aligned} B &= r^{-n-1}\iint_{E_r}[-4n\psi_r u_s + 4(\psi_r)_s\, y\cdot Du] \\ &= -4nr^{-n-1}\iint_{E_r}\psi_r u_s - 2nr^{-n-1}\iint_{E_r}\frac{y\cdot Du}{s} - A. \end{aligned}$$

The A terms cancel. Use $y/s = 2D\psi_r$ and integrate spatially once more:

$$\boxed{\phi'(r) = 4nr^{-n-1}\iint_{E_r}\psi_r(\Delta u - u_s).} \tag{R}$$

For a caloric function the right side is zero, so ϕ is constant.

D. Check both tips and the regularity assumption. Zero lateral values alone do not justify ignoring the singular tip. First truncate to $\varepsilon < \tau < a - \eta$. The lateral terms vanish because $\psi_r = 0$. At the top cap, $R = O(\sqrt{\tau \log(a/\tau)})$ and

$$\int_{B_R} |y| \psi_r \, dy \le C R^{n+1} \log(a/\tau) \longrightarrow 0.$$

This controls the time-boundary term containing $y \cdot Du$. At the lower cap, $R \to 0$ and $\psi_r \to 0$, so that term also vanishes. The weight remaining near the upper tip is bounded by a constant times

$$\int_0^{\varepsilon} \tau^{n/2-1} [\log(a/\tau)]^{n/2+1} \, d\tau \longrightarrow 0.$$

The differentiated terms in (AB) and those with ψ_r have the same integrable powers or better ones. Let $\varepsilon, \eta \downarrow 0$.

For only $C^{2,1}$ regularity, convolve in space-time on a slightly larger neighborhood. The constant-coefficient operator L commutes with convolution, so smooth approximations still solve the equation where needed. They converge uniformly on the compact region; the integrable weight passes the established mean identity to u. This avoids requiring mixed third derivatives in the original function. At a terminal cylinder time, take limits from interior target times.

E. Identify the constant by shrinking. The heat ball lies in $|y| \le C_n r$, $|s| \le r^2/(4\pi)$. Uniform continuity and the mass computation imply $\phi(r) \to 4u(0,0)$ as $r \downarrow 0$. Therefore $\phi(r) = 4u(0,0)$ for every admissible radius. Divide by four and restore the target (x,t). □

3. The precise connection to elliptic averages

For general smooth u let $M_r[u] = \phi(r)/4$. The preceding computation, before imposing the PDE, gives

$$\frac{d}{dr} M_r[u](x,t) = -\frac{n}{r^{n+1}} \iint_{E_r(x,t)} \log\left(r^n K(x-y, t-s)\right) Lu(y,s) \, dy \, ds. \tag{HR}$$

Thus $Lu \le 0$ implies $M_r[u]$ nondecreasing, and shrinking r gives $u(x,t) \le M_r[u](x,t)$. Reverse the signs for supersolutions. This parallels $\Delta v \ge 0$ and $\partial_r S_v \ge 0$ in (M).

对数核为何出现

ψ_r 的零边界值消掉侧边界项；空间梯度产生平均权重；时间导数让缩放导数的两项抵消。这三种作用共同决定热球的定义。

A counterexample to unweighted replacements. The polynomial $u(y,s) = |y-x|^2 + 2n(s-t)$ has $Lu = 0$ and $u(x,t) = 0$. At time t its uniform spatial ball average is $nr^2/(n+2) > 0$. On $B_r(x) \times (t-r^2, t)$ its uniform space-time average is $nr^2/(n+2) - nr^2 \neq 0$. A symmetric Euclidean space-time ball cancels the time term but still leaves a positive spatial square. None of these is the required mean.

Stationary solutions. If $u(y,s) = h(y)$, then $Lu = 0$ is exactly $\Delta h = 0$. At fixed τ, the heat-ball weight is radial in $y - x$. Integrate over spherical shells first: every spherical mean of h equals $h(x)$, and the remaining scalar integral is the total mass one. Thus heat-ball averaging of a stationary solution is an aggregate of elliptic sphere averages. A time-dependent caloric solution needs the past slices as well as the weight.

Moments and a converse: editorial supplement. There is an exact probability description of μ_r. Put $b = n/2$, $a = r^2/(4\pi)$. Let Q have Gamma density proportional to $q^{b+1}e^{-bq}$, $q > 0$, and set $\tau = ae^{-Q}$. Conditional on τ, choose a uniform direction and radius $\varrho = R_r(\tau)U^{1/(n+2)}$, with U uniform on $(0,1)$. Step A gives the time density after the logarithmic substitution. The conditional spatial radial density is proportional to ϱ^{n+1}, because the volume Jacobian contributes ϱ^{n-1} and the weight contributes ϱ^2. Hence this construction samples the actual weighted measure.

Gamma integration and symmetry yield

$$\mathbb{E}(Y_i - x_i) = 0, \qquad \mathbb{E}[(Y_i - x_i)(Y_j - x_j)] = 2\delta_{ij}\mathbb{E}\tau,$$

$$\mathbb{E}\tau = c_n r^2, \qquad c_n = \frac{1}{4\pi}\left(\frac{n}{n+2}\right)^{n/2+2} > 0.$$

For the second moment, conditional radial integration gives $\mathbb{E}(|Y-x|^2 \mid \tau) = [(n+2)/(n+4)]\,2n\tau Q$. Integrating the Gamma density then gives $\mathbb{E}|Y-x|^2 = 2n\mathbb{E}\tau$. Taylor expansion for $C^{2,1}$ functions therefore implies

$$M_r[u](x,t) = u(x,t) + c_n r^2(\Delta u - u_t)(x,t) + o(r^2). \qquad \text{(HT)}$$

The remainder is uniform on the shrinking region, and $\mathbb{E}(|Y-x|^2 + \tau) = O(r^2)$ controls its integral. An exact identity for all small radii at every center forces $Lu = 0$. Compare this with the elliptic expansion (T).

两个次解的符号

椭圆次调和是 $\Delta v \geq 0$，即 $-\Delta v \leq 0$；抛物次解是 $v_t - \Delta v \leq 0$。统一算子符号后，两者都是“中心不超过正权平均”。

最简单的公式自检

$|y-x|^2 + 2n(s-t)$ 的空间曲率与时间变化恰好平衡。正确权重必须把二者抵消；普通空间平均看不到时间补偿。

几何平均在检测算子

球面平均的二阶误差是 $r^2\Delta u/(2n)$；热球平均的二阶误差是 $c_n r^2(\Delta u - u_t)$。两种几何平均分别检测各自的 PDE。

Do not confuse this with Gaussian averaging. At a fixed earlier time s, the whole-space solution can be written $u(x,t) = \int K(x-y,t-s)u(y,s)\,dy$ in a class where the Cauchy problem is unique. That is a global spatial Gaussian average on one time slice. Heat-ball averaging is local in space-time and uses many past slices. The fundamental solution is the same; the two averaging measures differ.

2.3.3 Maximum principles and regularity (Evans §2.3.3)

Write $U_T = U \times (0,T]$ and $\Gamma_T = (\overline{U} \times \{0\}) \cup (\partial U \times [0,T])$. The top $U \times \{T\}$ is not part of this parabolic boundary.

Theorem 2.10 (Source Theorems 4–5)
On bounded U, a continuous classical solution satisfies $\max_{\overline{U}_T} u = \max_{\Gamma_T} u$. If U is connected and the global maximum occurs at (x_0,t_0) with $x_0 \in U$, $t_0 > 0$, then u is constant on $\overline{U} \times [0,t_0]$. The initial-boundary problem has at most one classical solution for given forcing and parabolic boundary data.

Expanded proof.

For the weak maximum assertion set $v = u - \varepsilon t$. At a maximum in U_T, including the terminal time, the left time derivative is nonnegative and the spatial Laplacian is nonpositive. Thus $Lv \geq 0$, contradicting $Lv = -\varepsilon$. A maximum must lie on Γ_T. Let $\varepsilon \downarrow 0$.

For the strong assertion put $M = \max u$. The heat-ball mean at an interior maximum averages $u \leq M$ against a probability density strictly positive except on a negligible axis. Equality forces $u = M$ almost everywhere in the heat ball; continuity makes it equal everywhere. This propagates equality backward locally.

To reach a distant past point, use a polygonal spatial path in U and assign strictly decreasing times to its finitely many vertices. A small backward portion of each space-time line lies in a heat ball about its later endpoint: the spatial displacement is $O(\tau)$, whereas the heat-ball radius is of order $\sqrt{\tau \log(1/\tau)}$, which is larger for small τ. On each segment the earliest time of an already-constant terminal portion is attained by continuity. If it were not the segment's earliest endpoint, a small heat ball would extend it, a contradiction. Propagate through the finite chain. Every earlier interior point therefore has value M; continuity reaches time zero and the boundary. Finally apply the weak principle to both signs of the difference of two solutions for uniqueness. □

强最大值结论只向过去

内部时刻 t_0 达到全局最大值，会强迫此前恒定；不自动强迫此后恒定。边界条件在未来改变时，解也可以随后改变。

Whole-space uniqueness needs a growth class (source 6–7). Suppose $Lu \le 0$, $u(\cdot,0) \le 0$ and $u(x,t) \le Ae^{a|x|^2}$ for $0 \le t \le T$. Choose $b > a$ and the positive caloric barrier

$$V(x,t) = (1-4bt)^{-n/2} \exp\left(\frac{b|x|^2}{1-4bt}\right) \quad (0 \le t < 1/(4b)).$$

Its PDE follows from direct Gaussian-type differentiation. For fixed $\varepsilon > 0$, $u - \varepsilon V < 0$ on the lateral boundary of a sufficiently large ball, uniformly on a shorter interval $[0,\delta]$, $\delta < 1/(4b)$; the exponent $b > a$ ensures domination. Its initial values are nonpositive. The bounded-cylinder principle gives $u \le \varepsilon V$. Let the radius tend to infinity, then $\varepsilon \downarrow 0$. Repeat on consecutive time intervals of length δ, using the same growth bound, to cover $[0,T]$. This proves the whole-space maximum principle and, for $|u - \tilde{u}| \le Ae^{a|x|^2}$, uniqueness. Without a growth restriction the source discusses nonphysical zero-initial-data solutions; this draft does not use their existence as a substitute for specifying a uniqueness class.

Interior smoothing and derivative estimates (source 8–9). For a backward cylinder $Q_r = B_r(x_0) \times (t_0 - r^2, t_0]$ inside the domain, choose a smooth cutoff ζ equal to one on $Q_{3r/4}$, vanishing near the bottom and lateral boundary of Q_r. Then

$$L(\zeta u) = (\zeta_s - \Delta\zeta)u - 2D\zeta \cdot Du.$$

Duhamel and spatial integration by parts in the last term yield, on $Q_{r/2}$,

$$\begin{aligned} u(x,t) = \iint_{Q_r} \big[& K(x-y,t-s)(\zeta_s + \Delta\zeta)(y,s) \\ & + 2D_y K(x-y,t-s) \cdot D\zeta(y,s) \big] u(y,s)\, dy\, ds. \end{aligned} \tag{C}$$

Set the kernel to zero for $s \ge t$. Where cutoff derivatives do not vanish, either $t - s$ stays separated from zero or $|x - y|$ stays separated from zero. In the latter case every derivative of the Gaussian tends to zero faster than any power as $s \uparrow t$. Thus the combined kernel in (C) is smooth with bounded derivatives on the relevant compact sets, despite the original fundamental solution's singularity. A preliminary local convolution proves (C) for merely $C^{2,1}$ functions, then passes to the limit. Differentiating (C) now gives $u \in C^\infty$ locally.

On a unit cylinder, its kernel derivatives are bounded by constants $C_{\alpha,l}$. Rescale $v(z,\tau) = u(x_0 + rz, t_0 + r^2\tau)$ to obtain

$$\sup_{Q_{r/2}} |D_x^\alpha \partial_t^l u| \le C_{\alpha,l,n} r^{-n-2-|\alpha|-2l} \|u\|_{L^1(Q_r)}.$$

One time derivative costs two powers of spatial scale. For completeness, spatial analyticity can also be read from (C), using the elementary complex-analytic parameter-integral theorem and Cauchy estimates as additional prerequisites. Fix a time and replace the spatial variable by $z = x + iv$ in the Gaussian, using the polynomial $\sum_j (z_j - y_j)^2$ in its exponent. Where $t - s$ is small and cutoff derivatives are nonzero, $|x - y|$ is bounded below by some $d > 0$. For $|v| < d/2$ the real part of that polynomial remains positive, bounded below by a fixed multiple of d^2. The resulting Gaussian and its spatial derivatives have an integrable bound uniform on a small complex neighborhood. Where $t - s$ is bounded below there is no singularity at all. Thus (C) extends holomorphically in z near every interior spatial point. Cauchy's formula supplies factorial derivative bounds, proving real analyticity in space. This argument says nothing comparable about analyticity in time; smoothness alone does not establish that.

2.3.4 Energy and backward uniqueness (Evans §2.3.4, source 10–11)

Let w be the difference of two sufficiently smooth solutions with the same lateral boundary values. Then $w = 0$ on ∂U and

$$\frac{1}{2}\frac{d}{dt}\int_U w^2 = \int_U w\Delta w = -\int_U |Dw|^2.$$

Zero initial difference gives zero energy at every later time. This proves forward uniqueness and displays dissipation.

Backward uniqueness requires more than this decreasing-energy identity. Assume enough boundary and mixed derivative regularity for the next identities (smooth solutions up to the closed cylinder suffice). Set $e(t) = \int_U w^2$. Since $w_t = 0$ on the fixed lateral boundary,

$$e' = -2\int |Dw|^2, \qquad e'' = -4\int Dw \cdot Dw_t = 4\int (\Delta w)^2.$$

Cauchy–Schwarz in $\int |Dw|^2 = -\int w\Delta w$ gives $(e')^2 \le ee''$. Wherever $e > 0$, this means $(\log e)'' \ge 0$. If e were positive at t_1 and first vanished at $t_2 \le T$, convexity on $[t_1, t_2)$ would give, for $0 < \theta < 1$ and $b < t_2$,

$$e((1-\theta)t_1 + \theta b) \le e(t_1)^{1-\theta} e(b)^{\theta}.$$

Let $b \uparrow t_2$. The right side vanishes, contradicting positivity at the earlier interior time. Thus $e(T) = 0$ forces $e \equiv 0$.

切断函数移走奇点

把导数转移给核后，核奇点附近的切断导数全为零；远离空间中心时高斯又快速衰减。因此才可任意多次求导，而不是直接对奇异积分硬求导。

2.4 Wave Equation

2.4.1 Spherical means and explicit solutions (Evans §2.4.1)

The wave equation $u_{tt} - \Delta u = 0$ needs two initial data: $u(\cdot,0) = g$, $u_t(\cdot,0) = h$. Unlike the heat equation, it scales with $t \mapsto \lambda t$ when $x \mapsto \lambda x$.

One dimension: d'Alembert (source Theorem 1). Write $\xi = x + t$, $\eta = x - t$. Then $\partial_t^2 - \partial_x^2 = -4\partial_\xi \partial_\eta$. Thus $u_{\xi\eta} = 0$, and integration gives $u = F(\xi) + G(\eta)$. At $t = 0$, $F + G = g$ and $F' - G' = h$, whereas $F' + G' = g'$. Solving for F', G' and integrating yields

$$u(x,t) = \frac{g(x+t) + g(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy. \tag{DA}$$

For $g \in C^2$, $h \in C^1$, direct differentiation verifies the equation, $u(\cdot,0) = g$, and $u_t(\cdot,0) = h$. Only initial points in $[x-t, x+t]$ matter.

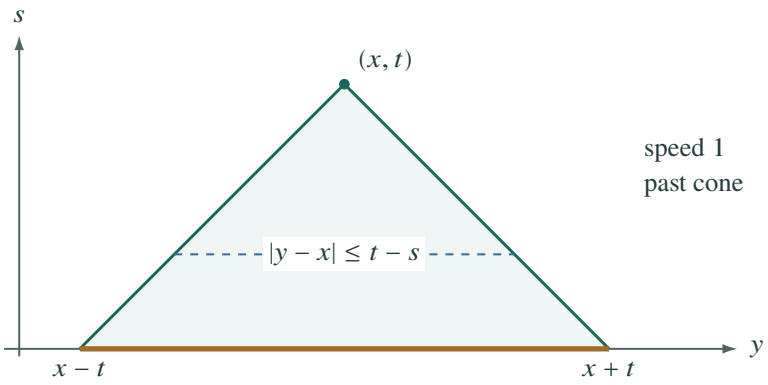


Diagram T2.4. The wave equation has a finite domain of dependence. For speed one in one dimension, the solution at (x,t) uses initial data only on $[x-t, x+t]$. At an intermediate time s, the relevant interval is $|y - x| \le t - s$. In higher dimensions this interval becomes a ball. A forced equation also accumulates sources from the past cone.

The common spherical-mean identity. Fix a spatial center x and set $M(r,t) = |\partial B_r|^{-1} \int_{\partial B_r(x)} u(y,t)\,dS$. The radius-flux computation used for harmonic means now gives

$$M_r = \frac{1}{n\omega_n r^{n-1}} \int_{B_r(x)} \Delta u, \qquad (r^{n-1}M_r)_r = r^{n-1} M_{tt}.$$

Thus M satisfies the Euler–Poisson–Darboux equation

$$M_{tt} = M_{rr} + \frac{n-1}{r} M_r, \qquad M(0,t) = u(x,t), \quad M_r(0,t) = 0. \tag{EPD}$$

逆向唯一不等于逆向稳定

知道最终状态可唯一确定过去，不代表微小最终误差只造成微小过去误差。热方程高频逆向增长仍会导致严重不稳定，第一章的例子依然适用。

两个初值各管什么

初始位移 g 生成两列反向平移的形状；初始速度 h 在区间内累积。热方程只需一个初值，而波动方程的二阶时间导数要求两个。

The second radius derivative differentiates the ball integral by shells; the wave equation changes Δu into u_{tt}.

Three dimensions: Kirchhoff. For $n = 3$, $W(r,t) = rM(r,t)$ satisfies $W_{tt} = W_{rr}$. Extend M evenly in r, hence W oddly. Its initial data are $rS_g(x,r)$ and $rS_h(x,r)$. Use (DA) for W and differentiate in r at zero. Since $W_r(0,t) = u(x,t)$, this gives

$$u(x,t) = \partial_t\big[tS_g(x,t)\big] + tS_h(x,t). \tag{KIR}$$

Equivalently

$$u(x,t) = \frac{1}{4\pi}\int_{\partial B_1}\left[g(x+t\theta) + tDg(x+t\theta)\cdot\theta + th(x+t\theta)\right]dS_\theta.$$

The latter fixed-sphere expression makes initial limits transparent. The formula solves the wave equation: by (EPD) applied to spatial spherical means of g or h, tS_h satisfies the wave equation in the variables (x,t); time differentiation handles the g term. At zero, $tS_h = th(x) + O(t^3)$ and $\partial_t(tS_g) = g(x) + O(t^2)$. These observations prove existence as well as the derivation from a hypothetical solution, for smooth data.

Two dimensions: descent from three dimensions. Make g, h independent of a third coordinate. Project the upper and lower hemispheres onto the disk. Each contributes surface Jacobian $t/\sqrt{t^2 - |y-x|^2}$. Formula (KIR) becomes

$$u(x,t) = \partial_t\left[\frac{1}{2\pi}\int_{B_t(x)}\frac{g(y)}{\sqrt{t^2-|y-x|^2}}\,dy\right] + \frac{1}{2\pi}\int_{B_t(x)}\frac{h(y)}{\sqrt{t^2-|y-x|^2}}\,dy. \tag{W2}$$

Use $y = x + tz$ before differentiating: the singularity $(1-|z|^2)^{-1/2}$ is integrable on the fixed unit disk. This justifies differentiation for sufficiently smooth data. Descent preserves the PDE because the third-coordinate derivative is zero; the three-dimensional initial limits give the desired two data.

All odd dimensions (source Theorem 2). For $n = 2k+1 \geq 3$, put $D_r = r^{-1}\partial_r$ and $\gamma_n = 1\cdot 3\cdots(n-2)$. Define

$$S_n(t)f = \frac{1}{\gamma_n}D_t^{k-1}\big[t^{n-2}S_f(x,t)\big].$$

Then

$$u(x,t) = \partial_t S_n(t)g + S_n(t)h. \tag{WO}$$

Here $S_n(t)$ denotes a solution operator, whereas $S_f(x,t)$ is the normalized spherical average defined earlier.

同一个通量公式三次出现

Laplace 方程令球面平均对半径不变；波动方程把同一半径变化联系到二阶时间导数；热方程则需要带方向的时空平均。不要因都出现球面而混为一谈。

降维为何出现权重

投影把球面压到圆盘，靠近圆周时球面近乎竖直，面积 Jacobian 变大，产生 $(t^2-|y-x|^2)^{-1/2}$。它是几何投影权重，不是热球的权重。

Expanded proof (the radial intertwining step).

Let $E = r\partial_r$ and $A_k F = D_r^{k-1}(r^{2k-1}F) = rP(E)F$, where $P(E) = \prod_{j=1}^{k-1}(E+2j+1)$; for $k=1$ the product is one. This factorization follows successively from $D_r(r^a F) = r^{a-2}(E+a)F$. The radial Laplacian is $r^{-2}E(E+2k-1)$. The polynomial identity

$$(E+1)P(E) = (E+2k-1)P(E-2)$$

therefore proves

$$(A_k F)'' = A_k\left(F'' + \frac{2k}{r}F'\right).$$

For $F(r) = S_f(x,r)$ the bracket equals $\Delta_x F$ by the divergence identity behind (EPD). Hence $A_k S_f$, now with $r = t$, solves the wave equation. It commutes with Δ_x since the operator acts only on r.

Spherical symmetry gives $S_f(x,t) = f(x) + O(t^2)$ with an even smooth extension at zero. The coefficient of $tf(x)$ in $A_k S_f$ is $P(0) = 3\cdot 5\cdots(2k-1) = \gamma_n$. All other terms start at t^3. Thus $S_n(t)f = tf + O(t^3)$, giving initial value zero and initial velocity f. Its time derivative has initial value f and initial velocity zero. This proves (WO) for smooth data and explains its constants. Finite differentiability follows by differentiating the fixed-sphere integrals; the source sufficient assumptions are $g \in C^{m+1}$, $h \in C^m$, $m = (n+1)/2$. □

All even dimensions (source Theorem 3). For even $n \geq 2$, put $\gamma_n = 2\cdot 4\cdots n$ and

$$B_f(x,t) = \frac{t^n}{\omega_n t^n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy = \frac{1}{\omega_n}\int_{B_t(x)} \frac{f(y)}{\sqrt{t^2-|y-x|^2}}\,dy.$$

Set $S_n(t)f = \gamma_n^{-1} D_t^{(n-2)/2} B_f(x,t)$. The same formula $u = \partial_t S_n(t)g + S_n(t)h$ holds. To derive it, extend the data constantly in one extra coordinate, apply (WO) in dimension $n+1$, and project both hemispheres. Their Jacobian gives $2t/\sqrt{t^2-|y-x|^2}$; the sphere and ball area constants simplify to the displayed γ_n. The inherited PDE and initial limits prove it. For smooth data there is no endpoint issue; for the source finite regularity statement take $m = (n+2)/2$, $g \in C^{m+1}$, $h \in C^m$. When $n = 2$, this reduces to (W2).

For odd $n \geq 3$, only data on the sphere at distance t enter the formula, the sharp Huygens phenomenon. For even n, the whole ball enters and a trailing influence remains. Dimension one is an exception to the odd-dimensional sharp-front statement: (DA) includes an interval integral of h.

2.4.2 Forcing (Evans §2.4.2, Theorem 4)

The operator $S_n(t)$ above solves the homogeneous equation with initial displacement zero and velocity f. For smooth forcing set

$$v(t) = \int_0^t S_n(t-s)f(s)\,ds.$$

Since $S_n(0) = 0$ and $\partial_t S_n(0) = I$, differentiate twice:

$$v_t = \int_0^t \partial_t S_n(t-s)f(s)\,ds, \qquad v_{tt} = f(t) + \int_0^t \partial_t^2 S_n(t-s)f(s)\,ds.$$

The integrals in v_{tt} and Δv cancel. Thus $v_{tt} - \Delta v = f$, with both initial data zero. Add the homogeneous solution for general initial data.

2.4.3 Energy and finite propagation (Evans §2.4.3, Theorems 5–6)

For a smooth solution with fixed zero Dirichlet boundary values,

$$\mathscr{E}(t) = \frac{1}{2}\int_U (u_t^2 + |Du|^2), \qquad \mathscr{E}'(t) = \int_{\partial U} u_t \partial_\nu u = 0.$$

The boundary term vanishes because $u = 0$ there for all times implies $u_t = 0$ there. Apply this to the difference for uniqueness; zero initial energy gives $Du = u_t = 0$, and zero initial displacement removes constants.

For local propagation, consider the shrinking ball $B_{R-t}(x_0)$ and the cone energy

$$e(t) = \frac{1}{2}\int_{B_{R-t}(x_0)} (u_t^2 + |Du|^2).$$

Differentiating the moving domain contributes a negative surface term. The PDE and spatial integration by parts yield

$$e'(t) = \int_{\partial B_{R-t}} \left[u_t \partial_\nu u - \tfrac{1}{2}(u_t^2 + |Du|^2)\right] dS \le 0.$$

The inequality follows from $2u_t\partial_\nu u \le u_t^2 + |\partial_\nu u|^2 \le u_t^2 + |Du|^2$. If both initial data vanish on $B_R(x_0)$, $e(0) = 0$ and hence $e(t) = 0$ for $t < R$. Along each vertical segment inside the cone $u_t = 0$ and its initial value is zero, so $u = 0$ throughout the cone. Continuity gives the apex value when applicable.

传播速度来自能量通量

球半径以速度 1 收缩时，移边界损失足以压住波的能量流入。热核虽处处为正，波的锥外影响却严格为零；这是两个方程的本质区别。

2.5 Problems

Source: Evans §2.5, Problems 1–24, printed pp. 81–86 (PDF pp. 98–103). Statements are concisely restated, and solutions are editorial. Smoothness is assumed wherever a displayed differentiation requires it. Problems 3–5 and 17 are especially useful checks on the mean-value arguments.

2.5.1 Problems 1–8: transport and elliptic averages

1. Transport with a zeroth-order term. Solve $u_t + b \cdot Du + cu = 0$, $u(\cdot, 0) = g$.

Solution. Along $X(s) = x + (s-t)b$, the quantity $e^{cs}u(X(s), s)$ has derivative zero. Evaluate at $s = 0, t$ to obtain $u(x,t) = e^{-ct}g(x - tb)$. □

2. Rotation invariance of the Laplacian.

Solution. For an orthogonal matrix O, the Hessian of $v(x) = u(Ox)$ is $D^2v = O^T(D^2u)(Ox)O$. Its trace equals $\operatorname{tr}((D^2u)(Ox)OO^T) = \Delta u(Ox)$. Thus a harmonic u gives a harmonic v. □

3. The center value for Poisson's equation. Derive the center formula for $-\Delta u = f$ on B_r, $u = g$ on ∂B_r.

Solution. Integrate (M) from zero to r:

$$S_u(0,r) - u(0) = -\frac{1}{n\omega_n}\int_0^r \rho^{1-n}\int_{B_\rho} f(y)\,dy\,d\rho.$$

Changing the order gives (P), with the nonnegative kernel $[|y|^{2-n} - r^{2-n}]/[n(n-2)\omega_n]$ when $n \geq 3$. Replace $S_u(0,r)$ by the boundary average of g. The kernel's singularity is locally integrable, justifying the change for bounded f; signed f is treated by absolute integrability. □

4. A direct maximum-principle proof.

Solution. For $\varepsilon > 0$, $v = u + \varepsilon|x|^2$ satisfies $\Delta v = 2n\varepsilon > 0$. At an interior maximum its Hessian would be negative semidefinite and $\Delta v \leq 0$, impossible. Thus v reaches its maximum on the boundary. Bound the added quadratic uniformly on bounded U and let $\varepsilon \downarrow 0$. □

5. Subharmonic functions. Show the mean inequality and boundary maximum principle for $\Delta v \geq 0$; prove that $\phi(u)$ and $|Du|^2$ are subharmonic for harmonic u and smooth convex ϕ.

Solution. Formula (M) makes spherical means nondecreasing, and (SB) gives $v(x) \leq A_v(x, r)$. For the boundary maximum argument use $v + \varepsilon|x|^2$, as in Problem 4. The chain rule gives $\Delta\phi(u) = \phi''(u)|Du|^2 + \phi'(u)\Delta u \geq 0$. Since harmonic functions are smooth, differentiating the PDE is valid:

$$\Delta|Du|^2 = 2|D^2u|^2 + 2Du \cdot D\Delta u = 2|D^2u|^2 \geq 0.$$

□

凸性与平均相容

调和函数经凸函数变换后不一定仍调和，却成为次调和。Jensen 不等式给直觉，链式法则中的 $\phi''(u)|Du|^2$ 给出可核验的原因。

6. A uniform Poisson bound.

Solution. Enclose U in $B_R(0)$, put $F = \|f\|_\infty$, $G = \|g\|_\infty$, and set $b(x) = G + F(R^2 - |x|^2)/(2n)$. Then $b \geq G$ on ∂U and $-\Delta b = F$. The function $u - b$ is subharmonic because $-\Delta(u - b) = f - F \leq 0$; its boundary values are nonpositive. Problem 5 yields $u \leq b$. Repeat for $-u$, giving $\|u\|_\infty \leq G + R^2F/(2n)$. □

7. Explicit Harnack bounds on a ball.

Solution. For positive harmonic u on B_R with continuous boundary values, use (PB), $R - |x| \leq |x - y| \leq R + |x|$, and $\int_{\partial B_R} u = n\omega_n R^{n-1} u(0)$:

$$R^{n-2}\frac{R - |x|}{(R + |x|)^{n-1}}u(0) \leq u(x) \leq R^{n-2}\frac{R + |x|}{(R - |x|)^{n-1}}u(0).$$

If u is only given on the open ball, first use radii $\rho < R$ with $|x| < \rho$, then let $\rho \uparrow R$. □

8. Verify the ball Poisson formula.

Solution. The full verification following (PB) establishes mass one using the known solution 1, harmonicity by direct differentiation, and boundary convergence by a near/far split. Positivity controls the near error, and the factor $R^2 - |x|^2$ kills the far error. This proves existence; the maximum principle supplies uniqueness. □

2.5.2 Problems 9–17: boundaries and the heat equation

9. Lipschitz boundary data can give an unbounded normal derivative. Let bounded continuous half-space data satisfy $g(z) = |z|$ for $|z| \le 1$, $n \ge 2$. Show the Poisson solution has unbounded gradient near zero.

Solution. The boundary value at zero is zero. For $\lambda > 0$,

$$\frac{u(\lambda e_n)}{\lambda} = \frac{2}{n\omega_n} \int_{\mathbb{R}^{n-1}} \frac{g(z)}{(|z|^2 + \lambda^2)^{n/2}}\, dz.$$

The portion $|z| > 1$ is uniformly bounded in absolute value, by boundedness of g and the integrable radial tail. On $\lambda < |z| < 1$, polar coordinates bound the remaining positive integral below by $c_n \int_\lambda^1 d\rho/\rho$, diverging. A bounded gradient would bound this difference quotient, a contradiction. □

10. Odd reflection across a flat zero boundary.

Solution. First assume u is C^2 up to the flat boundary of a half-ball, harmonic inside, and zero on the flat portion. Tangential first and second derivatives vanish there. The PDE then makes $u_{nn} = 0$ there. For the odd extension, tangential derivatives change sign below the plane, normal first derivatives match, mixed derivatives match, and normal second derivatives match because their boundary values are zero. Thus the extension is C^2 and harmonic.

For only continuous boundary regularity, take a small full ball centered on the flat plane. Prescribe its continuous odd-reflected spherical data and solve by (PB). Uniqueness makes this harmonic extension odd, hence zero on the plane. On the upper half-ball it and u have the same full boundary data, so the maximum principle makes them equal. Its lower half is the desired odd reflection. Such balls cover the interface. □

反射要检查到二阶

奇延拓的函数值匹配只是第一步。要直接证明 C^2，还需用零边界值消去切向二阶导数，再用方程推出法向二阶导数为零。

11. Kelvin transform. Show $\overline{u}(x) = |x|^{2-n} u(x/|x|^2)$ is harmonic when u is harmonic, on the inverted domain away from zero.

Solution. Put $r = |x|$, $z = x/r^2$ and $e = x/r$. Then $Dz = r^{-2}(I - 2ee^T)$, so $Dz(Dz)^T = r^{-4}I$, and $\Delta z_i = -2(n-2)x_i/r^4$. Also $\Delta r^{2-n} = 0$ off zero. In the Laplacian product and chain rule, the first-derivative terms cancel: the contribution from $r^{2-n}\Delta z_i$ is opposite to that from $2D(r^{2-n})\cdot Dz_i$. The remaining Hessian term is

$$\Delta_x\big[r^{2-n}u(z)\big] = r^{-n-2}(\Delta u)(z).$$

This proves the claim, including $n = 2$ with prefactor one. □

12. Parabolic scaling and its infinitesimal generator.

Solution. For $\lambda > 0$, define $u_\lambda(x,t) = u(\lambda x, \lambda^2 t)$. The chain rule gives

$$Lu_\lambda = \lambda^2 (Lu)(\lambda x, \lambda^2 t) = 0.$$

Differentiate this identity in λ at one. The result is $L(x \cdot Du + 2tu_t) = 0$. Nonzero negative λ also gives a solution by spatial reflection; $\lambda = 0$ requires data at time zero and is not needed for the argument. □

13. A self-similar profile. For $u(x,t) = v(x/\sqrt{t})$ in one space dimension, derive the kernel.

Solution. For $z = x/\sqrt{t}$, $u_t = -zv'/(2t)$ and $u_{xx} = v''/t$. Thus $v'' + (z/2)v' = 0$, so $v' = ce^{-z^2/4}$ and $v(z) = c\int_0^z e^{-q^2/4}\,dq + d$. Choose $c = (4\pi)^{-1/2}$ and $d = 1/2$ to obtain initial step data. Differentiating in x gives $u_x = (4\pi t)^{-1/2} e^{-x^2/(4t)} = K(x,t)$. The distributional derivative of the initial step is δ_0, so this construction indeed produces the fundamental solution. □

14. Heat flow with reaction and forcing.

Solution. For $u_t - \Delta u + cu = f$, let $v = e^{ct}u$. Then $Lv = e^{ct}f$ and $v(\cdot,0) = g$. Duhamel gives

$$u(x,t) = e^{-ct}(K_t * g)(x) + \int_0^t e^{-c(t-s)}(K_{t-s} * f(\cdot,s))(x)\,ds.$$

□

15. Heat equation on a half-line with boundary forcing. For zero initial data and boundary value $g(t)$, $g(0) = 0$, derive

$$u(x,t) = \int_0^t \frac{x\,e^{-x^2/[4(t-s)]}}{\sqrt{4\pi}(t-s)^{3/2}} g(s)\,ds, \qquad x > 0.$$

Solution. The zero-boundary half-line heat Green kernel is $K(x-y,\tau) - K(x+y,\tau)$. Its inward boundary flux gives the Poisson kernel $P(x,\tau) = -2K_x(x,\tau) = xe^{-x^2/(4\tau)}/(\sqrt{4\pi}\tau^{3/2})$. It satisfies the heat equation for $x, \tau > 0$. The substitution $z = x/(2\sqrt{\tau})$ shows $\int_0^\infty P(x,\tau)\,d\tau = 1$. As $x \downarrow 0$, the mass on $\tau > \delta$ tends to zero. Thus $P(x,\cdot)$ is an approximate identity at time zero. After extending g by zero to negative times, its convolution tends to $g(t)$; compatibility $g(0) = 0$ makes the corner continuous. For fixed $x > 0$, the exponential kills all singular powers at $\tau = 0$, justifying differentiation and the PDE. The time interval shrinks to zero at the initial limit, proving the zero initial datum. □

16. Direct parabolic maximum principle.

Solution. The argument $v = u - \varepsilon t$ in §2.3.3 supplies the proof. At a top-time maximum the left derivative is nonnegative; one does not need to pretend that the terminal point is a two-sided time maximum. □

17. Subsolutions and convex transformations.

Solution. For $Lv \le 0$, (HR) and its limit at zero imply $v(x,t) \le M_r[v]$. The perturbed maximum argument from Problem 16 applies to subsolutions, giving the boundary maximum principle on bounded cylinders. If ϕ is smooth and convex and $Lu = 0$, then

$$L\phi(u) = \phi'(u)Lu - \phi''(u)|Du|^2 \le 0.$$

For a smooth caloric u, spatial and time derivatives are caloric, and $L(w^2) = -2|Dw|^2$ whenever $Lw = 0$. Sum for $w = \partial_i u$ and $w = u_t$:

$$L(|Du|^2 + u_t^2) = -2|D^2u|^2 - 2|Du_t|^2 \le 0.$$

□

椭圆与热的凸性公式

椭圆侧是 $\Delta\phi(u) = \phi''(u)|Du|^2 \ge 0$；热算子侧是 $L\phi(u) = -\phi''(u)|Du|^2 \le 0$。符号不同，得到的平均不等式方向一致。

2.5.3 Problems 18–24: waves, systems, and energy

18. Stokes' rule. If u has wave data $(0,h)$, show $v = u_t$ has data $(h,0)$.

Solution. Time differentiation commutes with $\partial_t^2 - \Delta$, so v solves the same PDE. At zero, $v = h$ and $v_t = u_{tt} = \Delta u = \Delta 0 = 0$. This is also the reason the displacement part of the solution is a time derivative of the velocity solution operator. □

19. Null coordinates and one-way waves.

Solution. The equation $u_{xy} = 0$ first makes u_x independent of y; integrating gives $u = F(x) + G(y)$. For $\xi = x + t$, $\eta = x - t$, the wave operator is $-4\partial_\xi\partial_\eta$, so $u = F(x+t) + G(x-t)$. The initial conditions give $F' = (g' + h)/2$, $G' = (g' - h)/2$, producing (DA). A purely right-moving profile $G(x-t)$ requires $h = -g'$. A purely left-moving profile $F(x+t)$ requires $h = g'$. Additive constants can be allocated to either profile without changing these derivative conditions. □

20. Undistorted radial profiles. For nonzero attenuation $\alpha(r)$ and nonnegative delay $\beta(r)$, $\beta(0) = 0$, suppose $\alpha(r)\phi(t - \beta(r))$ solves the wave equation for every smooth profile ϕ.

Solution. Insert the form into $u_{tt} - u_{rr} - (n-1)u_r/r$. Independence of ϕ, ϕ', ϕ'' gives

$$\alpha(1 - \beta'^2) = 0, \qquad 2\alpha'\beta' + \alpha\beta'' + \frac{n-1}{r}\alpha\beta' = 0,$$

$$\alpha'' + \frac{n-1}{r}\alpha' = 0.$$

For a nontrivial smooth attenuation, the first and the nonnegative-delay condition give $\beta(r) = r$; the middle equation gives $\alpha = Cr^{-(n-1)/2}$. Substitution into the last equation yields $(n-1)(n-3)C = 0$. Thus $n = 1$, $\alpha = C$, or $n = 3$, $\alpha = C/r$. The identically zero attenuation is a trivial exception and has to be excluded to make the dimension claim true. □

量词“任意波形”很强

把 ϕ, ϕ', ϕ'' 的系数分别置零，依赖于要求对所有光滑 ϕ 成立。若只要求某一个特定波形，不能推出相同的维数限制。

21. Maxwell and elastic waves.

Solution. Differentiate $E_t = \operatorname{curl} B$ and use $B_t = -\operatorname{curl} E$:

$$E_{tt} = -\operatorname{curl}\operatorname{curl} E = -\nabla \operatorname{div} E + \Delta E = \Delta E.$$

Likewise $B_{tt} = \Delta B$, using the divergence constraints. For elasticity, $u_{tt} - \mu\Delta u - (\lambda + \mu)\nabla \operatorname{div} u = 0$. Taking divergence and curl respectively gives

$$w_{tt} - (\lambda + 2\mu)\Delta w = 0, \qquad \mathbf{w}_{tt} - \mu\Delta \mathbf{w} = 0,$$

where $w = \operatorname{div} u$ and $\mathbf{w} = \operatorname{curl} u$. When $\mu > 0$ and $\lambda + 2\mu > 0$, the longitudinal and transverse speeds are $\sqrt{\lambda + 2\mu}$ and $\sqrt{\mu}$. □

22. Telegraph equation from velocity switching. Given $u_t + u_x = d(v - u)$ and $v_t - v_x = d(u - v)$, show each component satisfies $w_{tt} + 2dw_t - w_{xx} = 0$.

Solution. Let $a = u+v$, $b = u-v$. Adding and subtracting gives $a_t + b_x = 0$ and $b_t + a_x = -2db$. Differentiate the first in time: $a_{tt} = -b_{xt} = a_{xx} + 2db_x = a_{xx} - 2da_t$. Differentiate the second in time and use $a_{xt} = -b_{xx}$: $b_{tt} = b_{xx} - 2db_t$. Linear combinations give the same telegraph equation for u, v. □

23. A compact space-time source and its outgoing shape. In the diamond $S = \{|x| + |t-2| < 1\}$, take $f = 1$ below $t = x+2$ and $f = -1$ above it, zero elsewhere. Solve with zero wave data and describe u for $t > 3$.

Solution. The one-dimensional Duhamel formula is

$$u(x,t) = \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} f(y,s)\,dy\,ds.$$

Set $\alpha = s + y$, $\beta = s - y$. Then S becomes $1 < \alpha < 3$, $1 < \beta < 3$, and $dy\,ds = \frac{1}{2}d\alpha\,d\beta$. The sign depends only on β: it is $+1$ for $\beta < 2$, -1 for $\beta > 2$. The backward cone imposes $\alpha \le t + x$, $\beta \le t - x$. Define

$$F(a) = \min\{2, \max\{0, a-1\}\},$$

$$H(b) = \begin{cases} 0, & b \le 1, \\ b-1, & 1 < b \le 2, \\ 3-b, & 2 < b < 3, \\ 0, & b \ge 3. \end{cases}$$

The integrals factor, giving $u(x,t) = \frac{1}{4}F(t+x)H(t-x)$. If $t > 3$ and $H(t-x) \neq 0$, then $t + x = 2t - (t-x) > 3$, so $F(t+x) = 2$. Therefore

$$u(x,t) = \tfrac{1}{2}H(t-x) \quad (t > 3).$$

It is a right-moving triangular pulse of height $1/2$, supported on $t - 3 < x < t - 1$, with peak at $x = t - 2$. The sign-balanced source cancels the other outgoing pulse. This piecewise forcing is interpreted through the integral formula; across its jump lines the PDE is read distributionally. □

特征坐标把菱形变方形

$\alpha = s + y$, $\beta = s - y$ 同时把波锥不等式变成坐标上界。二重积分才能分解成两个一维截断积分；漏掉 Jacobian 的 $1/2$ 会让最终脉冲高度错一倍。

24. Equipartition in one dimension. For compactly supported smooth g, h, show total wave energy is constant and kinetic and potential energies eventually coincide.

Solution. Write $u = F(x+t) + G(x-t)$ as in Problem 19. Although the primitives F, G need not be compactly supported, their derivatives $F' = (g' + h)/2$, $G' = (g' - h)/2$ are. Since $u_t = F' - G'$ and $u_x = F' + G'$,

$$k(t) + p(t) = \int_{\mathbb{R}} \left(F'(x+t)^2 + G'(x-t)^2\right) dx$$

is independent of t by translation invariance. Also

$$k(t) - p(t) = -2\int_{\mathbb{R}} F'(x+t)G'(x-t)\,dx.$$

For sufficiently large t the translated compact supports are disjoint, so this integral is zero. If the initial supports lie in $[-R, R]$, every $t > R$ suffices. □

2.6 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 2, printed pp. 17–87 (local PDF pp. 34–104). The exposition here is reconstructed for study; source theorem numbering is retained only as an explicit source identifier.

Evans's §2.6 attributes the heat-ball theorem and the associated strong maximum principle to N. Watson, *Proceedings of the London Mathematical Society* 26 (1973), pp. 385–417. The proof above follows the source's scaling/logarithm method and adds the full mass computation, tip estimates, scale-defect identity, and moment interpretation. The latter supplements are editorial deductions; no separate Watson-paper comparison is claimed. The English source omits the normalization calculation; the Chinese translator's footnote on printed p. 37 suggests polar coordinates and a Gamma substitution. The present proof spells out those constants.

The source points to Gilbarg–Trudinger, Chapters 2–4, for elliptic theory; John and Friedman for the heat equation; Mikhailov for regularity; Payne for backward uniqueness; and Folland and Strauss for wave representation formulas. These are source reading directions, not claims that those books were independently consulted for this draft.

The Chinese chapter opening (PDF p. 31) provides section starts at printed pp. 13, 14, 30, 43, 56, and 60 for §2.1–§2.6. Its mean-value and heat-ball pages were visually checked for terminology and formulas as recorded in the chapter log. This is not a claim of a line-by-line comparison of the entire Chinese chapter.

Scope and review. Elementary calculus, measure-theoretic limit theorems, the Gaussian integral and Gamma integral, and the explicitly invoked complex-analytic parameter-integral/Cauchy facts are prerequisites. General-domain existence of harmonic boundary correctors is left to later Dirichlet theory. All four model equations, their representation methods, and the 24 source exercises have study explanations here; mathematical self-checks and numerical moment checks are not independent mathematical review.

Chapter 3

Nonlinear First-Order PDE

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 3; printed pp. 89–158; PDF pages 106–175.

Chinese reference: 中译本：第三章，书内第 61 页起；章首页为 PDF 第 79 页。

Editorial study edition. This chapter reconstructs Evans's exposition and proofs in the source order; it is not a verbatim transcription. Source theorem identifiers are distinguished from our chapter-wide numbering. Expanded calculations, Chinese mathematical annotations, figures, and exercise solutions are editorial additions.

For $F = F(p, z, x)$ we study $F(Du, u, x) = 0$, with p denoting the gradient variable and z the function-value variable. The main route is

$$\begin{aligned} &\text{envelopes and characteristic ODE} \\ &\longrightarrow \text{ local smooth solutions} \\ \longrightarrow &\text{ variational formulas and admissible weak solutions.} \end{aligned}$$

从局部光滑到全局弱解

特征线在相空间中仍可光滑演化，但投影到物理空间后可能相交。单值光滑解因此失效；Hamilton–Jacobi 选择半凹分支，守恒律选择满足熵条件的跳跃。

Conventions and prerequisites. Spatial gradients are column vectors. Time is a distinguished extra coordinate for evolution equations. Hamiltonians and fluxes are real valued; convexity and superlinearity are stated wherever used. We use local ODE existence/uniqueness and smooth dependence, the implicit and inverse function theorems, standard integration and weak derivative calculus, Rademacher's theorem, and Gronwall's inequality as explicit prerequisites. Supporting hyperplanes for finite convex functions are invoked for the nonsmooth convex-duality statements.

3.1 Complete Integrals, Envelopes

3.1.1 Complete integrals (Evans §3.1.1)

An n-parameter family $u(x;a)$ is a complete integral for $F(Du,u,x)=0$ in n independent variables if every member solves the PDE and the $n\times(n+1)$ matrix

$$(D_a u,\ D^2_{ax}u)$$

has rank n. The condition concerns variation of the whole first jet, not only the gradient.

If this family factored through $n-1$ parameters, $u(x;a)=v(x;\psi(a))$, the chain rule would factor the displayed matrix through $D_a\psi$, whose rank is at most $n-1$. Thus full rank rules out that redundancy.

完全积分不是所有解

它是含足够独立参数的一族解，不能把“完全”理解成已经穷尽全部解。边界值通常是一整个函数，有限多个常数不能直接编码任意边界数据。

Three useful families. For Clairaut's equation $x\cdot Du+f(Du)=u$,

$$u(x;a)=a\cdot x+f(a)$$

solves the PDE and $D^2_{ax}u=I$, proving full rank. For the eikonal equation $|Du|=1$, use $u=a\cdot x+b$ with $|a|=1$. A local chart of the sphere has $n-1$ independent parameters and b adds one; the independent tangential gradient variations and value variation give rank n. This family is locally parametrized by an open subset of $\mathbb{R}^n$, rather than treating the sphere as an open set.

For $u_t+H(D_x u)=0$ in $n+1$ independent variables,

$$u(x,t;a,b)=a\cdot x-tH(a)+b.$$

There are $n+1$ parameters. The a derivatives change $D_x u$ by the standard basis, and the b derivative changes the value by one without changing the gradient. The corresponding jet matrix has rank $n+1$, as required.

3.1.2 Envelopes and singular integrals (Evans §3.1.2, Theorem 1)

Theorem 3.1
Let $U(x;a)$ be a smooth family of solutions and suppose $D_a U(x;\alpha(x))=0$ with α continuously differentiable. Then $v(x)=U(x;\alpha(x))$ is also a solution wherever this construction is defined.

Expanded proof.

For each coordinate, the full chain rule is

$$\partial_i v=\partial_{x_i}U(x;\alpha(x))+\sum_j \partial_{a_j}U(x;\alpha(x))\,\partial_i\alpha_j.$$

The sum vanishes by the stationary-parameter condition. Thus v and the selected family member have the same value and the same spatial gradient at x. Substitution into F proves the PDE. To obtain a smooth stationary parameter locally, one sufficient condition is invertibility of $D^2_{aa}U$, by the implicit function theorem. That condition is sufficient, not part of the abstract envelope identity. □

For $U(x,t;a) = a \cdot x - t|a|^2$, stationarity gives $a = x/(2t)$, and the envelope is $v(x,t) = |x|^2/(4t)$ for $t > 0$. Indeed $v_t = -|x|^2/(4t^2)$ and $|D_x v|^2 = |x|^2/(4t^2)$. This example uses $H(p) = |p|^2$; later the common choice $H = |p|^2/2$ has a different factor of two.

For $U(x;a) = \pm\sqrt{1-|x-a|^2}$, the PDE is $U^2(1+|DU|^2) = 1$. Stationarity yields $a = x$ and the envelopes $v = \pm 1$. The derivative with respect to a of the upper branch is $(x-a)/\sqrt{1-|x-a|^2}$; changing the branch changes the sign.

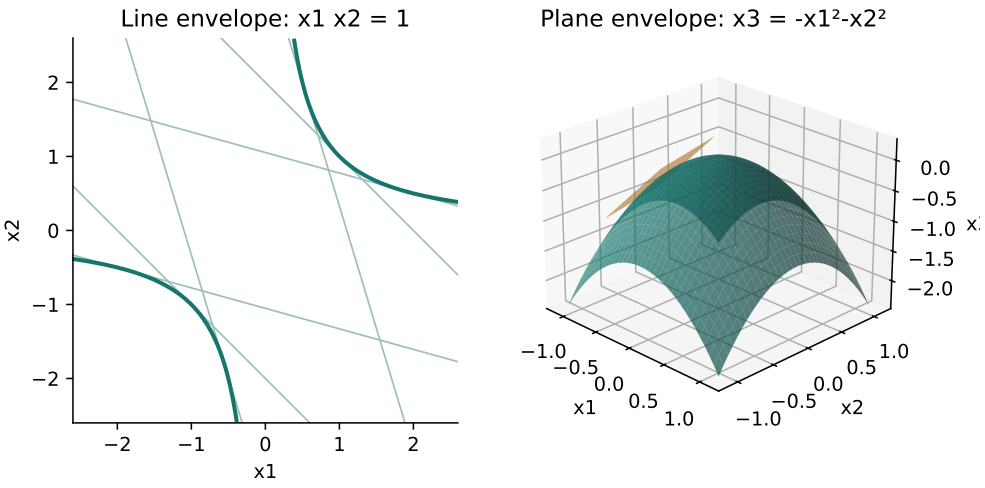
One can restrict the parameters to $a = (a', h(a'))$ and then form an envelope over a'. The stationary equations become $U_{a_i} + U_{a_n} h_{a_i} = 0$. The resulting general integral depends on the arbitrary function h, whenever a smooth envelope exists. It still need not describe every PDE solution: a factorized equation $F_1(Du,u,x)F_2(Du,u,x) = 0$ may have a branch belonging only to F_2.

为何参数导数消失

对 $U(x,a(x))$ 求导时，不能忘掉参数链式项。正是 $D_a U = 0$ 使它消失，包络与被选中的曲面才在该点相切。

二次 Hamiltonian 的系数

若 $H(p) = |p|^2$，其对偶为 $L(v) = |v|^2/4$；若 $H(p) = |p|^2/2$，则 $L(v) = |v|^2/2$。检查公式时，把 PDE、对偶与包络中的系数一起核对。



Editorial figure for Problem 2. A hyperbola is tangent to a family of lines; a negative paraboloid is tangent to the associated family of planes. Elimination and parameter-domain restrictions are worked out in the exercise solution.

3.2 Characteristics

3.2.1 Deriving the full characteristic system (Evans §3.2.1)

Suppose first that a C^2 solution exists. Along a curve $X(s)$ set $Z(s) = u(X(s))$ and $P(s) = Du(X(s))$. Then

$$\dot{P}_i = \sum_j u_{x_i x_j}(X)\dot{X}_j.$$

Differentiate the PDE itself in x_i:

$$\sum_j F_{p_j} u_{x_j x_i} + F_z u_{x_i} + F_{x_i} = 0.$$

Choosing $\dot{X}_j = F_{p_j}$ makes the unknown Hessian sum match the first expression. Hence

$$\boxed{\begin{aligned} \dot{X} &= F_p(P, Z, X), \\ \dot{P} &= -F_x(P, Z, X) - F_z(P, Z, X)P, \\ \dot{Z} &= P \cdot F_p(P, Z, X). \end{aligned}} \tag{CH}$$

All partial derivatives of F here hold its other arguments fixed. The third equation is the chain rule for $u(X(s))$.

选择路径是为了消去 Hessian

任意路径的梯度变化含 D^2u，并不封闭。取速度 F_p 后，恰好能用微分后的 PDE 替换这组二阶导数，得到只含 X, P, Z 的 ODE。

An invariant of the system. Even before knowing $P = Du$, solutions of (CH) satisfy

$$\frac{d}{ds}F(P, Z, X) = F_p \cdot (-F_x - F_z P) + F_z(P \cdot F_p) + F_x \cdot F_p = 0.$$

Initial data with $F(P(0), Z(0), X(0)) = 0$ therefore remain on the PDE constraint surface. This invariant alone does not yet prove that the transported vector P is the gradient of a single-valued function.

3.2.2 Reduced systems and explicit examples (Evans §3.2.2)

For $b(x) \cdot Du + c(x)u = f(x)$, only

$$\dot{X} = b(X), \qquad \dot{Z} = f(X) - c(X)Z$$

are needed to compute values. If $b = b(x, u)$, the value equation remains $\dot{Z} = -c(X, Z)$ for $b(x, u) \cdot Du + c(x, u) = 0$; the path now depends on the unknown value.

Rotation. For $x_1 u_{x_2} - x_2 u_{x_1} = u$ in the first quadrant with $u(r, 0) = g(r)$,

$$X(s) = (r\cos s, r\sin s), \qquad Z(s) = e^s g(r).$$

Thus $u(x_1, x_2) = e^{\arctan(x_2/x_1)} g(\sqrt{x_1^2 + x_2^2})$. This is also immediate from $u_\theta = u$ in polar coordinates.

Value blow-up without crossing. For $u_{x_1} + u_{x_2} = u^2$, $u(x_1, 0) = g(x_1)$,

$$X(s) = (r + s, s), \qquad Z(s) = \frac{g(r)}{1 - sg(r)},$$

so

$$u(x_1, x_2) = \frac{g(x_1 - x_2)}{1 - x_2 g(x_1 - x_2)}$$

on the connected region reached before a denominator vanishes. If $g(r) > 0$, the characteristic value blows up at $s = 1/g(r)$. Continuing the rational expression beyond its pole is not continuing a finite classical solution through the blow-up time.

两种失效机制

半线性例子可在路径不相交时因 Z 爆破失效；守恒律常见的是值仍有界、空间梯度先爆破。只检查特征线位置还不够。

A fully nonlinear example. For $u_{x_1} u_{x_2} = u$, $u(0, r) = r^2$, compatibility gives $P_2(0) = 2r$ and $P_1(0) = r/2$ when $r \neq 0$. The system is $\dot{P}_i = P_i$, $\dot{X}_1 = P_2$, $\dot{X}_2 = P_1$, $\dot{Z} = 2P_1 P_2$. It integrates to

$$X_1 = 2r(e^s - 1), \quad X_2 = \tfrac{r}{2}(e^s + 1), \quad Z = r^2 e^{2s}.$$

Since $4X_2 - X_1 = 4r$ and $4X_2 + X_1 = 4re^s$,

$$u(x_1, x_2) = \frac{(x_1 + 4x_2)^2}{16}.$$

Direct differentiation verifies the PDE and boundary data everywhere. The characteristic construction just given is locally invertible only where $r \neq 0$; the polynomial formula extends across the excluded locus. An explicit extension does not mean the local existence theorem was applicable at the characteristic boundary point $r = 0$.

3.2.3 Admissible and noncharacteristic data (Evans §3.2.3)

Flatten the initial hypersurface to $x_n = 0$ and write its coordinates as $y \in \mathbb{R}^{n-1}$. Prescribing $u(y, 0) = g(y)$ fixes

$$Z(0) = g(y), \quad P_i(0) = g_{y_i}(y) \ (i < n), \quad F(P(0), g(y), (y, 0)) = 0. \tag{IC}$$

Only the normal derivative $P_n(0)$ remains to be found. At an admissible root p^0, the sufficient condition $F_{p_n}(p^0, g(y_0), (y_0, 0)) \neq 0$ lets the implicit function theorem solve for a smooth branch $q_n(y)$. Different roots may produce different local solutions; uniqueness is relative to the chosen admissible branch.

For a curved surface, choose a smooth extension of g and write $p = D_{\tan} g + \lambda \nu$, where ν is its unit normal. Differentiating the scalar constraint in λ gives

$$\frac{d}{d\lambda} F(D_{\tan} g + \lambda \nu, g, x) = F_p \cdot \nu.$$

Thus the invariant geometric noncharacteristic condition is $F_p \cdot \nu \neq 0$: the characteristic velocity crosses the surface. Changing the normal's orientation changes the sign, not this condition.

For example $u_{x_2} = 0$ with data on $x_2 = 0$ is noncharacteristic. The equation $u_{x_1} = 0$ on the same boundary is characteristic: nonconstant data in x_1 are incompatible, while constant data leave the x_2 dependence undetermined. Failure of the hypothesis can mean either nonexistence or nonuniqueness.

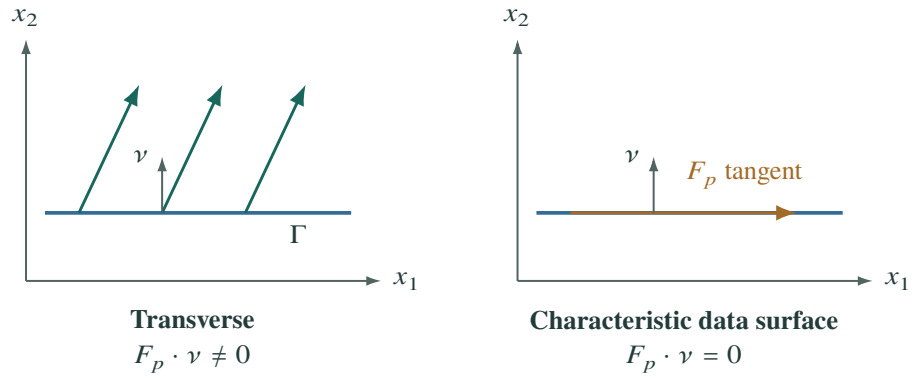


Diagram T3.1. Noncharacteristic data permit local reconstruction. The projected characteristic direction is F_p. Transversality makes the initial-surface parameters together with the flow parameter independent local coordinates. A tangent direction makes this coordinate construction singular; it does not by itself prove that no solution can exist.

3.2.4 Reconstructing a local solution (Evans §3.2.4, Theorem 2)

Theorem 3.2

For smooth F, g and a chosen admissible noncharacteristic triple, (CH) with data (IC) reconstructs a smooth local solution. The construction is unique locally among smooth solutions sharing that branch of boundary gradient.

Expanded proof (including the gradient compatibility step).

1. Solve the ODE with boundary parameters. Use the boundary-parameterized solution

$$(X, P, Z)(y, 0) = ((y, 0), q(y), g(y)).$$

边界值没有给出完整梯度

沿边界求导只能得到切向分量，法向分量要由 PDE 求出。代数方程可能无根或多根；非特征条件保证的是已选根附近的局部唯一分支。

非特征条件是局部条件

$F_p \cdot \nu \neq 0$ 让特征穿过初始曲面，从而给出局部坐标。它不保证以后投影仍可逆；法向速度的正负另用于区分流入、流出方向。

Local ODE theory with smooth dependence supplies a common parameter neighborhood. At $s = 0$, the columns X_{y_i} are e_i, $i < n$, and $X_s = F_p$. Hence

$$\det D_{(y,s)} X(y_0, 0) = F_{p_n}(p^0, z^0, x^0) \neq 0.$$

The inverse function theorem gives local coordinates $(y, s) = (y(x), s(x))$. Define $u(x) = Z(y(x), s(x))$.

2. Preserve the algebraic PDE constraint. The invariant above and initial compatibility give $F(P(y,s), Z(y,s), X(y,s)) = 0$. We still have to prove $P(y,s) = Du(X(y,s))$.

3. Propagate the contact identities. The s identity $Z_s = P \cdot X_s$ is in (CH). For every boundary parameter set $R_i = Z_{y_i} - P \cdot X_{y_i}$. Initially $R_i = 0$, by the tangential compatibility in (IC). Differentiate:

$$\begin{aligned}(R_i)_s &= P_{y_i} \cdot X_s - P_s \cdot X_{y_i} \\ &= F_p \cdot P_{y_i} + F_x \cdot X_{y_i} + F_z P \cdot X_{y_i}.\end{aligned}$$

Differentiating the preserved identity $F(P, Z, X) = 0$ in y_i yields $F_p \cdot P_{y_i} + F_x \cdot X_{y_i} = -F_z Z_{y_i}$. Therefore $(R_i)_s = -F_z R_i$ with zero initial value. Uniqueness for this scalar linear ODE gives $R_i = 0$.

4. Return to physical coordinates. Together the identities say $dZ = P \cdot dX$ for all parameter differentials. Since $D_{(y,s)}X$ is invertible, every physical differential occurs, so $Du = P$. The preserved constraint now becomes $F(Du, u, x) = 0$. At $s = 0$ we recover $u = g$. Any other smooth solution with the same boundary gradient branch generates the same characteristic ODE data, so ODE uniqueness and coordinate invertibility identify the solutions. □

3.2.5 Evolution equations and breakdown (Evans §3.2.5)

For $u_t + \mathbf{f}'(u) \cdot D_x u = 0$ with initial data g,

$$X(a,t) = a + t\mathbf{f}'(g(a)), \qquad u(X(a,t),t) = g(a).$$

Eliminating a gives $u = g(x - t\mathbf{f}'(u))$ while the map is invertible. Its Jacobian is $I + t\,\mathbf{f}''(g(a)) \otimes Dg(a)$. The rank-one determinant identity yields

$$\det D_a X = 1 + t\, Dg(a) \cdot \mathbf{f}''(g(a)).$$

In one dimension,

$$u_x(X(a,t),t) = \frac{g'(a)}{1 + tf''(g(a))g'(a)}.$$

不能提前把 P 叫作梯度

从 ODE 构造出来的 P 起初只是向量场。必须证明 $dZ = P \cdot dX$，再利用坐标可逆性，才能认定 $P = Du$；否则只保留 $F(P, Z, X) = 0$ 还没有解出 PDE。

If the denominator first vanishes with $g'(a) < 0$ and $f'' > 0$, gradients blow up although $u = g(a)$ remains finite. For Burgers flux $f(u) = u^2/2$ and bounded C^1 data with $m = \inf g' < 0$, the first loss of a uniform positive Jacobian is at $T_* = -1/m$. If the infimum is attained, that characteristic gives an actual zero Jacobian at T_*.

For $u_t + H(D_x u, x) = 0$, the full system becomes Hamilton's equations and an action equation:

$$\dot{X} = H_p(P, X), \qquad \dot{P} = -H_x(P, X), \qquad \dot{Z} = P \cdot H_p - H.$$

When $H = H(p)$, $P = Dg(a)$ stays constant and

$$X = a + tDH(Dg(a)), \qquad Z = g(a) + t[Dg(a) \cdot DH(Dg(a)) - H(Dg(a))].$$

This formula is local until the projection ceases to be invertible. The next section gives a variational continuation with a selection rule.

相空间不交不保证投影不交

扩展状态 (X, P, Z) 的 ODE 可保持唯一性，但只看 X 时不同状态可落在同一位置。物理空间的逆映射失效，正是特征构造中光滑解失效的关键。

3.3 Introduction to Hamilton–Jacobi Equations

3.3.1 Variations and Hamilton's ODE (Evans §3.3.1)

For a path $X : [0, t] \to \mathbb{R}^n$ with fixed endpoints consider

$$\mathcal{A}[X] = \int_0^t L(\dot{X}(s), X(s))\, ds.$$

Suppose a smooth minimizer exists. For every smooth variation V vanishing at both endpoints, differentiate the action at $X + \varepsilon V$:

$$0 = \int_0^t [L_v(\dot{X}, X) \cdot \dot{V} + L_x(\dot{X}, X) \cdot V]\, ds.$$

Integration by parts, with the endpoint terms zero, gives $\frac{d}{ds} L_v(\dot{X}, X) = L_x(\dot{X}, X)$ by the fundamental lemma of the calculus of variations. These are the Euler–Lagrange equations. They are necessary conditions; their derivation does not prove a minimizer exists or that every stationary path minimizes.

Assume now the velocity–momentum relation $P = L_v(v, x)$ can be inverted, and define $H(P, x) = P \cdot v(P, x) - L(v(P, x), x)$. In differentiating this identity the terms containing derivatives of v cancel because $P = L_v$. Thus $H_p = v$ and $H_x = -L_x$. The Euler–Lagrange equations become

$$\dot{X} = H_p(P, X), \qquad \dot{P} = -H_x(P, X).$$

For $L(v, x) = |v|^2/2 - V(x)$ this gives $P = v$, $H = |P|^2/2 + V(x)$ and $\ddot{X} = -DV(X)$. For an autonomous Hamiltonian, $\frac{d}{ds} H = H_p \cdot (-H_x) + H_x \cdot H_p = 0$.

3.3.2 Convex duality and Hopf–Lax (Evans §3.3.2)

A function A is superlinear if $A(v)/|v| \to +\infty$ as $|v| \to \infty$. For a finite convex function define

$$A^*(p) = \sup_v \{p \cdot v - A(v)\}.$$

Superlinearity of A makes the objective tend to $-\infty$, so the maximum is attained and A^* is finite. It is convex as a supremum of affine functions. For every $R > 0$,

$$A^*(p) \geq R|p| - \max_{|v|=R} A(v);$$

hence A^* is also superlinear when A is finite everywhere.

Theorem 3.3 (Convex duality; source Theorem 3)
For finite convex superlinear A, $A^{**} = A$ and

$$A(v) + A^*(p) \geq p \cdot v.$$

Equality holds precisely when p is a supporting slope of A at v. For differentiable A, this means $p = DA(v)$.

Expanded proof.

The inequality is the definition of the supremum. A supporting slope satisfies $A(w) \geq A(v) + p \cdot (w - v)$ for every w. Rearranging shows the supremum defining $A^*(p)$ is attained at v, giving equality. Conversely equality rearranges to that support inequality. Always $A^{**}(v) \leq A(v)$. Choose a supporting slope at v; then the equality just established gives the reverse inequality. Existence of supporting hyperplanes for finite convex functions is the invoked convex-analysis prerequisite. For a differentiable convex function the tangent-plane inequality provides this slope directly as $DA(v)$. □

The supporting-plane argument for $A^{**} = A$ also holds for every finite convex A without superlinearity; in that case A^* may take the value $+\infty$. Subdifferentials of such extended-valued functions are taken only at points where the function is finite.

两个来源得到同一组 ODE

变分法从作用量的驻值推出 Hamilton 方程；特征线法从 Hamilton–Jacobi PDE 推出同一方程。前者给出力学意义，后者解释它如何搬运函数与梯度。

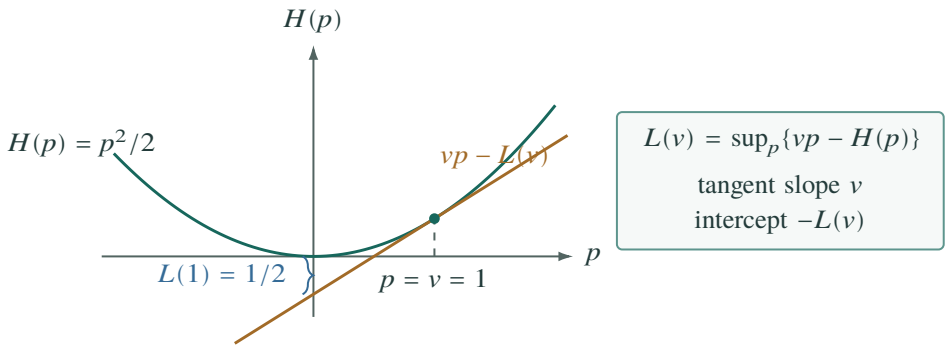


Diagram T3.2. Legendre duality is a supporting-line construction. For $H(p) = p^2/2$ and $v = 1$, maximizing $vp - H(p)$ finds the tangent point $p = 1$. The tangent's intercept is $-L(1) = -1/2$. Convexity makes the whole graph lie above this line, which is exactly the Fenchel inequality.

For the rest of this subsection assume H is finite, convex, smooth and superlinear on $\mathbb{R}^n$, set $L = H^*$, and let g be globally Lipschitz with constant K. Then L is finite, convex and superlinear, and $H = L^*$.

Why a straight path gives the formula (source Theorem 4). Since L no longer depends on position, Jensen's inequality gives for any absolutely continuous path with $X(0) = y$, $X(t) = x$,

$$\int_0^t L(\dot X(s))\,ds \geq tL\left(\frac{1}{t}\int_0^t \dot X(s)\,ds\right) = tL((x-y)/t).$$

The straight path $X(s) = y + s(x-y)/t$ attains equality. Minimizing over its starting point gives

$$\boxed{u(x,t) = \min_{y\in\mathbb{R}^n}\{tL((x-y)/t) + g(y)\}.} \qquad \text{(HL)}$$

The minimum exists: with $v = (x-y)/t$, its objective is at least $g(x) + t[L(v) - K|v|]$, coercive in v. Comparing with $y = x$ also proves that every minimizer satisfies $L(v) - K|v| \leq L(0)$, hence $|v| \leq R_K$ for a fixed finite R_K. This is the first finite-range localization of the minimization.

对偶变换中的极大值

求 $v \cdot p - H(p)$ 的驻点只是候选；凸性让该目标关于 p 凹，超线性增长使它在无穷远趋负无穷，才保证极大值确实取得。

超线性排除逃向远方的起点

g 的下降至多是线性的，而 $tL((x-y)/t)$ 在位移很大时增长更快。这不仅给出最小值存在，还把最优起点限制在 $|x-y| \leq Rt$ 内。

直线最优依赖哪些条件

这里用到 L 的凸性和无位置依赖。若 $L = L(v,x)$，不同路径经过不同势场，不能只用端点位移除以时间就替代整条作用量。

Lemma 3.4 (Dynamic programming; source Lemma 1)
For $0 < s < t$,

$$u(x,t) = \min_z\{(t-s)L((x-z)/(t-s)) + u(z,s)\}. \qquad \text{(DP)}$$

Expanded proof.

For any z and a minimizing starting point y for $u(z,s)$, convexity applied to $(x-y)/t = [(t-s)/t](x-z)/(t-s) + (s/t)(z-y)/s$ gives $u(x,t) \le (t-s)L((x-z)/(t-s)) + u(z,s)$. This proves one inequality for the infimum over z. Conversely take a minimizer y in (HL) and choose $z = y + (s/t)(x-y)$. Both subinterval velocities then equal $(x-y)/t$. Use this same y as a competitor for $u(z,s)$ to obtain the opposite inequality. Coercivity ensures the intermediate minimum exists. □

Lemma 3.5 (Lipschitz bounds and initial values; source Lemma 2)
The function in (HL) is globally Lipschitz in (x,t) *for* $t \ge 0$, *when defined by* $u(x,0) = g(x)$.

Expanded proof.

If y minimizes at x, use $y + h$ as competitor at $x + h$. The velocity is unchanged, so $u(x+h,t) - u(x,t) \le K|h|$. Exchange $x, x+h$ for the reverse inequality.

For time zero, the competitor $y = x$ gives $u(x,t) \le g(x) + tL(0)$, while duality gives

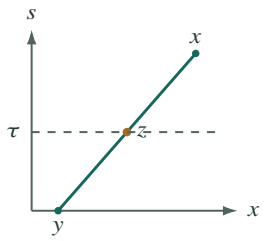
$$u(x,t) \ge g(x) + t \inf_v [L(v) - K|v|] = g(x) - t \max_{|p|\le K} H(p).$$

Thus $|u(x,t) - g(x)| \le Ct$, uniformly in x, with

$$C = \max\{|L(0)|, \max_{|p|\le K} |H(p)|\}.$$

Apply the same argument to (DP), using the spatial Lipschitz bound for $u(\cdot,s)$, to get $|u(x,t) - u(x,s)| \le C|t-s|$. □

图 T3.M：把路径在中途切开



中间点 z 也要优化。两段代价相加后再取下确界，才得到动态规划恒等式。

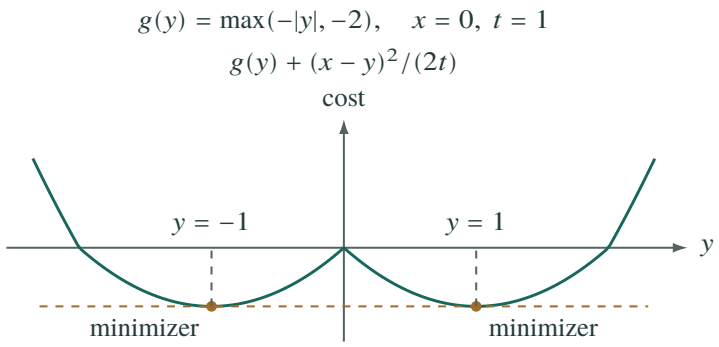


Diagram T3.3. Equal minimizing costs can produce a corner in the value. With $H(p) = p^2/2$, $t = 1$ and $g(y) = \max(-|y|, -2)$, the Hopf–Lax cost at $x = 0$ has two minimizers $y = \pm 1$. Their candidate endpoint slopes $(x-y)/t$ differ. Indeed near $x = 0$ the value is $u(x,1) = -1/2 - |x|$, so it is not differentiable there.

Theorem 3.6 (PDE at differentiability points; source Theorems 5–6)
The Hopf–Lax function satisfies $u_t + H(Du) = 0$ at every point where it is differentiable, hence almost everywhere by Rademacher's theorem.

Expanded proof.

At a differentiability point (x,t), (DP) with the competitor x over a short forward time gives $u(x+hv, t+h) \le hL(v) + u(x,t)$ for every fixed v. Divide by $h > 0$ and pass to zero: $u_t + v \cdot Du \le L(v)$. Supremizing in v yields $u_t + H(Du) \le 0$.

For the other sign choose a minimizing y in (HL) at (x,t) and put $v_* = (x-y)/t$. At time $t-h$, $z = x - hv_*$ lies on the straight minimizing path. Using y as competitor there yields $u(x,t) - u(z, t-h) \ge hL(v_*)$. Differentiability gives $u_t + v_* \cdot Du \ge L(v_*)$. Since $H(Du) \ge v_* \cdot Du - L(v_*)$, the reverse sign follows. Neither argument differentiates a potentially nonsmooth minimizing point $y(x,t)$. □

For $H(p) = |p|^2/2$, $L(v) = |v|^2/2$ and $u(x,t) = \min_y\{|x-y|^2/(2t) + g(y)\}$. At a smooth minimizing branch, $(x-y)/t = Dg(y)$, so $x = y + tDg(y)$ and $Du = Dg(y)$, recovering the local characteristic formulas. When branches compete, the minimum remains meaningful even though their smooth projection has broken down.

最小值不一定可微

多个最优起点可能同时出现，不能直接对 $y(x,t)$ 求导。动态规划用固定竞争者给一侧不等式，再沿一条最优路径给另一侧，避开了这个问题。

3.3.3 Semiconcavity and uniqueness (Evans §3.3.3)

An a.e. PDE alone is insufficient for uniqueness. For $H(p) = p^2$ and $g = 0$ in one dimension, both $u = 0$ and

$$v(x,t) = \min\{0, |x| - t\}$$

are Lipschitz and satisfy $v_t + v_x^2 = 0$ off the three lines $x = 0, \pm t$. At the central upward corner, $v(h,t) + v(-h,t) - 2v(0,t) = 2|h|$ for $|h| < t$. This cannot be bounded by a constant times h^2.

Definition 3.7

A function f is semiconcave with constant C if

$$f(x+z) - 2f(x) + f(x-z) \le C|z|^2.$$

For continuous f this is equivalent to concavity of $f(x) - C|x|^2/2$, by the midpoint criterion and continuity. For smooth f it is equivalent to $D^2 f \le CI$.

半凹只限制向上弯曲

允许向下尖角，禁止二阶差分过大的向上尖角。它不是要求 Hessian 的绝对值有界，也不是要求函数本身凹；先减去一个二次函数后才凹。

Theorem 3.8 (Source Lemmas 3–4)

If g is semiconcave with constant C, the Hopf–Lax function has the same spatial constant for all t. If instead $D^2 H \ge \theta I$, $\theta > 0$, it satisfies

$$u(x+z,t) - 2u(x,t) + u(x-z,t) \le \frac{|z|^2}{\theta t},$$

even for initial g that is merely Lipschitz.

Expanded proof (including the dual estimate of Problem 15).

For the first assertion choose a minimizer y at (x,t). Use $y+z$ and $y-z$ as competitors at $x+z$ and $x-z$. All three velocities are equal, so their L terms cancel. The remaining second difference is that of g at y.

For the second assertion, uniform convexity makes the maximizer $p(v)$ in $L(v) = \max_p(v \cdot p - H(p))$ unique, with $v = DH(p(v))$. Strong monotonicity gives

$$\theta |p(v) - p(w)|^2 \le (v - w) \cdot (p(v) - p(w)),$$

so $|p(v) - p(w)| \le |v - w|/\theta$. Comparing the optimizing expressions at v and $v + h$ shows $DL(v) = p(v)$; equivalently, apply the implicit function theorem to $DH(p) = v$. Hence DL is $1/\theta$-Lipschitz. Integrating its increment along a segment gives

$$L(v+h) \le L(v) + DL(v) \cdot h + \frac{|h|^2}{2\theta}.$$

Add the estimates for h and $-h$: $L(v+h)+L(v-h)-2L(v)\le |h|^2/\theta$. Now use the same minimizing y at all three target points. The initial terms cancel and $h=z/t$ gives the claimed constant.

In particular, taking $v=(v_1+v_2)/2$, $h=(v_1-v_2)/2$ proves

$$\tfrac12 L(v_1)+\tfrac12 L(v_2)\le L((v_1+v_2)/2)+\frac{|v_1-v_2|^2}{8\theta}.$$

This is the dual inequality omitted from the source lemma's proof. □

Theorem 3.9 (Uniqueness in the source weak class; source Theorems 7–8)
Let H be smooth, convex and superlinear. Among globally Lipschitz functions with the same initial datum, satisfying the HJ equation a.e. and the spatial semiconcavity bound

$$u(x+z,t)-2u(x,t)+u(x-z,t)\le C(1+t^{-1})|z|^2,$$

there is at most one solution. The Hopf–Lax function belongs to this class if g is semiconcave or H is uniformly convex.

Expanded proof (the source's regularized cone argument).

Let $w=u-\tilde u$. Subtract the a.e. PDE and integrate DH on the gradient segment:

$$w_t+b\cdot Dw=0,\qquad b=\int_0^1 DH(\lambda Du+(1-\lambda)D\tilde u)\,d\lambda.$$

Both gradients lie in a fixed ball, so $|b|\le R$ for a finite R.

1. Smooth the coefficients, preserving a one-sided bound. Locally convolve $u,\tilde u$ in space-time and define b_ε by the same expression with their smoothed spatial gradients. For $t>2\varepsilon$, semiconcavity implies

$$D^2u_\varepsilon,\ D^2\tilde u_\varepsilon\le C'(1+t^{-1})I.$$

To check this, integrate their second-difference inequalities against the nonnegative smoothing kernel, bound $1/s$ by $2/t$ on its time support, then divide by the squared increment and let it tend to zero. The gradients remain bounded and converge a.e. to the original ones.

Writing $A=D^2H\ge 0$ and B for the convex combination of smoothed Hessians gives $B\le C'(1+t^{-1})I$. Thus $\operatorname{tr}(AB)\le C'(1+t^{-1})\operatorname{tr}A$, by conjugating with $A^{1/2}$ and taking traces. Since A is bounded on the gradient ball, $\operatorname{div} b_\varepsilon\le C''(1+t^{-1})$. Also $|b_\varepsilon|\le R$ and $b_\varepsilon\to b$ a.e.

2. Integrate a nonnegative transform in a shrinking cone. Take a smooth nonnegative function ϕ, and put $v = \phi(w)$. Weak Lipschitz calculus gives $v_t + b \cdot Dv = 0$ a.e. On $B_{\rho(t)}(x_0)$, $\rho(t) = r + R(T - t)$ with $r > 0$, set $e(t) = \int v$. The moving-boundary term and $|b_\varepsilon| \le R$ make its boundary flux nonpositive. For a.e. t this yields

$$e'(t) \le C''(1 + t^{-1})e(t) + \int_{B_{\rho(t)}} (b_\varepsilon - b) \cdot Dv.$$

These integrations hold for Lipschitz v by approximation, or its Sobolev integration formula on the smooth ball. For each fixed $\delta > 0$, the error tends to zero in $L^1(\delta, T)$ by dominated convergence on the compact cone. Integrate against a nonnegative time test function and let $\varepsilon \to 0$, obtaining the same differential inequality without the error on $[\delta, T]$.

3. Start Gronwall away from the singular time zero. Let M be the sum of the time Lipschitz constants. Since $w(\cdot, 0) = 0$, $|w(x, \delta)| \le M\delta$. Choose $\phi = \phi_\delta$ zero on $[-M\delta, M\delta]$ and strictly positive outside. Then $e(\delta) = 0$. Gronwall on $[\delta, T]$ has a finite coefficient integral, and gives $e(T) = 0$. Hence $|w| \le M\delta$ on $B_r(x_0)$ at time T, using continuity to pass from a.e. points. Let $\delta \downarrow 0$. The ball and T were arbitrary, so $w = 0$. Existence in the stated class follows from the two semiconcavity estimates and the preceding Hopf–Lax theorem. □

This is the semiconcavity-based weak formulation used in this chapter. Viscosity solutions provide a broader comparison framework in Chapter 10; the a.e. equation and the present selection condition must not be silently interchanged.

凸性把单边 Hessian 界传给速度

两个半凹函数的 Hessian 只需有上界。因 D^2H 半正定，迹运算仍给出 $\operatorname{div} b$ 的上界；若失去凸性，这一步的符号便没有保证。

不能从零时刻硬套 Gronwall

$1/t$ 在零附近不可积。证明先用时间 Lipschitz 界把差值截掉，在正时刻 δ 得到零初始能量，再应用 Gronwall，最后令 $\delta \to 0$。

3.4 Introduction to Conservation Laws

3.4.1 Shocks and entropy (Evans §3.4.1)

We now work in one space dimension:

$$u_t + f(u)_x = 0, \qquad u(x, 0) = g(x).$$

The conservative form must be retained after smoothness breaks down. An integral solution is a locally bounded function satisfying

$$\int_0^\infty \int_{\mathbb{R}} [u\phi_t + f(u)\phi_x]\, dx\, dt + \int_{\mathbb{R}} g(x)\phi(x, 0)\, dx = 0 \qquad \text{(CL)}$$

for every smooth compactly supported test function on $\mathbb{R} \times [0, \infty)$. For smooth u, integration by parts proves equivalence with the PDE. For discontinuous u, the expression $f'(u)u_x$ is not an ordinary product of functions and cannot replace (CL) without additional theory.

Theorem 3.10 (Rankine–Hugoniot)
Across a smooth shock curve $x = \gamma(t)$ with traces u_l, u_r, an otherwise smooth integral solution must satisfy

$$\gamma'(t)(u_r - u_l) = f(u_r) - f(u_l). \tag{RH}$$

For unequal states, the speed is the secant slope of f.

Expanded proof.

Write locally $u = u_l + (u_r - u_l)\mathbf{1}_{\{x>\gamma(t)\}}$, with the smooth extensions on each side understood. The singular part of u_t is $-\gamma'[u]\delta_{x=\gamma(t)}$; that of $f(u)_x$ is $[f]\delta_{x=\gamma(t)}$. Here $[u] = u_r - u_l$ and $[f] = f(u_r) - f(u_l)$. The regular parts vanish by the PDE on each side. The remaining distribution is zero precisely when (RH) holds. Equivalently, integrate each smooth side of (CL) and add their oppositely oriented interface fluxes. □

For strictly convex f, an admissible shock has $u_l > u_r$. Its speed satisfies $f'(u_r) < \gamma' < f'(u_l)$, since secant slopes lie strictly between endpoint tangent slopes. Characteristics enter the shock from both sides. An upward jump also satisfies (RH) with its secant speed, but would send characteristics outward and is rejected by entropy.

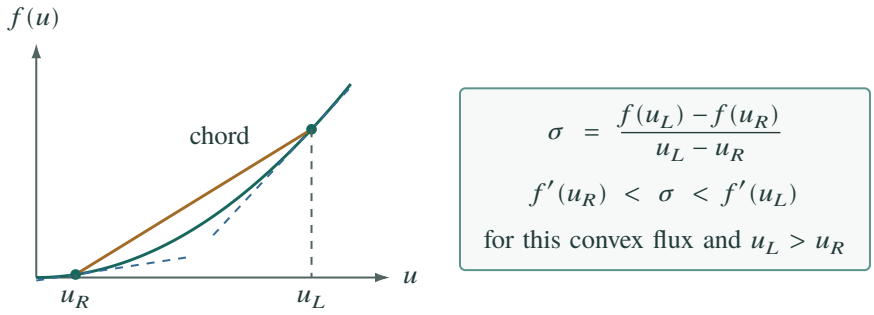


Diagram T3.4. Shock speed is a secant slope. For $f(u) = u^2/2$, the secant between $u_R = 0.2$ and $u_L = 1.4$ has slope $\sigma = 0.8$. The tangent slopes satisfy $0.2 < 0.8 < 1.4$, so characteristics from both sides enter the shock. Reversing the spatial states keeps the secant slope but violates the compression direction.

Why weak conservation alone is not unique. For Burgers flux $f(u) = u^2/2$ and data $g = 0$ on $x < 0$, $g = 1$ on $x > 0$, both the upward step

先守恒后求导

光滑时 $f(u)_x = f'(u)u_x$；跳跃时 u_x 含集中测度。守恒形式能直接对 $f(u)$ 作分布求导，避免给不连续函数与 Dirac 的乘积随意赋值。

moving at speed $1/2$ and

$$u(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t \end{cases}$$

satisfy (CL). In the fan the PDE is $-x/t^2 + (x/t)(1/t) = 0$, and both interfaces are continuous. Each candidate attains the same initial data in local L^1. Only the fan satisfies the one-sided bound derived below.

跳跃条件不负责筛选方向

Rankine–Hugoniot 只保证接口两侧通量守恒，上跳和下跳都可能满足。熵条件额外保留压缩型跳跃，排除向外散开的非物理解。

Convex entropy pairs: an editorial explanation. For smooth convex η, define q by $q' = \eta' f'$. A smooth solution satisfies $\eta(u)_t + q(u)_x = 0$. For the viscous equation $u_t + f(u)_x = \varepsilon u_{xx}$,

$$\eta(u)_t + q(u)_x = \varepsilon \partial_{xx}\eta(u) - \varepsilon \eta''(u)|u_x|^2 \le \varepsilon \partial_{xx}\eta(u).$$

If bounded viscous solutions converge strongly locally, their limit therefore satisfies the distributional entropy inequality. This is a conditional explanation of the selection rule, not a proof of viscous existence or convergence.

For a downward shock $l = u_l > u_r = r$ with secant speed s, let $C(v) = f(v) - f(r) - s(v-r) \le 0$ on $[r,l]$. The singular entropy coefficient is

$$[q] - s[\eta] = -\int_r^l \eta'(v)C'(v)\,dv = \int_r^l \eta''(v)C(v)\,dv \le 0.$$

Thus the convex entropy inequality selects the same direction. For general nonsmooth solutions this chapter uses the source's one-sided derivative condition as its explicit uniqueness class.

3.4.2 The Lax–Oleinik construction (Evans §3.4.2)

Assume f smooth and $f'' \ge \theta > 0$ on $\mathbb{R}$, and $g \in L^\infty$. An additive constant in f does not affect the PDE; take $f(0) = 0$. Let

$$P(x) = \int_0^x g(y)\,dy, \qquad L = f^*, \qquad G = (f')^{-1} = L'.$$

The potential

$$W(x,t) = \min_y \{tL((x-y)/t) + P(y)\} \tag{LOP}$$

is a Hopf–Lax solution of $W_t + f(W_x) = 0$. The candidate conservation-law solution is $u = W_x$ a.e.

Since f' is strictly increasing and onto, G exists and is $1/\theta$-Lipschitz. Differentiating the convex dual at its unique maximizer gives $L' = G$ and $L'' = 1/f''(G) > 0$.

Theorem 3.11 (Source Theorem 1)
For each $t > 0$, minimizing starting points in (LOP) are ordered with x. The minimizer is unique except at at most countably many x. At differentiability points of W,

$$\boxed{u(x,t) = G((x - y(x,t))/t).} \tag{LO}$$

Expanded proof.

1. Ordering and countability. Put $c(x,y) = tL((x-y)/t)$. Its mixed derivative is $c_{xy} = -L''((x-y)/t)/t < 0$. If $x_1 < x_2$ and proposed minimizers had $y_1 > y_2$, integration of this mixed derivative on the rectangle gives

$$c(x_1,y_2) + c(x_2,y_1) < c(x_1,y_1) + c(x_2,y_2).$$

Adding $P(y_1) + P(y_2)$ contradicts the sum of the two minimizing inequalities. Hence every minimizer at x_1 is at most every minimizer at x_2. The minimum sets are nonempty compact sets by coercivity. If a set has two distinct endpoints $y_- < y_+$, its interval (y_-, y_+) is disjoint from the corresponding intervals for other x. Assigning a rational point to each nonempty interval proves there are at most countably many such x.

2. Differentiate the value without differentiating the optimizer. For a fixed minimizing y, use that same point as a competitor at $x + h$. Then $W(x+h,t) - W(x,t) \le c(x+h,y) - c(x,y)$. For positive and negative h, differentiability of W gives $W_x = L'((x-y)/t) = G((x-y)/t)$. This works even if P is not differentiable at the minimizing y. Strict monotonicity of G also forces uniqueness of y at such a point. □

Theorem 3.12 (Integral solution and initial trace; source Theorem 2)
The construction gives a bounded integral solution of (CL). It satisfies $u(\cdot,t) \to g$ in L^1_{loc} as $t \downarrow 0$.

Expanded proof.

First W is spatially Lipschitz with constant $\|g\|_\infty$, hence $|u| \le \|g\|_\infty$ a.e. The a.e. HJ equation $W_t + f(W_x) = 0$ is an equality of locally integrable

HJ 解是守恒律解的原函数

$W_t + f(W_x) = 0$ 对空间作分布求导，就得到 $u_t + f(u)_x = 0$。给初始原函数 P 加常数只会给 W 加同一常数，不改变 u 或最优起点。

可数例外指固定的时间切片

每个多重最优点对应一个互不相交的起点开区间，再各选一个有理数，才得到可数性。这不表示整个时空只存在可数个激波点，也不等于全局光滑。

最优起点的跳跃产生激波

$y(x,t)$ 保持单调，但可发生跳跃。跨过竞争分支时，$(x-y)/t$ 随之下降，因此 $u = G((x-y)/t)$ 出现向下跳跃；这把变分选择和熵方向联系起来。

functions. Multiply it by ϕ_x. Integration by parts first in time and then in space gives

$$\iint W_t \phi_x = -\iint W \phi_{xt} - \int P\phi_x(\cdot,0) = \iint W_x \phi_t + \int g\phi(\cdot,0).$$

All supports are compact and W is Lipschitz; thus these are valid weak integration identities. Substituting $u = W_x$ proves (CL).

To strengthen initial recovery, fix a Lebesgue point x of g, with value g_0. Then $P(x+h) = P(x) + g_0 h + o(|h|)$. The coercive minimizer bound gives $|x-y|/t \le R$ uniformly. Writing $y = x - tv$, the minimizing objective becomes

$$P(x) + t[L(v) - g_0 v] + o(t)$$

uniformly for $|v| \le R$. The strictly convex function $L(v) - g_0 v$ has unique minimizer $v_* = f'(g_0)$. Compactness implies every minimizing velocity tends to v_*: otherwise a convergent subsequence away from v_* would minimize the limiting strictly convex objective. Formula (LO) therefore gives $u(x,t) \to G(v_*) = g_0$. The uniform L^∞ bound and dominated convergence on compact intervals give the asserted local L^1 convergence. □

3.4.3 A one-sided estimate and uniqueness (Evans §3.4.3)

For the Lax–Oleinik solution, choose a monotone minimizing representative $y(x,t)$ (for example the largest minimizer). For $z > 0$, $y(x+z,t) \ge y(x,t)$ and therefore

$$u(x+z,t) - u(x,t) \le \frac{z}{\theta t}. \tag{OS}$$

Indeed the corresponding velocity can increase by at most z/t; G is increasing and $1/\theta$-Lipschitz. If the velocity decreases, the displayed upper bound is automatic. Thus $u(x,t) - x/(\theta t)$ is nonincreasing at fixed t. It has one-sided limits and only downward jumps, although its decrease may be arbitrarily steep.

The source's entropy class consists of bounded integral solutions with a representative satisfying $u(x+z,t) - u(x,t) \le C(1+t^{-1})z$ for $z > 0$ a.e. The term 1 permits a more general bound than (OS).

Theorem 3.13 (Source Theorem 3)
For smooth convex f, an entropy solution in this one-sided class with a prescribed bounded initial datum is unique up to null sets. Uniform convexity is used for the explicit Lax–Oleinik construction, but plain convexity suffices for this uniqueness statement.

单边界允许激波

约束是 $u(x+z) - u(x) \le Cz/t$，不是绝对值的 Lipschitz 界。向下跳跃完全允许，向上跳跃则被禁止；这正适合包含压缩激波的解空间。

Expanded proof (dual transport, with two separate small-time scales).

Let $w = u - \tilde{u}$ and

$$b(x,t) = \int_0^1 f'(\lambda u + (1-\lambda)\tilde{u})\, d\lambda.$$

Then $f(u) - f(\tilde{u}) = bw$. Subtracting the two integral identities with the same initial datum gives

$$\iint w(\phi_t + b\phi_x) = 0 \tag{DU}$$

for tests compactly supported in $\mathbb{R} \times [0,\infty)$. The common data cancel exactly; no pointwise initial derivative is assumed.

1. Regularize the velocity and control its upper derivative. Smooth bounded extensions of $u, \tilde{u}$ in space-time by nonnegative kernels and set $b_\varepsilon = \int_0^1 f'(\lambda u_\varepsilon + (1-\lambda)\tilde{u}_\varepsilon)\, d\lambda$. The coefficients are uniformly bounded, say $|b_\varepsilon| \le R$, and converge a.e. to b. For $t > 2\varepsilon$, the one-sided bound, averaged against the kernel and divided by z, gives $(u_\varepsilon)_x, (\tilde{u}_\varepsilon)_x \le C'(1 + t^{-1})$. Because $f'' \ge 0$ and is bounded on the common bounded value range,

$$(b_\varepsilon)_x = \int_0^1 f''(\lambda u_\varepsilon + (1-\lambda)\tilde{u}_\varepsilon)[\lambda (u_\varepsilon)_x + (1-\lambda)(\tilde{u}_\varepsilon)_x]\, d\lambda \le C''(1+t^{-1}).$$

Negative derivatives need no lower bound.

2. Solve a smooth backward test equation. Fix $\psi \in C_c^\infty(\mathbb{R} \times (0,T))$. Let v_ε solve $(v_\varepsilon)_t + b_\varepsilon (v_\varepsilon)_x = \psi$ with $v_\varepsilon(\cdot, T) = 0$. The characteristic formula is

$$v_\varepsilon(x,t) = -\int_t^T \psi(X_\varepsilon(s;x,t), s)\, ds, \qquad \dot{X}_\varepsilon = b_\varepsilon(X_\varepsilon, s).$$

Bounded speed makes all its supports lie in a fixed spatial interval. It vanishes near T, so it is an admissible smooth test in (DU).

For any fixed $\delta > 0$, $\varepsilon < \delta/2$, differentiate the flow in x:

$$\partial_x X_\varepsilon(s;x,t) = \exp\left(\int_t^s (b_\varepsilon)_x(X_\varepsilon(\tau), \tau)\, d\tau \right).$$

The upper derivative bound for $t \ge \delta$ gives a uniform finite bound on this positive factor. Differentiating the test formula therefore yields $|(v_\varepsilon)_x| \le C_\delta$ for $\delta \le t \le T$.

3. A fixed early interval controls total variation. Choose a fixed $t_* > 0$ so that $\psi = 0$ for $0 < t < 2t_*$. For $t \le t_*$, v_ε is transported unchanged along characteristics to time t_*. A one-dimensional smooth flow preserves order and is a diffeomorphism of $\mathbb{R}$ (its speed is bounded). Hence composition with this flow preserves total variation:

$$\int_{\mathbb{R}} |(v_\varepsilon)_x(x,t)|\,dx = \operatorname{Var} v_\varepsilon(\cdot,t) = \operatorname{Var} v_\varepsilon(\cdot,t_*) \le D.$$

The final bound follows from the derivative bound at the fixed t_* and the fixed compact support. Crucially, D is independent of both ε and a later splitting time $\delta < t_*$.

4. Pass the dual identity to the limit. Use v_ε in (DU):

$$\iint w\psi = \iint w(b_\varepsilon - b)(v_\varepsilon)_x.$$

Split at $t = \delta$, with $0 < \delta < t_*$. On $[\delta, T]$ the integrand tends to zero by dominated convergence, using C_δ and common compact support. On $(0,\delta)$, boundedness of w, b_ε, b and the total-variation estimate bound its absolute integral by $CD\delta$. First let $\varepsilon \downarrow 0$ for this fixed δ, then let $\delta \downarrow 0$. Thus $\iint w\psi = 0$ for every interior test ψ, so $w = 0$ a.e. □

一维保序带来变差守恒

光滑流关于初始位置严格递增，因此复合流只是重新排列同一条曲线上的取值次序，不增加总变差。这里不需要流的导数在零时刻附近有统一上界。

两个小时间不能混用

先固定源项消失的时段 t_*，用它得到统一的变差界；再另取 $\delta < t_*$ 切开误差。若让估计常数也随 δ 发散，就不能仅凭区间长度趋零断言误差消失。

3.4.4 Riemann's problem (Evans §3.4.4, Theorem 4)

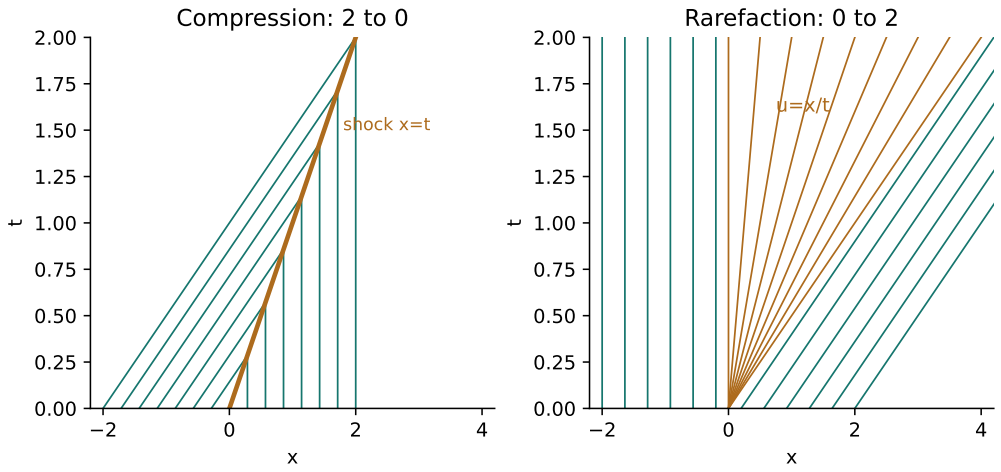
For $g = u_l$ on $x < 0$ and $g = u_r$ on $x > 0$, the entropy solution for a strictly convex flux is as follows. If $u_l > u_r$, it is a shock with speed $s = [f(u_l) - f(u_r)]/(u_l - u_r)$. If $u_l < u_r$, it is the rarefaction

$$u(x,t) = \begin{cases} u_l, & x/t \le f'(u_l), \\ G(x/t), & f'(u_l) < x/t < f'(u_r), \\ u_r, & x/t \ge f'(u_r). \end{cases} \tag{RF}$$

Equal states give a constant solution.

Expanded proof.

The shock satisfies the PDE off its interface, (RH) on it, and the downward entropy direction. In the fan, $u_t = -(x/t^2)G'(x/t)$ and $f'(u)u_x = (x/t)G'(x/t)/t$, which cancel. The fan joins continuously to both constant states and $u_x = G'(x/t)/t \le 1/(\theta t)$. Both candidates are integral solutions and approach their Riemann data locally in L^1; the transition region has width $O(t)$. The entropy uniqueness theorem just proved identifies them as the selected solutions. □



Editorial figure. For Burgers flow, a downward step compresses characteristics into a shock; an upward step opens a rarefaction fan. Both geometries use the same flux, but different data.

3.4.5 Long-time decay and N-waves (Evans §3.4.5)

For this subsection assume $f(0) = 0$, f smooth, and $0 < \theta \le f'' \le \Theta < \infty$. These convenient sufficient hypotheses ensure uniform convexity of both f and $L = f^*$, as required in the source asymptotic arguments. Write $\sigma = f'(0)$ and $d = f''(0)$.

Theorem 3.14 (Supremum decay; source Theorem 5)
For bounded integrable initial data, $\|u(\cdot,t)\|_\infty \le Ct^{-1/2}$.

Expanded proof.

The primitive P in (LOP) is bounded by $M = \|g\|_1$. Duality gives $L(\sigma) = 0, L'(\sigma) = 0$. Since $L'' \ge 1/\Theta$, $L(v) \ge (v-\sigma)^2/(2\Theta)$. Compare a minimizing y with $y_0 = x - \sigma t$:

$$\frac{|x - y - \sigma t|^2}{2\Theta t} - M \le tL((x-y)/t) + P(y) \le P(x - \sigma t) \le M.$$

Thus $|(x-y)/t - \sigma| \le C/\sqrt{t}$. As $G(\sigma) = 0$ and G is $1/\theta$-Lipschitz, formula (LO) gives the claim. □

Suppose additionally $\operatorname{supp} g \subset [-R, R]$. Use the equivalent primitive $P(x) = \int_{-\infty}^{x} g$, so $P_- = 0$, $P_+ = m = \int g$. Let $P_* = \min P$,

$$p = -2P_*, \qquad q = 2(m - P_*), \qquad p, q \geq 0.$$

The associated N-wave is

$$N(x,t) = \begin{cases} (x - \sigma t)/(dt), & -\sqrt{pdt} < x - \sigma t < \sqrt{qdt}, \\ 0, & \text{otherwise.} \end{cases} \tag{NW}$$

Its negative and positive signed areas are $-p/2$ and $q/2$, so $\int N = (q - p)/2 = m$.

Theorem 3.15 (Source Theorem 6)
If $p, q > 0$, then $\|u(\cdot,t) - N(\cdot,t)\|_1 \leq Ct^{-1/2}$.

Expanded proof (locating the edges before integrating the error).

Pass to the moving coordinate $\xi = x - \sigma t$ and put $\ell(v) = L(\sigma + v)$. Then $\ell(0) = \ell'(0) = 0$ and $\ell''(0) = 1/d$. The minimizing velocity bound just proved is $|(\xi - y)/t| \leq C/\sqrt{t}$. Taylor expansion on this fixed small-velocity range gives

$$t\ell((\xi - y)/t) = \frac{(\xi - y)^2}{2dt} + O(t^{-1/2}),$$

$$u(x,t) = \frac{\xi - y(x,t)}{dt} + O(t^{-1}). \tag{NT}$$

The constants are uniform for minimizers; the same Taylor bound holds for the bounded starting-point competitors used below when $|\xi| = O(\sqrt{t})$.

Choose $\varepsilon = A/\sqrt{t}$ with a fixed sufficiently large A. We work for large t, so $\varepsilon < 1$.

1. Exterior edges: the solution is zero beyond them. If $\xi < -R - \sqrt{pd(1+\varepsilon)t}$, the competitor $y = \xi$ has objective $P_- = 0$. A minimizing $y \geq -R$ would, by (NT), have objective at least

$$\frac{pd(1+\varepsilon)t}{2dt} + P_* + O(t^{-1/2}) = \frac{pA}{2\sqrt{t}} + O(t^{-1/2}) > 0.$$

This contradicts the competitor when A is large enough. Thus $y < -R$, where $P = 0$, and the strictly positive cost ℓ is minimized only by $y = \xi$. Hence $u = G(\sigma) = 0$. The right-tail argument is the same with the baseline $P_+ = m$ and q. Therefore $u = 0$ outside

$$-R - \sqrt{pd(1+\varepsilon)t} \leq \xi \leq R + \sqrt{qd(1+\varepsilon)t}. \tag{OUT}$$

衰减的是高度，不一定是质量

波的高度可按 $t^{-1/2}$ 降低，同时宽度按 $\sqrt{t}$ 增长。积分质量不一定趋零，所以长期形状应和一个展开的 N 波比较，而非只和零比较。

N 波参数由原函数最低点决定

p, q 不是初值正负质量各乘二。例如先取正脉冲再取等质量负脉冲，原函数始终非负且最终回零，此时 $p = q = 0$，即使初值两种符号都有。

2. Interior edges: the minimizing point stays in the initial support. At $\xi_l = R - \sqrt{pd(1-\varepsilon)t}$, any $y < -R$ has objective at least $P_- = 0$. Choose a point $z \in [-R, R]$ with $P(z) = P_*$. Since $|\xi_l - z| \le \sqrt{pd(1-\varepsilon)t}$ for large t, its objective is at most

$$\frac{p(1-\varepsilon)}{2} + P_* + O(t^{-1/2}) = -\frac{pA}{2\sqrt{t}} + O(t^{-1/2}) < 0.$$

Thus every minimizer at ξ_l is at least $-R$. Similarly at $\xi_r = -R + \sqrt{qd(1-\varepsilon)t}$ it is at most R. Monotonicity of minimizers then gives $|y(x,t)| \le R$ throughout $\xi_l < \xi < \xi_r$. There (NT) implies $|u(x,t) - \xi/(dt)| \le C/t$.

3. Integrate over the middle and the transition strips. The middle interval has length $O(\sqrt{t})$ and lies inside the ideal N-wave interval, so its L^1 error is $O(t^{-1/2})$. Each strip between the interior, ideal, and exterior edges has bounded length, since $\sqrt{(1 \pm A/\sqrt{t})t} - \sqrt{t} = O(1)$. Both $|u|$ and $|N|$ are $O(t^{-1/2})$, giving the same error there. Outside (OUT) both are zero. This proves the bound for large times.

For a bounded time interval the minimizing-velocity bound from (HL) gives finite propagation of the compact support and $|u| \le \|g\|_\infty$. Thus $\|u\|_1$ is uniformly bounded on that interval; $\|N\|_1 = (p+q)/2$. Enlarge C to cover all remaining $t > 0$. □

For Burgers flow with initial data $\mathbf{1}_{(0,1)}$, the fan is $u = x/t$. Until $t = 2$ it meets a height-one plateau ending at shock $x = 1 + t/2$. After $t = 2$ the shock is $x = \sqrt{2t}$ and $u = x/t$ on $0 < x < \sqrt{2t}$, zero elsewhere. This follows from the fan–shock matching equation $\gamma' = \gamma/(2t)$ and $\gamma(2) = 2$. It has height $O(t^{-1/2})$ and constant mass one. Here $p = 0$, $q = 2$; the explicit example illustrates decay, while the general L^1 theorem above was stated with $p, q > 0$.

边缘位置也要估计

只有波内 $O(1/t)$ 点态误差还不够做全空间积分。还要把两侧支集误差压进宽度 $O(1)$ 的条带，再乘以高度 $O(t^{-1/2})$，才能得到总误差。

3.5 Problems

Source: Evans §3.5, Problems 1–20, printed pp. 154–158 (PDF pp. 171–175). Statements below are concisely restated. All solutions are editorial, with smoothness assumed where required; extensions beyond the smooth class are identified explicitly.

3.5.1 Problems 1–8: envelopes and characteristics

1. A complete Hamilton–Jacobi integral.

Solution. For $U = a \cdot x - tH(a) + b$, $D_x U = a$ and $U_t = -H(a)$, proving the PDE. There are $n+1$ parameters for $n+1$ independent variables. The derivatives with respect to a_i independently vary $D_x U$; the derivative with respect to b changes the value by one without changing any gradient component. These $n+1$ jet variations are linearly independent, proving completeness. □

2. Envelopes of lines and planes.

Solution. For $G = x_1 + a^2 x_2 - 2a$, the stationary condition is $G_a = 2ax_2 - 2 = 0$. Thus $a \neq 0$, $x_2 = 1/a$, and $G = 0$ then gives $x_1 = a$. The envelope is $x_1 x_2 = 1$, with both branches represented. The line with $a = 0$ has no finite stationary contact. For $G = 2a_1 x_1 + 2a_2 x_2 - x_3 + a_1^2 + a_2^2$, stationarity yields $a_i = -x_i$ and elimination gives $x_3 = -x_1^2 - x_2^2$. The derivatives of this paraboloid equal the slopes $2a_i$ of the selected plane, directly verifying tangency. The figure in §3.1 shows both constructions. □

3. Eliminate parameters to obtain a PDE.

Solution. (a) For $\sum_i a_i x_i^2 + u^3 = 0$, implicit differentiation gives $2a_i x_i + 3u^2 u_{x_i} = 0$. Multiply by x_i, sum, and eliminate $\sum_i a_i x_i^2 = -u^3$:

$$u^2(3x \cdot Du - 2u) = 0.$$

On a nonzero branch this is $3x \cdot Du = 2u$. For a C^1 graph it extends to the zero set by continuity from nonzero points, while on the interior of $\{u = 0\}$ the gradient is zero. This avoids division by a possibly zero coordinate x_i.

(b) A unit sphere centered at $(a, 0)$ has $|x-a|^2+u^2 = 1$. Differentiation yields $x - a = -uDu$. Substitution eliminates a and gives $u^2(1 + |Du|^2) = 1$, on regions where the sphere is a differentiable graph. □

参数消元要记录分支

除以 u、x_i 或某个行列式前，先说明它不为零。得到的简化 PDE 可能丢掉零解或特殊分支；保留未约去的形式有助于检查。

4. Forced transport.

Solution. For $u_t + b \cdot Du = f(x,t)$, use time as the characteristic parameter: $\dot{X} = b, \dot{Z} = f(X, s)$. Starting from the target, $X(s) = x - (t-s)b$. Therefore

$$u(x,t) = g(x - tb) + \int_0^t f(x - (t-s)b, s)\, ds.$$

This is the same formula as in Chapter 2. In the full characteristic system the spatial gradient satisfies $\dot{P} = D_x f$ and the time-gradient component satisfies $\dot{Q} = f_t$. □

5. Three characteristic calculations.

Solution. (a) For $x_1 u_{x_1} + x_2 u_{x_2} = 2u$, start at $(a, 1)$. Then $X = (ae^s, e^s)$, $Z = e^{2s} g(a)$. For $x_2 > 0$, $s = \log x_2$ and $a = x_1/x_2$, so $u = x_2^2 g(x_1/x_2)$.

(b) For $x_1 u_{x_1} + 2x_2 u_{x_2} + u_{x_3} = 3u$, start at $(a, b, 0)$. Then $X = (ae^s, be^{2s}, s)$, $Z = e^{3s} g(a, b)$, giving $u(x) = e^{3x_3} g(x_1 e^{-x_3}, x_2 e^{-2x_3})$.

(c) For $u u_{x_1} + u_{x_2} = 1$, start at (a, a) with $Z(0) = a/2$. The equations integrate to

$$X_2 = a + s, \qquad Z = a/2 + s, \qquad X_1 = a + (a/2)s + s^2/2.$$

Eliminate $s = X_2 - a$: $X_1 = a(1 - X_2/2) + X_2^2/2$. Thus

$$a = \frac{2x_1 - x_2^2}{2 - x_2}, \qquad u(x_1, x_2) = x_2 - a/2$$

locally near boundary points (a, a) with $a \neq 2$. The parameter Jacobian is $1 - x_2/2$. At $(2, 2)$ the boundary conditions themselves are incompatible: Tangential differentiation gives $u_{x_1} + u_{x_2} = 1/2$, whereas the PDE and $u = 1$ give the same left side equal to one. There is no C^1 solution through that boundary point. □

6. Flow Jacobians and a continuity equation.

Solution. Let $X(s; x, t)$ solve $\dot{X} = b(X)$ with $X(t; x, t) = x$. On any interval where the smooth flow exists, its derivative matrix $M = D_x X$ satisfies $M_s = Db(X)M$ and $M(t) = I$. Jacobi's determinant formula gives

$$J_s = \operatorname{tr}(M^{-1} Db(X) M) J = (\operatorname{div} b)(X) J, \quad J(t) = 1.$$

In particular $J > 0$ and is the exponential of the divergence integral. For the forward map $Y(t; a)$, the desired density satisfies

$$\frac{d}{dt}\big[u(Y(t; a), t) \det D_a Y(t; a)\big] = 0.$$

Set the product equal to $g(a)$; differentiating proves $u_t + b \cdot Du + u \operatorname{div} b = 0$. Inverting the flow then gives $u(x, t) = g(X(0; x, t)) J(0; x, t)$. A smooth vector field need not have a global complete flow; the formula is asserted on its actual existence interval unless completeness is assumed. □

密度与被搬运的数值不同

连续性方程保持的是 $u(Y,t)\det D_a Y$，因为体积也在变化；纯传输保持 $u(Y,t)$。当速度无散时 Jacobian 为一，两者才在这一点上重合。

7. A curved noncharacteristic surface.

Solution. The tangential boundary gradient is fixed. Write admissible gradients as $D_{\tan} g + \lambda \nu$. The derivative of their scalar PDE constraint in λ is $F_p \cdot \nu$. Nonvanishing permits local implicit solution for the normal component. Equivalently, the characteristic column F_p and the boundary tangent columns form a basis exactly when this scalar product is nonzero. □

8. Verify the implicit conservation formula.

Solution. Let $a = x - t\mathbf{f}'(u)$ and $u = g(a)$. With $D = 1 + tDg(a)\cdot\mathbf{f}''(u) \neq 0$, differentiate:

$$Du = \frac{Dg(a)}{D}, \qquad u_t = -\frac{Dg(a)\cdot \mathbf{f}'(u)}{D}.$$

Hence $u_t + \mathbf{f}'(u)\cdot Du = 0$. The nonzero denominator is exactly the rank-one determinant from the characteristic projection. Its loss is not repaired by the implicit formula itself. □

3.5.2 Problems 9–16: variations and convex duality

9. A free initial endpoint.

Solution. Variations have $V(t) = 0$ but arbitrary $V(0)$. After integration by parts the first variation is

$$\int_0^t \left[L_x - \frac{d}{ds} L_v \right] \cdot V \, ds - L_v(\dot{X}(0), X(0)) \cdot V(0).$$

First choose $V(0) = 0$ to obtain the Euler–Lagrange equation. Then arbitrary $V(0)$ gives $L_v(\dot{X}(0), X(0)) = 0$. Adding the initial cost $g(X(0))$ adds $Dg(X(0))\cdot V(0)$, so the new boundary condition is $L_v(\dot{X}(0), X(0)) = Dg(X(0))$. □

初始代价决定初始动量

自由起点不代表可以忽略端点项。变分边界项的负号与 $g(X(0))$ 的变化抵消后，得到 $P(0) = Dg(X(0))$，恰好接上 HJ 特征初值。

10. Two Legendre transforms.

Solution. (a) For $H(p) = |p|^r/r$, $1 < r < \infty$, the maximizer of $v\cdot p - H(p)$ is parallel to v, with $|p|^{r-1} = |v|$. Writing $1/r + 1/s = 1$ gives $L(v) = |v|^s/s$, also valid at $v = 0$. (b) For $H(p) = \frac{1}{2}p^T A p + b\cdot p$, A symmetric positive definite, stationarity gives $p = A^{-1}(v - b)$. Substitution gives $L(v) = \frac{1}{2}(v-b)^T A^{-1}(v-b)$. Positive definiteness makes this stationary point the unique global maximum of the concave objective. □

11. Subdifferential reciprocity.

Solution. The condition $v \in \partial H(p)$ is $H(q) \geq H(p) + v \cdot (q-p)$ for every q. It is equivalent to p maximizing $v \cdot q - H(q)$, hence to $H(p) + L(v) = p \cdot v$. Using $H = L^*$, the same argument with the roles exchanged is equivalent to $p \in \partial L(v)$. For nonsmooth finite convex functions, this uses the supporting-hyperplane version of the duality established above, rather than an unjustified gradient. □

12. A finite-dimensional dual minimum. For smooth convex superlinear L_i and $H_i = L_i^*$, prove

$$\min_v [L_1(v) + L_2(v)] = \max_p [-H_1(p) - H_2(-p)].$$

Solution. For any v, p, Fenchel's inequalities give $L_1(v) + L_2(v) \geq -H_1(p) - H_2(-p)$. The convex superlinear sum has a minimizing v_*. Smoothness gives $DL_1(v_*) + DL_2(v_*) = 0$. Choose $p_* = DL_1(v_*) = -DL_2(v_*)$. Both Fenchel inequalities are equalities for this pair. Thus the minimum equals the stated dual maximum, which is attained at p_*. □

13. A sharper finite-speed radius.

Solution. Assume here the source's smooth initial datum, with bounded gradient. At a minimizing y in (HL), differentiation in y gives $DL((x-y)/t) = Dg(y)$ when L is smooth. More generally the convex optimality condition is $Dg(y) \in \partial L((x-y)/t)$. Reciprocity and differentiability of H give $(x-y)/t = DH(Dg(y))$. Thus $|x-y| \leq Rt$ for $R = \sup_y |DH(Dg(y))|$, proving the restricted minimum formula on the closed ball $\overline{B}_{Rt}(x)$. For merely Lipschitz g, a minimizing point need not be a differentiability point of g; the coarser coercive radius already proved for (HL) remains available without that assumption. □

14. Distance to a closed set.

Solution. For $H(p) = |p|^2$, $L(v) = |v|^2/4$. If the initial cost is zero on a nonempty closed set E and infinity elsewhere, the extended-value formula reduces to

$$\inf_{y \in E} \frac{|x-y|^2}{4t} = \frac{\operatorname{dist}(x,E)^2}{4t}.$$

Closedness and coercivity give a nearest point. The formula tends to zero on E and to infinity at each point outside E as $t \downarrow 0$. These are extended-value data, outside the globally Lipschitz-data theorem; this is a computation of the proposed formula rather than an invocation of that theorem. For empty E, both sides are infinity by convention. □

不要对不可微的初值硬求导

Lipschitz 函数几乎处处可微，但指定的最优起点仍可能落在不可微集合。习题 13 的驻点等式使用光滑初值；一般情形可先用不依赖求导的强制性估计。

15. Semiconcavity from uniform convexity.

Solution. The proof in §3.3.3 gives all steps: strong monotonicity of DH makes $DL = (DH)^{-1}$ Lipschitz with constant $1/\theta$; integrate its increment to obtain the quadratic upper bound on L; add opposite increments; use the same minimizing starting point and increments $\pm z/t$. The factor is exactly $1/(\theta t)$. □

16. Supremum contraction.

Solution. Let $\delta = \sup|g_1 - g_2| < \infty$. For every competitor y, the objective for u_1 is at most that for u_2 plus δ. Taking infima gives $u_1 \le u_2 + \delta$. Interchanging indices gives the reverse bound, so $\sup|u_1(\cdot,t) - u_2(\cdot,t)| \le \delta$. No differentiation or uniqueness theorem is needed. □

3.5.3 Problems 17–20: entropy and wave interactions

17. An unbounded Burgers entropy example. Consider

$$u(x,t) = \begin{cases} -\frac{2}{3}(t + \sqrt{3x + t^2}), & 4x + t^2 > 0, \\ 0, & 4x + t^2 < 0. \end{cases}$$

Solution. On the right smooth region write $A = \sqrt{3x + t^2}$. Then $u_x = -1/A$ and $u_t = -\frac{2}{3}(1 + t/A)$, whereas $uu_x = \frac{2}{3}(1 + t/A)$. Thus Burgers' equation holds there; it also holds in the zero region. The interface is $\gamma(t) = -t^2/4$. At that point $A = t/2$, so $u_l = 0$, $u_r = -t$. Its speed $\gamma' = -t/2$ equals $(u_l + u_r)/2$, proving (RH). The jump is downward and $u_x < 0$ on the right. Hence at every fixed $t > 0$, u is nonincreasing and satisfies the one-sided entropy bound even with constant zero. The formula has initial trace $g(x) = 0$ for $x < 0$ and $g(x) = -2\sqrt{3x}/3$ for $x > 0$, locally in L^1. It is an unbounded entropy example in this local sense, not an element of the globally bounded uniqueness class stated in §3.4.3. □

例子与定理的解空间不同

这题的熵解无界，不能直接套用要求全局 L^∞ 的唯一性定理。验证 PDE、跳跃条件、单边估计后，还需明确它实际属于哪个局部解类。

18. One-sided bounds survive smoothing.

Solution. If $u(x + h) - u(x) \le Eh$ for $h > 0$, integrate this inequality after translation against a nonnegative mollifier of mass one: $u_\varepsilon(x + h) - u_\varepsilon(x) \le Eh$. Divide by h and pass to $h \downarrow 0$, obtaining $(u_\varepsilon)'(x) \le E$. Only an upper derivative bound is obtained; there is no absolute-value bound. The same argument in each spatial direction gives the regularized Hessian bound used in the HJ uniqueness proof. □

19. Conservation of mass. Assume a continuous integral solution, $f(0) = 0$, and a common compact spatial support on each bounded time interval.

Solution. For a fixed time horizon choose a spatial cutoff equal to one on that common support and use its product with a time test in (CL). The flux term is zero: where the spatial cutoff changes, $u = 0$ and $f(u) = f(0) = 0$. Thus the distributional time derivative of $\int u(x,t)\,dx$ is zero, with initial value $\int g$ supplied by (CL). Continuity of u and the common support make this integral continuous in time, so the equality holds at every time, not only almost everywhere. □

20. Two shocks interacting with one rarefaction. For Burgers flux take $g = 1$ on $x < -1$, 0 on $(-1, 0)$, 2 on $(0, 1)$, and 0 on $x > 1$.

Solution. At -1, a $1 \to 0$ shock starts with speed $1/2$: $\gamma_l = -1 + t/2$. At 0, a $0 \to 2$ fan has $u = x/t$ on $0 < x < 2t$. At 1, a $2 \to 0$ shock starts with speed one: $\gamma_r = 1 + t$.

First stage: $0 < t < 1$**.** The regions from left to right are

x	$(-\infty, \gamma_l)$	$(\gamma_l, 0)$	$(0, 2t)$	$(2t, \gamma_r)$	(γ_r, ∞)
u	1	0	x/t	2	0.

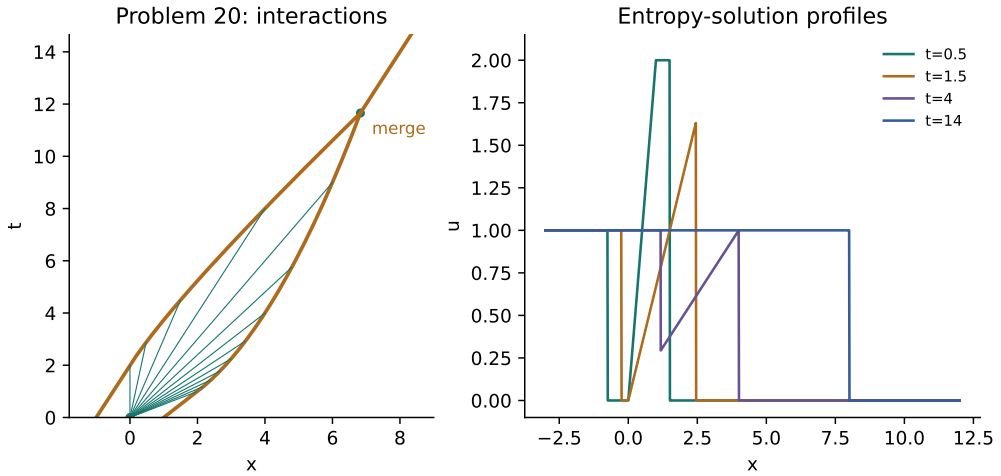
The right shock first meets the fan when $1 + t = 2t$, namely at $t = 1$.

Second stage: $1 \le t < 2$**.** The right shock now separates $u_l = \gamma_r/t$ from zero, so $\gamma_r' = \gamma_r/(2t)$. Matching $\gamma_r(1) = 2$ gives $\gamma_r = 2\sqrt{t}$. The height-two plateau has disappeared. The remaining regions have $u = 1$ for $x < \gamma_l$, $u = 0$ for $\gamma_l < x < 0$, $u = x/t$ for $0 < x < \gamma_r$, and zero to the right. At $t = 2$, the left shock reaches the fan's left edge $x = 0$.

Third stage: $2 \le t < T$**.** The left shock now separates one from γ_l/t, hence $\gamma_l' = (1 + \gamma_l/t)/2$. Multiply by $t^{-1/2}$ to integrate: $(\gamma_l/\sqrt{t})' = 1/(2\sqrt{t})$. Matching $\gamma_l(2) = 0$ gives $\gamma_l = t - \sqrt{2t}$. Between this curve and $\gamma_r = 2\sqrt{t}$, the state is x/t; it is one to the left and zero to the right. The shocks meet at

$$T = (2 + \sqrt{2})^2 = 6 + 4\sqrt{2}, \qquad X = 2\sqrt{T} = 4 + 2\sqrt{2}.$$

Final stage: $t \geq T$**.** Only a $1 \to 0$ shock remains, of speed $1/2$. Continuity of its trajectory at the interaction gives $\gamma(t) = X + (t-T)/2 = 1 + t/2$. Thus $u = 1$ for $x < 1 + t/2$ and $u = 0$ for $x > 1 + t/2$. In all stages each smooth piece solves the PDE, each discontinuity satisfies (RH), and each shock is downward. The fan has derivative $1/t$ and other smooth pieces have zero derivative; therefore the global one-sided bound holds. The initial trace and entropy uniqueness identify this as the required solution. Values assigned exactly on the shock curves do not change the weak solution. □



Editorial figure. The space-time plot records interactions at $t = 1$, $t = 2$ and T; the profiles show all four stages. Independent piecewise Hopf–Lax minimization checks the formulas numerically away from shock ties.

最终激波位置可用质量再检查

相对于 $x = 0$ 处的 $1 \to 0$ 阶跃，初值的积分差为 $-1 + 2 = 1$。最终阶跃应比速度 $1/2$ 的参考激波右移一，故为 $x = t/2 + 1$。

进入稀疏波后速度会变

激波两侧的值不再是两个固定常数，而包含 $\gamma(t)/t$，所以速度要重新由 Rankine–Hugoniot 求出。不能把初始的两条直线一直延长到所有时间。

3.6 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 3, printed pp. 89–158 (local PDF pp. 106–175). The section order and explicit source theorem identifiers follow that edition. The prose, expanded arguments, Chinese notes, plots and exercise solutions are editorial reconstructions.

Evans's §3.6 points to Courant–Hilbert for complete integrals; Carathéodory, Garabedian, John, Chester and Sneddon for characteristics; Lions, Rund and Benton for Hamilton–Jacobi theory; and Lax, Smoller and Whitham for conservation laws. The source attributes the HJ uniqueness argument to Douglis, the entropy uniqueness argument to Oleinik, and the long-time estimates to DiPerna, *Indiana University*

Mathematics Journal 24 (1975), pp. 1047–1071. These are source attributions and reading directions, not claims of independent consultation of each listed work.

The Chinese chapter opening (PDF p. 79) lists section starts at printed pp. 61, 64, 76, 90, 107, and 110 for §3.1–§3.6. Selected Chinese pages for the local characteristic construction, Hopf–Lax and shocks were checked as recorded in the chapter log. A line-by-line comparison of the whole Chinese chapter has not been made.

Review boundary. The chapter invokes ODE existence and smooth dependence, implicit and inverse function theorems, supporting hyperplanes for convex functions, integration and weak calculus, Rademacher, and Gronwall as prerequisites. The local existence proof includes the contact-compatibility step; the two uniqueness arguments explicitly retain their different selection classes. Long-time estimates are stated under convenient sufficient convexity hypotheses, not optimal assumptions. All 20 source exercises have solutions or exact pointers to full earlier arguments. Numerical agreement with selected models and successful compilation do not establish independent mathematical review.

Chapter 4

Other Ways to Represent Solutions

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 4; printed pp. 159–236; PDF pp. 176–253.

Chinese reference: 中译本：第四章，书内第 111 页起；章首页 PDF 第 129 页。

Editorial study edition. This chapter reconstructs the source exposition, not its wording. Source theorem identifiers are stated separately from our numbering. Detailed proofs, Chinese mathematical annotations, exercise solutions and figures are editorial additions.

The organizing question is how a PDE can be represented by simpler objects:

$$\text{modes and profiles} \longrightarrow \text{integral transforms},$$
$$\text{changes of unknown} \longrightarrow \text{asymptotics and power series}.$$

Reading conventions. Fourier transforms use the unitary normalization $(2\pi)^{-n/2}$. The diffusion exponent is denoted by m to distinguish it from multiindices. A formal asymptotic calculation is identified explicitly; its conclusion is not promoted to an unconditional convergence theorem. We invoke standard integration, ODE theory, implicit and inverse function theorems, and elementary complex analysis. Spectral completeness and elliptic solvability are identified where used, rather than proved by writing a series. Sobolev terminology is available in Chapter 5.

表示公式还需要合法性

猜到一个形式不等于已经解决问题。接着要检查初边值、可逆性、级数或积分的收敛，以及能否逐项求导；渐近展开还需另估余项。

4.1 Separation of Variables

4.1.1 Separated modes and the missing convergence question

For $u_t - \Delta u = 0$ with zero Dirichlet data on a bounded domain U, try $u(x,t) = T(t)X(x)$. On a product neighborhood where both factors are nonzero,

$$\frac{T'}{T} = \frac{\Delta X}{X} = -\lambda.$$

Fixing x and then t shows that the common expression is constant. The separated equations are

$$-\Delta X = \lambda X, \quad X|_{\partial U} = 0, \qquad T' = -\lambda T.$$

The identities without division continue across zeros of X. For a nonzero smooth eigenfunction, integration by parts gives

$$\lambda \int_U |X|^2 = \int_U |DX|^2 > 0.$$

Strict positivity follows since a zero-gradient function with zero boundary trace is zero. Thus $e^{-\lambda t}X$ is a heat mode.

分离与叠加是两步

乘积形式只构造单个模态；把多个模态相加依赖方程的线性。非线性方程即使能分离变量，也通常不能把分离解相加来拟合任意初值。

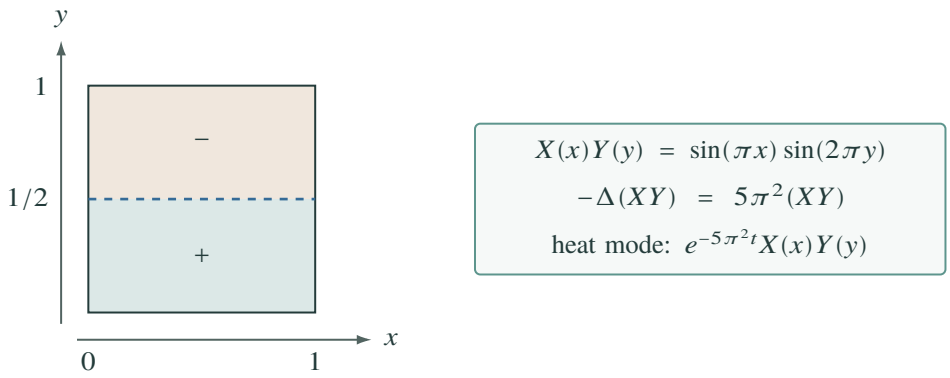


Diagram T4.1. A separated mode and its nodal line. On the unit square, this Dirichlet mode is the product of two one-dimensional modes. The dashed line $y = 1/2$ is a zero set inside the domain. Applying $-\Delta$ adds the two eigenvalues, giving $5\pi^2$; heat evolution then changes only the time factor.

Theorem 4.1 (A justified spectral representation, conditional on the spectral basis)

Assume the Dirichlet eigenfunctions X_j form an orthonormal basis of $L^2(U)$, with eigenvalues $\lambda_j > 0$. For $g \in L^2(U)$, set $g_j = (g, X_j)$ and

$$u(\cdot,t) = \sum_{j=1}^{\infty} e^{-\lambda_j t} g_j X_j. \tag{S4}$$

This converges in L^2, tends to g in L^2 at time zero, and solves the heat

> *equation in the spectral weak sense.*

Expanded proof.

Parseval gives $\sum |g_j|^2 = \|g\|_2^2$, so the tail norm squared is at most $\sum_{j>N} |g_j|^2$, uniformly for $t \geq 0$. Moreover

$$\|u(t) - g\|_2^2 = \sum_j |e^{-\lambda_j t} - 1|^2 |g_j|^2 \longrightarrow 0$$

by dominated convergence for a summable series. For $t \geq \tau > 0$, $\sup_{\lambda \geq 0} \lambda^k e^{-\tau\lambda} < \infty$. Hence time derivatives of every order converge in L^2 away from zero; the spectral Laplacian and time derivative agree termwise. Testing against any finite eigenfunction combination proves the weak equation, and density extends it to the corresponding energy test class. Interior classical smoothness requires the usual elliptic/local regularity facts; it does not follow simply from writing (S4). Existence and completeness of the basis are the later spectral theorem of Chapter 6, not a conclusion of separation itself. □

On $(0, \pi)$ one has $X_j = \sqrt{2/\pi}\sin(jx)$, $\lambda_j = j^2$. The coefficient formula is then an ordinary sine-series expansion. A finite sum is already a classical solution for its own initial datum.

4.1.2 Nonlinear and additive separation

For $u_t - \Delta(u^m) = 0$, $m > 1$, try $u = T(t)X(x) \geq 0$. Homogeneity gives $T'/T^m = \Delta(X^m)/X = \mu$ and therefore

$$T(t) = [c - (m-1)\mu t]^{-1/(m-1)}, \qquad \Delta(X^m) = \mu X.$$

Taking $X = |x|^\alpha$ and using $\Delta r^q = q(q+n-2)r^{q-2}$ for $r > 0$ yields

$$\alpha = \frac{2}{m-1}, \qquad \mu = \alpha m(\alpha m + n - 2) > 0.$$

The positive bracket restricts the solution interval to $t < c/((m-1)\mu)$. At the origin X need not be C^2, but $X^m = r^{2+2/(m-1)}$ is C^2 and its Laplacian extends continuously to zero. Thus the equation holds there in this natural sense. The profile grows at infinity and has infinite total mass.

For the Hamilton–Jacobi equation, additive separation is more natural: $u = X(x) + T(t)$ gives $H(DX) = \mu$ and $T = -\mu t + b$. The affine choice $X = a \cdot x$ recovers the complete family of Chapter 3.

4.1.3 Turing instability: diffusion can destabilize a coupled system

Let $J = D\mathbf{f}(0) = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$, and $D = \operatorname{diag}(d_1, d_2)$, $d_i > 0$. The reaction ODE is linearly asymptotically stable precisely when

$$\operatorname{tr} J = \alpha + \delta < 0, \qquad \det J = \alpha\delta - \beta\gamma > 0.$$

Indeed its two eigenvalues either are real with negative sum and positive product, or are conjugate with negative real part.

For $\mathbf{v}_t - D\Delta\mathbf{v} = J\mathbf{v}$, expand in Laplacian modes. The coefficient of X_j solves

$$s_j' = (J - \lambda_j D)s_j.$$

Its trace is still negative, but its determinant is

$$p(\lambda) = d_1 d_2 \lambda^2 - (d_1\delta + d_2\alpha)\lambda + \det J.$$

Theorem 4.2 (Linear mode instability)
A mode grows exponentially if and only if $p(\lambda_j) < 0$. For a continuous positive instability band it is necessary and sufficient that

$$B = d_1\delta + d_2\alpha > 0, \qquad B^2 > 4d_1 d_2 \det J.$$

Actual instability additionally requires a domain eigenvalue in (λ_-, λ_+), where

$$\lambda_\pm = \frac{B \pm \sqrt{B^2 - 4d_1 d_2 \det J}}{2d_1 d_2}.$$

Expanded proof.

Negative determinant gives two real eigenvalues of opposite signs. Positive determinant and negative trace give negative real parts; zero determinant is a neutral threshold, not strict exponential growth. The quadratic p opens upward and has positive roots exactly under the displayed conditions. Its negative set is the interval between them. The discrete spectrum of the chosen domain must meet this interval. □

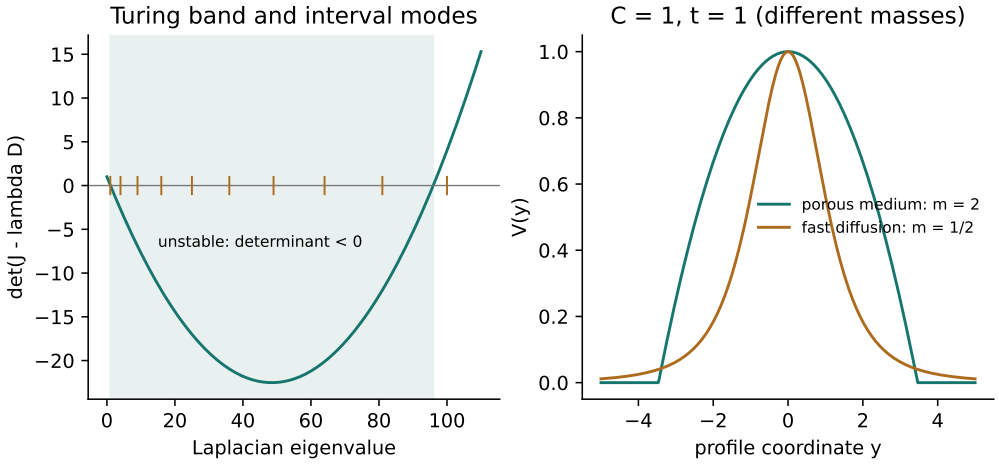
无穷远的条件影响物理解

这个分离解能在有限时间爆破，但它从一开始就有无限质量。不能用它否定有限质量、紧支集数据下的扩散结论；全空间问题也要记录增长类。

For example, take

$$J = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \qquad D = \operatorname{diag}(0.01, 1).$$

Then $\operatorname{tr} J = -2$, $\det J = 1$, while $p(4) = 0.16 - 3.88 + 1 = -2.72$. On $(0, \pi)$ the second sine mode has $\lambda_2 = 4$ and grows. If $D = dI$, by contrast, every eigenvalue is shifted by $-d\lambda_j$, so a stable J remains stable.



Editorial figure. The determinant band is compared with the actual interval spectrum; the other panel compares scaling profiles.

4.2 Similarity Solutions

4.2.1 Plane waves, phase velocity and group velocity

For $u = e^{i(\xi\cdot x - \omega t)}$, direct substitution gives:

equation	dispersion relation
$u_t - \Delta u = 0$	$\omega = -i\lvert\xi\rvert^2$
$u_{tt} - \Delta u = 0$	$\omega = \pm\lvert\xi\rvert$
$u_{tt} - \Delta u + M^2 u = 0$	$\omega = \pm\sqrt{\lvert\xi\rvert^2 + M^2}$
$iu_t + \Delta u = 0$	$\omega = \lvert\xi\rvert^2$
$u_t + u_{xxx} = 0$	$\omega = -\xi^3$

For instance $\partial_{xxx} e^{i\xi x} = -i\xi^3 e^{i\xi x}$, so the last sign is negative with this phase convention. A real ω gives oscillation; a negative imaginary part gives damping. For $\xi \neq 0$, the signed phase speed in direction $\xi/|\xi|$ is $\omega/|\xi|$.

To explain group velocity, set $x = tv$ in a Fourier superposition:

$$u(tv, t) = \int e^{it(\xi\cdot v - \omega(\xi))} a(\xi)\, d\xi.$$

连续波段不等于域内存在模态

代数条件先给出不稳定的 λ 区间；还要看具体区域的离散特征值是否落入其中。$p = 0$ 只是临界中性情形，严格增长需要 $p < 0$。

线性失稳不等于已证明斑图形成

这里证明的是线性化系统存在增长模态。非线性解是否饱和、形成何种稳定空间图案，需要额外的分岔与演化理论，不能由特征值计算直接得出。

Stationary points satisfy $v = D_\xi \omega$. The stationary-phase theorem below makes this observation precise for localized amplitudes and nondegenerate stationary points.

4.2.2 KdV solitary waves

For $u_t + 6uu_x + u_{xxx} = 0$, take $u = v(s)$, $s = x - ct$. The ODE is $-cv' + 6vv' + v''' = 0$.

Theorem 4.3 (The source solitary-wave calculation)
For $c > 0$ and $s_0 \in \mathbb{R}$,

$$u(x,t) = \frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct - s_0)\right)$$

is a decaying traveling wave.

Expanded proof.

One integration gives $v'' - cv + 3v^2 = A$. Multiply by v' and integrate again:

$$\frac{1}{2}(v')^2 = \frac{c}{2}v^2 - v^3 + Av + B.$$

The solitary-wave limits $v, v', v'' \to 0$ at both ends force $A = B = 0$. A positive hump therefore has maximum $c/2$ and obeys $v' = -v\sqrt{c - 2v}$ on its decreasing half. With $v = (c/2)\operatorname{sech}^2\theta$,

$$\frac{dv}{v\sqrt{c - 2v}} = -\frac{2}{\sqrt{c}}\,d\theta \quad (\theta > 0).$$

Thus $\theta = \sqrt{c}(s - s_0)/2$ on that half. Reflection supplies the other half. For a direct check use $(\operatorname{sech}^2 z)'' = 4\operatorname{sech}^2 z - 6\operatorname{sech}^4 z$, which gives $v'' = cv - 3v^2$ everywhere. Differentiating verifies the original PDE, including the crest. □

A solitary-wave formula does not by itself prove the stronger scattering properties often associated with the word soliton. Also the KdV sign in Problems 11–12 is the opposite nonlinear sign; there $u \mapsto -u$ relates the conventions.

4.2.3 Bistable fronts: a detailed shooting argument

Assume $f(0) = f(a) = f(1) = 0$, $0 < a < 1$, $f < 0$ on $(0,a)$, $f > 0$ on $(a,1)$, $f'(0), f'(1) < 0$, and $\int_0^1 f > 0$. An increasing front from zero to one satisfies

$$v' = w, \qquad w' = -cw - f(v),$$
$$(v,w)(-\infty) = (0,0), \quad (v,w)(+\infty) = (1,0).$$

相速不是信息传播速度

Klein–Gordon 的正频相速为 $\sqrt{|\xi|^2 + M^2}/|\xi| > 1$，群速度的大小却小于一。单一相位面的移动不能直接当作信号速度或支集传播速度。

不能把减号分支用在整个波峰两侧

一阶能量式含正负两个分支。上升半支与下降半支要在 $v = c/2$ 处拼合；最后直接核验二阶 ODE，才能排除分支选择造成的假解。

At each endpoint the linear eigenvalues are $(-c \pm \sqrt{c^2 - 4f'(j)})/2, j = 0, 1$. They have opposite signs. We use the standard local stable/unstable manifold theorem for these hyperbolic ODE equilibria, including continuous dependence on c. The following fills in the source's phase-plane shooting outline.

Put $F(v) = \int_0^v f$ and $E = w^2/2 + F(v)$. Then $dE/ds = -cw^2$. Since $F(1) > 0$, a front must have $c < 0$:

$$c \int_{\mathbb{R}} (v')^2 \, ds = -\int_0^1 f(v) \, dv. \tag{BF4}$$

Choose a fixed $b \in (a, 1)$ so close to a that $F(b) < 0$. For $c = 0$ the upper unstable branch from zero and upper stable branch into one meet the section $v = b$ at heights

$$w_L(b; 0) = \sqrt{-2F(b)}, \qquad w_R(b; 0) = \sqrt{2(F(1) - F(b))}.$$

The first is smaller.

Expanded proof (existence and uniqueness of the front speed).

Where $w > 0$, write each trajectory as a graph in v:

$$\frac{dw}{dv} = -c - \frac{f(v)}{w}. \tag{SH4}$$

The endpoint slopes are $w_L/v \to \lambda_0^+(c)$ and $w_R/(1 - v) \to -\lambda_1^-(c)$. As c decreases, the first slope strictly increases and the second strictly decreases. At a first contact between two left branches, their slope difference from (SH4) is the positive difference of $-c$; this prevents crossing. Integrating from the right reverses the comparison for right branches. Thus $w_L(b; c)$ increases strictly as c decreases, while $w_R(b; c)$ decreases strictly.

For $c \le 0$ the left branch stays above its $c = 0$ graph up to b, so cannot stop by hitting zero. The right branch also reaches b: on $(b, 1), f > 0$, and at a hypothetical earlier endpoint with $w = 0$ the equation $w_s = -f < 0$ would give negative w immediately to its right in v, contradicting the positive branch. There is no finite-v upward blowup in these intervals since (SH4) has bounded positive slope whenever w is large. Along the right branch $E(v) \le F(1)$, since energy is increasing in forward time; hence $w_R(b; c) \le \sqrt{2(F(1) - F(b))}$.

To force the left height arbitrarily large, fix $K > 0$. The quantity $|f(v)/v|$ is bounded on $(0, b]$, say by C. On the line $w = Kv$,

$$\frac{dw}{dv} = -c - \frac{f(v)}{Kv} \ge -c - C/K > K$$

when $-c$ is sufficiently large, and $\lambda_0^+(c) > K$ then. The left branch starts above this line and cannot cross downward; so $w_L(b;c) \geq Kb$. The two heights therefore exchange order for sufficiently negative c. Continuous dependence and the intermediate value theorem give a matching speed; strict comparison makes it unique. At the match, ODE uniqueness joins the two trajectories. The hyperbolic endpoint asymptotics take infinite s to reach the endpoints, giving a front on all $\mathbb{R}$, unique up to translation. Integration of the energy identity proves (BF4). □

速度符号取决于连接方向

本稿取 $v(-\infty) = 0$、$v(+\infty) = 1$，故 $\int_0^1 f > 0$ 推出速度为负。若把波形左右反转，速度也反号，不能脱离行波坐标与端态单记正负。

4.2.4 Scaling and the Barenblatt solution

For $u_t - \Delta(u^m) = 0$, try $u = t^{-\alpha}V(y)$, $y = xt^{-\beta}$. The chain rule gives

$$u_t = t^{-\alpha-1}(-\alpha V - \beta y \cdot DV), \qquad \Delta u^m = t^{-\alpha m-2\beta}\Delta_y V^m.$$

Balancing time powers requires $\alpha + 1 = \alpha m + 2\beta$. Finite nonzero time-independent mass requires $\alpha = n\beta$, since $\int u = t^{-\alpha+n\beta}\int V$. Consequently

$$\beta = \frac{1}{n(m-1)+2}, \qquad \alpha = n\beta.$$

For a radial profile $V = W(r)$, the equation integrates to

$$[r^{n-1}(W^m)' + \beta r^n W]' = 0.$$

Regularity and zero flux at $r = 0$ set the integration constant to zero. For $m > 1$, on the positive set,

$$(W^{m-1})' = -\frac{m-1}{m}\beta r, \quad V(y) = (C - k|y|^2)_+^{1/(m-1)}, \quad k = \frac{(m-1)\beta}{2m}.$$

Theorem 4.4 (Barenblatt profile, source §4.2.2)
For $C > 0$, $m > 1$, the above formula defines a nonnegative distributional solution on $\mathbb{R}^n \times (0,\infty)$, with fixed finite mass. Its support radius is $\sqrt{C/k}\,t^\beta$.

Expanded proof (including the free-boundary check).

Inside its positive set the preceding identities prove the PDE. At the moving interface u tends to zero, and so the time derivative of the zero extension has no jump-mass term. Furthermore $V^m = (C - kr^2)_+^{m/(m-1)}$ has zero normal derivative at the interface because $m/(m-1) > 1$. Thus the distributional Laplacian adds no surface delta. The interior time derivative and second spatial derivatives are locally integrable up to the interface: the worst boundary power is $d^{1/(m-1)-1}$, whose exponent is greater than -1. Integration by parts on interior truncated regions, followed by

this integrability and the zero traces, proves the weak PDE across the boundary. Mass conservation follows by the change $x = t^{\beta} y$. Finally for compactly supported continuous ϕ,

$$\int u(x,t)\phi(x)\,dx = \int V(y)\phi(t^{\beta}y)\,dy \longrightarrow \phi(0)\int V.$$

The initial datum is a point mass, not a fixed bounded function. □

The radius grows sublinearly for $m > 1$. Its derivative is finite for each $t > 0$ but diverges as $t \downarrow 0$; finite propagation here means compact support at each positive time, not a uniform speed bound down to the point-source initial time. The fast-diffusion counterpart in Problem 5 has algebraic tails.

自由边界处要检查通量

只在正集内代入公式还不够。这里 $u = 0$ 消去移动边界的时间接口项，$\partial_\nu(u^m) = 0$ 消去空间接口项；扩展后的 PDE 才在分布意义下成立。

4.3 Transform Methods

4.3.1 Fourier transform: normalization and Plancherel

For $f \in L^1(\mathbb{R}^n)$ define

$$\hat{f}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi} f(x)\,dx,$$
$$\mathscr{F}^{-1}h(x) = (2\pi)^{-n/2}\int e^{ix\cdot\xi} h(\xi)\,d\xi.$$

The integrals are absolutely convergent. For Schwartz functions integration by parts and Fubini yield

$$\widehat{\partial_j f} = i\xi_j \hat{f}, \qquad \widehat{f * h} = (2\pi)^{n/2}\hat{f}\,\hat{h}. \tag{F4}$$

For the second identity, set $x = z + y$ in the double integral. Both normalizations must be retained; $\mathscr{F}$ is unitary, whereas ordinary convolution has no normalizing factor.

卷积公式为何多一个常数

每个 Fourier 变换自带 $(2\pi)^{-n/2}$，两个变换相乘就出现两份。卷积只做一次变换，所以要乘回 $(2\pi)^{n/2}$；后面的核常数都由此决定。

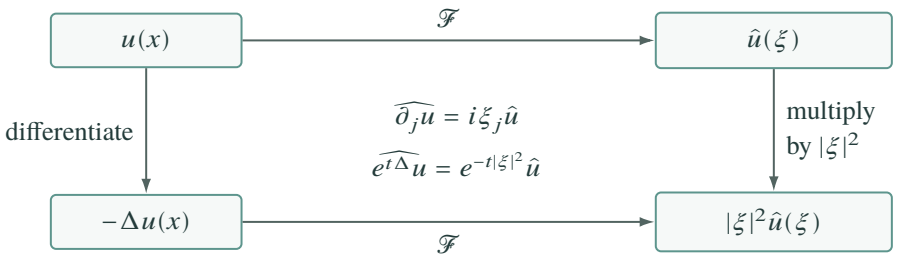


Diagram T4.2. Fourier transformation turns differentiation into multiplication. The square commutes with the unitary normalization used here. There is no 2π in the multiplier because the exponential is $e^{-ix\cdot\xi}$. The heat multiplier is small at large $|\xi|$, connecting this representation with the stability picture in Chapter 1.

Theorem 4.5 (Plancherel and inversion; source Theorems 1–2)
The Fourier transform extends uniquely to a unitary operator on L^2. It preserves inner products, has the displayed inverse, and $\widehat{D^\gamma f} = (i\xi)^\gamma \hat{f}$ whenever the weak derivative is in L^2.

Expanded proof (Gaussian regularization without circular inversion).

1. The Gaussian transform. For $a > 0$, $I(b) = \int_{\mathbb{R}} e^{-ax^2} e^{-ibx}\, dx$. Differentiate under the integral and integrate by parts to get $I'(b) = -bI(b)/(2a)$. Since $I(0) = \sqrt{\pi/a}$, $I(b) = \sqrt{\pi/a}\, e^{-b^2/(4a)}$. Products give the n-dimensional identity. This real-variable proof does not yet require contour deformation.

2. The norm identity. For $f \in L^1 \cap L^2$ and $\eta > 0$, Fubini gives

$$\int e^{-\eta|\xi|^2} |\hat{f}(\xi)|^2\, d\xi = \iint f(x)\overline{f(y)} G_\eta(x-y)\, dx\, dy,$$

where $G_\eta(z) = (4\pi\eta)^{-n/2} e^{-|z|^2/(4\eta)}$. Fubini is legitimate since $\|f\|_1^2 < \infty$ and the frequency Gaussian is integrable. The right side tends to $\|f\|_2^2$ because $G_\eta * f \to f$ in L^2. For completeness,

$$\|G_\eta * f - f\|_2 \le \int G_\eta(z) \|f(\cdot - z) - f\|_2\, dz \longrightarrow 0 :$$

translations are continuous in L^2 by approximation with compact continuous functions; split the last integral into a small ball and its Gaussian tail. Monotone convergence on the left gives the norm identity. Polarization gives the inner-product identity.

3. Inversion and extension. For Schwartz f, (F4) implies rapid decay of $\hat{f}$. Fubini and the Gaussian formula give

$$\mathscr{F}^{-1}(e^{-\eta|\xi|^2}\hat{f}) = G_\eta * f.$$

The right side tends to f; the left tends to $\mathscr{F}^{-1}\hat{f}$ by dominated convergence. Thus inversion holds on Schwartz functions. Approximate arbitrary L^2 data by Schwartz functions. The norm identity makes their transforms Cauchy and makes the limit independent of the approximation. Inversion extends by continuity, so the range is all of L^2. Finally integration by parts against Schwartz tests proves the derivative identity in distributions; if both sides belong to L^2, it is an a.e. equality of those representatives. □

4.3.2 Fourier multipliers and evolution equations

Bessel potential. For $-\Delta u + u = f, f \in L^2$, $\hat{u} = (1 + |\xi|^2)^{-1}\hat{f}$. This gives a unique L^2 distributional solution with $\int (1 + |\xi|^2)^2 |\hat{u}|^2 < \infty$. A rigorous kernel representation is

$$K(x) = \int_0^\infty e^{-s} G_s(x)\, ds, \qquad u = K * f.$$

Tonelli gives $K \geq 0$ and $\|K\|_1 = 1$; its transform is $(2\pi)^{-n/2}/(1 + |\xi|^2)$. The kernel called B in the source is $(2\pi)^{n/2}K$. The singularity of K at zero in some dimensions does not prevent its L^1 integrability or convolution action on L^2.

Heat and Schrödinger equations. Spatial transformation of $u_t - \Delta u = 0$ gives $\hat{u}_t + |\xi|^2 \hat{u} = 0$, hence $\hat{u} = e^{-t|\xi|^2}\hat{g}$. Gaussian inversion recovers $u = G_t * g$ from Chapter 2. For $iu_t + \Delta u = 0$ the sign is instead $\hat{u}_t = -i|\xi|^2\hat{u}$, so

$$\hat{u} = e^{-it|\xi|^2}\hat{g}, \qquad u(x,t) = (4\pi i t)^{-n/2} \int e^{i|x-y|^2/(4t)} g(y)\, dy. \tag{SC4}$$

For $t \neq 0$ use $(it)^{-n/2} = |t|^{-n/2} e^{-i\pi n\, \mathrm{sgn}(t)/4}$. The kernel identity follows by first replacing it with $\eta + it$, $\eta > 0$, and then letting $\eta \downarrow 0$. For $g \in L^1 \cap L^2$, the kernels converge under the spatial integral by domination, while the multipliers converge in L^2. Thus both descriptions agree. For L^2 data this defines a unitary group:

$$\|u(t)\|_2 = \|g\|_2, \qquad \|u(t) - g\|_2^2 = \int |e^{-it|\xi|^2} - 1|^2 |\hat{g}|^2 \longrightarrow 0.$$

If $g, |x|^2 g \in L^1$, differentiation of (SC4) on compact space-time sets away from $t = 0$ verifies the classical PDE. For arbitrary L^2 data the statement is distributional.

Wave and telegraph equations. For $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = h$,

$$\hat{u}(\xi, t) = \cos(t|\xi|)\hat{g} + \frac{\sin(t|\xi|)}{|\xi|}\hat{h}. \tag{WV4}$$

At $\xi = 0$ the quotient is defined as t. It is bounded by t, so the formula has no genuine zero-frequency pole. For $u_{tt} + 2du_t - u_{xx} = 0$ use $\hat{u} = e^{-dt}v$. Then $v'' + (\xi^2 - d^2)v = 0$ and

$$\hat{u} = e^{-dt} \begin{cases} \cosh(qt)\hat{g} + \dfrac{\sinh(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| < d,\ q = \sqrt{d^2 - \xi^2}, \\ \hat{g} + t(\hat{h} + d\hat{g}), & |\xi| = d, \\ \cos(qt)\hat{g} + \dfrac{\sin(qt)}{q}(\hat{h} + d\hat{g}), & |\xi| > d,\ q = \sqrt{\xi^2 - d^2}. \end{cases}$$

L^2 变换不必是逐点积分

一般 L^2 函数未必可积，其 Fourier 变换通过范数极限定义。反演和乘子公式首先是 L^2 等式；若要逐点表示，还须增加可积性或正则性。

乘子有界不表示核处处有限

$(1 + |\xi|^2)^{-1}$ 给出有界的 L^2 乘子。对应核可能在原点无界，但仍可积；不能把一个点的奇性误当作表示公式失效。

色散与耗散的范数表现不同

热乘子模长小于一，会压低高频；Schrödinger 乘子模长为一，保持 L^2。后者的点态衰减来自振荡抵消，不来自 L^2 质量损失。

These are the overdamped, critical and oscillatory regimes. The common limiting formula at $q = 0$ is essential.

Theorem 4.6 (Asymptotic energy equipartition; source §4.3.1)
If $g \in H^1(\mathbb{R}^n)$ and $h \in L^2(\mathbb{R}^n)$, the wave solution has

$$\|Du(t)\|_2^2 \longrightarrow E_0, \qquad \|u_t(t)\|_2^2 \longrightarrow E_0, \qquad E_0 = \tfrac{1}{2}(\|Dg\|_2^2 + \|h\|_2^2).$$

Expanded proof.

Write $A = |\xi|\hat{g}$ and $B = \hat{h}$ in L^2. Plancherel and (WV4) give

$$\|Du\|_2^2 = \int |A|^2 \cos^2(t|\xi|) + |B|^2 \sin^2(t|\xi|) + 2\,\mathrm{Re}(A\overline{B}) \sin(t|\xi|)\cos(t|\xi|).$$

All three coefficient functions are L^1, the cross term by Cauchy–Schwarz. For any $F \in L^1(\mathbb{R}^n)$, polar coordinates give $\int e^{2it|\xi|}F(\xi)\,d\xi = \int_0^\infty e^{2itr}H(r)\,dr$ with $\|H\|_1 \le \|F\|_1$. The latter tends to zero: approximate H in L^1 by smooth functions supported inside $(0,\infty)$ and integrate those by parts. Now use the double-angle identities to obtain the first limit. The analogous formula for u_t gives energy conservation and the second limit. This also handles $n = 1$ without discarding a possible endpoint term in a radial integration by parts. □

4.3.3 Radon transform and plane-wave decomposition

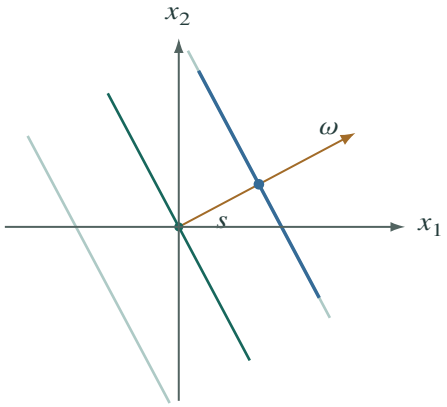
For $f \in C_c^\infty(\mathbb{R}^n)$, set

$$Rf(s,\omega) = \int_{x\cdot\omega = s} f(x)\,dS_x, \qquad \omega \in S^{n-1}.$$

Coordinates $x = s\omega + \sum y_j b_j$, where b_j are an orthonormal basis of $\omega^\perp$, make the surface measure dy. They immediately give

$$Rf(-s,-\omega) = Rf(s,\omega), \quad R(\partial_j f) = \omega_j \partial_s Rf, \quad R(\Delta f) = \partial_s^2 Rf.$$

For the middle identity decompose ∂_j into a normal and tangential derivative. Tangential integrals vanish by compact support. Induction gives $R(D^\gamma f) = \omega^\gamma \partial_s^{|\gamma|} Rf$. If $\mathrm{supp} f \subset B_R$, then $Rf = 0$ for $|s| \ge R$.



$$Rf(s,\omega) = \int_{x\cdot\omega=s} f\, dS$$

fix the unit normal ω;
vary s to scan parallel lines

Diagram T4.3. Radon data integrate along level hyperplanes. In dimension two the integration sets are lines perpendicular to ω. The scalar s is signed distance in the normal direction, not distance traveled along a line. The blue line is one member of the family; higher dimensions replace it by an $(n-1)$-plane.

Theorem 4.7 (Slice theorem and inversion; source Theorems 4–5)
With the unnormalized one-dimensional transform $B(r,\omega) = \int_{\mathbb{R}} e^{-irs} Rf(s,\omega)\, ds$,

$$B(r,\omega) = (2\pi)^{n/2}\hat{f}(r\omega).$$

For every dimension,

$$f(x) = \frac{1}{2(2\pi)^n}\int_{S^{n-1}}\int_{\mathbb{R}} B(r,\omega)|r|^{n-1}e^{irx\cdot\omega}\, dr\, dS_\omega.$$

If $n = 2k+1$ *is odd, this becomes the local formula*

$$f(x) = C_n\int_{S^{n-1}} \partial_s^{2k} Rf(x\cdot\omega,\omega)\, dS_\omega, \qquad C_n = \frac{(-1)^k}{2(2\pi)^{2k}}. \quad \text{(RI4)}$$

Expanded proof.

Fubini in the orthonormal coordinates gives

$$B(r,\omega) = \int f(x)e^{-ir\omega\cdot x}\, dx.$$

In Fourier inversion, polar coordinates with signed r double-count each nonzero frequency, giving the factor $1/2$ and $|r|^{n-1}$. All integrals converge since $\hat{f}$ is rapidly decreasing. For $n-1 = 2k$, one-dimensional inversion gives

$$\partial_s^{2k} Rf(s,\omega) = \frac{(-1)^k}{2\pi}\int_{\mathbb{R}} r^{2k}B(r,\omega)e^{irs}\, dr.$$

Substituting this in the general inversion formula proves (RI4). □

For odd n, define $\rho_f = C_n \partial_s^{n-1} Rf$. Then the plane-wave integral

$$u(x,t) = \int_{S^{n-1}} \rho_g(x\cdot\omega - t, \omega)\, dS_\omega$$

solves $u_{tt} - \Delta u = 0, u(0) = g, u_t(0) = 0$. The PDE holds because $|\omega| = 1$; the first initial condition is (RI4). The integrand for $u_t(0)$ is odd under $\omega \mapsto -\omega$, since an odd s derivative reverses the symmetry of Rg.

For the constant-coefficient system $\mathbf{u}_t + \sum B_j \mathbf{u}_{x_j} = 0$, suppose $B(\omega) = \sum \omega_j B_j$ is diagonalizable with real eigenvalues $\lambda_l(\omega)$. In Radon coordinates the diagonal components satisfy one-dimensional transport and equal

$$v_l(s,\omega,t) = h_l(s - \lambda_l(\omega)t, \omega).$$

After applying (RI4), use $\rho_l(s,\omega) = C_n \partial_s^{n-1} h_l(s,\omega)$ exactly once:

$$\mathbf{u}(x,t) = \int_{S^{n-1}} \sum_l a_l(\omega)\, \rho_l(x\cdot\omega - \lambda_l(\omega)t, \omega)\, dS_\omega.$$

Here a_l are the diagonalizing columns; the shift is not inserted again in the definition of ρ_l. For compact data in B_R, if $\min_{\omega,l} |\lambda_l(\omega)| = \sigma > 0$, then $\mathbf{u} = 0$ for $|x| < \sigma t - R$. Indeed each shifted argument has magnitude at least $\sigma t - |x| > R$. This is a conditional odd-dimensional Huygens statement; the diagonalization and nonzero-speed hypotheses cannot be removed.

奇偶维差别藏在绝对值里

带符号的极径在一般维数必须用 $|r|^{n-1}$；只有奇维时才等于多项式 r^{2k}，从而变成有限阶局部导数。偶维的反演通常带非局部算子。

4.3.4 Laplace transform, resolvents and the heat–wave link

If v and v_t have suitable exponential bounds, integration by parts gives

$$\mathscr{L}(v_t)(s) = s\mathscr{L}v(s) - v(0), \qquad \mathscr{L}v(s) = \int_0^\infty e^{-st} v(t)\, dt.$$

Thus for $v_t - \Delta v = 0$, $v(0) = f$,

$$(s - \Delta)\mathscr{L}v = f.$$

Use the same spatial boundary condition before and after transformation; for example a zero Dirichlet condition stays zero. In the whole space this recovers $(s-\Delta)^{-1} f = \int_0^\infty e^{-st}(G_t * f)\, dt$. For L^2 data the Bochner integral is justified by $\|G_t * f\|_2 \le \|f\|_2$.

平面支集条件怎样变成内部空区

若 $|x| < \sigma t - R$，所有方向的平面参数都移出初始支集区间，整个反演积分为零。零特征速度会留下不移动的分量，因此必须单独排除。

Theorem 4.8 (Gaussian averaging of the wave group; source Example 9)
Let g be smooth compactly supported, and let $U(x,s)$ be the wave solution with displacement g and zero initial velocity, extended evenly in s. Then

$$V(x,t) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-s^2/(4t)} U(x,s)\, ds$$

is the heat solution with datum g.

Expanded proof.

For smooth data the derivatives of U grow at most polynomially, as can also be checked from its Fourier representation. Gaussian weights justify differentiation and kill the boundary terms. Writing $q_t(s) = (4\pi t)^{-1/2} e^{-s^2/(4t)}$, observe $\partial_t q_t = \partial_{ss} q_t$. Hence

$$V_t = \int q_{ss} U = \int q U_{ss} = \int q \Delta U = \Delta V.$$

The Gaussian approximate identity in s gives $V(0) = g$. Alternatively the frequency multiplier is $\int q_t(s) \cos(s|\xi|)\, ds = e^{-t|\xi|^2}$, which proves the same statement in L^2. □

For $n = 2k+1 \geq 3$, let $M_g(x,r)$ be the spherical average of g. Equating the two Gaussian representations gives, with $\lambda = 1/(4t)$,

$$\int_0^\infty e^{-\lambda s^2} U(x,s)\, ds = \frac{|S^{n-1}|}{2\pi^k} \lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g(x,r)\, dr.$$

Use $\lambda e^{-\lambda r^2} = -(2r)^{-1} \partial_r e^{-\lambda r^2}$ and integrate by parts k times. The result is

$$\lambda^k \int_0^\infty e^{-\lambda r^2} r^{2k} M_g\, dr = 2^{-k} \int_0^\infty e^{-\lambda r^2} r (r^{-1}\partial_r)^k (r^{2k-1} M_g)\, dr.$$

There are no boundary terms at infinity by compact support in r. At zero, M_g is smooth and even; before the last differentiation the relevant odd power has not fallen below one, so the boundary terms are zero. After $s^2 = \tau$, uniqueness of the Laplace transform implies

$$U(x,t) = \frac{1}{(n-2)!!} \partial_t (t^{-1}\partial_t)^{(n-3)/2} (t^{n-2} M_g(x,t)).$$

Here $|S^{n-1}|/(2^{k+1}\pi^k) = 1/(n-2)!!$ follows from the Gaussian sphere area formula. The one-dimensional case is $U = (g(x+t) + g(x-t))/2$. One can justify Laplace uniqueness without a new PDE theorem: after multiplying by $e^{-a\tau}$, substitute $z = e^{-\tau}$; vanishing at all $s = a + j$ gives zero moments on $[0,1]$. Polynomial density in continuous functions makes the finite signed measure zero. This applies to the continuous, exponentially integrable functions above.

初值正是 Laplace 分部积分的边界项

时间导数并非只替换成乘以 s。零时刻留下 $-v(0)$，它成为椭圆 resolvent 方程的右端；漏掉这一项就把非零初值问题变成了齐次问题。

4.4 Converting Nonlinear into Linear PDE

4.4.1 Cole–Hopf: choose the change of unknown by cancellation

Consider $u_t - a\Delta u + b|Du|^2 = 0$, $a > 0$. For $w = \phi(u)$, the chain rule yields

$$w_t - a\Delta w = -(b\phi'(u) + a\phi''(u))|Du|^2.$$

To cancel the nonlinear term solve $a\phi'' + b\phi' = 0$. For $b \neq 0$ choose $\phi(z) = e^{-bz/a}$.

Theorem 4.9 (Cole–Hopf; source §4.4.1)
If g is bounded and continuous, define

$$w(x,t) = \int_{\mathbb{R}^n} G_{at}(x-y)e^{-bg(y)/a}\,dy, \qquad u = -\frac{a}{b}\log w. \tag{CH4}$$

Then u is smooth for $t > 0$, solves the nonlinear PDE, and has initial value g at continuity points. It is the unique bounded classical solution with these data in the corresponding heat uniqueness class.

Expanded proof.

Boundedness of g gives positive constants $0 < c \le w \le C$, since the heat kernel is positive and has mass one. Heat smoothing, differentiation under Gaussian integrals and $w_t = a\Delta w$ hold for $t > 0$. Thus the logarithm is well defined and differentiable; substitution or the preceding chain rule proves the nonlinear equation. The approximate-identity limit gives $w \to e^{-bg/a}$ and hence $u \to g$. Conversely any bounded classical solution transforms into a bounded heat solution. Heat uniqueness and injectivity of the exponential prove the assertion. If $b = 0$, the original equation is already the heat equation; division by b in (CH4) is not allowed. □

For viscous Burgers, $u_t + uu_x - au_{xx} = 0$, let $P' = g$. A potential W with $W_x = u$ satisfies $W_t + \frac{1}{2}W_x^2 - aW_{xx} = c(t)$. Subtracting $\int c$ removes this purely time-dependent term. With $Z = e^{-W/(2a)}$, $Z_t = aZ_{xx}$, so

$$W = -2a\log\int_{\mathbb{R}} G_{at}(x-y)e^{-P(y)/(2a)}\,dy,$$

$$u(x,t) = \frac{\int_{\mathbb{R}}(x-y)t^{-1}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}{\int_{\mathbb{R}}e^{-\{(x-y)^2/(2t)+P(y)\}/(2a)}\,dy}. \tag{VB4}$$

For bounded g, choose $P(x) = \int_0^x g$; it is Lipschitz and need not be bounded. The Gaussian dominates its at-most-linear growth, so the integrals and their derivatives on compact positive-time sets converge. Integrating the y derivative of the exponential shows also u is its weighted

正性同时保证变换可逆

指数变换总是正的，而热核保持严格正性。反变换要取对数，所以不能把任意带符号的热解代入；有界初值还给出远离零的统一下界。

average of $g(y)$, a.e.:

$$\int_{\mathbb{R}} \big((x-y)/t - g(y)\big) e^{-K(y)/(2a)}\, dy = 0.$$

The endpoint terms vanish by quadratic Gaussian decay. In particular $|u| \le \|g\|_\infty$.

4.4.2 Potential flow and Bernoulli's law

For incompressible Euler, $\mathbf{u}_t + (\mathbf{u}\cdot D)\mathbf{u} = -Dp + Dh$ and $\operatorname{div}\mathbf{u} = 0$, impose the additional ansatz $\mathbf{u} = Dv$. Then $\Delta v = 0$, and componentwise symmetry of $D^2 v$ gives

$$[(\mathbf{u}\cdot D)\mathbf{u}]_i = \sum_j v_{x_j} v_{x_i x_j} = \partial_i \left(\tfrac{1}{2}|Dv|^2\right).$$

Therefore

$$D(v_t + \tfrac{1}{2}|Dv|^2 + p - h) = 0, \qquad p = h - v_t - \tfrac{1}{2}|Dv|^2 + c(t)$$

on each connected spatial component. To match initial data one needs $\mathbf{g} = Dv_0$ as well as $\operatorname{div}\mathbf{g} = 0$. Locally curl-free data admit a potential; globally a topological condition or a prescribed global potential is required.

4.4.3 Hodograph: invert dependent and independent variables

Write the two velocity components as $p = u^1$, $q = u^2$. For steady irrotational compressible flow the equations are

$$(c^2 - p^2)p_{x_1} - pq(p_{x_2} + q_{x_1}) + (c^2 - q^2)q_{x_2} = 0, \qquad p_{x_2} = q_{x_1},$$

where $c = c(p,q) > 0$ is given. If $J = \det D_x(p,q) \neq 0$, invert to $x = X(p,q)$. The inverse matrix identity is

$$\begin{pmatrix} p_{x_1} & p_{x_2} \\ q_{x_1} & q_{x_2} \end{pmatrix} = J \begin{pmatrix} X^2_q & -X^1_q \\ -X^2_p & X^1_p \end{pmatrix}.$$

Substitution and division by J give the linear system

$$(c^2 - p^2)X^2_q + pq(X^1_q + X^2_p) + (c^2 - q^2)X^1_p = 0, \qquad X^1_q = X^2_p.$$

Locally $X = DV$ for a scalar potential $V(p,q)$, and

$$(c^2 - q^2)V_{pp} + 2pqV_{pq} + (c^2 - p^2)V_{qq} = 0.$$

The coefficient determinant is $c^2(c^2 - p^2 - q^2)$: the equation is elliptic in the subsonic region and hyperbolic in the supersonic region, degenerate at sonic speed.

原函数的常数与时间规范

给 P 加常数只会把热解 Z 乘一个正数，不改变 $u = -2aZ_x/Z$。从 Burgers 积分得到的时间函数也能吸收进势函数，不能无说明地直接丢掉。

无散初值还不足以成为势流

$\operatorname{div}\mathbf{g} = 0$ 是不可压条件，不能推出 $\mathbf{g} = Dv_0$。还需无旋及相应的全局条件；这里线性化的是势流子类，并未解出一般 Euler 方程。

4.4.4 A local Legendre transform for the minimal surface equation

In two dimensions the minimal surface equation is

$$(1+u_{x_2}^2)u_{x_1x_1} - 2u_{x_1}u_{x_2}u_{x_1x_2} + (1+u_{x_1}^2)u_{x_2x_2} = 0.$$

Assume $\det D^2u = J \neq 0$ locally. Use $p = Du(x)$ as independent variable, let $x = X(p)$, and set $v(p) = X(p)\cdot p - u(X(p))$.

Expanded proof (the Hessian inverse and the linear equation).

The chain rule gives

$$\partial_{p_i} v = X_i + \sum_j p_j \partial_{p_i} X_j - \sum_j u_{x_j}\partial_{p_i}X_j = X_i.$$

Hence $D^2v = D_pX = (D^2u)^{-1}$. For a symmetric 2×2 matrix the inverse formula implies

$$u_{x_1x_1} = Jv_{p_2p_2}, \quad u_{x_1x_2} = -Jv_{p_1p_2}, \quad u_{x_2x_2} = Jv_{p_1p_1}.$$

Substitute and divide by J to obtain

$$(1+p_1^2)v_{p_1p_1} + 2p_1p_2v_{p_1p_2} + (1+p_2^2)v_{p_2p_2} = 0.$$

Its coefficient matrix $I + p\otimes p$ is positive definite. Conversely, a solution with invertible Hessian yields a local inverse $x = Dv(p)$ and reconstructs $u = x\cdot p - v$ satisfying the original PDE. Boundary data need a separate transformation; a fixed boundary in x generally becomes an unknown surface in p. □

变线性没有消除退化

系数现在依赖独立变量，因而方程线性，但声速边界仍使主部退化。还要把 $X(p,q)$ 反解回去；Jacobian 变零处不能继续套用变换。

4.5 Asymptotics

The source presents several asymptotic constructions formally. We distinguish exact algebraic identities, proved local integral asymptotics, and conditional PDE expansions. No convergence theorem is inferred merely by matching powers.

局部 Legendre 变换不要求凸性

这里凭 $\det D^2u \neq 0$ 局部反演，Hessian 可以不定。第三章的凸共轭是全局上确界构造；两者在严格凸、可逆情形相接，但假设与用途不同。

4.5.1 A small-diffusion plume: geometry and transverse scale

Consider $\operatorname{div}(\mathbf{b}\,\mu) = \delta_0$ in two dimensions. Let $\dot X = \mathbf{b}(X)$, $X(0) = 0$, and suppose $|\mathbf{b}(X)| > 0$. If the forward trajectory leaves compact sets, the locally finite measure

$$\int \phi\, d\mu = \int_0^\infty \phi(X(t))\, dt$$

solves this equation, since

$$-\int \mathbf{b}\cdot D\phi\, d\mu = -\int_0^\infty \frac{d}{dt}\phi(X(t))\, dt = \phi(0).$$

This is an outgoing solution; an extra background homogeneous solution is not excluded by this calculation. With speed $\sigma = |\mathbf{b}(X)|$, tangent τ and unit normal ν, use $x = X(t) + y\nu(t)$ in a nonoverlapping tubular neighborhood. Frenet identities give $\nu' = -\sigma\kappa\tau$ and

$$x_t = \sigma(1-\kappa y)\tau, \quad x_y = \nu, \quad dx = H\, dt\, dy, \qquad H = \sigma(1-\kappa y) > 0.$$

Thus the same measure has density $\delta(y)/\sigma(t)$ with respect to spatial area.

Now perturb to $-\varepsilon\Delta u^\varepsilon + \operatorname{div}(\mathbf{b}u^\varepsilon) = \delta_0$. Away from the injection point, set $z = y/\sqrt{\varepsilon}$, $u^\varepsilon = \varepsilon^{-1/2}v^\varepsilon(t,z)$. The coordinate metric gives the exact identity

$$\Delta w = \frac{1}{H}\partial_t(H^{-1}w_t) + \frac{1}{H}\partial_y(Hw_y).$$

Also write $\mathbf{b} = (\sigma + \alpha y + O(y^2))\tau + (\beta y + O(y^2))\nu$. Then on compact $t > 0$, bounded z, provided the needed derivatives of v^ε are uniformly bounded,

$$\mathbf{b}\cdot Dv^\varepsilon = v_t^\varepsilon + \beta z v_z^\varepsilon + O(\sqrt{\varepsilon}), \qquad \varepsilon\Delta v^\varepsilon = v_{zz}^\varepsilon + O(\sqrt{\varepsilon}),$$

$$\operatorname{div}\mathbf{b} = \sigma'/\sigma + \beta + O(\sqrt{\varepsilon}).$$

For example the transverse first-derivative term in the Laplacian is $\varepsilon(H_y/H)\partial_y v^\varepsilon = O(\sqrt{\varepsilon})$; the tangential second-derivative terms are $O(\varepsilon)$. Thus the conditional leading equation is

$$v_t - v_{zz} + (\beta z v)_z + (\sigma'/\sigma)v = 0, \qquad v(0) = \delta_0/\sigma(0). \tag{PL4}$$

Its exact Gaussian solution, derived again in Problem 13, is

$$v(t,z) = \frac{e^{-z^2/\Gamma(t)}}{\sigma(t)\sqrt{\pi\Gamma(t)}}, \qquad \Gamma' = 4 + 2\beta\Gamma, \quad \Gamma(0) = 0.$$

Indeed $A = \sigma v$ satisfies $A_t = A_{zz} - (\beta z A)_z$; substituting the Gaussian gives the displayed ODE and mass $\int A = 1$. The transverse width is $\sqrt{\varepsilon\Gamma(t)}$. This does not by itself prove approximation of the original elliptic Green function, nor control the injection region or large times.

线上的密度必须带速度因子

注入量按行进时间计量，而面积元在曲线处含 $\sigma(t)$。因此横向 Dirac 前要乘 $1/\sigma$。若轨道封闭或反复回返，时间积分可能不是局部有限测度。

4.5.2 Laplace concentration and vanishing viscosity

Lemma 4.10 (Source asymptotics lemma, with tail control)
Let k, l be continuous on $\mathbb{R}$, $|l(y)| \le C(1+|y|)$, and $k(y) \ge cy^2 - C$, $c > 0$. If k has a unique minimum at y_0, then

$$\frac{\int l(y) e^{-k(y)/\varepsilon}\, dy}{\int e^{-k(y)/\varepsilon}\, dy} \longrightarrow l(y_0). \tag{LC4}$$

Expanded proof.

Subtract the minimum k_0 and put $h = k - k_0 \ge 0$. Fix a small neighborhood N of y_0 on which $|l - l(y_0)| < \eta$. On a large compact interval outside N, $h \ge d > 0$ by uniqueness. Increase the interval so that outside it $h(y) \ge d/2 + (c/2)y^2$, decreasing d if necessary. By continuity there is an interval of length $2r > 0$ near y_0 where $h < d/4$. The denominator is at least $2re^{-d/(4\varepsilon)}$. The numerator of the absolute error over the compact complement is at most $C_1 e^{-d/\varepsilon}$. The tail numerator is bounded by

$$C_2 e^{-d/(2\varepsilon)} \int (1+|y|) e^{-cy^2/(2\varepsilon)}\, dy,$$

whose ratio to the denominator tends to zero. Inside N the normalized error is at most η. Let $\varepsilon \downarrow 0$ and then $\eta \downarrow 0$. This proves concentration together with the required control of the unbounded observable l. □

For bounded initial g, let $P' = g$ and $K(y) = (x-y)^2/(2t) + P(y)$. The quadratic term dominates the Lipschitz primitive. Formula (VB4) and (LC4), with parameter 2ε, imply

$$u^\varepsilon(x,t) \longrightarrow \frac{x - y(x,t)}{t}$$

whenever the minimizing point is unique. By Chapter 3, at each $t > 0$ only countably many x have multiple minimizers, and the right side is the Lax–Oleinik entropy solution. Since $|u^\varepsilon| \le \|g\|_\infty$, this convergence is also L^1_{loc} in x for each fixed t. Joint a.e. convergence and the same bound give local space-time L^1 convergence away from zero. The multiple-minimizer set is measurable, for instance by expressing separation of two minimizers using rational compact neighborhoods; Fubini therefore applies to the countable time slices.

形式余项依赖尚未证明的统一界

把 y 换成 $\sqrt{\varepsilon} z$ 后，系数误差很小，但它仍乘着解的导数。只有这些导数在所考察区域有统一界，才能写成统一的 $O(\sqrt{\varepsilon})$。

逐点密度趋零不足以推出集中

归一化密度在每个非极小点趋零，还不能控制积分或无穷远的质量。证明需要分母下界、远离极小点的能量间隙，以及带权尾积分。

If k is smooth with finitely many nondegenerate global minima y_j, local Taylor expansion $y = y_j + \sqrt{\varepsilon} z$ and Gaussian domination give

$$\int l e^{-k/\varepsilon} = e^{-k_0/\varepsilon} \sqrt{2\pi\varepsilon} \left(\sum_j \frac{l(y_j)}{\sqrt{k''(y_j)}} + o(1) \right).$$

To justify domination, restrict to a small neighborhood where $k - k_0 \geq c_j |y - y_j|^2$, and use the energy gap outside. Thus at a tie the limit is a weighted average, not necessarily the velocity at one selected minimizing point. Values on shock curves do not affect the weak solution.

4.5.3 Geometric optics and stationary phase

For $u^\varepsilon = a e^{ip/\varepsilon}$ the exact wave-operator identity is

$$\begin{aligned} & e^{-ip/\varepsilon} (u^\varepsilon_{tt} - \Delta u^\varepsilon) \\ & \quad = \varepsilon^{-2} a (|Dp|^2 - p_t^2) \\ & \qquad + \frac{i}{\varepsilon} \{ 2 p_t a_t - 2 Dp \cdot Da + a(p_{tt} - \Delta p) \} + (a_{tt} - \Delta a). \end{aligned}$$

Matching the first two orders gives the eikonal and amplitude equations

$$p_t^2 = |Dp|^2, \qquad 2 p_t a_t - 2 Dp \cdot Da + (p_{tt} - \Delta p) a = 0.$$

On a smooth nonzero-frequency branch choose $p_t = \pm |Dp|$. The sign cannot be chosen independently at every point. For $u_{tt} - \sum a_{ij}(x) u_{x_i x_j} = 0$ the first equation becomes $p_t^2 = \sum a_{ij} p_{x_i} p_{x_j}$, with the corresponding amplitude expression obtained by the same differentiation. These identities define a candidate expansion. A theorem about exact solutions needs control of the residual and the evolution, especially near caustics.

Lemma 4.11 (Nonstationary phase; strengthening source Lemma 1)
If $a \in C_c^\infty$ and $|D\phi| > 0$ on its support, then for every integer $N \geq 1$, $\int e^{i\phi/\varepsilon} a = O(\varepsilon^N)$.

Expanded proof.

Near the support let $V = D\phi / |D\phi|^2$. Then $V \cdot D e^{i\phi/\varepsilon} = (i/\varepsilon) e^{i\phi/\varepsilon}$. Integration by parts gives

$$\int e^{i\phi/\varepsilon} a = -\frac{\varepsilon}{i} \int e^{i\phi/\varepsilon} \operatorname{div}(aV).$$

Iterate N times; the final smooth compactly supported amplitude has finite L^1 norm independent of ε. This also proves the source's linear-phase case. □

与第三章的最小化公式接上了

黏性解是最优速度的指数加权平均。黏性趋零时，权重集中到作用量最低点；最低点的分支竞争正对应守恒律的激波。

相位误差除以小参数会放大

即使 $p^\varepsilon - p = o(1)$，也不能直接推出两个指数相近；指数里的差是 $(p^\varepsilon - p)/\varepsilon$。振荡展开需要比普通逐点收敛更精确的误差尺度。

Lemma 4.12 (Quadratic phase; source Lemma 2)
For real symmetric invertible H and $a \in C_c^\infty(\mathbb{R}^d)$,

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = (2\pi\varepsilon)^{d/2} \frac{e^{i\pi\,\operatorname{sgn} H/4}}{|\det H|^{1/2}} [a(0) + O(\varepsilon)]. \quad \text{(QP4)}$$

Expanded proof (regularization, branch and error).

For $\operatorname{Re} q > 0$,

$$\int_{\mathbb{R}} e^{-qs^2+ibs}\,ds = \sqrt{\pi}\, q^{-1/2} e^{-b^2/(4q)}.$$

Choose the square root with positive real part. For real positive q this is the Gaussian identity; analytic continuation in the right half-plane proves it for complex q, since compact subsets admit an integrable Gaussian majorant. Equivalently the contour argument of Problem 15 proves it directly.

Diagonalize H orthogonally, multiply the phase by $e^{-\eta|z|^2}$, and apply this identity in each coordinate. As $\eta \downarrow 0$, an eigenvalue λ contributes the factor $|\lambda|^{-1/2} e^{i\pi\,\operatorname{sgn}\lambda/4}$ after the common $(2\pi\varepsilon)^{1/2}$ factor. Insert Fourier inversion for a before this calculation. Its transform is rapidly decreasing. The regularized Gaussian integral is uniformly bounded in frequency, so dominated convergence yields the exact oscillatory identity

$$\int e^{iz\cdot Hz/(2\varepsilon)} a(z)\,dz = C_{\varepsilon,H}(2\pi)^{-d/2} \cdot \int \hat{a}(\xi) e^{-i\varepsilon\,\xi\cdot H^{-1}\xi/2}\,d\xi,$$

where $C_{\varepsilon,H}$ is the prefactor in (QP4). Since $|e^{ir} - 1| \le |r|$, the bracket differs from $a(0)$ by at most

$$C\varepsilon \int |\xi|^2 |\hat{a}(\xi)|\,d\xi < \infty.$$

This proves both the constant and the error in (QP4). □

Lemma 4.13 (Morse coordinates; source Lemma 3(ii))
If $D\phi(0) = 0$ and $H = D^2\phi(0)$ is invertible, there is a local smooth diffeomorphism Φ with $\Phi(0) = 0$, $D\Phi(0) = I$, and $\phi(\Phi(z)) = \phi(0) + z\cdot Hz/2$.

Expanded proof (an editorial matrix proof).

Taylor's formula with integral remainder gives $\phi(x) = \phi(0) + x^T A(x) x/2$, where $A(x) = 2\int_0^1 (1-s) D^2\phi(sx)\,ds$ and $A(0) = H$. On symmetric matrices solve

$$H + 2S + SH^{-1}S = A(x).$$

The derivative in S at zero is $2I$, so the implicit function theorem gives a smooth symmetric $S(x)$ with $S(0) = 0$. Set $B = I + H^{-1}S$; then $B^T H B = A$. The map $z = B(x)x$ has derivative I at zero. Its local inverse Φ has the required properties. This works for indefinite H as well as positive definite H. □

For completeness the source's noncritical coordinate lemma is also elementary. Choose rows $r_1, \dots, r_{d-1}$ complementary to $D\phi(0)$, impose $r_j \cdot (y - x) = 0$ and $\phi(y) - \phi(0) = D\phi(0) \cdot x$. The y Jacobian is invertible; the implicit function theorem gives $y = \Phi(x)$ and differentiation gives $D\Phi(0) = I$.

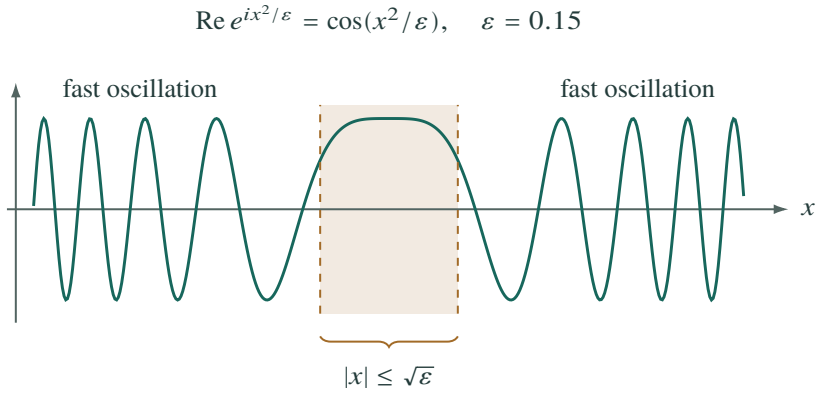


Diagram T4.4. Near a nondegenerate stationary point the scale is square-root. For phase x^2, oscillation is slow near its stationary point at zero and rapid farther away. The highlighted interval has width $2\sqrt{\varepsilon}$. Rescaling $x = \sqrt{\varepsilon}\, z$ explains the leading $\varepsilon^{1/2}$ factor in one dimension; cancellation, not pointwise smallness of the integrand, suppresses the outer region.

Theorem 4.14 (Nondegenerate stationary phase)
Suppose ϕ has just finitely many critical points z_j on supp a, *and all $H_j = D^2\phi(z_j)$ are invertible. Then*

$$\int e^{i\phi(z)/\varepsilon} a(z)\, dz = (2\pi\varepsilon)^{d/2} \sum_j \frac{e^{i\phi(z_j)/\varepsilon + i\pi\, \operatorname{sgn} H_j/4}}{|\det H_j|^{1/2}} [a(z_j) + O(\varepsilon)]. \tag{SP4}$$

Expanded proof.

Choose a partition of unity separating small neighborhoods of the critical points from the noncritical compact remainder. The latter is $O(\varepsilon^N)$ for arbitrary N by the first lemma. At each critical point change to Morse coordinates. The transformed amplitude is $a(\Phi(z))|\det D\Phi(z)|$ times its cutoff and equals $a(z_j)$ at the origin. Apply (QP4) to each localized integral and add them. Taking $N > d/2 + 1$ absorbs the noncritical remain-

der. □

Recovering the rays of Chapter 3. For high-frequency displacement $g^\varepsilon(z) = a(z)e^{ip_0(z)/\varepsilon}$ with $|Dp_0| > 0$ on $\operatorname{supp} a$ and zero initial velocity, Fourier representation splits the wave into two equal half-waves. After frequency rescaling their phases are

$$\phi_\pm(x, y, z, t) = (x - z) \cdot y \pm t|y| + p_0(z).$$

The critical equations are

$$y = Dp_0(z), \qquad x = z \mp t\,\frac{Dp_0(z)}{|Dp_0(z)|}. \tag{RAY4}$$

These are exactly the projected HJ rays for $p_t \mp |Dp| = 0$. The joint Hessian at a critical point is

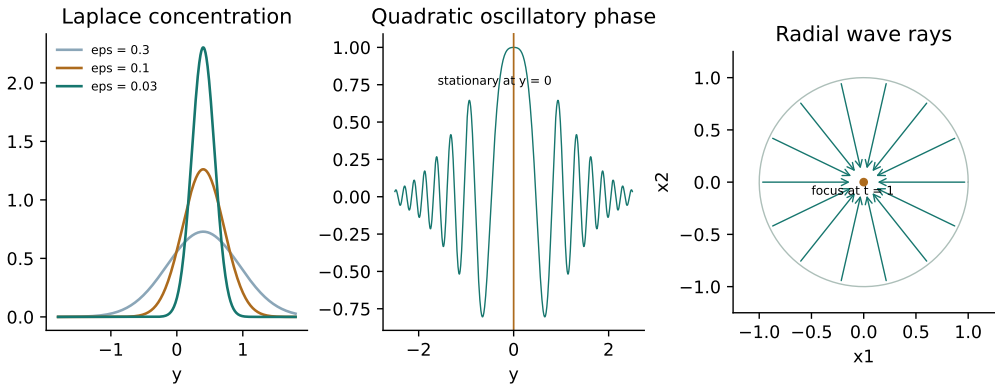
$$\begin{pmatrix} \pm\dfrac{t}{|y|}(I - \hat{y} \otimes \hat{y}) & -I \\ -I & D^2 p_0(z) \end{pmatrix}, \qquad \hat{y} = y/|y|.$$

Its determinant equals $(-1)^n$ times the Jacobian determinant of the ray map in (RAY4), by block elimination. For small time that map is locally invertible. The critical value is $p_0(z)$, since $(x-z)\cdot y \pm t|y| = 0$. Applying (SP4) in $2n$ variables cancels the half-wave prefactor $(2\pi\varepsilon)^{-n}$ and gives the amplitude $a(z)/|\det D_z x|^{1/2}$ and the Hessian signature phase. The two half-waves are then added with weight $1/2$.

The amplitude in y, z is initially not compact in y. To justify this application, first cut frequencies off to a compact neighborhood of $Dp_0(\operatorname{supp} a)$ disjoint from zero. Outside that neighborhood $D_z\phi = Dp_0 - y$ is nonzero. Repeated integration by parts in z makes the tail integrable in y, with arbitrarily high powers of ε; for large y each integration also supplies inverse powers of $|y|$. This reduces to the compactly supported theorem. The formula is local before caustics or summed over the finitely many nondegenerate preimages; it asserts no uniform approximation at a caustic.

行列式为零时不能继续用驻相公式

$|\det H|^{-1/2}$ 描述非退化临界点的集中强度。它发散时，表示这一阶近似失效；需研究更高阶相位，不能把发散系数直接当作精确解的无穷大。



Editorial figure. Laplace concentration is compared with oscillatory cancellation; the ray diagram displays a focusing map whose Jacobian eventually vanishes.

4.5.4 Periodic homogenization: expansion, cell problem, and a proof

We use the symmetric uniformly elliptic case, sufficient also for Problem 18: $A(y) = A(y)^T$ is smooth, Q-periodic, $Q = (0,1)^n$, and $\theta I \le A \le \Theta I$. Consider $-\operatorname{div}(A(x/\varepsilon)Du^\varepsilon) = f$ on a bounded Lipschitz domain with zero trace and $f \in L^2$. The source uses a formal expansion; we first derive it, then give a weak-convergence proof under these stated hypotheses.

Two-scale bookkeeping. Write $u^\varepsilon = u_0(x,y) + \varepsilon u_1(x,y) + \cdots$, $y = x/\varepsilon$, with all u_j periodic in y. Every spatial derivative becomes $D_x + \varepsilon^{-1} D_y$. Thus the three operator pieces are

$$L_1 = -D_y \cdot AD_y, \quad L_2 = -D_y \cdot AD_x - D_x \cdot AD_y, \quad L_3 = -D_x \cdot AD_x.$$

The coefficients of $\varepsilon^{-2}, \varepsilon^{-1}, 1$ give

$$L_1 u_0 = 0, \quad L_1 u_1 + L_2 u_0 = 0, \quad L_1 u_2 + L_2 u_1 + L_3 u_0 = f.$$

Testing the first identity by u_0 and integrating over Q gives $D_y u_0 = 0$; opposite periodic faces cancel. For zero-mean correctors χ_i, define

$$\operatorname{div}_y\{A(y)(e_i - D_y\chi_i)\} = 0, \qquad \int_Q \chi_i = 0. \tag{CELL4}$$

Then $u_1 = -\sum_i \chi_i(y) u_{x_i}$ solves the next equation. Averaging the order-one equation kills $L_1 u_2$ and yields

$$-\operatorname{div}(\overline{A}Du) = f, \qquad \overline{A}e_i = \int_Q A(y)(e_i - D\chi_i)\,dy. \tag{HOM4}$$

The minus sign in u_1 matches the source's corrector convention. Many other texts use $\varphi_i = -\chi_i$ and a plus sign instead.

Lemma 4.15 (Cell solvability and effective ellipticity)
(CELL4) has a unique zero-mean periodic weak solution. The matrix $\overline{A}$ is symmetric and satisfies $\theta I \le \overline{A} \le \Theta I$.

Expanded proof.

On the zero-mean periodic H^1 space minimize $\int_Q (e_i - Dv)\cdot A(e_i - Dv)$. Periodic Poincaré and ellipticity bound every minimizing sequence in H^1; weak compactness and weak lower semicontinuity give a minimizer. The first variation is $\int_Q A(D\chi_i - e_i)\cdot D\psi = 0$. Strict convexity in the gradient makes the difference of two minimizers constant, hence zero after normalization. These are standard Hilbert-space compactness and periodic Poincaré facts; the cell equation is not solved merely by asserting zero mean.

For $\xi \in \mathbb{R}^n$, let $\chi_\xi = \sum_i \xi_i \chi_i$. The cell identity allows any gradient $D\chi_\eta$ to be subtracted from the other factor, so

$$\eta \cdot \overline{A}\xi = \int_Q (\eta - D\chi_\eta)\cdot A(\xi - D\chi_\xi).$$

This proves symmetry. The minimum over all periodic v equals $\xi\cdot\overline{A}\xi$ by orthogonality. Its lower bound is $\theta\int_Q |\xi - Dv|^2 \ge \theta|\xi|^2$, since $\int_Q Dv = 0$ and $|Q| = 1$. The competitor $v = 0$ gives the upper bound. □

Theorem 4.16 (An editorial rigorous homogenization statement)
Under the symmetric hypotheses above, $u^\varepsilon \rightharpoonup u$ in H_0^1 and $u^\varepsilon \to u$ in L^2, where u is the unique weak solution of (HOM4).

Expanded proof (oscillating tests replace an unproved expansion).

Testing the original PDE by u^ε gives a uniform H_0^1 bound by ellipticity and Poincaré. Take a weakly convergent subsequence and use Rellich for strong L^2 convergence. The fluxes $\eta^\varepsilon = A(x/\varepsilon)Du^\varepsilon$ are bounded in L^2; take a weak limit η. The PDE gives $-\operatorname{div}\eta = f$.

For each i put

$$w_i^\varepsilon = x_i - \varepsilon\chi_i(x/\varepsilon), \qquad q_i^\varepsilon = A(x/\varepsilon)Dw_i^\varepsilon.$$

Periodic averaging implies $w_i^\varepsilon \to x_i$ in L^2_{loc} and $q_i^\varepsilon \rightharpoonup \overline{A}e_i$ in L^2_{loc}; also $\operatorname{div} q_i^\varepsilon = 0$. The averaging fact follows first by summing integrals on full ε-cells for continuous periodic functions and compact tests, then by $L^2(Q)$ approximation and uniform local cell bounds.

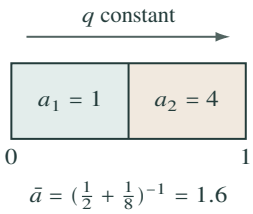
For $\phi \in C_c^\infty(U)$ symmetry gives $\eta^\varepsilon \cdot Dw_i^\varepsilon = Du^\varepsilon \cdot q_i^\varepsilon$. Integrate the two sides by parts in different ways:

$$\int \phi\, \eta^\varepsilon \cdot Dw_i^\varepsilon = \int f\phi w_i^\varepsilon - \int w_i^\varepsilon \eta^\varepsilon \cdot D\phi,$$

平均的是修正后的通量

有效系数通常不是直接对 A 求平均。微观梯度要先调整成 $e_i - D\chi_i$，使通量在单元中无散，再取平均；一维时结果是~~调和~~平均而非算术平均。

图 T4.M：一维有效系数



两个半格串联时通量相同，梯度与系数成反比。平均的是阻力 $1/a$，再取倒数。分段常系数可按弱形式理解。

$$\int \phi\, Du^\varepsilon \cdot q_i^\varepsilon = -\int u^\varepsilon q_i^\varepsilon \cdot D\phi.$$

Every product in these two right sides has a strong L^2 factor and a weak L^2 factor. The limits give $\eta_i = Du \cdot \overline{A}e_i$. By symmetry this says $\eta = \overline{A}Du$, proving (HOM4). Effective ellipticity gives uniqueness, so every subsequence has the same limit; the entire family converges. □

The expansion need not satisfy the original boundary condition termwise; boundary layers may be necessary for a quantitative error estimate. We have proved weak H^1 convergence, not strong convergence of the gradients or an $O(\varepsilon)$ global error. For nonsymmetric coefficients the source's formal cell calculation still applies with consistent index orientation; the rigorous testing argument uses adjoint correctors. That extension is not asserted here.

两个弱收敛量不能直接相乘

$A(x/\varepsilon)$ 与 Du^ε 都会振荡，不能分别取弱极限再相乘。振荡测试把乘积改写为一个强收敛因子和一个弱收敛因子的配对，才识别出有效通量。

4.6 Power Series

4.6.1 Noncharacteristic surfaces and determination of the full jet

For a quasilinear equation of order k, write

$$\sum_{|\gamma|=k} a_\gamma (D^{k-1}u, \dots, u, x) D^\gamma u + a_0(D^{k-1}u, \dots, u, x) = 0.$$

Its principal polynomial is $P(x, \zeta) = \sum_{|\gamma|=k} a_\gamma \zeta^\gamma$, with coefficients evaluated on the chosen lower-order jet. Cauchy data on a hypersurface Γ prescribe $u, \partial_\nu u, \dots, \partial_\nu^{k-1} u$. Here ∂_ν^j is the contraction of $D^j u$ with $\nu^{\otimes j}$ at the point, as in the source; iterating a variable normal field would also create lower-order curvature terms.

Theorem 4.17 (Source Theorem 1)
If $P(x, \nu) \neq 0$ along the specified data, then the PDE and the Cauchy data uniquely determine every derivative of any smooth solution along Γ. This is a determination statement, not an existence theorem.

Expanded proof.

On $\Gamma = \{x_n = 0\}$ the derivatives of the Cauchy functions in $x' = (x_1, \dots, x_{n-1})$ determine all derivatives of total order at most k, except $\partial_n^k u$. Its coefficient in the PDE is $a_{ke_n} = P(x, e_n) \neq 0$. Solving the PDE gives this missing derivative. Its tangential derivatives then give every derivative with exactly k normal differentiations. Differentiate the solved PDE once in x_n to find $\partial_n^{k+1} u$; all right-side normal orders are now at most k. Repeat, inducting on normal order and allowing arbitrary tangential differentiation at each stage.

For a curved surface use coordinates $y = \Psi(x)$ with $\Gamma = \{y_n = 0\}$. The chain rule shows that the coefficient of $\partial_{y_n}^k v$ in $u = v \circ \Psi$ is

$$\sum_{|\gamma|=k} a_\gamma (D\Psi_n)^\gamma .$$

On Γ, $D\Psi_n$ is a nonzero multiple of ν, so homogeneity and the noncharacteristic condition make this coefficient nonzero. The prescribed normal derivatives convert to y_n derivatives by a triangular system: at each order its leading coefficient is a nonzero power of $\partial_\nu \Psi_n$, while lower-order terms and tangential derivatives are already known. Apply the flat argument and transform the resulting jets back. □

For a surface $\{\rho = 0\}$, use $P(x, D\rho) \neq 0$. For the wave equation the principal polynomial is $\tau^2 - |\xi|^2$, giving characteristic light cones. For the heat equation, considered as a total-order-two PDE, the principal polynomial is $-|\xi|^2$: the time-zero surface has $(\xi, \tau) = (0, 1)$ and is characteristic. This convention does not negate forward parabolic well-posedness.

知道所有导数不等于知道函数

光滑函数可在一点具有全零 Taylor 系数而在邻域不为零，例如 e^{-1/x^2} 的零延拓。要由 jet 恢复解，需要额外的解析性或其他唯一性理论。

4.6.2 Analytic functions and coefficientwise majorants

An analytic function near zero is represented there by a convergent power series $f(x) = \sum_\gamma f_\gamma x^\gamma, f_\gamma = D^\gamma f(0)/\gamma!$. A formal series $F = \sum F_\gamma x^\gamma$ majorizes f if $F_\gamma \geq |f_\gamma|$ for every γ; write $F \gg f$.

Lemma 4.18 (Source majorant lemma, with the operations made explicit)
If F has nonnegative coefficients and converges near zero, then $F \gg f$ implies absolute convergence of f in a smaller neighborhood. Every analytic series has a majorant $C/(1 - (x_1 + \cdots + x_d)/r)$ after shrinking r. Majorization is preserved by addition, multiplication, partial differentiation, and composition with zero-constant-term series.

Expanded proof.

Coefficientwise comparison gives $\sum |f_\gamma x^\gamma| \leq F(|x_1|, \ldots, |x_d|)$. Analytic convergence on a smaller polydisc implies $|f_\gamma| \leq Cr^{-|\gamma|}$. But the coefficient of x^γ in the displayed rational series is $Cr^{-|\gamma|}|\gamma|!/\gamma! \geq Cr^{-|\gamma|}$. The multinomial expansion therefore proves the comparison. For products apply the triangle inequality to the finite coefficient convolution; for derivatives multiply each coefficient by its nonnegative integer exponent. Composition is a sum of such products. The zero constant term ensures that each fixed total-degree coefficient receives only finitely many composition contributions. These facts apply componentwise to vector and matrix series. □

4.6.3 Cauchy–Kovalevskaya: recurrence and actual convergence

After flattening an analytic noncharacteristic surface and subtracting analytic extensions of the Cauchy data, a quasilinear equation reduces to a first-order system for the derivatives of order below k. Write the normal coordinate as t and the $d = n - 1$ tangential coordinates as x':

$$U_t = \sum_{j=1}^{d} B_j(U, x')U_{x_j} + c(U, x'), \qquad U(x', 0) = 0. \tag{CK4}$$

Here U has m components. If coefficients originally depend on t, add a component $T_t = 1, T(x', 0) = 0$, and replace explicit t by T; this reduces to the displayed time-independent coefficient form.

Theorem 4.19 (Source Theorem 2)
For analytic B_j, c near $(0, 0)$, (CK4) has a unique analytic solution in a neighborhood of zero.

Expanded proof (the explicit majorant, fully verified).

1. Recursive formal solution. Write $U = \sum_{\ell \geq 1} t^\ell U_\ell(x')$. Equating the coefficient of t^ℓ gives

$$(\ell + 1)U_{\ell+1} = [t^\ell]\left\{\sum_j B_j(U, x')U_{x_j} + c(U, x')\right\}.$$

The right side only uses $U_1, \dots, U_\ell$ (and their tangential derivatives); $U_1 = c(0, x')$. Equivalently, each joint Taylor coefficient is determined recursively by smaller normal order. Composition is legitimate as a formal joint series since $U(0, 0) = 0$. No negative combinatorial coefficients occur in the operations on the right. This proves formal uniqueness, but not yet convergence.

2. Dominate the coefficient functions. Choose $r, C > 0$ so that every entry of every B_j and c is majorized by

$$F(Z, x') = \frac{Cr}{r - \sum_{j=1}^{d} x_j - \sum_{\ell=1}^{m} Z_\ell}.$$

The preceding lemma supplies such a common choice for the finite collection of analytic entries. The comparison system has all its components equal to a scalar v. With $s = \sum_j x_j$, its equation is

$$v_t = \frac{Cr}{r - s - mv}(md\, v_s + 1), \qquad v(s, 0) = 0. \tag{MAJ4}$$

控制函数值不等于控制系数

majorant 比较的是每个 Taylor 系数的绝对值。它之所以能控制递推解，是因为求导、乘积和复合只通过非负组合数来组合这些系数。

Each of the d matrix rows sums m identical derivatives, explaining the factor md.

3. Solve and check the majorant. Let $N = m(d+1)$ and set

$$v(s,t) = \frac{r - s - \sqrt{(r-s)^2 - 2NCr\,t}}{N}.$$

Use the square-root branch equal to $r - s$ near $(0,0)$. It is analytic near zero, is zero at $t = 0$, and obeys

$$(r-s)v - \frac{N}{2}v^2 = Cr\,t.$$

Implicit differentiation gives $v_t = Cr/(r-s-Nv)$ and $v_s = v/(r-s-Nv)$. Thus

$$\frac{Cr}{r-s-mv}(md\,v_s + 1) = \frac{Cr}{r-s-mv}\frac{r-s-Nv+md\,v}{r-s-Nv} = v_t,$$

because $N = md + m$. The expansion of $1 - \sqrt{1-z}$ has nonnegative coefficients; expanding $(r-s)^{-1}$ also does. Hence v is a nonnegative coefficient majorant in (x', t).

4. Compare recursively and obtain a PDE solution. At normal order zero both series vanish. Assume all coefficients up to normal order ℓ of U are dominated by those of $(v, \dots, v)$. The positive-operation rules and coefficient majorization of B_j, c give the same domination for the right side of the recurrence. Division by $\ell + 1 > 0$ preserves it. Thus the domination holds at every normal order. The convergent analytic series v bounds the absolute joint series of every component of U on a smaller polydisc. On still smaller polydiscs derivatives also converge normally. The analytic compositions are therefore genuine functions; their Taylor coefficients satisfy (CK4), so the analytic identity holds near zero. Analytic uniqueness follows from the coefficient recurrence and equality with the Taylor series. □

Returning to the scalar higher-order equation. Include every derivative $D^\gamma u$, $|\gamma| < k$, as a component. Normal derivatives of lower-order components are other components; a derivative of a top-order component either is a tangential derivative of another top-order component or, for the pure normal term, is obtained from the solved original PDE. This yields the quasilinear system above. The formal scalar jet determined by the noncharacteristic recursion satisfies these derivative identities and hence

优级数的作用是证明收敛

形式递推只确定候选解的系数。显式平方根解控制这些系数的绝对值，才把形式级数变成可求导的解析函数；数据解析本身还没有完成收敛证明。

the formal system. Its formally unique system solution is exactly the analytic series constructed above. Consequently all compatibility identities between the components hold as analytic identities. The zeroth component really is a scalar solution with the required derivatives; introducing a vector of independent unknowns did not silently discard compatibility.

For nonzero analytic initial data subtract a polynomial in t with analytic x' coefficients matching the prescribed normal jets, then incorporate its known terms into the coefficients. Analytic changes of coordinates preserve analyticity. This proves the quasilinear analytic noncharacteristic Cauchy theorem. A fully nonlinear version additionally solves its highest normal derivative by the analytic implicit function theorem and needs the corresponding nonzero derivative condition; that broader theorem is a further result, not proved by the quasilinear reduction alone.

解析存在不保证低正则范数下稳定

CK 给出非特征解析数据的局部解析解，未给出 C^0 或 Sobolev 数据到解的连续依赖。习题 2 的 Laplace Cauchy 问题正好说明存在与稳定必须分别检查。

4.7 Problems

Source: Evans §4.7, Problems 1–19. Statements are restated concisely; all solutions and domain clarifications below are editorial.

4.7.1 Separation, scaling and traveling profiles

1. Aronsson's infinity-harmonic example.

Solution. For $u = f(x) + g(y)$, the equation is $(f')^2 f'' + (g')^2 g'' = 0$. Set these terms equal to $64/81$ and $-64/81$ respectively. A resulting profile is

$$u(x, y) = |x|^{4/3} - |y|^{4/3}.$$

Off the axes, $(f')^2 f'' = 64/81$ and $(g')^2 g'' = -64/81$, so the PDE holds classically there. The function is C^1 on $\mathbb{R}^2$ but not C^2 on the axes. It is a global viscosity solution of $F(Du, D^2u) = -Du \cdot D^2u\, Du = 0$, as follows.

At $x = 0$, $y \neq 0$, no C^2 function touches u from above: the positive $|x|^{4/3}$ increment is larger than every quadratic with zero first derivative at zero. For a test ϕ touching from below, $D\phi = Du$ and $\phi_{yy} \leq u_{yy} < 0$, so $D\phi \cdot D^2\phi D\phi \leq 0$, giving $F \geq 0$. At $y = 0$, $x \neq 0$ the lower tests are absent, while an upper test has $\phi_{xx} \geq u_{xx} > 0$ and hence $F \leq 0$. At the origin every touching test has gradient zero, and $F = 0$. These are exactly the viscosity inequalities. An affine function supplies a global classical solution if that solution class is required; the nonaffine Aronsson example needs the explicit weaker interpretation. □

经典解与黏性解要分清

$|x|^{4/3} - |y|^{4/3}$ 不能在坐标轴上按经典二阶导数代入。本题用接触测试说明它的全局解意义；黏性解的一般理论留待第十章。

2. Hadamard instability of Laplace Cauchy data.

Solution. The separated ansatz $\sin(nx_1)T(x_2)$ gives $T'' = n^2 T$. The data $T(0) = 0, T'(0) = 1/n$ yield

$$u_n(x_1, x_2) = n^{-2} \sin(nx_1) \sinh(nx_2).$$

The Cauchy data tend uniformly to zero, but on any line $x_2 = h > 0$, $\sup_{x_1} |u_n| = n^{-2} \sinh(nh) \to \infty$. Thus the data-to-solution map is not continuous in these uniform data and solution norms. One must not claim pointwise divergence at every fixed x_1, since $\sin(nx_1)$ can vanish. Existence of analytic solutions does not restore this stability. □

3. A balanced bistable front.

Solution. For $f(v) = -2v^3 + 3v^2 - v$, $\int_0^1 f = 0$. The energy identity (BF4) forces speed $c = 0$ for a nonconstant front. Here $F(v) = -v^2(1 - v)^2/2$, so $(v')^2 = v^2(1 - v)^2$ and the increasing branch obeys $v' = v(1 - v)$. Hence $v(s) = (1 + e^{-(s-s_0)})^{-1}$. It has the required endpoints and vanishing derivatives. Its second derivative is $v(1 - v)(1 - 2v) = -f(v)$, verifying the ODE directly. □

4. Emden–Fowler coordinates.

Solution. Put $a = 2/(p - 1)$, $t = \log r$, and $X(t) = e^{at} v(e^t)$. Then

$$v' = r^{-a-1}(X' - aX), \quad v'' = r^{-a-2}\{X'' - (2a + 1)X' + a(a + 1)X\}.$$

Since $ap = a + 2$, the radial equation becomes the autonomous ODE

$$X'' + (n - 2 - 2a)X' + a(a - n + 2)X + X^p = 0.$$

This change treats $r > 0$; regularity at the center is an additional condition corresponding to $t \to -\infty$. □

5. Fast-diffusion scaling profile.

Solution. In the standard parabolic range $\max\{0, (n-2)/n\} < m < 1$, use $\beta = 1/(n(m-1)+2) > 0$, $\alpha = n\beta$ as in §4.2. Integrating $(W^m)' = -\beta r W$ gives

$$u(x, t) = t^{-\alpha} \left(C + \frac{(1 - m)\beta}{2m} |x|^2 t^{-2\beta} \right)^{-1/(1-m)}, \qquad C > 0.$$

It is positive and smooth at every $t > 0$. Its spatial tail is $|x|^{-2/(1-m)}$; integrability is equivalent to $2/(1-m) > n$, the stated mass-critical lower bound. Directly differentiating $W^{m-1} = C + (1-m)\beta r^2/(2m)$ verifies the profile ODE. Unlike the porous-medium profile it has no compact support. If $n = 1$, the source's written bound $(n-2)/n < m < 1$ alone also permits $m \le 0$. The above forward diffusion construction uses $m > 0$; it does not assert a decaying positive profile for those nonparabolic exponents. At $m = 0$ the positive equation is $u_t = 0$, a separate degeneracy. □

快扩散仍需正的扩散指数

扩散系数为 mu^{m-1}。$0 < m < 1$ 时在低密度处很大，产生代数尾；若 $m < 0$，最高阶符号反转，已不是同一类前向抛物扩散。

6. A boundary blowup solution.

Solution. Assume $n \ge 3$ and set $p = (n+2)/(n-2)$. For $u = A(1-r^2)^{-b}$, radial differentiation gives

$$\Delta u = 2bA(1 - r^2)^{-b-2}\{n + (2b + 2 - n)r^2\}.$$

Choose $b = (n-2)/2$ so the brace is n and $bp = b+2$. Then $-\Delta u + u^p = 0$ if

$$A = [n(n-2)]^{(n-2)/4}.$$

This is smooth and finite in the open ball but diverges at every boundary point, as required. □

7. Viscous shock profiles and the entropy direction.

Solution. For $u_t + F(u)_x - au_{xx} = 0$ put $u = v(x - ct)$. Integration yields

$$av' = F(v) - cv + b =: Q(v).$$

Separation gives $s = \int_{v(0)}^{v(s)} a/Q(z)\,dz$ wherever $Q \ne 0$. A connection from u_l to u_r forces both endpoint values of Q to vanish; otherwise v' stays away from zero near that endpoint and v cannot have the asserted limit. Subtracting the endpoint identities gives $c = (F(u_l) - F(u_r))/(u_l - u_r)$. Conversely, for $u_l > u_r$, strict convexity makes $Q < 0$ on (u_r, u_l) when b is chosen to make the endpoints zero. The zeros are simple: $Q'(u_r) < 0 < Q'(u_l)$. The integral therefore diverges to $+\infty$ as $v \downarrow u_r$ and to $-\infty$ as $v \uparrow u_l$. Its inverse is the decreasing heteroclinic profile, unique up to shift.

Normalize $v^\varepsilon(0) = (u_l + u_r)/2$. Then $v^\varepsilon(s) = v^1(s/\varepsilon)$. For $x - ct < 0$ its limit is u_l, for $x - ct > 0$ it is u_r, and on $x = ct$ it is the chosen midpoint. Boundedness gives local L^1 convergence to the Rankine–Hugoniot downward shock. The viscous connection thus selects the entropy direction. □

4.7.2 Transforms and linearization

8. Schrödinger dispersive bound.

Solution. For $g \in L^1$, the modulus of the kernel in (SC4) is $(4\pi|t|)^{-n/2}$. The triangle inequality gives $\|u(t)\|_\infty \le (4\pi|t|)^{-n/2}\|g\|_1$. The bound holds for both signs of $t \neq 0$. It is consistent with conservation of the L^2 norm, since the norms measure different features. □

9. Inverting a nonlinear heat equation.

Solution. For $y = u(x,t)$, $x = v(y,t)$ and $u_x > 0$, differentiate $u(v(y,t),t) = y$:

$$u_x = 1/v_y, \quad u_t = -v_t/v_y, \quad u_{xx} = -v_{yy}/v_y^3.$$

Substitution into $u_t = u_{xx}/u_x^2$ gives $v_t = v_{yy}$. This is local in the range of u; the inverse's spatial interval and boundary conditions may move with time. □

10. A local transformation to Liouville's equation.

Solution. Taken literally for every solution u with a smooth real $f : \mathbb{R}^3 \to \mathbb{R}$, the request is impossible: constant u would give constant $w = f(u,0,0)$, while $w_{xt} = e^w > 0$. The intended transformation must be local on a nondegenerate branch.

For $u_{xt} = 0$, write $u = A(x) + B(t)$. On $u \neq 0$, $u_x u_t > 0$, choose

$$w = f(u,u_x,u_t) = \log(2u_x u_t/u^2) = \log 2 + \log|u_x| + \log|u_t| - 2\log|u|.$$

The first two derivative terms have zero mixed derivative, while

$$(-2\log|u|)_{xt} = 2u_x u_t/u^2 = e^w.$$

This verifies the transformation on its real domain. More generally, for an additive ansatz $f = a(z) + b(p) + c(q)$, the mixed derivative equals $a''(u)u_x u_t$, which explains the logarithmic separation. □

变换的定义域不能隐去

本题对数公式只在 $u \neq 0$ 且 $u_x u_t > 0$ 的分支上成立。常数解给出一个直接检查：不存在对所有三元输入光滑、又能转换所有解的实函数。

11. Lax pairs preserve a smooth eigenvalue branch.

Solution. Differentiate $Lw = \lambda w$ and take the real Hilbert inner product with $w \neq 0$. Symmetry of L cancels the two w_t terms, leaving $\lambda' \|w\|^2 = (L_t w, w)$. If $L_t = BL - LB$, then

$$(L_t w, w) = \lambda (Bw, w) - (Bw, Lw) = 0.$$

Thus $\lambda' = 0$. For unbounded operators all these pairings require a common suitable domain and the differentiability stipulated by the exercise; formal operator symbols alone do not supply these hypotheses. □

12. The KdV commutator.

Solution. Let $D = \partial_x$, $L = -D^2 + u$, $B = -4D^3 + 6uD + 3u_x$. Using $D(av) = aDv + a_x v$ gives

$$\begin{aligned} [-4D^3, u] &= -12u_x D^2 - 12u_{xx} D - 4u_{xxx}, \\ [6uD, -D^2] &= 12u_x D^2 + 6u_{xx} D, \\ [3u_x, -D^2] &= 6u_{xx} D + 3u_{xxx}, \\ [6uD, u] &= 6uu_x. \end{aligned}$$

All derivative terms cancel, leaving $[B, L] = -u_{xxx} + 6uu_x$. Since L_t is multiplication by u_t, $(L_t - [B, L])v = (u_t + u_{xxx} - 6uu_x)v$. For a KdV solution, every eigenvalue branch meeting Problem 11's operator hypotheses is constant. This is a conditional spectral conclusion, not a domain-free proof about every possible realization of the differential expression. □

4.7.3 Asymptotics, homogenization and analytic obstructions

13. The plume's Gaussian width.

Solution. Set $A = \sigma v$ in (PL4). Then $A_t - A_{zz} + (\beta z A)_z = 0$ and $A(0) = \delta_0$. For $A = (\pi\Gamma)^{-1/2} e^{-z^2/\Gamma}$,

$$A_t/A = -\Gamma'/(2\Gamma) + z^2\Gamma'/\Gamma^2, \quad A_{zz}/A = -2/\Gamma + 4z^2/\Gamma^2.$$

Also $(\beta z A)_z/A = \beta - 2\beta z^2/\Gamma$. Both coefficients vanish precisely when $\Gamma' = 4 + 2\beta\Gamma$, with $\Gamma(0) = 0$ for the point-mass trace. Hence

$$\Gamma(t) = 4e^{2\int_0^t \beta(r)\,dr} \int_0^t e^{-2\int_0^s \beta(r)\,dr}\,ds > 0.$$

The normalized Gaussian has mass one and converges weakly to δ_0, verifying the initial condition after division by σ. □

14. A relaxation system produces logarithmic diffusion.

Solution. Write the equations as $u_t^\varepsilon + \varepsilon^{-1}u_x^\varepsilon = \varepsilon^{-2}((v^\varepsilon)^2 - (u^\varepsilon)^2)$ and the opposite equation for v^ε. Multiplying by ε^2 and using the assumed bounded remainders yields $(u^0)^2 = (v^0)^2$. Positivity forces $u^0 = v^0 = w$.

Define the bounded corrections $U^\varepsilon = (u^\varepsilon - w)/\varepsilon$ and $V^\varepsilon = (v^\varepsilon - w)/\varepsilon$; the source's remainder is only assumed to be order ε. The first equation, multiplied by ε, gives

$$(V^\varepsilon - U^\varepsilon)(u^\varepsilon + v^\varepsilon) = u_x^\varepsilon + \varepsilon u_t^\varepsilon.$$

On compact sets $w > 0$, so the denominator stays positive and $V^\varepsilon - U^\varepsilon \to w_x/(2w)$ locally uniformly. Adding the two original equations gives

$$u_t^\varepsilon + v_t^\varepsilon + \partial_x(U^\varepsilon - V^\varepsilon) = 0.$$

Pass to the limit in distributions, using the uniform derivative bounds in the hypotheses. The result is $2w_t - \partial_x(w_x/(2w)) = 0$, or

$$w_t - \tfrac{1}{4}(\log w)_{xx} = 0.$$

No separate limit of the two first-order corrections is needed. □

只能按实际余项阶数做比较

若余项仍是 ε 乘一个有界函数，就不能把固定的 u_1, v_1 自动认作唯一一阶系数。证明直接控制两者的差，再用两方程之和取极限。

15. Gaussian contour deformation.

Solution. For $q = \delta - i\lambda$, $\delta > 0$, the completed-square variable puts the contour on a translated line $e^{i\theta}\mathbb{R} + c$, where $\theta = \arg\sqrt{q}$ satisfies $|\theta| < \pi/4$. First remove the translation: close the two parallel line segments at parameters $\pm R$ by segments of bounded length. On those connectors $\operatorname{Re} z^2 \geq R^2\cos(2\theta) - CR - C$, so their integrals of e^{-z^2} tend to zero. Cauchy's theorem equates the two infinite line integrals.

Then rotate $e^{i\theta}\mathbb{R}$ onto $\mathbb{R}$. On connecting arcs of radius R within the fixed sector, $|e^{-z^2}| \leq e^{-R^2\cos(2\theta)}$; their length is $O(R)$, so they also vanish. The integral is $\sqrt{\pi}$ with the inherited orientation. This argument holds for each fixed $\delta > 0$; the limiting Fresnel branch is obtained afterward, not by claiming uniform decay on a sector boundary where the cosine is zero. □

16. Schrödinger initial recovery.

Solution. For $g \in C_c^\infty$, apply (SP4) to $\phi(y) = |x - y|^2/4$ with parameter $t > 0$. Its sole critical point is $y = x$ and Hessian is $I/2$. The factor $(4\pi t)^{n/2}e^{i\pi n/4}$ from stationary phase cancels the prefactor in (SC4), so $u(x,t) = g(x) + O(t)$, uniformly for x in compact sets. For general $g \in L^2$,

the multiplier and dominated convergence give $u(t) \to g$ strongly in L^2, as proved in §4.3. One cannot replace this by absolute-integral approximate-identity theory for a positive kernel: the Schrödinger kernel is oscillatory and is not in L^1 as a function of space. □

17. One-dimensional effective coefficient.

Solution. Set $F(x) = \int_0^x f(s)\,ds$ and $b = 1/a$. Integration of the flux equation gives

$$(u^\varepsilon)' = b(x/\varepsilon)(C_\varepsilon - F(x)), \qquad C_\varepsilon = \frac{\int_0^1 b(x/\varepsilon)F(x)\,dx}{\int_0^1 b(x/\varepsilon)\,dx}.$$

The boundary condition $\int_0^1 (u^\varepsilon)' = 0$ determines this constant. Periodic averaging gives $b(x/\varepsilon) \rightharpoonup^* \langle b\rangle$, $C_\varepsilon \to C = \int_0^1 F$. The formula is uniformly bounded in L^2, and its weak limit is $u' = \langle b\rangle(C - F)$. The zero traces and uniqueness identify the weak H_0^1 limit, with $-\overline{a}\,u'' = f$, $\overline{a} = \langle 1/a\rangle^{-1}$. For the cell problem, $a(1 - \chi')$ is constant; integrating $\chi' = 1 - \overline{a}/a$ over a period yields exactly the same harmonic mean. □

18. Primal and dual variational principles.

Solution. All averages use the unit cube, of volume one. For $\chi_\xi = \sum \xi_i \chi_i$, the cell orthogonality proves

$$\xi \cdot \overline{A}\xi = \min_{v \text{ periodic}} \int_Q (\xi + Dv) \cdot A(\xi + Dv),$$

with minimizer $v = -\chi_\xi$. For the dual problem fix η and set $\xi = \overline{A}^{-1}\eta$, $\sigma_* = A(\xi - D\chi_\xi)$. It is periodic, divergence free and has mean η. For any admissible $\sigma = \sigma_* + \tau$, $\int_Q \tau = 0$ and $\operatorname{div}\tau = 0$, so periodic integration by parts gives

$$\int_Q \tau \cdot A^{-1}\sigma_* = \int_Q \tau \cdot (\xi - D\chi_\xi) = 0.$$

Expansion of the square therefore shows that σ_* minimizes $\int_Q \sigma \cdot A^{-1}\sigma$, with minimum $\xi \cdot \overline{A}\xi = \eta \cdot \overline{A}^{-1}\eta$.

The primal competitor $v = 0$ gives $\overline{A} \le \int_Q A$. The constant dual competitor $\sigma = \eta$ gives $\overline{A}^{-1} \le \int_Q A^{-1}$. Inverting positive definite matrices reverses this order (conjugate by a square root and diagonalize), so

$$\left(\int_Q A^{-1}\right)^{-1} \le \overline{A} \le \int_Q A.$$

These are quadratic-form inequalities, not entrywise inequalities. □

19. Analytic initial data need not give a time-analytic heat solution.

Solution. The total-order-two principal polynomial of $u_t - u_{xx}$ is $-\xi^2$. At the normal to $\{t = 0\}$, $(\xi, \tau) = (0, 1)$, it vanishes. Thus this is a characteristic surface for the CK theorem.

If a solution with $u(x, 0) = 1/(1 + x^2)$ were analytic near $(0, 0)$, the PDE would give

$$\partial_t^k u(0, 0) = \partial_x^{2k}(1 + x^2)^{-1}|_{x=0} = (-1)^k (2k)!.$$

Its time Taylor series along $x = 0$ would be $\sum_{k\geq 0}(-1)^k (2k)! t^k/k!$. For every $t \neq 0$, successive term magnitudes have ratio $|t|(2k+2)(2k+1)/(k+1) \to \infty$, so terms do not even tend to zero. There is no analytic solution in a neighborhood of $(0, 0)$, and therefore none on all $\mathbb{R}^2$. The bounded forward heat solution still exists and is smooth for $t > 0$; it is time analyticity across the initial surface that fails. □

数据解析与跨初始面的解析性不同

空间初值在原点附近解析，却可能因连续两次空间求导反复产生过快的阶乘增长，使时间 Taylor 级数发散。这与正时间的热方程光滑化没有矛盾。

4.8 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 4, printed pp. 159–236 (local PDF pp. 176–253). Section order and explicit source theorem numbers follow that edition; exposition, proof expansions, Chinese annotations, figures and exercise solutions are editorial reconstructions.

The source's §4.8 directs readers to Pinsky, Strauss, Thoe and Zachmanoglou, and Weinberger for separation; Grindrod and Nishiura for Turing instability; Linares–Ponce, Tao and Olver for dispersive equations and symmetries; Stein–Weiss, Stein, Hörmander, Rauch, Treves and Zworski for Fourier methods; Helgason, John and Lax for Radon methods; and Courant–Hilbert for hodograph and Legendre transforms. Its asymptotics discussion credits Neu, Ralston and Bensoussan–Lions–Papanicolaou; its analytic Cauchy theory refers to Folland, John and DiBenedetto. These are source reading directions, not claims of independent consultation of every listed work. The Aronsson and Hadamard exercise attributions are retained; the source credits Kurtz for Problem 14 and Kohn for Problem 18.

The Chinese chapter opening (PDF p. 129) lists the section starts at printed pp. 111, 116, 124, 136, 140, 154, 162 and 165. Selected scanned pages and the exact review range are recorded in the chapter log. A line-by-line comparison of the whole Chinese chapter has not been made.

Scope of the proofs. Spectral basis existence, periodic Poincaré, weak Hilbert compactness, Rellich and standard local hyperbolic ODE manifold theory are explicitly invoked prerequisites. Formal plume and geometric-optics expansions are kept distinct from convergence theorems. Stationary phase is proved for smooth compact amplitudes and nondegenerate critical points. The homogenization proof uses the symmetric uniformly elliptic periodic setting; it makes no global boundary-layer error claim. CK is proved in its analytic quasilinear noncharacteristic setting, not for arbitrary smooth data. Problems 1, 5 and 10 explicitly record the relevant solution classes or transformation domains instead of silently using formulas outside their domains. Numerical model checks and typesetting checks do not constitute independent mathematical review.

Part II

Theory for Linear Partial Differential Equations

Chapter 5

Sobolev Spaces

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 5; printed pp. 239–292; PDF pages 256–309.

Chinese reference: 中译本：书内第 169–207 页；PDF 第 187–225 页。

Editorial study edition. This is an original expanded exposition following Evans, Chapter 5, rather than a verbatim transcription. Source section and theorem identifiers are given in headings. Proofs, worked solutions, and Chinese annotations are editorial reconstructions. Our theorem numbers run throughout the chapter and differ from the source. The 21 source problems are concisely restated with editorial solutions in §5.10.

The model is the Dirichlet problem $-\Delta u = f$. Its weak formulation is

$$\int_U Du \cdot D\phi \, dx = \langle f, \phi \rangle, \qquad \phi \in H_0^1(U).$$

Only one derivative of u is required. This chapter makes each symbol precise and explains which estimates survive limits of approximate solutions.

先看整条逻辑链

弱导数容纳能量有限但不光滑的函数；逼近让经典计算可用；延拓处理边界；迹解释边界值；嵌入与紧性负责估计和取极限。

Reading route and prerequisites. Read §5.2–§5.7 in order; use §5.8 as a toolbox and §5.9 for evolution equations. Unless stated otherwise, $U \subset \mathbb{R}^n$ is open, derivatives are weak, and functions equal almost everywhere are identified. The notation $V \Subset U$ means $\overline{V}$ is compact and contained in U. Constants C are independent of the functions being estimated. Basic integration theory, Hölder and Minkowski inequalities, completeness of L^p, smooth partitions of unity, and explicitly named functional analysis theorems are prerequisites; they are not proved anew here.

5.1 Hölder Spaces

5.1.1 Definitions and a complete function space

Definition 5.1 (Evans §5.1)
For $0 < \gamma \le 1$, define

$$[u]_{C^{0,\gamma}(\overline{U})} = \sup_{x \neq y \in U} \frac{|u(x) - u(y)|}{|x - y|^{\gamma}}.$$

The space $C^{k,\gamma}(\overline{U})$ consists of C^k functions on U whose derivatives through order k extend continuously to $\overline{U}$ and have finite norm

$$\|u\|_{C^{k,\gamma}} = \sum_{|\alpha| \le k} \|D^{\alpha} u\|_{\infty} + \sum_{|\alpha| = k} [D^{\alpha} u]_{C^{0,\gamma}}.$$

Here $C^{0,1}$ means Lipschitz continuous.

For $0 < \gamma < 1$, $|x|^{\gamma}$ on $[-1, 1]$ is γ-Hölder but not Lipschitz at zero. Concavity gives $|a^{\gamma} - b^{\gamma}| \le |a - b|^{\gamma}$ for $a, b \ge 0$, whereas $|h|^{\gamma}/|h| \to \infty$. This distinguishes a quantitative modulus of continuity from differentiability.

半范数为何不够

常数函数的 Hölder 半范数为零，故必须加上 $\|u\|_{\infty}$ 才成为范数。γ 控制的是振荡速度，不是函数大小。

Theorem 5.2 (Evans §5.1, Theorem 1)
$C^{k,\gamma}(\overline{U})$ is a Banach space.

Expanded proof (editorial; also source Problem 1).

Let (u_j) be Cauchy. Each $D^{\alpha} u_j$, $|\alpha| \le k$, converges uniformly on $\overline{U}$ to a continuous bounded function g_{α}. Put $u = g_0$. We must verify that these limits are its derivatives.

Fix a closed ball contained in U. For a coordinate segment in the ball,

$$D^{\alpha} u_j(x + h e_i) - D^{\alpha} u_j(x) = \int_0^h D^{\alpha + e_i} u_j(x + t e_i)\, dt \quad (|\alpha| < k).$$

Uniform convergence allows passage through the finite integral. The fundamental theorem of calculus yields $\partial_i g_{\alpha} = g_{\alpha + e_i}$. Induction in the derivative order proves $u \in C^k(U)$ with the required extensions.

Given $\varepsilon > 0$, choose N with $\|u_j - u_l\|_{C^{k,\gamma}} \le \varepsilon$ for $j, l \ge N$. Pass $l \to \infty$ in the difference quotients for each fixed pair x, y, then take suprema. Lower semicontinuity of the finite sum of norm components gives $\|u_j - u\|_{C^{k,\gamma}} \le \varepsilon$. Thus u belongs to the space and the convergence is in norm. The norm axioms follow from the triangle inequality and the included supremum norm. □

Interpolation (source Problem 2). For $0 < \beta < \gamma < 1$, set $\theta = (\gamma - \beta)/(1 - \beta)$. For every $x \neq y$,

$$\frac{|u(x) - u(y)|}{|x - y|^{\gamma}} = \left(\frac{|u(x) - u(y)|}{|x - y|^{\beta}} \right)^{1-\theta} \left(\frac{|u(x) - u(y)|}{|x - y|} \right)^{\theta}.$$

Hence $[u]_{\gamma} \leq [u]_{\beta}^{1-\theta} [u]_{1}^{\theta}$. The two-term Hölder inequality on the supremum and seminorm components also gives $\|u\|_{C^{0,\gamma}} \leq \|u\|_{C^{0,\beta}}^{1-\theta} \|u\|_{C^{0,1}}^{\theta}$. For $\gamma = 1$ this is an identity.

5.2 Sobolev Spaces

5.2.1 Weak derivatives: transfer derivatives to test functions

Definition 5.3 (Evans §5.2.1)
For $u, v \in L^1_{\mathrm{loc}}(U)$, write $D^{\alpha} u = v$ if

$$\int_U u D^{\alpha} \phi \, dx = (-1)^{|\alpha|} \int_U v \phi \, dx \qquad (\phi \in C_c^{\infty}(U)).$$

The function v is a weak derivative. A distributional derivative always exists as a functional; it need not be represented by a locally integrable function.

For a classical $C^{|\alpha|}$ function this identity follows by $|\alpha|$ integrations by parts, each contributing one minus sign. Thus the two notions agree whenever classical derivatives exist.

Lemma 5.4 (Evans §5.2.1: uniqueness)
A weak derivative is unique up to equality almost everywhere.

Expanded proof.

If v, w both work, $g = v - w$ satisfies $\int g\phi = 0$ for every test function. Take a nonnegative smooth kernel ρ, supported in the unit ball with integral one, and set $\rho_{\varepsilon}(x) = \varepsilon^{-n} \rho(x/\varepsilon)$. On $V \Subset U$, for sufficiently small ε, testing with $\rho_{\varepsilon}(x - \cdot)$ gives $g * \rho_{\varepsilon} = 0$. Approximate identities converge to g in $L^1(V)$, so $g = 0$ there. A countable exhaustion of U proves the assertion. The approximate-identity fact follows from

$$\|\rho_{\varepsilon} * g - g\|_{L^1(V)} \leq \int \rho(z) \|g(\cdot - \varepsilon z) - g\|_{L^1(V)} \, dz \to 0,$$

using a cutoff on a larger compact set, translation continuity in L^1, and domination by twice that cutoff's L^1 norm. □

导数的极限要认领

一致收敛保证候选导数连续；线段积分恒等式保证它确实是极限函数的导数。仅凭 $u_j \to u$ 一致，不能推出 $Du_j \to Du$。

为何没有边界项

试验函数的支集紧含于 U，所以在边界附近恒为零。弱导数的定义不规定 u 的边界值；边界条件要靠 $W_0^{1,p}$ 和迹来表达。

Corners, jumps, and singularities: editorial examples. On $(-1,1)$, splitting the integral at zero gives

$$\int_{-1}^{1} |x|\phi'(x)\,dx = -\int_{-1}^{1} \operatorname{sgn}(x)\phi(x)\,dx.$$

The interface terms vanish because $|0| = 0$. A corner therefore has a bounded weak first derivative. Its second distributional derivative is $2\delta_0$, which is not a locally integrable function.

For $H = \mathbf{1}_{(0,1)}$ on $(-1,1)$, $-\int H\phi' = \phi(0)$. If some locally integrable g represented this, choose $|\phi_\varepsilon| \le 1$, supported in $(-\varepsilon,\varepsilon)$, with $\phi_\varepsilon(0) = 1$. Then $|\int g\phi_\varepsilon| \le \int_{-\varepsilon}^{\varepsilon} |g| \to 0$, a contradiction.

On $B_1 \subset \mathbb{R}^n$, $n \ge 2$, put $u(x) = |x|^{-a}$, $a > 0$. Away from zero, $|Du| = a|x|^{-a-1}$. For finite p,

$$\int_{B_1} |u|^p < \infty \iff ap < n, \qquad \int_{B_1} |Du|^p < \infty \iff (a+1)p < n.$$

Under the second condition, integration by parts outside B_ε leaves an inner boundary term at most $C\varepsilon^{n-1-a}\|\phi\|_\infty \to 0$. This identifies the weak gradient. Equality of the exponent gives the divergent integral $\int_0^1 dr/r$.

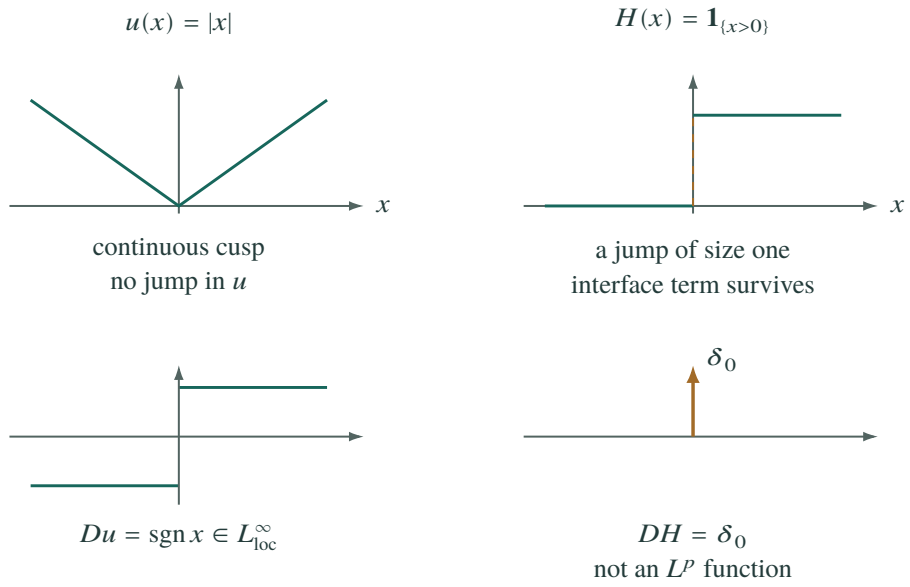


Diagram T5.1. A corner and a jump have different distributional derivatives. The continuous corner has a bounded weak first derivative. The step has a Dirac mass, so it has a distributional derivative but no weak first derivative representable by an L^p function. The vertical Dirac arrow denotes mass, not a graph of a finite-valued function.

尖角与跳跃

连续的分段光滑函数，其接口项可抵消；阶跃函数的跳跃留下 Dirac 质量。几乎处处有经典导数，不保证这个导数就是弱导数。

5.2.2 Spaces, norms, and convergence (Evans §5.2.2)

Definition 5.5

For an integer $k \geq 0$ and $1 \leq p \leq \infty$, $W^{k,p}(U)$ is the space of L^p functions whose weak derivatives through order k are all in $L^p(U)$. Use the norms

$$\|u\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_p^p \right)^{1/p} \qquad (p < \infty),$$

$$\|u\|_{W^{k,\infty}} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_\infty.$$

Write $H^k = W^{k,2}$. Define $W_0^{k,p}(U)$ as the closure of $C_c^\infty(U)$ in $W^{k,p}(U)$. Local membership means membership on every $V \Subset U$. Functions equal almost everywhere are identified.

Our $p = \infty$ norm is equivalent to the source's convention. Strong convergence means L^p norm convergence of every derivative. For $1 < p < \infty$, weak convergence means weak L^p convergence of every component. The inner product on H^k is $\sum_{|\alpha| \leq k} \int_U D^\alpha u D^\alpha v$.

5.2.3 Calculus and completeness (Evans §5.2.3)

Theorem 5.6 (Source Theorem 1: weak calculus)

Weak derivatives are linear, restrict to open subsets, and commute: $D^\beta D^\alpha u = D^{\alpha+\beta} u$ if the total order is at most k. For $\zeta \in C_c^\infty(U)$,

$$D^\alpha(\zeta u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \zeta D^{\alpha-\beta} u.$$

Expanded proof.

Add the defining identities for linearity. A test function on an open subset extends by zero to one on U, proving locality. For commutation,

$$\int D^\alpha u D^\beta \phi = (-1)^{|\alpha|} \int u D^{\alpha+\beta} \phi = (-1)^{|\beta|} \int D^{\alpha+\beta} u \phi.$$

For the product, test with $\zeta\phi$:

$$\int \zeta u \partial_i \phi = \int u \partial_i(\zeta\phi) - \int u(\partial_i \zeta)\phi = -\int (\zeta \partial_i u + u \partial_i \zeta)\phi.$$

Iterate. The coefficients combine by the multiindex Pascal identity, giving the formula. All terms lie in L^p because all cutoff derivatives are bounded. □

零下标是闭包

先用紧支光滑逼近定义 $W_0^{1,p}$，不能提前把它解释成逐点边界值为零。有界 C^1 域、$p < \infty$ 时，迹定理才证明两种表述等价。

Theorem 5.7 (Source Theorem 2)
$W^{k,p}(U)$ is complete, including $p = 1$ and $p = \infty$.

Expanded proof.

The norm axioms follow from those of the finite product of L^p spaces. If (u_j) is Cauchy, completeness of L^p gives $D^\alpha u_j \to g_\alpha$ in L^p for every $|\alpha| \le k$. Put $u = g_0$. For $\phi \in C_c^\infty(U)$,

$$\int u D^\alpha \phi = \lim_j \int u_j D^\alpha \phi = (-1)^{|\alpha|} \lim_j \int D^\alpha u_j \phi = (-1)^{|\alpha|} \int g_\alpha \phi.$$

Hölder justifies each limit on the test function's compact support; for $p = 1$ use its L^∞ norm, and for $p = \infty$ its L^1 norm. Thus $g_\alpha = D^\alpha u$ and the finite sum of component errors tends to zero. □

For $1 < p < \infty$, the product of L^p spaces is reflexive, hence its closed Sobolev subspace is reflexive. Bounded sequences have weakly convergent subsequences. This uses reflexivity and weak compactness from functional analysis; strong convergence still requires additional work.

闭图思想

把 u 映到 $(u, Du, \dots, D^k u)$，所得像是有限个 L^p 的乘积中的闭子空间。闭性靠弱导数恒等式取极限，而不是对极限逐点求导。

5.3 Approximation

5.3.1 Interior smoothing (Evans §5.3.1, Theorem 1)

Use the kernel ρ above. If $u \in W^{k,p}(U)$, $V \Subset U$, and $\varepsilon < \operatorname{dist}(V, \partial U)$, then

$$D^\alpha(\rho_\varepsilon * u) = \rho_\varepsilon * D^\alpha u \quad \text{on } V,$$

$$\rho_\varepsilon * u \to u \quad \text{in } W^{k,p}(V) \quad (1 \le p < \infty).$$

卷积不能越界

卷积会读取半径 ε 内的数据。任意补零可能在边界制造跳跃，因此不能直接把补零后的 Du 当成补零函数的导数。

Expanded proof.

For fixed $x \in V$, differentiate the kernel under the integral; its smooth compact support and local integrability of u justify this. Since $D_x^\alpha \rho_\varepsilon(x - y) = (-1)^{|\alpha|} D_y^\alpha \rho_\varepsilon(x - y)$, the weak identity in y transfers derivatives to u and cancels the signs.

For $f = D^\alpha u$, multiply by a cutoff equal to one near $\overline{V}$ and extend as an L^p function to $\mathbb{R}^n$. Minkowski gives

$$\|\rho_\varepsilon * f - f\|_{L^p(V)} \le \int \rho(z) \|f(\cdot - \varepsilon z) - f\|_{L^p(V)} \, dz.$$

Translations are continuous in L^p for finite p: prove this first for continuous compactly supported functions by uniform continuity, then use their L^p density and the translation isometry. The integrand is dominated by $2\|f\|_p$, so dominated convergence applies. Sum the finitely many derivative errors. □

5.3.2 Smooth in the interior (Evans §5.3.2, Theorem 2)

Theorem 5.8 (Meyers–Serrin approximation)
For any open U and finite p, $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$.

Expanded proof.

Choose a locally finite smooth partition (ζ_j) with $\sum_j \zeta_j = 1$ on U, each supported compactly in a relatively compact patch A_j. Slight enlargements of these patches can also be chosen locally finite, by using layers of an exhaustion of U. Extend $f_j = \zeta_j u$ by zero. There are no interface derivatives: the cutoff vanishes in a neighborhood of the edge of its patch.

Given $\delta > 0$, choose radii ε_j so small that

$$\operatorname{supp}(\rho_{\varepsilon_j} * f_j) \subset A_j, \qquad \|\rho_{\varepsilon_j} * f_j - f_j\|_{W^{k,p}(\mathbb{R}^n)} < \delta 2^{-j}.$$

Then $v = \sum_j \rho_{\varepsilon_j} * f_j$ is locally a finite sum of smooth functions. The error series converges absolutely in the Banach space $W^{k,p}$, with norm at most δ. Locally it equals $v - u$, since $\sum_j f_j = u$ locally. Uniqueness of distributions identifies its global sum with $v - u$. Thus $\|v - u\|_{W^{k,p}(U)} \le \delta$. □

5.3.3 Smooth up to the boundary (Evans §5.3.3, Theorem 3)

Theorem 5.9
If U is bounded with C^1 boundary and $p < \infty$, restrictions of functions smooth in a neighborhood of $\overline{U}$ are dense in $W^{k,p}(U)$.

Expanded proof.

Use a finite partition on $\overline{U}$, subordinate to interior patches and boundary graph patches. In a rotated boundary patch write $U = \{x_n > g(x')\}$ with $g \in C^1$. On the compact base, g is Lipschitz with a constant L. Localize to $f = \zeta u$ supported away from the lateral edges. Translate inward: $f_\delta(x) = f(x + \delta e_n)$.

For $x_n \ge g(x')$ and $|z| < c\delta$,

$$x_n + \delta - z_n - g(x' - z') \ge \delta - (1 + L)|z| > 0 \quad \text{if } c < 1/(1 + L).$$

Thus a small convolution of f_δ is defined in a neighborhood of the closed patch. The translated weak derivatives are $D^\alpha f(x + \delta e_n)$, by a change of test variables. Extend each $D^\alpha f$ by zero only for estimating L^p translations. Translation continuity proves $f_\delta \to f$ in $W^{k,p}$ on the original patch intersected with U. For each fixed δ, choose a smaller convolution radius

每块半径不同

靠近边界的块需要更小的 ε_j。误差预算 $\delta 2^{-j}$ 控制无穷多块的总误差；局部有限性保证内点附近只相加有限项。

giving any desired Sobolev error. The result is smooth across the original boundary. Approximate interior patches by ordinary convolution and sum the finitely many pieces. □

Why the endpoint $p = \infty$ is excluded. The derivative of $|x|$ jumps from -1 to 1. No continuous derivative can approximate it in essential supremum norm with error less than one: the two one-sided requirements contradict continuity at zero. Smooth functions need not be norm dense in $W^{1,\infty}$. Local smoothing does converge in every finite $W^{1,q}$ norm on bounded interior sets and remains uniformly bounded in $W^{1,\infty}$. Later endpoint arguments use those facts instead.

两种稠密性不同

$C^\infty(\overline{U})$ 稠密不等于 $C_c^\infty(U)$ 稠密。常数 1 可光滑逼近，但在有界光滑域中不属于 $W_0^{1,p}$；它的迹不是零。

5.4 Extensions

5.4.1 Reflection and flattening (Evans §5.4, Theorem 1)

Theorem 5.10 (Extension)
Let U be bounded with C^1 boundary, and fix V with $U \Subset V$. For $1 \le p \le \infty$ there is a linear operator

$$E : W^{1,p}(U) \longrightarrow W^{1,p}(\mathbb{R}^n), \qquad Eu = u \text{ on } U,$$

with support in a fixed compact subset of V and $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \le C\|u\|_{W^{1,p}(U)}$. The constant depends on U, V, n, p.

Expanded proof.

先比较两种延拓

反射延拓在接口两侧保持同一函数值；补零只有在迹为零时才不会制造跳跃。延拓算子要同时控制函数和梯度，不能只保持 L^p 范数。

1. A flat boundary. On the upper half-space, for a smooth localized function define

$$\tilde{u}(x', t) = u(x', |t|).$$

For $t < 0$, its tangential derivatives are $(\partial_i u)(x', -t)$ and its normal derivative is $-(\partial_n u)(x', -t)$. Integrating against a test function separately in $t > 0$ and $t < 0$, the interface terms cancel because the values at $t = 0$ agree. Thus these piecewise derivatives are the weak derivatives. For finite p, reflection doubles both the p-th power of the L^p norm and the gradient integral; for $p = \infty$ it preserves their bounds.

2. Flatten a graph. In a boundary chart let $U = \{x_n > g(x')\}$ and set

$$\Phi(y', y_n) = (y', y_n + g(y')).$$

The map and its inverse have bounded first derivatives on the smaller chart, and $\det D\Phi = 1$. For smooth u, $D(u \circ \Phi) = (D\Phi)^T (Du) \circ \Phi$, so $\|u \circ \Phi\|_{W^{1,p}} \le C\|u\|_{W^{1,p}}$, with an analogous inverse bound. For Sobolev u, interior smooth approximation proves the same formula locally in the chart; the global integrability bounds then identify its weak derivatives on the entire chart. Reflect $u \circ \Phi$ and compose with Φ^{-1}.

3. Localize and assemble. Take finitely many smaller boundary charts and an interior patch covering $\overline{U}$, with a smooth partition (ζ_j). Extend each $\zeta_j u$ by the preceding construction, multiplying by a fixed exterior cutoff supported in its chart. The interior term extends by zero. Sum the pieces. On U the sum equals u, because $\sum_j \zeta_j = 1$. The product rule and finitely many chart estimates give the asserted norm bound. Choose all exterior cutoffs inside V. This construction is linear and uses fixed charts and cutoffs.

For finite p, one can also construct first on smooth functions and complete using Theorem 5.9. For $p = \infty$, apply the same reflection formula directly. Locally approximate in finite $W^{1,q}$ norms and pass to the weak identity; the candidate derivatives have the uniform L^∞ bounds already calculated. No $W^{1,\infty}$ norm density is needed. □

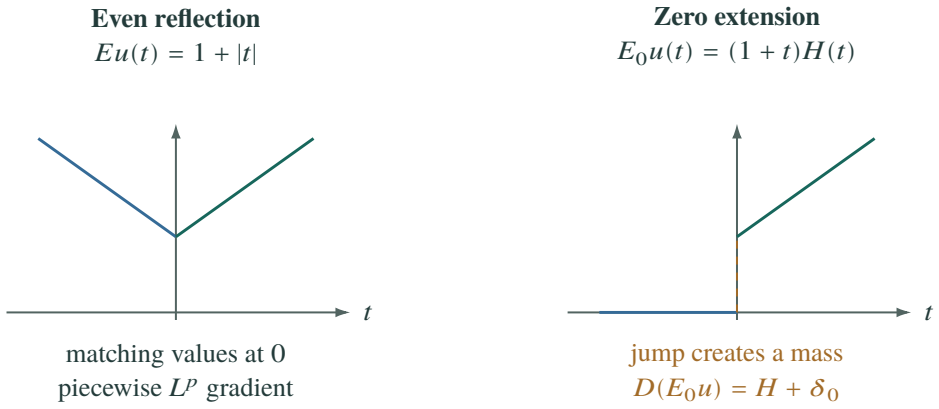


Diagram T5.2. Reflection and zero extension treat the interface differently. Take $u(t) = 1 + t$ for $0 < t < 1$ and inspect only the interface at zero. Reflection matches values and preserves first-order Sobolev regularity locally. Zero extension inserts a jump and a Dirac mass in the first derivative. Even reflection still has a corner, so higher-order extensions need additional matching.

边界正则性用在哪里

C^1 使边界可拉平，坐标变换及逆变换的一阶导数有界；这才保证 Sobolev 范数不被放大到无穷。尖裂缝域不能直接套用此结论。

5.4.2 Higher derivatives: match the jets

Ordinary even reflection need not preserve $W^{2,p}$: for $u(t) = t$ on $t > 0$ it gives $|t|$, whose second derivative has a Dirac mass. For an integer $k \ge 1$,

on $t < 0$ use instead

$$\tilde{u}(x', t) = \sum_{j=1}^{k} a_j u(x', -jt), \qquad \sum_{j=1}^{k} a_j (-j)^{\ell} = 1 \quad (0 \le \ell < k).$$

The Vandermonde matrix is invertible because $-1, \dots, -k$ are distinct. The derivatives of normal order $0, \dots, k-1$ match at zero, so repeated integration by parts has no interface distributions through order k. Each derivative on the lower side is bounded by a finite sum of scaled upper-side derivatives. This proves a $W^{k,p}$ bound on a sufficiently small strip. For $k = 2$ the formula is $3u(x', -t) - 2u(x', -2t)$.

If ∂U is C^k, flattening has bounded derivatives through order k on the chosen compact charts. The higher chain rule and the same partition construction produce an operator bounded simultaneously on $W^{j,p}$ for $0 \le j \le k$. For finite p, smooth boundary approximation passes the identities to the completion; for $p = \infty$, finite-exponent local convergence and the uniform derivative bounds give the same conclusion. We will use a single such operator on L^2 and H^2 in the time-dependent estimates.

高阶延拓不能照抄偶反射

一阶只需接口函数值匹配；二阶还需一阶导数匹配。系数 $3, -2$ 同时满足 $3 - 2 = 1$ 和 $-3 + 4 = 1$，正是为了消掉接口奇异项。

5.5 Traces

5.5.1 Boundary values as a continuous operator

Theorem 5.11 (Evans §5.5, Theorem 1)
If U is bounded with C^1 boundary and $1 \le p < \infty$, there is a unique bounded linear map

$$T : W^{1,p}(U) \longrightarrow L^p(\partial U)$$

agreeing with restriction for functions in $W^{1,p}(U) \cap C(\overline{U})$. In particular, $\|Tu\|_{L^p(\partial U)} \le C \|u\|_{W^{1,p}(U)}$.

Expanded proof.

边界不能随意赋值

改变 u 在零测集上的值不改变 Sobolev 元素；边界通常正是体积零测集。迹由域内逼近的极限决定，不能通过任意指定边界代表值来定义。

1. A flat strip estimate. For smooth v on a strip $Q' \times [0,a]$, the fundamental theorem gives

$$|v(x',0)|^p \le |v(x',t)|^p + p\int_0^a |v|^{p-1}|\partial_n v|(x',s)\,ds.$$

For $p = 1$, use the absolutely continuous function $|v|$ and its derivative bound $|(|v|)'| \le |v'|$. Average t over $(0,a)$ and integrate x'. For $p > 1$, Young's inequality bounds $p|v|^{p-1}|\partial_n v|$ by $C_p(|v|^p + |Dv|^p)$. Consequently

$$\int_{Q'} |v(x',0)|^p\,dx' \le C_{p,a}\int_{Q'\times(0,a)} (|v|^p + |Dv|^p)\,dx.$$

Boundary flattening introduces only bounded Jacobian and derivative factors, including $\sqrt{1+|Dg|^2}$ for surface measure. A finite partition yields the global estimate for smooth u.

2. Complete the restriction map. Choose u_j smooth near $\overline{U}$, with $u_j \to u$ in $W^{1,p}$. The estimate on $u_j - u_l$ shows that $u_j|_{\partial U}$ is Cauchy in $L^p(\partial U)$. Define Tu to be its limit. The same estimate on two approximating sequences proves independence of choice and boundedness. Linearity and uniqueness follow from density.

3. Agreement with continuous functions. For u continuous on $\overline{U}$, the inward-translation and smoothing construction in §5.3 can be chosen to converge uniformly as well as in $W^{1,p}$. Uniform continuity on the compact closure justifies this assertion for each localized piece. The boundary restrictions therefore converge to $u|_{\partial U}$ uniformly, identifying the trace. □

5.5.2 Zero trace and the closure definition

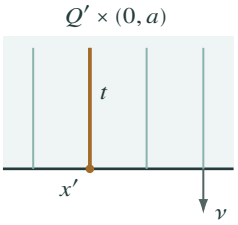
> **Theorem 5.12** (Evans §5.5, Theorem 2)
> *Under the same hypotheses,*
>
> $$W_0^{1,p}(U) = \{u \in W^{1,p}(U) : Tu = 0\}.$$

Expanded proof.

If $u_j \in C_c^\infty(U)$ tends to u in $W^{1,p}$, boundedness of T gives $Tu = \lim Tu_j = 0$.

For the reverse direction we supply the boundary-layer argument. In a flat chart the trace equals the value at $t = 0$ of the absolutely continuous

图 T5.M：沿法向控制边界值



先沿固定 x' 的线段比较内部值与边界值，再对高度和切向变量积分。图中外法向朝下。

迹估计的实际用处

只要 $u_j \to u$ 于 $W^{1,p}$，就自动有 $Tu_j \to Tu$ 于边界 L^p。边界条件因此可随强逼近传递，而不要求逐点追踪边界值。

representative on almost every vertical line. To justify this, use smooth approximations, Fubini, and the one-dimensional estimate

$$|w(0)| \le a^{-1}\|w\|_{L^1(0,a)} + \|w'\|_{L^1(0,a)}.$$

A subsequence converges in $W^{1,p}$ on almost every line, and this estimate identifies the endpoint limit with the trace limit.

Thus if $Tv = 0$, $v(x',t) = \int_0^t \partial_n v(x',s)\,ds$ on almost every such line. Hölder (or the integral triangle inequality for $p = 1$) gives

$$\int_{Q'}\int_0^{2\delta} |v(x',t)|^p\,dt\,dx' \le (2\delta)^p \int_{Q'}\int_0^{2\delta} |\partial_n v(x',t)|^p\,dt\,dx'.$$

Choose a smooth $\chi_\delta(t)$ equal to zero for $t < \delta$, one for $t > 2\delta$, with $|\chi_\delta'| \le C/\delta$. Then

$$D(\chi_\delta v) - Dv = (\chi_\delta - 1)Dv + vD\chi_\delta.$$

The first term tends to zero by absolute continuity of the integral. The p-th power of the second norm is bounded by $C\int_{0<t<2\delta}|Dv|^p$, hence also tends to zero. The function error tends to zero in the same way. Flattening and a finite partition transfer this argument to U. It gives functions supported a positive distance from ∂U converging to u in $W^{1,p}$. For each one, convolution with a smaller radius produces an element of $C_c^\infty(U)$ with arbitrarily small additional error. A diagonal choice completes the proof. □

截断产生的危险项

$D(\chi_\delta u)$ 中有 $uD\chi_\delta$，而 $D\chi_\delta$ 约为 $1/\delta$。零迹给出的法向积分公式补回一个 δ 因子，才使此项趋于零。

Zero extension. If $u \in W_0^{1,p}(U)$, extend compactly supported approximations by zero. They are Cauchy in $W^{1,p}(\mathbb{R}^n)$ with exactly the same norms, so completeness gives a zero extension $\tilde{u}$ satisfying $D\tilde{u} = \widetilde{Du}$. This argument works without regularity of U for finite p. A nonzero trace generally produces a boundary distribution instead.

5.6 Sobolev Inequalities

5.6.1 Scaling predicts the exponent

For $u_\lambda(x) = u(\lambda x)$ on $\mathbb{R}^n$,

$$\|u_\lambda\|_q = \lambda^{-n/q}\|u\|_q, \qquad \|Du_\lambda\|_p = \lambda^{1-n/p}\|Du\|_p.$$

An estimate $\|u\|_q \le C\|Du\|_p$ with a constant independent of scale must therefore have $1/q = 1/p - 1/n$. For $p < n$ put $p^* = np/(n-p)$.

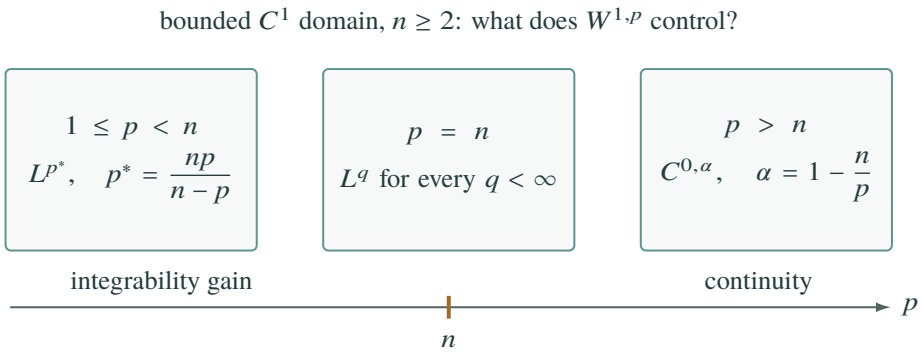


Diagram T5.3. The dimension separates three embedding regimes. The diagram records standard continuous embeddings on a bounded C^1 domain with $n \geq 2$. At $p = n$, the conclusion is valid for each finite q, with constants that may depend on q. It does not provide a uniform estimate as $q \to \infty$. Compactness is a further question addressed in the next section.

5.6.2 Gagliardo–Nirenberg–Sobolev (Evans §5.6.1)

We first give the integral inequality behind the endpoint $p = 1$. For $n \geq 2$ and nonnegative f_i on $\mathbb{R}^{n-1}$, with $\hat{x}_i$ denoting all coordinates except x_i,

$$\int_{\mathbb{R}^n} \prod_{i=1}^{n} f_i(\hat{x}_i)^{1/(n-1)}\, dx \leq \prod_{i=1}^{n} \left(\int_{\mathbb{R}^{n-1}} f_i \right)^{1/(n-1)}. \tag{LW}$$

Expanded proof (editorial integral lemma).

For $n = 2$ this is Fubini. Suppose it holds in dimension $n-1$. Integrate first in x_n and apply Hölder with $n-1$ equal exponents to the factors $i < n$. Put $F_i = \int_{\mathbb{R}} f_i(\hat{x}_i)\, dx_n$. What remains is

$$\int_{\mathbb{R}^{n-1}} f_n^{1/(n-1)} \prod_{i<n} F_i^{1/(n-1)}\, dx'.$$

Apply Hölder with exponents $n-1$ and $(n-1)/(n-2)$, obtaining

$$\left(\int f_n \right)^{1/(n-1)} \left(\int_{\mathbb{R}^{n-1}} \prod_{i<n} F_i^{1/(n-2)} \right)^{(n-2)/(n-1)}.$$

The induction hypothesis on the last integral gives (LW). Tonelli justifies the order changes, first for finite integrals and then by monotone truncation in general. □

Theorem 5.13 (Evans §5.6.1, Theorem 1)
For $1 \leq p < n$ and $u \in C_c^1(\mathbb{R}^n)$,

$$\|u\|_{p^*} \leq C_{n,p} \|Du\|_p.$$

Expanded proof.

先用缩放验算

$1/p^* = 1/p - 1/n$ 是量纲平衡，不是记忆技巧。若估计两侧缩放次数不同，就必须出现域尺度、低阶项或其他约束。

1. Start with $p = 1$. The fundamental theorem on the ith coordinate line gives

$$|u(x)| \le \int_{\mathbb{R}} |\partial_i u(\hat{x}_i, t)|\, dt =: f_i(\hat{x}_i).$$

Multiply the n inequalities and raise to the power $1/(n-1)$. By (LW),

$$\int |u|^{n/(n-1)} \le \prod_i \|\partial_i u\|_1^{1/(n-1)} \le \|Du\|_1^{n/(n-1)}.$$

Taking the power $(n-1)/n$ proves the endpoint.

2. Obtain $1 < p < n$ by a power substitution. Set $a = p(n-1)/(n-p) > 1$. Apply the endpoint to $|u|^a$, a compactly supported C^1 function. Hölder with $p' = p/(p-1)$ yields

$$\Big(\int |u|^{an/(n-1)}\Big)^{(n-1)/n} \le Ca \int |u|^{a-1}|Du| \le Ca\|Du\|_p \Big(\int |u|^{(a-1)p'}\Big)^{1/p'}.$$

Both exponents $an/(n-1)$ and $(a-1)p'$ equal p^*. The left side is $\|u\|_{p^*}^a$ and the last factor is $\|u\|_{p^*}^{a-1}$. Cancel it if nonzero; if zero the conclusion is immediate. □

Corollary 5.14 (Evans §5.6.1, Theorems 2–3)
On a bounded C^1 domain, $\|u\|_{p^*} \le C\|u\|_{W^{1,p}(U)}$. On any bounded open set, for $u \in W_0^{1,p}(U)$,

$$\|u\|_{p^*} \le C_{n,p}\|Du\|_p, \qquad \|u\|_p \le C_{n,p}|U|^{1/n}\|Du\|_p.$$

Expanded proof.

For the first statement extend by Theorem 5.10, smooth on $\mathbb{R}^n$ with fixed compact support, and apply Theorem 5.13. Applying it to differences shows the smooth sequence is Cauchy in L^{p^*}. Its L^p limit identifies the L^{p^*} limit with the extension, so the inequality passes to u. For zero boundary data use zero extensions of the defining compactly supported sequence instead. Finally Hölder gives $\|u\|_p \le |U|^{1/p-1/p^*}\|u\|_{p^*}$ and $1/p - 1/p^* = 1/n$. □

5.6.3 The critical case $p = n$

For $n \ge 2$, bounded C^1 domains satisfy $W^{1,n}(U) \hookrightarrow L^q(U)$ for every finite q. Indeed choose $1 \le r < n$ so close to n that $r^* \ge q$. Finite measure gives $W^{1,n} \subset W^{1,r}$, and the preceding inequality followed by Hölder proves the claim. The constant depends on q and need not stay bounded as $q \to \infty$.

幂次为何这样选

a 同时满足 $an/(n-1) = p^*$ 与 $(a-1)p' = p^*$，所以最后能约去同一个函数范数。随便取幂次，两个积分指数就对不上。

为何一般域估计有低阶项

常数 $u = 1$ 满足 $Du = 0$，所以对任意 $W^{1,p}$ 函数不能只有梯度控制。零迹排除非零常数，才可去掉右侧的 $\|u\|_p$。

A useful counterexample, equivalent in behavior to source Problem 14, is

$$u(x) = \log\log(e/|x|) \quad (0 < |x| < 1).$$

It is unbounded, but $|Du| = 1/(|x|\log(e/|x|))$ and

$$\int_{B_1} |Du|^n = C_n \int_0^1 \frac{dr}{r\log(e/r)^n} < \infty \quad (n > 1).$$

The function itself has finite L^n norm. Its weak derivative is identified by integration outside B_ε; the interface term is $O(\varepsilon^{n-1}\log\log(e/\varepsilon)) \to 0$. In dimension one, by contrast, $W^{1,1}$ on an interval has an absolutely continuous, hence bounded, representative.

5.6.4 Morrey's inequality (Evans §5.6.2, Theorems 4–5)

Theorem 5.15

If $n < p < \infty$, then $W^{1,p}(\mathbb{R}^n)$ has a representative satisfying

$$|u(x) - u(y)| \le C_{n,p}|x-y|^{1-n/p}\|Du\|_{L^p(\mathbb{R}^n)}, \qquad \|u\|_\infty \le C_{n,p}\|u\|_{W^{1,p}}.$$

On a bounded C^1 domain, $\|u\|_{C^{0,1-n/p}(\overline{U})} \le C\|u\|_{W^{1,p}(U)}$. For $p = \infty$, the corresponding exponent is 1.

Expanded proof.

1. A local kernel estimate. Let D be a ball of radius R and $x \in \overline{D}$. For smooth u, integrate the line identity $|u(x) - u(z)| \le \int_0^{|z-x|} |Du(x + t\omega)|\,dt$ over $z = x + s\omega \in D$. Convexity keeps each segment inside D. The s integration after changing order contributes at most CR^n. Dividing by $|D|$ gives

$$|u(x) - u_D| \le C_n \int_D \frac{|Du(z)|}{|x-z|^{n-1}}\,dz, \qquad u_D = |D|^{-1}\int_D u.$$

Hölder bounds the integral by

$$C\|Du\|_{L^p(D)} \left(\int_0^{2R} t^{n-1-(n-1)p'}\,dt\right)^{1/p'} \le CR^{1-n/p}\|Du\|_{L^p(D)}.$$

The radial integral is finite exactly when $p > n$ (for $n = 1$, the same computation is simply the integral triangle inequality and Hölder).

2. Compare two points. For $r = |x-y|$, choose $D = B(x, 2r)$ containing both. Bound $|u(x) - u_D|$ and $|u(y) - u_D|$ by step 1. This also yields the local estimate

$$|u(x) - u(y)| \le Cr^{1-n/p}\|Du\|_{L^p(B(x,2r))}$$

临界点不是无穷大嵌入

$p \uparrow n$ 时公式中的 p^* 趋于无穷，但常数也可能发散，不能把极限直接代入。$\log\log(e/|x|)$ 给出明确反例。

whenever that ball lies in the domain. With a unit ball in step 1, $|u(x)| \le |u_D| + C\|Du\|_p \le C(\|u\|_p + \|Du\|_p)$.

3. Pass to Sobolev functions. Smooth approximation on $\mathbb{R}^n$ and the inequalities on differences give uniform convergence to a Hölder representative, agreeing almost everywhere with the L^p limit. On U, use the extension operator. For $p = \infty$ on $\mathbb{R}^n$, local smoothing has gradient bounded by $\|Du\|_\infty$; the line identity gives a common Lipschitz constant. A locally uniform subsequential limit, identified by finite L^q convergence, proves the assertion. Extend first for the domain version. □

5.6.5 Higher-order embeddings (Evans §5.6.3, Theorem 6)

Let U be bounded with C^1 boundary, $k \ge 1$, and $1 \le p < \infty$. If $kp < n$,

$$W^{k,p}(U) \hookrightarrow L^q(U), \qquad \frac{1}{q} = \frac{1}{p} - \frac{k}{n}.$$

If $k > n/p$ and n/p is not an integer, put $l = \lfloor n/p \rfloor$, $m = k - l - 1$, and $\gamma = l + 1 - n/p$. Then $W^{k,p} \hookrightarrow C^{m,\gamma}(\overline{U})$. If $n/p = j$ is an integer and $k > j$, the conclusion is $C^{k-j-1,\gamma}$ for every $0 < \gamma < 1$. Each embedding has a norm bound depending on U, n, k, p and, where applicable, γ.

Expanded proof.

Apply the first-order estimate to each derivative of order at most $k-1$. This improves $W^{k,p}$ to W^{k-1,p_1} with $1/p_1 = 1/p - 1/n$. Iterate while the input exponent is below n. If $kp < n$, exactly k steps yield L^q. If n/p is nonintegral, after l steps the exponent r satisfies $1/r = 1/p - l/n$ and $r > n$. Apply Morrey to each derivative through order $k - l - 1$; $1 - n/r = l + 1 - n/p$. If $n/p = j$ is integral, after $j-1$ steps the exponent is n. The critical finite-q estimate gives $W^{k-j,q}$ for arbitrary finite q. Choose $q > n$ and apply Morrey, obtaining $C^{k-j-1,1-n/q}$. Any $\gamma < 1$ follows by choosing q sufficiently large. Multiplying the finitely many estimate constants gives the bounds. The continuous weak derivatives obtained here are classical derivatives: mollify locally and pass to the coordinate-segment integral identity, as in the completeness proof in §5.1. For $n = p = 1$, interval absolute continuity gives the needed bounded derivatives directly and hence the stated weaker Hölder conclusions. □

连续代表元的含义

原先 u 是几乎处处等价类。Morrey 证明其中存在连续函数；两个连续函数若几乎处处相同，便处处相同。因此此时谈逐点值才有唯一含义。

5.7 Compactness

5.7.1 Rellich–Kondrachov (Evans §5.7, Theorem 1)

Theorem 5.16
Let U be bounded with C^1 boundary. If $1 \le p < n$ and $1 \le q < p^*$, every bounded sequence in $W^{1,p}(U)$ has a subsequence converging in $L^q(U)$. We write $W^{1,p}(U) \Subset L^q(U)$. For $p = n$ the same conclusion holds for every finite q. For $p > n$, including $p = \infty$, bounded sequences have subsequences converging uniformly on $\overline{U}$.

Expanded proof.

1. Put every function on the same compact set. Extend a bounded sequence (u_j) by the fixed operator in Theorem 5.10. Call the extensions v_j. They have uniformly bounded $W^{1,p}(\mathbb{R}^n)$ norm and support in a fixed compact set K.

2. Uniform smoothing error. For smooth v, the line integral formula and Minkowski give

$$\|v(\cdot + h) - v\|_p \le |h| \int_0^1 \|Dv(\cdot + th)\|_p \, dt = |h| \|Dv\|_p.$$

Smooth approximation passes this to every finite-p Sobolev function. Consequently

$$\|\rho_\varepsilon * v_j - v_j\|_p \le C\varepsilon \|Dv_j\|_p \le C\varepsilon$$

uniformly in j. For $q \le p$, finite support gives the corresponding L^q bound. For $p < q < p^*$ choose $0 < \theta < 1$ with $1/q = \theta/p + (1-\theta)/p^*$. Hölder interpolation gives

$$\|\rho_\varepsilon * v_j - v_j\|_q \le \|\rho_\varepsilon * v_j - v_j\|_p^\theta \|\rho_\varepsilon * v_j - v_j\|_{p^*}^{1-\theta} \le C\varepsilon^\theta.$$

The second factor is uniformly bounded by the Sobolev inequality and convolution's L^{p^*} bound.

3. Compactness for a fixed smoothing radius. For fixed ε, the functions $v_{j,\varepsilon} = \rho_\varepsilon * v_j$ have support in $K + \overline{B}_\varepsilon$. Moreover

$$\|v_{j,\varepsilon}\|_\infty \le \|\rho_\varepsilon\|_\infty \|v_j\|_1, \qquad \|Dv_{j,\varepsilon}\|_\infty \le \|D\rho_\varepsilon\|_\infty \|v_j\|_1.$$

The right sides are independent of j by bounded support and Hölder. Arzelà–Ascoli, invoked here as a prerequisite, supplies a uniformly convergent subsequence on that compact set.

高阶指数逐层算

每使用一次一阶嵌入，导数阶数减一，$1/p$ 减去 $1/n$。到达临界指数时先转到任意有限 q，所以整数临界处不能擅自宣称 Lipschitz。

连续嵌入与紧嵌入

连续嵌入只把有界集送到有界集；紧嵌入还能从任意有界序列抽出强收敛子列。强收敛常用于通过非线性项的极限。

4. Make one subsequence work at every scale. Choose $\varepsilon_l \downarrow 0$ so the L^q smoothing error is at most 2^{-l}. Extract nested subsequences for $l = 1, 2, \ldots$, and take the diagonal sequence (v_{j_a}). For fixed l, its smoothed sequence is Cauchy, so

$$\limsup_{a,b\to\infty} \|v_{j_a} - v_{j_b}\|_q \le 2^{1-l}.$$

Let $l \to \infty$. Completeness of L^q gives convergence, and restriction to U proves the result.

5. The remaining exponents. If $p = n \ge 2$, choose $1 \le r < n$ with $r^* > q$ and apply the preceding result to the bounded $W^{1,r}$ sequence. If $p > n$, Morrey gives uniform boundedness and a common Hölder modulus on $\overline{U}$; Arzelà–Ascoli gives uniform convergence. For the exceptional case $n = p = 1$, extension to a fixed bounded interval and the translation/smoothing argument above prove compactness in L^1. The interval fundamental theorem gives a uniform L^∞ bound; interpolation then gives compactness in every finite L^q. □

5.7.2 Why the critical exponent is excluded

Fix nonzero $\psi \in C_c^\infty(B_1)$ and an interior point x_0. For $p < n$ define, for sufficiently large λ,

$$u_\lambda(x) = \lambda^{(n-p)/p}\psi(\lambda(x - x_0)).$$

Then $\|Du_\lambda\|_p = \|D\psi\|_p$, $\|u_\lambda\|_p = \lambda^{-1}\|\psi\|_p$, but $\|u_\lambda\|_{p^*} = \|\psi\|_{p^*} > 0$. The sequence tends to zero at every $x \ne x_0$. A strongly L^{p^*} convergent subsequence would have a further almost-everywhere convergent subsequence, so its limit would be zero, contradicting its constant nonzero norm.

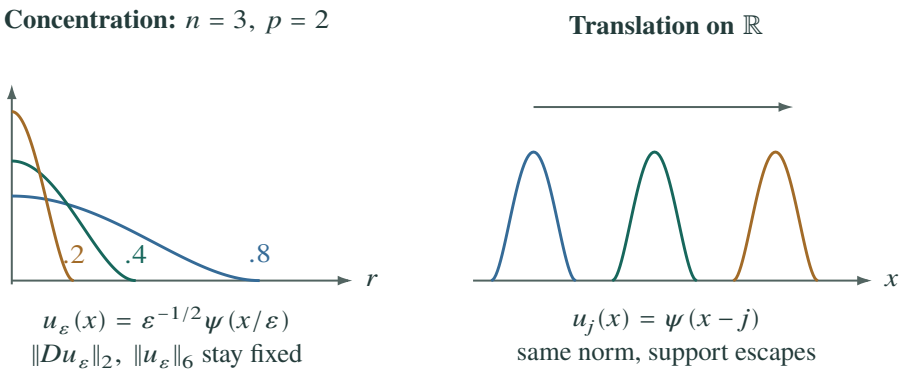


Diagram T5.4. Bounded norms do not prevent concentration or escape. The left panel uses radial bump profiles $\psi(x) = (1 - |x|^2)_+^2$ in $\mathbb{R}^3$, scaled with $\varepsilon = 0.8, 0.4, 0.2$; their gradient L^2 and critical L^6 norms stay fixed. The right panel illustrates translations of a compact bump on $\mathbb{R}$. Each mechanism obstructs strong compactness for a different reason.

对角线到底做什么

固定 ε 得到的子列可能随 ε 改变。先抽嵌套子列再取对角线，保证每个固定尺度最终都收敛；最后才令尺度趋零。

集中导致临界失紧

尖峰越来越窄、越来越高，梯度范数和临界范数都不变。次临界范数会衰减，因此 $q < p^*$ 留出的余量正好压住集中。

Zero boundary data on rough bounded domains. For finite p, zero extensions of a bounded sequence in $W_0^{1,p}(U)$ have a common compact support and bounded whole-space Sobolev norm. The proof above therefore gives compactness in $L^p(U)$ on any bounded open U, without regular boundary. On unbounded domains, translations of a fixed bump can escape to infinity and destroy global compactness.

5.8 Additional Topics

5.8.1 Poincaré inequalities (Evans §5.8.1, Theorems 1–2)

Lemma 5.17 (Zero gradient means constant)
If U is connected and $u \in W^{1,1}_{\mathrm{loc}}(U)$ satisfies $Du = 0$, then u is constant almost everywhere.

Expanded proof (also source Problem 11).

On a ball compactly contained in U, mollification has zero classical gradient on each smaller concentric ball and is constant there. As $\varepsilon \to 0$ it converges in L^1; the constants converge to the average of u on the smaller ball. Hence u is constant there almost everywhere. Constants agree on overlapping balls, because the overlap has positive measure. Any two points in an open connected subset of $\mathbb{R}^n$ can be joined by a finite chain of such balls: the set of points reachable from a fixed ball is both relatively open and relatively closed. Connectedness completes the proof. □

Theorem 5.18
On a bounded connected C^1 domain, for $1 \le p \le \infty$,

$$\|u - u_U\|_{L^p(U)} \le C_{U,n,p}\|Du\|_{L^p(U)}, \qquad u_U = |U|^{-1}\int_U u.$$

On a ball B_r the constant is $C_{n,p}r$.

连通性不可删

在两个不相交的球上分别取常数 0 和 1，梯度仍处处为零，却不是同一个常数。均值型 Poincaré 必须排除这种情况。

Expanded proof.

If the inequality failed, normalization would give

$$(v_j)_U = 0, \quad \|v_j\|_p = 1, \quad \|Dv_j\|_p < 1/j.$$

For finite p, compactness gives a subsequence converging strongly in L^p to v; for $p = \infty$, use the uniform compactness just proved. The mean and norm pass to the limit, giving $v_U = 0$ and $\|v\|_p = 1$. Testing the weak derivative identity and passing to the limit gives $Dv = 0$. Lemma 5.17 makes v constant, and its zero mean makes it zero, a contradiction. For

$B_r(x_0)$ use $w(y) = u(x_0 + ry)$: the function norm scales by $r^{-n/p}$ and the gradient norm by $r^{1-n/p}$, leaving precisely one factor r. □

For $u \in W^{1,n}(\mathbb{R}^n)$, $n \geq 2$, the $p = 1$ ball estimate and Hölder give

$$\frac{1}{|B_r|}\int_{B_r} |u - u_{B_r}| \leq Cr\frac{1}{|B_r|}\int_{B_r} |Du| \leq C\|Du\|_{L^n(\mathbb{R}^n)}.$$

Thus $W^{1,n}$ controls bounded mean oscillation (BMO), whose seminorm is the supremum of the left side over balls. BMO controls average oscillation rather than pointwise size.

5.8.2 Difference quotients (Evans §5.8.2, Theorem 3)

For $h \neq 0$ define

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h}.$$

Theorem 5.19

If $u \in W^{1,p}(U)$, $V \Subset U$, and $|h| < \frac{1}{2}\operatorname{dist}(V, \partial U)$, then $\|D_i^h u\|_{L^p(V)} \leq \|\partial_i u\|_{L^p(U)}$ for finite p. Conversely, if $1 < p < \infty$, $u \in L^p(U)$, and all these quotients are uniformly bounded in $L^p(V)$ for every coordinate, then $u \in W^{1,p}(V)$, with the corresponding derivative bounds.

Expanded proof.

For smooth u, $D_i^h u(x) = \int_0^1 \partial_i u(x + the_i)\, dt$. Jensen and Fubini give

$$\int_V |D_i^h u|^p \leq \int_0^1 \int_V |\partial_i u(x + the_i)|^p \, dx\, dt \leq \int_U |\partial_i u|^p.$$

Interior approximation on a slightly larger set passes this to u.

For the converse, choose $h_j \to 0$. Reflexivity of $L^p(V)$ gives, after a finite diagonal extraction over coordinates, $D_i^{h_j} u \rightharpoonup g_i$ in $L^p(V)$. For a test function ϕ and small h a change of variables yields

$$\int D_i^h u\, \phi = -\int u D_i^{-h}\phi.$$

There are no edge terms because the test support and its translates stay inside V. Since $D_i^{-h}\phi \to \partial_i \phi$ uniformly on a common compact support, the right side tends to $-\int u\partial_i\phi$. The left side tends weakly to $\int g_i\phi$. Thus $g_i = \partial_i u$. Weak lower semicontinuity proves the norm bound. □

消掉常数的两种方式

零迹型要求 $Tu = 0$；均值型减去 u_U。二者都排除梯度算子的常数核，但适用条件和边界信息不同。

For finite p, one also has $D_i^h u \to \partial_i u$ strongly on interior sets: subtract the derivative in the line-average formula and use L^p translation continuity. This gives a practical way to recognize weak derivatives in elliptic regularity estimates.

5.8.3 Lipschitz functions (Evans §5.8.2, Theorem 4)

Locally, $W^{1,\infty}$ functions have Lipschitz representatives, by smoothing on balls and using the gradient bound on segments. Conversely an L-Lipschitz function has bounded difference quotients. On a relatively compact ball, extract weak limits in L^2. The weak identity above gives weak derivatives; the limit still satisfies $|g_i| \le L$ almost everywhere, since the closed convex set $\{g : |g| \le L\}$ is weakly closed in L^2. This identifies locally bounded weak derivatives without appealing to Rademacher's theorem.

On bounded C^1 domains the extension theorem gives a global Lipschitz representative for $W^{1,\infty}(U)$, with constant controlled by the full $W^{1,\infty}$ norm. Global Lipschitz continuity need not follow on arbitrary domains: points across a narrow slit can be very close in Euclidean distance but far apart along paths inside the domain.

5.8.4 Differentiability a.e. (Evans §5.8.3, Theorems 5–6)

Theorem 5.20
For $p > n$, a function $u \in W^{1,p}_{\mathrm{loc}}(U)$, identified with its continuous representative, is differentiable almost everywhere, and its classical gradient equals its weak gradient there. Every locally Lipschitz function is therefore differentiable a.e.

Expanded proof.

First let $n < p < \infty$. The L^p version of the Lebesgue differentiation theorem, invoked here, says that for almost every x,

$$\frac{1}{|B_r|}\int_{B_r(x)} |Du(z) - Du(x)|^p \, dz \to 0.$$

Fix such an x and put $v(z) = u(z) - u(x) - Du(x)\cdot(z-x)$. For $r = |y-x|$ small enough that $B(x,2r) \Subset U$, the local Morrey estimate gives

$$\begin{aligned} |v(y) - v(x)| &\le C r^{1-n/p} \left(\int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} \\ &= C r \left(\frac{1}{|B_{2r}|} \int_{B(x,2r)} |Du(z) - Du(x)|^p \, dz \right)^{1/p} = o(r). \end{aligned}$$

$p = 1$ 的反向结论失效

阶跃函数的差商高度为 $1/|h|$、支集宽约 $|h|$，故 L^1 范数有界；极限却是 Dirac 质量。L^1 有界不保证弱子列极限仍是函数。

避免循环证明

先由差商得到 Lipschitz 函数的弱导数，再由 Morrey 和微分定理证明几乎处处可微。若先引用 Rademacher 来构造弱导数，又用它证明 Rademacher，就循环了。

This is precisely differentiability with derivative $Du(x)$. For $p = \infty$, work locally with any finite exponent greater than n. A locally Lipschitz function belongs locally to $W^{1,\infty}$ by the preceding argument, proving Rademacher's theorem as a consequence. □

5.8.5 Hardy's inequality (Evans §5.8.4, Theorem 7)

Theorem 5.21
If $n \geq 3$ and $u \in H^1(B_r)$, then

$$\int_{B_r} \frac{|u|^2}{|x|^2}\,dx \leq C_n \int_{B_r} \left(|Du|^2 + r^{-2}|u|^2\right)\,dx.$$

Expanded proof.

Start with u smooth on $\overline{B}_r$ and set $X = x/|x|^2$, so $\operatorname{div} X = (n-2)/|x|^2$. Apply the divergence theorem on $B_r \setminus \overline{B}_\varepsilon$. The inner boundary term is $O(\varepsilon^{n-2})$ and tends to zero. With $I = \int_{B_r} u^2/|x|^2$ and $A = \|Du\|_2$, the result is

$$(n-2)I = r^{-1}\int_{\partial B_r} u^2\,dS - 2\int_{B_r} u\,Du \cdot \frac{x}{|x|^2}.$$

Cauchy–Schwarz and Young yield $I \leq C_n A^2 + C_n r^{-1}\int_{\partial B_r} u^2$. To control the last term, apply divergence to xu^2:

$$r\int_{\partial B_r} u^2 = \int_{B_r} (nu^2 + 2u\,x\cdot Du) \leq C_n \int_{B_r} (u^2 + r^2|Du|^2).$$

Combine the estimates. For general u, use smooth boundary approximation. Applying the estimate to differences shows $u_j/|x|$ is Cauchy in L^2. An a.e. convergent subsequence identifies its limit with $u/|x|$, and the bound passes to the limit. □

5.8.6 Fourier characterization (Evans §5.8.5, Theorem 8)

Use the unitary convention $\hat{u}(\xi) = (2\pi)^{-n/2}\int e^{-ix\cdot\xi}u(x)\,dx$. For integer $k \geq 0$,

$$u \in H^k(\mathbb{R}^n) \iff (1+|\xi|^2)^{k/2}\hat{u} \in L^2(\mathbb{R}^n),$$

$$\|u\|_{H^k} \asymp \|(1+|\xi|^2)^{k/2}\hat{u}\|_2.$$

Expanded proof.

For compactly supported smooth functions integration by parts gives $\widehat{D^\alpha u} = (i\xi)^\alpha \hat{u}$. Whole-space cutoff and convolution approximate every

可微比 Hölder 更强

Morrey 单独给出 $O(r^{1-n/p})$。减去候选线性项后，梯度的局部平均误差趋零，才把余项改善为 $o(r)$。

奇点与边界都要处理

不能直接在含原点的球上对奇异向量场用散度定理；先挖去小球再估内边界项。外边界项不为零，所以一般球域估计保留 $r^{-2}\|u\|_2^2$。

H^k function by such functions. Plancherel and passage to the L^2 limit preserve the identity. Also

$$c_{n,k}(1+|\xi|^2)^k \le \sum_{|\alpha|\le k} |\xi^\alpha|^2 \le C_{n,k}(1+|\xi|^2)^k.$$

This follows by the multinomial expansion of $(1+\xi_1^2+\cdots+\xi_n^2)^k$ and comparison of its finitely many positive coefficients. It proves the forward bound. Conversely, the weighted condition makes $g_\alpha = \mathscr{F}^{-1}((i\xi)^\alpha \hat{u})$ an L^2 function. Plancherel against test functions gives the weak derivative identity, hence $g_\alpha = D^\alpha u$ and $u \in H^k$. The same polynomial comparison proves both norm bounds. □

For real $s > 0$ this suggests the definition

$$\|u\|_{H^s(\mathbb{R}^n)} = \|(1+|\xi|^2)^{s/2}\hat{u}\|_2.$$

It is equivalent to the source weight $1+|\xi|^s$; equivalent norms describe the same space.

频率中的光滑度

求导变成乘以 $i\xi$。高频尾部越小，可承受的频率权重越高，意味着可积的导数越多；分数阶空间沿用同一机制。

5.9 Other Spaces of Functions

5.9.1 The dual space H^{-1} (Evans §5.9.1, Theorem 1)

Define $H^{-1}(U) = (H_0^1(U))^*$ with

$$\|f\|_{H^{-1}} = \sup_{\|v\|_{H^1}\le 1} |\langle f, v\rangle|.$$

Here we use the full H^1 norm, rather than the equivalent gradient norm available on bounded domains.

Theorem 5.22

For every $f \in H^{-1}(U)$ there are $f^0, \dots, f^n \in L^2(U)$ with

$$\langle f, v\rangle = \int_U f^0 v + \sum_{i=1}^n f^i \partial_i v \, dx, \qquad v \in H_0^1(U).$$

Moreover

$$\|f\|_{H^{-1}} = \min\left\{ \left(\sum_{i=0}^n \|f^i\|_2^2 \right)^{1/2} : (f^0, \dots, f^n) \text{ represents } f \right\}.$$

In distributions, $f = f^0 - \sum_i \partial_i f^i$.

Expanded proof.

Apply the Hilbert-space Riesz representation theorem, an external functional analysis result, to $H_0^1(U)$ with inner product $\int (wv + Dw \cdot Dv)$. There is a unique $w \in H_0^1(U)$ such that

$$\langle f, v\rangle = \int_U (wv + Dw \cdot Dv), \qquad \|f\|_{H^{-1}} = \|w\|_{H^1}.$$

Set $f^0 = w$ and $f^i = \partial_i w$. Their product L^2 norm equals $\|f\|_{H^{-1}}$. Any other representation (g^i) satisfies $|\langle f, v\rangle| \le (\sum_i \|g^i\|_2^2)^{1/2}\|v\|_{H^1}$ by Cauchy–Schwarz. Take the supremum over unit v to prove minimality. The distributional sign follows from $\langle \partial_i g, v\rangle = -\int g\partial_i v$. □

The embedding $L^2(U) \to H^{-1}(U)$ is $\langle f, v\rangle = \int_U fv$, with norm at most $\|f\|_2$. It is injective because test functions separate locally integrable functions. Thus the useful pivot-space chain is

$$H_0^1(U) \subset L^2(U) \subset H^{-1}(U).$$

负一阶不是负次求导

H^{-1} 的元素可以是导数型分布。若 $F \in L^2(U;\mathbb{R}^n)$，则 $-\operatorname{div} F$ 作用在 v 上是 $\int F \cdot Dv$，由 Cauchy 不等式连续。

5.9.2 Banach-valued time functions (Evans §5.9.2, Theorem 2)

For a real Banach space X, $L^p(0,T;X)$ consists of strongly measurable functions with integrable pth power of their X norm (or essentially bounded norm when $p = \infty$). Integration below is Bochner integration. Its construction, norm inequality, and Fubini properties are prerequisites from measure theory. We say $u' = g$ weakly if

$$\int_0^T u(t)\phi'(t)\,dt = -\int_0^T g(t)\phi(t)\,dt \quad (\phi \in C_c^\infty(0,T)),$$

as an equality in X. The space $W^{1,p}(0,T;X)$ requires both u and g in $L^p(0,T;X)$.

不要抹去中间的 L^2

Riesz 可把 Hilbert 空间与其对偶抽象同构，但演化方程要保留 L^2 配对。u_t 可只在 H^{-1} 中，$\langle u_t, u\rangle$ 仍有意义。

Theorem 5.23 (Fundamental theorem in X)
Every $u \in W^{1,p}(0,T;X)$ has a continuous representative with

$$u(t) - u(s) = \int_s^t u'(\tau)\,d\tau.$$

$$\max_{[0,T]} \|u(t)\|_X \le T^{-1/p}\|u\|_{L^p(X)} + T^{1-1/p}\|u'\|_{L^p(X)}.$$

Interpret $1/p = 0$ for $p = \infty$.

Expanded proof.

Set $G(t) = \int_0^t u'(\tau)\,d\tau$. The Bochner integral norm inequality makes G absolutely continuous. Fubini against scalar test functions shows $G' = u'$ weakly, hence $w = u - G$ has weak derivative zero.

To show w is constant without making an uncountable exceptional-set argument over X^*, smooth w in time on interior intervals. Its smooth derivative is zero, so w_ε is an X-valued constant there. The convergence $w_\varepsilon \to w$ in local $L^1(0,T;X)$ forces these constants to converge in X on each fixed subinterval. Overlapping subintervals give the same constant c. Thus $u = c + G$ almost everywhere; take $c + G$ as the representative. The formula holds for all s, t, including endpoints. Its norm inequality gives

$$\|u(t)\|_X \le \|u(s)\|_X + \int_0^T \|u'(\tau)\|_X \, d\tau.$$

Average in s and apply Hölder, obtaining the stated bound. □

5.9.3 Energy-space continuity (Evans §5.9.2, Theorem 3)

Write $V = H_0^1(U)$, $H = L^2(U)$, and $V^* = H^{-1}(U)$.

Theorem 5.24
If $u \in L^2(0,T;V)$ and $u' \in L^2(0,T;V^*)$, then u has a representative in $C([0,T];H)$ and

$$\|u(t)\|_H^2 - \|u(s)\|_H^2 = 2\int_s^t \langle u'(\tau), u(\tau)\rangle \, d\tau.$$

The squared norm is absolutely continuous and its derivative is the integrand almost everywhere. Also

$$\max_t \|u(t)\|_H \le C_T\big(\|u\|_{L^2(V)} + \|u'\|_{L^2(V^*)}\big).$$

Expanded proof.

1. Construct time-smooth approximations without endpoint jumps. Since $V \subset V^*$ continuously, $u \in W^{1,2}(0,T;V^*)$. Theorem 5.23 gives a continuous V^* representative. Reflect it across 0 and T to a slightly larger interval. The reflected function remains in $L^2(V)$, with weak derivative in $L^2(V^*)$ and the appropriate reflected signs. There is no endpoint Dirac term because the V^* values match. Convolve in time. Translation continuity of Bochner L^2 gives

$$u_\varepsilon \to u \text{ in } L^2(V), \qquad u_\varepsilon' \to u' \text{ in } L^2(V^*).$$

Each smoothed function is smooth with values in V.

2. A uniform-in-time Cauchy estimate. For any smooth $w : [0,T] \to V$, $\frac{d}{dt}\|w\|_H^2 = 2\langle w', w\rangle$. Integrate between s and t, take absolute values of

初值需要时间代表元

L^p 时间函数可以在单个时刻任意修改，因此 $u(0)$ 原先没有确定意义。时间基本定理选出连续代表元，才允许严谨地指定初值。

the integral, and average s over $(0,T)$. It follows that

$$\sup_t \|w(t)\|_H^2 \le T^{-1}\|w\|_{L^2(H)}^2 + 2\|w'\|_{L^2(V^*)}\|w\|_{L^2(V)}.$$

Apply this to $w = u_\varepsilon - u_\delta$. Both graph-norm errors tend to zero, so (u_ε) is Cauchy in $C([0,T];H)$. Its uniform limit agrees with u almost everywhere by the $L^2(H)$ convergence. This proves continuity, including at both endpoints.

3. Pass the energy identity to the limit. The endpoint terms converge uniformly. The integrands converge in L^1 because

$$\begin{aligned}\|\langle u_\varepsilon', u_\varepsilon\rangle - \langle u', u\rangle\|_{L^1} &\le \|u_\varepsilon' - u'\|_{L^2(V^*)}\|u_\varepsilon\|_{L^2(V)} \\ &\quad + \|u'\|_{L^2(V^*)}\|u_\varepsilon - u\|_{L^2(V)} \to 0.\end{aligned}$$

The limiting integral identity gives absolute continuity and its a.e. derivative. Applying the uniform estimate to u_ε and then Young's inequality proves the norm bound. □

5.9.4 Continuity in a better spatial space (source Theorem 4)

Let U be bounded with smooth boundary and $m \ge 0$ an integer. If

$$u \in L^2(0,T;H^{m+2}(U)), \qquad u' \in L^2(0,T;H^m(U)),$$

then $u \in C([0,T];H^{m+1}(U))$, with the corresponding bound

$$\max_t \|u(t)\|_{H^{m+1}} \le C_{U,T,m}(\|u\|_{L^2(H^{m+2})} + \|u'\|_{L^2(H^m)}).$$

Expanded proof.

First $m = 0$. Choose the same compact-support extension operator E bounded both on L^2 and H^2, as constructed in §5.4. Set $v = Eu$. Bounded linear maps commute with Bochner integration, so testing in time gives $v' = Eu'$. For time-smooth v, integrate by parts in space on $\mathbb{R}^n$:

$$\frac{d}{dt}\|v\|_{H^1(\mathbb{R}^n)}^2 = 2(v - \Delta v, v')_{L^2}.$$

This is valid for H^2 functions by spatial approximation, with no boundary term on the whole space. Thus its absolute value is at most $C\|v\|_{H^2}\|v'\|_2$. Average an integrated identity in the initial time to obtain

$$\sup_t \|v(t)\|_{H^1}^2 \le C_T(\|v\|_{L^2(H^2)}^2 + \|v'\|_{L^2(L^2)}^2).$$

Reflect and convolve in time as in the preceding proof; now use continuity in L^2, which follows from $u \in W^{1,2}(0,T;L^2)$. The estimate on differences gives convergence in $C(H^1)$. Restrict to U and use the bounds for E.

能量计算的合法化

形式上把方程乘 u 会得到 $\frac{1}{2}(d/dt)\|u\|_2^2$。当 u_t 只有 H^{-1} 正则性时，必须先时间平滑、建立一致估计，再通过配对的 L^1 收敛取极限。

For general m, apply this result to every $D^\alpha u$, $|\alpha| \le m$. Spatial derivatives commute with the time weak derivative, as one checks against separated space-time test functions. Summing the finitely many H^1 estimates yields an equivalent H^{m+1} norm and the claimed continuity. □

空间与时间交换一阶

空间有 $m+2$ 阶、时间导数有 m 阶，得到时间连续的中间空间 H^{m+1}。空间分部积分把两个导数移到同一侧，正是这个阶数配平的来源。

5.10 Problems

Evans §5.10, Problems 1–21, printed pp. 289–292 (PDF pp. 306–309). Statements are concisely restated; solutions are editorial. Boundary smoothness is assumed where needed.

5.10.1 Problems 1–6: the basic language

1. Completeness of Hölder spaces. Prove $C^{k,\gamma}(\overline{U})$ is Banach.

Solution. The full proof in §5.1 identifies uniform limits of derivative components through coordinate-segment integration, then passes the Cauchy bound to the Hölder seminorm. Derivative compatibility and seminorm convergence are both needed, beyond uniform convergence of u_j. □

2. Hölder interpolation. For $0 < \beta < \gamma \le 1$, prove

$$\|u\|_{C^{0,\gamma}} \le \|u\|_{C^{0,\beta}}^{(1-\gamma)/(1-\beta)} \|u\|_{C^{0,1}}^{(\gamma-\beta)/(1-\beta)}.$$

Solution. The quotient factorization and two-term Hölder inequality in §5.1 give this formula. The weights sum to one and satisfy $\gamma = (1-\theta)\beta + \theta$ for $\theta = (\gamma-\beta)/(1-\beta)$. □

3. A pyramid on a square. On $U = (-1,1)^2$ the source's piecewise function is $u(x) = 1 - \max(|x_1|, |x_2|)$. Determine its $W^{1,p}$ membership.

Solution. On the four sectors where one signed coordinate is largest, Du is respectively $(-1,0), (1,0), (0,-1), (0,1)$. Integrate against a test function separately on the sectors. At each shared edge, values of u agree and outward normals are opposite, so the interface terms cancel. This bounded vector field is the weak gradient. Both u and Du are bounded on a finite-measure domain, so $u \in W^{1,p}(U)$ for all $1 \le p \le \infty$. □

4. One-dimensional Sobolev functions. Show $u \in W^{1,p}(0,1)$, $p < \infty$, has an absolutely continuous representative, and derive a Hölder estimate for $p > 1$.

Solution. Let g be its weak derivative and $G(x) = \int_0^x g(t)\,dt$. Fubini shows $G' = g$ weakly. Since $(u - G)' = 0$, Lemma 5.17 makes $u - G$ constant. The representative $c + G$ is absolutely continuous, with classical derivative g a.e. For every $x, y \in [0,1]$, Hölder gives

$$|u(x) - u(y)| \le \int_{\min(x,y)}^{\max(x,y)} |g| \le |x - y|^{1-1/p} \|g\|_{L^p(0,1)}.$$

For $p = 1$, absolute continuity persists but this argument supplies no fixed positive Hölder exponent. □

5. A smooth cutoff. For $V \Subset U$, construct $\zeta \in C_c^\infty(U)$ with $\zeta = 1$ on V.

Solution. Put $d = \min\{1, \operatorname{dist}(\overline{V}, \mathbb{R}^n \setminus U)\} > 0$ and $W = \{x : \operatorname{dist}(x, \overline{V}) < d/3\}$. For $\varepsilon < d/6$, $\zeta = \rho_\varepsilon * \mathbf{1}_W$ equals one on V, lies in $[0,1]$, and has compact support within distance $d/2$ of $\overline{V}$, hence inside U. □

6. A finite partition of unity. If $\overline{U}$ is compact and contained in $\bigcup_{i=1}^N V_i$, construct smooth ζ_i supported in V_i, with $0 \le \zeta_i \le 1$ and $\sum_i \zeta_i = 1$ on U.

Solution. Cover $\overline{U}$ by finitely many smaller balls with closures in the given cover. Choose nonnegative smooth bumps positive on these balls and group them by V_i, obtaining $\psi_i \in C_c^\infty(V_i)$. Their sum S is positive on a neighborhood O of $\overline{U}$. Choose $\chi \in C_c^\infty(O)$ equal to one near $\overline{U}$, with $0 \le \chi \le 1$. Set $\zeta_i = \chi\psi_i/S$ on O, zero elsewhere. The cutoff makes this smooth globally, and $\psi_i \le S$ gives $\zeta_i \le 1$. Their sum is χ, as required. □

5.10.2 Problems 7–10: boundary and interpolation estimates

7. Trace from a transverse vector field. If α is smooth and $\alpha \cdot \nu \ge 1$ on ∂U, prove the trace estimate for $1 \le p < \infty$.

分片求导还要查接口

每块导数有界仍不够，必须检查接口函数值相同。若有跳跃，弱导数就会多出沿接口集中的测度。

分母先保证为正

S 只在目标集邻域中确定为正；乘上 χ 后，在离开该邻域前商已为零，才可安全延拓。

Solution. For smooth u and $p > 1$, divergence and Young's inequality give

$$\begin{aligned}\int_{\partial U} |u|^p &\le \int_{\partial U} |u|^p \alpha \cdot \nu \\ &= \int_U |u|^p \operatorname{div} \alpha + p|u|^{p-2} u\, \alpha \cdot Du \\ &\le C \int_U (|u|^p + |Du|^p).\end{aligned}$$

The coefficients $\alpha, D\alpha$ are bounded on the compact closure. For $p = 1$, replace $|u|$ by $\sqrt{u^2 + \varepsilon^2}$; its gradient has magnitude at most $|Du|$. Let $\varepsilon \to 0$. Smooth boundary approximation gives the Sobolev estimate. □

8. No bounded trace on L^p alone. On a nonempty bounded C^1 domain, show no bounded linear map $L^p(U) \to L^p(\partial U)$, $p < \infty$, can agree with continuous boundary restriction.

Solution. Write $d(x) = \operatorname{dist}(x, \partial U)$. Choose continuous $\eta : [0, \infty) \to [0, 1]$ with $\eta(0) = 1$ and $\eta = 0$ on $[1, \infty)$. The functions $v_\delta(x) = \eta(d(x)/\delta)$ are continuous on $\overline{U}$, equal one on ∂U, and tend pointwise to zero in U. Dominated convergence gives $\|v_\delta\|_{L^p(U)} \to 0$. A bounded trace would force $0 < |\partial U|^{1/p} = \|T v_\delta\|_p \to 0$, a contradiction. □

边界薄层反例

函数支撑在越来越薄的边界层中，体积范数趋零，边界值却始终为 1。其梯度变大，说明迹估计为何需要导数信息。

9. An L^2 interpolation estimate. Prove $\|Du\|_2 \le C\|u\|_2^{1/2}\|D^2u\|_2^{1/2}$ for $C_c^\infty(U)$ and $H^2(U) \cap H_0^1(U)$.

Solution. For compactly supported smooth u,

$$\|Du\|_2^2 = -\int_U u\Delta u \le \|u\|_2 \|\Delta u\|_2 \le C_n \|u\|_2 \|D^2 u\|_2.$$

For $u \in H^2 \cap H_0^1$, the distributional identity $\int Du \cdot D\phi = -\int (\Delta u)\phi$ extends from tests to $\phi \in H_0^1$ by density, since both sides are continuous in that norm. Take $\phi = u$. No compactly supported H^2 approximation is needed. □

不能假设更强的逼近

紧支函数在 H^2 中的闭包还会限制法向导数，通常比 $H^2 \cap H_0^1$ 小。此处只在 H_0^1 中延伸测试恒等式，避免加上多余边界条件。

10. Nonlinear interpolation. For compactly supported smooth u, prove

$$\|Du\|_p \le C\|u\|_p^{1/2}\|D^2u\|_p^{1/2} \quad (2 \le p < \infty),$$

$$\|Du\|_{2p} \le C\|u\|_\infty^{1/2}\|D^2u\|_p^{1/2} \quad (1 \le p < \infty).$$

Solution. For $s \ge 2$, integrate by parts and differentiate $|Du|^{s-2}Du$ to get

$$\int |Du|^s \le C_s \int |u|\, |Du|^{s-2} |D^2 u|.$$

The vector map $z \mapsto |z|^{s-2}z$ is C^1 for $s \geq 2$, including the linear case $s = 2$. With $s = p > 2$, Hölder with exponents $p, p/(p-2), p$ gives $\|Du\|_p^p \leq C_p \|u\|_p \|Du\|_p^{p-2} \|D^2 u\|_p$. At $p = 2$ use Cauchy directly. Cancel the gradient factor unless zero, then take a square root. With $s = 2p$, use $\|u\|_\infty$ and exponents $p, p/(p-1)$ to get

$$\|Du\|_{2p}^{2p} \leq C_p \|u\|_\infty \|D^2 u\|_p \|Du\|_{2p}^{2p-2}.$$

For $p = 1$ the last factor is one. Cancellation gives the second result. □

5.10.3 Problems 11–16: the limits of the theory

11. Zero weak gradient. Prove on connected U that $Du = 0$ implies u constant a.e.

Solution. Lemma 5.17 gives the full argument: mollify on balls, identify constant limits, and propagate the common constant along a finite chain of overlapping balls. Connectedness enters the last step. □

12. Failure at $p = 1$. Show bounded L^1 difference quotients do not imply $W^{1,1}$ membership.

Solution. Take $U = (-2, 2)$, $V = (-1, 1)$, and $u = \mathbf{1}_{(0,2)}$ on U. For small h, $|D^h u| = 1/|h|$ on an interval of length $|h|$, and is zero elsewhere in V. Thus $\|D^h u\|_{L^1(V)} = 1$. But its distributional derivative is δ_0, so it is not in $W^{1,1}(V)$. □

13. Bounded gradient without global Lipschitz continuity.

Solution. Let U be the slit annulus $\{(r\cos\theta, r\sin\theta) : 1/2 < r < 1,\ 0 < \theta < 2\pi\}$ and put $u(r, \theta) = \theta$. This is smooth on U, with $|u| \leq 2\pi$ and $|Du| = 1/r \leq 2$. At $r = 3/4$, points of angles ε and $2\pi - \varepsilon$ approach each other, but their values differ by $2\pi - 2\varepsilon$. The function is in $W^{1,\infty}(U)$ and is not globally Lipschitz. The domain is connected but fails the regular boundary hypothesis at the slit. □

14. The source's unbounded critical example. For $n > 1$ verify $u(x) = \log\log(1 + 1/|x|) \in W^{1,n}(B_1)$.

Solution. For $r = |x|$,

$$|u'(r)| = \frac{1}{r(r+1)\log(1+1/r)} \asymp \frac{1}{r\log(1/r)} \quad (r \downarrow 0).$$

梯度控制路径长度

梯度界控制沿域内路径的变化；路径长度未必能由直线距离统一控制。裂缝反例不违背球内的局部 Lipschitz 结论。

The gradient integral near zero is comparable to $\int_0^{1/2} dr/(r\log(1/r)^n)$, finite for $n > 1$. The function integral has an exponentially decaying weight after $t = \log(1/r)$, and is also finite. Away from zero all quantities are bounded. The inner integration-by-parts term is bounded by $C\varepsilon^{n-1}|u(\varepsilon)| \to 0$, proving that the classical gradient off zero is its weak gradient. The function itself diverges as $r \downarrow 0$. □

15. A zero set of positive measure. If $u \in H^1(B_1)$ and $|\{u = 0\}| \geq \alpha > 0$, prove $\|u\|_2 \leq C_{n,\alpha}\|Du\|_2$.

Solution. Let $c = u_{B_1}$ and $Z = \{u = 0\}$. Poincaré gives

$$\alpha|c|^2 \leq \int_Z |u - c|^2 \leq \|u - c\|_2^2 \leq C_n\|Du\|_2^2.$$

Then $\|u\|_2 \leq \|u - c\|_2 + |B_1|^{1/2}|c|$ proves the estimate, with constant $C_n(1 + |B_1|^{1/2}\alpha^{-1/2})$. □

16. Whole-space Hardy inequality. For $n \geq 3$ prove

$$\int_{\mathbb{R}^n} \frac{|u|^2}{|x|^2} \leq \frac{4}{(n-2)^2}\int_{\mathbb{R}^n} |Du|^2, \qquad u \in H^1(\mathbb{R}^n).$$

Solution. First take compactly supported smooth u. Integrate the identity

$$\operatorname{div}\left(\frac{x}{|x|^2}\right) = \frac{n-2}{|x|^2}$$

outside B_ε. The inner boundary term is $O(\varepsilon^{n-2})$ and vanishes; there is no outer term. Thus

$$(n-2)\int \frac{u^2}{|x|^2} = -2\int u\,Du \cdot \frac{x}{|x|^2} \leq 2\left(\int \frac{u^2}{|x|^2}\right)^{1/2}\|Du\|_2.$$

Cancel and square. For general u, compactly supported smooth H^1 approximations are available by cutoff and convolution. The estimate on differences makes $u_j/|x|$ Cauchy in L^2; an a.e. convergent subsequence identifies its limit with $u/|x|$. □

5.10.4 Problems 17–19: nonlinear operations

17. Chain rule. On bounded U, let $F \in C^1(\mathbb{R})$ have bounded derivative. Prove $F(u) \in W^{1,p}$ and $D(F(u)) = F'(u)Du$.

Solution. The bound $|F(s)| \le |F(0)| + \|F'\|_\infty |s|$ gives integrability. For finite p, smooth interior approximations converge in local $W^{1,p}$ and, along a subsequence, a.e. The Lipschitz bound gives $F(u_j) \to F(u)$ in local L^p. Split the gradients:

$$F'(u_j)Du_j - F'(u)Du = F'(u_j)(Du_j - Du) + (F'(u_j) - F'(u))Du.$$

The first term tends to zero by boundedness of F'; the second by dominated convergence with majorant $2\|F'\|_\infty |Du|$. Pass to the weak identity locally. The candidate derivative is globally L^p, so this proves membership on U. For $p = \infty$, use finite local exponents and retain the resulting L^∞ bounds on function and gradient. □

18. Positive parts and level sets. Prove $|u|, u^+, u^- \in W^{1,p}(U)$ and

$$Du^+ = \mathbf{1}_{\{u>0\}}Du, \qquad Du^- = -\mathbf{1}_{\{u<0\}}Du, \qquad Du = 0 \text{ a.e. on } \{u = 0\}.$$

Solution. The function

$$F_\varepsilon(s) = \begin{cases} \sqrt{s^2 + \varepsilon^2} - \varepsilon, & s \ge 0, \\ 0, & s < 0 \end{cases}$$

is C^1, with $|F_\varepsilon'| \le 1$, $F_\varepsilon' \to \mathbf{1}_{\{s>0\}}$ pointwise, and $|F_\varepsilon(s) - s^+| \le \varepsilon$. Problem 17 and dominated convergence show for finite p that

$$F_\varepsilon(u) \to u^+ \text{ in } L^p, \qquad D(F_\varepsilon(u)) \to \mathbf{1}_{\{u>0\}}Du \text{ in } L^p.$$

The weak identity yields the positive-part formula. Apply it to $-u$ for the negative part. Differentiate $u = u^+ - u^-$ to obtain $Du = \mathbf{1}_{\{u\neq 0\}}Du$, proving the level-set assertion. Use $|u| = u^+ + u^-$ for the absolute value. For $p = \infty$ use a finite exponent on bounded U and the derivative bounds to recover $W^{1,\infty}$ membership. □

用常数检验边界项

球域 Hardy 需要低阶项，因为常数在球上属于 H^1；全空间非零常数不在 L^2，所以不反驳纯梯度型估计。

链式法则如何取极限

$F'(u_j) \to F'(u)$ 只是逐点信息；第二项需要由 $2\|F'\|_\infty |Du|$ 支配，第一项则依赖梯度的强收敛。

19. A second level-set argument for H^1. Let ϕ be smooth, bounded, nondecreasing, with bounded derivative, and $\phi(s) = s$ for $|s| \le 1$. Set $u_\varepsilon = \varepsilon\phi(u/\varepsilon)$.

Solution. Since $\phi(0) = 0$ and ϕ' is bounded, $|u_\varepsilon| \le \|\phi'\|_\infty |u|$ while $|u_\varepsilon| \le \varepsilon\|\phi\|_\infty \to 0$ pointwise. Dominated convergence gives $u_\varepsilon \to 0$ in L^2, even when U has infinite measure. Also $Du_\varepsilon = \phi'(u/\varepsilon)Du$ is bounded in L^2. Every sequence $\varepsilon_j \to 0$ has a weak H^1 subsequence by reflexivity; its limit is zero by strong L^2 convergence. Thus the whole family converges weakly to zero. Pairing its gradient with Du gives

$$0 \le \int_{\{u=0\}} |Du|^2 \le \int_U \phi'(u/\varepsilon)|Du|^2 = \int_U Du_\varepsilon \cdot Du \to 0.$$

We used $\phi' \ge 0$ and $\phi'(0) = 1$, without needing any pointwise limit of $\phi'(u/\varepsilon)$ away from $\{u = 0\}$. □

不能漏掉零水平集

$\{u = 0\}$ 可能有正测度，不能只在正负区域求导就结束。先证明正负部公式，再由 $u = u^+ - u^-$ 推出该集合上梯度为零。

5.10.5 Problems 20–21: Fourier estimates

20. Boundedness above half the dimension. Show $H^s(\mathbb{R}^n) \hookrightarrow L^\infty$ for $s > n/2$.

Solution. Put $w_s(\xi) = (1 + |\xi|^2)^{s/2}$. Cauchy–Schwarz gives

$$\|\hat{u}\|_1 \le \|w_s\hat{u}\|_2 \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{-s}\, d\xi \right)^{1/2} \le C_{n,s}\|u\|_{H^s}.$$

The radial integral at infinity is $\int_1^\infty r^{n-1-2s}\, dr$, finite exactly when $2s > n$. The inverse Fourier integral is bounded and continuous, with supremum at most $(2\pi)^{-n/2}\|\hat{u}\|_1$, and agrees with u in L^2 by Plancherel/inversion. □

半维数的来源

要让逆变换绝对可积，频率权重的倒数需属于 L^2；径向积分恰好给出条件 $2s > n$，而不是随意选的正则性门槛。

21. The algebra property. For $s > n/2$, prove $\|uv\|_{H^s} \le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}$.

Solution. For Schwartz functions the triangle inequality gives $w_s(\xi) \le C_s(w_s(\eta) + w_s(\xi - \eta))$. The product-convolution formula implies

$$w_s|\widehat{uv}| \le C_{n,s}\big[(w_s|\hat{u}|) * |\hat{v}| + |\hat{u}| * (w_s|\hat{v}|)\big].$$

Young's $L^2 * L^1 \to L^2$ inequality and Problem 20 give

$$\begin{aligned} \|uv\|_{H^s} &\le C_{n,s}\big(\|u\|_{H^s}\|\hat{v}\|_1 + \|\hat{u}\|_1\|v\|_{H^s}\big) \\ &\le C_{n,s}\|u\|_{H^s}\|v\|_{H^s}. \end{aligned}$$

For general u, v, approximate their transforms by smooth compactly supported functions in weighted L^2. Their inverse transforms are Schwartz. The bilinear estimate makes their products Cauchy in H^s. Problem 20 gives uniform convergence of the factors, and

$$\|u_j v_j - uv\|_2 \le \|u_j\|_\infty \|v_j - v\|_2 + \|u_j - u\|_\infty \|v\|_2 \to 0.$$

Thus the H^s limit is the actual product uv. □

乘积为何不丢阶

$s > n/2$ 提供频率的 L^1 控制，使卷积可用 $L^2 * L^1$ 估计。因此多项式非线性可以在同一个 H^s 空间内处理。

5.11 References

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 5, printed pp. 239–292 (local English PDF pp. 256–309). The chapter order and source theorem identifiers follow that edition. The Chinese scan has Chapter 5 at printed pp. 169–207 (PDF pp. 187–225); the chapter opening and its list of section starts were visually checked. It is a terminology reference, not the source of the English prose written here.

For Sobolev theory, Evans's §5.11 points to Gilbarg–Trudinger (Chapter 7), Lieb–Loss, Ziemer, and Evans–Gariepy. For BMO the source points to Stein, Chapter IV; for time-dependent spaces to Temam, pp. 248–273; and for whole-space Hardy to Tartar, Chapter 17. These are source pointers, not claims that those books have been independently consulted for this draft.

Proof dependencies and review boundary. This chapter invokes the measure-theoretic and functional-analytic prerequisites listed at its opening, including Lebesgue differentiation, Plancherel, Riesz representation, reflexivity, and Arzel–Ascoli. These foundational theorems are explicitly invoked rather than reproved. Other displayed chapter theorems have reconstructed proofs here. All 21 exercises have editorial solutions or exact pointers to full arguments earlier in the chapter. Source comparison, mathematical self-checking, compilation, and visual inspection have separate scopes. This study draft has not received independent mathematical review.

Chapter 6

Second-Order Elliptic Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 6; printed pp. 293–348; PDF pages 310–365.

Chinese reference: 中译本：书内第 208–246 页；PDF 第 226–264 页。

Editorial study edition. This chapter reconstructs Evans's exposition in original English, with expanded proofs and Chinese mathematical annotations. It is not a verbatim transcription. Headings identify the source results; our theorem numbers run throughout this chapter. All 17 source problems are restated concisely and solved. Foundational functional analysis is invoked explicitly; the Schaefer fixed-point theorem and Weyl's asymptotic law are identified where used or stated, rather than presented as proved here.

The central example is

$$-\operatorname{div}(ADu) + b \cdot Du + cu = f \quad \text{in } U, \qquad u = 0 \quad \text{on } \partial U.$$

There are four distinct questions: does a weak solution exist, how smooth is it, what controls its sign, and when does the homogeneous equation have a nonzero solution? The answers require different hypotheses.

弱解到经典解

存在性先在能量空间中完成，不要求二阶导数。正则性随后补足导数，才能使用经典最大值原理；特征值问题再把紧性与正性结合起来。

Conventions and reading route. Repeated indices are summed from 1 to n. Unless specified otherwise, U is a bounded open subset of $\mathbb{R}^n$, $A = (a^{ij})$ is real symmetric, and $a^{ij}\xi_i\xi_j \geq \theta|\xi|^2$ for some $\theta > 0$. The norm $\|\cdot\|_2$ is over the indicated domain, or over U if omitted. Begin with §6.1–§6.2, then follow the difference quotient argument in §6.3. Read §6.4 before the positivity arguments in §6.5. Chapter 5 supplies approximation, traces, Poincaré, difference quotients, and Rellich compactness. Riesz representation, weak subsequential compactness in Hilbert spaces, and basic finite-dimensional linear algebra are prerequisites.

6.1 Definitions

6.1.1 Two forms of an elliptic operator

The divergence form and nondivergence form are, respectively,

$$Lu = -\partial_j(a^{ij}\partial_i u) + b^i\partial_i u + cu, \tag{6.1}$$

$$Ku = -a^{ij}\partial_{ij}u + \tilde{b}^i\partial_i u + cu. \tag{6.2}$$

Uniform ellipticity means

$$\theta|\xi|^2 \le a^{ij}(x)\xi_i\xi_j \quad (x \in U,\ \xi \in \mathbb{R}^n).$$

For measurable coefficients this is required almost everywhere. Boundedness of the entries gives an upper bound $\Lambda|\xi|^2$. It is the principal part, not the lower-order terms, that defines ellipticity.

If A is C^1, the two formulas agree precisely when $\tilde{b}^i = b^i - \partial_j a^{ij}$. If A is merely bounded measurable, this conversion is not a pointwise identity between ordinary functions. Moreover, symmetry of A does not imply symmetry of the full operator: the drift $b \cdot Du$ generally destroys self-adjointness.

散度型与非散度型

散度型可把一个导数移到测试函数上，因而适合弱解。只有系数足够可微时，才能展开散度并把系数导数并入漂移项。

For example, $A = \operatorname{diag}(1,4)$ has $\theta = 1$ and $\Lambda = 4$; the principal quadratic form has elliptical level sets. In contrast, $u_{xx} - u_{yy}$ has a principal form of both signs and is not elliptic. If the smallest eigenvalue of $A(x)$ approaches zero, pointwise positive definiteness alone gives no uniform θ.

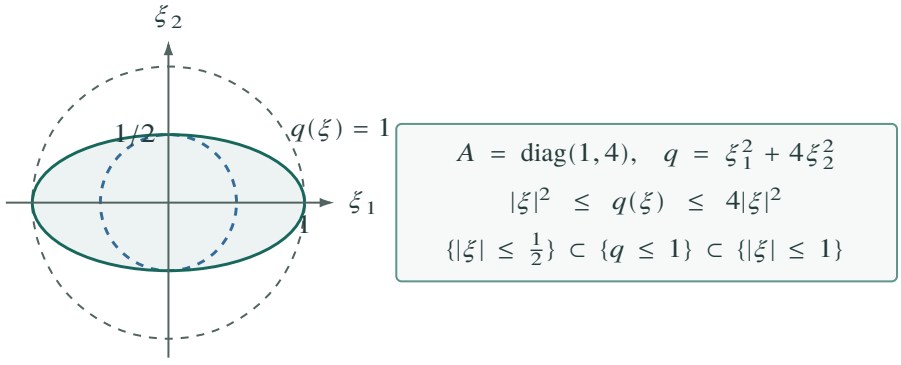


Diagram T6.1. Ellipticity bounds the shape of the principal quadratic form. Here $\theta = 1$ and $\Lambda = 4$. The sublevel ellipse contains the circle of radius $\Lambda^{-1/2}$ and lies inside the circle of radius $\theta^{-1/2}$. A uniform positive lower bound prevents this principal geometry from degenerating as the base point varies.

6.1.2 Weak solutions and the boundary condition

Assume $A, b, c \in L^\infty(U)$ and set

$$B[u,v] = \int_U (a^{ij}u_i v_j + b^i u_i v + cuv)\,dx, \qquad u, v \in H_0^1(U). \tag{6.3}$$

A weak solution of $Lu = f$, with zero Dirichlet boundary values, is $u \in H_0^1(U)$ such that

$$B[u, v] = \langle f, v \rangle \quad (v \in H_0^1(U)). \tag{6.4}$$

Here $f \in H^{-1}(U)$; in particular $f \in L^2(U)$ is allowed.

For smooth u with $u = 0$ on the boundary, integration by parts in the principal term gives (6.4). Conversely, testing against $C_c^\infty(U)$ shows that (6.4) implies $Lu = f$ as a distribution. If u subsequently belongs to $C^2(U)$ and the coefficients and f are continuous in the relevant form, the distributional equation becomes the classical equation.

A useful description of rough data is $f = f_0 - \operatorname{div} F$, with $f_0, F_i \in L^2(U)$. Its action is

$$\langle f, v \rangle = \int_U (f_0 v + F \cdot Dv)\, dx, \qquad |\langle f, v \rangle| \le C(\|f_0\|_2 + \|F\|_2)\|v\|_{H^1}.$$

The plus sign in the second term comes from distributional differentiation.

For prescribed trace $g = Tw$ with $w \in H^1(U)$, solve

$$B[z, v] = \langle f, v \rangle - B[w, v] \quad (v \in H_0^1(U)), \qquad z = u - w \in H_0^1(U).$$

Thus the homogeneous theory applies whenever a lift is available. The resulting u is independent of the chosen lift whenever the homogeneous problem is unique.

Adjoint and sign conventions. Define $B^*[v, u] = B[u, v]$. For smooth coefficients, the formal adjoint is

$$L^* v = -\partial_i(a^{ij}\partial_j v) - \partial_i(b^i v) + cv. \tag{6.5}$$

The adjoint drift differentiates the product $b^i v$. In particular, even if $c \ge 0$, the expanded zeroth-order coefficient of L^* can be negative. This matters in the nonsymmetric spectral theory.

零边界的准确含义

$H_0^1(U)$ 是紧支撑光滑函数在 H^1 中的闭包。光滑边界上它等价于零迹；一般弱解没有逐点边界值，更不能先假设法向导数存在。

非齐次边界先做提升

若边界数据是某个 $w \in H^1(U)$ 的迹，就写 $u = w + z$，再解 $z \in H_0^1(U)$。边界数据仅属于 $L^2(\partial U)$，并不自动保证存在这样的提升。

6.2 Existence of Weak Solutions

6.2.1 Lax–Milgram (Evans §6.2.1, Theorem 1)

Theorem 6.1

Let H be a real Hilbert space. Suppose a bilinear form B satisfies

$$|B[u,v]| \le M\|u\|_H\|v\|_H, \qquad B[u,u] \ge \beta\|u\|_H^2, \quad \beta > 0.$$

For each bounded linear functional ℓ on H there is a unique $u \in H$ with $B[u,v] = \ell(v)$ for all $v \in H$. Moreover, $\|u\|_H \le \beta^{-1}\|\ell\|_{H^*}$.

Expanded proof.

By Riesz representation there is a bounded linear operator $T : H \to H$ such that $B[u,v] = (Tu, v)_H$. Coercivity and Cauchy–Schwarz give

$$\beta\|u\|_H^2 \le (Tu,u)_H \le \|Tu\|_H\|u\|_H, \qquad \|Tu\|_H \ge \beta\|u\|_H.$$

Thus T is injective. Its range is closed: if Tu_j converges, the last inequality applied to $u_j - u_k$ makes (u_j) Cauchy, and its limit maps to the given range limit.

The range is also dense. If w is orthogonal to it, then $0 = (Tw, w)_H = B[w,w] \ge \beta\|w\|_H^2$, so $w = 0$. A closed dense subspace is all of H. Represent $\ell(v) = (g,v)_H$ and solve $Tu = g$. Uniqueness follows from injectivity; testing with $v = u$ gives the asserted bound. □

强制性排除失控方向

有界性保证双线性型连续；强制性则使 $B[u,u]$ 控制整个空间范数。这里不要求 B 对称，所以不能一律把证明改写成能量最小化。

6.2.2 Energy estimates and a shifted inverse (Evans §6.2.2)

Theorem 6.2 (Boundedness and Gårding inequality)

Under the hypotheses of §6.1, there are constants M, C_0 such that

$$|B[u,v]| \le M\|u\|_{H^1}\|v\|_{H^1}, \qquad B[u,u] \ge \tfrac{\theta}{2}\|Du\|_2^2 - C_0\|u\|_2^2.$$

Consequently, for sufficiently large $\kappa > 0$, $B_\kappa[u,v] = B[u,v] + \kappa(u,v)_{L^2}$ is coercive on $H_0^1(U)$. For every $f \in H^{-1}(U)$, $(L+\kappa)u = f$ has a unique weak solution, with $\|u\|_{H^1} \le C\|f\|_{H^{-1}}$.

Expanded proof.

Hölder bounds each integral defining B. For the diagonal,

$$\left|\int_U b\cdot Du\, u\right| \le \|b\|_\infty\|Du\|_2\|u\|_2 \le \tfrac{\theta}{2}\|Du\|_2^2 + \tfrac{\|b\|_\infty^2}{2\theta}\|u\|_2^2.$$

The potential contributes at worst $-\|c_-\|_\infty\|u\|_2^2$. Choose $C_0 = \|b\|_\infty^2/(2\theta) + \|c_-\|_\infty$ and $\kappa \ge C_0 + 1$. Then

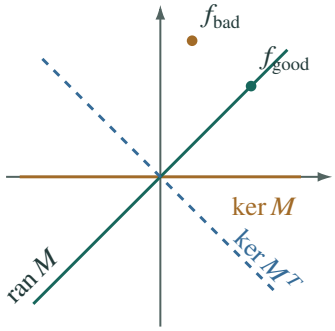
$$B_\kappa[u,u] \ge \min(\theta/2, 1)\|u\|_{H^1}^2.$$

Apply Theorem 6.1. All constants are independent of f. □

If $b = 0$ and $c \geq 0$, Poincaré gives coercivity without a shift. For general drift, $c \geq 0$ alone does not make this particular energy estimate coercive. Uniqueness may nevertheless follow from a maximum principle once the needed regularity is established.

6.2.3 The compact Fredholm mechanism (Evans §6.2.3)

Write $R = (L + \kappa)^{-1}$. As a map $L^2(U) \to H_0^1(U)$ it is bounded. The inclusion $H_0^1(U) \Subset L^2(U)$ is compact, so R is a compact operator on $L^2(U)$. For arbitrary bounded open U, this last compactness also follows by extending zero to a fixed enclosing ball and applying Chapter 5 there.



$$M = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad Mu = (u_2, u_2)$$

$$Mu = f \text{ is solvable iff } f_1 = f_2$$

$$\operatorname{ran} M = (\ker M^T)^{\perp}$$

$$\ker M \neq \ker M^T$$

Diagram T6.2. Compatibility uses the adjoint kernel. In this explicit nonsymmetric matrix example, the range is the diagonal line. A right side with $f_1 \neq f_2$ is unsolvable. Its obstruction is its component along $\ker M^T$, while the horizontal $\ker M$ describes nonuniqueness of solutions. The elliptic Fredholm alternative has the same distinction.

Lemma 6.3 (Compact Fredholm alternative)
For a compact operator T on a Hilbert space, $I - T$ has closed range, finite-dimensional kernel and cokernel of equal dimension, and

$$\operatorname{ran}(I - T) = \ker(I - T^*)^{\perp}.$$

In particular, injectivity is equivalent to surjectivity.

Expanded proof.

Compactness makes T a norm limit of finite-rank operators: choose a finite ε-net for the image of the unit ball, project onto the span of its centers, and set $F = PT$. Then $\|T - F\| \leq \varepsilon$. Choose $\varepsilon < 1$, put $E = T - F$, and invert $Q = I - E$ by its norm-convergent Neumann series. We have $I - T = Q(I - G)$, where $G = Q^{-1}F$ has finite rank.

平移算子而非平移解

加上 κu 是把算子换成 $L + \kappa I$，从而恢复强制性。原问题仍需另行研究；这个可逆的平移算子将成为 Fredholm 方法的起点。

紧性来自空间之间的往返

求解方程先把 L^2 数据送入更强的 H_0^1 空间，再忘掉导数回到 L^2。这第二步是紧嵌入；有界性本身不足以推出逆算子紧。

无限维问题的有限维障碍

紧性不能保证可逆，但能把不可逆的障碍压缩到有限维核中。右端必须与伴随核正交；非对称情形不能把伴随核随意换成原算子的核。

Let $M = \operatorname{ran} G$. Relative to $H = M \oplus M^{\perp}$,

$$I - G = \begin{pmatrix} D & -G|_{M^{\perp}} \\ 0 & I \end{pmatrix}, \qquad D = I_M - G|_M .$$

Its kernel is $\ker D \oplus \{0\}$. Its range consists of pairs (x, y) with $x + Gy \in \operatorname{ran} D$. This condition defines a closed subspace of codimension $\dim(M/\operatorname{ran} D) = \dim \ker D$. Multiplication by the isomorphism Q preserves closedness and both dimensions. Finally, for every bounded operator S, $(\operatorname{ran} S)^{\perp} = \ker S^{*}$; closedness now gives the formula. □

Theorem 6.4 (Solvability and resonance)
For each real λ, the weak Dirichlet equation $(L-\lambda)u = f, f \in L^2(U)$, has the following alternative. Either the homogeneous equation has only the zero solution and every f has a unique solution, or its kernel is nonzero and finite dimensional. In the latter case solvability is exactly

$$(f, v)_{L^2} = 0 \quad \text{for every weak solution } (L^{*} - \lambda)v = 0.$$

Whenever a solution exists, all solutions differ by a homogeneous solution.

Expanded proof.

For $\lambda \neq -\kappa$ the equation is equivalent to

$$[I - (\lambda + \kappa)R]u = Rf \quad \text{in } L^2(U). \tag{6.6}$$

Conversely, any L^2 solution of this identity satisfies $u = R((\lambda + \kappa)u + f)$, so lies in $H_0^1(U)$ and solves the original weak equation.

The adjoint inverse is $R^{*} = (L^{*} + \kappa)^{-1}$: if $u = Rf$ and $v = (L^{*} + \kappa)^{-1} g$, then $(u, g) = B_{\kappa}[u, v] = (f, v)$. Thus the kernel of $I - (\lambda + \kappa)R^{*}$ is precisely $\ker(L^{*} - \lambda)$. For v in this kernel,

$$(Rf, v) = (f, R^{*}v) = \frac{(f, v)}{\lambda + \kappa}.$$

Lemma 6.3 gives the required compatibility condition and the finite dimensional alternative. At $\lambda = -\kappa$ the shifted problem is already uniquely solvable. Subtraction describes all solutions. □

相容条件来自伴随恒等式

若伴随齐次解为 v，把原方程与它配对就得到 $(f, v) = 0$。Fredholm 定理补上反方向：所有这些正交条件满足时，解确实存在。

6.2.4 The exceptional set and bounds for the inverse

Theorem 6.5 (Evans §6.2.3, Theorems 5–6)
Let Σ be the set of real Dirichlet eigenvalues of L. It is bounded below and has only finitely many elements in any bounded interval. If infinite, its elements tend to $+\infty$. If $\lambda \notin \Sigma$, the solution of $(L-\lambda)u = f$ satisfies

$$\|u\|_{H^1(U)} \le C_\lambda \|f\|_2.$$

Expanded proof.

For an eigenfunction, Gårding gives $\lambda\|u\|_2^2 = B[u,u] \ge -C_0\|u\|_2^2$, hence $\lambda \ge -C_0$. Nonzero eigenvalues μ of the compact operator R have finite multiplicity and cannot accumulate away from zero. Here is the compactness argument for the latter assertion. If infinitely many distinct μ_j satisfy $|\mu_j| \ge \delta > 0$, let E_j be the span of eigenvectors for $\mu_1, \dots, \mu_j$, and choose $x_j \in E_j \cap E_{j-1}^{\perp}$ with $\|x_j\| = 1$. Then $(R-\mu_j I)x_j \in E_{j-1}$, whereas $Rx_i \in E_{j-1}$ for $i < j$. Consequently $\|Rx_j - Rx_i\| \ge |\mu_j| \ge \delta$, contradicting compactness. An infinite-dimensional eigenspace is excluded by the same argument with an orthonormal sequence. Since $\mu = (\lambda+\kappa)^{-1}$, the assertions about real Σ follow.

If an L^2 inverse bound failed at a fixed $\lambda \notin \Sigma$, there would be solutions u_j with $\|u_j\|_2 = 1$ and right sides $f_j \to 0$ in L^2. Gårding, applied to $B[u_j,u_j] = \lambda\|u_j\|_2^2 + (f_j,u_j)$, bounds (u_j) in H_0^1. A subsequence converges weakly in H_0^1 and strongly in L^2 to u of norm one. Passing to each fixed test function yields $(L-\lambda)u = 0$, a contradiction. The same energy inequality then upgrades the L^2 bound to the displayed H^1 estimate. □

For a nonsymmetric operator, complex eigenvalues can also occur after complexification, and the real exceptional set need not be infinite. An orthogonal eigenbasis or an exact inverse norm equal to a reciprocal spectral distance requires additional self-adjoint structure.

逼近共振时常数会变大

若 $Lw = \lambda_* w$ 且 $\|w\|_2 = 1$，则用右端 $(\lambda_* - \lambda)w$ 得到解 w。因此逆算子范数至少为 $|\lambda - \lambda_*|^{-1}$，不能给出跨越共振点的一致估计。

6.3 Regularity

6.3.1 Interior energy localization

Throughout the H^2 argument, $A \in C^1(U)$ is uniformly elliptic, $b, c \in L^\infty_{\mathrm{loc}}(U)$, $f \in L^2_{\mathrm{loc}}(U)$, and $u \in H^1(U)$ solves $Lu = f$ against compactly supported tests. All coefficient bounds below are on a fixed relatively compact region.

Lemma 6.6 (Caccioppoli estimate)
For a smooth cutoff $0 \le \eta \le 1$ with compact support in U,

$$\int_U \eta^2 |Du|^2 \le C \int_{\operatorname{supp}\eta} (|f|^2 + (1 + |D\eta|^2)|u|^2).$$

Expanded proof.

The admissible test $\eta^2 u$ gives

$$\int \eta^2 ADu \cdot Du = \int f \eta^2 u - 2 \int \eta u\, ADu \cdot D\eta - \int b \cdot Du\, \eta^2 u - \int c\eta^2 u^2.$$

The second and third terms are bounded, for any $\varepsilon > 0$, by

$$\varepsilon \int \eta^2 |Du|^2 + C_\varepsilon \int (|D\eta|^2 + \eta^2) u^2.$$

Use $|fu| \le (f^2 + u^2)/2$, bound the potential term, and choose the two ε's so their sum is less than $\theta/2$. Move these gradient terms to the left. □

6.3.2 Interior H^2 regularity (Evans §6.3.1, Theorem 1)

Theorem 6.7
For $V \Subset U$, a weak solution as above belongs to $H^2(V)$ and

$$\|u\|_{H^2(V)} \le C\big(\|f\|_{L^2(W)} + \|u\|_{L^2(W)}\big), \qquad V \Subset W \Subset U.$$

Here C depends on the nested regions, ellipticity and the local C^1 norm of A and L^∞ norms of b, c. Local Lipschitz bounds on A suffice for this proof.

Expanded proof.

1. Choose a legal discrete substitute for a second derivative. Fix a coordinate direction e_k and set $\delta_h z(x) = [z(x + he_k) - z(x)]/h$ for either sign of h. Choose $\eta = 1$ on V, compactly supported in an intermediate region, and $|h|$ small enough that all translated supports remain in W. For fixed h,

$$q = \eta^2 \delta_h u \in H_0^1(W), \qquad v = -\delta_{-h} q \in H_0^1(W).$$

The difference quotient is in H^1 for fixed h even though its H^1 norm has not yet been bounded uniformly. Thus v is a valid test.

截断函数的代价

局部估计中出现 $D\eta$，所以常数依赖内外区域的距离。把内区推向边界会使常数变大，不能把内部估计直接当成全局边界估计。

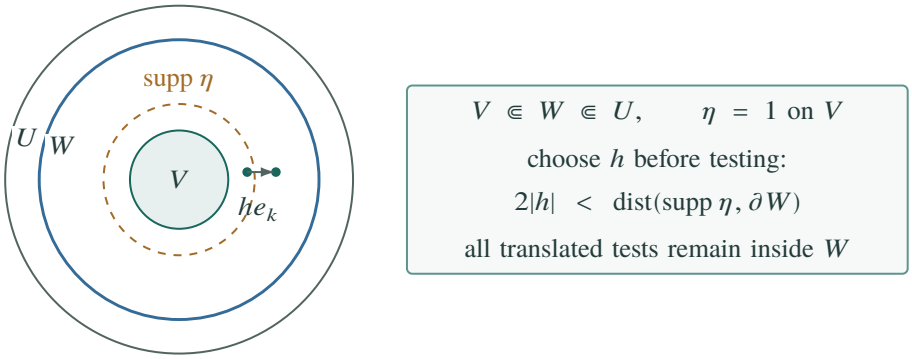


Diagram T6.3. Localize before taking differences. The cutoff equals one on V and is supported strictly inside W. A fixed geometric gap permits all sufficiently small positive and negative steps. The support condition is settled before the estimate; the estimate's constant must then stay independent of h as $h \to 0$.

2. Discrete integration by parts. A change of variables, with no support crossing the boundary, gives

$$\int g\, \delta_{-h} q = -\int (\delta_h g) q.$$

Absorb lower-order terms into $\tilde{f} = f - b \cdot Du - cu \in L^2(W)$. The weak equation becomes

$$\int \delta_h(a^{ij} u_i)\, \partial_j(\eta^2 \delta_h u) = -\int \tilde{f}\, \delta_{-h}(\eta^2 \delta_h u).$$

The exact product rule is

$$\delta_h(a^{ij} u_i) = a^{ij}(x + he_k)\delta_h u_i + (\delta_h a^{ij}) u_i.$$

The coefficient in the first term is translated and remains elliptic.

3. Estimate each error before absorbing it. Put $X = \|\eta\, \delta_h Du\|_2$ and $Y = \|Du\|_{L^2(W)} + \|u\|_{L^2(W)}$. The main term is at least θX^2. Since $\|\delta_h u\|_2 \le \|\partial_k u\|_{L^2(W)}$ on the relevant supports and $|\delta_h A| \le \|DA\|_\infty$, the other terms on the left are bounded by

$$CXY + CY^2 \le \varepsilon X^2 + C_\varepsilon Y^2.$$

For example the cutoff term is bounded by $C\|\eta\, \delta_h Du\|_2 \|D\eta\, \delta_h u\|_2$; the coefficient commutator is bounded by $C\|Du\|_2(X + \|\delta_h u\|_2)$.

For the right side, the difference quotient inequality applied to q gives

$$\|\delta_{-h} q\|_2 \le \|\partial_k q\|_2 \le C(X + Y).$$

Therefore it is at most $\varepsilon X^2 + C_\varepsilon(\|\tilde{f}\|_2^2 + Y^2)$. Since $\|\tilde{f}\|_2 \le C(\|f\|_2 + Y)$, absorption yields

$$\|\delta_h Du\|_{L^2(V)}^2 \le C\big(\|f\|_{L^2(W)}^2 + \|u\|_{H^1(W)}^2\big), \tag{6.7}$$

with C independent of h.

不能先假定二阶导数

直接测试 $-\partial_{kk} u$ 会预先要求尚未证明的正则性。固定步长的差商仍在 H^1 中，先合法计算，再用一致估计令步长趋零。

一致估计才能识别弱导数

差商判别准则作用于每个一阶导数 u_i：各方向的 $\delta_h u_i$ 一致有界，就得到 $u_{ik} \in L^2$。只证明每个固定步长有限并没有用。

4. Pass to weak derivatives and remove the first-order norm. The difference quotient criterion of Chapter 5 applied to each u_i gives $u_{ik} \in L^2(V)$. Repeating in every direction proves H^2. Apply (6.7) on slightly larger nested regions and use Lemma 6.6 there to bound $\|Du\|_2$ by f and u. This gives the stated estimate with only the L^2 norm of u. □

6.3.3 Higher interior regularity (Evans §6.3.1, Theorems 2–3)

Theorem 6.8

Let $m \geq 0$ be an integer, $A, b, c \in C^{m+1}(U)$, and $f \in H^m_{\mathrm{loc}}(U)$. Then $u \in H^{m+2}_{\mathrm{loc}}(U)$, with

$$\|u\|_{H^{m+2}(V)} \leq C\big(\|f\|_{H^m(W)} + \|u\|_{L^2(W)}\big), \qquad V \Subset W \Subset U.$$

Smooth coefficients and smooth f give $u \in C^\infty(U)$.

Expanded proof.

The case $m = 0$ is proved. Suppose $u \in H^{m+2}$ on intermediate regions and $f \in H^{m+1}$. For $|\alpha| = m+1$, differentiate the equation in distributions. With $z = D^\alpha u \in H^1$,

$$\begin{aligned} Lz = D^\alpha f &+ \sum_{0<\beta\leq\alpha} \binom{\alpha}{\beta} \partial_j\big((D^\beta a^{ij})\,\partial_i D^{\alpha-\beta} u\big) \\ &- \sum_{0<\beta\leq\alpha} \binom{\alpha}{\beta} \big((D^\beta b^i)\,\partial_i D^{\alpha-\beta} u + (D^\beta c) D^{\alpha-\beta} u\big). \end{aligned}$$

Expand the divergence in the first line. Its highest derivative of u has order $|\alpha| - |\beta| + 2 \leq m+2$, already known in L^2; coefficient derivatives have order at most $m+2$. Thus the entire right side lies in L^2. Interior H^2 regularity for z gives derivatives of u through order $m+3$. A finite chain of nested regions gives the norm bound inductively. For any fixed k, choose a Sobolev order $s > k + n/2$ and use the embedding $H^s_{\mathrm{loc}} \hookrightarrow C^k_{\mathrm{loc}}$. Since k is arbitrary, smoothness follows. □

6.3.4 Boundary H^2 regularity (Evans §6.3.2, Theorem 4)

Theorem 6.9
Suppose ∂U is C^2, $A \in C^1(\overline{U})$, $b, c \in L^\infty(U)$ and $f \in L^2(U)$. A weak Dirichlet solution $u \in H_0^1(U)$ belongs to $H^2(U)$ and satisfies

$$\|u\|_{H^2(U)} \le C(\|f\|_2 + \|u\|_2).$$

If the Dirichlet operator is invertible, the last term can be removed.

Expanded proof.

1. A flat boundary permits tangential differences. Work first in a half-ball $B_r^+ = \{x \in B_r : x_n > 0\}$, where the trace of u is zero on $x_n = 0$. Choose a cutoff vanishing near the curved boundary. For $k < n$, tangential translation preserves the half-space and the zero trace, so the test $-\delta_{-h}(\eta^2 \delta_h u)$ remains admissible. The preceding calculation bounds every u_{ik} with $k < n$ in L^2 on a smaller half-ball. This includes the mixed normal–tangential derivatives, since distributional derivatives commute.

2. Ellipticity recovers the missing derivative. Expand the divergence equation in distributions:

$$a^{nn} u_{nn} = - \sum_{(i,j) \neq (n,n)} a^{ij} u_{ij} + (b^i - \partial_j a^{ij}) u_i + cu - f. \tag{6.8}$$

Every term on the right is in L^2 by Step 1. Because $a^{nn} \ge \theta$, division bounds u_{nn}. One rigorous way to avoid assuming H^2 up to the boundary at this step is to use the already proved interior H^2 theorem: the identity holds almost everywhere away from $x_n = 0$, and its right side has a uniform L^2 bound on the whole smaller half-ball. The resulting L^2 function is the distributional derivative u_{nn} on this open half-ball. No boundary distribution is introduced because test functions are supported inside it.

3. Flatten a curved boundary without losing coefficients. After a rigid motion, write the boundary as $x_n = \gamma(x')$ and set

$$y = \Phi(x) = (x', x_n - \gamma(x')), \qquad v(y) = u(\Phi^{-1}(y)).$$

This particular map has determinant one. Writing $J = D\Phi$, the weak change of variables gives

$$\widehat{A}(y) = JAJ^T, \quad \hat{b}(y) = Jb, \quad \hat{c}(y) = c, \quad \hat{f}(y) = f,$$

提升正则性不是循环论证

先由已知阶数判断交换子右端属于 L^2，再把二阶正则性定理用于导函数。光滑系数得到光滑解，并不自动得到解析解。

正则性估计不等于可解性

估计以解已经存在为前提。右边保留 $\|u\|_2$ 是为容纳有限维齐次核；只有先证明算子可逆，才能用右端独自控制整个解。

法向差商为何失效

沿边界平移不会越出半空间，也保留零迹；法向平移会穿过边界，测试函数不再合法。因此先求所有带切向指标的导数，最后用方程解出纯法向导数。

where all right sides are evaluated at $\Phi^{-1}(y)$. Indeed $D_x u = J^T D_y v$ and $dx = dy$; the same gradient formula holds for each test function. Further,

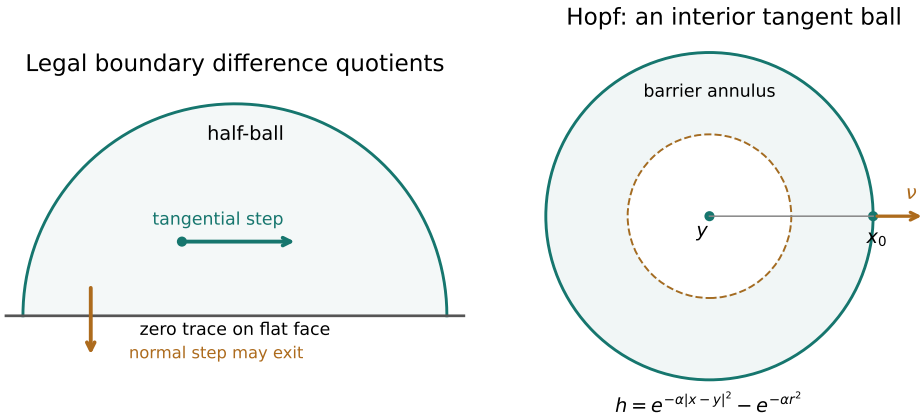
$$\xi^T \widehat{A}\xi \geq \theta |J^T \xi|^2 \geq \theta \|J^{-T}\|^{-2} |\xi|^2.$$

Bounded C^2 coordinate maps preserve the relevant H^2 estimates by the chain rule, including its lower-order terms. Since $\gamma \in C^2$ and $A \in C^1$, $\widehat{A} \in C^1$. The flat argument therefore applies.

4. Patch the local estimates. A finite boundary cover, together with interior patches, yields $\|u\|_{H^2} \leq C(\|f\|_2 + \|u\|_{H^1})$. For a solution, Gårding and $B[u,u] = (f,u)$ imply $\|Du\|_2^2 \leq C(\|f\|_2^2 + \|u\|_2^2)$. This removes the first-order term. At a nonresonant parameter, Theorem 6.5 removes $\|u\|_2$ as well. □

拉平边界也要变换算子

坐标变换后主系数是 JAJ^T。此处选用行列式为一的图形变换；一般坐标变换还会出现体积雅可比因子，不能把它漏掉。



Two boundary mechanisms. Tangential translations preserve the flat zero-trace face; normal translations need not. Hopf's exponential barrier is used only in the shaded annulus of an interior tangent ball. Diagrams are schematic.

6.3.5 Higher boundary regularity (Evans §6.3.2, Theorems 5–6)

Theorem 6.10

If ∂U is C^{m+2}, $A, b, c \in C^{m+1}(\overline{U})$ and $f \in H^m(U)$, then a weak zero-Dirichlet solution satisfies

$$u \in H^{m+2}(U), \qquad \|u\|_{H^{m+2}(U)} \leq C(\|f\|_{H^m(U)} + \|u\|_2).$$

Smooth boundary, coefficients and data give $u \in C^\infty(\overline{U})$.

Expanded proof.

The case $m = 0$ is Theorem 6.9. For the induction step assume $u \in H^{m+2}$ and $f \in H^{m+1}$, with coefficients C^{m+2} and boundary C^{m+3}.

In a flattened chart, differentiate $m+1$ times only tangentially. The resulting derivative has zero trace on the flat face, because tangential differentiation commutes with the trace at these orders. Its localized equation has an L^2 right side: expand the commutators exactly as in the interior proof. Cutoffs keep tests away from the curved part of the chart. Applying boundary H^2 gives all derivatives of total order $N = m+3$ having at most two normal differentiations.

Recover the others from (6.8). To estimate an order-N derivative with $q \geq 3$ normal indices, apply an order-$N-2$ derivative with $q-2$ normal indices to that identity. The desired term is multiplied by a^{nn}; the other order-N terms have at most $q-1$ normal indices, and coefficient commutators contain derivatives of u of order at most $N-1$, already controlled. Induct on q and divide by $a^{nn} \geq \theta$. These differentiated identities can be justified first by interior regularity and then by the uniform estimates up to the flat face, as in the H^2 case. Patching proves the global estimate. Finally extend Sobolev functions across the smooth boundary and use embedding at arbitrarily high orders to obtain $C^\infty(\overline{U})$. □

What the estimates actually say. The two-derivative gain is an a priori estimate for solutions, not an existence theorem by itself. At an eigenvalue, the term $\|u\|_2$ cannot be removed: a nonzero eigenfunction solves a homogeneous shifted equation. Nonzero boundary data require a sufficiently regular lift before the zero-boundary theorem can be used.

边界光滑度不能省略

内部正则性只看系数和右端，边界正则性还看区域形状。角点可能产生奇性；零边界条件本身并不能保证解在角点附近有平方可积的二阶导数。

6.4 Maximum Principles

Here we use nondivergence notation

$$Lu = -a^{ij}u_{ij} + b^i u_i + cu.$$

Assume U bounded, A uniformly elliptic, and coefficients bounded and continuous unless a stronger hypothesis is stated. Solutions are classical, $u \in C^2(U) \cap C(\overline{U})$. These are different assumptions from the measurable-coefficient weak theory.

6.4.1 Weak maximum principles (Evans §6.4.1, Theorems 1–2)

Theorem 6.11

If $c = 0$ and $Lu \leq 0$, then $\max_{\overline{U}} u = \max_{\partial U} u$. If $c \geq 0$ and $Lu \leq 0$, then

$$\max_{\overline{U}} u \leq \max_{\partial U} u^+.$$

In particular, $Lu \le Lv$ and $u \le v$ on ∂U imply $u \le v$ in U when $c \ge 0$.

Expanded proof.

First let $c = 0$. A strict subsolution $Lu < 0$ cannot have an interior maximum: at such a point $Du = 0$, D^2u is negative semidefinite, and $-A : D^2u \ge 0$, a contradiction. For a non-strict subsolution use $u_\varepsilon = u + \varepsilon e^{\tau x_1}$. Since

$$L(e^{\tau x_1}) = (-a^{11}\tau^2 + b^1\tau)e^{\tau x_1} < 0$$

for a fixed sufficiently large τ, u_ε has its maximum on the boundary. Let $\varepsilon \downarrow 0$.

For $c \ge 0$, on the open set $D = \{u > 0\}$ the operator $K = L - c$ satisfies $Ku = Lu - cu \le 0$. Apply the zero-potential result on D; boundary values on $\partial D \cap U$ are zero, while the remaining values come from ∂U. The proof uses compact maxima and does not require a smooth level-set boundary. If D is empty the conclusion is immediate. Apply this assertion to $u - v$ for comparison. □

Applying the theorem to $-u$ gives the corresponding minimum principles. The restriction on c is substantive: on $(0, \pi)$, $u = \sin x$ solves $-u'' - u = 0$ with zero boundary values but is positive inside.

For comparison, a direct weak energy argument is available in divergence form when $b \in C^1(\overline{U})$ and $c - \frac{1}{2}\operatorname{div} b \ge 0$. For a weak subsolution with $u^+ \in H_0^1(U)$, testing with u^+ gives

$$0 \ge B[u, u^+] = \int ADu^+ \cdot Du^+ + \int (c - \tfrac{1}{2}\operatorname{div} b)(u^+)^2 \ge \theta \|Du^+\|_2^2.$$

Thus $u \le 0$. The drift integral was integrated by parts after writing $u^+Du^+ = D((u^+)^2)/2$. This sufficient energy condition is distinct from $c \ge 0$ in the classical nondivergence argument.

6.4.2 Hopf's boundary lemma (Evans §6.4.2)

Lemma 6.12

Suppose an interior ball $B_r(y) \subset U$ touches ∂U at x_0, $u(x) < u(x_0) = M$ for all $x \in U$, and $Lu \le 0$. Assume either $c = 0$, or $c \ge 0$ and $M \ge 0$. If u is C^1 at x_0 in the inward-ball neighborhood, then

$$\partial_\nu u(x_0) > 0, \qquad \nu = (x_0 - y)/r.$$

At a smooth tangency this is the outward normal to U.

先固定算子符号

本章主部是负号 $-a^{ij}u_{ij}$。因此 $Lu \le 0$ 控制上界；在内部极大点，$Du = 0$、$D^2u \le 0$，主部反而非负。换符号时不等式也要一起换。

弱最大值原理中的“弱”

这里“弱”表示结论只控制边界极值，并非指 Sobolev 弱解。对弱解也有相应原理，但必须说明测试函数和系数条件，不能混用两个含义。

Expanded proof.

On $r/2 < |x-y| < r$, put $h(x) = e^{-\alpha|x-y|^2} - e^{-\alpha r^2}$. It is positive inside and zero on the outer sphere. A direct calculation, with $s = |x-y|$, gives

$$Lh \le e^{-\alpha s^2}\left(-4\theta\alpha^2 s^2 + 2\alpha \operatorname{tr} A + 2\alpha|b|s + c\right) \le 0$$

for sufficiently large α; for $c = 0$ the same bound applies. On the inner sphere $u - M$ has a strictly negative maximum, so choose $\varepsilon > 0$ with $u - M + \varepsilon h \le 0$ there. It is also nonpositive on the outer sphere. Moreover $L(u - M + \varepsilon h) \le -cM \le 0$. Comparison on the annulus gives $u - M \le -\varepsilon h$. Along the radius $x_0 - t\nu$ divide by $t > 0$ and let $t \downarrow 0$:

$$\partial_\nu u(x_0) \ge -\varepsilon \partial_\nu h(x_0) = 2\varepsilon\alpha r e^{-\alpha r^2} > 0.$$

□

6.4.3 Strong maximum principle (Evans §6.4.2, Theorems 3–4)

Theorem 6.13

Let U be connected. If $Lu \le 0$ and $c = 0$, attainment of a maximum at an interior point forces u to be constant. For $c \ge 0$ the same holds for a nonnegative interior maximum. Dually, a nonpositive interior minimum of a supersolution forces constancy.

Expanded proof.

Let $M = \max_U u$, attained inside, and suppose both sets $C = \{u = M\}$ and $V = \{u < M\}$ are nonempty. A path inside U from V to C shows that their common relative boundary contains a point in U. Choose $y \in V$ sufficiently close to this point so that the ball with radius $\operatorname{dist}(y, C)$ is contained in U. Its interior is in V and its boundary touches C at some $x_0 \in U$. Hopf's lemma on this ball gives a strictly positive radial derivative at x_0. But x_0 is an interior maximum of a differentiable function, so $Du(x_0) = 0$. This contradiction proves the assertion. The minimum statement follows by replacing u by $-u$. □

In particular, for $c \ge 0$, a nonzero nonnegative supersolution $Lu \ge 0$ is strictly positive in a connected domain.

Hopf 引理提供严格斜率

最大值原理只说内部不能超过边界。Hopf 引理进一步说：若内部严格更小，函数沿外法向到达边界极大值时必须有正斜率；内球条件保证可放入显式屏障。

6.4.4 Harnack's inequality (Evans §6.4.3, Theorem 5)

Theorem 6.14 (Smooth principal-part case proved here)
Suppose A is smooth and uniformly elliptic and $-a^{ij}u_{ij} = 0$, $u \geq 0$ in U. For connected $V \Subset U$,

$$\sup_V u \leq C \inf_V u.$$

The constant depends on the geometry of V inside U, ellipticity and local coefficient bounds, including the derivatives used below, but not on u.

Expanded proof.

1. Take a logarithm to estimate relative oscillation. First take $u > 0$, and put $v = \log u$ and $w = a^{ij}v_i v_j$. Then

$$-a^{ij}v_{ij} = w, \qquad \theta|Dv|^2 \leq w \leq \Lambda|Dv|^2. \tag{6.9}$$

Interior smoothness justifies the following differentiations: rewrite the smooth nondivergence equation in divergence form and apply §6.3. On a fixed compact region,

$$w_{kl} = 2a^{ij}v_{ikl}v_j + 2a^{ij}v_{ik}v_{jl} + R_{kl}, \quad |R_{kl}| \leq \varepsilon|D^2v|^2 + C_\varepsilon|Dv|^2.$$

The remainder consists of $D^2A\,Dv\,Dv$ and $DA\,D^2v\,Dv$. Differentiate (6.9) once to obtain $-a^{kl}v_{ikl} = w_i + (\partial_i a^{kl})v_{kl}$. Also $a^{ij}a^{kl}v_{ik}v_{jl} \geq \theta^2|D^2v|^2$; this is seen by diagonalizing the positive matrix A at the point. Absorbing coefficient remainders gives, for $\mathscr{M} = -a^{kl}\partial_{kl} - 2a^{ij}v_j\partial_i$,

$$\mathscr{M}w \leq -\gamma|D^2v|^2 + C|Dv|^2 \leq -\gamma_1 w^2 + Cw.$$

The second inequality uses $w = -A : D^2v \leq C|D^2v|$.

2. Localize and retain the drift–cutoff error. Take $\zeta \in C_c^\infty(U)$ with $0 \leq \zeta \leq 1$ and $\zeta = 1$ on a chosen ball V. Let $z = \zeta^4 w$. If its maximum is positive, it occurs where $\zeta > 0$, and $\zeta Dw = -4wD\zeta$. At this point $Dz = 0$ and $\mathscr{M}z = -A : D^2z \geq 0$. Expanding the product yields

$$0 \leq -\gamma_1\zeta^4 w^2 + C\zeta^2 w + C\zeta^3 w^{3/2}.$$

The last term comes from $-2ADv \cdot D(\zeta^4)\,w$ and uses $|Dv| \leq C\sqrt{w}$. It must be included because this drift depends on v. Set $Y = \zeta^2 w$ at this maximum point. Then

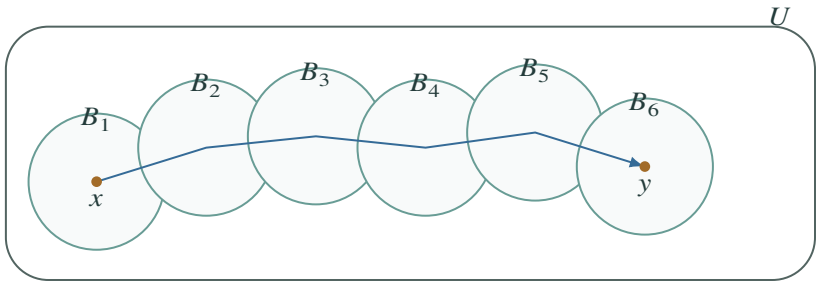
$$\gamma_1 Y^2 \leq C(Y + Y^{3/2}).$$

For $Y > 0$, divide by Y and solve the resulting quadratic inequality in $\sqrt{Y}$ to get $Y \leq C$. Hence $z \leq C$ at its maximum, and $w \leq C$, $|Dv| \leq C$ on V.

Harnack 是定量正性

强最大值原理说非负非零解处处为正；Harnack 还控制内区任意两点的比值。它需要方程等式，不能随意用于只有单边不等式的函数。

3. Integrate the logarithmic gradient. For x, y in the ball, integrate along their segment: $|v(x)-v(y)| \le C|x-y| \le C \operatorname{diam} V$. Exponentiation bounds $u(x)/u(y)$. For a general connected $V \Subset U$, cover it by finitely many balls whose slightly larger concentric balls remain inside U. A finite chain of overlapping balls connects their values; multiplying the finitely many comparison constants proves the result. Finally apply the estimate to $u+\delta$, which solves the same equation because there are no lower-order terms, and let $\delta \downarrow 0$. □



$$\sup_{B_i} u \le C_i \inf_{B_i} u \quad \Longrightarrow \quad u(x) \le (C_1 \cdots C_6)\, u(y)$$

Diagram T6.4. Local positivity comparisons propagate along a chain. Choose overlapping balls whose slightly larger balls still lie inside U. Comparing at overlap points transfers a local Harnack bound from x to y. The resulting constant depends on the finite chain and its distance from the boundary, which is why connected compact interior regions occur in the theorem.

Scope of the source theorem. Evans states Harnack for the full operator with lower-order terms and indicates extensions to bounded measurable coefficients. The proof supplied in the source, and reconstructed above, is restricted to smooth A and $b = c = 0$. The broader theorem is a further regularity result, cited there to Gilbarg–Trudinger; it is not proved by the logarithmic computation above. In the general case $u+\delta$ need not solve $Lu = 0$, and differentiating measurable coefficients is unavailable.

对数变换后的漂移项

新算子的漂移含有 $D\log u$，不能预先当成已有一致界的系数。截断计算保留三次项，再用四次项吸收，才能保证最终常数不依赖解。

6.5 Eigenvalues and Eigenfunctions

Assume in this section that U is bounded, connected and has smooth boundary, and that all coefficients are smooth on $\overline{U}$. This keeps the spectral arguments separate from technical boundary regularity issues. Eigenfunctions have zero Dirichlet trace.

6.5.1 Symmetric operators (Evans §6.5.1, Theorem 1)

First let $Lu = -\partial_j(a^{ij}u_i)$ with A symmetric uniformly elliptic. Then $B[u,v] = \int ADu \cdot Dv$ is a symmetric positive form.

Theorem 6.15

There are real eigenvalues, repeated according to multiplicity,

$$0 < \lambda_1 \le \lambda_2 \le \cdots, \qquad \lambda_j \longrightarrow \infty,$$

and a complete orthonormal basis (w_j) of $L^2(U)$ such that $Lw_j = \lambda_j w_j$. Each eigenspace is finite dimensional, and $w_j \in C^\infty(\overline{U})$.

Expanded proof.

Let $S = L^{-1} : L^2 \to L^2$. Lax–Milgram, Poincaré and Rellich show that S is bounded and compact. If $u = Sf$ and $v = Sg$, then

$$(Sf, g) = B[u,v] = (f, Sg), \qquad (Sf, f) = B[u,u] > 0 \quad (f \neq 0).$$

Thus S is self-adjoint, positive definite, and injective.

For completeness we give the compact spectral argument. Let $\mu_1 = \sup_{\|f\|_2=1}(Sf,f) > 0$. For a maximizing sequence f_k, pass to a weakly convergent subsequence $f_k \rightharpoonup f$ with $\|f\|_2 \le 1$. Compactness gives $Sf_k \to Sf$, so $(Sf,f) = \mu_1$. If $\|f\|_2 < 1$, scaling f to norm one increases this value, a contradiction. First variation on the unit sphere gives $Sf = \mu_1 f$.

Its orthogonal complement is invariant under S, by self-adjointness. Repeat the argument there, recording multiplicities, to obtain $\mu_1 \ge \mu_2 \ge \cdots > 0$ and orthonormal eigenvectors w_j. There are infinitely many, since an injective operator on infinite dimensional $L^2(U)$ cannot have finite-dimensional range. Compactness forces $\mu_j \to 0$: otherwise (Sw_j) has pairwise distances bounded below. If the closed span of the w_j had a nonzero orthogonal complement, the restriction of S to that complement would still be compact and positive definite and hence would have a positive maximal eigenvalue. That value would be at most every μ_j, contradicting $\mu_j \to 0$. This proves completeness. Set $\lambda_j = \mu_j^{-1}$. Finite multiplicity follows from compactness, and regularity bootstrapping applied to $Lw_j = \lambda_j w_j$ gives smoothness. □

6.5.2 Rayleigh quotient and positivity (Evans §6.5.1, Theorem 2)

Theorem 6.16

The first eigenvalue satisfies

$$\lambda_1 = \min_{\substack{u \in H_0^1(U) \\ \|u\|_2 = 1}} B[u,u].$$

A minimizer solves $Lu = \lambda_1 u$. There is a strictly positive first eigenfunction in U, and λ_1 is simple. Every eigenfunction for a larger eigenvalue changes sign.

Expanded proof.

The energy inner product is B, equivalent to the H_0^1 norm. The functions $w_j/\sqrt{\lambda_j}$ are orthonormal for this inner product, since $B[w_i, w_j] = \lambda_i (w_i, w_j)$. They are complete in the energy space: if $B[u, w_j] = 0$ for all j, then $(u, w_j) = 0$ for all j, hence $u = 0$ by the L^2 basis. Hilbert-space Parseval therefore gives

$$B[u,u] = \sum_j \lambda_j |(u, w_j)|^2, \qquad \|u\|_2^2 = \sum_j |(u, w_j)|^2. \tag{6.10}$$

The Rayleigh minimum follows, with equality at w_1. Alternatively, the direct method gives a minimizer: bounded energy gives weak H_0^1 convergence, Rellich preserves the L^2 normalization, and weak lower semicontinuity preserves minimality.

Differentiating the quotient $B[u + tv, u + tv]/\|u + tv\|_2^2$ at $t = 0$ yields $B[u, v] = \lambda_1 (u, v)$ for every v. If u is a minimizer, $|u| \in H_0^1$ has the same norm and energy by the Sobolev chain rule, so it is also a minimizer and hence smooth. It satisfies $L|u| = \lambda_1 |u| \geq 0$. The strong minimum principle, applied after rewriting L in nondivergence form with zero potential, shows $|u| > 0$ in U. Continuity and connectedness force u to have one fixed sign.

If two first eigenfunctions were independent, take a linear combination that vanishes at one interior point but is not identically zero. It is again a first eigenfunction, contradicting the preceding strict sign property. Thus the eigenspace is one dimensional. An eigenfunction for $\lambda > \lambda_1$ is orthogonal to the positive w_1, and consequently must take both signs. □

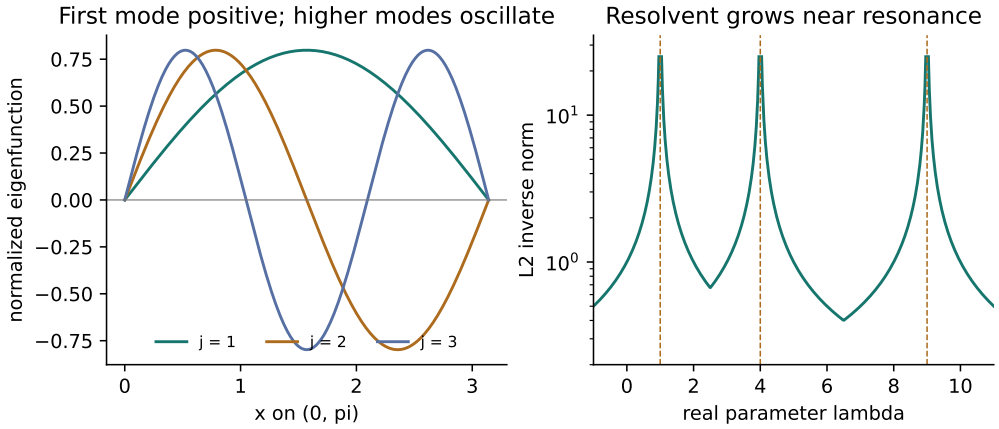
先研究紧的逆算子

L 在 L^2 上是无界算子，不宜直接套紧算子的谱定理。其逆先提升到 H_0^1，再由紧嵌入回到 L^2；逆的特征值趋零，对应 L 的特征值趋无穷。

能量空间中的归一化不同

w_j 在 L^2 中长度为一，但能量长度是 $\sqrt{\lambda_j}$。因此能量空间的正交归一基是 $w_j/\sqrt{\lambda_j}$，这决定了能量展开中的权重。

For $-\partial_{xx}$ on $(0,\ell)$, $w_j = \sqrt{2/\ell}\sin(j\pi x/\ell)$ and $\lambda_j = (j\pi/\ell)^2$. Their oscillations illustrate positivity of the first mode and sign changes of the higher modes.



Explicit interval model. On $(0,\pi)$ the Dirichlet eigenvalues are j^2. For real λ off the spectrum, the self-adjoint inverse norm is $\|(-\partial_{xx} - \lambda)^{-1}\|_{L^2\to L^2} = 1/\min_j |j^2 - \lambda|$. The poles mark loss of uniqueness.

Weyl's law: a source remark, not a proved theorem here. For the Dirichlet Laplacian on a smooth bounded domain,

$$N(\lambda) := \#\{j : \lambda_j \le \lambda\} \sim \frac{\omega_n |U|}{(2\pi)^n}\lambda^{n/2}.$$

Equivalently, $\lambda_j \sim 4\pi^2 (j/(\omega_n|U|))^{2/n}$. This asymptotic statement appears as a remark in Evans; its proof requires additional spectral estimates and is not part of the compactness argument above. Here ω_n is the volume of the unit ball.

6.5.3 A positive eigenfunction without symmetry (Evans §6.5.2)

Now write $L = -a^{ij}\partial_{ij} + b^i\partial_i + c$ and assume $c \ge 0$. The matrix A is symmetric, but the full operator need not be self-adjoint. For weak formulations and adjoints, first rewrite it in divergence form: the drift in B is $b^i + \partial_j a^{ij}$, rather than b^i itself.

> **Theorem 6.17**
> *There is a real eigenvalue $\lambda_1 > 0$ and an eigenfunction $\phi > 0$ in U, $\phi = 0$ on ∂U. Its eigenspace is one dimensional. Every complex eigenvalue λ satisfies* $\operatorname{Re}\lambda \ge \lambda_1$.

绝对值仍要通过变分方程

不能直接对 $|u|$ 做经典二阶求导。先用 Sobolev 链式法则证明能量不变，因而它仍是极小元；再由 Euler 方程与正则性得到光滑性和严格正性。

Expanded proof.

1. Construct a positive compact inverse. The homogeneous zero-Dirichlet problem has only the zero solution: regularity makes a weak homogeneous solution classical, and comparison applied to both signs gives zero. Fredholm therefore supplies $T = L^{-1}$. Fix an integer $m > n/2 + 2$ and let $X = H^m(U) \cap H_0^1(U)$ with its H^m norm. Boundary regularity makes $T : X \to H^{m+2}(U) \cap H_0^1(U)$ bounded, so $T : X \to X$ is compact. The closed convex cone $\mathscr{C} = \{u \in X : u \geq 0\}$ is preserved by T.

Fix $0 \neq w \in C_c^\infty(U)$, $w \geq 0$. The strong minimum principle and Hopf's lemma imply $Tw > 0$ inside and $\partial_\nu(Tw) < 0$ on the boundary. Since w has compact support, some $\mu > 0$ satisfies $\mu Tw \geq w$.

2. A compact fixed-point argument yields an eigenvector. We invoke the following *external* form of Schaefer's theorem: if a continuous compact map $F : \mathscr{C} \to \mathscr{C}$ has a bounded set $\{u : u = tF(u),\ 0 \leq t \leq 1\}$, then F has a fixed point. This topological theorem is a prerequisite here, not proved in this chapter.

For $\varepsilon > 0$, consider

$$u = \eta T(u + \varepsilon w), \qquad u \in \mathscr{C}.$$

No solution can have $\eta > \mu$. Indeed positivity gives $u \geq \varepsilon(\eta/\mu)w$; iterating $u \geq \eta Tu$ gives $u \geq \varepsilon(\eta/\mu)^k w$ for every k, impossible at a point where $w > 0$. If the solutions for $0 \leq \eta \leq 2\mu$ were bounded, Schaefer applied to $F(u) = 2\mu T(u + \varepsilon w)$ would give a solution at $\eta = 2\mu$, a contradiction. Hence that solution set is unbounded. Choose $\varepsilon_j \downarrow 0$ and solutions u_j, η_j with $\|u_j\|_X \geq \varepsilon_j^{-1}$ and $0 \leq \eta_j \leq 2\mu$. Normalize $v_j = u_j/\|u_j\|_X$. Then

$$v_j = \eta_j T\left(v_j + \frac{\varepsilon_j w}{\|u_j\|_X}\right).$$

Compactness gives a subsequence converging in X to v with $\|v\|_X = 1$, $v = \eta Tv$, and $\eta > 0$ (otherwise the right side would converge to zero). Thus $Lv = \eta v$, $v \geq 0$. Regularity, the strong minimum principle and Hopf give a smooth $\phi = v > 0$, $\partial_\nu \phi < 0$, with $\lambda_1 = \eta > 0$.

非对称时换一种结构

非对称算子通常没有正交特征基，Rayleigh 商也不再直接给出全部谱。主特征值的存在依赖紧逆算子保持非负性，极小实部则由正特征函数的商变换证明。

3. Exclude eigenvalues to the left of λ_1. For a complex eigenfunction z, smoothness and zero boundary values imply $|z(x)| \le Cd(x)$ near the boundary, with $d(x) = \operatorname{dist}(x, \partial U)$. Hopf and compactness of ∂U give $\phi(x) \ge ad(x)$. For $0 < \delta < 1$, put $h = \phi^{1-\delta}$ and $v = z/h$. Then v extends continuously as zero on ∂U.

A product calculation gives

$$L(hv) = hKv + vLh, \qquad K = -A : D^2 + (b - 2AD \log h) \cdot D,$$

and the chain rule gives

$$\frac{Lh}{h} = (1-\delta)\lambda_1 + \delta c + \delta(1-\delta)\frac{AD\phi \cdot D\phi}{\phi^2} \ge (1-\delta)\lambda_1.$$

Consequently, for $q = |v|^2$,

$$Kq = 2\operatorname{Re}(\bar{v}Kv) - 2ADv \cdot D\bar{v} \le 2\left(\operatorname{Re}\lambda - \frac{Lh}{h}\right)q.$$

If $\operatorname{Re}\lambda < \lambda_1$, choose δ small so that the coefficient on the right is strictly negative. The nonzero continuous function q, zero on ∂U, has a positive interior maximum, where $Dq = 0$ and $-A : D^2 q \ge 0$, a contradiction. This argument is pointwise at that maximum; it does not assume the drift $D \log h$ is bounded near the boundary.

4. Prove geometric simplicity by sliding positive functions. Let z be a real eigenfunction for λ_1, changing its sign if necessary so it is positive somewhere. The estimates $\phi \ge ad$ and $|z| \le Cd$, also valid after adjusting constants on the interior compact region, show that

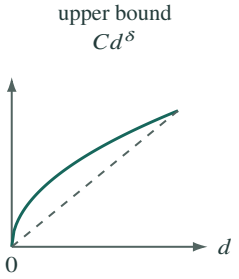
$$t_* = \sup\{t \ge 0 : \phi - tz \ge 0 \text{ in } U\}$$

is positive and finite. At t_* the residual $r = \phi - t_* z$ is nonnegative and satisfies $Lr = \lambda_1 r \ge 0$. If it were nonzero, the strong minimum principle and Hopf would give $r \ge a'd$. Since $|z| \le Cd$, a small further increase of t_* would preserve nonnegativity, a contradiction. Hence $r = 0$. Apply this separately to real and imaginary parts for complex eigenfunctions at the real value λ_1. □

归一化保住非零极限

紧性本身可能只给出零极限。把无界解列除以自身范数后，范数固定为一；扰动项同时趋零，极限才成为真正的非零特征函数。

图 T6.M：商为何在边界趋零



由 $|z| \le Cd$ 与 $\phi \ge ad$，得到 $|z|/\phi^{1-\delta} \le C'd^\delta \to 0$。图示上界取 $\delta = 1/2$。

6.5.4 The positive adjoint eigenfunction

Lemma 6.18
The formal Dirichlet adjoint has a positive eigenfunction ϕ^* for the same λ_1.

Expanded proof.

The inverse T^* is positive on L^2: for $f, g \geq 0$, $(T^*f, g) = (f, Tg) \geq 0$. Regularity makes its restriction to X compact. If $f \in \mathscr{C}$ is nonzero, $v = T^*f \geq 0$ is nonzero. Although the expanded potential c^* of L^* may be negative, choose M with $c^* + M \geq 0$; then $(L^* + M)v = f + Mv \geq 0$. The strong minimum principle and Hopf applied to $L^* + M$ give $v > 0$ and $\partial_\nu v < 0$. The compact-cone construction in Steps 1–2, with T^* in place of T, therefore gives $\phi^* > 0$ and $L^*\phi^* = \lambda^*\phi^*$, $\lambda^* > 0$. Both eigenfunctions have zero trace, so weak adjointness yields

$$\lambda_1(\phi, \phi^*) = B[\phi, \phi^*] = (\phi, L^*\phi^*) = \lambda^*(\phi, \phi^*).$$

The inner product is positive, hence $\lambda^* = \lambda_1$. □

If c has arbitrary sign but is bounded, apply the theorem to $L + \kappa$ with $c + \kappa \geq 0$. Translating its eigenvalues back preserves the positive principal eigenfunction, simplicity and the least-real-part property; the resulting λ_1 need not be positive. The word *simple* above describes the eigenspace. In fact there is also no generalized eigenvector at λ_1: an equation $(L - \lambda_1)z = \phi$ contradicts pairing with ϕ^*, since $(\phi, \phi^*) > 0$.

伴随的零阶项可能变号

原算子的 $c \geq 0$ 不推出伴随算子的零阶项非负。先由逆算子的对偶关系得到非负性，再加常数使用强最大值原理，才能严谨得到伴随正特征函数。

6.6 Problems and Worked Solutions

These are concise restatements of the 17 source problems, with editorial solutions. Unless a problem says otherwise, U is bounded with smooth boundary, coefficients are smooth and bounded, and ellipticity is uniform. Connectedness is specified where constants or positivity are involved. The same assumptions apply to each weak formulation below.

1. Removing a potential by a positive solution. Relate $-\Delta u + cu = 0$ to $-\operatorname{div}(aDv) = 0$.

Solution. If $w > 0$ solves the first equation, set $u = wv$. The product rule gives

$$-\Delta(wv) + cwv = -w\Delta v - 2Dw \cdot Dv + v(-\Delta w + cw) = -w^{-1} \operatorname{div}(w^2 Dv).$$

Thus u solves the first equation exactly when $v = u/w$ solves the second with $a = w^2$. Conversely, given a smooth positive a, set $w = \sqrt{a}$ and $c = \Delta w/w$. Then $-\Delta w + cw = 0$, so the same identity proves that $u = \sqrt{a}\, v$ solves the potential equation. These are local identities wherever $w > 0$; uniform bounds on a are additional requirements for a uniform elliptic theory. □

正解作为新的权重

除以一个正解后，零阶项被吸收到权重 w^2 中。关键是展开 $-\Delta(wv)$ 时保留两倍交叉项；反向构造的势是 $\Delta\sqrt{a}/\sqrt{a}$。

2. A small negative potential. For $L = -\partial_j(a^{ij}\partial_i) + c$, find $\mu > 0$ so that $c \geq -\mu$ ensures Lax–Milgram.

Solution. Let $\|u\|_2 \leq C_P\|Du\|_2$ on $H_0^1(U)$ and choose $\mu = \theta/(2C_P^2)$. Then

$$B[u,u] \geq \theta\|Du\|_2^2 - \mu\|u\|_2^2 \geq \tfrac{\theta}{2}\|Du\|_2^2 \geq C\|u\|_{H^1}^2.$$

Bounded coefficients give boundedness of B. Apply Theorem 6.1. The permitted negative part depends on both ellipticity and the domain's Poincaré constant. □

负势可以由梯度能量吸收

零阶系数略微为负并不立即破坏可解性。Poincaré 不等式把函数平方项转回梯度项；负势大小的阈值因此与区域几何有关。

3. The clamped biharmonic problem. For $f \in L^2(U)$, solve $\Delta^2 u = f$, with $u = \partial_\nu u = 0$, weakly in $H_0^2(U)$.

Solution. Use $B_2[u,v] = \int \Delta u\, \Delta v$ and $\ell(v) = \int fv$. For $u \in C_c^\infty(U)$, integration by parts twice gives

$$\int (\Delta u)^2 = \sum_{i,j} \int u_{ii}u_{jj} = \sum_{i,j} \int u_{ij}^2.$$

Approximation extends this to $H_0^2(U)$. Since u and each u_i belong to $H_0^1(U)$, Poincaré applied twice gives $\|u\|_{H^2} \leq C\|D^2u\|_2$. Thus B_2 is bounded and coercive on H_0^2 and ℓ is bounded. Lax–Milgram gives exactly one weak solution. On smooth boundaries the closure definition of H_0^2 encodes both zero traces. □

四阶方程用二阶能量空间

双调和方程分部积分两次后只需二阶弱导数。H_0^2 同时编码函数及一阶迹为零，不能用只有函数零迹的 $H^2 \cap H_0^1$ 随意替代。

4. Neumann compatibility. For connected U, prove that $\int Du \cdot Dv = \int fv$ for every $v \in H^1(U)$ is solvable exactly when $\int_U f = 0$.

Solution. Testing with $v = 1$ proves necessity. On $V = \{u \in H^1(U) : \int_U u = 0\}$, the mean-zero Poincaré inequality holds. To see this, a counterexample can be normalized to have $\|u_j\|_2 = 1$, mean zero and $\|Du_j\|_2 \to 0$. Rellich gives a strong L^2 limit, whose weak gradient is zero. It is constant by connectedness and zero by its mean, contradicting norm one. Lax–Milgram on V gives a unique mean-zero solution. For arbitrary v, subtract its mean; the removed constant contributes zero on both sides because $\int f = 0$. Thus the full weak equation holds. The difference of any two solutions has zero gradient, so all solutions differ by a constant. □

5. Robin boundary conditions. Formulate and solve $-\Delta u = f$, $u + \partial_\nu u = 0$.

Solution. Use $H^1(U)$ and its trace T, with

$$B_R[u,v] = \int_U Du \cdot Dv + \int_{\partial U} Tu\, Tv, \qquad \ell(v) = \int_U fv.$$

The plus boundary term follows from $\partial_\nu u = -u$ in Green's formula. Trace boundedness makes B_R bounded. Also

$$\|u\|_{H^1}^2 \le C\left(\|Du\|_2^2 + \|Tu\|_{L^2(\partial U)}^2\right).$$

Otherwise normalize a violating sequence to norm one. Its gradients and traces tend to zero; Rellich and weak H^1 convergence yield a constant on each component, with zero trace, hence zero. Strong L^2 convergence and vanishing gradients contradict norm one. This proves coercivity; Lax–Milgram yields a unique solution for every f. □

6. Mixed Dirichlet–Neumann data. Let connected U have $\partial U = \Gamma_1 \cup \Gamma_2$ with disjoint closed pieces. Impose $u = 0$ on Γ_1 and $\partial_\nu u = 0$ on Γ_2.

Solution. Set $V = \{u \in H^1(U) : Tu = 0 \text{ on } \Gamma_1\}$ and require $\int Du \cdot Dv = \int fv$ for every $v \in V$. The Dirichlet condition is built into V; the Neumann condition appears naturally after integration by parts. If Γ_1 has positive surface measure, Poincaré holds on V: a normalized counterexample converges to a constant by the argument in Problem 4, and its zero trace on Γ_1 forces that constant to vanish. Lax–Milgram gives existence and uniqueness for every f. Under the stated smoothness and disjoint closed partition, a nonempty Γ_1 is relatively open in ∂U and has positive measure. If Γ_1 is empty, Problem 4 gives the Neumann compatibility and constant ambiguity. If Γ_2 is empty, this is the usual Dirichlet problem. For more general partitions that meet along an interface, the weak formulation still makes sense, but global H^2 regularity is a separate issue. □

7. A monotone semilinear estimate. Let $u \in H^1(\mathbb{R}^n)$ have compact support and solve $-\Delta u + c(u) = f$ weakly, where c is smooth, $c(0) = 0$, $c' \ge 0$, and $c(u), f \in L^2(\mathbb{R}^n)$. Prove $\|D^2 u\|_2 \le C\|f\|_2$.

Solution. For fixed h in direction e_k, use $v = -\delta_{-h}\delta_h u \in H^1(\mathbb{R}^n)$, an admissible compactly supported weak test by density. Discrete integration by parts gives

$$\|\delta_h Du\|_2^2 + \int_{\mathbb{R}^n} \delta_h(c(u))\, \delta_h u = -\int_{\mathbb{R}^n} f\, \delta_{-h}\delta_h u.$$

边界项也能控制常数

Neumann 能量看不见常数，所以需要相容条件并允许加常数。Robin 能量多了边界平方项，常数方向也受到控制，于是无需零均值条件且解唯一。

自然边界与本质边界

Dirichlet 条件先限制解和测试函数所在空间，称为本质条件；Neumann 条件由分部积分后的边界项体现，称为自然条件。更换边界类型也就更换了变分空间。

The second term is nonnegative because $(c(s) - c(t))(s - t) \geq 0$. Moreover, $\|\delta_{-h}\delta_h u\|_2 \leq \|\partial_k \delta_h u\|_2 \leq \|\delta_h Du\|_2$. Hence $\|\delta_h Du\|_2 \leq \|f\|_2$ uniformly in h. The difference quotient criterion in each direction yields $u \in H^2$ and, for the Frobenius norm, $\|D^2 u\|_2 \leq \sqrt{n}\,\|f\|_2$. No upper bound on c' is required in this monotonicity argument. □

单调性替代导数上界

非线性项的差商乘积非负，因此可以留在左边后直接舍去。这里使用的是 c 的单调性，不需要把 $c'(u)$ 当作有界系数来估计。

8. Bernstein's gradient estimate. For $Lu = -a^{ij}u_{ij} = 0$, show that $v = |Du|^2 + Mu^2$ is a subsolution for large M and deduce a boundary estimate.

Solution. Differentiate the equation to get $Lu_k = (\partial_k a^{ij})u_{ij}$. Thus

$$L|Du|^2 = -2a^{ij}u_{ki}u_{kj} + 2u_k(\partial_k a^{ij})u_{ij} \leq -\theta|D^2u|^2 + C|Du|^2.$$

Also $L(u^2) = -2a^{ij}u_i u_j \leq -2\theta|Du|^2$. Choose $2M\theta \geq C$. Then $Lv \leq 0$. The weak maximum principle gives $\sup_U |Du|^2 \leq \sup_{\partial U}(|Du|^2 + Mu^2)$, hence

$$\|Du\|_{L^\infty(U)} \leq C\big(\|Du\|_{L^\infty(\partial U)} + \|u\|_{L^\infty(\partial U)}\big).$$

□

9. A boundary barrier estimate. For $Lu = -a^{ij}u_{ij} = f$, $u = 0$ on ∂U, suppose $Lw \geq 1$, $w \geq 0$ on ∂U, and $w(x_0) = 0$.

Solution. The minimum principle gives $w \geq 0$ in U. Let $F = \|f\|_\infty$. Since $L(\pm u - Fw) \leq 0$ and $\pm u - Fw \leq 0$ on the boundary, comparison gives $|u| \leq Fw$. Assuming the indicated boundary derivatives exist, evaluate at $x_0 - t\nu$ and divide by t to obtain $|\partial_\nu u(x_0)| \leq F|\partial_\nu w(x_0)|$. All tangential derivatives of the smooth zero-boundary function u vanish at x_0, so $|Du(x_0)| = |\partial_\nu u(x_0)|$. This is the desired estimate, with constant F. □

屏障把函数值变成法向斜率

先比较得到整个区域的 $|u| \leq \|f\|_\infty w$，再沿内法线取差商。边界上 u 恒为零，所以梯度只剩法向分量。

同一唯一性可有两种证明

能量法用积分把整个梯度消掉；最大值原理通过极值位置与边界斜率制造矛盾。前者更适合弱解，后者直接揭示正性与边界几何的作用。

10. Harmonic Neumann solutions are constant. Give energy and maximum-principle proofs on connected U.

Solution. Green's identity gives $\int_U |Du|^2 = -\int_U u\Delta u + \int_{\partial U} u\partial_\nu u = 0$, so $Du = 0$ and connectedness gives constancy. Alternatively, if u were nonconstant, its maximum would be on the boundary by the strong maximum principle and all interior values would be strictly smaller. Hopf's lemma at a boundary maximum would imply $\partial_\nu u > 0$, contradicting the homogeneous Neumann condition. The latter proof uses a smooth boundary, or at least the needed interior ball. □

11. Convex functions of weak solutions. If bounded $u \in H^1(U)$ satisfies $\int ADu \cdot Dv = 0$ for $v \in H_0^1(U)$, prove that $w = \varphi(u)$ is a weak subsolution for smooth convex φ.

Solution. The bounded range of u makes φ', φ'' bounded on that range, so $w \in H^1(U)$ and $Dw = \varphi'(u)Du$. First take bounded $0 \le v \in H_0^1(U)$. The product $\varphi'(u)v \in H_0^1(U)$ is an admissible test and gives

$$\int ADw \cdot Dv = -\int \varphi''(u)v\, ADu \cdot Du \le 0.$$

For arbitrary $0 \le v \in H_0^1(U)$, use $v_M = \min(v, M)$ and pass in the left side by $v_M \to v$ in H^1. This proves the required inequality for every nonnegative weak test. The function w need not have zero trace. □

测试函数的乘积要先检查

若 v 无界，$\varphi''(u)vDu$ 未必属于 L^2，不能直接宣布 $\varphi'(u)v$ 合法。先截断 v，证明不等式后再用 H^1 收敛去掉截断。

12. A positive supersolution implies comparison. Assume $h \in C^2(U) \cap C(\overline{U})$, $h > 0$ on *all of* $\overline{U}$, and $Lh \ge 0$. Prove the zero-boundary weak maximum principle without assuming $c \ge 0$.

Solution. For $Lu \le 0$ and $u \le 0$ on ∂U, set $z = u/h$. The product formula gives

$$Lu = hMz + zLh, \qquad M = -A : D^2 + (b - 2ADh/h) \cdot D.$$

On $\{z > 0\}$, $Mz \le 0$ and M has no potential. For each $\varepsilon > 0$, the closure of $\{z > \varepsilon\}$ is compact in U, because $z \le 0$ on the boundary and $\min_{\overline{U}} h > 0$. The drift is bounded there, so the zero-potential maximum principle gives $z \le \varepsilon$ on this set, which must therefore be empty. Let $\varepsilon \downarrow 0$. Positivity only in U would not suffice: $h = \sin x$ for $L = -\partial_{xx} - 1$ on $(0, \pi)$ is an interior positive solution but vanishes at the endpoints, and comparison fails. □

13. Courant's max–min principle. For the symmetric operator of §6.5, show

$$\lambda_k = \max_{\dim S = k-1} \min_{\substack{u \in H_0^1(U) \cap S^\perp \\ \|u\|_2 = 1}} B[u,u],$$

where $S \subset H_0^1(U)$ and orthogonality is in $L^2(U)$.

Solution. For any such S, linear algebra supplies a nonzero vector in $\operatorname{span}(w_1, \dots, w_k) \cap S^\perp$. Normalize it. Formula (6.10) gives its energy at most λ_k, so the inner minimum is at most λ_k. For $S = \operatorname{span}(w_1, \dots, w_{k-1})$, every admissible u has only modes $j \geq k$, and the same formula gives $B[u,u] \geq \lambda_k$. Equality occurs at w_k. For any fixed S, the minimum exists by the direct method; its orthogonality constraints are preserved by strong L^2 convergence. □

极小极大中的正交空间

这里去掉的是前若干模态在 L^2 中的方向。任意少于 k 个线性约束都不可能消灭前 k 个模态的全部组合，这就是一半不等式的有限维理由。

14. A principal-eigenvalue max–min formula. For the nonsymmetric operator, prove

$$\lambda_1 = \sup_{\substack{u \in C^\infty(\overline{U}),\ u>0 \text{ in } U \\ u=0 \text{ on } \partial U}} \inf_{x \in U} \frac{Lu(x)}{u(x)}.$$

Solution. Let $\phi^* > 0$ be the adjoint eigenfunction from Lemma 6.18. Both functions have zero trace, so Green's identity, or the weak adjoint identity, gives

$$\int_U \phi^* L u = \int_U u L^* \phi^* = \lambda_1 \int_U u \phi^*.$$

Thus the weighted average of Lu/u with positive weight $u\phi^*$ is λ_1. Its infimum cannot exceed this average. If the infimum is $-\infty$ the inequality is immediate; otherwise integrating its pointwise lower bound proves it directly. Taking $u = \phi$ achieves equality everywhere. A constant shift handles potentials of arbitrary bounded sign. □

非对称公式中的权重来自伴随

不能只对 Lu/u 做普通空间平均。与正的伴随特征函数配对后，正确权重是 $u\phi^*$；分部积分才把这个加权平均固定为主特征值。

15. Hadamard's domain-variation formula. Let $-\Delta w = \lambda w$ on a smoothly moving domain $U(\tau)$, $w = 0$ on its boundary and $\int_{U(\tau)} w^2 = 1$. Assume a smooth eigenvalue/eigenfunction branch, and denote boundary velocity by $\mathbf{v}$.

Solution. Use the Eulerian derivative $\dot{w}$ at a fixed spatial point. Differentiating $w(x(\tau), \tau) = 0$ on the moving boundary gives

$$\dot{w} = -Dw \cdot \mathbf{v} = -\partial_\nu w \, (\mathbf{v} \cdot \nu).$$

Reynolds' formula for normalization yields $2\int_U w\dot w + \int_{\partial U} w^2 \mathbf{v}\cdot\nu = 0$, hence $\int_U w\dot w = 0$. Since $\lambda = \int_U |Dw|^2$, differentiate once more:

$$\begin{aligned}\dot\lambda &= 2\int_U Dw\cdot D\dot w + \int_{\partial U} |Dw|^2 \mathbf{v}\cdot\nu \\ &= 2\lambda\int_U w\dot w + 2\int_{\partial U} \partial_\nu w\,\dot w + \int_{\partial U} (\partial_\nu w)^2 \mathbf{v}\cdot\nu \\ &= -\int_{\partial U} (\partial_\nu w)^2 \mathbf{v}\cdot\nu.\end{aligned}$$

Tangential derivatives vanish because w is zero on the boundary. The smooth-branch hypothesis is essential; a multiple eigenvalue need not admit a single arbitrarily chosen differentiable eigenfunction branch. □

扩张区域使特征值下降

若边界向外移动，$\mathbf{v}\cdot\nu > 0$，公式给出 $\dot\lambda < 0$。负号来自移动边界条件 $\dot w = -\partial_\nu w\,\mathbf{v}\cdot\nu$，不是凭直觉添加。

16. Helmholtz waves and radiation. Let real $\sigma \neq 0$ and $\lambda = \sigma^2$.

Solution. (a) For $|\omega| = 1$, $-\Delta e^{i\sigma\omega\cdot x} = \sigma^2 e^{i\sigma\omega\cdot x}$. Thus $e^{i\sigma(\omega\cdot x - t)}$ solves the wave equation.

(b) In dimension three put $\Phi(x) = e^{i\sigma r}/(4\pi r)$, $r = |x|$. The radial identity $\Delta g = g'' + 2g'/r$ shows $(-\Delta - \sigma^2)\Phi = 0$ for $r > 0$. At $r = \varepsilon$,

$$-\int_{\partial B_\varepsilon} \partial_r \Phi\, dS = e^{i\sigma\varepsilon}(1 - i\sigma\varepsilon) \longrightarrow 1.$$

Integration by parts against a smooth test function outside B_ε therefore gives $(-\Delta - \sigma^2)\Phi = \delta_0$; the other shrinking-sphere term and the volume integral of $\sigma^2\Phi$ tend to zero.

(c) For Φ, $r(\Phi_r - i\sigma\Phi) = -e^{i\sigma r}/(4\pi r) \to 0$ uniformly in angle. For the plane wave at $x = r\vartheta$,

$$r(w_r - i\sigma w) = i\sigma r(\omega\cdot\vartheta - 1)e^{i\sigma r\omega\cdot\vartheta},$$

whose modulus diverges on every ray with $\omega\cdot\vartheta \neq 1$. Thus the plane wave does not satisfy the Sommerfeld condition. □

辐射条件选择向外传播

时间因子取 $e^{-i\sigma t}$ 时，$e^{i\sigma r}/r$ 描述向外传播的球面波。条件中的减号与这一时间约定配套；平面波不会在所有方向都向外辐射。

17. Uniqueness of an entire radiating solution. Suppose w solves $-\Delta w = \sigma^2 w$ throughout $\mathbb{R}^3$ and satisfies $r(w_r - i\sigma w) \to 0$ *uniformly in angle*. Prove $w = 0$.

Solution. Green's identity and the real coefficient σ^2 imply

$$0 = \int_{\partial B_R} (\overline{w}\, w_r - w\,\overline{w_r})\, dS.$$

The imaginary flux is zero, so expansion of the square gives

$$\int_{\partial B_R} (|w_r|^2 + \sigma^2|w|^2)\, dS = \int_{\partial B_R} |w_r - i\sigma w|^2\, dS = o(1).$$

Uniform radiation justifies the last step, since the sphere has area $4\pi R^2$ and the integrand is $o(R^{-2})$ uniformly. For a fixed x_0 and $R > |x_0|$, Green's representation with the outgoing kernel $\Phi(x - x_0)$ gives

$$w(x_0) = \int_{\partial B_R} (\Phi\, w_r - w\, \partial_r \Phi)\, dS.$$

Here ∂_r on the kernel means the outward derivative on ∂B_R, not radial differentiation about x_0. For large R, both $\Phi(x-x_0)$ and this derivative are $O(R^{-1})$, so their $L^2(\partial B_R)$ norms remain bounded. Cauchy–Schwarz and the vanishing flux norm above imply $w(x_0) = 0$. Since x_0 was arbitrary, $w \equiv 0$. □

辐射条件还包含一致性

从逐方向极限跳到球面积分极限需要控制角变量。这里明确采用对方向一致的 Sommerfeld 条件；基本解在原点有奇性，所以并不违背“全空间光滑辐射解只能为零”。

6.7 References and Proof Dependencies

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 6, printed pp. 293–348 (local English PDF pp. 310–365). The Chinese scan corresponds to printed pp. 208–246 (PDF pp. 226–264); selected page comparisons are recorded in `docs/CHAPTER-06.md`. The English prose, proof expansions, solutions and Chinese annotations here are editorial reconstructions.

Evans's references point to Gilbarg–Trudinger for weak elliptic theory, regularity, maximum principles and Harnack; Krylov and Ladyzhenskaya–Uraltseva for regularity; Protter–Weinberger and Pucci–Serrin for maximum principles; and Smoller for spectral theory. For domain variations the source cites Garabedian; for radiation it cites Colton. These are source bibliography pointers, not claims that those books have been independently consulted for this draft.

Dependencies and boundaries. Riesz representation, Hilbert weak compactness and Hilbert Parseval are prerequisites. Chapter 5 supplies the Sobolev tools. Lax–Milgram, the compact Fredholm lemma, regularity, maximum principles, the smooth principal-part Harnack inequality, compact self-adjoint spectral decomposition and the principal-eigenvalue arguments are expanded above. The nonsymmetric existence proof explicitly invokes Schaefer's fixed-point theorem. Weyl's law is stated as an unproved source remark. The general measurable-coefficient Harnack theorem is described as an extension, not claimed as proved. All 17 source problems have editorial solutions. Compilation, model calculations and layout checks do not constitute independent mathematical review; this chapter remains an author-checked study draft.

Chapter 7

Linear Evolution Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 7; printed p. 349; PDF page 366.

Chinese reference: 中译本对应章：线性发展方程，书内第 248 页起。

Study edition / 英文重构学习稿。 The exposition follows Evans's section order. Source theorem identifiers are section-local; the displayed theorem numbers belong to these notes. All proofs, Chinese mathematical annotations, exercise solutions, and TikZ diagrams are editorial reconstructions or supplements, not verbatim transcriptions. The chapter has not received independent mathematical review.

We pass from elliptic boundary problems to trajectories in function spaces. Parabolic equations dissipate energy; hyperbolic equations transport it. Galerkin approximation treats time-dependent coefficients, Fourier analysis detects matrix structure, and semigroups organize autonomous evolution through elliptic resolvents. Constants may depend on the domain, final time, ellipticity, and stated coefficient norms, never on the approximation index.

7.1 Second-Order Parabolic Equations

7.1.1 The evolution space and the initial trace

Let $U \subset \mathbb{R}^n$ be bounded with smooth boundary, $T > 0$, and set $Q_T = U \times (0,T)$, $V = H_0^1(U)$, $H = L^2(U)$, $V' = H^{-1}(U)$. Identify $H \subset V'$ by $\langle h, v\rangle = (h,v)_H$. Consider

$$u_t + L(t)u = f, \qquad u|_{\partial U} = 0, \qquad u(0) = g, \tag{7.1}$$

where repeated spatial indices are summed and

$$L(t)u = -\partial_j(a^{ij}(x,t)\partial_i u) + b^i(x,t)\partial_i u + c(x,t)u.$$

Initially assume real bounded measurable coefficients, $a^{ij} = a^{ji}$, and $a^{ij}\xi_i\xi_j \geq \theta|\xi|^2$ almost everywhere, uniformly in time. The bilinear form

$B_t[u,v] = \int_U (a^{ij}u_i v_j + b^i u_i v + cuv)\,dx$ satisfies

$$\begin{aligned} |B_t[u,v]| &\le M\|u\|_V\|v\|_V, \\ B_t[v,v] &\ge \alpha\|v\|_V^2 - \gamma\|v\|_H^2 \end{aligned} \tag{7.2}$$

with $M, \alpha > 0$, $\gamma \ge 0$, uniform in t. Indeed $|\int b^i u_i u| \le \varepsilon\|Du\|_2^2 + C_\varepsilon\|u\|_2^2$; a negative potential is absorbed into γ.

时间导数的对偶空间

散度项把 H_0^1 函数送入 H^{-1}。弱解通常只有 $u_t \in L^2(0,T;H^{-1})$，不能先把 u_t 当成 $L^2(U)$ 函数。

Lemma 7.1 (Time trace and energy identity; cf. source §5.9.2, Theorem 3)
If $u \in L^2(0,T;V)$ and $u' \in L^2(0,T;V')$, then u has a unique representative in $C([0,T];H)$, and

$$\|u(t)\|_H^2 - \|u(s)\|_H^2 = 2\int_s^t \langle u', u\rangle\,dr. \tag{7.3}$$

The analogous product rule holds for two such functions.

Expanded proof (editorial reminder of time regularization).

For smooth V-valued curves this is differentiation of an inner product. Such curves are dense in $\|u\|_{L^2V} + \|u'\|_{L^2V'}$ by the time approximation theorem in §5.9: extend by reflection at the endpoints in V', then mollify in time. For a smooth curve z, choose s with $\|z(s)\|_H^2 \le T^{-1}\|z\|_{L^2H}^2$ and integrate its derivative:

$$\sup_t \|z(t)\|_H^2 \le T^{-1}\|z\|_{L^2H}^2 + 2\|z'\|_{L^2V'}\|z\|_{L^2V}.$$

Applied to differences, this gives uniform convergence in H. The products converge in L^1, so the identity passes to every pair of endpoints. Polarization gives the product rule. Equality almost everywhere makes the continuous representative unique. □

Definition 7.2 (Weak parabolic solution; source §7.1.1)
For $f \in L^2(0,T;V')$, $g \in H$, a weak solution belongs to $L^2(0,T;V)$, has $u' \in L^2(0,T;V')$, satisfies $u(0) = g$ in H, and obeys

$$\langle u'(t), v\rangle + B_t[u(t), v] = \langle f(t), v\rangle \quad (v \in V) \quad \text{for a.e. } t.$$

Allowing V'-valued forcing is an editorial extension of the source's $L^2(Q_T)$ forcing. Regularity statements below specify stronger data.

7.1.2 Galerkin construction, estimates, and uniqueness

Let w_k be the H-orthonormal Dirichlet eigenfunctions of $-\Delta$. They are orthogonal for $(u,v)_V = \int_U (Du\cdot Dv + uv)$. The H-projection P_m onto $V_m = \operatorname{span}(w_1,\dots,w_m)$ satisfies $\|P_m v\|_V \le \|v\|_V$ and $P_m v \to v$ in V. Seek $u_m = \sum_{k\le m} d_k(t) w_k$ with

$$\begin{aligned} d_k' + \sum_{\ell\le m} B_t[w_\ell, w_k] d_\ell &= \langle f, w_k\rangle, \\ d_k(0) &= (g, w_k). \end{aligned} \tag{7.4}$$

This linear system with measurable bounded coefficients has a unique absolutely continuous solution: solve the integral equation by contraction on short intervals where $\int \|E(t)\|dt < 1$, then concatenate them.

Theorem 7.3 (Weak well-posedness; source §7.1.2, Theorems 1–4)
There is exactly one weak solution, and

$$\sup_{0\le t\le T} \|u(t)\|_H^2 + \|u\|_{L^2V}^2 + \|u'\|_{L^2V'}^2 \le C\big(\|g\|_H^2 + \|f\|_{L^2V'}^2\big). \tag{7.5}$$

The same estimate holds for differences of solutions and their data.

Expanded proof (uniform bounds and recovery of the initial value).

Multiply the finite equations by d_k and sum. Using (7.2) and Young's inequality gives

$$\frac{1}{2}\frac{d}{dt}\|u_m\|_H^2 + \frac{\alpha}{2}\|u_m\|_V^2 \le \gamma\|u_m\|_H^2 + C\|f\|_{V'}^2.$$

Gronwall and integration bound u_m in $L^\infty H \cap L^2 V$. For arbitrary $v \in V$, since $u_m' \in V_m$,

$$\langle u_m', v\rangle = (u_m', P_m v) = \langle f, P_m v\rangle - B_t[u_m, P_m v].$$

Thus $\|u_m'\|_{V'} \le \|f\|_{V'} + M\|u_m\|_V$, uniformly in m. Stability of P_m in V is essential here.

Extract weak limits $u_m \rightharpoonup u$ in L^2V and $u_m' \rightharpoonup z$ in L^2V'. For $\phi \in C_c^\infty(0,T)$, $v\in V$, integration by parts identifies $\int\langle z,v\rangle\phi = -\int (u,v)\phi'$, hence $z = u'$. For each fixed k, the equation tested with $\phi(t)w_k$ passes to the limit. Bounded measurable coefficients act as bounded linear maps on these spaces; weak convergence suffices. Density extends the identity to V. Separability permits one common full-measure set of times.

To recover the initial value take $\phi \in C^1([0,T])$, $\phi(T) = 0$, and first $v \in V_N$, $m \ge N$. Then

$$\int_0^T \{-(u_m, v)\phi' + B_t[u_m, v]\phi\}\,dt = \int_0^T \langle f, v\rangle\phi\,dt + (P_m g, v)\phi(0).$$

初值不要求零迹

$g \in L^2(U)$ 无须有边界迹。零边界在正时间的 $L^2(0,T;H_0^1)$ 意义下成立；若要在角点 $\partial U \times \{0\}$ 光滑，才需要相容条件。

Pass to the limit and compare with the integration-by-parts identity for u, whose H trace exists by Lemma 7.1. Thus $(u(0), v) = (g, v)$ on a dense set, and $u(0) = g$. Lower semicontinuity gives the integral estimates. The uniform-in-time bound follows by testing on measurable time sets (Problem 6), followed by H continuity. Finally apply (7.3) to a difference of solutions and repeat the estimate. Zero data force zero difference. □

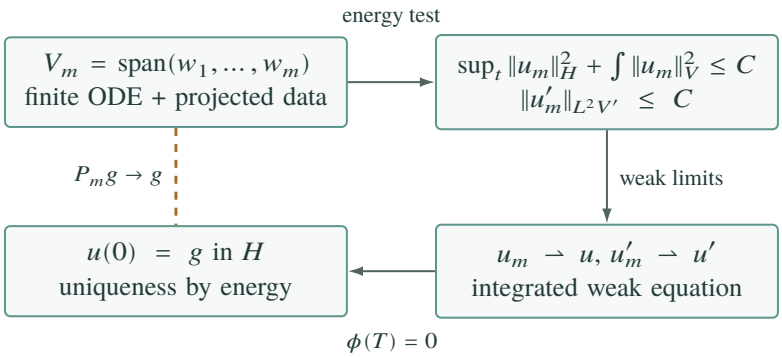


Diagram T7.1. Galerkin approximation preserves the energy estimate. The Dirichlet spectral projection is stable in H_0^1, controlling the time derivative in H^{-1}. Integrated tests retain the initial value in the limit. Arrows describe proof steps, not numerical error bounds.

7.1.3 Improved regularity and compatibility at the corner

Here assume A, b, c are smooth on $\overline{U}$ and independent of time. Write $B_0[u, v] = \int_U a^{ij} u_i v_j$ and $B = B_0 + B_1$; then $|B_1[u, v]| \le C\|u\|_V \|v\|_H$.

Theorem 7.4 (Improved regularity; source §7.1.3, Theorem 5)
If $g \in V$ and $f \in L^2(0, T; H)$, then

$$u \in L^\infty V \cap L^2 H^2(U), \qquad u' \in L^2 H,$$

with norm bound $C(\|g\|_V + \|f\|_{L^2 H})$. If also $g \in H^2(U)$ and $f \in H^1(0, T; H)$, then

$$u \in L^\infty H^2, \quad u' \in L^\infty H \cap L^2 V, \quad u'' \in L^2 V',$$

with bound $C(\|g\|_{H^2} + \|f\|_{H^1(0,T;H)})$. All unlabeled time spaces here and below use the interval $(0, T)$.

Expanded proof (legal tests and reconstruction of a time derivative).

Test the finite system with $u_m' \in V_m$. Symmetry and time indepen-

弱极限与初值

L^2 时间弱收敛不控制 $t = 0$。时间分部积分保留 $(g, v)\phi(0)$，再与极限方程比较，才恢复初值；不能直接把 $u_m(0)$ 换成 $u(0)$。

dence give

$$\|u_m'\|_H^2 + \tfrac{1}{2}\tfrac{d}{dt}B_0[u_m,u_m] = (f,u_m') - B_1[u_m,u_m'] \le \tfrac{1}{2}\|u_m'\|_H^2 + C(\|f\|_H^2 + \|u_m\|_V^2).$$

The initial projections are bounded in V; the basic energy estimate bounds $\int \|u_m\|_V^2$. Weak limits yield $u' \in L^2H$ and $u \in L^\infty V$. For a.e. t, $Lu = f - u' \in H$; the elliptic boundary estimate of §6.3 yields

$$\|u(t)\|_{H^2} \le C(\|f(t)\|_H + \|u'(t)\|_H + \|u(t)\|_H).$$

Integrate to prove the first assertion. For the second, construct v from $v' + Lv = f'$, $v(0) = g_1 = f(0) - Lg \in H$. Set $U(t) = g + \int_0^t v(s)\,ds$ in V. The integrated equation in V' gives $v(t) + LU(t) - f(t) = g_1 + Lg - f(0) = 0$. Thus $U = u$ by uniqueness and $u' = v$. The weak estimate bounds v in $L^\infty H \cap L^2 V$, v' in L^2V'. Since $f \in C([0,T];H)$, the elliptic estimate gives $u \in L^\infty H^2$. This argument avoids differentiating a weak solution prematurely. □

Theorem 7.5 (Higher regularity; source Theorems 6–7)
For an integer $m \ge 0$, suppose

$$g \in H^{2m+1}(U), \quad \partial_t^k f \in L^2H^{2m-2k}(U) \quad (0 \le k \le m).$$

Define $g_0 = g$, $g_j = \partial_t^{j-1} f(0) - Lg_{j-1}$ for $1 \le j \le m$. If $g_j \in V$ for $0 \le j \le m$, then

$$\sum_{k=0}^{m+1} \|\partial_t^k u\|_{L^2H^{2m+2-2k}} \le C\left(\|g\|_{H^{2m+1}} + \sum_{k=0}^{m} \|\partial_t^k f\|_{L^2H^{2m-2k}}\right). \tag{7.6}$$

Smooth data satisfying every compatibility condition give $u \in C^\infty(\overline{U} \times [0,T])$.

Expanded proof (the two-derivative induction).

The case $m = 0$ is the preceding theorem. In the step from m to $m+1$, reconstruct $v = u'$ as above. Its compatibility sequence is $g_1, \dots, g_{m+1}$, so induction controls every positive time derivative of u. Recover the missing term from $Lu = f - u'$ and

$$\|u\|_{H^{2m+4}} \le C(\|f - u'\|_{H^{2m+2}} + \|u\|_H).$$

The Sobolev time trace estimate from §5.9, using spatial extension, reads

$$\|F(0)\|_{H^{r+1}} \le C(\|F\|_{L^2H^{r+2}} + \|F'\|_{L^2H^r}) \quad (r \ge 0).$$

系数的时间独立性

若 $A = A(x,t)$，主部能量求导多出 $\frac{1}{2}\int a_t^{ij} u_i u_j$。仅有可测时间依赖时不能使用此计算；基本存在性仍允许时间依赖。

It controls the initial data at each induction step. Combining estimates proves the assertion. For smooth data apply the result for every m; all mixed derivatives then have arbitrarily high Sobolev order on the cylinder. Extension and embedding give continuous derivatives up to its boundary, including the initial corner. □

7.1.4 Weak maximum principles and comparison

Use the nondivergence operator $Pu = u_t - a^{ij}u_{ij} + b^i u_i + cu$, with continuous coefficients on the closed cylinder, symmetric uniformly positive A, and classical $u \in C^{2,1}(U \times (0,T]) \cap C(\overline{U} \times [0,T])$. The parabolic boundary is

$$\Gamma_T = (\overline{U} \times \{0\}) \cup (\partial U \times [0,T]).$$

The top interior $U \times \{T\}$ is not part of Γ_T.

Theorem 7.6 (Weak maximum principle; source Theorems 8–9)
If $Pu \le 0$ and $c \ge 0$, then $\max_{\overline{Q}_T} u \le \max_{\Gamma_T} u^+$. When $c = 0$ the positive part may be removed. Changing signs gives the minimum principle. Hence $Pu \le Pv$, $u \le v$ on Γ_T imply $u \le v$ when $c \ge 0$.

Expanded proof (including a maximum on the final slice).

Set $M = \max_{\Gamma_T} u^+$ and $w = u - M - \varepsilon t$. Then $Pw \le -cM - \varepsilon - c\varepsilon t < 0$. At a positive maximum outside Γ_T, one would have $Dw = 0$, $D^2 w \le 0$, and $w_t \ge 0$ (the left derivative at $t = T$), giving $Pw \ge 0$, a contradiction. Let $\varepsilon \downarrow 0$. When $c = 0$ any constant M may be subtracted. Apply the result to $u - v$ for comparison. If c is only bounded below, write $u = e^{\lambda t} z$; the new potential is $c + \lambda \ge 0$ for large λ. Sign preservation and uniqueness therefore still hold. □

7.1.5 Harnack's inequality and positivity in earlier times

We prove the smooth principal-part case of the source's Harnack theorem: $Pu = u_t - a^{ij}(x,t)u_{ij}$ with smooth symmetric uniformly positive A. A general measurable-coefficient theorem is not proved here.

相容条件是角点条件

$u = 0$ 在侧边界上迫使 $u_t = 0$，而方程给出 $u_t(0) = f(0) - Lg$。热方程 $g = 0, f = 1$ 的一阶相容失败，不能在整个闭柱上声称光滑。

时间方向决定边界

末时刻的内点仍受最大值原理约束。指数代换 $u = e^{\lambda t} z$ 把零阶系数改为 $c + \lambda$，符号不可写反。

Theorem 7.7 (Harnack inequality; source Theorem 10, smooth principal case)
Let smooth $u \geq 0$ solve $Pu = 0$. If $V \Subset U$ is open and connected and $0 < t_1 < t_2 \leq T$, then

$$\sup_V u(\cdot, t_1) \leq C \inf_V u(\cdot, t_2).$$

The constant depends on the interior sets and connecting paths, times, ellipticity, and local coefficient derivative bounds, not on u.

Expanded proof (logarithmic estimate, including cutoff remainders).

First use $u + \varepsilon$ and set $v = \log(u + \varepsilon)$, $p = Dv$, $H_v = D^2 v$, $z = a^{ij} p_i p_j$. Then $v_t = a^{ij} v_{ij} + z$. Put $\mathscr{D} = \partial_t - a^{kl}\partial_{kl} - 2a^{kl} p_l \partial_k$. Differentiating the equation in time gives

$$\mathscr{D} v_t = a_t^{ij}(v_{ij} + p_i p_j).$$

Differentiation of z gives the explicit identity

$$\begin{aligned}\mathscr{D} z = {} & (a_t^{ij} - a^{kl} a_{kl}^{ij}) p_i p_j + 2a^{ij} p_i a_j^{kl} v_{kl} \\ & - 4a^{kl} a_k^{ij} v_{il} p_j - 2a^{ij} a^{kl} v_{ik} v_{jl}.\end{aligned}$$

Here $a_k^{ij} = \partial_k a^{ij}$. The cubic terms in p cancel by symmetry and relabeling the dummy indices. Thus all remaining errors are bounded by $C(|H_v| + |p|^2 + |p||H_v|)$. Choose, for example, $\kappa = 1/2$, and set $F = v_t - (1 - \kappa) z = a^{ij} v_{ij} + \kappa z$. Ellipticity and Young's inequality now give

$$\mathscr{D} F \geq c_0 |H_v|^2 - C|p|^2 - C \qquad (c_0 > 0).$$

Choose a smooth cutoff $0 \leq \zeta \leq 1$, vanishing near $t = 0$ and the spatial boundary, equal to one on a fixed interior ball during $[t_1, t_2]$. At a negative minimum of $\zeta^4 F + \mu t$, $F < 0$ and $DF = -4F D\zeta / \zeta$. Also $F < 0$ implies $|p|^2 \leq C|H_v|$, $|F| \leq C|H_v|$. Applying $\mathscr{D}$ at that minimum produces cutoff errors at most

$$C\zeta^2 |F| + C\zeta^3 |p||F| \leq \delta \zeta^4 |H_v|^2 + C_\delta.$$

The term $C\zeta^4 |p|^2$ is absorbed likewise. A small δ and then large μ contradict $\mathscr{D}(\zeta^4 F + \mu t) \leq 0$. Consequently $F \geq -C$ on the smaller cylinder, and

$$v_t \geq a_0 |Dv|^2 - C_0 \qquad (a_0 > 0). \tag{7.7}$$

Along a path $\eta : [t_1, t_2] \to U$ in that cylinder,

$$\frac{d}{dt} v(\eta(t), t) \geq a_0 |Dv|^2 + Dv \cdot \eta' - C_0 \geq -\frac{|\eta'|^2}{4a_0} - C_0.$$

Integrate along the straight path in a ball and exponentiate. For general V, connect a finite ball covering of its compact closure by finitely many interior paths, divide the time gap into positive increments, and concatenate the estimates. Let $\varepsilon \downarrow 0$; the constants are independent of it. □

Harnack 的时间方向

早时刻的上界受晚时刻的下界控制。时间差趋零时常数通常发散，不能把它读成同一时间切片上的椭圆 Harnack 估计。

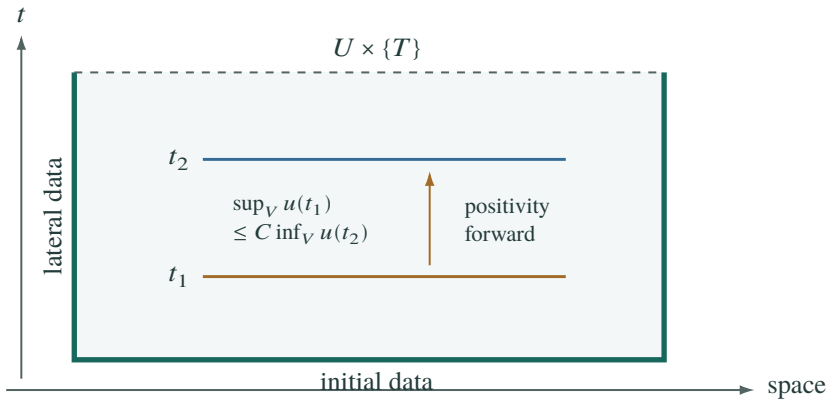


Diagram T7.2. The parabolic boundary and Harnack comparison. The thick bottom and side edges form Γ_T. The top interior belongs to the evolution region. Harnack compares an earlier interior slice to a later one; the arrow indicates forward propagation of positivity.

Theorem 7.8 (Strong maximum principle; source Theorems 11–12, smooth case)

Assume additionally that A is time independent. For $P = \partial_t - a^{ij}\partial_{ij}$ as above on connected U, if $Pu \le 0$ and u attains its global maximum M at $(x_0, t_0) \in U \times (0, T]$, then $u = M$ on $U \times [0, t_0]$. The minimum version follows by changing sign. For $P + c$, $c \ge 0$, the conclusion holds at a nonnegative maximum.

Expanded proof (a local comparison argument).

Set $w = M - u \ge 0$. For the principal case $Pw \ge 0$; in the case $c \ge 0$, $M \ge 0$, one has $(P + c)w \ge cM \ge 0$. Suppose $w(y, s) > 0$ for some $s < t_0$. Choose a smooth connected $W \Subset U$ containing y, x_0, and a nonzero smooth $0 \le \psi \le w(\cdot, s)$ compactly supported near y. Solve $Ph = 0$ on $W \times (s, t_0]$, with zero lateral data and $h(s) = \psi$. This solution is supplied by the preceding existence and regularity theory; compact support ensures corner compatibility. Let $C \ge \sup c$ (take $C = 0$ in the principal case). Then $\tilde{h} = e^{-C(t-s)}h$ satisfies $(P + c)\tilde{h} = (c - C)\tilde{h} \le 0$. Comparison gives $w \ge \tilde{h} \ge 0$. By continuity h is positive somewhere at a time just after s; Harnack then gives $h(x_0, t_0) > 0$, contradicting $w(x_0, t_0) = 0$. Thus $w = 0$ at all earlier interior times; continuity supplies the endpoints. This proof covers the smooth principal coefficients

just treated, with an additional smooth nonnegative potential, not the full rough theory. □

7.2 Second-Order Hyperbolic Equations

7.2.1 Two initial data and the natural energy

Keep $V = H_0^1(U)$, $H = L^2(U)$ and the divergence-form $L(t)$ from §7.1. Here assume $a^{ij}, b^i, c \in C^1(\overline{Q}_T)$, $a^{ij} = a^{ji}$, with the same uniform ellipticity bound. Study

$$u'' + L(t)u = f, \quad u|_{\partial U} = 0, \quad u(0) = g,\ u'(0) = h, \tag{7.8}$$

where $g \in V$, $h \in H$, $f \in L^2(0,T;H)$. A weak solution has

$$u \in L^2V, \quad u' \in L^2H, \quad u'' \in L^2V', \qquad \langle u'', v\rangle + B_t[u,v] = (f,v).$$

The initial values are meaningful in H and V', respectively: $u \in H^1(0,T;H) \subset C([0,T];H)$ and $u' \in H^1(0,T;V') \subset C([0,T];V')$. The estimates below in fact put u, u' in $L^\infty V, L^\infty H$.

Theorem 7.9 (Energy bound and existence; source §7.2.2, Theorems 1–3)
There exists a weak solution satisfying

$$\operatorname*{ess\,sup}_{0<t<T}(\|u(t)\|_V^2 + \|u'(t)\|_H^2) + \|u''\|_{L^2V'}^2 \le C(\|g\|_V^2 + \|h\|_H^2 + \|f\|_{L^2H}^2). \tag{7.9}$$

Expanded proof (the second-order finite system).

Use the same V_m and solve

$$(u_m'', w_k) + B_t[u_m, w_k] = (f, w_k), \quad u_m(0) = P_m g, \quad u_m'(0) = P_m h.$$

This finite second-order system becomes a linear first-order ODE in (d, d'), so has a unique solution. Test with u_m' and put

$$E_m = \tfrac{1}{2}(\|u_m'\|_H^2 + \int_U a^{ij}(u_m)_i(u_m)_j + \|u_m\|_H^2).$$

A direct calculation gives

$$E_m' = (f, u_m') - \int_U (b^i(u_m)_i + cu_m)u_m' + \tfrac{1}{2}\int_U a_t^{ij}(u_m)_i(u_m)_j + (u_m, u_m').$$

Thus $E_m' \le CE_m + C\|f\|_H^2$. Gronwall gives the uniform energy bound; E_m is equivalent to $\|u_m\|_V^2 + \|u_m'\|_H^2$. Using P_m in an arbitrary test gives $\|u_m''\|_{V'} \le C\|u_m\|_V + C\|f\|_H$. Weak compactness yields the equation and

强原理只追溯过去

在 t_0 取内点极值只能推出直到 t_0 的常值性。之后的边界输入或外力可能变化，不能据此断言整个未来为常值。

位移和速度的不同空间

能量包含 $\|Du\|_2^2 + \|u_t\|_2^2$，故自然数据为 $g \in H_0^1$、$h \in L^2$。基本弱解定义中速度初值先在 H^{-1} 识别，不能无证明地宣称 u_t 在 L^2 连续。

identifies successive weak derivatives by scalar time tests, exactly as in the parabolic construction.

For $v \in V_N$, $m \geq N$, and $\phi(T) = \phi'(T) = 0$, twice integrating in time gives

$$\int_0^T \{(u_m, v)\phi'' + B_t[u_m, v]\phi\}\, dt = \int_0^T (f, v)\phi\, dt + (P_m h, v)\phi(0) - (P_m g, v)\phi'(0).$$

Both endpoint coefficients survive the weak limit. Comparing with the trace formula for u, u' identifies $u(0) = g$, $u'(0) = h$. Lower semicontinuity, including the time-set argument of Problem 6, gives (7.9). □

7.2.2 Uniqueness without testing by an unavailable derivative

A weak solution need not have $u' \in V$, so the formal test $v = u'$ is not legitimate. The source's integrated test repairs this gap.

Theorem 7.10 (Weak uniqueness; source Theorem 4)
The weak solution of (7.8) *is unique in the class just defined.*

Expanded proof (the integrated test).

Subtract two solutions, so $f = g = h = 0$. Fix s and use $v(t) = \int_t^s u(r)\, dr$ for $0 \leq t \leq s$. Then $v' = -u$ in V, $v(s) = 0$, and v is a legal time-dependent V test by approximation with finite sums of scalar time tests. Write $B_t = S_t + K_t$, with S_t symmetric and K_t skew-symmetric. Integration by parts in space gives

$$S_t[z, z] = \int_U a^{ij} z_i z_j + (c - \tfrac{1}{2}\operatorname{div} b) z^2,$$
$$K_t[u, v] = -\int_U u\, b \cdot Dv - \tfrac{1}{2}(\operatorname{div} b) uv.$$

In particular $|K_t[u, v]| \leq C\|u\|_H \|v\|_V$, $|\dot{S}_t[v, v]| \leq C\|v\|_V^2$ (one may use the original symmetric formula, avoiding mixed coefficient derivatives). Also $S_t[v, v] \geq \alpha\|v\|_V^2 - \gamma\|v\|_H^2$. Time integration by parts, with $u'(0) = 0$ in V', gives

$$\tfrac{1}{2}\|u(s)\|_H^2 + \tfrac{1}{2} S_0[v(0), v(0)] = -\int_0^s (\tfrac{1}{2}\dot{S}_t[v, v] + K_t[u, v])\, dt.$$

Here $\int (u', u) = \frac{1}{2}\|u(s)\|_H^2$ is valid since $u \in H^1(0, T; H)$; all other pairings involve V', V. Thus

$$\|u(s)\|_H^2 + \alpha\|v(0)\|_V^2 \leq C \int_0^s (\|u\|_H^2 + \|v\|_V^2)\, dt + \gamma\|v(0)\|_H^2.$$

Put $W(t) = \int_0^t u$, so $v(t) = W(s) - W(t)$. The last term is bounded by $\gamma s \int_0^s \|u\|_H^2$. Consequently, for a fixed sufficiently short interval,

$$\|u(s)\|_H^2 + \tfrac{\alpha}{2}\|W(s)\|_V^2 \leq C \int_0^s (\|u(t)\|_H^2 + \|W(t)\|_V^2)\, dt.$$

Gronwall forces $u = 0$ there. Restart at its right endpoint: the H and V' traces are zero, and all constants are uniform on $[0, T]$. Finitely many repetitions give $u = 0$ throughout. □

7.2.3 Higher regularity and the absence of smoothing

For the remainder of the bounded-domain regularity discussion, coefficients are smooth and time independent.

Theorem 7.11 (Improved wave regularity; source §7.2.3, Theorem 5)
If $g \in H^2(U) \cap V$, $h \in V$, and $f \in H^1(0, T; H)$, then

$$u \in L^\infty H^2, \quad u' \in L^\infty V, \quad u'' \in L^\infty H, \quad u''' \in L^2 V',$$

with norm bound $C(\|g\|_{H^2} + \|h\|_V + \|f\|_{H^1(0,T;H)})$.

Expanded proof (reconstruction instead of formal differentiation).

Construct v solving $v'' + Lv = f'$ with initial data $v(0) = h$, $v'(0) = f(0) - Lg \in H$. Set $U = g + \int_0^t v$. In distributions, $(U'' + LU - f)' = v'' + Lv - f' = 0$, while the residual at 0 is zero in V'. Hence U solves the original equation and has data g, h. Uniqueness gives $U = u$, $u' = v$. Estimate v by (7.9) and recover $u \in L^\infty H^2$ from $Lu = f - u''$ and the elliptic estimate. This proves every stated norm bound. □

Theorem 7.12 (Higher wave regularity; source Theorems 6–7)
Let $m \geq 0$, $g \in H^{m+1}(U)$, $h \in H^m(U)$, and $\partial_t^k f \in L^2 H^{m-k}(U)$ for $0 \leq k \leq m$. Set $q_0 = g$, $q_1 = h$, and

$$q_{j+2} = \partial_t^j f(0) - Lq_j \quad (0 \leq j \leq m-2).$$

Require $q_j \in V$ for $0 \leq j \leq m$ (for $m = 0$ only $g \in V$). Then

$$\operatorname*{ess\,sup}_{0<t<T} \sum_{k=0}^{m+1} \|\partial_t^k u(t)\|_{H^{m+1-k}} \leq C\left(\|g\|_{H^{m+1}} + \|h\|_{H^m} + \sum_{k=0}^{m} \|\partial_t^k f\|_{L^2 H^{m-k}}\right).$$

Smooth data compatible to every order give a smooth solution on the closed cylinder.

Expanded proof (one time derivative at each induction step).

The case $m = 0$ is the basic wave bound. In passing to $m + 1$, the preceding reconstruction gives $v = u'$ with data $h, f(0) - Lg$ and forcing

积分测试恢复合法性

$v(t) = \int_t^s u$ 属于 H_0^1，而 $v' = -u$。它将不可用的 u_t 测试转成合法的时间分部积分；短时间吸收后再拼接，覆盖整个时间区间。

f'. Its compatibility sequence is shifted by one place. Apply induction to v. The trace inequality $\|F(0)\|_{H^r} \le C(\|F\|_{L^2H^r} + \|F'\|_{L^2H^r})$ controls the forcing traces required at this step. Then $Lu = f - u''$ and elliptic regularity recover the missing highest spatial norm of u. This proves the estimate and the induction. Arbitrarily high mixed Sobolev regularity and embedding prove the smooth conclusion. □

7.2.4 Finite propagation and the geometry of the cone

On $\mathbb{R}^n$ now take the source's nondivergence model $u_{tt} - a^{ij}(x)u_{ij} = 0$ with smooth symmetric positive A. Assume on the region under consideration that an optical distance q is smooth away from x_0, positive there, $q(x_0) = 0$, and

$$ADq \cdot Dq = 1.$$

Assume its relevant sublevel sets are bounded and shrink to x_0. Set $K_t = \{q < t_0 - t\}$ for $0 \le t < t_0$ and $K = \{(x,t) : 0 \le t < t_0,\ x \in K_t\}$.

> **Theorem 7.13** (Finite propagation; source §7.2.4, Theorem 8)
> *If a smooth solution has $u = u_t = 0$ on K_0, it vanishes throughout K, and at its vertex by continuity.*

Expanded proof (coarea, normal speed, and the energy flux).

The outward normal is $\nu = Dq/|Dq|$ and the moving boundary has outward normal velocity $-1/|Dq|$. Hence for $E(t) = \frac{1}{2}\int_{K_t}(u_t^2 + ADu \cdot Du)$, differentiation gives

$$E'(t) = -\int_{K_t} u_t(\partial_j a^{ij})u_i + \int_{\partial K_t} u_t ADu \cdot \nu - \frac{1}{2}\int_{\partial K_t} \frac{u_t^2 + ADu \cdot Du}{|Dq|}.$$

The volume error is at most $CE(t)$ by ellipticity and bounded first coefficient derivatives. Also

$$|ADu \cdot \nu| \le (ADu \cdot Du)^{1/2}(A\nu \cdot \nu)^{1/2} = \frac{(ADu \cdot Du)^{1/2}}{|Dq|}.$$

Young's inequality makes the boundary flux no larger than the negative moving-boundary term. Thus $E' \le CE$ and $E(0) = 0$, giving $E = 0$. For fixed $x \in K_t$, that same x belongs to every earlier K_s; integrating $u_t = 0$ from its zero initial value proves $u(x,t) = 0$. □

波动不会自动磨光

$\sin(kx)$ 模态的位移为 $\cos(kt)\sin(kx)$；高频振幅没有热方程的 e^{-k^2t} 衰减。提高解的正则性需要提高初值正则性，不能从能量估计推导任意正时刻光滑。

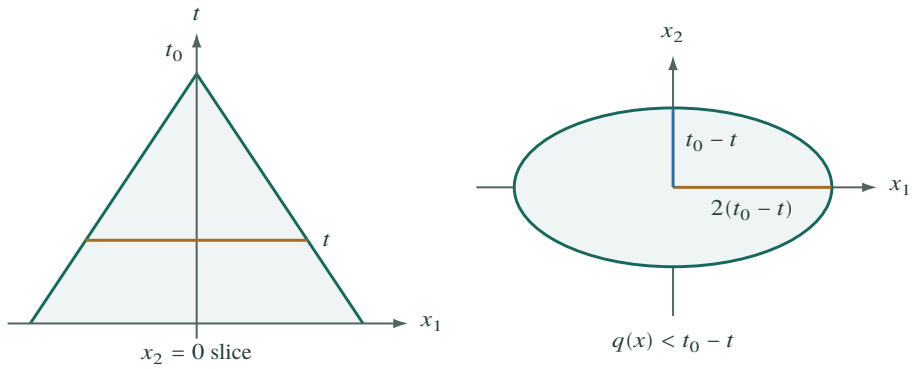


Diagram T7.3. Anisotropy changes the cross-section of a wave cone. For $A = \operatorname{diag}(4, 1)$, $q(x) = \sqrt{x_1^2/4 + x_2^2}$. At time t, the dependency cross-section is an ellipse with semiaxes $2(t_0 - t)$ and $t_0 - t$. The cone sketch and the spatial slice use separate coordinate scales.

光学距离的局部限制

度量是 A^{-1}，不是 A。变系数的距离函数可能有割迹，不能默认全空间光滑；此处光滑 q 是明确假设。常系数时 $q = \sqrt{(x - x_0)^T A^{-1} (x - x_0)}$，波前为椭球。

7.2.5 Characteristic coordinates in two variables

For $a u_{xx} + 2b u_{xy} + d u_{yy} + \text{lower terms} = 0$, assume smooth coefficients and $ad - b^2 < 0$. A coordinate function ξ must solve

$$a \xi_x^2 + 2b \xi_x \xi_y + d \xi_y^2 = 0.$$

After a fixed linear coordinate change one can arrange $a \neq 0$ near the point: an indefinite quadratic form cannot vanish on every direction. Writing $D = b^2 - ad > 0$ factors this equation into the two distinct fields

$$a \xi_x + (b + \sqrt{D}) \xi_y = 0, \qquad a \eta_x + (b - \sqrt{D}) \eta_y = 0.$$

Expanded proof (local invertibility and both canonical forms).

The two smooth nonzero vector fields are transverse, since their determinant is $-2a\sqrt{D} \neq 0$. The local flow-box construction (solving the characteristic ODE from a transverse curve) gives first integrals ξ, η with nonzero independent gradients. Thus $\Phi = (\xi, \eta)$ is locally invertible. By the chain rule the transformed principal matrix is $\widetilde{A} = D\Phi \, A \, D\Phi^T$, whose two diagonal entries vanish. Its determinant is $(\det D\Phi)^2 \det A < 0$, so the off-diagonal entry is nonzero. Division yields $v_{\xi\eta} + \text{lower terms} = 0$. For $z_1 = \xi + \eta$, $z_2 = \xi - \eta$, this becomes $v_{z_1 z_1} - v_{z_2 z_2} + \text{lower terms} = 0$. For example $u_{tt} - c^2 u_{xx}$, $c > 0$, becomes $-4c^2 v_{\xi\eta}$ with $\xi = x - ct$, $\eta = x + ct$. All these coordinates are local; no global characteristic chart is asserted. □

7.3 Systems of Hyperbolic First-Order Equations

7.3.1 The symbol and the meaning of hyperbolicity

For vector-valued $u : \mathbb{R}^n \times [0,T] \to \mathbb{R}^m$ study

$$u_t + \sum_{j=1}^{n} B^j(x,t)u_{x_j} = f, \qquad u(0) = g. \tag{7.10}$$

Its spatial symbol is $B(x,t;\xi) = \sum_j \xi_j B^j(x,t)$. Following the source, the system is hyperbolic if this matrix has a basis of real eigenvectors and real eigenvalues for each nonzero ξ; strictly hyperbolic means all eigenvalues are real and distinct. It is symmetric hyperbolic if every B^j is real symmetric. A plane wave $u = r\,F(\xi\cdot x - \sigma t)$ solves the constant coefficient homogeneous equation if $B(\xi)r = \sigma r$. For $|\xi| = 1$, σ is its signed speed. Repeated eigenvalues are permitted in symmetric systems.

More generally, $B^0 u_t + \sum B^j u_{x_j} = f$ is symmetrizable in this energy argument when all B^j, including B^0, are symmetric and B^0 is uniformly positive. Its energy is $\int u \cdot B^0 u$. For $v_{tt} - a^{ij}v_{ij} = 0$ set $u = (Dv, v_t)$, $B^0 = \operatorname{diag}(A, 1)$, and

$$B^j = \begin{pmatrix} 0 & -Ae_j \\ -e_j^{T}A & 0 \end{pmatrix}.$$

This gives a symmetric first-order system, since $(Dv)_t = Dv_t$ and $v_{tt} = a^{ij}v_{ij}$. Recovering v also requires the initial gradient constraint; arbitrary first-order data need not be gradients.

7.3.2 Symmetry, energy, and vanishing viscosity

Assume $B^j \in C_b^2(\mathbb{R}^n \times [0,T]; \mathbb{R}^{m\times m})$ are symmetric, $g \in H^1(\mathbb{R}^n;\mathbb{R}^m)$, and $f \in H^1(\mathbb{R}^n \times (0,T);\mathbb{R}^m)$. Here C_b^2 means that the matrices and all derivatives through order two are bounded. For smooth functions, spatial integration by parts yields

$$\frac{1}{2}\frac{d}{dt}\|u\|_2^2 = (f,u) + \frac{1}{2}\int_{\mathbb{R}^n} u\cdot\Big(\sum_j \partial_j B^j\Big)u. \tag{7.11}$$

The identity follows from $2u\cdot B^j u_j = \partial_j(u\cdot B^j u) - u\cdot(\partial_j B^j)u$. For Sobolev functions on $\mathbb{R}^n$, insert expanding compact cutoffs; the errors vanish by L^2 integrability and bounded coefficients.

双曲标准形是局部结论

判别式给出两个横截特征方向；逆函数定理还需要 $\det D\Phi \neq 0$。先处理 $a \neq 0$ 的局部坐标，避免在 $a = 0$ 处直接除法。

实谱与一致对角化

每个方向可对角化，并不自动给出方向上一致有界的对角化矩阵。L^2 适定性需要额外的一致控制；对称矩阵可用正交对角化，条件数恒为 1。

Theorem 7.14 (Symmetric hyperbolic well-posedness; source Theorems 1–4) *There is a unique solution of (7.10) with $u \in L^2(0,T;H^1)$, $u_t \in L^2(0,T;L^2)$, $u(0) = g$ in L^2. It satisfies*

$$\|u\|_{L^\infty H^1} + \|u_t\|_{L^2L^2} \le C(\|g\|_{H^1} + \|f\|_{H^1(\mathbb{R}^n\times(0,T))}).$$

The weak identity uses H^1 tests and the ordinary L^2 inner product.

Expanded proof (viscosity construction).

For $0 < \varepsilon \le 1$ solve

$$u_t^\varepsilon - \varepsilon\Delta u^\varepsilon + \sum_j B^j u_j^\varepsilon = f, \qquad u^\varepsilon(0) = g_\varepsilon = \rho_\varepsilon * g.$$

For fixed ε, Duhamel's formula with the heat semigroup gives an elementary construction. The estimate $\|De^{\varepsilon r\Delta}h\|_2 \le C(\varepsilon r)^{-1/2}\|h\|_2$ follows from its Fourier multiplier. Thus the integral operator is a contraction on $C([0,\delta];H^1)$ when $C(\delta + \sqrt{\delta/\varepsilon}) < 1$. Iterate finitely many intervals; the forcing term is continuous in H^1 because $f \in L^1(0,T;H^1)$. Smooth approximations to coefficients and data justify the following computations; their bounds depend only on the indicated norms.

Testing with u^ε gives (7.11) with the additional positive term $\varepsilon\|Du^\varepsilon\|_2^2$ on the left. Differentiate in x_k. The new forcing is $f_k - \sum_j (\partial_k B^j)u_j^\varepsilon$, bounded by $\|Df\|_2 + C\|Du^\varepsilon\|_2$. Summing the energy estimates and applying Gronwall yields

$$\sup_t \|u^\varepsilon(t)\|_{H^1}^2 + \varepsilon\int_0^T \|D^2u^\varepsilon\|_2^2\,dt \le C(\|g\|_{H^1}^2 + \|f\|_{L^2H^1}^2).$$

The equation then gives

$$\|u_t^\varepsilon\|_{L^2L^2} \le \varepsilon\|\Delta u^\varepsilon\|_{L^2L^2} + C\|u^\varepsilon\|_{L^2H^1} + \|f\|_{L^2L^2} \le C.$$

In fact the viscosity term tends to zero in L^2L^2 at rate at most $C\sqrt{\varepsilon}$ by the previous bound. Weak compactness now yields $u^\varepsilon \rightharpoonup u$ in L^2H^1 and $u_t^\varepsilon \rightharpoonup u_t$ in L^2L^2. All terms pass to the limit in integrated tests. Taking tests vanishing at T and integrating the time derivative recovers $u(0) = g$; $H^1(0,T;L^2)$ supplies a continuous L^2 representative. Finally use (7.11) for a difference of two solutions. The identity is legal at this regularity by spatial approximation and the Hilbert-valued chain rule. Gronwall proves uniqueness and continuous dependence. □

对称性的精确用途

对称性把 $u \cdot B^j u_j$ 写成通量散度加零阶项。若 B^j 不对称，会留下含 Du 的项，无法仅靠 $\|u\|_2$ 封闭估计。

7.3.3 Constant coefficients: real spectrum and derivative loss

Assume now constant real matrices and only that $B(\xi)$ has real spectrum; it may have Jordan blocks. Fourier transformation in space gives

$$\hat{u}(\xi, t) = e^{-itB(\xi)}\hat{g}(\xi). \tag{7.12}$$

This is a weaker hypothesis than the hyperbolicity definition above.

Lemma 7.15 (Polynomial multiplier bound; source proof of Theorem 5)
For each finite T, $\|e^{-itB(\xi)}\| \le C_T(1 + |\xi|^{m-1})$ for $0 \le t \le T$.

Expanded proof (triangular multiplier bound).

By complex Schur triangularization choose a unitary matrix Q such that $Q^*B(\xi)Q = D + N$, where D is real diagonal and N strictly upper triangular. Both have norm at most $C|\xi|$, uniformly in direction. Variation of constants around D expresses the matrix exponential as an ordered integral series. A product with r factors of N, interspersed with diagonal exponentials, has zero entries below the rth upper diagonal; it therefore vanishes for $r \ge m$. Diagonal exponentials have norm one, and the time simplex has volume $t^r/r!$. Consequently

$$\|e^{-itB(\xi)}\| \le \sum_{r=0}^{m-1} \frac{(Ct|\xi|)^r}{r!}.$$

Unitary conjugation preserves the norm. This proves a direction-uniform bound without assuming uniformly conditioned eigenvectors. □

Theorem 7.16 (Fourier solution; source Theorem 5, specified uniqueness class)
If $g \in H^s(\mathbb{R}^n;\mathbb{R}^m)$, $s > n/2 + m$, the inverse transform of (7.12) is C^1 in (x,t) on every finite time strip and solves the homogeneous system with initial data g. It belongs to $C([0,T];H^{s-m+1})$, with time derivative in $C([0,T];H^{s-m})$, and is unique among solutions with these properties.

Expanded proof (integrability, differentiation, and uniqueness).

The multiplier bound gives the two asserted Sobolev bounds. For pointwise differentiation, both space and time derivatives add at most one factor $C|\xi|$. Cauchy–Schwarz shows

$$\int_{\mathbb{R}^n}(1 + |\xi|)^m|\hat{g}(\xi)|\,d\xi \le C\|g\|_{H^s}\left(\int_{\mathbb{R}^n}(1 + |\xi|)^{-2(s-m)}d\xi\right)^{1/2} < \infty.$$

黏性极限的统一常数

固定 ε 的抛物正则性常数可能随 $\varepsilon \downarrow 0$ 发散。这里另证一致 H^1 估计，并由 $\varepsilon \int \|D^2u^\varepsilon\|_2^2 \le C$ 得到黏性项消失。

Thus dominated convergence proves C^1 regularity and justifies substitution into the equation. Continuity in the stated Sobolev spaces follows from the same multiplier bound. For uniqueness, Fourier localize a difference to a compact frequency set; multiplication by $B(\xi)$ is then a bounded operator on L^2 of that set. The resulting Banach-space ODE with zero initial value is zero. An increasing sequence of compact frequency cutoffs proves uniqueness. No unrestricted uniqueness assertion for arbitrary classical solutions with unspecified growth is needed. □

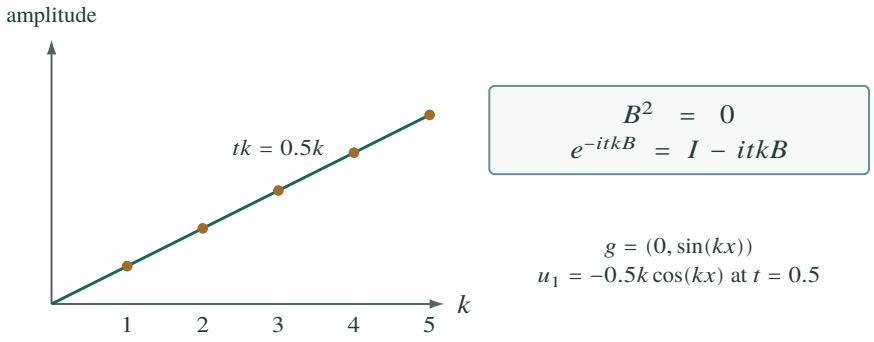


Diagram T7.4. A Jordan block amplifies high frequencies. For $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and input $g = (0, \sin(kx))$ on a periodic line, $u = (-tk\cos(kx), \sin(kx))$. At $t = 0.5$ the first component has amplitude $0.5k$. This exact periodic model illustrates the Fourier multiplier mechanism; the whole-space theorem above uses Sobolev data on $\mathbb{R}^n$.

Jordan 块造成导数损失

对 $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$，$u_2 = g_2$、$u_1 = g_1 - tg_2'$。谱全为实数，仍需要多一个空间导数；实谱本身不保证 $L^2 \to L^2$ 有界。

7.4 Semigroup Theory

7.4.1 Strong continuity, the generator, and its domain

Let X be a real Banach space. A strongly continuous semigroup is a family $S(t) \in \mathscr{L}(X)$, $t \geq 0$, satisfying

$$S(0) = I, \quad S(t+s) = S(t)S(s), \quad S(t)x \longrightarrow x \text{ in } X \text{ as } t \downarrow 0.$$

We first assume $\|S(t)\| \leq 1$ (a contraction semigroup). Its generator is the generally unbounded operator

$$D(G) = \left\{ x : \lim_{h \downarrow 0} \frac{S(h)x - x}{h} \text{ exists in } X \right\}, \qquad Gx = \lim_{h \downarrow 0} \frac{S(h)x - x}{h}.$$

We use G instead of the source's A, to distinguish it from the principal coefficient matrix. For $u_t + Lu = 0$, the generator is $G = -L$.

Theorem 7.17 (Generator properties; source §7.4.1, Theorems 1–2)
The domain $D(G)$ is dense, and G is closed. For $x \in D(G)$, $S(t)x \in D(G)$, $GS(t)x = S(t)Gx$, and

$$\frac{d}{dt}S(t)x = GS(t)x = S(t)Gx.$$

At $t = 0$ this is a right derivative. In particular $S(t)x - x = \int_0^t S(s)Gx\,ds$.

Expanded proof (domain invariance, density, and graph closure).

Commuting $S(t)$ with $S(h)$ and passing to the defining difference quotient proves invariance and commutation. The right derivative follows likewise; for the left derivative at $t > 0$ use

$$\frac{S(t)x - S(t-h)x}{h} = S(t-h)\frac{S(h)x - x}{h} \longrightarrow S(t)Gx.$$

Strong continuity and the contraction bound justify the last limit. For arbitrary $x \in X$, let $x_t = t^{-1}\int_0^t S(s)x\,ds$. A change of integration interval gives

$$x_t \in D(G), \qquad Gx_t = \frac{S(t)x - x}{t}, \qquad x_t \to x.$$

Thus the domain is dense. To prove closedness, take $x_k \to x$, $Gx_k \to y$ in X and pass to the limit in $S(t)x_k - x_k = \int_0^t S(s)Gx_k\,ds$. The contraction bound makes convergence of the integrals uniform on bounded time intervals. Divide the limit identity by t and let $t \downarrow 0$: its right side tends to y. Hence $x \in D(G)$ and $Gx = y$. □

7.4.2 Resolvents as Laplace transforms

For a closed densely defined operator G, let $\rho(G)$ be the set of real λ for which $\lambda I - G : D(G) \to X$ is bijective, and $R_\lambda = (\lambda I - G)^{-1}$. The closed graph theorem makes $R_\lambda : X \to X$ bounded. Its range is $D(G)$.

Theorem 7.18 (Resolvent identities and representation; source Theorem 3)
For $\lambda, \mu \in \rho(G)$,

$$R_\lambda - R_\mu = (\mu - \lambda)R_\lambda R_\mu, \qquad R_\lambda R_\mu = R_\mu R_\lambda. \tag{7.13}$$

For a contraction generator every $\lambda > 0$ is in $\rho(G)$, and

$$R_\lambda x = \int_0^\infty e^{-\lambda t}S(t)x\,dt, \qquad \|R_\lambda\| \le \lambda^{-1}. \tag{7.14}$$

The integral is a norm-convergent Bochner integral.

强连续不等于范数连续

要求每个固定 x 的 $\|S(t)x - x\| \to 0$。若要求 $\|S(t) - I\| \to 0$，就排除了许多无界生成元，例如 L^2 上的热半群。

闭算子与有界算子的区别

闭性是 $x_k \to x$ 且 $Gx_k \to y$ 时才推出 $Gx = y$，不表示 $\|Gx\| \le C\|x\|$。热方程的高频特征函数说明后一不等式不成立。

Expanded proof (the inverse on both sides).

For the first identity insert $(\mu I - G) - (\lambda I - G) = (\mu - \lambda)I$ between R_λ and R_μ. These compositions are defined because each resolvent maps into $D(G)$. Interchanging λ, μ gives commutation (equal parameters are immediate). Let $J_\lambda x$ denote the integral in (7.14). It converges since its integrand has norm at most $e^{-\lambda t}\|x\|$. Changing variables gives

$$\frac{S(h)J_\lambda x - J_\lambda x}{h} = \frac{e^{\lambda h} - 1}{h} J_\lambda x - \frac{e^{\lambda h}}{h} \int_0^h e^{-\lambda t} S(t)x \, dt.$$

The limit is $\lambda J_\lambda x - x$. Thus $J_\lambda x \in D(G)$ and $(\lambda I - G)J_\lambda x = x$. For $x \in D(G)$, differentiate $e^{-\lambda t}S(t)x$ and integrate:

$$\int_0^\infty e^{-\lambda t} S(t)\,(\lambda x - Gx)\, dt = x.$$

The endpoint at infinity vanishes by the contraction bound. Therefore J_λ is also a left inverse, proving bijectivity and the formula. The norm bound follows by integration. This proof never passes an unbounded operator through an integral without a domain argument; Problem 13 gives the equivalent graph-norm justification. □

7.4.3 Constructing the semigroup from resolvent bounds

Theorem 7.19 (Hille–Yosida, contraction case; source §7.4.2, Theorem 4)
A closed densely defined linear G generates a contraction semigroup if and only if

$$(0, \infty) \subset \rho(G), \qquad \|R_\lambda\| \le \lambda^{-1} \quad (\lambda > 0).$$

Expanded proof (Yosida approximation).

Necessity is Theorem 7.18. Conversely define the bounded operators

$$G_\lambda = \lambda^2 R_\lambda - \lambda I = \lambda G R_\lambda.$$

For $x \in D(G)$, $\lambda R_\lambda x - x = R_\lambda G x \to 0$. The bound $\|\lambda R_\lambda\| \le 1$ and density extend this convergence to every $x \in X$. Consequently $G_\lambda x = \lambda R_\lambda G x \to Gx$ for $x \in D(G)$. Put $S_\lambda(t) = e^{tG_\lambda}$. Its power series gives

$$\|S_\lambda(t)\| \le e^{-\lambda t} \sum_{k=0}^{\infty} \frac{(\lambda^2 t)^k \|R_\lambda\|^k}{k!} \le 1.$$

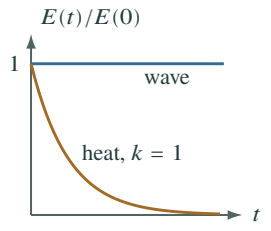
The resolvent identity shows that all G_λ commute. Differentiate $S_\mu(t - s)S_\lambda(s)x$ in s and integrate to obtain

$$\|S_\lambda(t)x - S_\mu(t)x\| \le t\|(G_\lambda - G_\mu)x\| \longrightarrow 0 \quad (x \in D(G)).$$

预解算子的符号

$G = -L$ 时 $R_\lambda = (\lambda I + L)^{-1}$。椭圆能量估计给出的正参数正好对应发展方程的稳定方向；若误写成 $L - \lambda I$，会引入不应出现的共振。

图 T7.M：同一模态的两种衰减



热方程取 $E = \frac{1}{2}\|u\|_2^2$，图示 $k = 1$ 模态。波动方程取 $E = \frac{1}{2}(\|u_t\|_2^2 + \|u_x\|_2^2)$，同一模态总能量守恒。

Thus $S_\lambda(t)x$ converges uniformly on bounded time intervals. Approximation of any $x \in X$ by points of $D(G)$ extends this convergence to all X. The limit $S(t)$ is a contraction, is strongly continuous, and satisfies the semigroup law: in the product limit use the uniform operator bound to control one factor and strong convergence for the other.

Let K be its generator. For $x \in D(G)$ pass to the limit in

$$S_\lambda(t)x - x = \int_0^t S_\lambda(s)G_\lambda x\,ds$$

to get $S(t)x - x = \int_0^t S(s)Gx\,ds$. Dividing by t proves $G \subset K$. This inclusion alone is insufficient. Fix $\lambda > 0$ and $y \in D(K)$, and define $x = R_\lambda(\lambda I - K)y \in D(G)$. Since $Kx = Gx$, $(\lambda I - K)x = (\lambda I - K)y$. The operator $\lambda I - K$ is injective by Theorem 7.18, so $y = x \in D(G)$. Thus $D(K) = D(G)$ and $K = G$. □

For $\|S(t)\| \le e^{\omega t}$, apply the theorem to $G - \omega I$. The resolvent condition becomes $\|R_\lambda\| \le (\lambda - \omega)^{-1}$ for $\lambda > \omega$. The theorem here has leading constant one; a general bound $Me^{\omega t}$ requires the corresponding bounds on all resolvent powers.

不能只证生成元的包含关系

极限积分公式只给出 $G \subset K$。再用共同正参数下的预解算子满射和单射，才能排除 K 有更大定义域，完成 $G = K$。

7.4.4 Parabolic and wave generators

Assume again smooth time-independent elliptic coefficients and smooth bounded U. Set $D(G) = H^2(U) \cap V$, $Gu = -Lu$ on H.

Theorem 7.20 (Parabolic semigroup; source §7.4.3, Theorem 5)
This G generates a semigroup on H with $\|S(t)\| \le e^{\gamma t}$, where $B[u,u] \ge \alpha\|u\|_V^2 - \gamma\|u\|_H^2$.

Expanded proof (the elliptic resolvent and graph estimate).

Density follows from $C_c^\infty(U) \subset D(G)$. If $u_k \to u$ and $Gu_k \to z$ in H, the elliptic estimate

$$\|u_k - u_\ell\|_{H^2} \le C(\|G(u_k - u_\ell)\|_H + \|u_k - u_\ell\|_H)$$

shows convergence in H^2. The trace is still zero and $Gu = z$; thus G is closed. For $\lambda > \gamma$, Lax–Milgram solves $B[u,v] + \lambda(u,v) = (f,v)$, and elliptic regularity puts u in $D(G)$. Testing with u gives $(\lambda - \gamma)\|u\|_H^2 \le \|f\|_H\|u\|_H$. Therefore $\|R_\lambda\| \le (\lambda - \gamma)^{-1}$, and the shifted Hille–Yosida theorem applies. For $g \in D(G)$ the orbit is a classical H-valued solution. For $g \in H$ approximate by $D(G)$ data and use the energy estimate; the semigroup orbit agrees with the weak parabolic solution. Domain regularity is not claimed at $t = 0$ for arbitrary H data. □

For the wave application require $b = 0, A = A^T, c \geq 0$. The symmetric form B is coercive on V. On $X = V \times H$ use

$$\|(u,v)\|_X^2 = B[u,u] + \|v\|_H^2, \qquad \mathcal{G}(u,v) = (v, -Lu),$$

with $D(\mathcal{G}) = (H^2(U) \cap V) \times V$.

Theorem 7.21 (Wave semigroup; source Theorem 6)
The operator $\mathcal{G}$ generates a contraction semigroup on X. In fact it extends to an isometric group.

Expanded proof (wave resolvent).

The domain is dense. Closedness follows as above: convergence of (u_k, v_k) and $(v_k, -Lu_k)$ in $V \times H$ implies $v \in V$ and $u \in H^2 \cap V$, with the correct graph limit. For $\lambda > 0$, solving $(\lambda I - \mathcal{G})(u,v) = (F, H_0)$ is equivalent to

$$(L + \lambda^2)u = H_0 + \lambda F, \qquad v = \lambda u - F.$$

Lax–Milgram and elliptic regularity give the unique domain solution. Moreover $\langle \mathcal{G}(u,v), (u,v)\rangle_X = B[v,u] - (Lu, v) = 0$ by symmetry. Hence

$$\lambda \|(u,v)\|_X^2 = \langle (F, H_0), (u,v)\rangle_X,$$

which proves $\|R_\lambda\| \leq \lambda^{-1}$. Hille–Yosida gives the contraction semigroup. The same argument works for $-\mathcal{G}$. The two semigroups are inverses: on domain data differentiate their product, then use density. Both are contractions, so each preserves the norm. This proves the group assertion as an editorial supplement, and explains energy conservation. □

7.4.5 Forcing and mild solutions: an editorial bridge

For $f \in L^1(0,T;X)$ the variation-of-constants formula is

$$u(t) = S(t)g + \int_0^t S(t-s)f(s)\,ds.$$

It defines a continuous X-valued mild solution; continuity follows from strong continuity and the integrable norm bound. For domain-valued data with continuous graph-norm forcing, differentiation and Theorem 7.17 verify $u' = Gu + f$. Approximation extends the formula as a mild solution for general g, f. Uniqueness for classical domain-valued solutions follows by differentiating $S(t-s)u(s)$. For parabolic forcing in $L^2(0,T;H)$ the energy estimate and approximation identify this formula with the weak solution. A general mild orbit is not asserted to be differentiable for arbitrary Banach-space forcing.

自主与非自主发展

单参数半群依赖系数不含时间。$L = L(t)$ 时通常要用双参数传播族 $U(t,s)$；不能把 $e^{-tL(t)}$ 当成解算子。Galerkin 法仍可直接处理此情形。

波动的压缩实际是等距

无阻尼、对称且时间独立时，反向演化也压缩，因此正向演化只能保持能量。不能由“压缩半群”一词推断波的能量严格衰减。

7.5 Problems and Worked Solutions

The following are concise restatements of all 16 source exercises, with editorial solutions. Unless specified otherwise, U is bounded with smooth boundary and all classical solutions have enough regularity for the indicated integrations. The source problem numbers are retained in the paragraph labels.

1. Neumann heat uniqueness. Prove uniqueness for $u_t - \Delta u = f$, $\partial_\nu u = 0$, $u(0) = g$.

Solution. A difference w has zero initial value and homogeneous equation and boundary data. Multiplication by w yields $\frac{1}{2}\frac{d}{dt}\|w\|_2^2 + \|Dw\|_2^2 = 0$; the boundary term is $\int_{\partial U} w\partial_\nu w = 0$. Thus $w = 0$. Neumann constants do not obstruct evolutionary uniqueness once the initial value is fixed. □

2. Sharp Dirichlet heat decay. Show $\|u(t)\|_2 \le e^{-\lambda_1 t}\|g\|_2$ for homogeneous heat flow.

Solution. The energy identity and the Rayleigh characterization from §6.5 give $\frac{d}{dt}\|u\|_2^2 = -2\|Du\|_2^2 \le -2\lambda_1\|u\|_2^2$. Multiply by $e^{2\lambda_1 t}$, integrate, then take square roots. Equality holds for a first Dirichlet eigenfunction; this fixes the factor two. □

平方范数的衰减率

$\|u\|_2^2$ 以 $2\lambda_1$ 衰减，而 $\|u\|_2$ 以 λ_1 衰减。Neumann 问题有常数模态，不能直接使用正的 Dirichlet 谱隙。

3. Adjoint dynamics. For $u_t + Lu = 0$, $u(0) = g$, and $v_t - L^*v = 0$, $v(T) = h$, both with zero Dirichlet data, prove $(v(0), g) = (u(T), h)$.

Solution. The formal adjoint is defined by $(Lu, v) = (u, L^*v)$ on the zero-boundary smooth functions, including the coefficient derivative in the adjoint drift. Therefore $\frac{d}{dt}(u, v) = -(Lu, v) + (u, L^*v) = 0$. Evaluate at $0, T$. The backward adjoint problem is prescribed terminal data; it is not a forward heat problem with the diffusion sign reversed arbitrarily. □

4. Galerkin approximation for Poisson's equation. For nested spans of a complete V-orthogonal basis, let $\int Du_m \cdot Dw_k = \int f w_k$ for $k \le m$. Show weak convergence to the zero-Dirichlet Poisson solution.

Solution. Testing with u_m and Poincaré bound $\|u_m\|_V$ uniformly. Each weak subsequential limit satisfies the equation on every fixed span, hence on V by density. Uniqueness identifies every subsequence. In fact one has

stronger convergence: if u is the weak solution, $\int D(u-u_m)\cdot Dv_m = 0$, so

$$\|D(u-u_m)\|_2 \le \inf_{v_m\in V_m} \|D(u-v_m)\|_2 \longrightarrow 0.$$

The finite systems are invertible since their gradient Gram matrices are positive definite. No L^2 orthonormality of this exercise's basis is needed. □

5. Identifying a weak time derivative. If $u_k \rightharpoonup u$ in L^2V and $u_k' \rightharpoonup z$ in L^2V', show $z = u'$.

Solution. For $\phi \in C_c^1(0,T)$, $w \in V$, $\int \langle u_k', w\rangle \phi = -\int (u_k, w)\phi'$. Both sides converge by the stipulated weak convergences. The resulting identity is exactly the definition of the V'-valued distributional derivative. □

6. The essential supremum survives a weak limit. If $u_k \rightharpoonup u$ in $L^2(0,T;H)$ and $\|u_k(t)\|_H \le C$ a.e., show $\|u(t)\|_H \le C$ a.e.

Solution. For any measurable $E \subset (0,T)$, $\mathbf{1}_E u \in L^2(0,T;H)$ is a valid fixed weak test. Thus

$$\int_E \|u\|_H^2 = \lim_k \int_E (u_k, u) \le C \int_E \|u\|_H.$$

Choosing $E = \{\|u\|_H > C + \delta\}$ makes its integrand difference at least $\delta\|u\|_H$, so E has measure zero. Take $\delta = 1/j$. This argument works even without assuming H separable. □

一致上界与弱极限

逐时刻弱收敛并非已知。用固定测试函数 $\mathbf{1}_E u$ 把问题化为每个时间集合上的积分不等式，直接得到极限的本质上确界。

7. Pointwise decay from a positive potential. For $u_t - \Delta u + cu = 0$, zero Dirichlet data and $c \ge \gamma > 0$, prove $|u(x,t)| \le Ce^{-\gamma t}$.

Solution. Set $z = e^{\gamma t}u$. Then $z_t - \Delta z + (c-\gamma)z = 0$ and the potential is nonnegative. Comparison with the constants $\pm\|g\|_\infty$ yields $|z| \le \|g\|_\infty$ on every finite cylinder. The constant is independent of its final time, proving the assertion. □

8. Positivity with a bounded signed potential. For the same equation, $g \ge 0$ and bounded c, show $u \ge 0$.

Solution. Choose $\lambda \ge \max(0, -\inf c)$ and $z = e^{-\lambda t}u$. Then $z_t - \Delta z + (c+\lambda)z = 0$. The minimum principle gives $z \ge 0$, and multiplication by $e^{\lambda t} > 0$ proves the claim. □

9. The second-order coercive estimate, including its boundary terms. Prove the source inequalities (54) in §7.1.3 and (59) in §7.2.3:

$$\beta\|u\|_{H^2}^2 \le (Lu, -\Delta u) + \gamma\|u\|_2^2 \qquad (u \in H^2(U) \cap V). \tag{7.15}$$

Solution. Take first smooth $u = 0$ on ∂U and smooth symmetric A. Expand the divergence operator as $Lu = -a^{ij}u_{ij} + \tilde{b}^i u_i + cu$. For the principal term put $I = \int_U a^{ij}u_{ij}u_{kk}$. Integrate first in k and then in j:

$$\begin{aligned} I = &\int_U a^{ij}u_{ik}u_{jk} + \int_U \big((\partial_j a^{ij})u_{ik}u_k - (\partial_k a^{ij})u_{ij}u_k\big) \\ &+ \int_{\partial U} \big((A : D^2u)\,\partial_\nu u - (D^2u\,A\nu)\cdot Du\big). \end{aligned}$$

The volume errors are bounded by $\delta\|D^2u\|_2^2 + C_\delta\|Du\|_2^2$. The first volume term is at least $\theta\|D^2u\|_2^2$. The boundary expression must not be discarded.

Write $q = \partial_\nu u$, $\Pi = I - \nu\otimes\nu$, and $a_\tau = \Pi A\nu$. Since the tangential derivatives of u vanish, $Du = q\nu$, $\Pi D^2u\,\nu = \nabla_\tau q$, and the tangential Hessian is $q\mathcal{S}$, where $\mathcal{S} = D_\tau\nu$ is the shape operator with this sign convention. Cancellation of the pure normal second derivative leaves precisely

$$(A : D^2u)q - q\nu\cdot D^2uA\nu = q^2\,\mathrm{tr}(\Pi A\Pi\mathcal{S}) + q\,a_\tau\cdot\nabla_\tau q.$$

Tangential integration by parts on the closed boundary turns its integral into

$$\int_{\partial U} q^2\left(\mathrm{tr}(\Pi A\Pi\mathcal{S}) - \tfrac{1}{2}\,\mathrm{div}_\tau\, a_\tau\right).$$

The coefficient is bounded by the C^1 norm of A and boundary curvature. A tubular-neighborhood trace estimate gives, for any $\delta > 0$,

$$\|Du\|_{L^2(\partial U)}^2 \le \delta\|D^2u\|_2^2 + C_\delta\|Du\|_2^2.$$

For completeness this follows by writing the boundary value of $|Du|^2$ as its value at an interior normal offset plus the normal integral of its derivative, averaging offsets, and applying Young to $|Du||D^2u|$; a finite collar cover has bounded Jacobians. Thus the boundary term costs only the same small highest-order norm and a lower-order norm. The drift and potential paired with $-\Delta u$ cost $\delta\|D^2u\|_2^2 + C_\delta\|u\|_{H^1}^2$. It follows that

$$(Lu, -\Delta u) \ge c_1\|D^2u\|_2^2 - C_1\|u\|_{H^1}^2.$$

Finally $\|Du\|_2^2 = -(u, \Delta u)$, so $\|Du\|_2^2 \le \delta\|D^2u\|_2^2 + C_\delta\|u\|_2^2$. Absorb this term and add a sufficiently large multiple of $\|u\|_2^2$ to get (7.15). Smooth zero-trace functions are dense in $H^2\cap V$ on a smooth domain; all final volume pairings are continuous in H^2, extending the estimate. Both source inequalities are the same estimate. Our earlier reconstruction proofs avoid needing it inside the Galerkin regularity passage, but it remains a useful independent result. □

10. Telegraph equation uniqueness. For $u_{tt} + du_t - u_{xx} = f$ on $(0,1)$, with zero endpoint values and two prescribed initial data, prove uniqueness for any real constant d.

Solution. For a difference w let $E = \frac{1}{2}\int_0^1 (w_t^2 + w_x^2)$. The endpoint term vanishes since $w_t = 0$ there. Then $E' = -d\int w_t^2 \le 2|d|E$. Zero initial energy and Gronwall imply $E = 0$; zero endpoint values give $w = 0$. For $d < 0$ energy can grow, but uniqueness still holds. □

11. Clamped beam uniqueness. For $u_{tt} + u_{xxxx} = 0$, $u = u_x = 0$ at $0, 1$, prove uniqueness.

Solution. For a difference let $E = \frac{1}{2}\int_0^1 (w_t^2 + w_{xx}^2)$. Twice integrating gives $\int w_{xxxx} w_t = \int w_{xx} w_{txx}$: the boundary terms are $[w_{xxx} w_t - w_{xx} w_{tx}]_0^1 = 0$ since both clamped data are constant in time. Thus $E' = 0$, $E(0) = 0$, and $w_{xx} = 0$. An affine function with zero clamped data is zero. □

12. Resolvent identities. Establish the source identities (12)–(13) in §7.4.1.

Solution. On $D(G)$ the difference of the two operators is $(\mu I - G) - (\lambda I - G) = (\mu - \lambda)I$. Multiplication on the left by R_λ and on the right by R_μ gives $R_\lambda - R_\mu = (\mu - \lambda) R_\lambda R_\mu$. Interchange parameters to get commutation when they differ; equality of parameters is automatic. The intermediate vectors are in $D(G)$, so no multiplication of an undefined unbounded expression occurs. □

13. Passing a closed generator through an integral. For $x \in D(G)$ justify $G\int_0^\infty e^{-\lambda t} S(t)x\,dt = \int_0^\infty e^{-\lambda t} GS(t)x\,dt$, $\lambda > 0$.

Solution. Equip $D(G)$ with $\|y\|_G = \|y\| + \|Gy\|$; closedness makes this a Banach space. The integrand is graph-norm continuous because $GS(t)x = S(t)Gx$, and its graph norm is at most $e^{-\lambda t}(\|x\| + \|Gx\|)$. Riemann sums on finite intervals therefore converge in graph norm, and the tails converge there as well. Applying the continuous map $G : (D(G), \|\cdot\|_G) \to X$ proves the identity. □

零边界不消去全部边界项

$u = 0$ 只消去切向一阶导数，法向导数 q 可不为零。纯法向二阶项抵消后，仍有曲率项和 $q\nabla_\tau q$；须切向分部积分并用带小常数的迹估计吸收。

14. Heat convolution on two different spaces. Let $S(t)g = \Phi_t * g$, $S(0) = I$. Show it is a contraction semigroup on $L^2(\mathbb{R}^n)$, but not a strongly continuous semigroup on $L^\infty(\mathbb{R}^n)$.

Solution. The Gaussian is nonnegative of mass one, so Young's inequality gives $\|S(t)\|_{2\to 2} \le 1$. Gaussian convolution gives $S(t+s) = S(t)S(s)$. On C_c^∞, approximate identities give $S(t)g \to g$ in L^2; density and contraction extend this to all L^2. On L^∞ the same convolution still contracts the norm. The failed axiom is strong continuity: take $g = \mathbf{1}_{\{x_1>0\}}$. Its smoothed value tends to $1/2$ as $x_1 \downarrow 0$, so for every $t > 0$, $\|S(t)g - g\|_\infty = 1/2$ (in the essential-supremum sense). The source's phrase "not a contraction semigroup" refers to this continuity failure, not to failure of the norm inequality. □

图范数使积分合法

只知道被积函数在 X 中可积不足以交换无界算子。同时控制函数及其 G 像，等于在完备图范数空间中积分。

15. Invariance of powers of the generator domain. Define $D(G^k) = \{x \in D(G^{k-1}) : G^{k-1}x \in D(G)\}$. Show $S(t)D(G^k) \subset D(G^k)$.

Solution. For $k = 1$ this is Theorem 7.17. Inductively, for $x \in D(G^k)$, one has $S(t)x \in D(G^{k-1})$ and $G^{k-1}S(t)x = S(t)G^{k-1}x \in D(G)$ by the $k = 1$ case. This proves invariance and also $G^k S(t)x = S(t)G^k x$. □

同一公式依赖所选空间

热卷积在 L^∞ 仍满足 $\|S(t)\| \le 1$，但阶跃函数阻止强连续。不能仅核对卷积律与范数，就宣布得到 C_0 半群。

16. Smooth spatial heat orbits from compactly supported data. For the Dirichlet heat semigroup and $g \in C_c^\infty(U)$, show $S(t)g \in C^\infty(U)$ for every $t \ge 0$.

Solution. Here $G = \Delta$, $D(G) = H^2 \cap V$. Every $\Delta^j g$ is again compactly supported smooth, so $g \in \bigcap_{k\ge 1} D(G^k)$. Problem 15 preserves these domains. Repeated elliptic regularity gives $D(G^k) \subset H^{2k}(U)$: apply the elliptic estimate to $Gu \in D(G^{k-1})$ inductively. For any spatial derivative order r, choose $2k > r + n/2$ and use Sobolev embedding. This proves the claim, including $t = 0$. Abstract domain invariance alone would not prove smoothing of arbitrary L^2 data; that stronger heat property needs additional estimates. □

7.6 References and Proof Dependencies

The primary mathematical source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 7, printed pp. 349–423 (local English PDF pp. 366–440). The Chinese scan corresponds to printed pp. 248–299 (PDF pp. 266–317). Selected comparisons and the source theorem map are recorded in `docs/CHAPTER-07.md`. All English exposition, proof reconstructions, solutions, annotations and diagrams in this chapter are editorial writing.

Evans points to Ladyzhenskaya, J.-L. Lions, Lions–Magenes and Wloka for evolution equations; Davies for Harnack's calculation; Krylov, Ladyzhenskaya–Solonnikov–Uraltseva and Lieberman for further regularity; John and Treves for the Fourier approach; Friedman and Yosida for semigroups. Problem 9 is linked to Brezis–Evans, *Archive for Rational Mechanics and Analysis* 71 (1979), 1–13, and Ladyzhenskaya–Uraltseva. These source bibliography pointers were not independently consulted.

What is proved and what is assumed. The Galerkin constructions, initial-trace passage, integrated-test wave uniqueness, regularity inductions, smooth principal-part Harnack inequality, finite propagation under a smooth optical-distance hypothesis, vanishing viscosity construction, Fourier multiplier bound, generator properties, contraction Hille–Yosida theorem, and both PDE applications have expanded proofs. All 16 source exercises have editorial solutions; Problem 9 retains and estimates the boundary term explicitly. Hilbert weak compactness, Banach contraction, the closed graph theorem, Bochner integration, Fourier Plancherel and finite-dimensional Schur triangularization are prerequisites. Sobolev approximation, traces, extension and embedding come from Chapter 5; the Dirichlet spectral basis, Lax–Milgram and elliptic boundary estimates come from Chapter 6. The local flow-box construction uses classical smooth ODE theory.

Scope boundaries. General rough-coefficient Harnack theory, global smooth optical distances, nonsymmetric L^2 well-posedness under real spectrum alone, and analytic semigroup smoothing are not proved. Our strong maximum proof additionally assumes time-independent principal coefficients. The Fourier uniqueness class is specified explicitly; autonomous semigroups require time-independent coefficients. This chapter has not received independent mathematical review.

Part III

Theory for Nonlinear Partial Differential Equations

Chapter 8

The Calculus of Variations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 8; printed p. 427; PDF page 444.

Chinese reference: 中译本对应章：变分法，书内第 303 页起。

Study edition / 英文重构学习稿。 This chapter follows the source's section order. Parenthetical theorem identifiers refer to Evans's section-local numbering; the displayed numbers belong to these notes. Exposition, proof reconstructions, Chinese annotations, solutions and TikZ drawings are editorial contributions, not verbatim text. The chapter has not received independent mathematical review.

The central question is how an energy produces a PDE. First variations give the equation; coercivity and lower semicontinuity produce minimizers; constraints replace equations by multipliers or variational inequalities. Minimax methods locate further critical points, while invariance yields conservation and monotonicity laws. Unless stated otherwise, U is bounded with smooth boundary and $1 < q < \infty$. A prescribed trace is represented by a fixed Sobolev lift, and all existence claims below require a finite-energy competitor. External regularity and divergence-solving theorems are identified where used.

8.1 Introduction

8.1.1 First variation and the Euler–Lagrange equation

For a smooth Lagrangian $L(p, z, x)$ define

$$I[u] = \int_U L(Du, u, x)\, dx, \qquad u - g \in W_0^{1,q}(U). \tag{8.1}$$

Here $g \in W^{1,q}(U)$ is a lift of the boundary data. An energy is a functional on functions; its derivative is initially a linear functional on variations, not automatically a pointwise differential operator. For a smooth minimizer u and $\phi \in C_c^\infty(U)$, the curve $i(s) = I[u+s\phi]$ has $i'(0) = 0$. Differentiation and integration by parts give

$$\delta I[u](\phi) = \int_U L_p(Du, u, x) \cdot D\phi + L_z(Du, u, x)\phi = 0, \tag{8.2}$$

so the Euler–Lagrange equation is

$$-\operatorname{div} L_p(Du,u,x) + L_z(Du,u,x) = 0. \tag{8.3}$$

For smooth functions and compactly supported variations, the derivatives are bounded on the compact ranges involved; this justifies differentiating inside the integral. The Sobolev justification appears in §8.2.

Expanding the equation shows its quasilinear principal part:

$$-L_{p_ip_j}u_{ij} - L_{p_iz}u_i - L_{p_ix_i} + L_z = 0.$$

Thus convexity of L in the gradient variable determines ellipticity. The standard examples, in the source's order, are

$L(p,z,x)$	Euler–Lagrange equation
$\frac{1}{2}\lvert p\rvert^2$	$-\Delta u = 0$
$\frac{1}{2}a^{ij}(x)p_ip_j - f(x)z$	$-\partial_i(a^{ij}u_j) = f$
$\frac{1}{2}\lvert p\rvert^2 - F(z),\ F' = f$	$-\Delta u = f(u)$
$(1+\lvert p\rvert^2)^{1/2}$	$\operatorname{div}\dfrac{Du}{\sqrt{1+\lvert Du\rvert^2}} = 0$

For the second example $A = A^T$; no arbitrary drift is generated by a quadratic unweighted Dirichlet energy. The last integrand measures graph area. Its Hessian has eigenvalues $(1+|p|^2)^{-1/2}$ transverse to p and $(1+|p|^2)^{-3/2}$ parallel to p: it is strictly convex, but not uniformly convex on all gradient space.

8.1.2 Second variation and the Legendre condition

For a smooth local minimizer, $i''(0) \geq 0$, hence

$$Q_u[\phi] = \int_U L_{p_ip_j}\phi_i\phi_j + 2L_{p_iz}\phi_i\phi + L_{zz}\phi^2\,dx \geq 0. \tag{8.4}$$

All coefficients are evaluated at (Du,u,x).

Theorem 8.1 (Necessary gradient convexity; source §8.1.3)
For every $x \in U$ and $\xi \in \mathbb{R}^n$, $L_{p_ip_j}(Du(x),u(x),x)\xi_i\xi_j \geq 0$.

Expanded proof (small amplitude, oscillatory gradient).

Let ρ be the periodic triangular function of period one, with $|\rho'| = 1$ a.e., and take $\phi_\varepsilon = \varepsilon\rho(x\cdot\xi/\varepsilon)\zeta(x)$, $\zeta \in C_c^\infty(U)$. Smooth approximation makes (8.4) valid for this compactly supported Lipschitz function. Its value is $O(\varepsilon)$ and $D\phi_\varepsilon = \rho'\xi\zeta + O(\varepsilon)$, uniformly on the fixed support except at the harmless corners. The mixed and zeroth-order terms vanish as $\varepsilon \to 0$, while $(\rho')^2 = 1$ a.e. gives $\int L_{p_ip_j}\xi_i\xi_j\zeta^2 \geq 0$. If this continuous

偏导数与全导数

L_{x_i} 只对显式的 x 变量求偏导；$\partial_i[L_{p_i}(Du,u,x)]$ 还包含 D^2u 与 Du 的链式项。Noether 公式中也必须保持这个区别。

极小曲面的退化

面积密度在大梯度方向的最小曲率约为 $|p|^{-3}$。严格凸不能代替一致椭圆；面积能量还只有线性增长，不属于后面的超线性强制性框架。

coefficient were negative at a point, choosing ζ in a small neighborhood would contradict the inequality. This is positivity along the actual gradient of u, not global convexity of L at all (p, z, x), and it is necessary rather than sufficient for a minimum. □

小振幅仍有大梯度

$\phi_\varepsilon = O(\varepsilon)$，但 $D\phi_\varepsilon = O(1)$。高频测试分离出二阶变分的主部；仅看函数值的接近不能忽略梯度振荡。

8.1.3 Systems, null Lagrangians, and the Piola identity

For $u : U \to \mathbb{R}^m$ write $P_{\alpha i} = \partial_i u^\alpha$. The first variation uses the Frobenius product $A : B = \sum_{\alpha,i} A_{\alpha i} B_{\alpha i}$:

$$\delta I[u](\phi) = \int_U L_P : D\phi + L_z \cdot \phi, \qquad -\partial_i L_{P_{\alpha i}} + L_{z_\alpha} = 0.$$

A null Lagrangian is one whose Euler–Lagrange expression vanishes for every smooth map. It need not have constant density or zero integral.

Theorem 8.2 (Boundary dependence; source §8.1.4, Theorem 1)
If L is null and $u, v \in C^2(\overline{U}; \mathbb{R}^m)$ have the same boundary values, then $I[u] = I[v]$.

Expanded proof (a homotopy with fixed trace).

Put $u_s = v + s(u - v)$ for $0 \le s \le 1$. Its energy derivative is $\int L_P(Du_s, u_s, x) : D(u - v) + L_z(Du_s, u_s, x) \cdot (u - v)$. Integration by parts has zero boundary term because $u - v = 0$ there. The remaining Euler–Lagrange expression vanishes by nullity, so $\frac{d}{ds} I[u_s] = 0$. Integrating in s proves the claim. □

Lemma 8.3 (Piola identity; source divergence-free rows lemma)
For smooth $u : U \to \mathbb{R}^n$, with the cofactor matrix in the same row-column convention as Du,

$$\partial_i (\operatorname{cof} Du)_{\alpha i} = 0. \tag{8.5}$$

Consequently $L(P) = \det P$ is a null Lagrangian (source Theorem 2).

Expanded proof (antisymmetry rather than invertibility).

The entry $(\operatorname{cof} Du)_{\alpha i}$ is

$$\frac{1}{(n-1)!} \epsilon_{\alpha \alpha_2 \cdots \alpha_n} \epsilon_{i i_2 \cdots i_n} u^{\alpha_2}_{i_2} \cdots u^{\alpha_n}_{i_n}.$$

Differentiate in i and sum. In each resulting term the second derivative is symmetric in i, i_j, whereas the spatial alternating symbol is antisymmetric in those indices; paired terms cancel. For $n = 1$ the cofactor is the constant 1. Since $\partial(\det P)/\partial P_{\alpha i} = (\operatorname{cof} P)_{\alpha i}$, (8.5) is precisely the Euler–Lagrange identity. No division by $\det Du$ occurs, so singular gradients are included. □

In two dimensions it reads $\operatorname{cof} Du = \begin{pmatrix} u^2_y & -u^2_x \\ -u^1_y & u^1_x \end{pmatrix}$; each row divergence is equality of mixed partial derivatives. All minors enjoy analogous cancellation, a fact used in the weak continuity argument below.

余子式的转置约定

这里 Du 的行是分量、列是空间方向；Piola 恒等式对每行取散度。矩阵恒等式为 $P(\operatorname{cof} P)^T = (\det P)I$，不能混用伴随矩阵与余子式矩阵。

8.1.4 An analytic route to Brouwer's fixed point theorem

Theorem 8.4 (Brouwer; source §8.1.4, Theorem 3)
Every continuous map $F : \overline{B} \to \overline{B}$, $B = B(0,1) \subset \mathbb{R}^n$, has a fixed point.

Expanded proof (no retraction, including the smoothing step).

A smooth retraction $r : \overline{B} \to \partial B$ would have the same boundary data as the identity. Nullity gives $\int_B \det Dr = |B|$. But $|r|^2 = 1$ implies $(Dr)^T r = 0$, so $\det Dr = 0$, a contradiction. If a continuous retraction existed, extend it by $r(x) = x$ outside B. It is nonzero everywhere and equals the identity outside a compact set. Convolution with an even radial mollifier gives smooth r_ε that remains nonzero for sufficiently small ε: uniform approximation on a fixed compact ball preserves the positive minimum of $|r|$, and farther away convolution reproduces x exactly. For $|x| \geq 2$, $r_\varepsilon(x) = x$ if $\varepsilon < 1$. Thus $2r_\varepsilon/|r_\varepsilon|$ is a smooth retraction from $\overline{B(0,2)}$ onto its boundary, the same contradiction after rescaling. Finally, if F had no fixed point, the ray from $F(x)$ through x would meet the sphere in a continuous point $r(x)$ with $r(x) = x$ on ∂B. Explicitly, for $a = F(x)$ and $d = x - a \neq 0$, take

$$r(x) = a + \lambda d, \qquad \lambda = \frac{-a \cdot d + \sqrt{(a \cdot d)^2 + |d|^2(1 - |a|^2)}}{|d|^2}.$$

The forward exit parameter is at least one; the formula is continuous because $d \neq 0$. It supplies the forbidden continuous retraction. □

8.2 Existence of Minimizers

8.2.1 Coercivity and the direct method

Fix $g \in W^{1,q}(U;\mathbb{R}^m)$ and the affine class $\mathscr{A} = g + W^{1,q}_0(U;\mathbb{R}^m)$, $1 < q < \infty$. The direct method separates three tasks: boundedness of minimizing sequences, closure of the admissible class under the chosen convergence, and a lower bound for the energy at the limit. Suppose

$$L(P,z,x) \geq \alpha|P|^q - \beta, \qquad \alpha > 0,\ \beta \geq 0. \tag{8.6}$$

A bounded energy sequence then has bounded Du_k in L^q. Poincaré applied to $u_k - g$ bounds u_k in $W^{1,q}$. Reflexivity gives a weakly convergent subsequence. The affine class is weakly closed: the continuous trace map has weakly closed kernel, or use that a norm-closed linear subspace is weakly closed by Hahn–Banach. The missing issue is weak lower semicontinuity:

$$u_k \rightharpoonup u \text{ in } W^{1,q} \quad \Longrightarrow \quad I[u] \le \liminf_k I[u_k].$$

Strong norm continuity alone does not imply this inequality.

三个条件各做什么

强制性给出有界极小化序列；反射性给出弱子列；弱闭性保住边界和约束；弱下半连续性才保证极限达到下确界。四步不能互相替代。

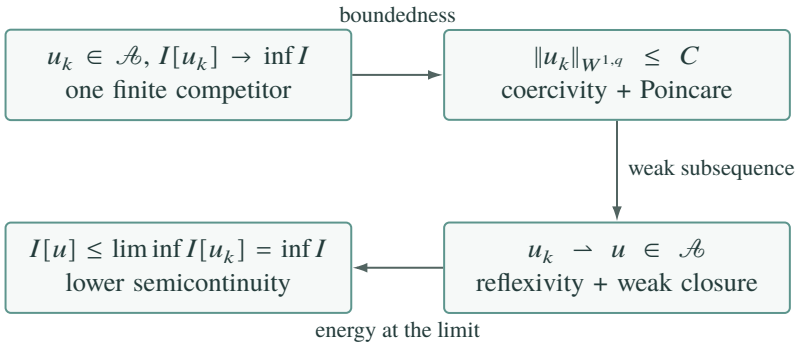


Diagram T8.1. The direct method keeps feasibility and energy in the limit. The arrows represent separate arguments. A finite-energy competitor makes the infimum meaningful; coercivity and Poincare give boundedness. Weak closure preserves constraints, while lower semicontinuity supplies the minimizing inequality.

8.2.2 Convexity in the gradient and weak lower semicontinuity

Theorem 8.5 (Lower semicontinuity; source §8.2.2, Theorem 1)
Let L be smooth on gradient and value space times $\overline{U}$, bounded below, and convex in P for each (z, x). Then I is sequentially weakly lower semicontinuous on $W^{1,q}(U;\mathbb{R}^m)$. No convexity in z is required. The scalar case is the source Theorem 1; the vector version is used in Theorem 5.

Expanded proof (linearizing only the weakly convergent variable).

Add a constant so $L \ge 0$. Pass to a subsequence realizing the liminf; if it is infinite there is nothing to prove. Compact embedding gives $u_k \to u$ strongly in L^q, and a further subsequence converges a.e. For $\varepsilon > 0$, Egoroff gives a set E_ε where this convergence is uniform, with $|U \setminus E_\varepsilon| < \varepsilon$. Intersect it with $\{|u| + |Du| \le \varepsilon^{-1}\}$ to obtain G_ε, whose complement has measure tending to zero. Convexity gives

$$\begin{aligned} I[u_k] \ge & \int_{G_\varepsilon} L(Du, u_k, x) \\ & + \int_{G_\varepsilon} L_P(Du, u_k, x) : (Du_k - Du). \end{aligned}$$

On this set all arguments except Du_k lie in a fixed compact range. The first integral converges uniformly in its integrand. In the second, $L_P(Du, u_k, x)$ converges uniformly to $L_P(Du, u, x)$. Its difference pairs with a uniformly bounded L^q sequence and tends to zero by Hölder; its fixed limit pairs to zero by weak convergence. Thus $\liminf I[u_k] \geq \int_{G_\varepsilon} L(Du, u, x)$. To exhaust U without assuming these sets are nested, fix $R < \infty$:

$$\int_{G_\varepsilon} L(Du, u, x) \geq \int_U \min\{L(Du, u, x), R\} - R|U \setminus G_\varepsilon|.$$

Let $\varepsilon \downarrow 0$, then $R \uparrow \infty$ by monotone convergence. This also covers an infinite limit energy. □

Theorem 8.6 (Existence and uniqueness; source Theorems 2–3, 5–6)
Under (8.6) and Theorem 8.5, if $\mathcal{A}$ contains w_0 with $I[w_0] < \infty$, then I has a finite minimizer in $\mathcal{A}$. If $L = L(P, x)$ is independent of z and uniformly convex in P, this minimizer is unique. These statements apply to scalar and vector maps.

Expanded proof (the limit, and the midpoint deficit).

Choose $I[u_k] \downarrow \inf_{\mathcal{A}} I$, bounded above by $I[w_0] + 1$ and below by $-\beta|U|$. Coercivity and Poincaré give a weakly convergent subsequence, with limit in $\mathcal{A}$. Lower semicontinuity gives $I[u] \leq \inf_{\mathcal{A}} I$; the reverse inequality follows from feasibility. This proves existence. For two minimizers u, v, uniform convexity with constant $\theta > 0$ gives

$$I[(u + v)/2] + \frac{\theta}{8} \int_U |Du - Dv|^2 \leq \frac{1}{2}(I[u] + I[v]).$$

This follows by applying the supporting quadratic inequality at the midpoint and adding the two endpoint inequalities; the linear terms cancel. Since the midpoint is feasible, minimality forces $Du = Dv$. Poincaré for $u - v \in W_0^{1,q}$ then gives $u = v$. A merely convex dependence on P with arbitrary z dependence need not give uniqueness: the value variable may introduce several wells. □

The finite-competitor condition is stated explicitly here; otherwise a formal minimizer with energy $+\infty$ says nothing about the PDE.

为什么只对梯度要求凸

Rellich 给出 u_k 的强收敛，而 Du_k 通常只有弱收敛。支撑超平面把 Du_k 变成线性出现，正好能使用弱收敛；不能假定梯度有逐点收敛子列。

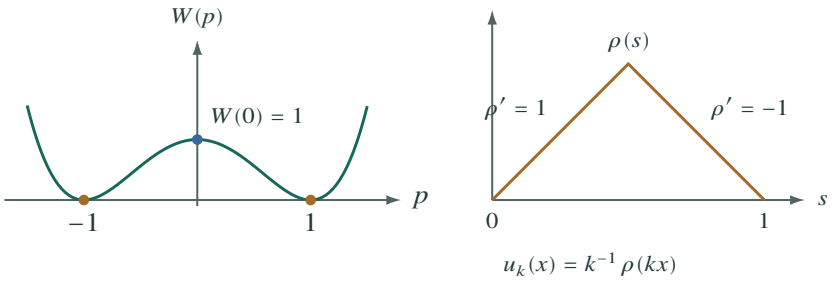


Diagram T8.2. Gradient oscillations test the need for convexity. The left graph is $W(p) = (p^2 - 1)^2$. A zero-trace triangular profile on $(0,1)$ can have slopes $+1, -1$ while its amplitude tends to zero. Its energy is zero, whereas the limit profile has energy $W(0) = 1$. The right profile is one exact period, before multiplying its horizontal and vertical scales by $1/k$.

8.2.3 A Sobolev minimizer solves the Euler–Lagrange equation

Assume the additional growth bounds, with $q' = q/(q-1)$,

$$\begin{aligned} |L(P,z,x)| &\le C(1 + |P|^q + |z|^q), \\ |L_P| + |L_z| &\le C(1 + |P|^{q-1} + |z|^{q-1}). \end{aligned} \tag{8.7}$$

Theorem 8.7 (Weak Euler–Lagrange equation; source Theorems 4 and 7)
A minimizer $u \in \mathcal{A}$ satisfies

$$\int_U L_P(Du,u,x) : D\phi + L_z(Du,u,x) \cdot \phi = 0 \quad (\phi \in W_0^{1,q}(U;\mathbb{R}^m)).$$

Expanded proof (a difference quotient under an integrable bound).

For $|s| \le 1$, the fundamental theorem of calculus expresses $[L(Du + sD\phi, u + s\phi, x) - L(Du,u,x)]/s$ as an average of derivatives paired with $(D\phi, \phi)$. Its absolute value is bounded by

$$C(1 + |Du|^{q-1} + |u|^{q-1} + |D\phi|^{q-1} + |\phi|^{q-1})(|D\phi| + |\phi|),$$

which lies in L^1 by Hölder. Dominated convergence proves the first variation formula for every admissible ϕ. Both signs of s are allowed, so minimality makes the derivative zero. The coefficients $L_P(Du,u,x), L_z(Du,u,x)$ belong to $L^{q'}$, making this a continuous weak identity. Without the growth bound, differentiability of the energy on the whole Sobolev class has not been justified. □

8.2.4 Weak continuity of minors and polyconvexity

Lemma 8.8 (Weak determinant continuity; source §8.2.4)
If $q > n$ and $u_k \rightharpoonup u$ in $W^{1,q}(U;\mathbb{R}^n)$, then $\det Du_k \rightharpoonup \det Du$ in $L^{q/n}(U)$. More generally every r-minor converges weakly in $L^{q/r}$, $1 \le r \le n$.

Expanded proof (induction on minor order).

The gradients are bounded in L^q and Morrey plus compact embedding gives $u_k \to u$ uniformly after extraction. An r-minor is bounded in $L^{q/r}$ by the polynomial estimate $|M_r(Du_k)| \le C|Du_k|^r$. Its Laplace expansion along one fixed component can be written in divergence form, with that component undifferentiated and its $(r-1)$-cofactors inside the flux. The other derivative terms cancel by the same alternating-index argument as Piola. For a compactly supported smooth scalar test ψ, integration by parts therefore pairs $u_k^\alpha D\psi$, converging uniformly, with $(r-1)$-minors. By induction the latter converge weakly in $L^{q/(r-1)}$. Thus the r-minors converge in distributions; the base $r = 1$ is weak gradient convergence. Since $q/r > 1$, weak compactness and density of smooth tests identify every weak subsequential limit with $M_r(Du)$. This identifies the limit of the original sequence as well. The same divergence identity extends to Sobolev maps by local smoothing and Hölder, so it has not been used outside its legitimate class. □

The source uses the restricted polyconvex form

$$L(P,z,x) = F(P,\det P,z,x), \qquad (P,r) \mapsto F(P,r,z,x) \text{ convex.}$$

The standard broader definition permits all minors. We use the displayed restricted form, not an unmentioned change of definition in higher dimensions.

Theorem 8.9 (Polyconvex direct method; source Theorems 8–9)
If $q > n$, L is bounded below, and the smooth representation above holds, then I is weakly lower semicontinuous. If also (8.6) holds and a finite competitor exists, there is a minimizer.

Expanded proof (a supporting plane in an enlarged space).

Use Lemma 8.8 to get simultaneous weak convergence of Du_k in L^q and $\det Du_k$ in $L^{q/n}$, and uniform convergence of u_k along a subsequence. On a good set where $|u| + |Du|$ is bounded, apply the convex supporting inequality to F in the independent variables (P,r) at $(Du,\det Du)$. The

解方程与达到极小

一阶变分为零只给临界点。若 L 对 (P,z) 联合凸，则支撑不等式配合弱方程可推出全局极小；仅对 P 凸而对 z 非凸时，这个逆命题会失败。

非线性行列式的弱连续

一般弱收敛不能交换非线性映射。行列式的散度结构把一个因子变为强收敛的 u_k，剩余余子式逐阶归纳；$q > n$ 还保证 $L^{q/n}$ 反射。

two linear errors pair with the respective weakly convergent sequences; the coefficients converge uniformly and are bounded there. The argument of Theorem 8.5, including its truncated exhaustion of the domain, proves the inequality. Coercivity and weak closure finish the direct method. For example, $L(P) = |P|^4 + (\det P)^2$ for 2×2 matrices is of this form with $q = 4$. Polyconvexity does not mean convexity of the nonlinear map $P \mapsto (P, \det P)$. □

8.2.5 Local minimizers embedded in a field of extremals

Assume a smooth family $u(x, \lambda)$ solves the scalar Euler–Lagrange equation for each λ in an interval containing zero, with $u(x, 0) = u_0(x)$. Suppose L is convex in p. For a smooth $\vartheta : \overline{U} \to \mathbb{R}$ taking values in that interval, with $\vartheta = 0$ on ∂U, put $w(x) = u(x, \vartheta(x))$.

Theorem 8.10 (Field of extremals; source §8.2.5, Theorem 10)
For every such w, $I[w] \geq I[u_0]$. If u_λ is bounded away from zero near $\lambda = 0$, the inverse function theorem represents all sufficiently C^0-close smooth competitors with the same trace in this form.

Expanded proof (a calibration flux).

Here $D_x u$ denotes differentiation with λ fixed. The chain rule gives $Dw = D_x u + u_\lambda D\vartheta$ at $\lambda = \vartheta(x)$. Convexity bounds $L(Dw, w, x)$ below by $L(D_x u, w, x) + u_\lambda L_p(D_x u, w, x) \cdot D\vartheta$. Define the vector field

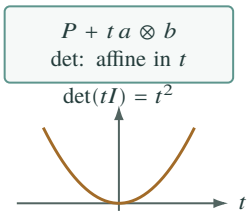
$$b_i(x) = \int_0^{\vartheta(x)} u_\lambda(x, s) L_{p_i}(D_x u(x, s), u(x, s), x)\, ds.$$

Differentiating the upper endpoint and using the Euler–Lagrange equation inside the integral gives

$$\begin{aligned} \operatorname{div} b &= u_\lambda L_p \cdot D\vartheta + \int_0^{\vartheta} (L_p \cdot D_x u_\lambda + L_z u_\lambda)\, ds \\ &= u_\lambda L_p \cdot D\vartheta + L(D_x u(x, \vartheta), w, x) - L(Du_0, u_0, x). \end{aligned}$$

Thus $I[w] \geq I[u_0] + \int_U \operatorname{div} b = I[u_0]$, since $b = 0$ on the boundary. The last assertion follows by a uniform local inverse in λ on the compact closed domain. Without that transversality, the theorem compares only competitors of the stated form. □

图 T8.M：秩一方向与全矩阵方向



对 2×2 矩阵，$t \mapsto \det(P + t\,a \otimes b)$ 是仿射函数，但 $t \mapsto \det(tI) = t^2$。秩一方向上的条件不等于整个矩阵空间中的凸性。

8.3 Regularity

8.3.1 Nonlinear difference quotients

Consider a scalar weak solution $u \in H^1(U)$ of

$$-\operatorname{div} DL(Du) = f, \qquad f \in L^2(U), \tag{8.8}$$

where $L \in C^2(\mathbb{R}^n)$ and

$$\theta|\xi|^2 \le D^2L(p)\xi\cdot\xi \le \Lambda|\xi|^2 \quad (p,\xi\in\mathbb{R}^n), \qquad 0<\theta\le\Lambda. \tag{8.9}$$

Subtracting the affine function $DL(0)\cdot p$ from L leaves the weak equation unchanged, because $\int_U DL(0)\cdot D\phi = 0$ for $\phi \in H^1_0$. We may therefore assume $DL(0)=0$, giving $|DL(p)| \le \Lambda|p|$.

Theorem 8.11 (Second derivatives; source §8.3.1, Theorem 1)
For $V \Subset U$, $u \in H^2(V)$ and

$$\|D^2u\|_{L^2(V)} \le C\bigl(\|f\|_{L^2(U)} + \|Du\|_{L^2(U)}\bigr).$$

If $u \in H^1_0(U)$ and ∂U is C^2, then $u \in H^2(U)$ and $\|u\|_{H^2(U)} \le C\|f\|_2$. Constants depend on θ, Λ, the domains and their indicated geometry.

Expanded proof (uniformly elliptic averaged Hessians).

Choose $V \Subset W \Subset U$ and a cutoff ζ supported in W, equal to one on V. For a small step in direction e_k let $\delta_h v = (v(x+he_k)-v(x))/h$ and use the legal test $\phi = -\delta_{-h}(\zeta^2\delta_h u)$. It belongs to $H^1_0(U)$ because its support stays away from the boundary. Discrete integration by parts gives

$$\int_U \delta_h DL(Du)\cdot D(\zeta^2\delta_h u) = -\int_U f\,\delta_{-h}(\zeta^2\delta_h u).$$

The exact nonlinear linearization is

$$\delta_h DL(Du) = A_h\,\delta_h Du, \quad A_h = \int_0^1 D^2L(Du + s(Du(x+he_k)-Du))\,ds.$$

Hence $\theta I \le A_h \le \Lambda I$ independently of h. Expanding the cutoff derivative, the principal integral bounds $\theta\|\zeta\,\delta_h Du\|_2^2$. Its cross term costs $\varepsilon\|\zeta\,\delta_h Du\|_2^2 + C_\varepsilon\|\delta_h u\|_2^2$. The forcing is controlled without differentiating f:

$$\begin{aligned}\Bigl|\int f\,\delta_{-h}(\zeta^2\delta_h u)\Bigr| &\le C\|f\|_2\|D(\zeta^2\delta_h u)\|_2\\ &\le \varepsilon\|\zeta\,\delta_h Du\|_2^2 + C_\varepsilon(\|f\|_2^2 + \|Du\|_2^2).\end{aligned}$$

Use $\|\delta_h u\|_2 \le \|Du\|_2$ on a slightly larger set and absorb with $2\varepsilon<\theta$. The difference-quotient criterion from §5.8 then gives the interior estimate. □

平均 Hessian 保留椭圆性

A_h 是沿两点梯度连线的 Hessian 平均；一致上下界在平均后不变。不能只在某个未经控制的中间点套用局部凸性。

Expanded proof (boundary and normal estimates).

This expands the boundary argument left to the reader in the source. In a boundary chart write $x = \Phi(y) = (y', y_n + \gamma(y'))$, $v = u \circ \Phi$. Its Jacobian is one and $v = 0$ on $y_n = 0$. The transformed flux is

$$a(y,p) = D\Phi^{-1}\, DL(D\Phi^{-T}p), \qquad -\partial_{y_i} a_i(y, Dv) = \tilde{f}.$$

Bounded $D\Phi, D\Phi^{-1}$ preserve uniform monotonicity in p; bounded $D^2\Phi$ gives $|D_y a(y,p)| \le C|p|$, since $DL(0) = 0$. Tangential difference quotients now work on the half-chart. Their tests have zero trace because v and its tangential translates have zero trace. Besides the averaged Hessian term, differencing a produces a coefficient error bounded by $C|Dv|$, which is absorbed as above. Thus all $v_{i\alpha}$, $\alpha < n$, are in L^2, with bounds by $\|\tilde{f}\|_2 + \|Dv\|_2$.

To recover v_{nn} without assuming it exists, let $w = a_n(y, Dv)$. The known tangential derivatives and the flux chain rule give $\partial_\alpha a_i(y, Dv) \in L^2$. The equation gives $\partial_n w = -\tilde{f} - \sum_{\alpha<n} \partial_\alpha a_\alpha \in L^2$. For fixed (y, p'), $p_n \mapsto a_n(y, p', p_n)$ is strictly increasing with derivative at least $c > 0$ and at most C; it maps $\mathbb{R}$ onto $\mathbb{R}$. Its inverse $p_n = \Psi(y, p', w)$ satisfies

$$|\Psi_w| + |\Psi_{p'}| \le C, \qquad |\Psi_y| \le C(|p'| + |\Psi|).$$

Apply normal difference quotients to $v_n = \Psi(y, D'v, w)$. The L^2 bounds for the normal derivatives of $D'v$ and w give an L^2 bound for the normal difference quotients of v_n. The criterion proves $v_{nn} \in L^2$; this step avoids a circular classical expansion of an equation before H^2 has been established. Cutoffs, a finite chart cover and the interior estimate yield $\|u\|_{H^2} \le C(\|f\|_2 + \|u\|_{H^1})$. Finally testing the original equation by u, monotonicity and Poincaré give

$$\theta\|Du\|_2^2 \le \int_U DL(Du)\cdot Du = (f,u) \le C\|f\|_2\|Du\|_2.$$

This removes the lower-order norm and proves the global estimate. □

8.3.2 Why further smoothness needs a new theorem

For $f = 0$, interior H^2 and the Lipschitz chain rule imply that $v_k = u_{x_k}$ weakly solves

$$-\partial_i(A^{ij}(x)\partial_j v_k) = 0, \qquad A^{ij}(x) = L_{p_i p_j}(Du(x)).$$

These coefficients are only bounded measurable at this stage. The smooth coefficient estimates of Chapter 6 cannot be applied to them. The scalar De Giorgi–Nash–Moser theorem, used here as an external regularity result, states that H^1_{loc} solutions of a uniformly elliptic divergence equation with bounded measurable coefficients are locally Hölder continuous. It gives $Du \in C^{0,\alpha}_{\mathrm{loc}}$. For smooth L, A is then Hölder. The external Schauder estimates for linear divergence equations improve v_k to $C^{1,\alpha}_{\mathrm{loc}}$; iteration improves coefficient and solution regularity together, yielding $u \in C^\infty(U)$. This is a conditional bootstrap using those two named theorems, not a proof of either deep estimate. For systems the scalar Hölder theorem is unavailable in general; smoothness cannot be asserted by the same argument.

法向导数的非循环恢复

先证明切向二阶导数，再由方程得到法向通量的弱导数，最后利用 $p_n \mapsto a_n$ 的一致单调可逆性恢复 u_{nn}。不能在已知 $u \in H^1$ 时直接展开二阶链式法则。

8.4 Constraints

8.4.1 Integral constraints and a scalar multiplier

On connected U, minimize $I[u] = \frac{1}{2}\int_U |Du|^2$ over

$$\mathcal{A} = \{u \in H^1_0(U) : J[u] = 0\}, \qquad J[u] = \int_U G(u).$$

Assume $g = G'$ satisfies $|g(z)| \le C(1 + |z|)$. Then $|G(z)| \le C(1 + z^2)$, so the constraint is well defined.

Theorem 8.12 (Constrained existence; source §8.4.1, Theorem 1)
If $\mathcal{A} \neq \emptyset$, it contains an energy minimizer.

Expanded proof (compactness preserves the nonlinear integral).

An energy-bounded minimizing sequence is bounded in H^1_0. After extraction $u_k \rightharpoonup u$ in H^1_0 and $u_k \to u$ in L^2. The mean value theorem and Cauchy–Schwarz give

$$|J[u_k] - J[u]| \le C\|u_k - u\|_2(|U|^{1/2} + \|u_k\|_2 + \|u\|_2) \longrightarrow 0.$$

Thus u is feasible, and weak lower semicontinuity of the Dirichlet norm makes it minimizing. Every member of this particular class has finite energy. □

线性化不意味着光滑系数

$A(x) = D^2L(Du(x))$ 的正则性由未知梯度决定。先得到梯度的 Hölder 连续性，才能启动 Schauder 提升；多次形式求导不能绕过这一步。

Theorem 8.13 (Lagrange multiplier; source Theorem 2)
For a minimizer u there is $\lambda \in \mathbb{R}$ such that

$$\int_U Du \cdot D\phi = \lambda \int_U g(u)\phi \quad (\phi \in H_0^1(U)). \tag{8.10}$$

Expanded proof (an admissible curve and the degenerate case).

First suppose $g(u) \not\equiv 0$. By density choose $\psi \in H_0^1$ with $b = \int_U g(u)\psi \neq 0$. For any $\phi \in H_0^1$, set $j(s,t) = J[u + s\phi + t\psi]$. Its derivatives are continuous by the linear growth bound on g and dominated convergence on this finite parameter plane. Since $j_t(0,0) = b$, the ordinary implicit function theorem supplies $t = \eta(s)$ with $j(s,\eta(s)) = 0$, $\eta(0) = 0$, and $\eta'(0) = -\int g(u)\phi/b$. Minimality on this curve gives $\int Du \cdot D\phi + \eta'(0) \int Du \cdot D\psi = 0$. Thus (8.10) holds with $\lambda = (\int Du \cdot D\psi)/b$, independent of ϕ.

If $g(u) = 0$ a.e., the chain rule with truncation gives $G(u) \in W^{1,1}$ and $D(G(u)) = g(u)Du = 0$. The product is integrable by Cauchy–Schwarz. Connectedness makes $G(u)$ constant, and $J[u] = 0$ makes it zero. To identify its boundary value, approximate u in H_0^1 by compactly supported smooth functions: the linear growth of g implies convergence of their compositions in $W^{1,1}$, so $G(u) - G(0)$ has zero trace. Hence $G(0) = 0$ and zero is feasible. Minimality gives $I[u] = 0$, whence $u = 0$. Both sides of (8.10) vanish for every λ. This final conclusion is special to this energy and constraint, not a universal multiplier theorem without a constraint qualification. □

退化约束不能直接除法

只有 $J'[u] \neq 0$ 时才能用隐函数定理并除以 $\int g(u)\psi$。本例退化时另证 $u = 0$；一般约束问题不保证退化点存在普通 Lagrange 乘子。

8.4.2 The obstacle problem and a variational inequality

Let $f \in L^2(U)$, let h be a smooth obstacle, and assume $K = \{w \in H_0^1(U) : w \geq h \text{ a.e.}\} \neq \emptyset$. Minimize $I[w] = \frac{1}{2}\|Dw\|_2^2 - (f, w)$.

Theorem 8.14 (Obstacle minimizer; source §8.4.2, Theorems 3–4)
There is a unique minimizer $u \in K$, characterized equivalently by

$$\int_U Du \cdot D(w - u) \geq \int_U f(w - u) \qquad (w \in K). \tag{8.11}$$

Expanded proof (one-sided variations and the converse).

Young and Poincaré imply $I[w] \geq \frac{1}{4}\|Dw\|_2^2 - C\|f\|_2^2$. Weak limits of minimizing sequences stay in K: strong L^2 convergence has an a.e. convergent subsequence preserving $w \geq h$. The direct method gives existence. The midpoint identity gives strict convexity on H_0^1, hence uniqueness. For $w \in K$, the segment $u + s(w - u) \in K$ for $0 \leq s \leq 1$. Its

right energy derivative at zero is nonnegative, giving (8.11). Conversely, expand the quadratic energy exactly:

$$I[w] - I[u] = \int_U Du \cdot D(w-u) - f(w-u) + \tfrac{1}{2}\|D(w-u)\|_2^2 \geq 0.$$

This proves equivalence, including sufficiency of the inequality. □

For every nonnegative $\phi \in C_c^\infty(U)$, $u + \phi \in K$ gives $\mu = -\Delta u - f \geq 0$ in distributions. If u is continuous, then $O = \{u > h\}$ is open; on a compact subset of O both signs of a small variation are allowed, so $-\Delta u = f$ there. If also $u \in H^2_{\mathrm{loc}}$, then μ is a function and

$$u - h \geq 0, \quad \mu \geq 0, \quad (u-h)\mu = 0 \quad \text{a.e.}$$

Interior continuity and $C^{1,1}_{\mathrm{loc}}$ regularity for smooth obstacle and forcing require external obstacle regularity theory; they are not proved by the direct method above. No global $W^{2,\infty}$ estimate is claimed without specifying its boundary compatibility hypotheses. The free boundary is $\partial\{u > h\} \cap U$.

接触集与反力

约束只允许向上变化，故反力 $\mu = -\Delta u - f$ 非负。在非接触区可双向变化，反力为零。可行集、反力和互补条件要使用同一符号约定。

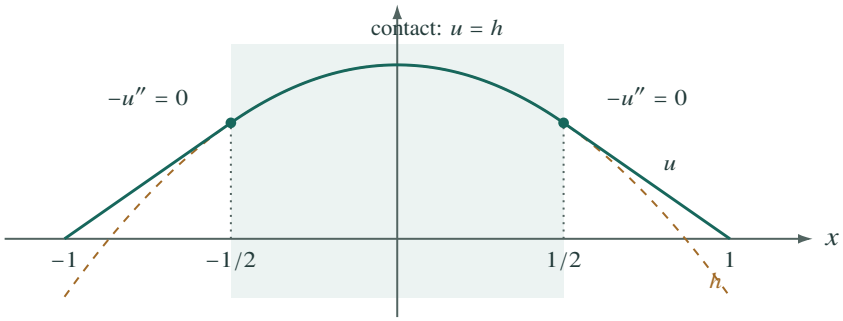


Diagram T8.3. An exact obstacle solution with two free-boundary points. On $(-1,1)$ take $f = 0$ and $h(x) = 1 - 4x^2/3$. The minimizer is $u = h$ on $|x| \leq 1/2$, and $u = 4(1-|x|)/3$ outside. Values and first derivatives match at $\pm 1/2$, so $-u'' = 8/3$ on the contact interval and zero elsewhere, with no point masses.

8.4.3 Pointwise constraints: harmonic maps

Consider $\mathcal{A} = \{u \in H^1(U;\mathbb{R}^m) : u - g \in H^1_0,\ |u| = 1 \text{ a.e.}\}$, and $I[u] = \frac{1}{2}\int_U |Du|^2$. A minimizing sequence is bounded in H^1 ($|u| = 1$ fixes its L^2 norm). A strong L^2 subsequence preserves the sphere constraint a.e.; the trace condition is weakly closed, so a minimizer exists if the class is nonempty.

Theorem 8.15 (Harmonic map equation; source §8.4.3, Theorem 5)
A minimizer satisfies

$$\int_U Du : D\phi = \int_U |Du|^2 u \cdot \phi \quad (\phi \in H_0^1(U;\mathbb{R}^m) \cap L^\infty(U;\mathbb{R}^m)). \tag{8.12}$$

Expanded proof (normalization generates tangent variations).

For bounded ϕ, $u_s = (u + s\phi)/|u + s\phi|$ is well defined for $|s|\|\phi\|_\infty < 1/2$. Sobolev composition and traces show $u_s \in \mathcal{A}$. Its derivative at zero in H^1 is $\dot{u}_0 = \phi - (u \cdot \phi)u$. To justify this derivative, derivatives of the normalization map are uniformly bounded on $|u + s\phi| \geq 1/2$; its spatial derivative and parameter derivative are dominated in L^2 by $C(|D\phi| + |Du|\,\|\phi\|_\infty)$. Thus differentiating the energy is legitimate. Since $u \cdot \partial_i u = 0$ a.e.,

$$Du : D((u \cdot \phi)u) = |Du|^2 (u \cdot \phi).$$

The vanishing derivative is precisely (8.12). The multiplier is the function $|Du|^2 \in L^1$, explaining why bounded variations appear in the weak formulation. □

约束曲率进入方程

球面约束的切向投影是 $\phi - (u \cdot \phi)u$。对 $|u|^2 = 1$ 求导给 $u \cdot u_i = 0$；再次求导产生 $|Du|^2 u$，这里乘子是函数而非一个常数。

8.4.4 Divergence constraints and pressure

For $f \in L^2(U;\mathbb{R}^n)$ minimize $I[v] = \frac{1}{2}\int |Dv|^2 - \int f \cdot v$ on the closed linear space $V_\sigma = \{v \in H_0^1(U;\mathbb{R}^n) : \operatorname{div} v = 0\}$. Coercivity and strict convexity give a unique u, with $\ell(v) = \int Du : Dv - f \cdot v = 0$ for $v \in V_\sigma$. To recover pressure we use the following external divergence theorem: on a bounded connected smooth domain there is a bounded right inverse

$$\mathcal{B} : L_0^2(U) \to H_0^1(U;\mathbb{R}^n), \quad \operatorname{div} \mathcal{B}q = q, \quad \|\mathcal{B}q\|_{H^1} \leq C\|q\|_2.$$

Here L_0^2 denotes zero mean. This is the divergence-solving estimate also invoked without proof in the source, often realized by a Bogovskii operator. Its construction is not proved here.

Theorem 8.16 (Pressure multiplier; source §8.4.4, Theorem 6)
There is a unique mean-zero $p \in L^2(U)$ satisfying

$$\int_U Du : Dv = \int_U f \cdot v + p \operatorname{div} v \quad (v \in H_0^1(U; \mathbb{R}^n)).$$

Thus $-\Delta u + Dp = f$, $\operatorname{div} u = 0$ in distributions. The global conclusion uses the external right inverse just stated; the source proves a local pressure statement.

Expanded proof (factoring a functional through divergence).

The functional $q \mapsto \ell(\mathcal{B}q)$ is bounded on L_0^2. Riesz representation gives mean-zero p with $\ell(\mathcal{B}q) = \int pq$. For any $v \in H_0^1$, $q = \operatorname{div} v$ has zero mean and $v - \mathcal{B}q \in V_\sigma$. Hence $\ell(v) = \ell(\mathcal{B}q) = \int p \operatorname{div} v$. Surjectivity of divergence gives uniqueness in L_0^2 and $\|p\|_2 \le C(\|Du\|_2 + \|f\|_2)$. Without mean normalization constants are undetermined, since $\int \operatorname{div} v = 0$. This argument needs connectedness and the right inverse, not a claim that every curl-free field is globally a gradient. □

8.4.5 Nonlinear incompressibility

In three-dimensional elasticity, u is the deformation map (the new position), not the displacement $u - x$. Local incompressibility is $\det Du = 1$. For an actual displacement $v = u - x$ it is $\det(I + Dv) = 1$, whose first-order linearization is $\operatorname{div} v = 0$.

Theorem 8.17 (Determinant constraint; source Theorem 7)
For $q > 3$, let $L(P, x)$ be smooth, convex in P, and satisfy (8.6). If

$$\mathcal{A} = \{u \in g + W_0^{1,q}(U; \mathbb{R}^3) : \det Du = 1 \text{ a.e.}\}$$

has a finite-energy member, then it has an energy minimizer.

Expanded proof (weak closure of the determinant condition).

Coercivity bounds a minimizing sequence in $W^{1,q}$. Its weak limit keeps the trace, and Lemma 8.8 gives $\det Du_k \rightharpoonup \det Du$ in $L^{q/3}$. Since every determinant equals one, the limit does too. Convex lower semicontinuity completes existence. The feasible set is not convex, so convexity of L alone supplies no uniqueness assertion here. The determinant condition is local and does not by itself prove global injectivity or prevent self-interpenetration. □

压力只确定到常数

测试速度具有零边界，故散度积分为零；压力加常数不改变弱式。右逆估计将消失在无散度空间上的泛函表示为 $p \operatorname{div} v$，并给出压力范数控制。

8.5 Critical Points

8.5.1 Palais–Smale compactness and a deformation

Let H be a real Hilbert space and $I \in C^1(H;\mathbb{R})$. Write $\nabla I(u) \in H$ for the Riesz representative of $DI(u)$; the source writes this representative as $I'[u]$. Assume ∇I is Lipschitz on bounded subsets, the source's class $\mathscr{C}$. A critical point has $\nabla I(u) = 0$. The Palais–Smale condition says that every sequence with bounded $I(u_k)$ and $\nabla I(u_k) \to 0$ has a strongly convergent subsequence in H. Boundedness of the sequence alone would not suffice in infinite dimension.

Theorem 8.18 (Deformation theorem; source §8.5.1, Theorem 1)
Suppose $I \in \mathscr{C}$ satisfies Palais–Smale and has no critical point at level c. For sufficiently small $\varepsilon > 0$ there are $0 < \delta < \varepsilon$ and a continuous flow $\eta : [0,1] \times H \to H$ with $\eta(0,u) = u$, fixing points outside $\{|I - c| < \varepsilon\}$, not increasing I, and satisfying

$$\eta(1, \{I \le c + \delta\}) \subset \{I \le c - \delta\}.$$

Expanded proof (a bounded descending vector field).

There exist $\varepsilon_0 > 0, 0 < \sigma \le 1$ such that $\|\nabla I(u)\| \ge \sigma$ whenever $|I(u) - c| \le \varepsilon_0$. Otherwise choose $|I(u_k) - c| \le 1/k$ and $\|\nabla I(u_k)\| \le 1/k$; Palais–Smale and continuity give a critical point at c, a contradiction. Choose $0 < \varepsilon \le \varepsilon_0$, a Lipschitz scalar cutoff χ equal to one on $[c - \varepsilon/2, c + \varepsilon/2]$ and zero outside $(c - \varepsilon, c + \varepsilon)$, and define

$$V(u) = -\chi(I(u)) \frac{\nabla I(u)}{\max\{1, \|\nabla I(u)\|\}}.$$

It is Lipschitz on bounded sets and has norm at most one. The Banach-space ODE $\dot\eta = V(\eta)$ has unique global solutions: Picard iteration gives local existence, and bounded speed makes trajectories Cauchy at a finite endpoint, allowing continuation. Continuous dependence gives the required continuous flow, and $V = 0$ outside the energy band fixes those points. The Hilbert chain rule gives

$$\frac{d}{dt} I(\eta) = -\chi(I(\eta)) \frac{\|\nabla I(\eta)\|^2}{\max\{1, \|\nabla I(\eta)\|\}} \le 0.$$

Choose $0 < \delta < \varepsilon/2$ with $2\delta < \sigma^2$. An orbit starting at energy at most $c + \delta$ is already below $c - \delta$, exits the narrow band downward, or stays in it for time one. In the last case $\chi = 1$ and the derivative is at most $-\sigma^2$, so it too ends below $c - \delta$. This proves every asserted property. □

8.5.2 The mountain pass level

Theorem 8.19 (Mountain pass; source Theorem 2)
Let $I \in \mathscr{C}$ satisfy Palais–Smale, $I(0) = 0$, and suppose $I(u) \geq a > 0$ when $\|u\| = r$, while some e satisfies $\|e\| > r$, $I(e) \leq 0$. For

$$\Gamma = \{\gamma \in C([0,1]; H) : \gamma(0) = 0,\ \gamma(1) = e\},$$
$$c = \inf_{\gamma \in \Gamma} \max_{s \in [0,1]} I(\gamma(s)),$$

there is $u \in H$ with $I(u) = c \geq a$ and $\nabla I(u) = 0$.

Expanded proof (deforming a nearly optimal path).

Every path crosses the radius-r sphere by continuity of its norm, so $c \geq a$. The straight path shows c is finite. If c were not critical, use Theorem 8.18 with $\varepsilon < a/2$. Choose a path with maximum at most $c + \delta$. The deformed path has maximum at most $c - \delta$; its endpoints remain $0, e$, because their energies lie below $c - \varepsilon$. It still belongs to Γ, contradicting the definition of c. The conclusion is existence at a minimax level; it does not assert nondegeneracy, a particular Morse index, or uniqueness of the critical point. □

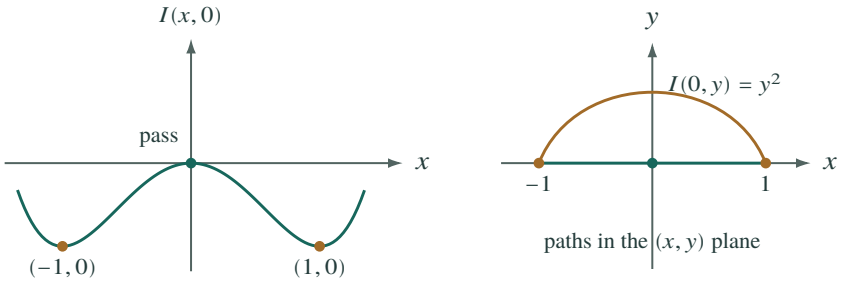


Diagram T8.4. A minimax path must cross an energy barrier. This is an exact two-dimensional model, $I(x,y) = (x^2 - 1)^2 + y^2 - 1$, connecting $(1,0)$ to $(-1,0)$. Both endpoints have energy -1 and every path crosses $x = 0$, where $I \geq 0$. The horizontal path attains maximum zero at the saddle $(0,0)$. Translation and addition of a constant put the first endpoint at the theorem's 0 with energy zero.

Palais–Smale 的用途

若临界层附近不存在梯度下界，就能构造能量趋于 c、梯度趋零的序列。PS 将它变为强收敛子列，从而得到被假设排除的临界点。

8.5.3 A nonzero solution of a semilinear Dirichlet equation

Let $n \geq 3$, $1 < p < (n+2)/(n-2)$, $f \in C^1(\mathbb{R})$, and $F(s) = \int_0^s f(\tau)\,d\tau$. Assume for some $a, A > 0$ and $\mu > 2$,

$$\begin{aligned} |f(s)| &\leq C(1+|s|^p), \quad |f'(s)| \leq C(1+|s|^{p-1}), \\ a|s|^{p+1} &\leq F(s) \leq A|s|^{p+1}, \qquad \mu F(s) \leq s f(s). \end{aligned} \tag{8.13}$$

These are the source assumptions with $\mu = 1/\gamma > 2$.

Theorem 8.20 (Semilinear application; source §8.5.2, Theorem 3)
The problem $-\Delta u = f(u)$, $u \in H_0^1(U)$, has a nonzero weak solution.

Expanded proof (differentiability and compactness).

Use $H = H_0^1(U)$ with $\|u\|_H = \|Du\|_2$ and define $I(u) = \frac{1}{2}\|u\|_H^2 - \int F(u)$. Sobolev embedding into L^{p+1} makes I finite. The mean value bound on f' gives

$$\|f(u) - f(v)\|_{L^{(p+1)/p}} \leq C(1 + \|u\|_{p+1}^{p-1} + \|v\|_{p+1}^{p-1})\|u - v\|_{p+1}.$$

Together with the quadratic Taylor remainder for F, this proves Fréchet differentiability and a gradient Lipschitz on bounded H sets. If $K : H^{-1} \to H_0^1$ is the inverse Dirichlet Laplacian, Riesz representation gives $\nabla I(u) = u - Kf(u)$. The nonlinear functional $v \mapsto \int f(u)v$ is in H^{-1} by Hölder and the L^{p+1} embedding; no pointwise identification of the Hilbert gradient with $-\Delta u - f(u)$ is made.

For a Palais–Smale sequence, write $\epsilon_k = \|\nabla I(u_k)\| \to 0$. The superquadratic inequality yields

$$\left(\tfrac{1}{2} - \tfrac{1}{\mu}\right)\|u_k\|_H^2 \leq I(u_k) - \tfrac{1}{\mu} DI(u_k)[u_k] \leq C + \tfrac{1}{\mu}\epsilon_k \|u_k\|_H.$$

Thus u_k is bounded in H. Since $p + 1 < 2^* = 2n/(n-2)$, a subsequence converges strongly in L^{p+1}; the preceding estimate gives $f(u_k) \to f(u)$ in H^{-1}. Then $u_k = Kf(u_k) + o_H(1)$ converges strongly in H. This verifies Palais–Smale, including its strong compactness conclusion.

Finally $I(u) \geq \frac{1}{2}\|u\|_H^2 - C\|u\|_H^{p+1}$ is positive on a small sphere, since $p + 1 > 2$. For any nonzero $\phi \in C_c^\infty(U)$,

$$I(t\phi) \leq \tfrac{1}{2}t^2\|\phi\|_H^2 - at^{p+1}\|\phi\|_{p+1}^{p+1} \longrightarrow -\infty.$$

The mountain pass theorem supplies a positive-energy critical point, which is nonzero because $I(0) = 0$. Its weak equation follows from $DI(u)[v] = 0$. In dimensions one and two the same proof works for finite $p > 1$ using the corresponding finite-exponent embeddings. □

8.6 Invariance, Noether's Theorem

8.6.1 Domain variations and a sign convention for the current

Let Φ_s be smooth local diffeomorphisms near $\overline{U}$, with $\Phi_0(x) = x$, and w_s smooth functions with $w_0 = u$. Write $X = \partial_s \Phi_s|_0$ and $m = \partial_s w_s|_0$. The source's invariance convention is

$$\int_V L(Dw_s, w_s, x)\,dx = \int_{\Phi_s(V)} L(Du, u, x)\,dx \tag{8.14}$$

for every relatively compact subdomain V, not just the whole U. The variations need not preserve a particular prescribed boundary value; Noether's conclusion is a local identity for smooth solutions.

Theorem 8.21 (Noether identity; source §8.6.2, Theorem 1)
Under (8.14)*,*

$$\operatorname{div}(mL_p - LX) = m(\operatorname{div} L_p - L_z). \tag{8.15}$$

Thus $J = mL_p - LX$ is divergence free on Euler–Lagrange solutions.

Expanded proof (local invariance and the current).

The derivative of the left side is $\int_V L_p \cdot Dm + L_z m$. For the right side, changing variables $y = \Phi_s(x)$ and using $\frac{d}{ds} \det D\Phi_s|_0 = \operatorname{div} X$ gives $\int_V \operatorname{div}(LX)$; here differentiation of the composed $L(Du, u, x)$ is a total spatial derivative. Since this holds for every V, the smooth integrands coincide pointwise: $L_p \cdot Dm + L_z m = \operatorname{div}(LX)$. Expanding $\operatorname{div}(mL_p)$ proves (8.15). Invariance on only one fixed domain would yield only an integral identity, not this pointwise conservation law. □

8.6.2 Translations, scaling, and monotonicity

If $L = L(p, z)$ is independent of x, translations give $X = e_k$, $m = u_k$. Therefore the stress tensor

$$T_{ik} = L_{p_i} u_k - L\delta_{ik} \quad \text{satisfies} \quad \partial_i T_{ik} = 0.$$

For L with explicit x dependence, direct differentiation instead gives $\partial_i T_{ik} = -L_{x_k}$ on a solution. Explicit inhomogeneity is a source term, not a broken algebraic identity. For $I[u] = \int |Du|^p$, the scaling

$$\Phi_s(x) = e^s x, \quad w_s(x) = e^{(n-p)s/p} u(e^s x)$$

obeys (8.14): the gradient contributes e^{ns}, which cancels the change-of-variables Jacobian. Thus $X = x$ and $m = x \cdot Du + (n-p)u/p$.

严格次临界保证紧性

$p + 1 < 2^*$ 不仅保证能量可积，更让 u_k 在 L^{p+1} 强收敛。临界指数处连续嵌入仍在，但紧性丢失，不能照搬这个 PS 证明。

守恒流的符号约定

本书用主动变化 $w_s(x) = u(\Phi_s(x))$，对应 $J = mL_p - LX$。若换成逆映射的被动约定，公式会换号；不能混用两种定义。

Theorem 8.22 (Scaled energy monotonicity; source §8.6.2, Example 2)
For a smooth p-harmonic critical point for which the preceding calculation is classical, or a weak local minimizer of $\int |Du|^p$, $p > 1$, *set* $E(r) = \int_{B_r} |Du|^p$ *with* $B_R \Subset U$. *For a.e.* $0 < r < R$,

$$\frac{d}{dr}\left(r^{p-n}E(r)\right) = pr^{p-n}\int_{\partial B_r} |Du|^{p-2}u_r^2\, dS \geq 0. \tag{8.16}$$

The integrand is defined as zero where $Du = 0$.

Expanded proof (from an inner variation to a radial flux).

For smooth solutions the Noether current above has zero boundary flux on B_r. The term containing u integrates to zero because $\int_{\partial B_r} u|Du|^{p-2}u_r = \int_{B_r} |Du|^p$, by the p-Laplace equation. Rearrangement gives

$$(n-p)E(r) = r\int_{\partial B_r} (|Du|^p - p|Du|^{p-2}u_r^2).$$

Since $E'(r) = \int_{\partial B_r} |Du|^p$ a.e., differentiation gives (8.16) with the displayed power of r. For a weak local minimizer avoid differentiating Du: take compactly supported inner variations $u \circ (\mathrm{id} + sY)$. After changing variables, only the matrix and Jacobian are differentiated; the derivative is dominated by $C|Du|^p$ and yields

$$\int_U |Du|^p \operatorname{div} Y - p|Du|^{p-2}Du \cdot (DY)Du = 0.$$

Use radial cutoffs $Y(x) = \chi(|x|)x$ approaching the indicator of B_r. Coarea and Lebesgue differentiation give the same radial identity for a.e. r. This extends the proof without assuming L is C^2 at zero gradient when $1 < p < 2$. The scaled energy has its natural nondecreasing representative. □

8.6.3 Time translation and wave energy

For the action with kinetic minus potential density

$$\mathscr{L}(D_x u, u_t, u) = \tfrac{1}{2}u_t^2 - \tfrac{1}{2}|D_x u|^2 - F(u),$$

the Euler–Lagrange equation is $u_{tt} - \Delta u + f(u) = 0$, $F' = f$. Time translation gives $m = u_t$ and the space–time current $(-u_t D_x u, e)$, where

$$e = \tfrac{1}{2}(u_t^2 + |D_x u|^2) + F(u).$$

Hence $e_t - \operatorname{div}_x(u_t D_x u) = 0$. For compact spatial support, or zero boundary flux, $\int e$ is constant. A signed F still gives conservation; nonnegativity is an additional requirement if this energy is to control norms.

单调量含有尺度因子

单调的是 $r^{p-n}\int_{B_r}|Du|^p$。右侧仅含径向导数的非负平方；这个恒等式比“积分区域增大所以能量增大”包含更多信息。

For the linear wave action, simultaneous scaling of (x,t) has multiplier $m = tu_t + x\cdot Du + (n-1)u/2$ and gives $P_t - \operatorname{div} Q = 0$, with

$$P = te_0 + (x\cdot Du)u_t + \tfrac{n-1}{2}uu_t,$$
$$Q = mDu + \tfrac{1}{2}(u_t^2 - |Du|^2)x, \qquad e_0 = \tfrac{1}{2}(u_t^2 + |Du|^2).$$

The exponent $(n-1)/2$ comes from the space–time dimension $n+1$.

8.6.4 Conformal multiplier and local energy decay

The source motivates a further multiplier by hyperbolic inversion $(x,t) \mapsto (x,t)/(|x|^2 - t^2)$ and the wave Kelvin transform

$$u \mapsto |\,|x|^2 - t^2\,|^{-(n-1)/2} u\left(\frac{x}{|x|^2-t^2}, \frac{t}{|x|^2-t^2}\right),$$

away from its singular cone. Conjugating a time translation suggests

$$M = (r^2+t^2)u_t + 2t\,x\cdot Du + (n-1)tu, \qquad r = |x|.$$

We do not assume the action is invariant under this transformation. Instead the useful conclusion follows from the direct multiplier identity

$$C_t - \operatorname{div} R = M(u_{tt} - \Delta u), \tag{8.17}$$

where

$$C = (r^2+t^2)e_0 + 2t(x\cdot Du)u_t + (n-1)tuu_t - \tfrac{n-1}{2}u^2,$$
$$R = MDu + t(u_t^2 - |Du|^2)x.$$

Expanded proof (conformal current and its sign).

Expand $M(u_{tt} - \Delta u)$, integrating each second derivative once by the product rule. The second-derivative terms give $C_t - \operatorname{div} R$. The remaining coefficients of u_t^2, $|Du|^2$ and uu_t cancel using $\partial_t(r^2+t^2) = 2t$, $D(r^2+t^2) = 2x$ and $\operatorname{div} x = n$, proving (8.17) off shell. For a more useful sign, put $\alpha = (n-1)/2$ and $u_r = Du\cdot x/r$. Direct expansion gives $C = S - \frac{\alpha}{2}\operatorname{div} Z$, with

$$\begin{aligned} S = {} & \frac{(t+r)^2}{4}(u_t + u_r + \alpha u/r)^2 \\ & + \frac{(t-r)^2}{4}(u_t - u_r - \alpha u/r)^2 \\ & + \frac{r^2+t^2}{2}\left(|Du|^2 - u_r^2 + \frac{(n-3)(n-1)}{4r^2}u^2\right), \\ Z = {} & \frac{r^2+t^2}{r^2}u^2 x. \end{aligned}$$

Indeed expanding the two squares produces the cross term $2\alpha tuu_t$ and an extra $\alpha(r^2+t^2)uu_r/r$. The latter and the remaining u^2 terms are exactly $\frac{\alpha}{2}\operatorname{div} Z$, with $\operatorname{div} Z = 2(r^2+t^2)uu_r/r + [n + (n-2)t^2/r^2]u^2$. For $n \geq 3$, every term of S is nonnegative. The pointwise completion is used away from $r = 0$; the exterior application below has the origin inside the excluded obstacle. □

Theorem 8.23 (Exterior local energy decay; source application, pp.487–489)
Let $n \geq 3$ and let a smooth bounded obstacle O contain a neighborhood of the origin and be star-shaped about it. A smooth solution of the wave equation on $\Omega = \mathbb{R}^n \setminus \overline{O}$, with zero Dirichlet boundary values and compatible compactly supported initial data, satisfies

$$\int_{B_R \setminus \overline{O}} (u_t^2 + |Du|^2)\, dx \leq C_R t^{-2} \quad (t \geq 2R), \qquad \overline{O} \subset B_R.$$

Expanded proof (the obstacle flux and the radial correction).

Finite propagation keeps all fields compactly supported at each finite time. The outer normal ν to Ω points into O. On ∂O, $u = u_t = 0$ and $Du = (\partial_\nu u)\nu$, so $R \cdot \nu = t|Du|^2 x \cdot \nu \leq 0$ by star-shapedness. Integrating (8.17) gives $\int_\Omega C(t) \leq \int_\Omega C(0)$. The divergence correction has zero total flux, since $u = 0$ on the obstacle and is compactly supported. Thus $\int_\Omega S(t) \leq C$. For $r \leq R$, $t \geq 2R$, both $(t \pm r)^2 \geq t^2/4$; hence

$$\int_{B_R \setminus \overline{O}} u_t^2 + (u_r + \alpha u/r)^2 + (|Du|^2 - u_r^2) + (n-3)u^2/r^2 \leq C_R t^{-2}.$$

To extract u_r^2, integrate the radial cross term on this bounded region:

$$\int (u_r + \alpha u/r)^2 = \int u_r^2 + \alpha \int_{\partial B_R} u^2/R + \frac{(n-1)(3-n)}{4} \int u^2/r^2.$$

There is no inner boundary contribution because $u = 0$ there. For $n = 3$ the last coefficient vanishes; for $n > 3$ its absolute value is controlled by the already bounded $(n-3)u^2/r^2$ term. The outer boundary term is nonnegative. This gives the claimed bound on u_r^2, and combining with the tangential part controls $|Du|^2$. Total wave energy is conserved; the estimate concerns energy remaining in a fixed bounded spatial region. □

8.7 Problems and Worked Solutions

These are concise restatements of all 20 source exercises, with editorial solutions. Source problem numbers are retained. Smoothness and boundary assumptions are stated where they affect a conclusion; in Problem 17 connectedness is essential.

猜出乘子仍需验证

Kelvin 变换启发 M，但本稿不据此宣称作用量不变。直接核对 $C_t - \operatorname{div} R = M(u_{tt} - \Delta u)$ 才建立守恒律；负项须用散度项与平方和重写。

1. Oscillatory weak limits. Show $\sin(kx) \rightharpoonup 0$ in $L^2(0,1)$, and that a periodic mixture with proportions $\lambda, 1-\lambda$ and values a, b converges weakly to $\lambda a + (1-\lambda)b$.

Solution. For a C^1 test ϕ, integration by parts bounds $|\int \sin(kx)\phi| \le C_\phi/k$. Uniform L^2 boundedness and density extend convergence to all L^2 tests. For the mixture, its integral over each cell $[j/k, (j+1)/k)$ is its mean divided by k. Subtract the value $\phi(j/k)$ in each cell; uniform continuity bounds the resulting error by $C \sup_{|x-y|\le 1/k} |\phi(x) - \phi(y)| \to 0$. Again use density and uniform boundedness. Neither result gives strong convergence unless the oscillatory variance vanishes. □

2. A drift with a variational weight. Find a Lagrangian for $-\Delta u + D\varphi \cdot Du = f$.

Solution. Take $L(p,z,x) = e^{-\varphi(x)}(\frac{1}{2}|p|^2 - f(x)z)$. Its equation is $-\operatorname{div}(e^{-\varphi}Du) = e^{-\varphi}f$. Division by the positive weight gives the stated equation. The positive exponential factor and its sign are fixed by the drift. □

3. Elliptic regularization of heat flow. Find an energy for $u_t - \Delta_x u - \varepsilon u_{tt} = 0$.

Solution. Use the space–time Lagrangian $L_\varepsilon(p,s,z,x,t) = \frac{1}{2}e^{-t/\varepsilon}(|p|^2 + \varepsilon s^2)$. Its equation is $-\operatorname{div}_x(e^{-t/\varepsilon}Du) - \partial_t(\varepsilon e^{-t/\varepsilon}u_t) = 0$. Canceling the weight gives precisely the asserted equation. This is a variational identity for fixed endpoint variations; a choice of temporal boundary conditions is additional data, not determined by the PDE alone. □

4. Weighted null Lagrangians. Show $L(P,z) = \eta(z)\det P$ is null and its integral depends only on the trace.

Solution. By Piola,

$$\partial_i[\eta(u)(\operatorname{cof} Du)_{\alpha i}] = \eta_{z_\beta}(u)u_i^\beta(\operatorname{cof} Du)_{\alpha i} = \eta_{z_\alpha}(u)\det Du = L_{z_\alpha}.$$

Thus the Euler–Lagrange expression vanishes. The homotopy argument in §8.1 proves boundary dependence. The calculation needs no invertibility or fixed sign of $\det Du$. □

5. The degree is an integer. Let $B = B(x_0, r)$ avoid $u(\partial U)$ and let η be supported in B with integral one. Show $\int_U \eta(u) \det Du \in \mathbb{Z}$.

Solution. Define the distribution $\nu(\psi) = \int_U \psi(u) \det Du$ on B. For $\psi \in C_c^\infty(B)$, the preceding cofactor identity gives

$$\int_U (\partial_\alpha \psi)(u) \det Du = \int_U \partial_i [\psi(u) (\operatorname{cof} Du)_{\alpha i}] = 0,$$

because $\psi(u)$ vanishes near ∂U. Thus all distributional derivatives of ν vanish. A distribution with zero derivatives on a connected ball is constant: mollification makes it a constant smooth function on each smaller ball, and overlapping balls identify the constants. Therefore $\nu(\psi) = d \int_B \psi$.

Use Sard's theorem (an external finite-dimensional theorem) to choose a regular value $y \in B$; a value outside the image is permitted. Its preimages in U form a finite set, since they are isolated by the inverse function theorem and lie in a compact set away from the boundary. For a sufficiently small neighborhood of y, all preimages lie on these finitely many inverse branches; compactness excludes extra branches. Signed change of variables gives $d = \sum_{x \in u^{-1}(y)} \operatorname{sgn} \det Du(x) \in \mathbb{Z}$. Hence $\nu(\eta) = d$. Approximation extends the formula to the stated C^1 density. This proof also shows independence from the chosen normalized η within the same ball. □

度数是带符号计数

正负 Jacobian 的原像分别计 +1 与 −1。先证推前分布在避开边界像的球上为常数，再在正则值附近换元；不能对可能多对一的映射直接套全局一一换元。

6. The curvature integral is a boundary quantity. Show $\int_U (1 + |Du|^2)^{-3/2} \det D^2 u$ depends only on $Du|_{\partial U}$.

Solution. Apply Problem 4 to the map $v = Du : U \subset \mathbb{R}^2 \to \mathbb{R}^2$ and $\eta(z) = (1 + |z|^2)^{-3/2}$. The resulting weighted determinant integral is exactly the displayed expression. The relevant boundary datum is the full gradient of u, not merely u itself. □

7. A quadratic null Lagrangian. Show $L(P) = \operatorname{tr}(P^2) - (\operatorname{tr} P)^2$ is null.

Solution. $L_{P_{\alpha i}} = 2P_{i\alpha} - 2(\operatorname{tr} P)\delta_{\alpha i}$. With $P = Du$, its row divergence is $2 \sum_i \partial_i \partial_\alpha u^i - 2\partial_\alpha \sum_i \partial_i u^i = 0$. Notice $\operatorname{tr}(P^2)$ is not the Frobenius norm squared for an arbitrary nonsymmetric matrix. □

8. Why the area functional escapes this direct method. Explain the obstruction for $I[u] = \int \sqrt{1 + |Du|^2}$ in $W^{1,q}$.

Solution. For $q > 1$ there is no coercive lower bound $\alpha|p|^q - \beta$ because the density grows only linearly as $|p| \to \infty$. For $q = 1$, $W^{1,1}$ is not reflexive and an energy bound gives no automatic weakly convergent $W^{1,1}$ subsequence: gradients may concentrate. The functional is convex, so the missing step is compactness/coercivity, not gradient convexity. A BV relaxation and its boundary terms belong to a different theory, not to the $q > 1$ theorem above. □

9. Legendre–Hadamard condition for systems. Derive the rank-one second-variation condition, and exhibit a nonconvex example.

Solution. Use $\phi_\varepsilon = \varepsilon a \rho(x \cdot b/\varepsilon)\zeta(x)$, so $D\phi_\varepsilon = \rho'(a \otimes b)\zeta + O(\varepsilon)$. The scalar triangular-wave proof gives

$$L_{P_{\alpha i} P_{\beta j}}(Du, u, x) a_\alpha a_\beta b_i b_j \geq 0.$$

For 2×2 matrices, $L(P) = \det P$ is affine on every rank-one line, so this expression is identically zero. Yet with $P = \operatorname{diag}(1, -1)$ and $Q = -P$, one has $L(P) = L(Q) = -1$ and $L((P + Q)/2) = 0$, violating convexity. For larger dimensions use a fixed 2×2 minor. □

10. A nonzero semilinear solution by constrained minimization. For $n > 2$ and $1 < q < (n + 2)/(n - 2)$, solve $-\Delta v = |v|^{q-1} v$ with zero trace.

Solution. Minimize $\frac{1}{2}\|Du\|_2^2$ with $\int |u|^{q+1} = 1$. The class is nonempty, and compact $H_0^1 \hookrightarrow L^{q+1}$ preserves its constraint under a minimizing sequence. The first derivative of the constraint evaluated on u equals $q + 1 \neq 0$, so the implicit-curve argument gives $-\Delta u = \lambda |u|^{q-1} u$. Testing by u gives $\lambda = \|Du\|_2^2 > 0$. Then $v = \lambda^{1/(q-1)} u$ is the desired nonzero solution. The constraint is superquadratic, so here its differentiability and compactness use the subcritical embedding rather than the linear-growth hypothesis of §8.4.1. □

11. A monotone nonlinear Robin condition. Assume $0 < a \leq \beta'(s) \leq b$ and $f \in L^2(U)$. Prove weak existence and uniqueness for $-\Delta u = f$, $\partial_\nu u + \beta(u) = 0$.

Solution. Let $B(s) = \int_0^s \beta(t)\,dt$ and minimize on $H^1(U)$

$$E[u] = \tfrac{1}{2}\|Du\|_2^2 + \int_{\partial U} B(Tu)\, dS - \int_U f u.$$

The trace map is continuous to $L^2(\partial U)$ and B has quadratic growth, so the energy is finite. Also $B(s) \geq as^2/4 - C$, and $\|u\|_{H^1}^2 \leq C(\|Du\|_2^2 + \|Tu\|_{L^2(\partial U)}^2)$. The latter follows by a compactness contradiction: a normalized sequence with vanishing gradient tends to a constant on each connected component, and vanishing trace makes every such constant zero. Thus E is coercive. Its convex gradient and boundary terms are weakly lower semicontinuous (a continuous convex functional is weakly lower semicontinuous by separation), giving a minimizer. The weak equation is

$$\int_U Du \cdot D\phi + \int_{\partial U} \beta(Tu)T\phi = \int_U f\phi \quad (\phi \in H^1(U)).$$

For two solutions test their difference; monotonicity gives $\|D(u-v)\|_2^2 + a\|T(u-v)\|_2^2 \leq 0$, hence uniqueness. □

12. Fixed volume gives constant mean curvature. For a smooth area minimizer with fixed trace and $\int_U u = 1$, show its graph has constant mean curvature.

Solution. The integral constraint has nonzero derivative, so the same implicit-curve argument yields $\int Du \cdot D\phi / \sqrt{1 + |Du|^2} = \lambda \int \phi$. Thus

$$-\operatorname{div}\left(\frac{Du}{\sqrt{1 + |Du|^2}}\right) = \lambda.$$

With $nH = \operatorname{div} N$, where $N = Du/\sqrt{1 + |Du|^2}$, $H = -\lambda/n$ is constant. Reversing normal orientation reverses H. □

13. The dual Dirichlet principle. Prove the minimum of $\frac{1}{2}\|Dw\|_2^2 - (f, w)$ over H_0^1 equals $\max_{\operatorname{div}\xi = f} -\frac{1}{2}\|\xi\|_2^2$ over $L^2(U; \mathbb{R}^n)$.

Solution. Distributional divergence and zero trace give $(f, w) = -\int \xi \cdot Dw$. Consequently

$$\tfrac{1}{2}\|Dw\|_2^2 - (f, w) = \tfrac{1}{2}\|Dw + \xi\|_2^2 - \tfrac{1}{2}\|\xi\|_2^2 \geq -\tfrac{1}{2}\|\xi\|_2^2.$$

For the Poisson solution u, take $\xi = -Du$. It is feasible since $\operatorname{div}\xi = f$ and the square vanishes. Thus both extrema are attained and equal; the dual sign is fixed by this choice of ξ. □

对偶通量的正负

原方程 $-\Delta u = f$ 配合对偶约束 $\operatorname{div}\xi = f$，最优通量应取 $\xi = -Du$。完成平方是 $|Dw + \xi|^2$，不是减号。

14. A multivalued obstacle equation. Relate the variational inequality to $-\Delta u + \beta(u - h) \ni f$, where $\beta(s) = 0$ for $s > 0$, $\beta(0) = (-\infty, 0]$, and $\beta(s) = \emptyset$ for $s < 0$.

Solution. Under the local H^2 and continuity regularity described in §8.4.2, let $\mu = -\Delta u - f \geq 0$, supported on $\{u = h\}$. Then $\eta = -\mu \in \beta(u - h)$ and $-\Delta u + \eta = f$ a.e. Conversely these relations give $\int Du \cdot D(w - u) - f(w - u) = \int \mu(w - u) \geq 0$ for feasible w, whenever these function pairings are justified; globally H^2 and $\mu \in L^2$ suffice. For the bare H^1 formulation the precise equivalent is $0 \in -\Delta u - f + N_K(u)$ in H^{-1}, where $\langle \eta, w - u \rangle \leq 0$ defines the dual normal cone. A pointwise multiplier is not asserted for a general distributional reaction. □

15. A gradient constraint. Minimize the obstacle-type quadratic energy on $K = \{u \in H_0^1(U) : |Du| \leq 1 \text{ a.e.}\}$ and derive its inequality.

Solution. This nonempty set is convex and norm closed: strong H^1 convergence has an a.e. convergent gradient subsequence. Hence it is weakly closed. The quadratic energy is coercive and weakly lower semicontinuous, so it has a unique minimizer by strict convexity. Its segment variations give $\int Du \cdot D(w - u) \geq \int f(w - u)$ for all $w \in K$, with the same converse by completing the energy difference. Pointwise gradient bounds are preserved by convex weak closure, not by assuming weak gradients converge pointwise. □

16. A singular harmonic map. For $n \geq 3$ show $u = x/|x| \in H^1(U; \mathbb{R}^n)$ is a weak harmonic map into the sphere.

Solution. Away from zero, $Du = (I - u \otimes u)/r$, $|Du|^2 = (n - 1)/r^2$, and $-\Delta u = (n - 1)u/r^2$. The integral $\int_0^\varepsilon r^{n-3} dr$ is finite exactly for $n > 2$. Deleting a small ball and passing to the limit proves these are the weak first derivatives; its boundary error is $O(\varepsilon^{n-1})$. For the equation, the new spherical normal derivative is zero because u is independent of r. Thus integration by parts has no inner flux, and the integrable right side converges, proving the distributional equation. Uniformly bounded smooth approximations to a bounded H_0^1 test extend the identity to the weak class in (8.12). No energy-minimizing property has been claimed merely from this equation. □

17. Uniqueness of a positive normalized ground state. On connected U, let positive $u, v \in H_0^1$ both minimize $\int |Dw|^2$ with $\int w^2 = 1$. Show $u = v$.

Solution. The Lipschitz composition $w = \sqrt{(u^2 + v^2)/2}$ belongs to H_0^1 and is normalized. In the interior positivity permits $s = u^2/(u^2 + v^2)$ and

$$|Dw|^2 = \frac{u^2 + v^2}{2} \left| s\frac{Du}{u} + (1 - s)\frac{Dv}{v} \right|^2 \leq \tfrac{1}{2}(|Du|^2 + |Dv|^2).$$

Both endpoint energies are minimal, so equality holds a.e. Strict convexity of the square gives $Du/u = Dv/v$ a.e. The minimizers solve the first-eigenvalue equation and are smooth and strictly positive locally, so $D\log(u/v) = 0$ and u/v is constant on connected U. Normalization fixes that constant at one. Without connectedness the conclusion fails: two identical disjoint components allow different positive masses on their ground states. □

正特征函数的单性与连通性

对数比值梯度为零只保证每个连通分支上为常数。不连通区域的多个分支若有相同最低特征值，可重新分配归一化质量，不能推出全域唯一。

18. Ordered conductivities and identical Cauchy data. Assume $0 < a_1 \leq a_2$, $\operatorname{div}(a_i Du_i) = 0$, and on the boundary $u_1 = u_2$, $\partial_\nu u_1 = \partial_\nu u_2$, $a_1 = a_2$. If $Du_2 \neq 0$ a.e., show both coefficients and solutions agree.

Solution. Boundary integration gives $\int a_1|Du_1|^2 = \int a_2|Du_2|^2$. Each u_i minimizes its strictly convex weighted Dirichlet energy with the common trace. Hence

$$\int a_1|Du_1|^2 \leq \int a_1|Du_2|^2 \leq \int a_2|Du_2|^2 = \int a_1|Du_1|^2.$$

Equality gives $\int (a_2 - a_1)|Du_2|^2 = 0$. The integrand is nonnegative and $Du_2 \neq 0$ a.e., so $a_1 = a_2$ a.e., then everywhere by continuity. Uniqueness for the common Dirichlet problem gives $u_1 = u_2$. □

19. Wave momentum. For $u_{tt} - \Delta u + f(u) = 0$, find p^k, j^k with $(p^k)_t - \operatorname{div} j^k = 0$.

Solution. Use the space–time Lagrangian $\mathscr{L} = \frac{1}{2}u_t^2 - \frac{1}{2}|Du|^2 - F(u)$. Spatial translation in direction k gives multiplier u_k. The time component of its Noether current is $p^k = u_t u_k$; the spatial component is $-j^k$, where

$$j^k = u_k Du + \mathscr{L} e_k.$$

Thus Noether's space–time divergence identity gives the desired law. As a direct check, differentiating and substituting the wave equation cancels $u_t u_{tk}$, the spatial Hessian terms and $f(u)u_k$. □

20. Almgren's frequency formula. Let u be harmonic on a neighborhood of $\overline{B_R}$, nonzero on B_R, and $u(0) = 0$. Set $H(r) = \int_{\partial B_r} u^2$, $D(r) = \int_{B_r} |Du|^2$, $a = r^{1-n}H$, $b = r^{2-n}D$, $N = b/a$.

Solution. First $a(r) > 0$: if $u = 0$ on a sphere, the Dirichlet principle makes it zero inside; analyticity of harmonic functions then makes it zero on B_R, a contradiction. Differentiating surface integrals and using $D = \int_{\partial B_r} uu_r$ gives

$$a' = 2r^{1-n} \int_{\partial B_r} uu_r = 2b/r.$$

The $p = 2$ formula (8.16) gives $b' = 2r^{2-n} \int_{\partial B_r} u_r^2$. Cauchy–Schwarz gives $b^2 \le (r/2)ab'$, hence

$$N' = \frac{ab' - ba'}{a^2} = \frac{ab' - 2b^2/r}{a^2} \ge 0.$$

Let $\beta = 2N(R)$ and $\gamma = a(R)/R^\beta$. Then $a'/a = 2N(r)/r \le \beta/r$; integration from r to R yields $a(r) \ge \gamma r^\beta$. For a homogeneous harmonic polynomial of degree ℓ, $N = \ell$ and $a(r)$ is a constant times $r^{2\ell}$, checking the normalization. We explicitly assume harmonicity beyond $\overline{B_R}$ so all endpoint quantities at R are defined; one may instead use any smaller radius. □

8.8 References and Proof Dependencies

The primary source is Lawrence C. Evans, *Partial Differential Equations*, second edition, Chapter 8, printed pp. 427–494 (local English PDF pp. 444–511). The Chinese scan covers printed pp. 303–349 (PDF pp. 321–367). The exposition and all proofs, solutions, Chinese mathematical annotations and diagrams here are editorial reconstructions. The source theorem numbering is section-local, unlike these notes.

Evans points to Giaquinta and Giaquinta–Hildebrandt for variational calculus, Struwe and Zeidler for variational methods, Ladyzhenskaya–Uraltseva for the lower-semicontinuity proof, and Ball–Murat for fields of extremals. Further pointers include Kinderlehrer–Stampacchia and Friedman for variational inequalities, Hélein and Lin–Wang for harmonic maps, and Rabinowitz for minimax theory. The mountain pass theorem is attributed to Ambrosetti–Rabinowitz (*Journal of Functional Analysis* 14 (1973), 349–381). Noether and wave applications are linked to Reed–Simon, Strauss, Morawetz, Lax–Phillips and Lax; the conformal-square identity is attributed to Costa–Strauss. Exercise 17 carries the source's Belloni–Kawohl attribution, and exercise 18 its Alessandrini attribution. These are reported source pointers, not independently consulted works.

Expanded here. First and second variations; null Lagrangians and Piola cancellation; Brouwer via no retraction; convex lower semicontinuity and direct methods; Sobolev first variations; weak minors and restricted polyconvexity; fields of extremals; interior and boundary H^2 estimates; constrained minimization and multipliers; obstacle inequalities and harmonic maps; pressure recovery conditional on a divergence right inverse; deformation and mountain pass; the subcritical Palais–Smale argument; Noether currents, scaled-energy monotonicity, conformal multiplier and exterior local wave-energy decay. All 20 exercises have editorial solutions.

External inputs and limits. Chapter 5 supplies traces, compactness, extension, difference quotients, Morrey and Sobolev embedding; Chapter 6 supplies elliptic and spectral tools. Weak compactness, Hahn–Banach separation, Riesz representation, Egoroff, dominated convergence, finite-dimensional inverse/implicit function theorems and Banach-space Picard iteration are prerequisites. The De Giorgi–Nash–Moser and Schauder estimates in the smoothness bootstrap are quoted, not proved. The bounded divergence right inverse used for pressure is also an explicit external theorem. Obstacle $C^{1,1}_{\mathrm{loc}}$ regularity is not proved; its pointwise complementarity interpretation is conditional on the stated regularity. Problem 5 invokes Sard's theorem and the elementary fact that distributions with zero derivatives are constant. Problem 20 uses harmonic analyticity from Chapter 2. Our polyconvex discussion uses $q > n$ and the source's restricted $(P, \det P)$ representation; no full elasticity existence theory, self-injectivity, general system smoothness, or critical-exponent Palais–Smale theorem is claimed. Problem 17 explicitly requires connectedness. This chapter has not received independent mathematical review.

Chapter 9

Nonvariational Techniques

> Reading source
> **Evans, *Partial Differential Equations*, second edition**
> English source: Chapter 9; printed p. 495; PDF page 512.
> Chinese reference: 中译本对应章：非变分技术，书内第 350 页起。

This editorial study chapter reconstructs Evans's arguments in English, with Chinese mathematical annotations and additional proof steps. Source theorem numbers are separate from the continuous numbering of these notes. Monotonicity, compactness, order, scaling, and convex dissipation provide different mechanisms. Unless specified otherwise, $U \subset \mathbb{R}^n$ is bounded with smooth boundary, $H_0^1(U)$ has norm $\|Du\|_2$, and C_P is its Poincaré constant. This study draft has not undergone independent mathematical review.

9.1 Monotonicity Methods

9.1.1 Structure without a potential

Consider $-\operatorname{div} a(Du) = f$, $u \in H_0^1(U)$, where $f \in L^2(U)$ and $a : \mathbb{R}^n \to \mathbb{R}^n$ is continuous. The source assumes smoothness; existence needs only continuity and

$$\begin{aligned} (a(p) - a(q)) \cdot (p - q) &\geq 0, \\ |a(p)| &\leq C(1 + |p|), \\ a(p) \cdot p &\geq \alpha |p|^2 - \beta, \end{aligned} \tag{9.1}$$

with $\alpha > 0$, $\beta \geq 0$. A weak solution satisfies $\int_U a(Du) \cdot Dv = \int_U fv$ for every $v \in H_0^1(U)$. For $a = DF$ with F convex, monotonicity follows by adding its two supporting-plane inequalities. The converse fails: $a(p) = (I + K)p$ with $K^T = -K \neq 0$ is strongly monotone but has no scalar potential, since its Jacobian is not symmetric. For constant K its skew part cancels from the scalar second-order operator; this example tests the vector-field structure.

9.1.2 Finite-dimensional solvability and energy bounds

Lemma 9.1 (Zeros of a vector field; source §9.1 lemma)
If $V : \overline{B}_r \to \mathbb{R}^m$ is continuous and $V(d) \cdot d \geq 0$ on $|d| = r > 0$, it has a zero in the closed ball.

Expanded proof (the closed ball matters).

If V never vanishes, $T(d) = -rV(d)/|V(d)|$ is a continuous self-map of the closed ball. Brouwer gives $d = T(d)$, hence $|d| = r$ and $r^2 = -rV(d)\cdot d/|V(d)| \leq 0$, a contradiction. Brouwer is the finite-dimensional input from §8.1; an interior zero is not asserted. □

Choose an orthonormal basis (w_j) of $H^1_0(U)$, with dense finite spans X_m. Seek $u_m = \sum_{j=1}^m d_j w_j$ with

$$\int_U a(Du_m) \cdot Dv = \int_U fv \qquad (v \in X_m). \tag{9.2}$$

Theorem 9.2 (Approximation and bounds; source Theorems 1–2)
For every m there is such a u_m, and $\|Du_m\|_2 \leq C_0(1 + \|f\|_2)$, where C_0 depends only on $\alpha, \beta, |U|, C_P$, not on m.

Expanded proof (one coercive estimate does both jobs).

Set $V_j(d) = \int a(\sum d_i Dw_i) \cdot Dw_j - \int f w_j$. Continuity follows by dominated convergence on bounded coefficient sets: the growth bound dominates the integrand by an L^1 function built from the finitely many Dw_i. Orthonormality and Young's inequality give

$$V(d) \cdot d \geq \alpha|d|^2 - \beta|U| - C_P\|f\|_2|d| \geq \tfrac{\alpha}{2}|d|^2 - \beta|U| - \frac{C_P^2}{2\alpha}\|f\|_2^2.$$

A sufficiently large sphere satisfies the lemma, giving a zero and hence (9.2). Test with u_m and repeat the estimate: $\frac{\alpha}{2}\|Du_m\|_2^2 \leq \beta|U| + C_P^2\|f\|_2^2/(2\alpha)$. This also bounds $a(Du_m)$ in $L^2(U;\mathbb{R}^n)$. □

单调不等于梯度

对 $a(p) = (I + K)p$，单调性二次型为 $|p - q|^2$。反对称部分不影响它，却阻止 Da 成为 Hessian。

强制性与单调性分工

强制性先给有限维零点和一致范数界；单调性随后识别极限。弱紧性本身不能推出非线性通量的收敛。

9.1.3 Minty's limit argument

Theorem 9.3 (Weak existence; source Theorem 3)
Under (9.1)*, the Dirichlet problem has a weak solution.*

Expanded proof (eliminate the product of two weak limits).

Extract $u_m \rightharpoonup u$ in H_0^1 and $a(Du_m) \rightharpoonup \xi$ in L^2. For each fixed w_j, the Galerkin identity holds for all sufficiently large m. Passing to the limit and then using density yields $\int \xi \cdot Dv = \int fv$ for all $v \in H_0^1$. Moreover $\int a(Du_m) \cdot Du_m = \int fu_m \to \int fu = \int \xi \cdot Du$. This is an energy identity, not a general rule for products of weak limits.

For fixed $w \in H_0^1$, integrate monotonicity and expand the product. Use the energy identity for its only problematic term to obtain

$$\int_U (\xi - a(Dw)) \cdot (Du - Dw) \geq 0.$$

Set $w = u - tv$ with $t > 0$ and divide by t. As $t \downarrow 0$, $a(Du - tDv) \to a(Du)$ strongly in L^2: continuity gives pointwise convergence, and for $t \leq 1$ its square is dominated by $C(1 + |Du|^2 + |Dv|^2)$. Thus $\int (\xi - a(Du)) \cdot Dv \geq 0$. Replace v by $-v$ for equality. Together with the identity for ξ this proves the weak equation. Only equality on gradients is required; testing solely with gradients has not shown pointwise $\xi = a(Du)$. □

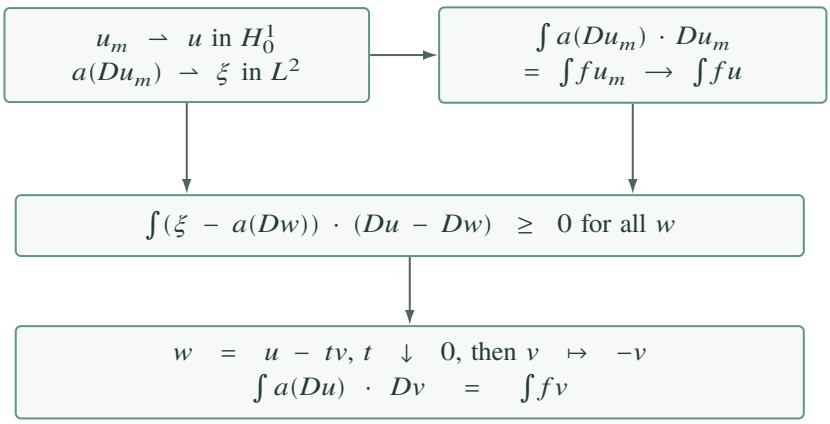


Diagram T9.1. Energy closes the weak-limit argument. The flux limit is initially named ξ. The Galerkin energy identity removes the weak–weak product; monotonicity with arbitrary w, followed by $w = u - tv$, identifies the weak equation. Each arrow requires its own justified limit.

9.1.4 Uniqueness and the regularity hypothesis

Theorem 9.4 (Strong monotonicity; source Theorem 4 and regularity remark)
If $(a(p)-a(q))\cdot(p-q)\ge\theta|p-q|^2$ *with* $\theta>0$, *the weak solution is unique. If also* $a\in C^1$, $|Da|\le M$, *and* ∂U *is* C^2, *then* $u\in H^2(U)$.

Expanded proof (uniqueness and difference quotients).

Subtract the equations for $u,\tilde u$ and test by $u-\tilde u$. Then $\theta\|D(u-\tilde u)\|_2^2\le 0$, and Poincaré gives equality. For regularity, difference quotients of $a(Du)$ have the form

$$D_h a(Du)=B_h D_h Du,\qquad B_h=\int_0^1 Da(Du+s(Du(\cdot+he)-Du))\,ds.$$

Here $B_h\eta\cdot\eta\ge\theta|\eta|^2$ and $|B_h|\le M$. Testing by $-D_{-h}(\zeta^2 D_h u)$ gives, after Cauchy–Young,

$$\int\zeta^2|D_hDu|^2\le C(\|f\|_2^2+\|Du\|_2^2).$$

The forcing term is bounded by $\|f\|_2\|D(\zeta^2D_hu)\|_2$; no derivative of f is taken. The difference-quotient criterion gives interior second derivatives. In a C^2 boundary chart the tangential estimate is the boundary argument of §6.3 (or §8.3 with nonsymmetric B_h). The transformed equation recovers the normal second derivative by division by its coefficient, bounded below by ellipticity. Bounded chart derivatives produce lower-order terms controlled by $\|Du\|_2$. A finite covering proves global H^2. This last step explicitly uses the boundary difference-quotient result developed in those chapters. □

只需识别散度

测试梯度只能确定两通量的散度相同。相差无散度场仍给相同弱方程，不应直接写成逐点通量相等。

9.2 Fixed Point Methods

9.2.1 Banach iteration and reaction–diffusion

Theorem 9.5 (Banach; source §9.2.1, Theorem 1)
If X *is Banach and* $A:X\to X$ *satisfies* $\|Au-Av\|\le q\|u-v\|$ *for* $0\le q<1$, *it has a unique fixed point. For* $u_{k+1}=Au_k$, $\|u_k-u\|\le q^k\|u_1-u_0\|/(1-q)$.

Expanded proof (completeness and an error estimate).

The successive differences are at most $q^k\|u_1-u_0\|$. Summing from k to $l-1$ shows that (u_k) is Cauchy with the stated tail bound. Completeness gives u, continuity gives $Au=u$, and two fixed points satisfy $\|u-v\|\le$

增长界不是导数界

线性增长不能单独给出 $|Da|\le M$。本稿在 H^2 命题中补明导数上界，以控制差商系数和误差项；存在性不需要它。

$q\|u-v\|$, hence coincide. The same proof works on any nonempty closed invariant subset of X. □

Let $F:\mathbb{R}^m\to\mathbb{R}^m$ be globally Lipschitz with constant L. Consider the zero-boundary system $u_t-\Delta u=F(u)$ with initial $g\in L^2(U;\mathbb{R}^m)$. A weak solution belongs to

$$u\in L^2(0,T;H_0^1),\qquad u_t\in L^2(0,T;H^{-1}),$$

obeys $\langle u_t,v\rangle+\int Du:Dv=\int F(u)\cdot v$ a.e., and has $u(0)=g$ in L^2. The time-continuity theorem gives $u\in C([0,T];L^2)$. The source starts with $g\in H_0^1$; the L^2 formulation below makes restart times unambiguous.

Theorem 9.6 (Reaction–diffusion; source Theorem 2)
There is a unique weak solution on every finite time interval. For two initial data its L^2 distance at time t is at most $e^{Lt}\|g-\tilde g\|_2$.

Expanded proof (short intervals with a uniform length).

On $X=C([0,h];L^2)$ let $Av=w$ solve the linear heat equation with forcing $F(v)$ and initial g. The linear theory of §7.1 applies since $\|F(v)\|_2\le L\|v\|_2+|F(0)||U|^{1/2}$. For $z=Av-A\tilde v$, zero initial data and the weak energy identity give

$$\tfrac12\frac{d}{dt}\|z\|_2^2+\|Dz\|_2^2\le LC_P\|v-\tilde v\|_2\|Dz\|_2\le\tfrac12\|Dz\|_2^2+\tfrac12L^2C_P^2\|v-\tilde v\|_2^2.$$

Thus $\|Av-A\tilde v\|_X\le LC_P\sqrt{h}\|v-\tilde v\|_X$. Choose h so this factor is less than one. It depends on L,U, not on g. Banach gives a solution on $[0,h]$; its endpoint is in L^2, so repeat the same argument on $[h,2h]$ and finitely many subsequent intervals. The matching L^2 traces eliminate time-interface jumps in the weak formulation. The finite sum of the interval energy norms remains finite.

For two solutions the difference satisfies $\tfrac12(\|z\|_2^2)'+\|Dz\|_2^2\le L\|z\|_2^2$. The Hilbert-triple chain rule justifies testing by z; Gronwall gives continuous dependence and uniqueness. □

全局 Lipschitz 的作用

短时间长度只依赖统一的 L，所以有限次拼接覆盖任意有限 T。对 $F(u)=u^2$，局部 Lipschitz 常数随振幅增长，不能用同一论证排除爆破。

9.2.2 Compact maps: Schauder and Schaefer

Theorem 9.7 (Schauder; source §9.2.2, Theorem 3)
If $K \subset X$ is nonempty, compact and convex and $A : K \to K$ is continuous, A has a fixed point.

Expanded proof (finite-dimensional approximation).

Cover K by finitely many balls $B(x_i, \varepsilon)$ with $x_i \in K$. Set $\rho_i(x) = (\varepsilon - \|x - x_i\|)_+$ and

$$P_\varepsilon x = \frac{\sum_i \rho_i(x) x_i}{\sum_i \rho_i(x)}.$$

The denominator is positive, P_ε is continuous, its range lies in $K_\varepsilon = \operatorname{conv}\{x_i\} \subset K$, and $\|P_\varepsilon x - x\| < \varepsilon$. The finite-dimensional convex-set form of Brouwer gives $x_\varepsilon = P_\varepsilon A x_\varepsilon$. Compactness provides a converging subsequence as $\varepsilon \to 0$; $\|x_\varepsilon - A x_\varepsilon\| < \varepsilon$ and continuity give $x = Ax$. □

A map is called compact here when it sends bounded sets to relatively compact sets. Continuity is an additional assumption.

Lemma 9.8 (A compact convex hull; editorial supplement)
In a Banach space the closed convex hull of a relatively compact set is compact.

Expanded proof (total boundedness survives convexification).

For a bounded set E with finite ε-net $\{x_i\}$, every convex combination of points of E is within ε of $\operatorname{conv}\{x_i\}$. The latter is compact, being the continuous image of a finite simplex, and has a finite ε-net. Thus the closed convex hull is totally bounded; it is complete as a closed subset of a Banach space, hence compact. □

Theorem 9.9 (Schaefer; source Theorem 4)
Let $A : X \to X$ be continuous and compact. If $\mathcal{E} = \{u : u = \lambda A u,\ 0 \le \lambda \le 1\}$ is bounded, then A has a fixed point.

Expanded proof (radial truncation and a strict radius).

Choose $R > \sup_{u \in \mathcal{E}} \|u\|$ and let $T(u) = Au \min\{1, R/\|Au\|\}$, taking $T(u) = 0$ if $Au = 0$. It is continuous and compact and maps $\overline{B}_R$ into itself. The preceding lemma makes $K = \overline{\operatorname{conv}} T(\overline{B}_R)$ compact; also $K \subset \overline{B}_R$ and $T(K) \subset K$. Schauder gives $u = T(u)$. If $\|Au\| > R$, then $u = \lambda A u$ with $\lambda = R/\|Au\| < 1$ and $\|u\| = R$, contradicting the strict choice of R. Therefore $u = Au$. The proof also works on a closed convex invariant set containing 0; closedness ensures its closed convex hull remains in the admissible set. □

9.2.3 An elliptic application with gradient dependence

Let $b : \mathbb{R}^n \to \mathbb{R}$ be globally Lipschitz, with constant L and $B = |b(0)|$. For $\mu > 0$ let $Av = w$ solve $-\Delta w + \mu w = -b(Dv)$ with zero trace, in $X = H_0^1(U)$.

Theorem 9.10 (Source Theorem 5, with an explicit sufficient bound)
If $\mu > L^2/2$, the problem $-\Delta u + b(Du) + \mu u = 0$, $u|_{\partial U} = 0$, has a solution $u \in H^2(U) \cap H_0^1(U)$.

Expanded proof (smoothing and a homotopy estimate).

Lax–Milgram and the H^2 estimate of §6.3 give

$$\|Av\|_{H^2} \le C_\mu (L\|Dv\|_2 + B|U|^{1/2}).$$

The Lipschitz bound on b gives continuity, even into H^2. Rellich's embedding $H^2(U) \Subset H^1(U)$ gives compactness in X. If $u = \lambda Au$ for $0 < \lambda \le 1$, then $-\Delta u + \mu u = -\lambda b(Du)$. Testing with u yields

$$\tfrac{1}{2}\|Du\|_2^2 + (\mu - L^2/2)\|u\|_2^2 \le B|U|^{1/2}\|u\|_2.$$

Writing $d = \mu - L^2/2 > 0$, Young bounds the right side by $d\|u\|_2^2/2 + B^2|U|/(2d)$. This uniformly bounds u in H_0^1; $\lambda = 0$ gives $u = 0$. Schaefer applies. The fixed point is in H^2 by its definition, so the PDE holds a.e. □

先验界必须沿同伦统一

只证明 $u = Au$ 的解有界还不够。截断固定点可能满足 $u = \lambda Au$；需对所有 $\lambda \in [0,1]$ 使用同一个界，才能排除它落在截断球面。

9.3 Method of Subsolutions and Supersolutions

9.3.1 Weak order and boundary traces

Let $f : \mathbb{R} \to \mathbb{R}$ be globally Lipschitz, with constant L. For $\underline{u}, \overline{u} \in H^1(U)$, call them a subsolution and a supersolution of $-\Delta u = f(u)$ when, for all nonnegative $v \in H_0^1(U)$,

$$\int D\underline{u} \cdot Dv \le \int f(\underline{u})v, \qquad \int D\overline{u} \cdot Dv \ge \int f(\overline{u})v.$$

The boundary conditions required below are inequalities of Sobolev traces, not pointwise values of arbitrary representatives.

存在性与迭代收敛

Schauder、Schaefer 给出固定点存在性，不自动保证 $u_{k+1} = Au_k$ 收敛。要证明这条序列收敛，还须另证压缩性、单调性或其他稳定结构。

Theorem 9.11 (Ordered existence; source Theorem 1)
If $\underline{u} \le \overline{u}$ a.e. and $\underline{u}|_{\partial U} \le 0 \le \overline{u}|_{\partial U}$, there is a weak solution $u \in H_0^1(U)$ between them. The increasing and decreasing iterations give respectively the least and greatest weak solution in this order interval.

Expanded proof (comparison, domination, and strong convergence).

Choose $\lambda \ge L$, $\lambda > 0$, so $h(s) = f(s) + \lambda s$ is nondecreasing. Starting from $u_0 = \underline{u}$, solve

$$(-\Delta + \lambda)u_{k+1} = h(u_k), \qquad u_{k+1} \in H_0^1(U). \tag{9.3}$$

The forcing is in L^2 and Lax–Milgram gives every iterate. Subtract the equation for u_1 from the subsolution inequality after adding the same $\lambda\underline{u}$ term. Test by $v = (\underline{u} - u_1)_+$. The trace inequality makes $v \in H_0^1$, and the Sobolev truncation rule gives $\|Dv\|_2^2 + \lambda\|v\|_2^2 \le 0$. Hence $u_0 \le u_1$.

If $u_{k-1} \le u_k$, subtract successive equations and test by $(u_k - u_{k+1})_+$. Since $h(u_{k-1}) \le h(u_k)$, the same reasoning gives $u_k \le u_{k+1}$. Comparing with the supersolution using $(u_{k+1} - \overline{u})_+$ gives $u_{k+1} \le \overline{u}$ whenever $u_k \le \overline{u}$. Thus

$$\underline{u} = u_0 \le u_1 \le \cdots \le u_k \le \overline{u}.$$

The pointwise limit u exists a.e. Also $|u_k| \le |\underline{u}| + |\overline{u}| \in L^2$; dominated convergence gives $u_k \to u$ in L^2. The Lipschitz property gives $h(u_k) \to h(u)$ in L^2. The bounded inverse $(-\Delta + \lambda)^{-1} : L^2 \to H_0^1$ now shows that $u_{k+1} \to (-\Delta + \lambda)^{-1}h(u)$ strongly in H_0^1. Its L^2 limit was u, hence u is this inverse image and solves the nonlinear equation after cancellation of λu.

If v is any solution in the interval, induction by the same comparison gives $u_k \le v$, hence $u \le v$. Starting instead at $\overline{u}$ gives a decreasing sequence and the greatest solution. These extremal properties are editorial consequences; uniqueness is not asserted. □

正部测试的合法性

当边界迹满足 $a \le b$ 时，$(a-b)_+$ 的迹为零，因此属于 H_0^1。没有边界序，积分中的非负项不能用来推出全域比较。

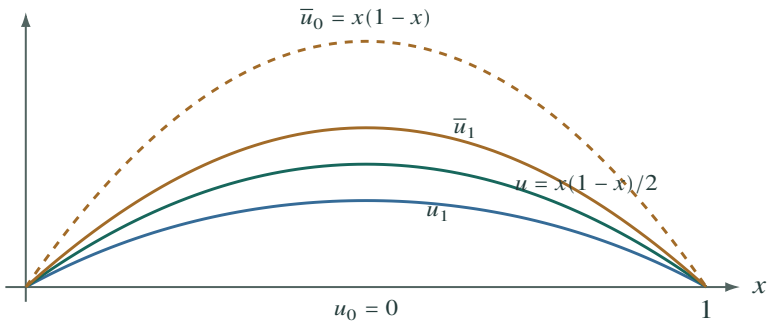


Diagram T9.2. Two ordered iterations trap the solution. For $-u'' = 1$ on $(0,1)$ with zero endpoint data, take $u_0 = 0$, $\overline{u}_0 = x(1-x)$ and shift $\lambda = 4$. The curves show the exact first lower and upper iterates and the solution $u = x(1-x)/2$. Later lower iterates increase and upper iterates decrease in the pointwise order.

9.3.2 Local nonlinearities and the limits of comparison

If the barriers are bounded and f is only Lipschitz on their common value interval $[a,b]$, extend $f|_{[a,b]}$ to a globally Lipschitz function (for example by constant continuation outside). The iteration stays between the barriers, so the resulting equation uses the original f. This is a useful editorial localization of the source's hypothesis.

If f is nonincreasing, two solutions with the same trace are identical: testing their difference gives $\|D(u-v)\|_2^2 = \int (f(u)-f(v))(u-v) \le 0$. More generally uniqueness follows when $\mathrm{Lip}(f)C_P^2 < 1$. This uses a quantitative bound; mere smoothness is insufficient.

上下解不保证唯一

对 $-\Delta u = \lambda_1 u$，零解和任意倍数的第一特征函数都是解。顺序构造给出区间内极小、极大解；唯一性需要额外结构。

9.4 Nonexistence of Solutions

9.4.1 Large data: a weighted moment becomes infinite

Assume U is connected and let $\phi_1 > 0$ be the first Dirichlet eigenfunction, $-\Delta\phi_1 = \lambda_1\phi_1$, normalized by $\int_U \phi_1 = 1$. For the problem $u_t - \Delta u = u^2$, $u|_{\partial U} = 0$, $u(0) = g \ge 0$, take a classical solution with enough boundary regularity for Green's identity. The maximum principle gives $u \ge 0$ as long as it exists.

Theorem 9.12 (Large-data obstruction; source §9.4.1, Theorem 1)
If $\eta_0 = \int g\phi_1 > \lambda_1$, a classical solution cannot persist through the time

$$T_* = \frac{1}{\lambda_1} \log \frac{\eta_0}{\eta_0 - \lambda_1}.$$

This is an upper bound for its lifespan, not necessarily its actual blow-up time.

Expanded proof (normalization gives Jensen's inequality).

Set $\eta(t) = \int u(t)\phi_1$. Both u and ϕ_1 vanish on the boundary, so the two boundary terms in Green's second identity vanish. Consequently

$$\eta' = -\lambda_1 \eta + \int u^2 \phi_1 \geq -\lambda_1 \eta + \eta^2.$$

The last step uses the probability measure $\phi_1 dx$. Comparison with the scalar ODE, or integration of $\xi = e^{\lambda_1 t}\eta$, gives

$$\eta(t) \geq \frac{\lambda_1 \eta_0 e^{-\lambda_1 t}}{\lambda_1 - \eta_0(1 - e^{-\lambda_1 t})} \qquad (0 \leq t < T_*).$$

The denominator stays positive until T_* and tends to zero there. If the PDE solution existed smoothly through that time, its finite moment would contradict this bound. It might cease to exist earlier. Moreover $\eta(t) \leq \|u(t)\|_\infty$, so moment divergence implies amplitude divergence when the solution persists up to the comparison time. □

9.4.2 Subcritical Fujita blow-up in all of space

Theorem 9.13 (Small-data obstruction; source Theorem 2)
Let $g \in C_c^\infty(\mathbb{R}^n)$ be nonnegative and nonzero. If $1 < p < 1 + 2/n$, there is no global nonnegative classical solution of $u_t - \Delta u = u^p$ in the class $C([0,T]; L^1 \cap L^\infty)$ for every finite T. This class supplies the decay and integrability for the calculation.

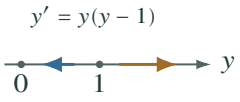
Expanded proof (a scale selected after the initial datum).

Fix $s > 0$, put $\Phi_s(x) = (4\pi s)^{-n/2} e^{-|x|^2/(4s)}$, and set $\eta(t) = \int u(t)\Phi_s$. Since

$$\Delta \Phi_s = \left(\frac{|x|^2}{4s^2} - \frac{n}{2s} \right) \Phi_s, \qquad \int \Phi_s = 1,$$

Green's identity and Jensen give $\eta' \geq -\lambda\eta + \eta^p$ with $\lambda = n/(2s)$. To justify the identity on $\mathbb{R}^n$, test first with $\chi_R \Phi_s$. After integrating by parts onto this compactly supported test, the errors $u(2D\chi_R \cdot D\Phi_s + \Phi_s \Delta \chi_R)$ tend to zero. On each finite time interval, the assumed L^1, L^∞ bounds also

图 T9.M：反应与扩散的阈值



归一化为 $\lambda_1 = 1$ 后，比较 ODE 为 $y' = y(y-1)$。箭头只表示这个 ODE 的相线；$y_0 \leq 1$ 时不能据此断言 PDE 全局存在。

control u^p; dominated convergence gives the integral differential identity. Thus no unproved decay of Du is required.

The comparison ODE blows up if $\eta(0)^{p-1} > \lambda$. This condition is equivalent to

$$(4\pi)^{-n/2} \int g(x) e^{-|x|^2/(4s)} dx > (n/2)^{1/(p-1)} s^{n/2-1/(p-1)}.$$

As $s \to \infty$, the left side tends to a positive constant and the right side tends to zero, since $p < 1 + 2/n$. Choose one such finite s; the ODE comparison then forbids a global solution. At $p = 1 + 2/n$ the right side is constant: this argument does not establish the endpoint. □

9.4.3 The full Derrick–Pohozaev identity

Let $n \geq 3, f \in C(\mathbb{R})$, $F(s) = \int_0^s f(r)dr$, and $u \in C^2(\overline{U})$ solve $-\Delta u = f(u)$ with zero boundary data.

Theorem 9.14 (Identity and nonexistence; source §9.4.2, Theorem 3)
One has

$$\frac{n-2}{2} \int_U |Du|^2 + \frac{1}{2} \int_{\partial U} (x \cdot \nu)(\partial_\nu u)^2 = n \int_U F(u). \tag{9.4}$$

If U is star-shaped about 0 and $f(u) = |u|^{p-1}u$ with $p > (n+2)/(n-2)$, then $u \equiv 0$.

Expanded proof (every boundary term retained).

Introduce the vector field $Q = (x \cdot Du)Du - x|Du|^2/2$. Direct differentiation cancels the Hessian cross terms and yields

$$\operatorname{div} Q = (x \cdot Du)\Delta u + \frac{2-n}{2}|Du|^2.$$

On ∂U, all tangential derivatives of u vanish, so $Du = (\partial_\nu u)\nu$ and $Q \cdot \nu = (x \cdot \nu)(\partial_\nu u)^2/2$. Also $f(u)x \cdot Du = x \cdot DF(u)$, and $\int x \cdot DF(u) = -n \int F(u)$ since $F(0) = 0$ on the boundary. The divergence theorem gives (9.4) with the displayed signs.

For a C^1 star-shaped boundary, choose a local defining function ρ with $U = \{\rho < 0\}$. Since $(1-t)x \in \overline{U}$ for small $t > 0$, $\rho((1-t)x) \leq 0 = \rho(x)$, so $-D\rho(x) \cdot x \leq 0$; hence $x \cdot \nu \geq 0$. Testing the PDE by u gives $\int |Du|^2 = \int |u|^{p+1}$. Substitution in the identity implies

$$\left(\frac{n-2}{2} - \frac{n}{p+1}\right) \int |u|^{p+1} \leq 0.$$

The coefficient is strictly positive at a supercritical exponent, so $u = 0$. At equality of the exponents it is zero: the displayed inequality alone

两个临界指数

热方程这里出现 $1 + 2/n$，来自质量与扩散尺度；椭圆 Pohozaev 中出现 $(n+2)/(n-2)$，来自 Dirichlet 能量与幂次能量的尺度。两者不能混同。

gives no conclusion. On a strictly star-shaped domain a positive nonzero solution at the critical exponent is also ruled out by the boundary term and Hopf's lemma, but this is a separate argument and does not extend the stated theorem to arbitrary sign-changing critical solutions. □

9.5 Geometric Properties of Solutions

9.5.1 Dilation derivatives and star-shaped level sets

Let $V \Subset W$ be smooth bounded star-shaped domains about 0, with $0 \in V$, and assume $U = W \setminus \overline{V}$ is connected. Suppose u is harmonic in U, smooth up to the boundary, equal to 1 on ∂V and 0 on ∂W.

Theorem 9.15 (Source §9.5.1, Theorem 1)
For $0 < \ell < 1$, the level surface $\{u = \ell\}$ is smooth and bounds a star-shaped set. In fact $x \cdot Du < 0$ throughout U.

Expanded proof (the sign on the inner boundary).

The maximum principle gives $0 < u < 1$. Set $v = x \cdot Du$. The commutator formula $\Delta(x \cdot Du) = 2\Delta u + x \cdot D\Delta u$ shows that v is harmonic. On the outer boundary Du is a negative multiple of the outward normal of W, and $x \cdot \nu_W \geq 0$, so $v \leq 0$. On the inner boundary Du points into V, hence is a negative multiple of its outward normal ν_V; once more $v \leq 0$. The strong maximum principle gives either $v < 0$ or $v \equiv 0$ in the connected shell. The latter is impossible: on each ray joining the two boundaries it would make u constant, contrary to the values $1, 0$. Thus $v < 0$, so $Du \neq 0$ in U and the implicit function theorem makes every intermediate level smooth.

Extend $u = 1$ inside V. On a ray $r\omega$, star-shapedness of the two domains makes their intersections initial radial intervals. In the shell, $\partial_r u(r\omega) = v(r\omega)/r < 0$. Therefore each ray crosses an intermediate level exactly once, and its superlevel set contains every segment from 0 to one of its points. This proves star-shapedness directly, including the global radial geometry. □

9.5.2 Maximum principles on a small set

For moving planes the zero-order coefficient need not have a sign. We record two precise substitutes for an unrestricted maximum principle.

边界几何进入哪一步

星形性只用来保证 $x \cdot \nu \geq 0$。Pohozaev 恒等式本身对一般光滑区域成立；若区域不是星形，不能直接丢掉边界积分。

内边界法向易混

相对于壳域 U，内边界外法向指向 V；相对于 V 则方向相反。证明中用 ν_V，于是 Du 为其负倍数，两层边界都给 $x \cdot Du \leq 0$。

Lemma 9.16 (Nonnegative supersolutions; source §9.5.2, Lemma 1)
Let D be connected, $c \in L^\infty(D)$, and $v \geq 0$, $v \not\equiv 0$ satisfy $-\Delta v + cv \geq 0$ classically. Then $v > 0$ in D. At a zero boundary point with an interior tangent ball and C^1 regularity, $\partial_\nu v < 0$.

Expanded proof (remove the zero-order term).

Put $v = e^{kx_1} w$ with $k^2 \geq \|c\|_\infty$. Then $w \geq 0$ and $-\Delta w - 2k w_{x_1} \geq (k^2 - c) w \geq 0$. The strong maximum principle and Hopf's lemma for this operator, whose zero-order coefficient is zero, apply. At a zero of v, $\partial_\nu w = e^{-kx_1} \partial_\nu v$, giving the asserted sign. □

Lemma 9.17 (Small-support comparison; editorial supplement)
Fix a ball B and $L \geq 0$. There is $\delta > 0$ such that if $D \subset B$, $w_- \in H_0^1(D)$, $-\Delta w = c(x) w$, $|c| \leq L$, and $|\{w < 0\}| < \delta$, then $w \geq 0$.

Expanded proof (the quantitative continuation step).

Choose $q > 2$ for which $\|z\|_{L^q(B)} \leq C_B \|Dz\|_2$ on $H_0^1(B)$ (take $q = 2n/(n-2)$ if $n \geq 3$, and $q = 4$ if $n = 1, 2$). Extend w_- by zero. Testing the equation with $-w_-$ gives

$$\|Dw_-\|_2^2 \leq L \|w_-\|_2^2 \leq L C_B^2 |\{w < 0\}|^{1-2/q} \|Dw_-\|_2^2.$$

Choose $L C_B^2 \delta^{1-2/q} < 1$ and absorb. Poincaré gives $w_- = 0$. The constant depends on the containing ball, not on the shape of D. □

9.5.3 Moving planes and radial symmetry

Let $B = B_1(0)$, let f be Lipschitz on $[0, \max u]$, and suppose $u \in C^2(\overline{B})$, $-\Delta u = f(u)$, $u > 0$ in B, $u = 0$ on ∂B. For $0 \leq \lambda < 1$, set

$$\Sigma_\lambda = B \cap \{x_n > \lambda\}, \quad x^\lambda = (x', 2\lambda - x_n), \quad w_\lambda(x) = u(x^\lambda) - u(x).$$

Reflection keeps the cap inside the ball. The difference satisfies $-\Delta w_\lambda = c_\lambda w_\lambda$ with $|c_\lambda| \leq L$, using the difference quotient of f (set it to zero where the two values agree). On the plane $w_\lambda = 0$; on the spherical cap it is nonnegative, and strictly positive away from the rim when $\lambda > 0$.

小区域替代系数符号

比较差满足 $-\Delta w = cw$，c 可正可负。负部支集足够小时，Sobolev 与 Hölder 给出小因子，才能吸收 $L\|w_-\|_2^2$；普通最大值原理不能直接套用。

Theorem 9.18 (Radial symmetry; source Theorem 2)
There is a radial profile $u(x) = v(|x|)$ and $v'(r) < 0$ for $0 < r < 1$.

Expanded proof (start, stop, and continue the plane).

For λ close to 1 the cap has measure less than δ from the small-support lemma. Its boundary values imply $(w_\lambda)_- \in H_0^1(\Sigma_\lambda)$; the lemma gives $w_\lambda \geq 0$. The previous strong principle gives $w_\lambda > 0$. Let λ_0 be the infimum of positions such that all planes further out have this comparison. Continuity gives $w_{\lambda_0} \geq 0$.

If $\lambda_0 > 0$, the spherical boundary values exclude $w_{\lambda_0} \equiv 0$, hence $w_{\lambda_0} > 0$ in the cap. Choose a compact set $K \Subset \Sigma_{\lambda_0}$ whose complement in the cap has measure less than $\delta/2$. On K the minimum of w_{λ_0} is positive. Uniform continuity therefore gives $w_\lambda > 0$ on K for all λ slightly below λ_0. The extra thin slice of the cap has measure less than $\delta/2$. Thus any negative set of w_λ has measure less than δ; the small-support lemma excludes it. The strong principle restores strict positivity, contradicting the stopping definition. Hence $\lambda_0 = 0$.

At $\lambda = 0$ one inequality follows. Repeating from the opposite pole gives the reverse inequality, so u is symmetric in the central plane. Rotate the argument to every direction: invariance under all reflections through 0 makes u radial. For $0 < \lambda < 1$, Hopf at an interior point of the flat face gives $\partial_\nu w_\lambda < 0$ with $\nu = -e_n$. Since $(w_\lambda)_{x_n} = -2u_{x_n}$ there, $u_{x_n} < 0$ on that face. Taking $x = re_n$, $\lambda = r$, proves $v'(r) < 0$. At the center the gradient is zero by symmetry. This proof replaces the source's boundary-derivative continuation with a fully quantified small-support argument; it covers the same positive ball problem. □

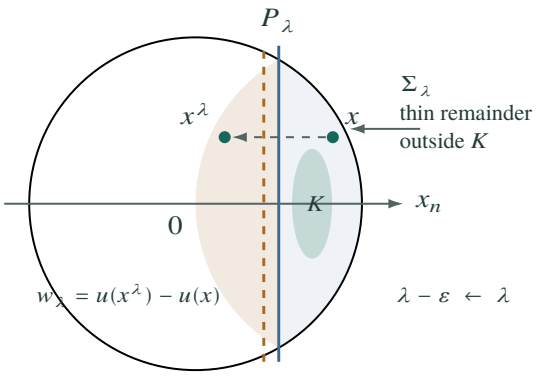


Diagram T9.3. Reflection and the region available for continuation. The cap Σ_λ reflects inside the unit ball. At a proposed positive stopping position, comparison is uniformly strict on a compact core K. Any negative part for a nearby plane must lie in the thin remainder, where the small-support estimate excludes it. Geometry is schematic; the equation is scalar and $u > 0$ is essential.

9.6 Gradient Flows

9.6.1 Extended energies and subgradients

Let H be a real Hilbert space and $I : H \to (-\infty, +\infty]$ be proper, convex and lower semicontinuous in the norm topology. Its effective domain is $D(I) = \{u : I(u) < \infty\}$. For $u \in D(I)$ set

$$\partial I(u) = \{a \in H : I(w) \geq I(u) + (a, w - u) \text{ for every } w \in H\};$$

outside $D(I)$ the set is empty. Write $A = \partial I$ and $D(A) = \{u : A(u) \neq \emptyset\}$. Adding the subgradient inequalities shows $(a - b, u - v) \geq 0$ for $a \in A(u)$, $b \in A(v)$. Also $0 \in A(u)$ exactly when u minimizes I.

Theorem 9.19 (Resolvent; source §9.6.1, Theorem 1)
For each $\lambda > 0$ and $w \in H$ there is exactly one u with $u + \lambda A(u) \ni w$. It uniquely minimizes

$$I(z) + \frac{1}{2\lambda}\|z - w\|^2.$$

Expanded proof (coercivity and the actual subgradient inequality).

The closed convex epigraph of I can be strictly separated from a point $(u_0, I(u_0) - 1)$ below it, where $I(u_0) < \infty$. The separating functional has a nonzero vertical component: otherwise its values at these two points with the same first coordinate could not be strictly ordered. Its sign follows from upward closure of the epigraph. Dividing gives an affine minorant $I(z) \geq (b, z) + d$. This uses the geometric Hahn–Banach theorem as an explicit functional analysis input. Therefore the quadratic functional above is coercive and bounded below. Norm-closed convex sublevel sets are weakly closed by the same separation theorem, so I is weakly lower semicontinuous. A bounded minimizing sequence has a weakly convergent subsequence in H; the limit minimizes. The quadratic term makes the minimizer unique.

Let that minimizer be u. Compare it with $u + t(z - u)$ for $0 < t < 1$. Convexity bounds its energy above by $(1 - t)I(u) + tI(z)$. Expansion of the quadratic, division by t, and $t \downarrow 0$ give $I(z) - I(u) \geq ((w - u)/\lambda, z - u)$. Thus $(w - u)/\lambda \in A(u)$. Conversely this inequality, together with expansion of the square, proves minimality, with a strict excess $\|z - u\|^2/(2\lambda)$ if $z \neq u$. □

Define the resolvent $J_\lambda = (\mathrm{Id} + \lambda A)^{-1}$ and the Yosida approximation $A_\lambda = (\mathrm{Id} - J_\lambda)/\lambda$.

正性和球形的用途

球形保证反射帽仍在定义域内；正性给球面边界上的严格比较，从而排除正停止位置。一般区域或变号解不能据此推出径向对称。

允许无穷能量

把不满足边界或约束的函数赋值 $+\infty$，能把限制条件写进定义域。$\partial I(u)$ 是 Hilbert 空间中的向量集合，依赖所选内积，不能只看形式变分。

Theorem 9.20 (Source Theorem 2, with sharpened Lipschitz bound)
For $u, v \in H$, writing $j = J_\lambda u - J_\lambda v$ and $a = A_\lambda u - A_\lambda v$, one has

$$\|j\|^2 + \lambda^2 \|a\|^2 \le \|u - v\|^2, \qquad (a, u - v) \ge 0.$$

In particular J_λ is nonexpansive, A_λ is $1/\lambda$- Lipschitz, and $A_\lambda u \in A(J_\lambda u)$. For $u \in D(A)$ and $a_0 \in A(u)$, $\|A_\lambda u\| \le \|a_0\|$ and $\|J_\lambda u - u\| \le \lambda \|a_0\|$. The last convergence extends to $u \in \overline{D(A)}$.

Expanded proof (firm nonexpansiveness).

The resolvent identity gives $u - v = j + \lambda a$, with $(a, j) \ge 0$. Squaring proves the first inequality; taking inner product with a proves monotonicity of A_λ. The inclusion is the defining resolvent equation. Comparing $(J_\lambda u, A_\lambda u)$ with (u, a_0) by monotonicity gives $\lambda \|A_\lambda u\|^2 \le \lambda (a_0, A_\lambda u)$, which gives the bound and convergence on $D(A)$. Approximate any point of its closure by $v \in D(A)$ and use $\|J_\lambda u - u\| \le 2\|u - v\| + \|J_\lambda v - v\|$. The source states the weaker $2/\lambda$ Lipschitz estimate; the squared identity supplies the sharper bound used here. □

Lemma 9.21 (Maximality and graph closure; editorial supplement)
If (u, a) satisfies $(a - b, u - v) \ge 0$ for every $b \in A(v)$, then $a \in A(u)$. Consequently the graph is closed for strong convergence of u and weak convergence of bounded a.

Expanded proof (one resolvent test).

Take $v = J_1(u + a)$ and $b = u + a - v \in A(v)$. The inequality becomes $-\|u - v\|^2 \ge 0$, so $u = v$, $a = b$. For a strong–weak limit of graph pairs, monotonicity passes to the limit against every fixed pair (v, b), and the first assertion applies. □

9.6.2 Constructing the nonlinear flow

Theorem 9.22 (Gradient flow; source §9.6.2, Theorem 3)
For $g \in D(A)$ there is a unique $u \in C([0, \infty); H)$, $u' \in L^\infty(0, \infty; H)$ such that $u(0) = g$, $u(t) \in D(A)$ for every $t \ge 0$, and $-u'(t) \in A(u(t))$ a.e. For any $a_0 \in A(g)$, $\|u'\|_{L^\infty} \le \|a_0\|$. If $\overline{D(A)} = H$, these solutions extend uniquely by contraction to a strongly continuous nonlinear semigroup $S(t)$ on H. For general $g \in H$ this extension initially asserts a continuous semigroup solution, without the global derivative bound above.

图闭性不是弱连续性

这里用“输入强收敛、输出弱收敛”和单调图的最大性。并没有假设多值算子 A 连续，也不能把 $A(u_k)$ 直接取极限写为 $A(u)$。

Expanded proof (Yosida approximation and a common graph test).

For $\lambda > 0$ solve the globally Lipschitz Hilbert-space ODE $u_\lambda' = -A_\lambda u_\lambda, u_\lambda(0) = g$. Picard iteration gives a global solution (linear growth prevents finite-time escape). Monotonicity gives contraction between two such trajectories. Compare one trajectory with its time translate, divide by the shift and let the shift vanish to obtain

$$\|u_\lambda'(t)\| \le \|A_\lambda g\| \le M := \|a_0\|.$$

Write $a_\lambda = A_\lambda u_\lambda$, $z_\lambda = J_\lambda u_\lambda$ so $u_\lambda = z_\lambda + \lambda a_\lambda$ and $a_\lambda \in A(z_\lambda)$. For two parameters, monotonicity gives

$$\tfrac{1}{2}(\|u_\lambda - u_\mu\|^2)' \le -(a_\lambda - a_\mu, \lambda a_\lambda - \mu a_\mu) \le 2(\lambda + \mu)M^2.$$

Thus the trajectories are Cauchy uniformly on every $[0,T]$, with $\|u_\lambda - u_\mu\|^2_{C([0,T];H)} \le 4T(\lambda + \mu)M^2$. They converge strongly to a Lipschitz u; after extracting on finite intervals, $a_\lambda \rightharpoonup a = -u'$ in $L^2(0,T;H)$. Also $z_\lambda \to u$ uniformly and $\|a\|_{L^\infty} \le M$.

To identify the graph without exceptional sets depending on arbitrary test vectors, define the measurable pair

$$v(t) = J_1(u(t) + a(t)), \qquad b(t) = u(t) + a(t) - v(t) \in A(v(t)).$$

Both functions are in $L^2(0,T;H)$ since the resolvent is Lipschitz. Integrate $(a_\lambda - b, z_\lambda - v) \ge 0$ and pass to the limit using strong convergence of z_λ and weak convergence of a_λ. The result is $\int (a - b, u - v) = -\int \|u - v\|^2 \ge 0$. Hence $u = v$ a.e., and $a = b \in A(u)$ a.e. This is the essential limit identification; pointwise convergence of a_λ was not assumed.

For any fixed t, choose times $t_j \to t$ outside the null set so $(u(t_j), a(t_j))$ are graph pairs and $\|a(t_j)\| \le M$. A weakly convergent subsequence of the second components and the strong–weak graph closure give $u(t) \in D(A)$. Two solutions satisfy $\frac{1}{2}(\|u - v\|^2)' \le 0$ by monotonicity, proving uniqueness and contraction. Uniqueness also gives $S(t + s) = S(t)S(s)$ on $D(A)$.

For $g \in \overline{D(A)}$, approximate by $g_j \in D(A)$. Contraction makes $S(t)g_j$ Cauchy uniformly even on all $t \ge 0$. The limit is independent of the approximating sequence, continuous at $t = 0$, and retains contraction and the semigroup law. It need not have a derivative uniformly bounded down to $t = 0$. This distinguishes the strong solution theorem from its continuous extension. □

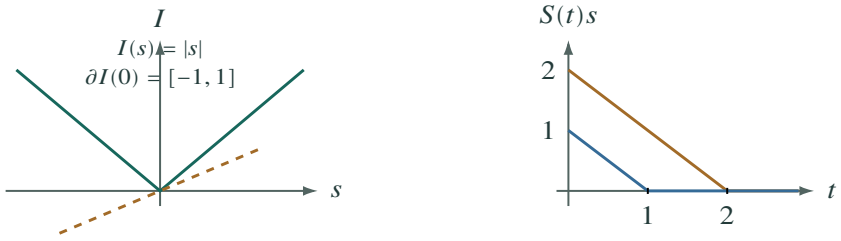


Diagram T9.4. A nonsmooth energy can merge trajectories. For $H = \mathbb{R}$ and $I(s) = |s|$, $J_\lambda(s) = \operatorname{sgn}(s)(|s| - \lambda)_+$ and $S(t)s = \operatorname{sgn}(s)(|s| - t)_+$. The supporting slopes at zero form $[-1, 1]$; different positive initial values reach zero at different times and thereafter coincide. These are exact graphs, not a discretization of another model.

9.6.3 Dissipation and a quasilinear parabolic equation

Lemma 9.23 (Convex chain rule; editorial supplement)
If $u \in W^{1,2}(0,T;H)$ and $a \in L^2(0,T;H)$ with $a(t) \in \partial I(u(t))$ a.e., then $I(u(t))$ has an absolutely continuous representative and $(I(u))' = (a, u')$ a.e. For the flow,

$$I(u(t)) + \int_s^t \|u'(r)\|^2 dr = I(u(s)).$$

For the solution above with $g \in D(A)$ the identity holds for all $0 \le s \le t$.

Expanded proof (two subgradient inequalities squeeze the derivative).

At times where the inclusions hold, for $h > 0$ the increment of I between $u(t)$ and $u(t+h)$ lies between $(a(t), u(t+h) - u(t))$ and $(a(t+h), u(t+h) - u(t))$. On compact subintervals these bounds also make $I(u)$ integrable: use one fixed graph point as an affine lower bound, and test the subgradient at $u(t)$ against a fixed finite-energy point for an upper bound in L^1. Multiply the increment inequalities by a nonnegative smooth test in time and integrate. Difference quotients of u converge to u' in L^2 and translations of a converge to a in L^2. Both bounds therefore converge to (a, u') in distributions. This L^1 derivative gives the asserted absolutely continuous representative. For our flow a is bounded. The subgradient inequalities and continuity of u then identify this representative with the actual $I(u(t))$ at every time, including 0, by graph closure and lower semicontinuity. Finally $a = -u'$ gives dissipation. □

Let $L \in C^2(\mathbb{R}^n)$ satisfy $\theta|\xi|^2 \le D^2L(p)\xi \cdot \xi$ and $|D^2L(p)| \le M$. On $H = L^2(U)$ define $I(u) = \int_U L(Du)$ for $u \in H_0^1(U)$, and $I(u) = +\infty$ otherwise.

收缩不代表严格收缩

这里 $\|S(t)g - S(t)h\| \le \|g - h\|$，因子是 1，不是 Banach 定理所需的小于 1。半群还可能把不同初值合并，见习题 11。

Theorem 9.24 (PDE realization; source §9.6.3, Theorem 4)
This energy is proper, convex and lower semicontinuous in L^2. Its subdifferential is single-valued with

$$D(\partial I) = H^2(U) \cap H_0^1(U), \qquad \partial I(u) = -\operatorname{div} DL(Du).$$

For g in this domain the flow solves $u_t - \operatorname{div} DL(Du) = 0$, with zero boundary data, and $u \in L^\infty(0,\infty; H^2 \cap H_0^1)$, $u_t \in L^\infty(0,\infty; L^2)$. For $g \in L^2$ its continuous semigroup extension is defined by approximation.

Expanded proof (the topology of the energy matters).

Taylor's inequality and $\int_U Du = 0$ for $u \in H_0^1$ give $I(u) \geq L(0)|U| + \theta\|Du\|_2^2/2$; the affine term integrates to zero by density of C_c^∞. A sequence converging in L^2 with bounded energy is thus bounded in H_0^1; a subsequence converges weakly there to the same limit. Convex integral lower semicontinuity from §8.2 gives the L^2 lower semicontinuity, including the case of infinite limit energy. Properness follows from $I(0) < \infty$.

If $u \in H^2 \cap H_0^1$, the Lipschitz bound on DL makes $-\operatorname{div} DL(Du) \in L^2$. The convex supporting inequality, integrated and integrated by parts against $w - u \in H_0^1$, shows it belongs to $\partial I(u)$; tests outside H_0^1 have infinite energy. Conversely, if $a \in \partial I(u)$, use $w = u \pm t\varphi$ for $\varphi \in H_0^1$. The linear growth of DL justifies differentiation under the integral, giving $\int a\varphi = \int DL(Du) \cdot D\varphi$. The H^2 result of §9.1 applies to DL, whose derivative is bounded and uniformly elliptic. Therefore $u \in H^2 \cap H_0^1$, and density of test functions identifies a uniquely. The same estimates, testing first by u, give $\|u\|_{H^2} \leq C\|a\|_2$; the constant vector $DL(0)$ has zero pairing with zero-trace gradients.

The domain is dense in L^2 because it contains $C_c^\infty(U)$. Apply the flow theorem; its speed bound and this elliptic estimate give the claimed uniform H^2 bound. In weak form the spatial divergence is exactly the L^2 subgradient identified above. The statement for rough g uses contraction extension, not the uniform-in-time H^2 claim made for domain data. □

Hilbert 梯度依赖度量

这里取 $H = L^2$，所以梯度为 $-\operatorname{div} DL(Du)$。若在 H_0^1 的梯度内积中取 Riesz 代表，还需再作用逆 Laplace 算子；所得流不同。

9.7 Problems

The following short problem descriptions preserve the source numbers 1–14. All solutions are editorial additions. Domain, decay, and regularity conditions needed by a computation are stated explicitly.

1. A flow proof of the finite-dimensional zero lemma

Solution. Let V be smooth near $\overline{B}_r$ and solve $x' = -V(x)$. The inequality $V(x) \cdot x \geq 0$ on the sphere makes the closed ball forward invariant. One justification uses a Lipschitz bound near the ball: for $d(x) = \operatorname{dist}(x, \overline{B}_r)$, projection to the sphere gives $(d(x)^2)' \leq Cd(x)^2$ outside, so an orbit starting with $d = 0$ cannot leave by Gronwall. Boundedness in the compact ball permits global continuation. For every $h > 0$ its continuous time-h map has a Brouwer fixed point y_h. Take $h_j \downarrow 0$ and $y_{h_j} \to y$. Uniformly for initial points in the ball, $x(h, y_h) = y_h - hV(y_h) + O(h^2)$, since V and DV are bounded there. The fixed-point identity gives $V(y_h) = O(h)$, hence $V(y) = 0$. A fixed point of just one time map could be a periodic orbit; the small-time limit is necessary to complete this argument. □

2. Weakly continuous scalar nonlinearities are affine

Solution. Write G for the continuous scalar nonlinearity. Fix $s, t \in \mathbb{R}$, $0 < \theta < 1$. On each interval of length $1/k$ let f_k equal s on the first fraction θ and t on the remainder. For continuous test functions, cellwise averaging gives $f_k \rightharpoonup m = \theta s + (1-\theta)t$; boundedness and density extend this to all L^2 tests. The same argument gives $G(f_k) \rightharpoonup \theta G(s) + (1-\theta)G(t)$. The assumed weak continuity forces $G(\theta s + (1-\theta)t) = \theta G(s) + (1-\theta)G(t)$. A continuous function satisfying this identity on every segment is $G(s) = \alpha s + \beta$: set $\beta = G(0)$, compare first on $[0, 1]$, then use segments containing $0, 1$ and any real s. Conversely affine maps preserve weak convergence by linearity. □

3. Penalty approximation of the nonnegative obstacle

Solution. Let $\beta_\varepsilon(s) = \min(s, 0)/\varepsilon$ and solve $-\Delta u_\varepsilon + \beta_\varepsilon(u_\varepsilon) = f$. Existence follows by minimizing the strictly convex functional

$$\tfrac{1}{2}\|Dv\|_2^2 + \frac{1}{2\varepsilon}\|v_-\|_2^2 - \int fv \quad \text{on } H_0^1(U).$$

Coercivity and convex lower semicontinuity give a minimizer, and its derivative gives the equation. Testing by u_ε bounds $\|Du_\varepsilon\|_2$ and $\varepsilon^{-1}\|(u_\varepsilon)_-\|_2^2$. Testing by $-(u_\varepsilon)_-$ gives more precisely

$$\|D(u_\varepsilon)_-\|_2^2 + \varepsilon^{-1}\|(u_\varepsilon)_-\|_2^2 \leq \|f_-\|_2 \|(u_\varepsilon)_-\|_2,$$

so $\|(u_\varepsilon)_-\|_2 \leq \varepsilon\|f_-\|_2$. A weakly convergent subsequence has limit $u \geq 0$ by compact L^2 convergence. For each $w \geq 0$ in H_0^1, $\beta_\varepsilon(u_\varepsilon)(w - u_\varepsilon) \leq 0$, hence

$$\int Du_\varepsilon \cdot D(w - u_\varepsilon) \geq \int f(w - u_\varepsilon).$$

Rearrange with $\|Du_\varepsilon\|_2^2$ on the left and use weak lower semicontinuity to obtain the desired variational inequality for u. Two such solutions, tested against one another, give equality by strict convexity. Thus every subsequential limit is u, giving convergence of the full family weakly in H_0^1. Taking $w = u$ in the last inequality also gives a matching upper bound for the gradient norm, so convergence is in fact strong in H_0^1. □

4. Periodic forcing selects one periodic initial datum

Solution. For $f(t+\tau) = f(t), f \in L^\infty$, define the period map $P(g) = u(\tau; g)$ on $L^2(U)$. The difference of two solutions has zero forcing, and its energy estimate gives $\|P(g) - P(h)\|_2 \le e^{-\lambda_1 \tau}\|g - h\|_2$. Banach provides a unique g_*. The functions $u(t+\tau; g_*)$ and $u(t; g_*)$ solve the same problem with the same initial trace; linear uniqueness proves periodicity. Every periodic solution supplies a fixed point, proving uniqueness of the initial datum as well. □

5. A small gradient nonlinearity

Solution. Assume $f \in L^2$ and b is globally Lipschitz with constant L. Let Av solve $-\Delta Av = f - b(Dv)$, zero trace. Testing the difference of two outputs gives

$$\|D(Av - Aw)\|_2^2 \le LC_P\|D(v-w)\|_2\|D(Av - Aw)\|_2.$$

Thus $LC_P < 1$ makes A a contraction on H_0^1. Its fixed point is unique and belongs to H^2 by the linear L^2 regularity theorem. The value $b(0)$ affects the forcing bound but not the contraction factor. □

6. A positive solution above the first eigenvalue

Solution. Assume U connected, f bounded and Lipschitz, $f(0) = 0$, and $f'(0) > \lambda_1$. Normalize $\|\phi_1\|_\infty = 1$. For small $\varepsilon > 0$, differentiability at 0 gives $f(\varepsilon\phi_1) \ge \lambda_1\varepsilon\phi_1$, so $\underline{u} = \varepsilon\phi_1$ is a subsolution. Let $-\Delta\psi = 1$, $\psi = 0$ on the boundary, and take $\overline{u} = M\psi$ with $M \ge \sup|f|$. It is a supersolution and dominates $\underline{u}$ after increasing M: indeed $-\Delta[(\lambda_1\|\phi_1\|_\infty)\psi - \phi_1] \ge 0$, so comparison bounds ϕ_1 by a multiple of ψ. The ordered theorem yields $u \ge \varepsilon\phi_1 > 0$ in U. This pointwise statement uses the continuous representative, supplied by elliptic regularity with bounded right side. On a disconnected domain, $f'(0) > \lambda_1(U)$ alone need not give positivity on every component; connectedness is the needed qualification here. □

7. Classical comparison for the ordered iterates

Solution. With $h(s) = f(s) + \lambda s$ nondecreasing, each difference satisfies $(-\Delta + \lambda)(u_{k+1} - u_k) = h(u_k) - h(u_{k-1}) \geq 0$ once $u_k \geq u_{k-1}$. Its boundary data are zero, except at the first comparison where they are nonnegative by the subsolution trace. The maximum principle starts and propagates the induction. Likewise $(-\Delta + \lambda)(\overline{u} - u_{k+1}) \geq h(\overline{u}) - h(u_k) \geq 0$. Thus the entire chain lies between the barriers. The weak proof in §9.3 is its positive-part counterpart. □

8. Critical bubbles and lost compactness

Solution. For $n \geq 3$, $U(r) = (1 + r^2)^{-(n-2)/2}$ has

$$U' = -(n-2)r(1+r^2)^{-n/2},$$
$$-U'' - \frac{n-1}{r}U' = n(n-2)(1+r^2)^{-(n+2)/2}.$$

Thus $c = n(n-2)$. Scaling gives $U_\lambda(x) = \lambda^{-(n-2)/2}U(x/\lambda)$ and the same equation. For $2^* = 2n/(n-2)$, changing variables $x = \lambda y$ proves invariance of both $\|U_\lambda\|_{2^*}$ and $\|DU_\lambda\|_2$. These norms are finite: at infinity the respective integrands have radial orders r^{-n-1} and r^{1-n}, and they are regular at the origin. As $\lambda \downarrow 0$, $U_\lambda(x) \to 0$ for $x \neq 0$. Any strongly convergent L^{2^*} subsequence would have limit zero (by a further a.e. convergent subsequence), contrary to its constant nonzero norm. This proves noncompactness along a specified sequence; constant norms alone would not prove it. □

集中需要极限证据

尺度不变范数只给有界性。还须取 $\lambda \downarrow 0$，证明几乎处处极限为零，再用强收敛保持范数得到矛盾。

9. Energy and a weighted parabolic Pohozaev identity

Solution. Assume sufficient rapid decay for every displayed integral and boundary flux, and $F' = f$, $F(0) = 0$. Set $E = \int(|Du|^2/2 - F(u))$. Differentiate, integrate by parts, and use $u_t = \Delta u + f(u)$: $E' = -\int u_t^2$. For the second moment E_2 of the same density,

$$E_2' = -\int |x|^2 u_t^2 - 2\int u_t(x \cdot Du).$$

The two cross terms are $\int \Delta u(x \cdot Du) = (n-2)\int |Du|^2/2$ and $\int f(u)(x \cdot Du) = -n\int F(u)$. Therefore

$$E_2' = -\int(|x|^2 u_t^2 - 2nF(u) + (n-2)|Du|^2).$$

The signs differ because the two densities have different scaling degrees. Weighted integrability is essential; finite unweighted energy alone does not license differentiating E_2. □

10. The normal cone and metric projection

Solution. For a nonempty closed convex $K \subset \mathbb{R}^n$, its indicator energy has

$$\partial I(x) = N_K(x) = \{v : v \cdot (y - x) \le 0 \ (y \in K)\} \quad (x \in K),$$

and empty subdifferential outside K. Minimizing $I(x)+|x-z|^2/(2\lambda)$ gives $J_\lambda z = P_K z$, independent of λ, and $A_\lambda z = (z - P_K z)/\lambda$. The projection condition $z - P_K z \in N_K(P_K z)$ follows by varying along each segment to $y \in K$. At an interior point the normal cone is $\{0\}$; on a smooth boundary it is the outward normal ray. □

11. An irreversible flow

Solution. Take $H = \mathbb{R}$, $I(s) = |s|$. Then $\partial I(s) = \{\operatorname{sgn} s\}$ for $s \neq 0$ and $\partial I(0) = [-1, 1]$. Directly, $S(t)s = \operatorname{sgn}(s)(|s| - t)_+$ solves the inclusion a.e. For example $S(2)1 = S(2)2 = 0$ although $1 \neq 2$. This semigroup is contractive and continuous but its time maps need not be injective; a backward inverse flow cannot be defined on all states. □

12. A constrained sandpile as a normal-cone flow

Solution. Let $K = \{v \in L^2(\mathbb{R}^2) : |Dv| \le 1 \text{ a.e.}\}$, with the inequality understood in the weak-gradient sense. It is convex and closed in L^2: distributional limits preserve the gradient bound, by weak-* compactness of bounded gradients on compact subsets. Its indicator I is proper, convex and lower semicontinuous. Assume the proposed fields are regular and integrable enough that $f - u_t = -\operatorname{div}(aDu) \in L^2$ and integration by parts against $v - u$ is justified, for instance by compactly supported flux and approximation. For $v \in K$,

$$(f - u_t, v - u) = \int aDu \cdot (Dv - Du) \le 0.$$

Indeed where $a > 0$ one has $|Du| = 1$, so $Du \cdot Dv - |Du|^2 \le |Dv| - 1 \le 0$; elsewhere the integrand is zero. This is precisely $f - u_t \in N_K(u) = \partial I(u)$. The calculation is conditional on the asserted flux and regularity; it does not construct a from a solution of the abstract inclusion. □

13. Exponential decay of the speed

Solution. For a smooth solution with compatible boundary data, put $v = u_t$. Differentiating $u_t - \operatorname{div} DL(Du) = 0$ gives $v_t - \operatorname{div}(D^2L(Du)Dv) = 0$, with $v = 0$ on the boundary. Testing by v, uniform convexity and Poincaré give

$$\tfrac{1}{2}(\|v\|_2^2)' \le -\theta\|Dv\|_2^2 \le -\theta\lambda_1\|v\|_2^2.$$

Hence $\|u_t(t)\|_2^2 \le \|u_t(0)\|_2^2 e^{-2\theta\lambda_1 t}$ when the initial speed is finite. Otherwise the same estimate starts at any positive regular time, with a constant depending on that time. The uniform coefficient θ is the source of the decay rate. □

14. Total variation under nonlinear diffusion

Solution. Write $v = \phi(u)$ with $\phi' > 0$. Under the source's nonvanishing-gradient assumption and sufficient integrability and vanishing flux at infinity, $Du/|Du| = Dv/|Dv|$. Thus

$$\frac{d}{dt}\int |Du| = \int \frac{Dv}{|Dv|}\cdot D\Delta v.$$

Differentiating $|Dv|$ twice gives the pointwise Bochner identity

$$\Delta|Dv| = \frac{Dv}{|Dv|}\cdot D\Delta v + \frac{|D^2 v|^2}{|Dv|} - \frac{|D^2 v Dv|^2}{|Dv|^3}.$$

Its Laplacian integrates to zero under the stated flux hypothesis, so

$$\frac{d}{dt}\int |Du| = -\int \left(\frac{|D^2 v|^2}{|Dv|} - \frac{|D^2 v Dv|^2}{|Dv|^3}\right) \le 0.$$

The inequality uses $|D^2 v Dv| \le |D^2 v||Dv|$. This is the requested smooth calculation where denominators are defined. At critical points one needs a regularization and a limiting argument; no general BV contraction theorem is proved here, and nonvanishing and whole-space decay should not be silently imposed on arbitrary data. □

9.8 References

Source directions

Evans's §9.8 points to J.-L. Lions [L2] and Zeidler [Zd, Vol. 2] for monotonicity, and to [E1] for nonlinear weak-convergence methods. For fixed points the source recommends Zeidler [Zd, Vol. 1] and Gilbarg–Trudinger [G-T, Chapter 11]; for ordered methods, Smoller [S, Chapter 10]. The blow-up discussion follows Payne [Pa, §9]; the small-data theorem is attributed there to H. Fujita, J. Fac. Sci. U. Tokyo Sect. I 13 (1966), 109–124. The radial symmetry result is attributed to Gidas, Ni and Nirenberg, Comm. Math. Phys. 68 (1979), 209–243. The convex-flow material follows Brezis [Br2]; Barbu [Ba] and Zeidler [Zd, Vol. 2] are further directions for nonlinear semigroups. These are the source's attributions and bibliography keys, not a claim that all of those works have been independently consulted for this draft.

Proof dependencies and scope of this reconstruction

The finite-dimensional input is Brouwer from §8.1. Functional analysis inputs include weak compactness of bounded Hilbert-space sequences, Hahn–Banach separation, Sobolev and Rellich embeddings, and the Hilbert-space Picard theorem. PDE inputs are the linear elliptic and parabolic theories of Chapters 6–7, Sobolev truncation and time chain rules from Chapter 5, and convex integral lower semicontinuity from Chapter 8. The global boundary H^2 step in §9.1 explicitly uses the boundary difference-quotient argument of Chapters 6 and 8.

Editorial changes include a bounded-Jacobian hypothesis for that H^2 claim, an L^2 restart formulation for reaction–diffusion, a complete compact-convex-hull argument in Schaefer, extremal ordered solutions, a quantified small-support proof of moving-plane continuation, and a space–time resolvent test identifying the gradient-flow limit. The critical Fujita endpoint, general critical sign-changing Pohozaev nonexistence, arbitrary-data H^2 bounds down to time zero, general BV contraction, and reconstruction of a sandpile flux are not proved. English exposition, proofs, Chinese mathematical annotations, solutions, and TikZ diagrams are editorial reconstructions or additions. Compilation and explicit model checks do not constitute independent mathematical review.

Chapter 10

Hamilton–Jacobi Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 10; printed pp. 543–570; PDF pages 560–587.

Chinese reference: 中译本书内第 384–403 页；扫描 PDF 第 402–421 页。

This chapter reconstructs the source in English with editorial proof details, Chinese mathematical annotations, and native TikZ diagrams. The source's theorem numbers are identified separately from the numbering of these notes. We distinguish the initial-value viscosity convention from the terminal-control convention, and state the extra comparison argument needed for a superlinear Hamiltonian. This is an editorial study draft; it has not undergone independent mathematical review.

10.1 Introduction, Viscosity Solutions

10.1.1 Touching replaces integration by parts

We study

$$u_t + H(D_x u, x) = 0, \qquad u(x,0) = g(x), \tag{10.1}$$

where $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is continuous. Characteristics may intersect, so a globally smooth solution is generally unavailable. The nonlinear expression $H(Du, x)$ has no general distributional interpretation for merely continuous u. At a local extremum of $u - \phi$, however, the derivative of a smooth test function can replace that of u. This is a pointwise contact argument, not integration by parts. Write $\mathrm{BUC}(\mathbb{R}^n \times [0,T])$ for bounded uniformly continuous functions on the whole closed slab; the uniformity includes spatial infinity.

Definition 10.1 (Initial-value viscosity convention; source §10.1.1)
A continuous u is a viscosity subsolution in $\mathbb{R}^n \times (0,T)$ if, whenever $\phi \in C^\infty$ and $u - \phi$ has a local maximum at (x,t),

$$\phi_t(x,t) + H(D\phi(x,t),x) \le 0.$$

A supersolution uses a local minimum and the reverse inequality. A solution is both. For the source's initial-value class we additionally require $u \in \mathrm{BUC}$ on each finite slab and $u(\cdot,0) = g$. The contact need not be strict and its value need not be zero, since a constant can be added to ϕ. All contacts in this definition have $t > 0$.

Lemma 10.2 (Persistence of strict contacts; editorial detail for §10.1.1)
Suppose $u_j \to u$ uniformly on a closed ball B in space–time, $u - \phi$ has a strict maximum at its center z_0, and B lies in the open PDE domain. Maximizers z_j of $u_j - \phi$ on B lie in its interior eventually and converge to z_0.

Expanded proof (the boundary gap and all cluster points).

Continuity and compactness give a positive gap between the central value and the maximum over ∂B. Uniform convergence preserves half that gap, so $z_j \notin \partial B$. Any convergent subsequence of z_j has a limit maximizing $u - \phi$ on B, by uniform convergence. Strictness makes this limit z_0. Compactness then gives convergence of the entire sequence. The minimum version follows by changing signs. For a nonstrict maximum, replace ϕ by $\phi + \delta|z - z_0|^2$: the contact is strict and the first derivatives at z_0 are unchanged. For second-order equations one can instead use $\delta|z - z_0|^4$ to preserve the Hessian as well. □

Theorem 10.3 (Vanishing viscosity; source motivation, made conditional)
Let $\varepsilon_j \downarrow 0$ and let smooth u_j satisfy $(u_j)_t + H(Du_j, x) - \varepsilon_j \Delta u_j = 0$. If $u_j \to u$ locally uniformly, then u solves the first-order equation in the viscosity sense in the open domain. If the convergence includes $t = 0$, the initial values converge to g, and u is in the BUC class, then it is a viscosity solution of the initial-value problem.

Expanded proof (the sign of the positive diffusion).

At a strict upper contact, choose $z_j \to z_0$ by the lemma. There $Du_j = D\phi$, $(u_j)_t = \phi_t$, and $D_x^2 u_j \le D_x^2 \phi$ as symmetric matrices. Hence

$$\phi_t(z_j) + H(D\phi(z_j), x_j) = \varepsilon_j \Delta u_j(z_j) \le \varepsilon_j \Delta \phi(z_j).$$

测试方向与极值

局部最大点对应测试函数从上接触，要求残差不大于零；局部最小点从下接触，要求残差不小于零。导数放在测试函数上，定义不要求 Du 处处存在。

The right side tends to zero because the test Hessian is bounded near z_0. Continuity of H gives the subsolution inequality. At a lower contact the Hessian order reverses and gives the supersolution inequality. Strictification treats arbitrary contacts. Convergence at $t = 0$ gives the trace. Local uniform convergence by itself does not prove global boundedness or uniform continuity, and this theorem assumes the smooth approximants; it does not construct them or establish their uniform estimates. □

10.1.2 Smooth tests, differentiability, and consistency

Lemma 10.4 (C^1 tests suffice; editorial expansion of source Theorem 1)
The first-order definition is unchanged if C^∞ test functions are replaced by C^1 test functions.

Expanded proof (mollify the test, not the unknown).

Only one implication needs proof. Strictify a C^1 upper contact, choose a ball compactly inside the PDE domain, and mollify an extension of ϕ on a slightly larger ball. Then $\phi_j \to \phi$ in C^1 on the contact ball. Maxima of $u - \phi_j$ are interior and converge to the contact point by the boundary-gap argument. Apply the smooth-test inequality and pass to the limit in the first derivatives and H. For a lower contact repeat with minima. A smooth cutoff extends each local test to the ambient domain without changing it near the contact. □

Lemma 10.5 (A differentiable function has a C^1 touching function; source §10.1.2 lemma)
If continuous u is differentiable at z_0, it has a C^1 strict upper test and a C^1 strict lower test there, both with derivative $Du(z_0)$.

Expanded proof (a radial majorant of the differentiability error).

Subtract the tangent affine function and translate so $z_0 = 0$, $u(0) = 0$, $Du(0) = 0$. On a small ball write $|u(z)| \le |z|\rho(|z|)$, where ρ is continuous, nondecreasing and $\rho(0) = 0$. One choice is the maximum over $|w| \le r$ of $|u(w)|/|w|$, extending that quotient by zero at $w = 0$. Set for $r = |z|$

$$\psi(z) = \int_r^{2r} \rho(s)\, ds + r^2.$$

Then $\psi(z) \ge r\rho(r) + r^2 > u(z)$ for $z \ne 0$. For $z \ne 0$ its radial derivative is $2\rho(2r) - \rho(r) + 2r \to 0$, while $\psi(z)/|z| \to 0$. Thus ψ extends as a C^1 function with $D\psi(0) = 0$. It is a strict upper test. The same construction applied to $-u$ gives a lower test for u. Add back the affine function and use a cutoff if a test on the whole ambient domain is desired. □

消黏法的条件式结论

不必控制 D^2u_j 本身：在接触点只需用 $D^2\phi$ 的单侧界。但要得到存在性，还须另证正则化方程可解及紧性，不能由这个极限论证直接宣布一般 H 的存在性。

Theorem 10.6 (Consistency; source Theorem 1)
A C^1 classical solution is a viscosity solution locally. Conversely, a viscosity solution differentiable at (x_0, t_0) satisfies $u_t + H(Du, x_0) = 0$ at that point. A locally Lipschitz viscosity solution therefore solves the equation a.e.

Expanded proof (two contacts recover equality).

For a classical solution, derivatives of u and its test agree at any interior contact, so both test inequalities follow with equality. Conversely, apply the preceding lemma in all $n + 1$ variables. The C^1 test equivalence gives the PDE residual at most zero from the upper test, and at least zero from the lower test. Their derivatives are those of u, proving equality. Rademacher's theorem supplies a.e. differentiability of a locally Lipschitz function. The converse implication from a.e. solvability to viscosity solvability is false: corners carry additional information, as the next diagram and Problem 5 demonstrate. □

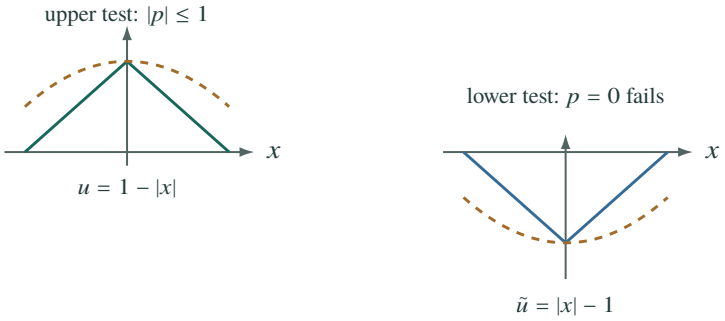


Diagram T10.1. A corner distinguishes a.e. and viscosity solutions. On $(-1, 1)$ both $1 - |x|$ and $|x| - 1$ solve $|u'| = 1$ away from zero and vanish at the endpoints. The upper test $1 - x^2/2$ is allowed for the downward corner, with slope zero and residual $-1 \le 0$. For the upward corner, the lower test $-1 + x^2/2$ has the same residual and violates the required supersolution inequality. No smooth lower test exists at the downward corner.

几乎处处满足还不够

尖点集合可以只有一个点、测度为零，却决定可接受的解。黏性定义在尖点也检查所有可能接触斜率；只在可微点代入方程会漏掉这个选择机制。

10.2 Uniqueness

10.2.1 Tests at the upper time boundary

Lemma 10.7 (Terminal extrema for an initial-value solution; source §10.2 lemma)
Let u be a continuous viscosity subsolution on $0 < t < T$. If $u - \phi$ has a local maximum at (x_0, T) relative to $t \le T$, then $\phi_t + H(D\phi, x_0) \le 0$ there. A supersolution has the reverse inequality at a relative local minimum.

Expanded proof (discard a positive penalty derivative).

Strictify the maximum and restrict to a small closed backward cylinder. For $t < T$ set $\phi_\delta = \phi + \delta/(T - t)$. The quantity $u - \phi_\delta$ tends to $-\infty$ as $t \uparrow T$. A comparison at $(x_0, T - \sqrt{\delta})$ shows its maximum tends to $(u - \phi)(x_0, T)$. The strict gap on the other cylinder boundaries therefore puts maximizing points (x_δ, t_δ) in the open PDE domain, and all cluster points equal (x_0, T). At those points,

$$\phi_t + \frac{\delta}{(T - t_\delta)^2} + H(D\phi, x_\delta) \le 0.$$

Discard the nonnegative middle term and pass to the limit. It need not tend to zero. For a lower contact use $\phi_\delta = \phi - \delta/(T - t)$, and discard its nonpositive derivative from the reverse inequality. This lemma does not reverse the viscosity convention: it only extends the initial-value convention to a one-sided contact at the top of its slab. □

10.2.2 Comparison by doubling space and time

Assume H satisfies, with a single constant C,

$$\begin{aligned} |H(p,x) - H(q,x)| &\le C|p - q|, \\ |H(p,x) - H(p,y)| &\le C|x - y|(1 + |p|). \end{aligned} \tag{10.2}$$

Theorem 10.8 (Comparison; source Theorem 1, strengthened to sub/supersolutions)
Let $u, v \in \mathrm{BUC}(\mathbb{R}^n \times [0,T])$ be a subsolution and a supersolution respectively, with the initial-value convention. If $u(x,0) \le v(x,0)$, then $u \le v$ on the slab. In particular there is at most one BUC viscosity solution with prescribed initial data. More generally,

$$\sup_x (u(x,t) - v(x,t)) \le \sup_x (u(x,0) - v(x,0)).$$

终端接触与终值问题不同

本引理研究初值解在 $t = T$ 的单侧接触，不改变不等式方向。下一节的终值控制问题采用反向时间演化，才会改变方向；这两个“终端”不能混用。

Expanded proof (localization, small penalty, and two touching tests).

1. A positive maximum after penalization. Suppose $u(x_*, t_*) - v(x_*, t_*) = a > 0$. Fix $\eta > 0$ with $2\eta T < a/4$. For $\varepsilon > 0$ maximize on $\mathbb{R}^{2n} \times [0, T]^2$

$$\begin{aligned}\Phi_\varepsilon(x, y, t, s) = u(x, t) - v(y, s) - \eta(t + s) \\ - \frac{|x - y|^2 + |t - s|^2}{2\varepsilon} - \varepsilon(|x|^2 + |y|^2).\end{aligned} \tag{10.3}$$

Boundedness and the quadratic confinement make the maximum attained; call its point $(x_\varepsilon, y_\varepsilon, t_\varepsilon, s_\varepsilon)$. For small ε, testing on (x_*, x_*, t_*, t_*) shows that this maximum is at least $a/2$. Below all variables refer to this point.

2. First bounds. Comparing with a fixed point, and using boundedness of u, v, gives

$$r := |x - y| = O(\sqrt{\varepsilon}), \quad d := |t - s| = O(\sqrt{\varepsilon}), \quad \varepsilon(|x| + |y|) = O(\sqrt{\varepsilon}).$$

These bounds allow x, y to escape to infinity. This is why uniform continuity on the full slab, rather than just on compact sets, is used.

3. The stronger small-penalty estimate. Compare the maximum with $\Phi_\varepsilon(x, x, t, t)$. If ω_v is a modulus for v, then

$$\frac{r^2 + d^2}{2\varepsilon} \le \omega_v(r + d) + \eta d + \varepsilon(|x| + |y|)r = o(1).$$

In particular $r^2/\varepsilon \to 0$. Merely knowing $r = O(\sqrt{\varepsilon})$ would only bound this expression and would not complete the proof.

4. Neither time is initial. Since the penalty is nonnegative, $u(x, t) - v(y, s) \ge a/2$. If a subsequence had $t \to 0$, then $s \to 0$ as well, and initial ordering plus uniform continuity would give

$$u(x, t) - v(y, s) \le \omega_u(t) + \omega_v(r + s) \longrightarrow 0.$$

This is a contradiction. The same reasoning excludes $s \to 0$. Thus both contact times are positive for all small ε. They may equal T, in which case the preceding lemma applies.

5. Freeze the other pair of variables. At the maximum, the penalty with (y, s) frozen is an upper test for u after adding a constant. Its time derivative and spatial gradient are

$$b_u = \eta + \frac{t - s}{\varepsilon}, \qquad p_u = \frac{x - y}{\varepsilon} + 2\varepsilon x.$$

The negative penalty with (x, t) frozen is a lower test for v, with

$$b_v = -\eta + \frac{t - s}{\varepsilon}, \qquad p_v = \frac{x - y}{\varepsilon} - 2\varepsilon y.$$

Consequently $b_u + H(p_u, x) \le 0$ and $b_v + H(p_v, y) \ge 0$. Subtracting, the uncontrolled time difference cancels:

$$2\eta \le H(p_v, y) - H(p_u, x).$$

6. Estimate the Hamiltonians. By (10.2),

$$\begin{aligned} 2\eta &\le C|p_u - p_v| + Cr(1 + |p_v|) \\ &\le C\sqrt{\varepsilon} + Cr + C\frac{r^2}{\varepsilon} + Cr\sqrt{\varepsilon} \longrightarrow 0. \end{aligned}$$

This contradicts the fixed $\eta > 0$, proving $u \le v$. Adding a constant to v leaves its supersolution property unchanged; apply ordering to $v + \sup_x(u(x,0) - v(x,0))$ for the last assertion. Applying the result in both directions yields uniqueness and sup-norm contraction for solutions with different initial values. □

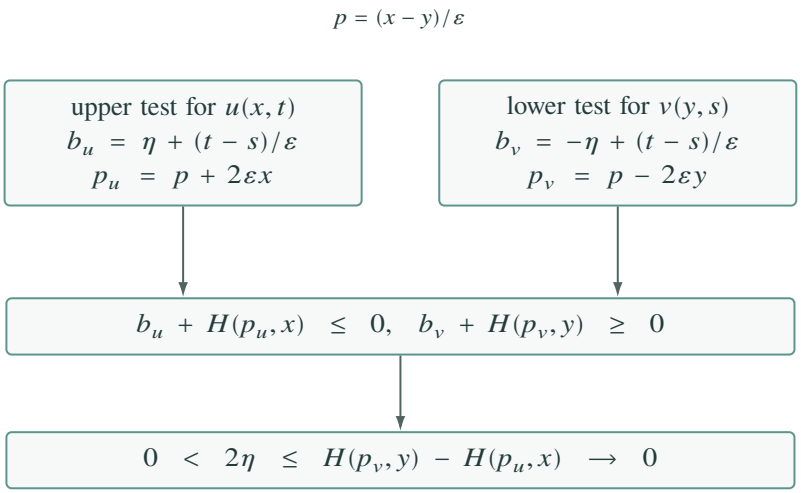


Diagram T10.2. Doubling variables produces two independent smooth tests. The common vector $(x - y)/\varepsilon$ need not be bounded. Global Lipschitz control in momentum handles the small difference $p_u - p_v$, while $|x - y|^2/\varepsilon \to 0$ handles the spatial dependence of H. The opposite strict time shifts leave the positive gap 2η after subtraction.

小量必须足够强

若只得到 $|x-y| = O(\sqrt{\varepsilon})$，则 $|x - y|^2/\varepsilon$ 可能不趋零。对角比较给出整个罚项 $o(1)$，正好消除 Hamiltonian 的空间误差。

10.2.3 A comparison extension for Hopf–Lax

The quadratic Hamiltonian $H(p) = |p|^2/2$ is not globally Lipschitz. The source's Hopf–Lax uniqueness statement therefore needs a separate comparison observation, supplied here.

Theorem 10.9 (One spatially Lipschitz competitor; editorial supplement)
Suppose $H(p)$ is continuous and independent of x. The comparison conclusion above still holds if at least one of u, v is uniformly Lipschitz in x on the slab. The other function may be merely BUC.

Expanded proof (a contact bounds the momentum).

If a K-Lipschitz function has a differentiable upper or lower touching test at a point, its spatial test gradient has norm at most K. Indeed restrict to increments he, use the touching inequality and the Lipschitz bound, divide by h with both signs, and conclude $|D\phi \cdot e| \le K$ for all unit e.

Use the same doubled maximum and first four steps above. If u is Lipschitz then $|p_u| \le K$; if v is Lipschitz then $|p_v| \le K$. Since $|p_u - p_v| = O(\sqrt{\varepsilon})$, both momenta lie in a fixed compact ball. Uniform continuity of H on that ball gives $H(p_v) - H(p_u) \to 0$, contradicting $2\eta > 0$ exactly as before. No global Lipschitz bound on H, convexity, or boundedness of the other function's gradient is needed for this particular extension. □

超线性 Hamiltonian 的衔接

$H(p) = |p|^2/2$ 的梯度无界，不能直接套用全局动量 Lipschitz 定理。若一方解在空间中为 K-Lipschitz，接触斜率被限制在有界球内，就只需该球上的一致连续性。

10.3 Control Theory, Dynamic Programming

10.3.1 Measurable controls and deterministic responses

Let $A \subset \mathbb{R}^m$ be nonempty and compact. Suppose $f : \mathbb{R}^n \times A \to \mathbb{R}^n$ and $r : \mathbb{R}^n \times A \to \mathbb{R}$ are continuous, and uniformly in $a \in A$ satisfy

$$|f(x,a)| \le M, \quad |f(x,a) - f(y,a)| \le L_f|x-y|,$$
$$|r(x,a)| \le R, \quad |r(x,a) - r(y,a)| \le L_r|x-y|.$$

Let g be bounded and G-Lipschitz. Fix a horizon $T > 0$. An admissible control $\alpha : [t,T] \to A$ is measurable. Its state is an absolutely continuous solution of

$$X'(s) = f(X(s), \alpha(s)) \text{ a.e.}, \qquad X(t) = x. \tag{10.4}$$

Lemma 10.10 (Response estimates; details for source §10.3.1)
Every admissible control has a unique response, and

$$|X(s) - x| \le M(s-t), \qquad |X^{x,\alpha}(s) - X^{y,\alpha}(s)| \le e^{L_f(s-t)}|x-y|.$$

Expanded proof (the integral equation accepts measurable controls).

For fixed α use the map $Z \mapsto x + \int_t^s f(Z(q), \alpha(q))dq$ on continuous paths. The integrand is measurable and bounded; the map returns continuous paths and is a contraction on intervals with $L_f h < 1$. Iterating these intervals gives an absolutely continuous solution on $[t, T]$. Its displacement bound follows by integrating $|f| \le M$. For two responses with the same control, subtract the integral equations and apply Gronwall. This gives the second estimate and uniqueness. No differentiability in time of the control is required. □

The cost and value are

$$\begin{aligned} J_{x,t}(\alpha) &= \int_t^T r(X(s), \alpha(s))ds + g(X(T)), \\ V(x,t) &= \inf_{\alpha} J_{x,t}(\alpha). \end{aligned} \tag{10.5}$$

An infimum does not itself assert existence of an optimal control. The following arguments use controls within an arbitrarily prescribed positive error of the infimum.

10.3.2 The dynamic programming principle

Theorem 10.11 (DPP; source §10.3.2, Theorem 1)
For $0 < h \le T - t$,

$$V(x,t) = \inf_{\alpha|_{[t,t+h]}} \left\{ \int_t^{t+h} r(X(s), \alpha(s))ds + V(X(t+h), t+h) \right\}. \tag{10.6}$$

Expanded proof (restriction and concatenation give opposite bounds).

Call the right side W. Any full control has a prefix up to $t + h$ and a continuation thereafter. Its continuation cost is at least the value at its reached state. Thus every full cost is at least W, giving $V \ge W$.

Conversely, fix any prefix, with reached state z. For $\delta > 0$ choose a continuation whose cost is at most $V(z, t+h) + \delta$. Concatenate the two measurable controls. Uniqueness of the state equation makes its trajectory exactly the concatenation of their responses. Therefore $V(x,t)$ is at most the displayed prefix cost plus δ. Take the infimum over prefixes and let $\delta \downarrow 0$, obtaining $V \le W$. A continuation is chosen for a single reached

紧控制集保证什么

连续性与紧性保证 Hamiltonian 的逐点最小值能取到。但这并不自动给出整个时间区间的最优控制，也不自动保证反馈场的闭环 ODE 可解；动态规划证明只使用近似最优控制。

state at a time; no measurable selection depending simultaneously on every state is needed. □

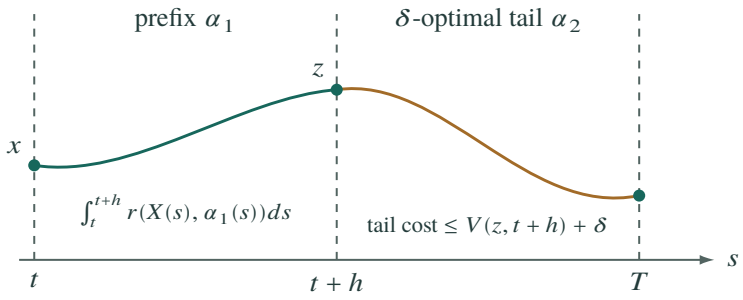


Diagram T10.3. A prefix and its continuation form one admissible control. Restricting a full control gives a continuation cost at least $V(z, t+h)$. Conversely, for a fixed prefix reaching z, choose one δ-optimal continuation and concatenate it. The two comparisons prove the equality in DPP. The curves are schematic state trajectories, not a claim of optimality for the drawn paths.

Lemma 10.12 (Regularity of the value; source §10.3.3 lemma)
The value is bounded by $RT + \|g\|_\infty$. Uniformly in t it is K_T-Lipschitz in x, where

$$K_T = Ge^{L_f T} + L_r \int_0^T e^{L_f s} ds.$$

It is $(R + K_T M)$-Lipschitz in time and $V(x, T) = g(x)$.

Expanded proof (compare identical controls before taking infima).

For the same control started at x, y, the response estimate bounds the cost difference by

$$\left(Ge^{L_f (T-t)} + L_r \int_0^{T-t} e^{L_f s} ds \right) |x - y|.$$

This holds for every control, so taking infima yields the same bound for $|V(x,t) - V(y,t)|$. In DPP every reached state satisfies $|X(t+h) - x| \le Mh$ and every running integral has absolute value at most Rh. The expression inside its infimum is therefore between $V(x, t+h) - (R + K_T M)h$ and $V(x, t+h) + (R + K_T M)h$. Taking its infimum gives the time estimate without assuming a smooth value function or an optimal control. At $t = T$ the running interval is empty and $X(T) = x$, so every control gives cost $g(x)$. □

10.3.3 The terminal HJB equation and its time orientation

Define the control Hamiltonian

$$H_c(p,x) = \min_{a\in A}\{f(x,a)\cdot p + r(x,a)\}. \tag{10.7}$$

It is concave in p, since it is a minimum of affine functions, and

$$|H_c(p,x) - H_c(q,x)| \le M|p-q|,$$
$$|H_c(p,x) - H_c(p,y)| \le (L_f|p| + L_r)|x-y|.$$

For a terminal problem written as $V_t + H_c(DV,x) = 0$, the source uses the following convention: an upper contact has residual $\phi_t + H_c(D\phi,x) \ge 0$, and a lower contact has residual at most zero. Equivalently this is the usual viscosity convention applied to $-V_t - H_c(DV,x) = 0$.

Lemma 10.13 (Time reversal; source remark after Theorem 2)
Set $w(x,\tau) = V(x,T-\tau)$. Then the terminal convention for V is exactly the initial-value convention for

$$w_\tau + K(Dw,x) = 0, \qquad K(p,x) = -H_c(p,x), \qquad w(x,0) = g(x).$$

In particular K is convex in p and satisfies (10.2).

Expanded proof (transform the test as well as the equation).

If $w-\psi$ has an upper contact at (x_0,τ_0), then $V-\phi$, $\phi(x,t) = \psi(x,T-t)$, has an upper contact at $(x_0,T-\tau_0)$. Now $\phi_t = -\psi_\tau$, $D\phi = D\psi$. The terminal inequality $-\psi_\tau + H_c(D\psi,x) \ge 0$ is precisely $\psi_\tau - H_c(D\psi,x) \le 0$. Lower contacts give the other direction. The implication is reversible. Negating the minimum of affine functions makes a maximum of affine functions, proving convexity of K. □

先用同一控制比较

估计两个初值的价值差时，对它们使用相同控制，先用 Gronwall 控制轨道差，再对代价取下确界。直接把两个未知最优控制相减没有这样的稳定性依据。

Theorem 10.14 (Value function solves HJB; source Theorem 2)
The value in (10.5) *is the unique BUC solution of* $V_t + H_c(DV, x) = 0$, $V(x,T) = g(x)$, *with the terminal convention just specified. Equivalently its time reversal solves the corresponding initial-value problem with Hamiltonian* $K = -H_c$.

Expanded proof (all constant controls versus an h^2-optimal prefix).

Normalize an upper test by $V(x_0,t_0) = \phi(x_0,t_0)$. For any fixed $a \in A$, use the constant control on $[t_0, t_0 + h]$. The displacement bound keeps its path in the contact neighborhood when h is small, independently of a. DPP and upper touching give

$$\begin{aligned} 0 &\le \int_{t_0}^{t_0+h} r(X(s),a)ds + V(X(t_0+h), t_0+h) - V(x_0,t_0) \\ &\le \int_{t_0}^{t_0+h} (\phi_t + f(X(s),a)\cdot D\phi + r(X(s),a))ds. \end{aligned}$$

Divide by h and let $h \downarrow 0$. The limit inequality holds for every a, hence $\phi_t(x_0,t_0) + H_c(D\phi(x_0,t_0),x_0) \ge 0$.

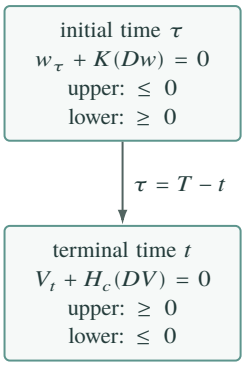
At a normalized lower contact choose a prefix within h^2 of the DPP infimum. The lower touching inequality now yields

$$h^2 \ge \int_{t_0}^{t_0+h} (\phi_t + f(X(s),\alpha(s))\cdot D\phi + r(X(s),\alpha(s)))ds.$$

Continuity in (x,t,a) on a compact neighborhood times A makes replacement of $(X(s),s)$ by (x_0,t_0) have a uniform $o(h)$ error. The resulting integrand is bounded below by $\phi_t(x_0,t_0) + H_c(D\phi(x_0,t_0),x_0)$ for every control value. After division by h and passage to the limit we obtain a residual at most zero. The error must be $o(h)$: a fixed δ-optimal control would leave the uncontrolled quotient δ/h.

The value is BUC and takes the terminal data by the regularity lemma. Time reversal converts it to an initial-value solution for K, whose structural estimates allow comparison in §10.2. This proves uniqueness. □

图 T10.M：时间方向与测试符号



图中残差均按方程所写的时间变量计算。令 $\tau = T - t$ 后，$\partial_t = -\partial_\tau$，Hamiltonian 同时变为 $-H_c$；不能只改变时间端点而保留所有符号。

近似最优误差要随步长缩小

在下接触证明中取 h^2-最优前缀，再除以 h，误差才趋零。上接触使用每个常值控制；下接触使用近似最优控制，二者的量词不同。

10.3.4 Verification and the limits of feedback synthesis

Theorem 10.15 (Smooth verification; editorial completion of source discussion) *Suppose $v \in C^1(\mathbb{R}^n \times [0,T])$ satisfies $v_t + H_c(Dv, x) = 0$ and $v(x,T) = g(x)$. Then $v(x,t) \le J_{x,t}(\alpha)$ for every admissible control. If an admissible α_* with response X_* attains the Hamiltonian minimum along its path a.e., then its cost equals $v(x,t)$ and it is optimal; in particular $v = V$ at that starting point.*

Expanded proof (integrate the nonnegative verification residual).

The chain rule for an absolutely continuous path gives

$$\frac{d}{ds}v(X(s),s)+r(X(s),\alpha(s)) = v_t+f(X(s),\alpha(s))\cdot Dv+r(X(s),\alpha(s)) \ge 0$$

a.e. Integrate to T and use the terminal value g to obtain $J_{x,t}(\alpha) \ge v(x,t)$. If the minimum is attained along X_*, the residual vanishes a.e. and equality holds. All evaluations are along a bounded path on a finite interval, where the derivatives of v are bounded. This verifies an admissible path; it does not establish solvability of a closed-loop ODE for an arbitrary minimizing feedback field. □

The pointwise prescription $a_*(x,t) \in \operatorname{argmin}_{a\in A}\{f(x,a)\cdot DV(x,t) + r(x,a)\}$ is conditional on the gradient and on an admissible resulting trajectory. At nonsmooth points DV may not exist. Even with a pointwise minimizer, measurability and closed-loop existence require additional arguments.

10.3.5 Hopf–Lax: compact minimizers and viscosity inequalities

Return now to an initial-value Hamiltonian $H : \mathbb{R}^n \to \mathbb{R}$, finite, convex and superlinear, with g bounded and G-Lipschitz. Let

$$L(q) = \sup_p\{p\cdot q - H(p)\}, \qquad Q_t g(x) = \min_y \left\{tL\left(\frac{x-y}{t}\right) + g(y)\right\}.$$

Convex duality from §3.3 gives finite continuous convex superlinear L and $H(p) = \sup_q(p\cdot q - L(q))$. In particular all displayed minima exist by coercivity. These assumptions include $H(p) = |p|^2/2$, which is outside the global momentum Lipschitz hypothesis of §10.2.

验证定理与控制构造

验证定理说：若已有合适的最小化轨道，则它最优。它没有证明任意逐点选出的反馈都可积分成轨道；尤其在价值函数不可微处，直接写 DV 没有定义。

Lemma 10.16 (Semigroup, speed bound, and regularity; details for source Theorem 3)
One has $Q_h Q_t g = Q_{h+t} g$. Each $Q_t g$ is G-Lipschitz in space. All minimizing velocities $q = (x-y)/t$ lie in a ball independent of x, t. With $Q_0 g = g$, the function $U(x,t) = Q_t g(x)$ is BUC on every finite slab and is Lipschitz in time.

Expanded proof (Jensen and the competitor with zero velocity).

For a two-stage path from y through z to x, Jensen gives

$$tL\left(\frac{z-y}{t}\right) + hL\left(\frac{x-z}{h}\right) \geq (t+h)L\left(\frac{x-y}{t+h}\right).$$

Equality is obtained by $z = (tx+hy)/(t+h)$, so infimizing proves the semigroup identity. Translating x and y by the same increment changes only $g(y)$, proving the spatial Lipschitz estimate.

At a minimizer compare with $y = x$. Since g is G-Lipschitz, $L(q) - G|q| \leq L(0)$. Superlinearity makes this sublevel bounded, say $|q| \leq R_G$. The bound is uniform even as $t \downarrow 0$. For any G-Lipschitz v, set $m_G = \inf_q (L(q) - G|q|) > -\infty$. Then

$$v(x) + hm_G \leq Q_h v(x) \leq v(x) + hL(0).$$

Together with the semigroup this gives a uniform time Lipschitz constant $\max\{|m_G|, |L(0)|\}$ and convergence to g at $t = 0$. The same estimate with $v = g$ bounds U on finite slabs. □

Theorem 10.17 (Hopf–Lax viscosity solution; source §10.3.4, Theorem 3)
Under the preceding assumptions $U(x,t) = Q_t g(x)$ is the unique BUC viscosity solution of $U_t + H(DU) = 0$, $U(x,0) = g(x)$.

Expanded proof (bounded minimizing velocities close the lower test).

At an upper contact at (x,t), choose any fixed velocity q and $0 < h < t$. The semigroup gives $U(x,t) \leq hL(q) + U(x-hq, t-h)$. Upper touching then gives $\phi(x,t) - \phi(x-hq, t-h) \leq hL(q)$. Divide by h and let $h \downarrow 0$ to get $\phi_t + D\phi \cdot q - L(q) \leq 0$ for every q. Taking the supremum in q yields $\phi_t + H(D\phi) \leq 0$.

At a lower contact, let q_h minimize the short-step semigroup formula, so $U(x,t) = hL(q_h) + U(x - hq_h, t-h)$. The spatial Lipschitz estimate for $U(\cdot, t-h)$ bounds $|q_h| \leq R_G$ by the preceding lemma. Thus the earlier point stays in the contact neighborhood and Taylor's first-order remainder is $o(h)$ uniformly in q_h. Lower touching gives

$$hL(q_h) \leq \phi(x,t) - \phi(x - hq_h, t-h) = h(\phi_t + D\phi \cdot q_h) + o(h).$$

Since $D\phi \cdot q_h - L(q_h) \le H(D\phi)$, $\phi_t + H(D\phi) \ge -o(1)$, which proves the lower-test inequality. Initial data and BUC regularity were proved in the lemma.

For uniqueness compare U with any other BUC solution in both directions. The extension at the end of §10.2 applies because U is uniformly spatially Lipschitz and H is continuous and independent of x. This explicitly supplies the comparison step without assuming a global Lipschitz bound for the superlinear H. □

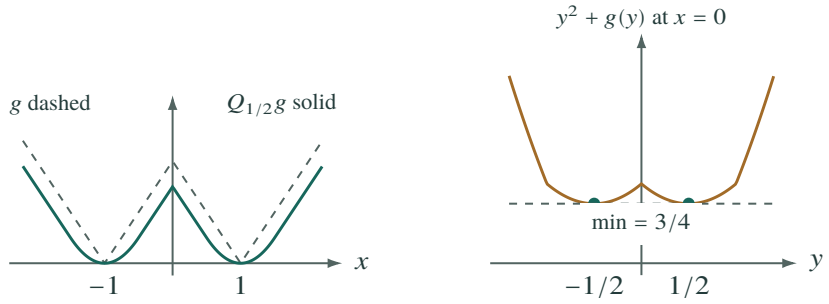


Diagram T10.4. Two minimizing origins can create a viscosity corner. Take $H(p) = p^2/2$, $t = 1/2$, and $g(y) = \min\{|y-1|, |y+1|, 2\}$. The exact value is $\min\{h_t(|x-1|), h_t(|x+1|), 2\}$, where $h_t(d) = d^2/(2t)$ for $d \le t$ and $d - t/2$ otherwise. At $x = 0$ the two minimizing origins are $y = \pm 1/2$, both of cost $3/4$. The left panel shows g and its value; the right panel shows the minimized cost at $x = 0$.

10.4 Problems

The descriptions below retain the source numbers 1–14. All solutions are editorial reconstructions. The viscosity convention is stated whenever the time direction or the sign of the equation changes.

1. Negating the unknown changes the Hamiltonian

Solution. Let $w = -u$ and $\widehat{H}(p,x) = -H(-p,x)$. If $w - \phi$ has an upper contact, $u - (-\phi)$ has a lower contact. The supersolution inequality for u reads $-\phi_t + H(-D\phi, x) \ge 0$, exactly $\phi_t + \widehat{H}(D\phi, x) \le 0$. Lower contacts reverse the reasoning. Boundedness and uniform continuity are preserved, as are initial traces with g replaced by $-g$. Negating the unknown interchanges upper and lower contacts; multiplying a PDE expression by -1 while leaving the unknown and the test inequalities unchanged does not do this. □

最小化速度为何有界

由与 $y = x$ 比较得 $L(q) - G|q| \le L(0)$。超线性把 q 限在固定球内，因此短步最小化点满足 $|x-y| = O(h)$，下接触证明中的 Taylor 余项才一致为 $o(h)$。

2. Uniform stability for a first-order equation

Solution. At a strict upper contact for the limit, maximize $u_k - \phi$ on a small closed contact ball. Uniform convergence puts the maxima in its interior and makes their points converge to the original contact point. Apply $\phi_t + H(D\phi, x) \le 0$ there and use continuity of H. Strictification and lower contacts give the full PDE. A uniform limit on each full closed slab of BUC functions remains BUC and inherits the limiting initial trace. If only local uniform convergence is known, the local PDE conclusion still holds, but membership in the global BUC class must be checked separately. The same contact proof works for $H_k \to H$ locally uniformly in (p, x): the test gradients stay in a compact set even when the unknown functions have no gradient bounds. □

3. Anisotropic positive diffusion selects the same notion

Solution. Let the fixed smooth matrix $A = (a_{ij})$ have nonnegative symmetric part (the source assumes uniform ellipticity). At a smooth upper contact, $D^2 u^\varepsilon \le D^2\phi$, and hence

$$\phi_t + H(D\phi, x) = \varepsilon \operatorname{tr}(AD^2 u^\varepsilon) \le \varepsilon \operatorname{tr}(AD^2\phi).$$

To see the trace order, for positive semidefinite symmetric A, B, $\operatorname{tr}(AB) = \operatorname{tr}(A^{1/2}BA^{1/2}) \ge 0$; the antisymmetric part has zero contraction with a symmetric Hessian. The contact points converge, A and $D^2\phi$ are bounded locally, and the right side tends to zero. Lower contacts yield the reverse inequality. Boundary traces and the BUC class are treated as in Problem 2. Uniform ellipticity helps construct approximants but is stronger than the positivity needed in this limit argument. □

4. Sup-norm contraction

Solution. Set $m = \sup_x |g_1 - g_2|$. Since H contains no zeroth-order value variable, $u_2 + m$ remains a supersolution and $g_1 \le g_2 + m$. Comparison gives $u_1 \le u_2 + m$. Interchange indices to obtain the opposite inequality. Thus $\|u_1(\cdot, t) - u_2(\cdot, t)\|_\infty \le m$ on every finite slab. There is no requirement that either supremum be attained at a finite spatial point. □

5. The sign of the eikonal equation matters

Solution. For $u = 1 - |x|$, every nonzero point has slope ± 1, so the equation holds classically there. If a smooth ϕ touches from above at 0, the right and left one-sided inequalities give $-1 \le \phi'(0) \le 1$. Thus $|\phi'(0)| - 1 \le 0$. There is no smooth lower test at 0: it would require both $\phi'(0) \le -1$ from the right and $\phi'(0) \ge 1$ from the left. Hence the lower-test condition is vacuous there. This proves (a).

For $\tilde{u} = |x| - 1$, the constant lower test $\phi = -1$ at zero violates $|\phi'(0)| - 1 \geq 0$, proving (b). For (c), use instead $F(p) = 1 - |p|$. The lower-test slopes for $\tilde{u}$ lie in $[-1, 1]$, so $F \geq 0$, and it has no upper test at its upward corner. Away from zero, $F = 0$. Thus $\tilde{u}$ solves the sign-reversed equation in its own viscosity sense. For (d), the two residuals have the same classical zero set, but their upper- and lower-test inequalities are different. They encode opposite choices at a corner, exactly as Diagram T10.1 illustrates. □

方程乘负号会改变选择

经典方程 $F(Du) = 0$ 与 $-F(Du) = 0$ 等价，但黏性定义包含残差的方向。若未知函数不变，乘负号会交换应满足的单侧条件，因此尖点解可能不同。

6. Distance to the boundary is an eikonal solution

Solution. For any nonempty bounded open U, $d(x) = \operatorname{dist}(x, \partial U)$ is 1-Lipschitz by the triangle inequality; no boundary smoothness is required. At an upper smooth contact its gradient has norm at most one by the Lipschitz contact estimate proved in §10.2. At a lower contact $x_0 \in U$, choose a nearest boundary point y, possible because ∂U is closed and bounded. Set $e = (y - x_0)/d(x_0)$. For $0 < h < d(x_0)$ the point $x_0 + he$ is inside U, and

$$d(x_0 + he) = d(x_0) - h.$$

The upper bound uses the same y; the lower bound uses the Lipschitz estimate. Lower touching gives $\phi(x_0 + he) - \phi(x_0) \leq -h$, hence $D\phi(x_0) \cdot e \leq -1$. Therefore $|D\phi(x_0)| \geq 1$, proving the supersolution condition. The continuous extension of d vanishes at the boundary. □

7. Admissible gradient jumps at a smooth interface

Solution. Continuity of u and smoothness of its traces force the tangential components of Du^+ and Du^- to agree on Γ. At a fixed point use smooth local coordinates (z, ρ), where ρ increases into V^+ and $\partial_\rho = \nu$ at that point. If $u_\nu^- < u_\nu^+$, choose any number a strictly between them. For a sufficiently small neighborhood the smooth function

$$\phi(z, \rho) = u(z, 0) + a\rho$$

is a lower test along the interface: on $\rho > 0$ the normal derivative of u exceeds a, while on $\rho < 0$ it is less than a. Integrating those derivative inequalities from zero gives $u \geq \phi$ on both sides. Its gradient at the point is the common tangential gradient plus $a\nu$. The viscosity lower-test condition gives $H(\lambda Du^- + (1 - \lambda)Du^+) \geq 0$ for $0 < \lambda < 1$. The endpoint values of H are zero by the classical equation on each side and continuity up to Γ; thus the closed interval follows. If the normal derivatives agree, both full gradients agree and the same endpoint argument suffices. For a downward jump the function ϕ is an upper test, giving $H \leq 0$ on the

segment. This proves both source inequalities, including equality cases. □

界面斜率形成一个线段

连续性固定切向梯度，只有法向分量能跳跃。上凸或下凸尖角的测试斜率填充两侧梯度之间的线段，因此要检查整段上的 H，而非只检查两端的 $H = 0$。

8. Shape from shading

Solution. For the graph $(x_1, x_2, u(x))$, the upward unit normal is $\nu = (-Du, 1)/\sqrt{1 + |Du|^2}$. Hence the normalized Lambertian intensity is $i = e_3 \cdot \nu = (1 + |Du|^2)^{-1/2}$. For $0 < i(x) \le 1$ this gives the eikonal equation

$$|Du| = n(x) := \sqrt{i(x)^{-2} - 1}.$$

If $i = 1$ the slope is zero; if $i \downarrow 0$ the required finite-graph slope diverges. Values outside $(0, 1]$ do not fit this normalized graph model. The PDE determines a slope magnitude, not a surface without boundary data and a solution-selection condition. □

9. Pure pursuit and an exact interception time

Solution. Use relative coordinates x_1 for the target's horizontal displacement from the pursuer and x_2 for the pursuer's height above the target. Writing $r = \sqrt{x_1^2 + x_2^2}$, the relative dynamics before interception are

$$x_1' = b_1 - b_2 x_1/r, \qquad x_2' = -b_2 x_2/r.$$

The remaining-time identity $u(x) = h + u(X(h))$ gives

$$(b_1 - b_2 x_1/r)u_{x_1} - (b_2 x_2/r)u_{x_2} = -1, \qquad u(0,0) = 0.$$

As an editorial check the exact solution is

$$u(x_1, x_2) = \frac{b_2\sqrt{x_1^2 + x_2^2} + b_1 x_1}{b_2^2 - b_1^2}.$$

Indeed $r' = b_1 x_1/r - b_2$, so differentiating the numerator along a trajectory gives $-(b_2^2 - b_1^2)$. Also $r' \le b_1 - b_2 < 0$ ensures finite interception, and the displayed expression tends to zero there. Thus its constant rate of decrease -1 identifies it as the actual time. On $x_2 = 0$, $x_1 > 0$ it reduces to $x_1/(b_2 - b_1)$, as expected. The transport field is singular at interception; the PDE is used only away from $r = 0$ and the zero-time value is imposed by continuity. □

10. Discount and regularity on an infinite horizon

Solution. For $\lambda > 0$ define $V(x) = \inf_\alpha \int_0^\infty e^{-\lambda s} r(X(s), \alpha(s))ds$. Since $|r| \le R$, $|V| \le R/\lambda$. For the same control at x, y, the running-cost difference is at most $\min\{L_r e^{L_f s}|x - y|, 2R\}$. If $\lambda > L_f$, integrate its first

bound to get $|V(x) - V(y)| \le L_r|x-y|/(\lambda - L_f)$. If $0 < \lambda \le L_f$, choose any $0 < \alpha < \lambda/L_f$ (and hence $\alpha < 1$). For nonnegative a, b, $\min(a,b) \le a^\alpha b^{1-\alpha}$, so

$$|V(x) - V(y)| \le \frac{L_r^\alpha (2R)^{1-\alpha}}{\lambda - \alpha L_f}|x-y|^\alpha.$$

The estimates are uniform in the control, so passing to infima preserves them. The case $R = 0$ or $L_r = 0$ is immediate. If $L_f = 0$, every $\lambda > 0$ is in the Lipschitz case. At $\lambda = L_f$ the stated argument gives every $\alpha < 1$, without asserting Lipschitz continuity. □

11. The discounted stationary Bellman equation

Solution. Restriction and concatenation, with the discount factored out of the tail, give

$$V(x) = \inf_\alpha \left\{ \int_0^h e^{-\lambda s} r(X(s), \alpha(s)) ds + e^{-\lambda h} V(X(h)) \right\}.$$

At a normalized upper contact $V - \phi$, choose a constant control a. The expression on the right for this control is at least $V(x)$; replace $V(X(h))$ by its upper test and expand to first order: $0 \le r(x,a) + f(x,a) \cdot D\phi(x) - \lambda V(x)$. This holds for every a, so

$$\lambda V(x) - \min_a \{ f(x,a) \cdot D\phi(x) + r(x,a) \} \le 0.$$

At a lower contact choose an h^2-optimal prefix, use the lower test, and the uniform continuity in a argument of §10.3. Division by h now gives the reverse inequality. Thus V is a viscosity solution of $\lambda V - H_c(DV, x) = 0$ with the usual stationary convention. The positive coefficient of V is produced by differentiating the discount factor, not by inserting an additional running cost. □

12. Degenerate ellipticity and extrema of operators

Solution. Write $R \ge S$ when $R - S$ is positive semidefinite. Then $-\operatorname{tr} R \le -\operatorname{tr} S$. For symmetric $A \ge 0$, $\operatorname{tr}(A(R-S)) = \operatorname{tr}(A^{1/2}(R-S)A^{1/2}) \ge 0$, so $F(R) = -A : R$ is elliptic in the source's decreasing-matrix convention. If every $F_k(R,p,x) \le F_k(S,p,x)$, taking a finite maximum or a finite minimum preserves that inequality. Consequently both Bellman-type maxima and minima of such linear operators are degenerate elliptic. Strict positive definiteness is not needed for these order statements. □

13. Classical consistency for second-order equations

Solution. Let $u \in C^2$ solve $F(D^2u, Du, x) = 0$ with F elliptic as in Problem 12. At an upper smooth contact, $Du = D\phi$ and $D^2u \le D^2\phi$, so

$$F(D^2\phi, D\phi, x) \le F(D^2u, Du, x) = 0.$$

At a lower contact both the matrix order and the resulting inequality reverse. This proves viscosity consistency. The ellipticity sign is essential: it is exactly what connects Hessian order at a contact to the direction of the PDE inequality. □

14. Uniform stability for second-order equations

Solution. Assume, as in the preceding problem, that F is continuous and elliptic, and that $u_k \to u$ locally uniformly. Strictify an upper contact by adding $\delta|x - x_0|^4$ to ϕ. This preserves both the gradient and Hessian at x_0. Interior maxima of $u_k - \phi_\delta$ converge to x_0. At those points use the viscosity subsolution inequality and pass to the limit in the continuous F, including the derivatives of the fixed test. At x_0 the quartic perturbation contributes no first or second derivative, giving the desired original-test inequality. Strictify a lower contact by subtracting the quartic and repeat. This is a stability proof; it does not prove existence or comparison for general fully nonlinear second-order equations. □

10.5 References

Source attributions

Evans's §10.5 attributes the test-function formulation to Crandall, Evans and Lions, Trans. AMS 282 (1984), 487–502, following the foundational Crandall–Lions paper, Trans. AMS 277 (1983), 1–42. The uniqueness argument follows the first paper, with improvements acknowledged there to M. Crandall. The control connection is attributed to P.-L. Lions [Li]. Further directions in the source are Tran [Tn], Fleming–Soner [F-S], and Bardi–Capuzzo Dolcetta [B-CD]. These are source attributions and bibliography keys. The source credits D. Ostrov with Problem 8 and directs readers to Cabré–Caffarelli [C-C] and Gilbarg–Trudinger [G-T] for the fully nonlinear elliptic exercises. The present reconstruction does not claim separate consultation of all those works.

Dependencies and proof boundaries

The proofs use compactness on contact balls, mollification, elementary convex duality from Chapter 3, Rademacher's theorem, the integral form of the ODE and Gronwall. The comparison proof is first-order: freezing the other variables produces smooth tests directly. No second-order matrix jet lemma is being assumed or proved. The source's uniqueness theorem is expanded to comparison of BUC sub- and supersolutions. An additional comparison theorem with one spatially Lipschitz competitor explicitly supplies the superlinear Hopf–Lax uniqueness step.

The vanishing-viscosity argument is conditional on smooth approximants and locally uniform convergence; existence and uniform estimates for a general nonconvex Hamiltonian are not proved here. Value-function existence is proved for the stated bounded continuous coefficients and compact control set; its terminal Hamiltonian is concave, and the time-reversed initial Hamiltonian is convex. Smooth verification is conditional on an admissible minimizing trajectory. General nonsmooth feedback synthesis, existence of an optimal ordinary control under additional hypotheses, second-order comparison, Perron's method, and nonlocal or state-constrained boundary theories are outside this chapter's completed proofs. The 14 problem solutions and all diagrams are editorial additions. Source checking, symbolic model checking, compilation, and visual QA are recorded separately; none is an independent mathematical review.

Chapter 11

Systems of Conservation Laws

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 11; printed p. 571; PDF page 588.

Chinese reference: 中译本对应章：守恒律组，书内第 404 页起。

Editorial reconstruction. This chapter develops wave geometry, the local Riemann solver, Riemann invariants, and entropy selection. Exposition, expanded proofs, Chinese annotations, problem solutions, and diagrams are reconstructed study material. Source theorem numbers are stated separately. External input and the scope of each existence assertion are specified explicitly.

11.1 Introduction

11.1.1 Conserved variables and the weak formulation

Let $\Omega \subset \mathbb{R}^m$ be an open state domain and let $F \in C^3(\Omega; \mathbb{R}^m)$. We study

$$u_t + F(u)_x = 0, \qquad u(x,0) = g(x), \quad x \in \mathbb{R}. \tag{11.1}$$

For a smooth solution, integration over $[a,b]$ gives $\frac{d}{dt}\int_a^b u\,dx = F(u(a,t)) - F(u(b,t))$. This is a vector balance: each component is a conserved density. In several space dimensions the flux is an $m \times n$ matrix and its divergence is taken row by row.

Definition 11.1 (Integral solution; source §11.1.1)

A bounded measurable u with essential range in a compact subset of Ω is an integral solution if, for every $\varphi \in C_c^\infty(\mathbb{R} \times [0,\infty); \mathbb{R}^m)$,

$$\int_0^\infty \int_{\mathbb{R}} [u \cdot \varphi_t + F(u) \cdot \varphi_x]\,dx\,dt + \int_{\mathbb{R}} g \cdot \varphi(x,0)\,dx = 0. \tag{11.2}$$

A strong L^1_{loc} initial trace, when required, is an additional property beyond this distributional initial condition.

守恒变量的选择

浅水取 $u = (h, q)$、$q = hv$。速度 v 是原始变量而不是动量密度。光滑时可作变量替换，跨激波时必须回到原守恒律的跳跃条件。

Theorem 11.2 (Jump condition; source §11.1.1, equation (20))
Suppose u is piecewise C^1 across $x = s(t)$, $s \in C^1$, with one-sided traces, and solves the PDE on each side. Put $[u] = u_r - u_l$, $[F] = F(u_r) - F(u_l)$. The interior weak equation holds if and only if

$$[F] = s'(t)[u]. \tag{11.3}$$

Together with the correct initial trace this implies (11.2).

Expanded proof (editorial distribution calculation).

Integrate separately on $x < s(t)$ and $x > s(t)$ on a compact time strip. The singular part of u_t is $-s'[u]\delta_{x=s(t)}$; that of $F(u)_x$ is $[F]\delta_{x=s(t)}$. All regular terms vanish by the classical equations. Arbitrary vector tests on the curve force every component of $[F]-s'[u]$ to vanish. Integrating from $t = \delta$ and sending $\delta \downarrow 0$ supplies the initial term if the trace converges in L^1_{loc}. Values assigned on the curve itself do not change any integral. □

11.1.2 Physical models and characteristic speeds

For the elastic p-system use strain a and velocity b:

$$a_t - b_x = 0, \qquad b_t - p(a)_x = 0, \qquad A = DF = \begin{pmatrix} 0 & -1 \\ -p'(a) & 0 \end{pmatrix}.$$

This follows from $a = y_x$, $b = y_t$ for $y_{tt} - (p(y_x))_x = 0$. If $p' > 0$, the speeds are $\lambda_1 = -c$, $\lambda_2 = c$, $c = \sqrt{p'}$, with right vectors $(1, c)$ and $(1, -c)$. For shallow water with gravity scaled to one,

$$h_t + q_x = 0, \qquad q_t + (q^2/h + h^2/2)_x = 0, \qquad h > 0. \tag{11.4}$$

The speeds are $v \mp \sqrt{h}$, $v = q/h$. The strict gap $2\sqrt{h}$ closes at vacuum. For Euler, the conserved variables are $(\rho, \rho v, \rho E)$, $E = e + v^2/2$, and the flux is $(\rho v, \rho v^2 + p, (\rho E + p)v)$. On $\rho > 0$, smooth solutions obey

$$\rho_t + v\rho_x + \rho v_x = 0, \quad v_t + vv_x + \rho^{-1}p_x = 0, \quad e_t + ve_x + (p/\rho)v_x = 0.$$

Subtracting the kinetic-energy balance from total energy gives the last equation. Their matrix is $vI + C$, where

$$C = \begin{pmatrix} 0 & \rho & 0 \\ p_\rho/\rho & 0 & p_e/\rho \\ 0 & p/\rho & 0 \end{pmatrix}, \qquad c^2 = p_\rho + pp_e/\rho^2.$$

Its characteristic polynomial is $\det(\mu I - C) = \mu(\mu^2 - c^2)$. Thus $c^2 > 0$ gives speeds $v-c, v, v+c$. Evans's assumptions $p, p_\rho, p_e > 0$ are sufficient; division by ρ requires positivity.

同一个激波速度

若某分量 $[u^i] = 0$，则须有 $[F^i] = 0$；不能在该分量中作除法。所有非零分量给出的 $[F^i]/[u^i]$ 必须相同，故系统中并非任意两状态都能接激波。

本章的应力符号

这里 $p(a)$ 随应变增加，故 $p' > 0$。气体的 Lagrange 比容形式常采用另一种符号约定，不能把那里的压强导数符号直接搬来。

11.1.3 Strict hyperbolicity and smooth eigenframes

Definition 11.3 (Source §11.1.2)
The system $u_t + B(u)u_x = 0$ is strictly hyperbolic if $B(z)$ has real distinct eigenvalues $\lambda_1 < \cdots < \lambda_m$. Write $Br_k = \lambda_k r_k$, $l_k B = \lambda_k l_k$ and choose $l_j \cdot r_k = \delta_{jk}$. Right eigenvectors are unit when discussing arc length; the dual left eigenvectors need not also be unit.

For $j \neq k$, $(\lambda_j - \lambda_k) l_j \cdot r_k = 0$. The right eigenvectors form a basis, so $l_k \cdot r_k \neq 0$ before normalization. A smooth inviscid traveling wave $V(x - \sigma t)$ satisfies $(B(V) - \sigma I)V' = 0$. Where $V' \neq 0$, its speed must be an eigenvalue along its path. A viscous profile instead uses the scaled variable $(x - \sigma t)/\varepsilon$.

Theorem 11.4 (Coordinate invariance; source §11.1.2, Theorem 1)
Let H be a C^2 diffeomorphism of state domains. For smooth u, $\tilde{u} = H(u)$ satisfies a system with matrix

$$\widetilde{B}(H(z)) = DH(z)B(z)DH(z)^{-1}.$$

Strict hyperbolicity and genuine nonlinearity are preserved. Conservation form need not be preserved.

Expanded proof (chain rule and similarity).

Multiply the PDE by $DH(u)$ and use $\tilde{u}_x = DH(u)u_x$. Similarity preserves eigenvalues. Right vectors become DHr_k; left vectors become $l_k DH^{-1}$. Moreover $D_{\tilde{z}}\widetilde{\lambda}_k \cdot (DHr_k) = D_z \lambda_k \cdot r_k$. Rescaling a vector by a nonzero factor preserves whether this expression is zero. This proof uses a classical chain rule. Through a jump, the corresponding product of a discontinuous coefficient and a measure is not justified. □

Theorem 11.5 (Eigenframe dependence; source §11.1.2, Theorem 2)
A smooth strictly hyperbolic matrix field has smooth ordered eigenvalues and local smooth eigenframes. On state domain $\mathbb{R}^m$, unit eigenvectors can be chosen globally and smoothly. On a general open state domain the eigenframe assertion made here is local.

Expanded proof (including the global orientation step).

At a fixed state take $Br_0 = \lambda_0 r_0$, $|r_0| = 1$. Apply the implicit function theorem to $(B(z)r - \lambda r, (|r|^2 - 1)/2)$. The kernel of its derivative in (r, λ) satisfies $(B - \lambda_0 I)h - \mu r_0 = 0$, $r_0 \cdot h = 0$. Multiplication by l_0 gives $\mu = 0$; simplicity gives $h = cr_0$, then $h = 0$. The square derivative is invertible. Ordered eigenvalues agree on overlaps, so they are global.

局部与全局特征向量

隐函数定理先给局部选择。一般有孔状态域的特征直线丛可能无法连续定向；本章波曲线和 Riemann 拼接只需要一个小邻域。

For global eigenvectors on $\mathbb{R}^m$, form the smooth projection

$$P_k(z) = \prod_{j \neq k} \frac{B(z) - \lambda_j(z) I}{\lambda_k(z) - \lambda_j(z)}.$$

Put $P(s) = P_k(sz)$ and solve $R_s = [P_s, P]R$, $R(0) = r_k(0)$, $0 \le s \le 1$. Differentiating $P^2 = P$ gives $PP_sP = 0$ and $[[P_s, P], P] = P_s$. If $U_s = [P_s, P]U$, $U(0) = I$, then $P(s)U(s) = U(s)P(0)$: subtract the two sides, differentiate, and use this commutator identity and uniqueness for a homogeneous linear ODE. Hence R remains in the range of P, and never vanishes since U is invertible. Smooth dependence on z and normalization at $s = 1$ give a global unit vector. Apply the same construction to the transpose and rescale the left vectors to be dual. This radial construction uses the contractibility of $\mathbb{R}^m$. □

11.2 Riemann's Problem

The initial data are u_l for $x < 0$ and u_r for $x > 0$. All constructions below take place in a sufficiently small strictly hyperbolic neighborhood of a fixed state. Distinguish the state-space curves R_k, S_k from their space-time realizations: the former parameterize possible states, whereas the latter have slopes x/t.

11.2.1 Simple waves and genuine nonlinearity

Seek $u = V(w(x,t))$ with $V' = r_k(V)$. Substitution gives

$$u_t + DF(u)u_x = r_k(V)\{w_t + \lambda_k(V(w))w_x\}.$$

Thus a smooth scalar solution of $w_t + f_k(w)_x = 0$, where $f_k'(s) = \lambda_k(V(s))$, produces a smooth system solution. The integral curve V through z is the k-rarefaction curve $R_k(z)$.

Definition 11.6 (Source §11.2.1)
The k-field is genuinely nonlinear if $\gamma_k = D\lambda_k \cdot r_k \neq 0$ throughout the neighborhood; it is linearly degenerate if $\gamma_k \equiv 0$ there.

For a genuinely nonlinear field orient the unit vector so that $\gamma_k > 0$. Then $f_k'' = \gamma_k(V) > 0$ locally. Reversing the orientation changes the parameter but not the geometric wave.

Theorem 11.7 (Centered rarefaction; source §11.2.2, Theorem 1)
Suppose the k-field is genuinely nonlinear and $u_r = V(s_r)$, $u_l = V(s_l)$ with $s_l < s_r$. On this short curve let $L(s) = \lambda_k(V(s))$. Then the

标量化的适用范围

简单波代入使用链式法则。连续稀疏波可逐片验证再拼接；若标量 w 跳跃，$V(w)$ 未必满足系统 Rankine–Hugoniot 条件，不能直接搬用标量激波。

continuous integral solution is

$$u(x,t) = \begin{cases} u_l, & x/t \le L(s_l), \\ V(L^{-1}(x/t)), & L(s_l) < x/t < L(s_r), \\ u_r, & x/t \ge L(s_r). \end{cases} \tag{11.5}$$

Expanded proof (PDE, interfaces, and the initial trace).

Since $L' = \gamma_k > 0$, its inverse exists and is smooth. In the fan, $U(\xi) = V(L^{-1}(\xi))$ satisfies $(DF(U) - \xi I)U' = 0$, so $u_t + DF(u)u_x = 0$ for $t > 0$. On a strip $t \ge \delta > 0$ the profile is piecewise C^1 and continuous at both fan edges; therefore there is no interface measure in its first derivatives. The fan and all differing initial states occupy an interval of length at most Ct near the origin. Boundedness gives $\|u(\cdot,t) - g\|_{L^1(K)} \le C_K t$ for each compact K. Integration by parts above δ followed by $\delta \downarrow 0$ proves the initial weak identity. □

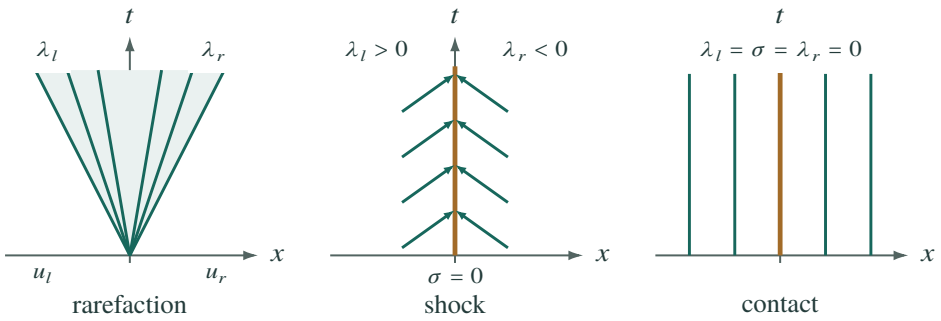


Diagram T11.1. Wave patterns in physical space The left panel is a centered rarefaction: characteristic speeds increase from λ_l to λ_r. The middle panel shows a compressive shock with $\lambda_r < \sigma < \lambda_l$; forward characteristics terminate on it. The right panel is a contact, with equal characteristic and interface speed. Only slopes and incidence are represented; these are schematic panels for a single characteristic family.

11.2.2 The Hugoniot locus and its second-order contact

The shock set at z consists of states v for which $F(v) - F(z) = \sigma(v - z)$ for some common speed σ.

Theorem 11.8 (Local shock curves; source §11.2.3, Theorem 2)
For $F \in C^3$ strictly hyperbolic near z, the shock set is locally the union of m curves $S_k(z)$. In a signed unit-speed parameter a, choose $S_k(0;z) = z$, $\partial_a S_k(0;z) = r_k(z)$. Writing $v = S_k(a;z)$, the shock

speed extends smoothly along this branch to $\sigma_k(0;z) = \lambda_k(z)$*, and*

$$\partial_a \sigma_k(0;z) = \tfrac{1}{2}\gamma_k(z), \qquad \sigma_k(a;z) = \frac{\lambda_k(v) + \lambda_k(z)}{2} + O(a^2). \tag{11.6}$$

The shock and rarefaction curves have equal second derivatives at zero when both use oriented arc length.

Expanded proof (construction and all Taylor coefficients).

Define $\overline{A}(v,z) = \int_0^1 DF(z+\theta(v-z))\,d\theta$. The jump equation becomes $(\overline{A}(v,z) - \sigma I)(v-z) = 0$. Near $v = z$ this averaged matrix is strictly hyperbolic; its real simple eigenvalues and dual eigenvectors are smooth. Reality follows by continuation of the real simple roots of the characteristic polynomial, and the spectral gaps persist after shrinking the ball. Let these left vectors be $\bar{l}_j(v,z)$. The $m-1$ equations

$$\bar{l}_j(v,z)\cdot(v-z) = 0, \qquad j \neq k,$$

have derivative at $v = z$ with independent rows $l_j(z)$. The implicit function theorem gives a curve tangent to $r_k(z)$. Every nearby nontrivial jump vector is an eigenvector of the averaged matrix, hence belongs to exactly one of these branches. The speed on the branch is its kth eigenvalue; at the common base point there are m distinct branch limits, not a single joint speed function.

Set $r = r_k(z)$, $A = DF(z)$, $\lambda = \lambda_k(z)$ and $H = D^2F(z)[r,r]$. Twice differentiating $F(S(a)) - F(z) = \sigma(a)(S(a)-z)$ at zero gives

$$(A - \lambda I)S''(0) = 2\sigma'(0)r - H.$$

Differentiate $DF\,r_k = \lambda_k r_k$ in direction r:

$$(A - \lambda I)Dr_k(z)r = \gamma_k(z)r - H.$$

Left multiplication by l_k yields $2\sigma'(0) = \gamma_k$. Subtracting the equations shows that $S''(0) - Dr_k r$ is parallel to r. Both terms are orthogonal to r, because both curves have unit speed; hence they are equal. Taylor's formula now gives $\lambda_k(S(a)) = \lambda + \gamma_k a + O(a^2)$ and the stated speed estimate. Local equivalence $|S(a)-z| \asymp |a|$ also permits the remainder to be written $O(|v-z|^2)$. □

Theorem 11.9 (Contacts; source §11.2.3, Theorem 3)
If the k-field is linearly degenerate on a neighborhood, its local shock and rarefaction curves coincide and their speed is $\lambda_k(z) = \lambda_k(v)$*. Two such states are joined by a contact at that speed.*

激波速度不是全域函数

每一条 S_k 上的速度可光滑延拓到基点，但不同分支极限是不同的 $\lambda_k(z)$。不能在所有分支的交点假定一个共同的光滑 $\sigma(v,z)$。

Expanded proof (inclusion and local equality).

Along $V' = r_k(V)$, $\frac{d}{ds}\lambda_k(V) = 0$. Integration gives $F(V(s)) - F(z) = \int_0^s \lambda_k(z)V'(a)\,da = \lambda_k(z)(V(s) - z)$. Thus the rarefaction curve lies in the shock set. Its tangent places it on branch k; the implicit-function construction gives the unique local curve on that branch. The two germs coincide, and (11.3) gives the contact solution. This is a local equality, not a statement about arbitrary distant branches of the shock set. □

11.2.3 Admissible shocks and the local solver

For a genuinely nonlinear field with $\gamma_k > 0$, (11.6) gives for small $a < 0$

$$\lambda_k(S_k(a;z)) < \sigma_k(a;z) < \lambda_k(z).$$

These are the compressive Lax shocks. For $a > 0$ the inequalities reverse and give expansion shocks, which are rejected. Spectral separation also gives $\lambda_{k-1}(z) < \sigma_k < \lambda_{k+1}(S_k(a;z))$ when the neighboring indices exist. The strict inequalities follow because the leading gaps have size $|\gamma_k a|/2$, whereas the error has size $O(a^2)$.

Define $T_k(a;z)$ as the rarefaction curve for $a \geq 0$ and the shock curve for $a \leq 0$, each with compatible oriented parameters. For a linearly degenerate field use its contact curve for both signs. The common value and tangent, as well as smooth base-state dependence, make T_k jointly C^1 near $(0,z)$; the second-order contact proved above gives additional matching in the wave parameter.

Theorem 11.10 (Local Riemann solver; source §11.2.4, Theorem 4)
If each field is genuinely nonlinear or linearly degenerate on a neighborhood of u_l, every sufficiently close u_r can be joined to u_l by at most m elementary waves, ordered by family. The resulting integral solution is self-similar. Its small wave-strength vector is unique within this construction.

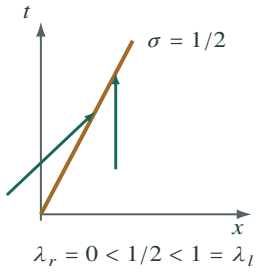
Expanded proof (state matching, speed ordering, weak gluing).

Let $z_0 = u_l$ and $z_k = T_k(a_k; z_{k-1})$. The endpoint map $\mathscr{E}(a) = z_m$ is C^1, $\mathscr{E}(0) = u_l$, and

$$D\mathscr{E}(0) = (r_1(u_l), \dots, r_m(u_l)).$$

This is invertible. The inverse function theorem gives a unique small a with $\mathscr{E}(a) = u_r$. Choose disjoint intervals around the ordered numbers $\lambda_k(u_l)$. For states and strengths sufficiently small, every fan edge, shock speed, and contact speed in family k remains in its own interval. Consequently the elementary waves do not overlap in $\xi = x/t$, and constant

图 T11.M：平均速度与压缩



本图取 Burgers 通量 $f(u) = u^2/2$、$u_l = 1, u_r = 0$。激波速度恰为 $1/2$，两侧速度为 1 和 0；一般系统的平均公式只在小振幅下成立到二阶余项。

intermediate states fill the gaps. Integrate on each smooth sector: continuous fan boundaries have no jump measure, and each shock or contact satisfies (11.3). All interior interface terms cancel. The difference from the initial step is bounded and confined to $|x| \le Ct$, giving the initial trace as in the rarefaction proof. The inverse theorem proves uniqueness of the small parameters; it does not prove uniqueness among all bounded weak or entropy solutions of the system. □

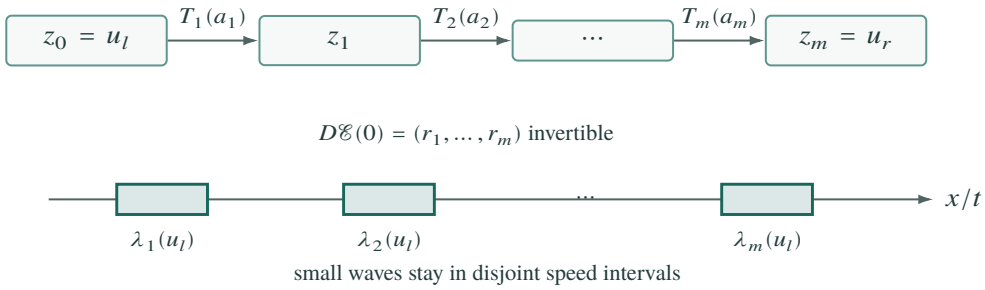


Diagram T11.2. Matching states before placing waves The endpoint map composes the family curves in state space. Its derivative at zero consists of the independent eigenvectors. Disjoint speed intervals then place those waves in increasing order in x/t. The unique small strength vector is a local finite-dimensional conclusion; no general system uniqueness theorem is inferred.

波曲线的方向

本稿始终写右状态 $z_k \in T_k(z_{k-1})$。原书 §11.2.4 的式 (43) 与随后 (45) 的左右方向写法不一致；这里以构造式 (45) 为准，明确每次从左状态走向右状态。

11.3 Systems of Two Conservation Laws

11.3.1 Local Riemann coordinates

In this section $m = 2$. Evans calls w_i an ith Riemann invariant when $Dw_i \cdot r_i = 0$: it is constant on the state-space curve R_i. A useful coordinate must also have $Dw_i \neq 0$.

Lemma 11.11 (Existence of local invariant coordinates; editorial detail)
Near a strictly hyperbolic state there are smooth w_1, w_2 satisfying $Dw_i \cdot r_i = 0$ and $\det D(w_1, w_2) \neq 0$.

Expanded proof (flow-box construction).

Choose a short curve $\Gamma(a)$ transverse to r_i at the base point. Let $X(s,a)$ solve $X_s = r_i(X)$, $X(0,a) = \Gamma(a)$. At the base point the two columns r_i, Γ' are independent, so the inverse function theorem makes (s,a) local coordinates. Define $w_i(X(s,a)) = a$. Then $Dw_i r_i = 0$ and $Dw_i \neq 0$. In two dimensions the annihilator of r_i is spanned by $l_j, j \neq i$. Thus Dw_1 and Dw_2 are nonzero multiples of independent left eigenvectors. After intersecting their neighborhoods, the two functions together are coordinates. □

Theorem 11.12 (Diagonal form; source §11.3.1, Theorem 1)
For a smooth solution, set $v_i = w_i(u)$ and express $u = z(v_1, v_2)$. Then

$$(v_1)_t + \lambda_2(v)(v_1)_x = 0, \qquad (v_2)_t + \lambda_1(v)(v_2)_x = 0. \tag{11.7}$$

Here $\lambda_i(v)$ means $\lambda_i(z(v))$.

Expanded proof (why the indices cross).

Because Dw_i is parallel to l_j, $j \neq i$, $Dw_i(-DF + \lambda_j I)u_x = 0$. The chain rule gives the displayed transport equation. Along a curve $X' = \lambda_j(u(X,t))$ its v_i value is constant. Also z_{v_j} is a nonzero multiple of r_i for $i \neq j$: it lies in the kernel of Dw_i and cannot be zero since Dz is invertible. Therefore genuine nonlinearity of family i is equivalent to $\partial_{v_j}\lambda_i \neq 0$. The transport speeds still depend on both invariants, so the diagonal equations are coupled, not independent scalar equations. □

For barotropic gas in conserved variables (ρ, q),

$$F = (q, q^2/\rho + p(\rho)), \qquad \rho > 0, \quad p' > 0.$$

In primitive variables (ρ, v), $v = q/\rho$, the matrix is $\left(\begin{smallmatrix} v & \rho \\ p'/\rho & v \end{smallmatrix}\right)$, with speeds $v \mp c$, $c = \sqrt{p'}$. Put

$$I(\rho) = \int_{\rho_*}^{\rho} \frac{c(s)}{s}\, ds, \qquad w_1 = I(\rho) + v, \quad w_2 = I(\rho) - v. \tag{11.8}$$

The Jacobian determinant in (ρ, v) is $-2c/\rho \neq 0$. Direct substitution of $\rho_t + v\rho_x + \rho v_x = 0$ and $v_t + vv_x + (c^2/\rho)\rho_x = 0$ proves $D_+ w_1 = 0$, $D_- w_2 = 0$, where $D_\pm = \partial_t + (v \pm c)\partial_x$. These equations use only the barotropic mass–momentum model. It is not necessary to impose the full Euler energy equation with constant internal energy.

11.3.2 Compression and a weighted Riccati equation

We give a version of source §11.3.2, Theorem 2 with its uniformity assumptions explicit. Suppose the characteristic variables exist on an open set containing the closed rectangle $K = [A_1, B_1] \times [A_2, B_2]$. Assume the coefficients are C^2 there,

$$\lambda_2 - \lambda_1 \geq d > 0, \qquad \partial_{v_1}\lambda_2 \geq c_0 > 0, \qquad \partial_{v_2}\lambda_1 \geq c_0 > 0 \quad \text{on } K. \tag{11.9}$$

The signs can be obtained from genuine nonlinearity by suitable orientations of the two invariant coordinates on a connected chart.

两种“不变”曲线

按 Evans 的编号，w_1 在状态空间沿 R_1 不变，但 $w_1(u(x,t))$ 沿时空的 λ_2 特征线不变。这两个陈述分别使用右特征向量与另一族左特征向量，并不矛盾。

远离真空的条件

当 $\rho \downarrow 0$，c/ρ 可能发散，特征间隙也可能消失。局部坐标可逆不保证整个解的值域始终留在同一张坐标图中；下面的爆破定理显式规定紧矩形值域。

Theorem 11.13 (Finite-time loss of smoothness)
Suppose the smooth initial invariants range in K, and a classical solution remains in this chart with range in K while it exists. If either initial derivative $(v_1)_x$ or $(v_2)_x$ is negative somewhere, it cannot remain a classical solution for all positive time. More precisely a weighted derivative diverges to $-\infty$ in a finite bounded time unless the assumed smooth regime ends sooner.

爆破的是梯度

输运使 v_i 保持初值范围，而 $e^H(v_1)_x$ 满足负二次 ODE。证明使用的是系数的一致正下界；只有逐点正性而没有紧值域控制，不能据此断言积分必在有限时刻达到临界值。

Expanded proof (removing the mixed derivative term).

Set $a = (v_1)_x$, $b = (v_2)_x$ and $D_2 = \partial_t + \lambda_2\partial_x$. Differentiating the first equation in (11.7) gives

$$D_2 a = -(\partial_1\lambda_2)a^2 - (\partial_2\lambda_2)ab, \qquad D_2 v_2 = (\lambda_2 - \lambda_1)b.$$

Choose H with

$$H(v_1, v_2) = \int_{A_2}^{v_2} \frac{\partial_2\lambda_2(v_1, s)}{\lambda_2(v_1, s) - \lambda_1(v_1, s)}\, ds.$$

Along a λ_2 characteristic, v_1 is constant, so $D_2 H = (\partial_2\lambda_2)b$. For $y = e^H a$ the mixed term cancels:

$$\frac{dy}{dt} = -k(t)y^2, \qquad k = e^{-H}\partial_1\lambda_2 \geq \kappa > 0. \tag{11.10}$$

The lower bound follows from compactness of K and (11.9), not merely pointwise positivity. If $y(0) < 0$, uniqueness preserves its sign before blow-up, and

$$\frac{1}{y(t)} = \frac{1}{y(0)} + \int_0^t k(s)\, ds.$$

The denominator reaches zero no later than $T_* = -1/(\kappa y(0))$. Thus a globally finite derivative is impossible. For b, use $D_1 = \partial_t + \lambda_1\partial_x$ and a weight J satisfying $\partial_1 J = \partial_1\lambda_1/(\lambda_1 - \lambda_2)$; then $D_1(e^J b) = -e^{-J}\partial_2\lambda_1(e^J b)^2$.

On a smooth solution confined to the chart, the characteristic speeds are bounded and locally Lipschitz in space, so the characteristics exist on every finite smooth time strip. Each invariant is transported from its initial range; this explains invariance of the rectangle provided the chart covers that entire rectangle. It does not supply a missing global inverse chart. Smooth compactly supported perturbations of a constant state have bounded invariant ranges. Any nonconstant one of these invariants, with equal limits at $\pm\infty$, has a negative derivative somewhere, so the criterion applies under the stated chart and uniformity assumptions. □

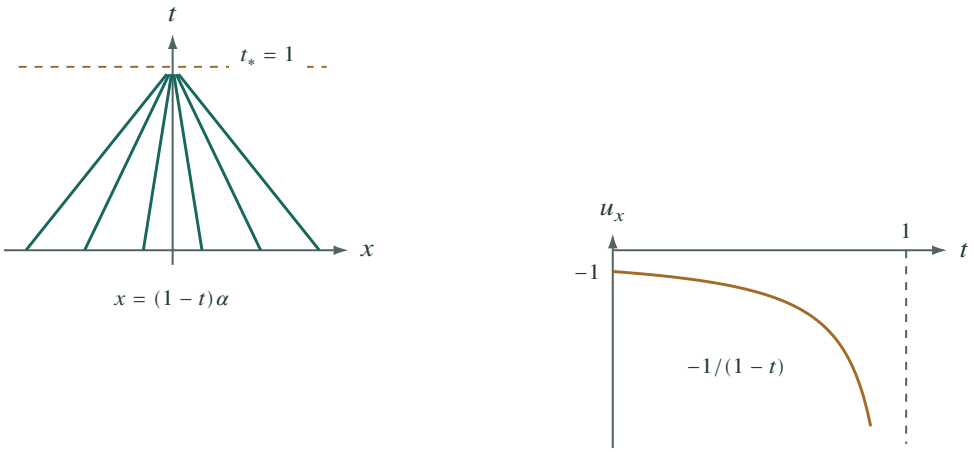


Diagram T11.3. Compression reaches a vertical slope An exact simple-wave example is the scalar Burgers characteristic map $x = \alpha + t\,g(\alpha)$ for $g(\alpha) = -\alpha$ on a local interval. Its Jacobian is $1 - t$, and $u_x = -1/(1 - t)$ until $t = 1$. The second panel plots this derivative. For two-field systems, the weight in (11.10) removes the mixed interaction and recovers the same Riccati mechanism; the drawn map is the scalar model.

11.4 Entropy Criteria

11.4.1 Viscous profiles and the limits of dimension counting

For identity viscosity, $u_t^\varepsilon + F(u^\varepsilon)_x = \varepsilon u_{xx}^\varepsilon$. The ansatz $u^\varepsilon = V((x-\sigma t)/\varepsilon)$ gives

$$\begin{aligned} V' &= F(V) - F(u_l) - \sigma(V - u_l) = G(V;\sigma), \\ V(-\infty) &= u_l, \qquad V(+\infty) = u_r. \end{aligned} \tag{11.11}$$

Here integration used $V'(-\infty) = 0$; the right limit then forces Rankine–Hugoniot. For an autonomous smooth ODE, convergence of V to an equilibrium also implies $V' \to 0$.

We use the following *external finite-dimensional theorem*, not proved here: a sufficiently smooth autonomous vector field with a spectral splitting into center, stable, and unstable subspaces has a local center manifold of any fixed finite differentiability allowed by the field. It is tangent to the center space and contains every complete orbit staying in a sufficiently small neighborhood, in particular all nearby equilibria. Parameters can be included as variables with zero derivative. The conclusion is local; it does not assert that a distant excursion lies on the manifold.

稳定维数不等于轨道存在

线性化特征值是 $\lambda_j(u_l) - \sigma$。某个方向不稳定只说明允许离开平衡点，不能单凭维数判断一条轨道必到达指定的右状态；一般系统须另用中心流形等工具。

Theorem 11.14 (Small local viscous profiles)
This is a local version of source §11.4.1, Theorem 1. Assume F is smooth and the k-field is genuinely nonlinear near u_l. For a sufficiently small nonzero k-shock state u_r, a profile whose entire image

> *stays near u_l exists if and only if it is on the compressive branch $S_k^-(u_l)$. The proof below uses the stated external center-manifold theorem.*

Expanded proof (center-manifold reduction).

Fix u_l, orient $r = r_k(u_l)$ so $\gamma = D\lambda_k(u_l)r > 0$, and write $\beta = \sigma - \lambda_k(u_l)$. Adjoin $\beta' = 0$ to (11.11). At $(V, \beta) = (u_l, 0)$ the center space has dimension two, spanned by $(r, 0)$ and $(0, 1)$; all other eigenvalues are separated from zero by strict hyperbolicity. A local center manifold can be parameterized by $a = l_k(u_l)(V - u_l)$ and β:

$$V = u_l + ar + W(a, \beta), \quad l_k W = 0, \quad W(0, \beta) = 0, \quad W = O(a^2 + |a\beta|).$$

The equality $W(0, \beta) = 0$ follows because $V = u_l$ is an equilibrium for every small β. Project the vector field by l_k. Taylor expansion, $l_k(DF - \lambda_k I) = 0$, and $l_k r = 1$ give

$$a' = g(a, \beta) = -\beta a + \tfrac{1}{2}\gamma a^2 + O(|a|(|a| + |\beta|)^2).$$

The coefficient identity $l_k D^2 F[r, r] = \gamma$ follows by differentiating the eigenvalue equation. The nearby right equilibrium lies on the manifold. Write its coordinate $a_r \neq 0$; by the shock speed expansion, $\beta = \gamma a_r/2 + O(a_r^2)$. For this fixed β, both 0 and a_r are roots of g. Divide smoothly first by a and then by $a - a_r$, using the integral form of the mean value theorem. Uniformly for a between the roots,

$$g(a, \beta) = a(a - a_r)h(a, \beta), \qquad h = \gamma/2 + O(|a_r|) > 0.$$

If $a_r < 0$, the vector field is negative between a_r and 0. The solution through any interior point decreases, stays between the equilibria, and converges to 0 as $s \to -\infty$ and to a_r as $s \to +\infty$. The simple roots make the times to both endpoints infinite (the separated integral has logarithmic divergences). Lifting gives the desired profile, with its full image small. If $a_r > 0$, the field is negative on $(0, a_r)$ and cannot connect 0 to a_r forward in s. Every complete sufficiently small profile lies on the center manifold, so this also proves the local necessity. The equivalence $a_r < 0 \iff u_r \in S_k^-(u_l)$ follows from (11.6). No assertion about profiles making large excursions is obtained from this local argument. □

For a non-genuinely nonlinear branch, the Liu criterion compares shock speeds along the full intervening Hugoniot arc:

$$\sigma(z, u_l) \geq \sigma(u_r, u_l) \quad \text{for intermediate states } z \in S_k(u_l).$$

The source writes a strict version for nontrivial intermediate states. The local general existence theory requires additional hypotheses and is not

proved by the preceding genuinely nonlinear reduction. In the scalar profile calculation below, strictness has a precise meaning: it excludes an extra equilibrium that would block passage.

11.4.2 A complete profile test for the viscous elastic system

Use viscosity only in the velocity equation: $a_t - b_x = 0$, $b_t - p(a)_x = \varepsilon b_{xx}$. For end strains $a_l < a_r$ choose

$$\sigma = \sqrt{\frac{p(a_r) - p(a_l)}{a_r - a_l}} > 0, \qquad b_r = b_l - \sigma(a_r - a_l), \qquad p' > 0.$$

Theorem 11.15 (Source §11.4.1, elastic example)
A traveling profile joining these states exists if and only if

$$p(a) - p(a_l) > \sigma^2(a - a_l), \qquad a_l < a < a_r. \tag{11.12}$$

It is unique up to translation among such monotone profiles.

Expanded proof (separation and infinite endpoint time).

The first integrated equation gives $B = b_l - \sigma(A - a_l)$. The second yields

$$A' = g(A) = \frac{p(A) - p(a_l) - \sigma^2(A - a_l)}{\sigma}.$$

Thus $g(a_l) = g(a_r) = 0$. If $g > 0$ in the open interval, the inverse of $s - s_0 = \int_{A_0}^{A} d\zeta / g(\zeta)$ is a strictly increasing profile. Since g is C^1 and vanishes at each endpoint, $|g(a)| \le C|a - a_l|$ near a_l and similarly near a_r; the endpoint integrals diverge, even if the roots are not simple. The inverse is defined for all real s with the required limits. Conversely any connecting scalar orbit must pass every intermediate value. Uniqueness for a scalar Lipschitz ODE prevents passage through an interior zero and prevents a change of direction. Hence $g > 0$ throughout the interval. Separation also proves translation uniqueness. Condition (11.12) is precisely strict decrease of the secant speed from a_l to an intermediate point relative to the final positive speed. This proof concerns the stated partial viscosity, not an unmentioned change of viscosity matrix. □

11.4.3 Entropy pairs, symmetrizers, and viscous limits

Definition 11.16 (Source §11.4.2)
An entropy pair (η, Q) consists of C^2 functions with η convex and $DQ = D\eta\, DF$. An integral solution is an entropy solution if for every

such pair and $0 \le \varphi \in C_c^\infty(\mathbb{R} \times (0,\infty))$,

$$\iint [\eta(u)\varphi_t + Q(u)\varphi_x]\,dx\,dt \ge 0. \tag{11.13}$$

Thus the distribution $\eta(u)_t + Q(u)_x$ is nonpositive.

For a smooth solution the entropy balance is equality. At a shock it reduces to $[Q] - \sigma[\eta] \le 0$ by the same interface calculation as before. The sign refers to convex mathematical entropy, often the negative of physical entropy.

Lemma 11.17 (Entropy compatibility; editorial supplement)
Locally, $D\eta\,DF$ is the gradient of a flux if and only if $D^2\eta\,DF$ is symmetric. If $D^2\eta$ is positive definite, DF is similar to a symmetric matrix and has real eigenvalues. This does not force distinctness.

Expanded proof (integrability and symmetrization).

For $b_j = \sum_a \eta_a F_j^a$, differentiate with respect to z_i. In $\partial_i b_j - \partial_j b_i$ the terms involving F_{ij}^a cancel. The remaining expression is the skew part of $D^2\eta DF$. A closed C^1 one-form has a potential on a small ball: integrate along radial segments and differentiate, using equality of the cross partials. This proves local sufficiency; global sufficiency requires, for example, a simply connected domain. If $S = D^2\eta > 0$, then $S^{1/2}DFS^{-1/2}$ is symmetric because $SDF = (DF)^T S$. It is similar to DF and hence has the same real spectrum. □

For the elastic system, a strictly convex pair is $\eta(a,b) = b^2/2 + P(a)$, $P' = p$, and $Q = -bp(a)$. Indeed $D^2\eta = \operatorname{diag}(p', 1) > 0$.

Theorem 11.18 (Viscous limit; source §11.4.2, Theorem 2)
Suppose smooth identity-viscosity solutions with initial data g are uniformly bounded with range in a fixed compact state set and converge a.e. to u on $\mathbb{R} \times (0,\infty)$. Then u is an integral solution and satisfies (11.13) for all smooth convex entropy pairs. This is conditional on the convergence; it is not a compactness or existence theorem for general systems.

Expanded proof (moving derivatives onto the compact test).

The chain rule yields

$$\eta(u^\varepsilon)_t + Q(u^\varepsilon)_x = \varepsilon\eta(u^\varepsilon)_{xx} - \varepsilon(u_x^\varepsilon)^T D^2\eta(u^\varepsilon)u_x^\varepsilon.$$

The last term is nonpositive. Multiply by $\varphi \ge 0$ and integrate:

$$\iint (\eta(u^\varepsilon)\varphi_t + Q(u^\varepsilon)\varphi_x) \ge -\varepsilon \iint \eta(u^\varepsilon)\varphi_{xx}.$$

系统熵对的相容性

标量情形每个凸 η 都能积分得到 Q；系统须满足超定条件 $D^2\eta\,DF$ 对称。并不是任意凸函数都可充当某个系统的熵，更不能由一个能量不等式直接得到所有解的唯一性。

The error is $O(\varepsilon)$ because the test support and the range are compact. Dominated convergence handles the other terms. For the weak PDE, test the viscous equation directly, including $t = 0$:

$$\iint (u^\varepsilon \cdot \varphi_t + F(u^\varepsilon) \cdot \varphi_x + \varepsilon u^\varepsilon \cdot \varphi_{xx}) + \int g \cdot \varphi(\cdot, 0) = 0.$$

The same argument proves (11.2). No integration over spatial infinity, energy decay assumption, or gradient bound is needed for these compactly supported tests. An arbitrary nonsmooth system entropy is not automatically covered by scalar convolution: that convolution need not preserve the system flux compatibility. □

11.4.4 Scalar entropy and the full contraction argument

Let $m = 1, f \in C^2(\mathbb{R})$, and work in $u \in C([0,\infty); L^1(\mathbb{R})) \cap L^\infty$ with initial trace g. For each $k \in \mathbb{R}$, the Kruzhkov pair is

$$\eta_k(u) = |u - k|, \qquad q_k(u) = \operatorname{sgn}(u - k)(f(u) - f(k)).$$

It follows from the smooth pairs by replacing $|s-k|$ by $\sqrt{(s-k)^2 + \delta^2}$ and taking $Q_\delta(s) = \int_k^s \eta'_\delta(r) f'(r)\, dr$. On a bounded range both components converge uniformly, so entropy inequalities pass to the limit.

Theorem 11.19 (Scalar contraction; source §11.4.3, Theorem 3)
If u, v are scalar entropy solutions in this class with data g, h, then for all $0 \le s \le t$,

$$\|u(\cdot, t) - v(\cdot, t)\|_1 \le \|u(\cdot, s) - v(\cdot, s)\|_1. \tag{11.14}$$

In particular, the solution with a prescribed initial trace is unique. No convexity of f is required.

Expanded proof (doubling, diagonal limit, and endpoint times).

For fixed (y, s) use $k = v(y, s)$ in the entropy inequality for u, and for fixed (x, t) use $k = u(x, t)$ in that for v. The pair $q(a, b) = \operatorname{sgn}(a - b)(f(a) - f(b))$ is symmetric in a, b. Add and integrate the inequalities to get, for nonnegative smooth compactly supported $\zeta(x, t, y, s)$ away from $t = 0, s = 0$,

$$\iiiint \{|u - v|(\partial_t + \partial_s)\zeta + q(u, v)(\partial_x + \partial_y)\zeta\} \ge 0.$$

Fubini is legitimate because all variables are bounded on the compact support; the inequalities hold for every constant k and smooth test. Take

$$\zeta = \rho_\delta(x - y)\rho_\delta(t - s)\phi\big((x + y)/2, (t + s)/2\big), \qquad \phi \ge 0.$$

For small δ the support remains away from both initial times. The difference-kernel derivatives cancel under the summed derivatives; the derivatives of ϕ have coefficient one. Use mean and difference variables; their constant positive Jacobian can be divided out. Translations are continuous in L^1 on compact space-time sets. The functions $|a-b|$ and $q(a,b)$ are jointly Lipschitz on any bounded square: for fixed b, the piecewise derivative of q is $\pm f'(a)$ and the two pieces meet continuously at $a=b$; then repeat in b. Thus the diagonal limit gives

$$\iint (A\phi_t + B\phi_x) \geq 0, \quad A = |u-v|, \quad B = q(u,v), \quad |B| \leq LA, \tag{11.15}$$

where $L = \max_{|r|\leq M} |f'(r)|$ and M bounds both solutions.

Choose $\phi = \alpha_R(x)\beta(\tau)$, with $0 \leq \alpha_R \leq 1$, $\alpha_R = 1$ on $[-R,R]$, zero outside $[-2R,2R]$, $|\alpha_R'| \leq C/R$, and $\beta \geq 0$ compact in positive time. The flux contribution tends to zero, since its absolute value is at most $(CL/R)\|\beta\|_\infty \int \|u-v\|_1 d\tau$. The time integral is finite by L^1 continuity on compact intervals. Dominated convergence yields $\int \|u-v\|_1 \beta' \geq 0$. Hence the distributional time derivative of this continuous function is nonpositive. Equivalently, approximate the indicator of $[s,t]$ by trapezoidal Lipschitz functions and then mollify; the two end averages tend to the values at s,t by continuity. This proves contraction first for $s>0$, then for $s=0$ by the strong initial traces. Continuity also upgrades an a.e.-time inequality to every time. □

The same Kato inequality gives a local finite-speed estimate:

$$\int_{|x|\leq R} |u(x,t) - v(x,t)|\,dx \leq \int_{|x|\leq R+Lt} |g(x) - h(x)|\,dx.$$

To verify it, use cutoffs of the backward interval $|x| < R + L(t-\tau)$ in (11.15); they satisfy $\phi_\tau + L|\phi_x| \leq 0$ between the two time cutoffs. The bound $|B| \leq LA$ makes the lateral flux nonpositive. Smooth approximation and the strong local traces supply the two end sections. This local version also proves uniqueness for bounded nonintegrable Riemann data with L^1_{loc} time continuity.

初始迹与全时刻唯一性

消黏的分布初值不自动给强 L^1 连续性。本定理把 $C_tL^1_x$ 与强初始迹作为解类条件，再把几乎处处时刻的收缩提升到每个时刻。一般系统没有相同的序结构。

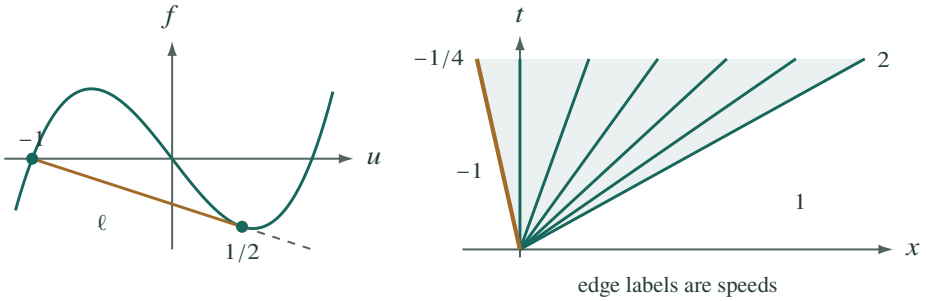


Diagram T11.4. The entropy chord for a nonconvex flux For $f(u) = u^3 - u$, the chord from $u_l = -1$ to $u_* = 1/2$ has slope $-1/4 = f'(1/2)$ and lies below f on that interval. The entropy solution for right state 1 joins this shock to a rarefaction $u = \sqrt{(x/t+1)/3}$ for $-1/4 < x/t < 2$. A single jump from -1 to 1 violates the chord criterion. The exact states and speeds are derived in Problems 14–15.

11.5 Problems

The following are abbreviated descriptions of the 17 source problems. All solutions and additional qualifications are editorial reconstructions.

1. Generalized elastic system

For $F = (-q(b), -p(a))$ with $p', q' > 0$,

$$DF = \begin{pmatrix} 0 & -q'(b) \\ -p'(a) & 0 \end{pmatrix}, \qquad \lambda_{\pm} = \pm\sqrt{p'(a)q'(b)}.$$

They are real and distinct. Corresponding vectors are $(q'(b), -\lambda_{\pm})$; substitution verifies the eigenvalue equation.

2. Shallow water and a change of conserved variables

In (h, q), $DF = \begin{pmatrix} 0 & 1 \\ h - q^2/h^2 & 2q/h \end{pmatrix}$, whose characteristic polynomial is $(\lambda - v)^2 - h$. For smooth $h > 0$, expansion of $q = hv$ and use of $h_t + (hv)_x = 0$ give $v_t + vv_x + h_x = 0$. Hence the alternative smooth system has flux $\widetilde{F}(h, v) = (hv, v^2/2 + h)$ and matrix $\begin{pmatrix} v & h \\ 1 & v \end{pmatrix}$, again with speeds $v \pm \sqrt{h}$. This is equivalence for smooth solutions only. For example, the original jump from $(h, v) = (1, 0)$ to $(2, \sqrt{3}/2)$ has speed $\sigma = \sqrt{3}$ and satisfies both original conservation laws. For the alternative velocity law its flux jump is $11/8$, whereas $\sigma[v] = 3/2$; the jump fails. Nonlinear coordinate changes can therefore alter admissible weak solutions even when both resulting smooth systems happen to have conservation form.

3. Smooth matrices with rotating eigenspaces

For $z \neq 0$ put $\theta = 1/z$ and

$$B(z) = e^{-1/z^2} \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}, \qquad B(0) = 0.$$

Every derivative is a finite sum of a bounded sine or cosine times $z^{-N} e^{-1/z^2}$, which tends to zero. Induction proves $B \in C^\infty$. The eigenvalues are $\pm e^{-1/z^2}$ for $z \neq 0$, but coincide at zero. The positive eigenline is spanned by $(\cos\theta, \sin\theta)$; the negative line is its perpendicular. Along sequences $\theta = n\pi$ and $\theta = \pi/2 + n\pi$, even the positive line has incompatible limits. Sign choices cannot repair this, so no continuous unit frame extends through zero. This failure concerns loss of strictness.

4. Shallow-water rarefaction curves

In (h, v) use $r_1 = (1, -1/\sqrt{h})$, $r_2 = (1, 1/\sqrt{h})$. Integrating $dv/dh = \mp h^{-1/2}$ gives $2\sqrt{h} + v = C$ on R_1 and $2\sqrt{h} - v = C$ on R_2. The gradient of each displayed invariant annihilates its own vector.

5. Shallow-water shock curves

Write the base state as $(h_0, h_0 v_0)$. Eliminating σ from $\sigma[h] = [hv]$, $\sigma[hv] = [hv^2 + h^2/2]$ gives, for $h \neq h_0$,

$$v = v_0 \pm (h - h_0)\sqrt{\frac{h + h_0}{2hh_0}}, \qquad q = hv_0 \pm (h - h_0)\sqrt{\frac{h(h + h_0)}{2h_0}}. \tag{11.16}$$

One way to see the elimination is to introduce the relative mass flux $j = h(v - \sigma) = h_0(v_0 - \sigma)$; momentum gives $j^2 = (h^2 - h_0^2)/(2(1/h_0 - 1/h)) = hh_0(h + h_0)/2$. The nontrivial branch speeds are $\sigma = v_0 \pm \sqrt{h(h + h_0)/(2h_0)}$. If $h = h_0$, the mass jump forces $q = q_0$, so there is no missing nonzero jump. For $q_0 = 0$, the formula is exactly $q = \pm(h - h_0)h\sqrt{(1/h + 1/h_0)/2}$.

6. Dam-break Riemann data with positive depths

Let $h_l > h_r > 0$ and both initial velocities be zero. The intermediate depth H is the unique root in (h_r, h_l) of

$$2(\sqrt{h_l} - \sqrt{H}) = (H - h_r)\sqrt{\frac{H + h_r}{2Hh_r}} =: V. \tag{11.17}$$

The left member strictly decreases and the right strictly increases: for the latter square it and differentiate $(H - h_r)^2(H + h_r)/(2Hh_r)$, or note both

positive factors $H-h_r$ and $\sqrt{1+h_r/H}$ have combined positive derivative. The endpoint signs are opposite, proving existence and uniqueness. Set $q_i = HV$, $\sigma = HV/(H-h_r)$. With $\xi = x/t$,

$$(h,v) = \begin{cases} (h_l, 0), & \xi \le -\sqrt{h_l}, \\ ((2\sqrt{h_l} - \xi)^2/9,\ 2(\sqrt{h_l} + \xi)/3), & -\sqrt{h_l} < \xi < V - \sqrt{H}, \\ (H, V), & V - \sqrt{H} \le \xi < \sigma, \\ (h_r, 0), & \xi > \sigma. \end{cases}$$

The fan preserves $v + 2\sqrt{h} = 2\sqrt{h_l}$ and has $\xi = v - \sqrt{h}$. Its right edge precedes the shock. Indeed

$$\sigma^2 = \frac{H(H+h_r)}{2h_r} > h_r, \qquad (\sigma - V)^2 = \frac{h_r(H+h_r)}{2H} < H.$$

Thus $\sqrt{h_r} < \sigma < V + \sqrt{H}$, the Lax inequalities for a 2-shock; also $V - \sqrt{H} < \sigma$. The fan is continuous, the shock satisfies both vector jumps, and the $O(t)$ disturbed interval proves the initial trace. For all smooth entropy pairs the shock inequality can also be checked without assuming small amplitude. Hold the left state (H, HV) fixed and follow its 2-shock branch toward smaller h. Parameterize by $s = H - h > 0$. Formula (11.16) gives $\sigma(s) = V + \sqrt{h(h+H)/(2H)}$, strictly decreasing in s. For $E(s) = [Q] - \sigma(s)[\eta]$, differentiated Rankine–Hugoniot gives

$$E'(s) = \sigma'(s)\{D\eta(u_r(s)) \cdot (u_r(s) - u_l) - \eta(u_r(s)) + \eta(u_l)\} \le 0$$

by convexity, and $E(0) = 0$. Thus this is an entropy solution in the system sense, including for large positive depth ratios.

7. Barotropic invariants in the proper variables

For $z = (\rho, q)$, use $r_1 = (1, v-c)$ and $r_2 = (1, v+c)$. The gradients of (11.8) are

$$Dw_1 = ((c-v)/\rho, 1/\rho), \qquad Dw_2 = ((c+v)/\rho, -1/\rho).$$

Their respective products with r_1, r_2 vanish. This verifies the source definition in conserved, rather than merely primitive, variables.

8. Compatibility for the alternative shallow-water system

For $\widetilde{F} = (hv, v^2/2 + h)$ the entropy relation reads $Q_h = v\eta_h + \eta_v$, $Q_v = h\eta_h + v\eta_v$. Equality of their cross derivatives gives $\eta_{vv} = h\eta_{hh}$. This is necessary for a flux; convexity of η is a separate part of being an entropy pair.

水坝问题不必小振幅

局部 Riemann 定理只保证邻近状态。本题通过单调求根、精确跳跃条件及沿激波支的熵耗散，单独处理任意 $h_l > h_r > 0$；不把局部定理扩大到真空或任意大数据。

9. Mechanical entropy for barotropic gas

In conserved variables (ρ, q) put $\eta = q^2/(2\rho) + P(\rho)$ with $P'' = p'/\rho$. A compatible flux is

$$Q = \frac{q^3}{2\rho^2} + qP'(\rho).$$

Direct differentiation gives $DQ = D\eta DF$. If the affine freedom in P is chosen so $\rho P' - P = p$, this is $Q = v(\eta + p)$; without that normalization these expressions differ by a constant multiple of the mass flux q and one must use the compatible expression above. In fact $\rho P' - P - p$ is a constant, so choosing the additive constant in P removes it. The Hessian quadratic form is

$$D^2\eta[(a,b),(a,b)] = \frac{p'(\rho)}{\rho}a^2 + \frac{1}{\rho}(b - va)^2 > 0 \quad ((a,b) \neq 0).$$

Convexity here is in (ρ, q), not asserted in (ρ, v).

10. Gradient flux

For $F = D\phi$ take $\eta = |z|^2/2$ and $Q = z \cdot D\phi - \phi$. The derivative of $z \cdot D\phi$ is $D\phi + z^T D^2\phi$; cancellation leaves $DQ = z^T D^2\phi = D\eta DF$. The Hessian of η is I.

11. Antigradient flux and the generalized elastic energy

For $F = (\phi_2, \phi_1)$ and convex ϕ, $\eta = \phi$, $Q = \phi_1\phi_2$. Componentwise, $D\eta DF = (\phi_1\phi_{12} + \phi_2\phi_{11}, \phi_1\phi_{22} + \phi_2\phi_{12}) = DQ$. For Problem 1 set $P' = p$, $R' = q$, and use

$$\eta(a,b) = P(a) + R(b), \qquad Q(a,b) = -p(a)q(b).$$

Then $D^2\eta = \operatorname{diag}(p', q') > 0$ and the compatibility identity follows directly. This energy is obtained by calculation; the flux itself need not come from a convex antigradient potential with the same sign as in the first part.

12. Several space dimensions

For $F = (F_1, \dots, F_n)$ with $F_j : \Omega \to \mathbb{R}^m$, an entropy pair is a convex η and $Q = (Q_1, \dots, Q_n)$ such that $DQ_j = D\eta DF_j$ for each j. Smooth solutions obey $\eta(u)_t + \operatorname{div} Q(u) = 0$; the entropy inequality for weak solutions is tested against every nonnegative compactly supported scalar space-time function.

13. Nonlinear dielectric Maxwell energy

With $B_t = -\operatorname{curl} E$, $D_t = \operatorname{curl} H$, $E = D_D\eta$, $H = D_B\eta$, the chain rule gives

$$\eta_t = H \cdot (-\operatorname{curl} E) + E \cdot \operatorname{curl} H = -\operatorname{div}(E \times H).$$

The identity $\operatorname{div}(E \times H) = H \cdot \operatorname{curl} E - E \cdot \operatorname{curl} H$ fixes the sign. To verify the algebraic entropy compatibility, the jth fluxes for the state (B, D) have components $F_j^{B_i} = \epsilon_{ijk}E_k$ and $F_j^{D_i} = -\epsilon_{ijk}H_k$. Hence $D\eta\, dF_j = H_i\epsilon_{ijk}dE_k - E_i\epsilon_{ijk}dH_k = d(E \times H)_j$ (summation over i, k). Convexity of the given energy supplies the entropy condition.

14. The entropy chord criterion for a scalar shock

Let $a = u_l$, $b = u_r$, $a \neq b$, and $\sigma = (f(b) - f(a))/(b - a)$. Write $\ell(k) = f(a) + \sigma(k - a)$. For $a < k < b$, the Kruzhkov jump defect is

$$[q_k] - \sigma[\eta_k] = 2(\ell(k) - f(k)).$$

For $b < k < a$, the defect is $2(f(k) - \ell(k))$. Outside the interval it is zero by Rankine–Hugoniot. Thus for an increasing jump entropy requires $f \geq \ell$ on $[a, b]$; for a decreasing jump it requires $f \leq \ell$ on $[b, a]$. This also suffices for every smooth convex entropy: on a bounded interval any such entropy is an affine function plus a positive superposition of $|u - k|$, with weights $\eta''(k)/2$; integrate the corresponding jump inequalities. The affine part contributes zero by the vector jump identity. Diagram T11.4 shows the increasing-jump geometry. A uniformly convex f lies strictly below every nontrivial chord, so only decreasing shocks are allowed.

15. Exact cubic-flux Riemann solution

For $f(u) = u^3 - u$, $a = -1$, seek a shock to a positive state b whose speed equals the left edge of a rarefaction:

$$\frac{f(b) - f(-1)}{b + 1} = b^2 - b = 3b^2 - 1 = f'(b).$$

The nontrivial root is $b = 1/2$ and $\sigma = -1/4$. The entropy solution is

$$u(x,t) = \begin{cases} -1, & x/t < -1/4, \\ \sqrt{(x/t + 1)/3}, & -1/4 < x/t < 2, \\ 1, & x/t \geq 2. \end{cases}$$

On $[-1, 1/2]$, $f(u) - \ell(u) = (u + 1)(u - 1/2)^2 \geq 0$. The shock satisfies the chord criterion and the fan is continuous at $x/t = 2$. Rankine–Hugoniot and the $O(t)$ trace argument prove the weak initial problem. The local scalar contraction theorem proves uniqueness in the bounded entropy class with strong local traces; these Riemann states are not in $L^1(\mathbb{R})$.

16. The perturbed sonic shock is curved

Take $0 < \lambda < 1/4$. Label left characteristics by $\alpha < 0$:

$$a(\alpha) = \begin{cases} -1 + \lambda, & \alpha \le -1, \\ -1 - \lambda\alpha, & -1 < \alpha < 0, \end{cases} \qquad x = \alpha + (3a(\alpha)^2 - 1)t.$$

Their Jacobian in the sloping part is $1 - 6\lambda a t > 0$, so this left region has no characteristic crossings. The initial shock from Problem 15 now encounters increasing left states. Let its position be $s(t)$, its left trace $a(t) < 0$, and its right trace $b(t) = -a(t)/2$. Rankine–Hugoniot and tangency give

$$s' = 3a^2/4 - 1 = f'(b), \qquad s = \alpha + (3a^2 - 1)t, \qquad a = -1 - \lambda\alpha. \tag{11.18}$$

Differentiating the second equation and substituting the first yields

$$\alpha' = -\frac{9a^2}{4(1 - 6\lambda a t)}, \qquad a' = \frac{9\lambda a^2}{4(1 - 6\lambda a t)} > 0.$$

Thus $s'' = (3/2)aa' < 0$: the shock bends to the left relative to its initial line. An explicit parameterization until it reaches the left plateau is, with $r = -a \in [1 - \lambda, 1]$,

$$t(r) = \frac{4}{15\lambda}(r^{-8/3} - r^{-1}), \quad \alpha(r) = \frac{r - 1}{\lambda}, \quad s(r) = \alpha(r) + (3r^2 - 1)t(r).$$

Indeed the linear ODE for $t(a)$ has integrating factor $(-a)^{8/3}$. The plateau is reached at $t_0 = t(1 - \lambda)$; after that $a = -1 + \lambda$ and the shock continues linearly with speed $3(1 - \lambda)^2/4 - 1$, matching its previous position and tangent.

The right-hand fan is no longer entirely centered at the origin. For each emission time $0 \le \theta \le \min(t, t_0)$, the tangent ray

$$x = s(\theta) + s'(\theta)(t - \theta), \qquad u = b(\theta)$$

fills the region from the shock to $x = -t/4$. Since $\partial_\theta x = s''(\theta)(t - \theta) < 0$ before t_0, these rays do not cross in their domain and define the new fan uniquely. For $-t/4 < x < 2t$ retain the original centered fan of Problem 15; for $x > 2t$ retain $u = 1$. On the left use the noncrossing initial characteristics up to the shock. The match at $x = -t/4$ has $u = 1/2$. Each shock jump $a \to -a/2$ obeys $f(u) - \ell(u) = (u - a)(u + a/2)^2 \ge 0$ on that interval, so it is admissible even though the fan derivative can be unbounded on the sonic boundary. The solution is continuous away from the shock, piecewise smooth away from fan edges, and its bounded fluxes permit exhaustion by such regions in the weak identity. Equivalently one

may use the scalar characteristic construction and its continuous fan limit. The initial transition region is bounded with an $O(t)$ perturbation near the origin, yielding the required local trace. For a quantitative early-time check,

$$s(t) = -\tfrac{1}{4}t - \tfrac{27}{16}\lambda t^2 + O(t^3).$$

Keeping the old fan $\sqrt{(x/t+1)/3}$ directly against a curved shock would violate the entropy tangency condition. The emitted rays are an essential part of the perturbation.

17. Smooth scalar contraction in several dimensions

Let $w = u - \hat{u}$ and define the smooth vector field $b = \int_0^1 F'(\hat{u} + \theta w)\, d\theta$. Then $w_t + \operatorname{div}(bw) = 0$. To avoid a discontinuous terminal adjoint, use $\eta_\delta(w) = \sqrt{w^2 + \delta^2} - \delta$. The chain rule gives

$$\partial_t \eta_\delta(w) + \operatorname{div}(b\eta_\delta(w)) = \{\eta_\delta(w) - w\eta'_\delta(w)\} \operatorname{div} b.$$

The expression in braces has absolute value at most δ and is zero where $w = 0$. On a finite time strip with common compact spatial support and smooth bounded derivatives, integrate and use $\delta \int |\operatorname{div} b|$ on that compact set as an error. It tends to zero. Dominated convergence gives $\|w(t)\|_1 = \|w(0)\|_1$, which is stronger than contraction as long as both solutions remain smooth. Uniform compact support on a finite strip follows from bounded speeds and initially compact support in the smooth regime, or may be assumed explicitly in this calculation. After shocks form the smooth renormalization argument no longer applies, and only entropy contraction remains available.

11.6 References

Source attributions

Evans states that T.-P. Liu wrote the initial chapter draft and cites Liu [Lu] as a major reference. The local Riemann construction is attributed to Lax, Comm. Pure Appl. Math. 10 (1957), 537–566. The scalar entropy uniqueness theorem is attributed to S. Kruzhkov, Mat. Sbornik 123 (1970), 228–255, using the source's bibliographic numbering. Further sources listed there include Bressan [Bs], Courant–Friedrichs [C-F], Dafermos [D], Majda [Mj], Xiao–Zhang [X-Z], Zheng [Zh], Logan [Lo], Neu [Ne1], and Smoller [S]. The source acknowledges P. Colella for calculations and W. Han for the eigenvector-dependence proof. Problem 3 is credited to Kato [K]; Problems 4–6 to Holden–Risebro [H-R]; Problems 10–11 to Smoller; Problem 13 to Dafermos. These are attributions read in Evans, not a claim of separate consultation of all the cited works.

Proof dependencies and boundaries

The proofs use inverse/implicit function theorems, smooth ODE theory, spectral algebra, dominated convergence, L^1 translation continuity, and cutoff approximation. Global eigenframes are constructed on $\mathbb{R}^m$; on other state domains the wave geometry is local. The Riemann solver has local amplitude and small-strength uniqueness. The blow-up proof requires a chart covering the entire compact invariant rectangle, a uniform spectral gap, and positive uniform nonlinearity coefficients.

The source refers general viscous-profile sufficiency to Majda–Pego, J. Diff. Eq. 56 (1985), 229–262. Our local proof supplies the scalar reduction but invokes an external center-manifold theorem, which is not proved here. It does not cover distant excursions. The general non-genuinely nonlinear Liu/profile result, referred to Conlon, Adv. Math. 35 (1980), 1–18, remains unproved here. The partially viscous elastic model is proved directly by a scalar ODE.

The viscous limit assumes bounded smooth approximants and a.e. convergence; general-system compactness is not proved. Scalar contraction includes strong initial traces and a local version for Riemann data. General system existence/uniqueness, Glimm/front-tracking theory, compensated compactness, vacuum, multidimensional systems, and nonsmooth system-entropy approximation remain outside the proofs. The 17 solutions and five diagrams are editorial additions. Source comparison, self-check, compilation, and visual QA are distinct from independent mathematical review, which has not been performed.

Chapter 12

Nonlinear Wave Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 12; printed p. 617; PDF page 634.

Chinese reference: 中译本对应章：非线性波动方程，书内第 437 页起。

Editorial reconstruction. We develop energy, causality, local solvability, defocusing global existence, and focusing blow-up. The exposition, expanded arguments, Chinese mathematical notes, problem solutions, and diagrams are reconstructed study material. Source identifiers are recorded separately from our numbering. In particular, a weak energy inequality is not smooth energy conservation, and a noninteger power does not automatically give a smooth nonlinearity. The critical proof states its external linear estimate explicitly.

12.1 Introduction

12.1.1 The equation and its conserved energy

Write $\Box = \partial_t^2 - \Delta_x$. The main equation is

$$\Box u + f(u) = 0, \qquad u(0) = g, \quad u_t(0) = h \quad \text{on } \mathbb{R}^n. \tag{12.1}$$

We also study $\Box u + N(Du, u_t, u) = 0$, with $N(0,0,0) = 0$. Its highest-order part is still the constant wave operator. Although Evans calls it mildly quasilinear, its nonlinearity involves only lower derivatives; equations with solution-dependent principal coefficients require additional analysis. Throughout $D = D_x$. For a scalar f set $F(z) = \int_0^z f(a)\,da$, so $F(0) = 0$.

Theorem 12.1 (Energy; source §12.1.1, Theorem 1)

For a smooth solution with support in one common compact spatial set on each finite time strip, or sufficient decay to justify the same integrations,

$$E(t) = \int_{\mathbb{R}^n} \left\{ \frac{1}{2}(u_t^2 + |Du|^2) + F(u) \right\} dx = E(0). \tag{12.2}$$

非线性项的符号

本章把 f 放在左侧。$f(u) = |u|^{p-1}u$ 给非负势能，$f(u) = -|u|^{p-1}u$ 给负势能。后者才可能满足负能量爆破条件；不能只比较幂次而忽略符号。

Expanded proof (local identity before integration).

Put $e = \frac{1}{2}(u_t^2 + |Du|^2) + F(u)$. The product rule gives

$$e_t - \operatorname{div}(u_t Du) = u_t(\Box u + f(u)) = 0.$$

Integrate on a ball containing the support over a fixed time strip; its boundary term is zero. Under decay hypotheses use cutoffs with radius tending to infinity and check that $u_t Du$ is integrable. This also explains why control of both kinetic and gradient energy is sufficient to eliminate the flux at infinity. For weak solutions the product $u_t \Box u$ may be undefined; conservation must then be established by approximation, not by this formal multiplication. □

12.1.2 Backward cones and the exact flux

For apex (x_0, T) put

$$\begin{aligned} B_t &= B(x_0, T-t), \\ K_{s,\tau} &= \{(x,t) : s < t < \tau,\ x \in B_t\}, \\ \Gamma_{s,\tau} &= \{(x,t) : s < t < \tau,\ |x - x_0| = T - t\}. \end{aligned}$$

On the mantle let $\nu = (x - x_0)/|x - x_0|$. Write $E_K(t) = \int_{B_t} e(x,t)\,dx$.

Theorem 12.2 (Cone flux; source §12.1.2, Theorem 2)
For a solution smooth up to every truncated cone,

$$E_K(s) - E_K(\tau) = \frac{1}{\sqrt{2}} \int_{\Gamma_{s,\tau}} \left\{ \tfrac{1}{2} |u_t \nu - Du|^2 + F(u) \right\} dS. \tag{12.3}$$

If $F \geq 0$, the local cone energy is nonincreasing. If its initial section has $g = h = 0$, then $u = 0$ in that cone.

Expanded proof (moving boundary and the surface factor).

Reynolds' formula for the ball shrinking at unit speed gives

$$E_K'(t) = \int_{\partial B_t} (u_t \partial_\nu u - e)\, dS_x = -\int_{\partial B_t} \{ \tfrac{1}{2} |u_t \nu - Du|^2 + F(u) \}\, dS_x.$$

The square is $u_t^2 + |Du|^2 - 2u_t \partial_\nu u$; its tangential part must not be dropped. Integrating in time proves the result, because the space-time mantle measure satisfies $dS = \sqrt{2}\, dS_x dt$. If the initial local energy is zero, positivity forces $u_t = Du = 0$ in every smaller section. Integrating along vertical segments from the zero initial data gives $u = 0$ there, even if F has other zeroes. When u is smooth at the apex, $E_K(\tau) \to 0$ as $\tau \uparrow T$. We retain this term whenever the apex might be singular: smoothness only for $t < T$ does not justify dropping it. □

锥面上的两种测度

锥面 $t = T - |x - x_0|$ 的诱导测度为 $\sqrt{2}\,dx$，分时切片则为 $\sqrt{2}\,dS_x dt$。因此式中的 $1/\sqrt{2}$ 是几何因子。临界点可能奇异时，顶端剩余能量不能预先设为零。

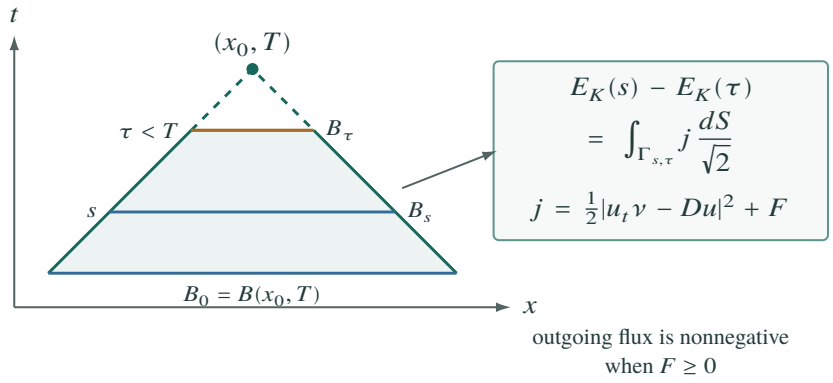


Diagram T12.1. The shrinking sections and the null flux A spatial section of the backward unit-speed cone has radius $T-t$. Energy leaves through the mantle with density $\frac{1}{2}|u_t \nu - Du|^2 + F(u)$, measured using $dS/\sqrt{2}$. The top section at $\tau < T$ is retained. The drawing is a one-space-dimensional section of the n-dimensional cone, not a reduction of the PDE.

12.1.3 Finite propagation without a positive potential

Theorem 12.3 (Domain of dependence; source §12.1.2, Theorem 3)
Let N be locally Lipschitz, $N(0) = 0$, and let two classical solutions of $\Box u + N(Du, u_t, u) = 0$ have bounded first-order state on a compact backward cone. If their two initial data agree on its base, they agree inside the cone. In particular compactly supported data yield support in $B(0, R + t)$ while a classical solution exists.

Expanded proof (difference energy and local Lipschitz constants).

For $w = u - v$, write $Z_u = (Du, u_t, u)$ and define Z_v similarly. The local Lipschitz bound is

$$|N(Z_u) - N(Z_v)| \leq L(|Dw| + |w_t| + |w|),$$

where L depends only on a bound for the two states in this cone. Add $w^2/2$ to the usual wave energy and integrate on its shrinking sections. The lateral boundary contribution is $-\frac{1}{2}|w_t \nu - Dw|^2 - w^2/2 \leq 0$. The interior contribution is $w_t\{w - (N(u) - N(v))\}$ and is at most $C_L(w_t^2 + |Dw|^2 + w^2)$. Thus $Y'(t) \leq C_L Y(t)$ and $Y(0) = 0$; Gronwall gives $Y = 0$. Comparison with zero uses $N(0) = 0$. For energy solutions with globally Lipschitz N, the same inequality is obtained by time/spatial regularization and smooth cone cutoffs, then passage to the limit in L^2. Thus it is also available for the local-data gluing in the next section. A cone is characteristic for the fixed principal wave operator; adding lower-order derivative terms does not change its speed. □

12.2 Existence of Solutions

12.2.1 The linear estimate and the energy solution class

For an integer $k \geq 1$, write

$$\|(u, u_t)\|_{\mathscr{E}_k(I)} = \sup_{t \in I} \{\|u(t)\|_{H^k} + \|u_t(t)\|_{H^{k-1}}\}.$$

The inhomogeneous H^k norm includes the low-frequency L^2 term. We use the whole-space linear wave construction from Chapters 2 and 7: Fourier multipliers $\cos(t|D|)$ and $\sin(t|D|)/|D|$ give, for $u_{tt} - \Delta u = G$,

$$\|(u, u_t)\|_{\mathscr{E}_k([0,T])} \leq C_T \left(\|g\|_{H^k} + \|h\|_{H^{k-1}} + \|G\|_{L^1(0,T;H^{k-1})} \right). \tag{12.4}$$

Here C_T is bounded for $T \leq 1$.

Expanded proof (the energy estimate with its missing low mode).

For smooth data apply D^α, $|\alpha| \leq k-1$, and multiply by $D^\alpha u_t$. The square root of $\|D^\alpha u_t\|_2^2 + \|DD^\alpha u\|_2^2$ grows at most by $\|D^\alpha G\|_2$. Replace this square root by $\sqrt{\varepsilon + \cdots}$ if it vanishes, then let $\varepsilon \downarrow 0$. The remaining $\|u(t)\|_2$ is bounded by $\|g\|_2 + \int_0^t \|u_t(s)\|_2 ds$. Summing gives (12.4); density gives a solution in $C_t H^k \cap C_t^1 H^{k-1}$. Duhamel's formula gives continuity also for $G \in L_t^1 H^{k-1}$. Thus no Poincare constant depending on an expanding spatial ball is needed. □

An energy solution below has $u \in C([0,T]; H^1)$, $u_t \in C([0,T]; L^2)$ and satisfies the PDE in distributions. If $N(Du, u_t, u) \in L^2_{\mathrm{loc}}$, then $u_{tt} \in L^2_{\mathrm{loc}}(0,T; H^{-1}_{\mathrm{loc}})$ and the usual weak formulation against compactly supported H^1 tests follows. The two time traces specify g, h separately.

12.2.2 Globally Lipschitz nonlinearities

Theorem 12.4 (Source §12.2.1, Theorem 1)
Assume $N : \mathbb{R}^{n+2} \to \mathbb{R}$ is globally Lipschitz, $N(0) = 0$. For $g \in H^1$, $h \in L^2$ there is a unique global energy solution of $\Box u + N(Du, u_t, u) = 0$. For $g \in H^2$, $h \in H^1$ it also has $u \in L^\infty(0,T;H^2)$, $u_t \in L^\infty(0,T;H^1)$, $u_{tt} \in L^\infty(0,T;L^2)$ on every finite strip. With $H^1_{\mathrm{loc}}, L^2_{\mathrm{loc}}$ data the corresponding statements hold locally in space, with uniqueness in the local energy class.

Expanded proof (contraction, restart, and local gluing).

On the complete space $C_t H^1 \cap C_t^1 L^2$ with the prescribed initial traces, define $A[v]$ by the linear equation with forcing $-N(Dv, v_t, v)$.

Since $N(0) = 0$, this forcing belongs to $C_t L^2$. For two inputs, (12.4) gives

$$\|A[v] - A[w]\|_{\mathscr{E}_1} \le C_1 LT \|v - w\|_{\mathscr{E}_1}.$$

The homogeneous solution makes the affine space nonempty. Choose $T \le 1$ so $C_1 LT < 1/2$ and apply Banach's theorem. The same step size works for every new initial state, so repeated restart constructs a solution on every finite interval. Its norm is finite there, for example by the integral estimate and Gronwall; no finite accumulation of restart times occurs.

For the additional derivative, apply the spatial difference quotient δ_ε to the PDE. Pointwise Lipschitz continuity gives

$$|\delta_\varepsilon N(Du, u_t, u)| \le L(|D\delta_\varepsilon u| + |\delta_\varepsilon u_t| + |\delta_\varepsilon u|).$$

The difference energy and Gronwall bound the quotient in $L^\infty_t H^1$ and its time derivative in $L^\infty_t L^2$, uniformly in ε, by the $H^2 \times H^1$ initial norm. Weak compactness and the difference-quotient characterization give $D^2 u, Du_t \in L^\infty_t L^2$; the equation then gives the asserted bound for u_{tt}. Differentiability of a merely Lipschitz N is not needed for this argument.

For local data, cut them off outside $B(0, j)$ and solve the global problem. Two cutoff solutions coincide in the backward domains where their data agree, by the regularized difference-cone estimate from §12.1. For every fixed compact space-time cylinder they therefore agree for all sufficiently large j. Their common values define a solution independent of the cutoff. The same cone argument proves uniqueness and localizes the higher-regularity estimates. □

12.2.3 Sobolev products and composition

Theorem 12.5 (Source §12.2.2, Theorem 2; Problem 12)
Let $r > n/2$ be an integer, $u_1, \dots, u_m \in H^r(\mathbb{R}^n)$. For multiindices with $\sum_j |\beta_j| \le r$,

$$\left\| \prod_j D^{\beta_j} u_j \right\|_2 \le C_{n,r,m} \prod_j \|u_j\|_{H^r}. \tag{12.5}$$

If $P : \mathbb{R}^m \to \mathbb{R}$ is smooth and $P(0) = 0$, then $P(u) \in H^r$, with a bound depending on the H^r norms and derivatives of P on the resulting bounded range.

Expanded proof (including the exponent allocation).

Put $b_j = |\beta_j|$. Sobolev embedding gives $D^{\beta_j} u_j \in L^{2r/b_j}$ for $0 < b_j \le r$ and $u_j \in L^\infty$ for $b_j = 0$. Indeed for $b_j < r$ the choice $1/p = b_j/(2r)$ lies

原点条件与全空间

若 $N(0) \neq 0$，零状态也会受到常量强迫，$N(Dv, v_t, v)$ 不一定在 $L^2(\mathbb{R}^n)$。$N(0) = 0$ 同时保证零解、有限传播中的比较基准与全空间能量构造。

strictly above $1/2 - (r - b_j)/n$, since $2r > n$; $b_j = r$ gives $p = 2$. Each derivative also lies in L^2. By interpolation choose $1/p_j \in [b_j/(2r), 1/2]$ so that $\sum_j 1/p_j = 1/2$. This is possible because the sum of the lower endpoints is at most $1/2$ and that of the upper endpoints is at least $1/2$. Holder's inequality proves (12.5). On the infinite measure space one must make this sum exact, rather than infer an L^2 estimate from a product lying only in a larger L^p space.

For $0 < |\alpha| \le r$, the chain rule expresses $D^\alpha P(u)$ as a finite sum of bounded coefficients $D^j P(u)$ times products of derivatives of the u_i with total order $|\alpha|$. Apply (12.5). For the undifferentiated term use $P(0) = 0$ and $|P(u)| \le C|u|$ on the bounded range. Approximation by smooth functions proves the Sobolev chain rule and the assertion for all H^r inputs. The same estimates applied to $P(u) - P(v) = \int_0^1 DP(v + \theta(u - v))(u - v)\,d\theta$ show local Lipschitz dependence in H^r; subtract the constant $DP(0)$ before using the algebra estimate on its coefficient. □

12.2.4 Local existence and the role of the two norms

Theorem 12.6 (Source §12.2.2, Theorem 3)
Assume N is smooth with $N(0) = 0$, and $k > n/2 + 1$ is an integer. For $g \in H^k$, $h \in H^{k-1}$ there are $T > 0$ and a unique solution

$$u \in C([0,T]; H^k), \qquad u_t \in C([0,T]; H^{k-1}).$$

The lifetime depends on these norms and the nonlinearity on a bounded state range. Classical C^2 regularity follows if $k > n/2 + 2$.

Expanded proof (closed strong ball with the weaker metric).

Fix $T \le 1$ and let Y_R consist of pairs (v, v_t) continuous in $H^1 \times L^2$, with the prescribed traces, and with $\|(v, v_t)\|_{\mathscr{E}_k} \le R$ in the essential-supremum sense. Use the $\mathscr{E}_1$ metric. This space is complete: a Cauchy sequence has a strong low-norm limit; the high-norm bounded sequence has a weak-star subsequence in $L^\infty(H^k \times H^{k-1})$. Both limits agree as distributions. Weak lower semicontinuity gives the bound R, and low-norm convergence preserves the derivative relation and both initial traces. The uniform high bound can also be assigned at each time by weak limits of nearby good time slices.

For $A[v]$ as above, composition at order $k - 1 > n/2$ gives

$$\|N(Dv, v_t, v)\|_{H^{k-1}} \le \Phi(R).$$

Take $R > 2C_1(\|g\|_{H^k} + \|h\|_{H^{k-1}}) + 1$. The free wave belongs to Y_R, and

$$\|A[v]\|_{\mathscr{E}_k} \le C_1 D_0 + C_1 T \Phi(R) \le R$$

两个范数各做什么

高阶界使 (Du, u_t, u) 的值域有界，从而给出 N 的局部 Lipschitz 常数。低阶能量范数用于收缩。只说迭代在弱范数收敛，还需证明强范数球在该度量下闭合。

for small T. Sobolev embedding bounds all first-order states by CR, so the local Lipschitz constant L_R gives $\|A[v]-A[w]\|_{\mathscr{E}_1} \le C_1 T L_R \|v-w\|_{\mathscr{E}_1}$. Shorten T to make this less than one half of the input distance. Banach's theorem gives a fixed point in the closed space Y_R. The resulting forcing is in $L^1_t H^{k-1}$; linear uniqueness and (12.4) then upgrade the solution to the asserted strong time continuity. Differences of two bounded high-regularity solutions obey the same low-energy Gronwall estimate, proving uniqueness without restricting them to one preselected ball. If $k > n/2+2$, D^2u, Du_t, u_{tt} are continuous by embedding and the PDE. □

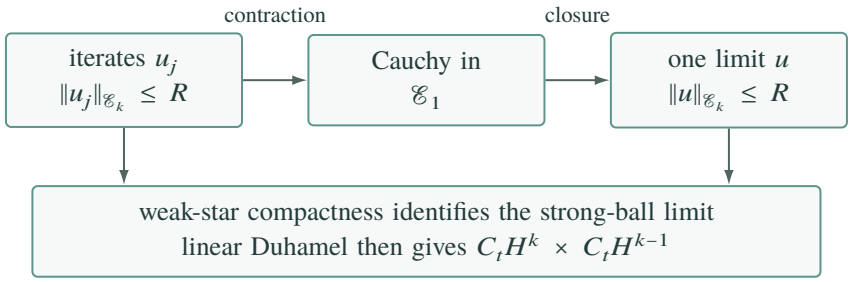


Diagram T12.2. A complete ball for the weak-metric iteration Every iterate stays in a bounded high Sobolev ball. Contraction is measured in the lower energy norm; weak-star compactness identifies its limit with the low-norm limit, preserving the high bound. The final linear equation restores strong time continuity. The arrows describe a proof, not a numerical convergence rate in the high norm.

12.3 Semilinear Wave Equations

12.3.1 A sign condition gives weak existence

Assume f is smooth and $zf(z) \ge 0$. Then $F \ge 0$. The resulting energy controls Du, u_t, but it does not directly control $\|u\|_2$ uniformly for all positive time.

Theorem 12.7 (Sign condition; source §12.3.1, Theorem 1)
If $g \in H^1$, $h \in L^2$ and $F(g) \in L^1$, there is a global distributional solution with $f(u) \in L^1_{\mathrm{loc}}$ and

$$\|Du(t)\|_2^2 + \|u_t(t)\|_2^2 + 2\int F(u(t)) \le 2E(0).$$

One can choose $u \in C([0,\infty); L^2)$ with $\|u(t)\|_2 \le \|g\|_2 + t\sqrt{2E(0)}$, and Du, u_t weakly continuous into L^2. The energy inequality holds at every time for these representatives. The initial energy traces are strong. No uniqueness or energy equality is asserted in this general class.

能量不控制全时间质量

这里 F 可为零，能量只控制 Du, u_t。由 $u(t) = g + \int_0^t u_t$ 得到的是 $\|u(t)\|_2 \le \|g\|_2 + t\sqrt{2E(0)}$。因此一般只能写每个有限时间条带上的 $L^\infty_t H^1_x$ 界。

Expanded proof (truncation, uniform integrability, and time slices).

Choose even smooth cutoffs $0 \le \chi_j \le 1$, equal to one on $[-j,j]$ and zero outside $[-2j,2j]$. Put $f_j = \chi_j f$ and $F_j(z) = \int_0^z f_j(a)da$. Then f_j is globally Lipschitz, $zf_j(z) \ge 0$, $0 \le F_j \le F$, and $F_j = F$ on $[-j,j]$. The previous section gives an energy solution u_j. Conservation for this Lipschitz problem follows by approximating the data with smooth data in $H^1 \times L^2$ and passing through the linear energy estimate; $F_j(z) \le C_j z^2$ makes the potential continuous in L^1 under L^2 convergence on bounded energy sets. Consequently $E_j(t) = E_j(0) \le E(0)$ and

$$\|Du_j(t)\|_2 + \|(u_j)_t(t)\|_2 \le C_E, \qquad \|u_j(t)\|_2 \le \|g\|_2 + tC_E.$$

Dominated convergence also gives $E_j(0) \to E(0)$. Rellich compactness on bounded balls and the equicontinuity $\|u_j(t) - u_j(s)\|_2 \le C_E|t-s|$ give, by a diagonal Arzela–Ascoli argument, $u_j \to u$ in $C([0,T]; L^2_{\mathrm{loc}})$ for every finite T, and a.e. in space-time along a subsequence. The first derivatives converge weak-star in $L^\infty_t L^2_x$.

Multiplying the approximate PDE by u_j and integrating yields

$$\int_0^T \int u_j f_j(u_j) = \int_0^T \int ((u_j)_t^2 - |Du_j|^2) - \Big[\int u_j (u_j)_t\Big]_0^T \le C_{E,g,T}.$$

All terms are justified first for smooth data and then by the same energy approximation. The integrand on the left is nonnegative. For a measurable subset A of a compact space-time cylinder and $M > 0$,

$$\int_A |f_j(u_j)| \le C_{E,g,T}/M + |A| \max_{|z|\le M} |f(z)|.$$

The first term comes from $|u_j| > M$; the second uses $|f_j| \le |f|$. This is uniform integrability. Also $f_j(u_j) \to f(u)$ a.e., because at any point with finite limit u, eventually $|u_j| < j$. Egoroff on the compact cylinder, followed by the preceding small-set bound, proves $f_j(u_j) \to f(u)$ in L^1 there. This is a space-time L^1_{loc} statement, not a claim of convergence at every fixed time. The weak equation now passes to the limit. For a smooth test reaching $t = 0$ its first-order form is

$$\iint \{-u_t\varphi_t + Du \cdot D\varphi + f(u)\varphi\} = \int h\varphi(\cdot,0), \qquad u(0) = g.$$

To obtain time representatives, for each compactly supported smooth ψ use $\frac{d}{dt}\int (u_j)_t\psi = \int u_j\Delta\psi - f_j(u_j)\psi$. Uniform integrability makes these scalar functions equicontinuous, including at zero. Diagonal extraction for a dense collection of ψ, and the uniform L^2 bounds, give weak L^2 convergence of the velocities at every time and weak time continuity of u_t. The gradients converge weakly at every time by the strong local convergence of u_j. At a fixed time, extract a further subsequence with a.e. spatial

convergence on expanding balls. Fatou for $F_j(u_j)$ and weak lower semicontinuity of the two quadratic terms give $E(t) \le E(0)$, independently of the chosen subsequence. Thus this inequality holds for the chosen representatives at all times. The identity $u(t) = g + \int_0^t u_t(s)ds$ gives the asserted global L^2 continuity and growth bound. As $t \downarrow 0$, lower semicontinuity of the potential and weak convergence of (Du, u_t), together with $E(t) \le E(0)$, force convergence of their squared norms; weak convergence plus convergence of Hilbert norms is strong. This proves both strong initial energy traces. A diagonal extraction in $T = 1, 2, \dots$ completes global existence. □

12.3.2 Three-dimensional retarded potentials and continuation

For smooth compactly supported g, h let v be their free wave. In three dimensions Duhamel's formula is

$$u(x,t) = v(x,t) - \frac{1}{4\pi} \int_{B(x,t)} \frac{f(u(y, t-|y-x|))}{|y-x|}\, dy. \tag{12.6}$$

Theorem 12.8 (Source §12.3.2, Theorem 2)
If f is smooth, $f(0) = 0$, this problem has a unique local smooth solution. If its maximal smooth lifetime $T_* < \infty$, then

$$\sup_{0 \le t < T_*} \|u(t)\|_\infty = \infty.$$

Equivalently, a bound for u on a finite strip permits continuation. We use this strip-bound criterion rather than infer a pointwise limit from the mere failure of a uniform bound.

Expanded proof (the kernel norm and derivative persistence).

On $0 \le t \le T \le 1$, take the closed sup-norm ball $\|u - v\|_\infty \le 1$ of bounded continuous functions with initial trace g. Its range is in $[-M, M]$ for a fixed M. The right side of (12.6) defines a map with

$$\|A[u] - A[w]\|_\infty \le \tfrac{1}{2} L_M T^2 \|u - w\|_\infty, \qquad \|A[u] - v\|_\infty \le \tfrac{1}{2} T^2 \max_{[-M,M]} |f|,$$

because $(4\pi)^{-1} \int_{B(0,T)} |y|^{-1} dy = T^2/2$. It is a contraction of the ball for small T.

The fixed point is smooth. For instance spatial difference quotients satisfy a linear retarded equation with coefficient $\int_0^1 f'(\theta u(x + \varepsilon e_j, t) + (1-\theta)u(x,t))d\theta$. It is uniformly bounded on a bounded range. The retarded operator has the estimate $\|R[G](t)\|_\infty \le \int_0^t (t-s)\|G(s)\|_\infty ds$. Gronwall therefore bounds the first spatial derivatives. The time derivative satisfies the differentiated wave equation with initial data $(h, \Delta g - f(g))$

and is bounded in the same way. Successive differentiation gives an equation with leading term $f'(u)D^\alpha u$ and forcing made from lower derivatives, already bounded by induction. Difference quotients justify each differentiation and convergence; the corresponding linear retarded equation has a unique continuous solution. This bounds every derivative on a finite strip if u is bounded there. Finite propagation provides a common support ball. Uniform bounds on one further derivative give limiting smooth data at T_*; applying local existence to them extends the solution, a contradiction. Alternatively higher-order energy limits provide the same restart. Uniqueness follows either from contraction on successive short strips or the difference-cone energy estimate. □

12.3.3 Subcritical defocusing powers

Consider $f(u) = |u|^{p-1}u$, $1 < p < 5$, in $n = 3$.

Lemma 12.9 (Cone-trace bounds; expanded source §12.3.3)
For a smooth defocusing solution and $t \le T_0 < \infty$, put $w(y) = u(y, t-|y-x|)$ on $B(x,t)$. Then

$$\|Dw\|_2 + t^{-1}\|w\|_2 + \|w/|y-x|\|_2 + \|w\|_6 \le C(E(0)^{1/2} + T_0^{1/2}\|g\|_\infty). \tag{12.7}$$

Constants are independent of the apex and the cone radius.

Expanded proof (the nonzero boundary value in Hardy's inequality).

The chain rule gives $Dw = Du - \nu u_t$ on the mantle. Thus (12.3) gives $\int |Dw|^2 \le 2E(0)$. Let $\overline{w}$ be the ball average. Integrating $|u_t|$ along vertical lines in the cone gives

$$\int_{B(x,t)} |w| \le \int_{B(x,t)} |g| + \iint_{K_{0,t}} |u_t|.$$

Indeed each vertical segment from time zero to $t - |y-x|$ lies in the cone. Since its space-time volume is Ct^4 and $\iint_K u_t^2 \le 2tE(0)$, $|\overline{w}| \le Ct^{-1/2}E(0)^{1/2} + \|g\|_\infty$. Poincare's inequality yields $t^{-2}\|w\|_2^2 \le C\|Dw\|_2^2 + Ct|\overline{w}|^2$. The scaled ball inequalities are

$$\|w\|_6 + \|w/|y-x|\|_2 \le C(\|Dw\|_2 + t^{-1}\|w\|_2).$$

They follow by scaling the unit-ball Sobolev and Hardy inequalities. For the latter, extend the unit-ball function by a bounded H^1 extension and cutoff to $\mathbb{R}^3$, then apply whole-space Hardy. One cannot apply the zero-trace version directly to this cone trace. These inequalities and the average bound prove the lemma. □

实数幂与光滑性

当 p 不是正奇整数，$|z|^{p-1}z$ 在零点通常不是 C^∞。下面对所有 $1 < p < 5$ 声称有限能量有界 mild 解；只有非线性本身光滑时，才由导数持续性得到光滑解。

Theorem 12.10 (Subcritical existence; source §12.3.3, Theorem 3)
For compactly supported smooth data and $1 < p < 5$, there is a unique global finite-energy mild solution, bounded on each finite time strip, with $u \in C_t H^1$, $u_t \in C_t L^2$ and conserved energy. When the nonlinearity is smooth, the solution is smooth.

Expanded proof (the sublinear bootstrap and the power qualification).

First work in a smooth regime. Formula (12.6) and Cauchy–Schwarz give

$$|u(x,t)| \le \|v\|_\infty + C\|w/|y-x|\|_2 \left(\int_{B(x,t)} |w|^{2(p-1)} dy \right)^{1/2}.$$

For $1 < p \le 4$, the last exponent is at most six. Holder on the ball and (12.7) give a uniform bound for $t \le T_0$; this uses integrals, not an invalid norm triangle inequality when $2(p-1) < 1$. For $4 < p < 5$, put $M(\tau) = \sup_{x, 0 \le t \le \tau} |u(x,t)|$. Then

$$\left(\int |w|^{2(p-1)} \right)^{1/2} \le M(\tau)^{p-4} \|w\|_6^3, \quad M(\tau) \le C_0 + C_1 M(\tau)^{p-4}.$$

Since $0 < p - 4 < 1$, Young's inequality absorbs the last term into $M/2$ plus a constant. Thus M is bounded independently of the current smooth lifespan below T_0. The continuation theorem proves global smooth existence whenever f is smooth.

For an arbitrary real p, approximate f by smooth sign-preserving nonlinearities, cut off at large amplitudes, converging in C^1 on every bounded interval and with $|f_j(z)| \le C(|z| + |z|^p)$. For example smooth $z(z^2 + \varepsilon_j^2)^{(p-1)/2}$ first and then truncate far out. Their nonnegative potentials and compact initial data have uniformly bounded energy. The preceding argument gives a uniform strip bound; the extra linear term is bounded by the same Hardy estimate times the square root of the ball volume. On this bounded range the nonlinearities have one uniform Lipschitz constant. Difference energy estimates, using uniform convergence of f_j on that range and the common finite support ball, show Cauchy convergence in $C_t H^1 \times C_t L^2$. The retarded Volterra difference estimate also gives Cauchy convergence in the supremum norm. The retarded formula passes to the limit, and its sup-norm contraction gives the same mild solution. The potentials converge uniformly on the bounded range, so their integrals converge and energy equality passes to the limit. The difference estimate also proves uniqueness among bounded energy solutions on each finite strip. Smoothness across zeros is not claimed for the noninteger powers. This supplies the regularity qualification absent from a literal all-power smoothness statement. □

12.4 Critical Power Nonlinearity

12.4.1 Scaling and the external linear estimate

In $\mathbb{R}^3$ the defocusing equation

$$\Box u + u^5 = 0, \qquad E = \int \{(u_t^2 + |Du|^2)/2 + u^6/6\} \tag{12.8}$$

is energy critical. For the general power, $u_\lambda(x,t) = \lambda^{2/(p-1)} u(\lambda x, \lambda t)$ solves the same equation, while its energy scales by $\lambda^{4/(p-1)-1}$. At $p = 5$ this factor is one: a small spatial scale cannot be excluded by total energy alone.

Theorem 12.11 (Linear estimate; source §12.4, Theorem 1)
For $\Box v = G$ on a time interval $I = [s,\tau]$ in $\mathbb{R}^3$, assume $v(s) \in \dot H^1$, $v_t(s) \in L^2$, and $G \in L^1(I;L^2)$. Then

$$\begin{aligned} &\|Dv, v_t\|_{L^\infty(I;L^2)} + \|v\|_{L^\infty(I;L^6)} + \|v\|_{L^4(I;L^{12})} \\ &\quad \le C\{\|Dv(s)\|_2 + \|v_t(s)\|_2 + \|G\|_{L^1(I;L^2)}\}. \end{aligned} \tag{12.9}$$

We explicitly use the homogeneous three-dimensional Strichartz estimate for the $(4,12)$ pair as an external theorem*, with constant independent of the interval. It is not proved here.*

Expanded proof (the proved energy part and the external input).

The energy argument of §12.2, without the L^2 position term, gives the first norm on the left. The homogeneous Sobolev inequality $\|v\|_6 \le C\|Dv\|_2$ gives the second. The external homogeneous estimate bounds the third norm for free waves by their $\dot H^1 \times L^2$ data. Apply this to the free evolution starting at each forcing time and use Minkowski in Duhamel's integral. Its $\dot H^1 \times L^2$ size is $\|G(t)\|_2$, giving the forcing term in (12.9). Restriction of the homogeneous estimate on the whole time line explains uniformity under interval shortening. The dispersive proof of that homogeneous estimate remains external; energy conservation alone does not imply an $L_t^4 L_x^{12}$ bound. □

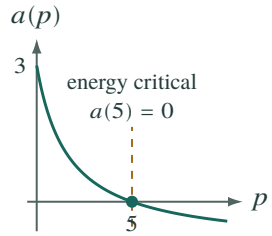
12.4.2 Morawetz identity and potential nonconcentration

Fix an apex (x_0, T) before which u is smooth and has finite conserved energy. Set $x_0 = 0$ by translation and $r = |x|$. The local energy $E_K(t)$ decreases, so it has a limit $E_* \ge 0$ as $t \uparrow T$. Define the tail flux

$$\phi(s) = E_K(s) - E_* = \frac{1}{\sqrt{2}} \int_{\Gamma_{s,T}} \{\tfrac{1}{2}|u_t \nu - Du|^2 + u^6/6\}\, dS. \tag{12.10}$$

Monotone convergence in the truncated flux identity gives this formula and $\phi(s) \to 0$. At this stage $E_* = 0$ is *not* known.

图 T12.M：能量临界的缩放指数



图中 $a(p) = 4/(p-1) - 1$，且 $E[u_\lambda] = \lambda^{a(p)} E[u]$。$p = 5$ 是三维退聚焦波方程的能量临界指数；它与后面聚焦小数据爆破的 $1 + \sqrt{2}$ 阈值属于不同问题。

Lemma 12.12 (Multiplier identity; source equations (6)–(8))
Put $\theta = t - T$, $m = \theta u_t + x \cdot Du + u$ *and*

$$P = \theta\{(u_t^2 + |Du|^2)/2 + u^6/6\} + (x \cdot Du + u)u_t,$$

$$Q = mDu + x\{(u_t^2 - |Du|^2)/2 - u^6/6\}.$$

Then $P_t - \operatorname{div} Q = m(\Box u + u^5) - u^6/3$.

Expanded proof (an identity also valid off solutions).

Use $\theta_t = 1$, $\operatorname{div} x = 3$ and $D(x \cdot Du) = Du + D^2u\,x$. The mixed terms $\theta Du \cdot Du_t$ cancel, as do $x \cdot D(u_t)u_t$ and $x \cdot D^2u\,Du$ after differentiation. The remaining second derivatives form $m(u_{tt} - \Delta u)$. The potential terms are $mu^5 - u^6/3$: their residual coefficient is $1/6 + 3/6 - 1 = -1/3$. The coefficients of u_t^2 and $|Du|^2$ are both zero. This proves the identity without presupposing the PDE. □

先证明通量尾部趋零

正确的关系是 $\phi(s) = E_K(s) - \lim_{t\uparrow T} E_K(t)$。单调性保证尾部趋零，不要求顶端剩余能量预先为零。后面通过线性 Duhamel 的强能量极限再证明全部能量不集中。

Theorem 12.13 (Potential nonconcentration; source §12.4, Theorem 2)
For $R = T - s$,

$$\int_{B(x_0,R)} |u(x,s)|^6 dx \le C\{\phi(s) + \phi(s)^{1/3}\} \longrightarrow 0. \tag{12.11}$$

Since $\phi \le E$, *the right side can also be written* $C_E\phi(s)^{1/3}$, *making its energy dependence explicit.*

Expanded proof (top term, boundary cancellation, and the endpoint).

Translate s to zero, so R replaces T. Integrate the multiplier identity on $K_{0,\tau}$ and let $\tau \uparrow R$. The top integral tends to zero even if the apex is singular: with $\rho = R - \tau$,

$$\left|\int_{B(0,\rho)} P(x,\tau)dx\right| \le C\rho E + C\|u\|_{L^2(B_\rho)}\|u_t\|_2 \le C\rho(E + E^{2/3}).$$

Here Holder gives $\|u\|_{L^2(B_\rho)} \le C\rho\|u\|_6$. Thus, putting $A = \frac{1}{\sqrt{2}}\int_\Gamma (P - Q \cdot \nu)dS$ and $B = \int_{B_R} P(x,0)dx$, we obtain $A - B = -\iint_K u^6/3 \le 0$.

On the mantle $r = R - t$, $P - Q \cdot \nu = -r(u_t - u_r)^2 + u(u_t - u_r)$. Set $w(y) = u(y, R - |y|)$. Then $w_r = u_r - u_t$ and

$$A = -\int_{B_R} r\left(w_r + \frac{w}{r}\right)^2 dy + \frac{1}{2}\int_{\partial B_R} u(x,0)^2 dS_x. \tag{12.12}$$

Indeed $A = -\int (rw_r^2 + ww_r)$ and $\int (w^2/r + ww_r) = \frac{1}{2}\int_{\partial B_R} w^2$, from $\operatorname{div}(y/r) = 2/r$. This calculation is valid for the possibly singular cone trace: flux gives $Dw \in L^2$ and $w \in L^6$, so $w \in H^1(B_R)$ (the point at

zero is removable for H^1 in dimension three). The weighted term w^2/r is integrable by Holder. Approximate in H^1 with the outer smooth trace fixed, or use radial cutoffs and a vanishing inner-boundary subsequence, to justify the integration.

For the bottom term use Young's inequality and $r \le R$:

$$\begin{aligned} B &\le -\frac{R}{6}\int_{B_R} u^6 + \frac{R}{2}\int_{B_R}(|Du + xu/r^2|^2 - |Du|^2) \\ &= -\frac{R}{6}\int_{B_R} u^6 + \frac{1}{2}\int_{\partial B_R} u^2. \end{aligned}$$

The last equality follows from $\operatorname{div}(x/r^2) = 1/r^2$; the origin contributes no boundary term for the smooth bottom data. The boundary terms cancel between this bound and (12.12). Therefore

$$R\int_{B_R} u(x,0)^6 dx \le C\int_{B_R} r(w_r + w/r)^2 \le CR\phi(0) + C\int_{B_R} w^2/r.$$

Finally

$$\int_{B_R} w^2/r \le \left(\int_{B_R} w^6\right)^{1/3}\left(\int_{B_R} r^{-3/2}\right)^{2/3} \le CR\phi(0)^{1/3}.$$

Divide by R and undo the time translation. Every estimate is on truncated smooth cones before a justified limit, so no regularity at the candidate blow-up point has been assumed. □

12.4.3 Localizing Strichartz and closing the global proof

Lemma 12.14 (Localized estimate and the bootstrap barrier)
Let $C = \{s < t < \tau,\ |x - x_0| < R - (t - s)\}$ be a backward cone. For a smooth quintic solution set $X(\tau) = \|u\|_{L^4(s,\tau;L^{12}(C_t))}$. Then

$$X(\tau) \le C_0 + C\sup_{s<t<\tau}\|u(t)\|_{L^6(C_t)} X(\tau)^4, \qquad C_0 \le C\sqrt{E}. \quad (12.13)$$

If the displayed L^6 coefficient is sufficiently small depending only on E, then $X(\tau)$ stays bounded up to the cone apex.

Expanded proof (no unsupported restriction of a global estimate).

Solve the linear equation with data $(u(s), u_t(s))$ on all space and forcing $-\mathbf{1}_C u^5$, zero outside C. Finite propagation for the linear equation implies its solution equals u inside C: the backward cone of any point of C remains inside C down to s. Apply (12.9) to this linear solution. Holder in space and time gives

$$\|u^5\|_{L^1L^2(C)} \le \sup_t \|u(t)\|_{L^6(C_t)}\int_s^\tau \|u(t)\|^4_{L^{12}(C_t)} dt.$$

This proves the estimate using a globally defined forcing and data; no boundary Strichartz theorem is being assumed. Choose $B = 2\max(C_0, 1)$ and require the nonlinear coefficient to be less than $1/(4B^3)$. The continuous function X starts at zero. At a first crossing of B, the inequality would give $B \le C_0 + B/4 \le 3B/4$, impossible. It therefore remains below B. The small coefficient alone is insufficient unless this continuity argument excludes the large branch of $X \le C_0 + \varepsilon X^4$. □

Theorem 12.15 (Global smooth quintic solution; source §12.4, Theorem 3)
For $g, h \in C_c^\infty(\mathbb{R}^3)$, the defocusing quintic problem has a unique global smooth solution. The proof uses the stated external homogeneous Strichartz estimate.

Expanded proof (from potential flux to an actual continuation bound).

Assume a finite maximal time T. For each x_0, potential nonconcentration makes the L^6 coefficient in (12.13) small on a sufficiently short tail of the cone with apex (x_0, T). Thus $X(T) < \infty$ on that cone, and $G = -\mathbf{1}_K u^5 \in L^1(s, T; L^2)$. The linear solution with this forcing and full data at s agrees with u inside the cone, as in the lemma. Because its forcing is integrable in time, Duhamel gives a strong $\dot{H}^1 \times L^2$ limit at T. Strong L^2 convergence of its gradient and velocity, together with absolute continuity of their squared integrals at the limiting time, now implies

$$\int_{B(x_0, T-t)} (|Du|^2 + u_t^2)\,dx \longrightarrow 0.$$

Together with (12.11) this proves $E_K(t) \to 0$. This argument replaces an unproved differentiation estimate on a cone with a characteristic boundary.

Fix x_0. Choose $s < T$ so its local energy is smaller than a chosen $\varepsilon > 0$. At this fixed smooth time, absolute continuity in the radius allows a $\delta > 0$ such that the energy on $B(x_0, T - s + \delta)$ is less than 2ε. Positive cone flux propagates this bound into the enlarged cone $|x - x_0| < T - t + \delta$. Its L^6 norm is at most $(12\varepsilon)^{1/6}$. The localized lemma, still using the full initial energy E for C_0, gives a finite L^4L^{12} norm on this enlarged cone. Choose ε small enough for the fixed barrier.

Finite propagation bounds all supports for $t < T$ by the fixed closed ball $\overline{B}(0, R_0 + T)$. Cover that ball by finitely many of the balls $B(x_j, \delta_j/2)$ just constructed. These balls lie inside the corresponding enlarged sections for every $t < T$. Start after the largest of their finitely many start times, and use the triangle inequality in L^{12} and then L^4. Adding the earlier smooth strip proves $u \in L^4(0, T; L^{12}(\mathbb{R}^3))$. No object $\operatorname{spt} u(\cdot, T)$ is needed before existence at T is known.

Finally $v = D_j u$ solves $\Box v = -5u^4 v$. The homogeneous energy estimate and Sobolev give, for $A_j(t) = \|Dv(t)\|_2 + \|v_t(t)\|_2$,

$$A_j(t) \le C_j + C\int_0^t \|u(s)\|_{12}^4 A_j(s)\,ds, \qquad \|u^4 v\|_2 \le \|u\|_{12}^4 \|v\|_6.$$

Gronwall bounds A_j up to T. Thus Du is bounded in L^6; energy already bounds u in L^6. The embedding $W^{1,6}(\mathbb{R}^3) \hookrightarrow L^\infty$ supplies a uniform amplitude bound. The continuation theorem of §12.3 contradicts maximality. Local uniqueness extends to all time. □

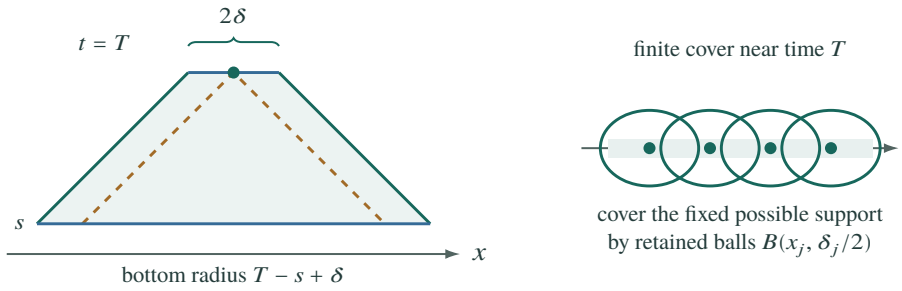


Diagram T12.3. Why the cone must be enlarged The flux tail first makes the potential energy small on shrinking cones. Localized Strichartz and a linear Duhamel limit then exclude concentration of the kinetic and gradient energy. At a fixed time s, the small-energy ball can be enlarged by $\delta > 0$. The enlarged sections retain a ball of radius δ near T, allowing a finite cover of the fixed possible support. A cone with zero terminal radius alone would not provide that cover.

12.5 Nonexistence of Solutions

12.5.1 Negative energy and the concavity method

Consider a classical solution of (12.1) with compactly supported smooth data and sufficient regularity for the following identities. Finite propagation bounds its support on every finite strip. This section allows a negative potential.

Theorem 12.16 (Source §12.5.1, Theorem 1)
Suppose $f(0) = 0$, $F' = f$, $F(0) = 0$, and for some $\lambda > 2$,

$$zf(z) \le \lambda F(z) \quad (z \in \mathbb{R}).$$

If the conserved energy $E(0) < 0$, no such solution can persist for all $t \ge 0$. In particular, the concavity argument forces breakdown in a finite time.

Expanded proof (obtaining a decreasing concave quantity).

Put $I(t) = \frac{1}{2}\int u^2$. Differentiating twice and integrating by parts gives

$$I' = \int u u_t, \qquad I'' = \int (u_t^2 - |Du|^2 - uf(u)).$$

Set $\alpha = (\lambda - 2)/4 > 0$ and $\beta = -\lambda E(0) > 0$. Energy conservation and the structural assumption yield

$$I'' \geq (2 + 2\alpha)\int u_t^2 + 2\alpha \int |Du|^2 + \beta, \qquad (I')^2 \leq 2I\int u_t^2.$$

Therefore $II'' - (1+\alpha)(I')^2 \geq \beta I$. The function I never vanishes: if $u = 0$ on a time slice, its energy there would be $\frac{1}{2}\int u_t^2 \geq 0$. For $J = I^{-\alpha} > 0$,

$$J'' = -\alpha I^{-\alpha-2}\{II'' - (1+\alpha)(I')^2\} \leq -\alpha\beta I^{-\alpha-1} < 0. \tag{12.14}$$

If a global solution existed, $I'' \geq \beta$ would make I' positive at some finite t_0. Then $J'(t_0) < 0$, and concavity implies $J(t) \leq J(t_0) + (t - t_0)J'(t_0)$ for $t \geq t_0$. The right side becomes zero in finite time, contradicting $J > 0$. No positivity assumption on the initial correlation $\int gh$ is needed; $I'' \geq \beta$ creates the required sign later. □

12.5.2 Small-data blow-up below the three-dimensional threshold

Now consider the different focusing equation

$$\Box u = |u|^p \quad \text{in } \mathbb{R}^3, \qquad \int g > 0, \quad \int h > 0, \quad \operatorname{spt}(g,h) \subset B(0,R_0). \tag{12.15}$$

The data themselves need not be nonnegative. For a noninteger p the statement concerns a classical solution with the regularity needed for the representation and integrations, not an automatic C^∞ solution of a nonsmooth nonlinearity.

Theorem 12.17 (Source §12.5.2, Theorem 2)
If $1 < p < 1 + \sqrt{2}$, the solution in (12.15) cannot exist classically for all positive time. The assumptions allow arbitrarily small positive multiples of fixed admissible data. The endpoint $p = 1 + \sqrt{2}$ is not established by this proof.

Expanded proof (the shell gain and a divergent time integral).

Assume a global solution and put $I(t) = \int u(x,t)dx$. Finite propagation and integration of the PDE give $I'' = \int |u|^p$, hence $I, I' > 0$. Holder on $B(0, R_0 + t)$ gives

$$I'' \geq c(1+t)^{-3(p-1)} I^p. \tag{12.16}$$

负能量不需要初速度向外

聚焦幂 $f(z) = -|z|^{p-1}z$ 满足 $zf = (p+1)F$。取 $h = 0, g = A\psi$，A 足够大就使 $E = A^2\|D\psi\|_2^2/2 - A^{p+1}\|\psi\|_{p+1}^{p+1}/(p+1) < 0$。证明不依赖 $I'(0) > 0$。

Let v be the free wave with the same data. Positivity of the three-dimensional retarded kernel gives $u \geq v$, despite possible sign changes of v. Huygens' principle puts v in the shell $A_t = \{\max(t - R_0, 0) \leq |x| \leq t + R_0\}$, whose volume is at most $C(1 + t)^2$. Also $\int v = \int g + t \int h$. Consequently

$$c(1 + t) \leq \int_{A_t} u \leq \Big(\int |u|^p \Big)^{1/p} |A_t|^{1-1/p}, \qquad I'' \geq c(1 + t)^{2-p}.$$

Since $p < 3$, two integrations and the positive initial values yield $I(t) \geq c(1 + t)^{4-p}$ (after adjusting the constant for small t). This stronger lower bound comes from shell volume, not ball volume.

Choose $0 < \varepsilon < p-1$ so that $\mu = (p-1)^2 + \varepsilon(4-p) < 2$. Combining the lower bound with (12.16) gives $I'' \geq cI^{1+\varepsilon}(1 + t)^{-\mu}$. Multiply by $I' > 0$ and integrate. The product rule

$$(I^{2+\varepsilon}(1 + t)^{-\mu})' = (2 + \varepsilon)I^{1+\varepsilon}I'(1 + t)^{-\mu} - \mu I^{2+\varepsilon}(1 + t)^{-\mu-1}$$

shows $(I')^2 \geq cI^{2+\varepsilon}(1+t)^{-\mu} - C$. The positive term tends to infinity: its power lower bound has exponent $2(4 - p) - (p - 1)^2 > 0$ on the stated range. Thus for large t,

$$I' \geq cI^{1+\varepsilon/2}(1 + t)^{-\mu/2}, \qquad (I^{-\varepsilon/2})' \leq -c(1 + t)^{-\mu/2}.$$

Because $\mu/2 < 1$, the time integral on the right diverges. It makes the positive left quantity negative at a finite time, a contradiction. Scaling the initial data down changes constants and the possible lifetime, but does not remove this contradiction. □

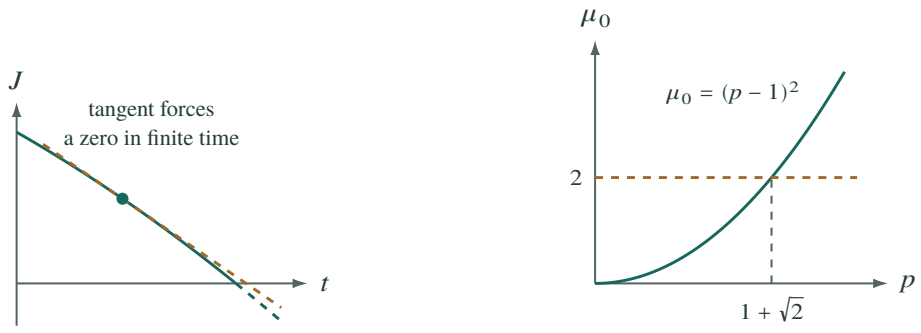


Diagram T12.4. Two different obstructions to a global solution On the left, a positive concave quantity with negative derivative lies below its tangent and must reach zero. The displayed curve $J = 1 - t - t^2/4$ is only an exact concavity model, not a PDE solution. On the right, $(p - 1)^2 < 2$ leaves room for the positive ε in the shell-based proof; the endpoint is excluded. Neither diagram asserts a quantitative lifespan formula for the PDE.

两个临界指数不可混用

这里 $1 + \sqrt{2}$ 来自 $(p - 1)^2 < 2$ 和球壳的 t^2 体积增长；三维退聚焦的能量临界 $p = 5$ 来自缩放。本节既没有证明 Strauss 临界端点，也没有讨论阈值以上的小数据全局理论。

12.6 Problems

These are abbreviated descriptions of all 18 source problems, followed by reconstructed solutions. Regularity, decay, and parameter qualifications are included where the source wording suppresses them.

1. Energy for a gradient-dependent principal part

For $u_{tt} - \operatorname{div} DL(Du) = 0$, use

$$E_L(t) = \int \{u_t^2/2 + L(Du) - L(0)\} dx.$$

The subtraction makes the integral finite for compactly supported u. Its derivative is $\int u_t u_{tt} + DL(Du) \cdot Du_t = \int u_t \{u_{tt} - \operatorname{div} DL(Du)\} = 0$. This needs no convexity of L for the identity; positivity or hyperbolicity would require further assumptions.

2. Klein–Gordon energy and equipartition

For $\Box u + m^2 u = 0$, take $m \geq 0$ and $E = \frac{1}{2} \int (u_t^2 + |Du|^2 + m^2 u^2)$. Multiplication by u_t proves conservation. For finite-energy data let $\omega(\xi) = \sqrt{|\xi|^2 + m^2}$ and use a unitary Fourier transform:

$$\hat{u} = \hat{g} \cos(\omega t) + \hat{h} \sin(\omega t)/\omega.$$

The difference between potential and kinetic terms is

$$\begin{aligned} D(t) &= \int (\omega^2 |\hat{u}|^2 - |\hat{u}_t|^2) d\xi \\ &= \int (\omega^2 |\hat{g}|^2 - |\hat{h}|^2) \cos(2\omega t) + 2\omega \operatorname{Re}(\hat{g}\overline{\hat{h}}) \sin(2\omega t)\, d\xi. \end{aligned}$$

Both amplitudes are L^1 by Cauchy–Schwarz and finite energy. In polar coordinates change the radial variable from r to $\omega = \sqrt{r^2 + m^2}$. The resulting amplitudes are L^1 on (m, ∞); the possible critical point $r = 0$ has no atom. Riemann–Lebesgue gives $D(t) \to 0$. Since potential plus kinetic is $2E$, each tends to E, including the claimed $\int (|Du|^2 + m^2 u^2) \to E$. A finite-energy approximation removes any auxiliary smoothness used for the Fourier calculation.

3. Descent and the one-dimensional massive kernel

Put $\overline{u}(x, y, t) = u(x, t) \cos(my)$. Then $(\partial_t^2 - \Delta_x - \partial_y^2)\overline{u} = \cos(my)(\Box u + m^2 u) = 0$. For $n = 1$, apply the two-dimensional Poisson formula and integrate out the additional variable. With $J_0(z) = \pi^{-1} \int_0^\pi \cos(z \sin\theta) d\theta$, the result is

$$\begin{aligned} u(x,t) = \frac{1}{2} \partial_t \int_{x-t}^{x+t} g(y) J_0\Big(m\sqrt{t^2 - (x-y)^2}\Big)\, dy \\ + \frac{1}{2} \int_{x-t}^{x+t} h(y) J_0\Big(m\sqrt{t^2 - (x-y)^2}\Big)\, dy. \end{aligned}$$

The integrated kernel identity is $\int_{-a}^{a} \cos(my)(a^2-y^2)^{-1/2}dy = \pi J_0(ma)$, obtained by $y = a\sin\theta$. If desired, differentiating the first integral gives endpoint term $\{g(x+t)+g(x-t)\}/2$ and interior kernel $-mtJ_1(m\sqrt{t^2-(x-y)^2})/(2\sqrt{t^2-(x-y)^2})$. Its endpoint limit is finite because $J_1(z) \sim z/2$. At $m = 0$ this reduces exactly to d'Alembert's formula.

4. Removing constant damping

For $\Box u + du_t = 0$, put $v = e^{dt/2}u$. Direct substitution gives $\Box v - (d^2/4)v = 0$. Thus the constant zero-order coefficient is negative when $d > 0$; the damping has not turned into a positive Klein–Gordon mass.

5. Phase speed and the domain of dependence

The plane wave $e^{i(x\cdot\xi-\omega t)}$ has dispersion $\omega^2 = |\xi|^2 + m^2$. For $m > 0$ its phase speed is $\omega/|\xi| > 1$, but its group velocity is $D_\xi\omega = \xi/\omega$, of magnitude less than one. A plane wave has no compact support and carries no newly created front. Causal propagation of a localized disturbance is the cone statement of §12.1, which remains true for the massive lower-order term. Group velocity describes narrow-band packets; the finite propagation proof is what controls an actual front. For $m = 0$ both speeds have magnitude one.

6. Uniform three-dimensional decay

Kirchhoff gives

$$u(x,t) = \frac{1}{4\pi}\int_{S^2}[g(x+t\omega) + tDg(x+t\omega)\cdot\omega + th(x+t\omega)]d\omega.$$

If the data are supported in $B(0,R)$ and $t \geq 2R$, the part of S^2 where $x+t\omega \in B(0,R)$ has area at most CR^2/t^2. This follows by intersecting the sphere of radius t centered at x with that fixed ball and using the spherical-cap area formula; if there is no intersection the integral is zero. The displayed formula is therefore bounded by C/t uniformly in x. For $t \leq 2R$, the same formula with the full sphere bounds u by a fixed constant, which is also at most a larger C/t.

7. Uniform two-dimensional decay

Differentiate the rescaled Poisson formula to write

$$u(x,t) = \frac{1}{2\pi}\int_{|x-y|<t} \frac{h(y) + \{g(y) + (y-x)\cdot Dg(y)\}/t}{\sqrt{t^2-|x-y|^2}}dy.$$

For t larger than a fixed multiple of the support radius, its absolute value is bounded by a data constant times $J = \int_{B_R \cap B(x,t)} (t^2 - |x-y|^2)^{-1/2} dy$. On $|x-y| \le t/2$ the contribution is C_R/t. For the remaining radii use coarea about x. The length of a circle of radius $r > t/2$ intersected with B_R is at most CR; the possible radii form an interval of length at most $2R$. Since the kernel increases toward $r = t$, this contribution is at most

$$CR \int_{\max(t-2R,t/2)}^{t} \frac{dr}{\sqrt{t^2 - r^2}} \le C_R t^{-1/2}.$$

The bounded small-time strip follows directly from the rescaled Poisson formula and the smooth data. Hence $|u(x,t)| \le Ct^{-1/2}$ for all $x, t > 0$. Unlike the three-dimensional case, the interior of the light cone generally carries a tail.

8. Bounded displacement data can amplify

For the given radial g^ε, $0 \le g^\varepsilon \le 1$ and $g^\varepsilon(r) \to \mathbf{1}_{(1,3)}(r)$ a.e. as $\varepsilon \downarrow 0$. At $x = 0$ and fixed $t > 3$, Poisson gives

$$u^\varepsilon(0,t) = -t \int_1^3 \frac{r g^\varepsilon(r)}{(t^2 - r^2)^{3/2}} dr \longrightarrow t \left\{ \frac{1}{\sqrt{t^2 - 1}} - \frac{1}{\sqrt{t^2 - 9}} \right\}.$$

Dominated convergence is valid for each such fixed t. The limit tends to $-\infty$ as $t \downarrow 3$. Given any M, first choose $3 < t < 4$ making its magnitude exceed $2M$, then take ε small enough. This proves that the supremum in the problem tends to infinity. No maximum principle for wave displacement follows from a bound on g alone.

9. Hyperbolic Kelvin transform

Let $d = n + 1$, $X = (x,t)$, $q = |x|^2 - t^2 \neq 0$, $Y = X/q$, $a = (d-2)/2$ and $w(X) = |q|^{-a}$. For the Lorentz matrix $G = \operatorname{diag}(-1, \dots, -1, 1)$, direct differentiation gives

$$DY\, G\, (DY)^T = q^{-2} G, \qquad \Box w = 0, \qquad w \Box Y^j + 2\langle Dw, DY^j \rangle_G = 0.$$

For example $\Box q = -2d$, $\langle Dq, Dq \rangle_G = -4q$, $\Box Y^j = 2(d-2)X^j/q^2$, and $\langle Dw, DY^j \rangle_G = -2awX^j/q^2$. The product and chain rules therefore cancel every first-derivative term and leave

$$\Box\{w\, u(Y)\} = |q|^{-(d+2)/2} (\Box u)(Y).$$

It follows that the transform preserves free waves on each region where $q \neq 0$. Absolute values allow both signs of q; no extension through the characteristic cone is asserted.

10. Compatibility of the sine–Gordon system

Set $A = (u+v)/2$, $B = (u-v)/2$. The given equations are $B_t = a\sin A$, $A_x = a^{-1}\sin B$. Differentiating gives $B_{xt} = \cos A \sin B$ and $A_{xt} = \cos B \sin A$. Adding and subtracting yield $u_{xt} = \sin(A+B) = \sin u$ and $v_{xt} = \sin(A-B) = \sin v$. These x, t are characteristic coordinates. With new variables $T = x + t$, $X = x - t$, the mixed derivative becomes $\partial_x \partial_t = \partial_T^2 - \partial_X^2$. Thus the equation has the usual wave-operator form in (X, T), with no missing factor when these summed coordinates are used.

11. A Bäcklund kink from the zero solution

With $v = 0$, the equations become $u_t = 2a\sin(u/2)$, $u_x = 2a^{-1}\sin(u/2)$. They imply $u = U(s)$, $s = at + x/a$, and $U' = 2\sin(U/2)$. Since $(\tan(U/4))' = \tan(U/4)$, nonconstant solutions are

$$u(x,t) = 4\arctan(Ce^{at+x/a}), \qquad C \neq 0,$$

with compatible 4π shifts; the constant solutions $u = 2\pi k$ are included separately. Differentiation gives $U'' = \sin U$, so $u_{xt} = U'' = \sin u$. For $a > 0, C > 0$ the profile connects 0 to 2π as s increases. The parameter C translates the kink.

12. The Sobolev product estimate

For $b_j = |\beta_j|$, the proof of (12.5) gives $D^{\beta_j} u_j \in L^{p_j}$ for every inverse exponent between $b_j/(2k)$ and $1/2$, when $k > n/2$. Choose these inverse exponents with sum exactly $1/2$; the endpoint sums bracket $1/2$ because $\sum b_j \le k$. Holder then gives $\| \prod D^{\beta_j} u_j \|_2 \le C \prod \|u_j\|_{H^k}$. This handles also zero multiindices and the case of a derivative of order exactly k, without using the false endpoint embedding $H^{n/2} \hookrightarrow L^\infty$.

13. Sphere-valued wave maps

Since $|u|^2 = 1$, differentiation gives $u \cdot u_{tt} = -|u_t|^2$ and $u \cdot \Delta u = -|Du|^2$. The normal space of the sphere is spanned by u itself. Thus $\Box u = \gamma u$ with $\gamma = u \cdot \Box u = |Du|^2 - |u_t|^2$, as asserted. This is a genuine derivative nonlinearity, so the scalar potential energy theory is not directly applicable.

14. Wave-map energy and a support correction

A unit-length map cannot itself vanish outside a compact set. Interpret the source's support wording as u equal to a fixed unit vector outside a compact set, or impose decay on Du, u_t sufficient for integration by parts.

Then

$$\frac{d}{dt}\frac{1}{2}\int(|u_t|^2+|Du|^2)=\int u_t\cdot\Box u=\int(|Du|^2-|u_t|^2)u_t\cdot u=0,$$

because $u_t\cdot u=0$. This states the nonvacuous finite-energy version of the exercise.

15. An elementary small-energy quintic proof

Modify the average estimate in (12.7) using $\int_{B_t}|g|\le Ct^{5/2}\|g\|_6$. Then

$$|\overline{w}|\le Ct^{-1/2}(E^{1/2}+\|g\|_6),\qquad \|w/r\|_2\le C(E^{1/2}+E^{1/6}),$$

since $\|g\|_6^6\le 6E$. Cone flux also gives $\int w^6\le 6E$. For $M(\tau)=\sup_{t\le\tau,x}|u(x,t)|$,

$$\int\frac{|w|^5}{r}\le\|w/r\|_2\left(\int|w|^8\right)^{1/2}\le C(E+E^{2/3})M(\tau).$$

The retarded formula gives $M(\tau)\le\|v\|_\infty+C(E+E^{2/3})M(\tau)$. Choose a universal small initial-energy threshold making the last coefficient at most $1/2$. On every finite strip the free wave is bounded, so the amplitude is bounded and the smooth solution continues globally. This argument needs neither the Morawetz identity nor Strichartz, and makes no smallness assumption on a high Sobolev norm of the data.

16. Galilean invariance for NLS

For $iu_t+\Delta u=f(|u|^2)u$, set $\theta=\xi\cdot x/2-|\xi|^2t/4$, $y=x-\xi t$, $w=e^{i\theta}u(y,t)$. Then

$$\begin{aligned}iw_t&=e^{i\theta}(|\xi|^2u/4+iu_t-i\xi\cdot Du),\\ \Delta w&=e^{i\theta}(\Delta u+i\xi\cdot Du-|\xi|^2u/4).\end{aligned}$$

The extra terms cancel, and $|w|=|u(y,t)|$ makes the nonlinear term transform with the same phase. This proves the stated invariance.

17. NLS mass, energy, and momentum

Assume the solution and the needed derivatives decay rapidly. Let $\rho=|u|^2$, $F(0)=0$, $F'=f$, with f real. The mass identity follows from $\rho_t=-2\operatorname{div}\operatorname{Im}(\overline{u}Du)$. For the energy $\mathscr{H}=\int(|Du|^2+F(\rho))$,

$$\mathscr{H}'=2\operatorname{Re}\int(-\Delta\overline{u}+f(\rho)\overline{u})u_t=2\operatorname{Re}\int(-i\overline{u}_t)u_t=0.$$

For $P_j=\operatorname{Im}\int\overline{u}\,u_{x_j}$, integration by parts in one of the two time-derivative terms gives

$$P_j'=2\operatorname{Im}\int\overline{u}_tu_{x_j}=-2\operatorname{Re}\int\Delta\overline{u}\,u_{x_j}+2\operatorname{Re}\int f(\rho)\overline{u}\,u_{x_j}=0.$$

球面值映射不能紧支为零

若 $|u|=1$，则 $u\not\equiv 0$ 于任何外部区域。题 14 的紧支应理解成 $u-u_\infty$ 紧支或导数充分衰减。能量只积分 Du,u_t，不积分处处为 1 的 $|u|^2$。

The last two integrals are respectively integrals of $\partial_j |Du|^2$ and $\partial_j F(\rho)$, hence vanish. This agrees with the source's momentum integrand $(\overline{u}Du - uD\overline{u})/(2i)$.

18. NLS virial identity and focusing cubic blow-up

In addition assume finite variance $V(t) = \int |x|^2 |u|^2$ and sufficient decay for the weighted integrations. The mass continuity identity gives $V' = 4\,\mathrm{Im} \int x \cdot \overline{u} Du$. Differentiate this expression, substitute $u_t = i\Delta u - if(\rho)u$, and integrate twice in the Laplacian terms. They give $2\int |Du|^2$ in the derivative of $V'/4$. The nonlinear terms give

$$-\int x \cdot \rho Df(\rho) = -\int x \cdot D\{\rho f(\rho) - F(\rho)\} = n \int \{\rho f(\rho) - F(\rho)\}.$$

Thus

$$V'' = 8\int |Du|^2 + 4n \int \{\rho f(\rho) - F(\rho)\}. \tag{12.17}$$

For focusing cubic NLS, $f(\rho) = -\rho$, $F = -\rho^2/2$ and $\mathcal{H} = K - P/2$, with $K = \int |Du|^2$, $P = \int |u|^4$. Hence $V'' = 8K - 2nP = 8\mathcal{H} + (4 - 2n)P \le 8\mathcal{H}(0) < 0$ when $n \ge 2$ and the initial energy is negative. Integrating gives $0 \le V(t) \le V(0) + V'(0)t + 4\mathcal{H}(0)t^2$, which fails at a finite time. This proves nonexistence of a global solution in the stated regularity and finite-variance class. The variance hypothesis is essential to this particular proof; negative energy alone is not a license to differentiate an infinite weighted integral.

12.7 References

Source attributions

Evans acknowledges D. Tataru's comments. His source for §12.2 and parts of §12.3 is Shatah–Struwe [S-S]. He attributes the sign-condition weak-existence argument to I. Segal, Bull. Soc. Math. France 91 (1963), 129–135, and the three-dimensional/subcritical arguments to K. Jörgens, Math. Z. 77 (1961), 295–308. The critical flux and global-existence theorems are attributed to M. Grillakis, Ann. of Math. 132 (1990), 485–509, with Smith–Sogge simplifications, and reference to Sogge [So] and Shatah–Struwe. Strauss [St1] and John [J3] are cited for blow-up. Problem 8 is credited to Protter–Weinberger [P-W]; small-energy quintic existence to J. Rauch; J. Holmer is thanked for help with Problems 16–18. The cubic NLS blow-up exercise cites R. Glassey, J. Math. Physics 18 (1977), 1794–1797. These are source attributions, not a claim to have separately consulted every listed publication.

Proof boundaries and editorial changes

The chapter uses the earlier linear wave, Sobolev, Hardy, Rellich, Gronwall, Banach fixed-point, Fourier, and measure-theoretic tools. The product exponent allocation and completeness of the strong ball in the weak metric are supplied. Sign-condition weak existence uses space-time uniform integrability; its energy inequality is distinguished from smooth conservation and supplies only finite-strip L^2 bounds. For noninteger powers, global bounded finite-energy mild solutions are asserted instead of unqualified smoothness. Sphere-valued maps are interpreted as constant outside a compact set, not zero there.

The homogeneous $(4, 12)$ three-dimensional Strichartz estimate is an explicit external theorem. Its dispersive proof is not supplied; the energy and Duhamel parts, localization by cone-supported forcing, Morawetz identity, singular-top limit, nonconcentration, cone enlargement, finite cover, and continuation are expanded. No solution at the candidate singular time is assumed in advance. The critical proof does not claim a general supercritical theorem or scattering. The focusing small-data proof covers $1 < p < 1 + \sqrt{2}$, not the critical endpoint or the range above it. The NLS virial argument assumes finite variance and enough decay. All 18 solutions, Chinese notes, and five diagrams are editorial reconstructions. Source comparison, symbolic checks, compilation, and visual review are separate from independent mathematical review, which has not been performed.

Appendices

Appendix

A

Notation

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix A; printed p. 653; PDF page 670.

Chinese reference: 中译本对应章：记号，书内第 463 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

A.1 Notation for matrices

Outline only. Text, annotations, and proof details are pending.

A.2 Geometric notation

Outline only. Text, annotations, and proof details are pending.

A.3 Notation for functions

Outline only. Text, annotations, and proof details are pending.

A.4 Vector-valued functions

Outline only. Text, annotations, and proof details are pending.

A.5 Notation for estimates

Outline only. Text, annotations, and proof details are pending.

A.6 Some comments about notation

Outline only. Text, annotations, and proof details are pending.

Appendix

B

Inequalities

> Reading source
> **Evans, *Partial Differential Equations*, second edition**
> English source: Appendix B; printed p. 661; PDF page 678.
> Chinese reference: 中译本对应章：不等式，书内第 471 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

B.1 Convex functions

Outline only. Text, annotations, and proof details are pending.

B.2 Useful inequalities

Outline only. Text, annotations, and proof details are pending.

Appendix

C

Calculus

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix C; printed p. 665; PDF page 682.

Chinese reference: 中译本对应章：微积分，书内第 475 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

C.1 Boundaries

Outline only. Text, annotations, and proof details are pending.

C.2 Gauss–Green Theorem

Outline only. Text, annotations, and proof details are pending.

C.3 Polar coordinates, coarea formula

Outline only. Text, annotations, and proof details are pending.

C.4 Moving regions

Outline only. Text, annotations, and proof details are pending.

C.5 Convolution and smoothing

Outline only. Text, annotations, and proof details are pending.

C.6 Inverse Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.7 Implicit Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.8 Uniform convergence

Outline only. Text, annotations, and proof details are pending.

Appendix

D

Functional Analysis

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix D; printed p. 674; PDF page 691.

Chinese reference: 中译本对应章：泛函分析，书内第 482 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

D.1 Banach spaces

Outline only. Text, annotations, and proof details are pending.

D.2 Hilbert spaces

Outline only. Text, annotations, and proof details are pending.

D.3 Bounded linear operators

Outline only. Text, annotations, and proof details are pending.

D.4 Weak convergence

Outline only. Text, annotations, and proof details are pending.

D.5 Compact operators, Fredholm theory

Outline only. Text, annotations, and proof details are pending.

D.6 Symmetric operators

Outline only. Text, annotations, and proof details are pending.

Appendix

E

Measure Theory

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix E; printed p. 684; PDF page 701.

Chinese reference: 中译本对应章：测度论，书内第 490 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

E.1 Lebesgue measure

Outline only. Text, annotations, and proof details are pending.

E.2 Measurable functions and integration

Outline only. Text, annotations, and proof details are pending.

E.3 Convergence theorems for integrals

Outline only. Text, annotations, and proof details are pending.

E.4 Differentiation

Outline only. Text, annotations, and proof details are pending.

E.5 Banach space-valued functions

Outline only. Text, annotations, and proof details are pending.

Bibliography

The source bibliography starts on English printed p. 689 (PDF page 706). Bibliographic entries and any additional references remain to be entered.

Outline only. Text, annotations, and proof details are pending.

Index

The source index starts on English printed p. 703 (PDF page 720). An index for these notes will be prepared as the text is written; page numbers must refer to this edition.

Outline only. Text, annotations, and proof details are pending.

Bilingual Glossary
英汉术语表

The entries below cover the annotated material in Chapters 1–6.

English	中文及正文位置
multiindex	多重指标；§1.1
principal part	主部；§1.1
linear / homogeneous	线性 / 齐次；§1.1
semilinear	半线性；§1.1
quasilinear	拟线性；§1.1
fully nonlinear	完全非线性；§1.1
divergence form	散度形式；§1.2
formal adjoint	形式伴随；§1.2
uniform ellipticity	一致椭圆性；§1.2
well-posedness	适定性；§1.3
continuous dependence	连续依赖；§1.3
classical solution	经典解；§1.3
weak solution / regularity	弱解 / 正则性；§1.3
shock / entropy condition	激波 / 熵条件；§1.3
gradient blow-up	梯度爆破；§1.3
viscosity solution	黏性解；§1.4
uniform remainder estimate	一致余项估计；§1.5

Classical model equations / 经典模型方程

English	中文及正文位置
characteristic	特征线；§2.1
fundamental solution	基本解；§2.2
harmonic function	调和函数；§2.2
spherical mean	球面平均；§2.2.2
ball mean	球体平均；§2.2.2
Poisson kernel	Poisson 核；§2.2
Green function	Green 函数；§2.2
heat ball	热球；§2.3.2
parabolic scaling	抛物缩放；§2.3.2
backward heat kernel	向后热核；§2.3.2
parabolic boundary	抛物边界；§2.3
subsolution	次解；§2.3
Duhamel’s principle	Duhamel 原理；§2.3
method of descent	降维法；§2.4
finite propagation speed	有限传播速度；§2.4

Nonlinear first-order equations / 非线性一阶方程

English	中文及正文位置
complete integral	完全积分；§3.1
envelope / singular integral	包络 / 奇异积分；§3.1
first jet	一阶 jet（函数值与梯度）；§3.1
noncharacteristic data	非特征数据；§3.2
characteristic projection	特征投影；§3.2
Legendre transform	Legendre 变换；§3.3
convex conjugate	凸共轭；§3.3
superlinear growth	超线性增长；§3.3
dynamic programming	动态规划；§3.3
semiconcavity	半凹性；§3.3.3
Rankine–Hugoniot condition	跳跃条件；§3.4
one-sided entropy bound	单边熵估计；§3.4.3
rarefaction wave	稀疏波；§3.4
Riemann problem	Riemann 问题；§3.4
N-wave	N 波；§3.4

Representation methods / 解的表示方法

English	中文及正文位置
separation of variables	分离变量法；§4.1
eigenfunction expansion	特征函数展开；§4.1
Turing instability	Turing 失稳；§4.1
phase / group velocity	相速度 / 群速度；§4.2
solitary wave	孤立波；§4.2
Barenblatt solution	Barenblatt 解；§4.2.4
Plancherel theorem	Plancherel 定理；§4.3
Radon transform	Radon 变换；§4.3.3
resolvent	预解算子；§4.3
Cole–Hopf transformation	Cole–Hopf 变换；§4.4
hodograph transform	速度图变换；§4.4
vanishing viscosity	消失黏性；§4.5.2
stationary phase	驻相法；§4.5.3
caustic	焦散；§4.5.3
homogenization	均匀化；§4.5.4
cell problem / corrector	单元问题 / 修正子；§4.5.4
noncharacteristic surface	非特征曲面；§4.6
coefficientwise majorant	优级数；§4.6.3

Sobolev theory / Sobolev 理论

English	中文及正文位置
Hölder seminorm	Hölder 半范数；§5.1
weak derivative	弱导数；§5.2
test function	试验函数；§5.2
Sobolev space	Sobolev 空间；§5.2
mollification	磨光 / 光滑化；§5.3
partition of unity	单位分解；§5.3
extension operator	延拓算子；§5.4
boundary flattening	边界拉平；§5.4
trace / zero trace	迹 / 零迹；§5.5
Sobolev conjugate	Sobolev 共轭指数；§5.6
critical exponent	临界指数；§5.6
continuous representative	连续代表元；§5.6
compact embedding	紧嵌入；§5.7
concentration	集中；§5.7
difference quotient	差商；§5.8
mean oscillation	平均振荡；§5.8
negative Sobolev space	负阶 Sobolev 空间；§5.9
Bochner integral	Bochner 积分；§5.9
duality pairing	对偶配对；§5.9
absolutely continuous	绝对连续；§5.9

Elliptic theory / 椭圆方程理论

English	中文及正文位置
uniform ellipticity	一致椭圆性；§6.1
divergence form	散度型；§6.1
nondivergence form	非散度型；§6.1
bilinear form	双线性型；§6.1
weak Dirichlet solution	弱 Dirichlet 解；§6.1
coercivity	强制性；§6.2
Gårding inequality	Gårding 不等式；§6.2
Fredholm alternative	Fredholm 二择一定理；§6.2
formal adjoint	形式伴随；§6.1
compatibility condition	相容条件；§6.2
Caccioppoli estimate	Caccioppoli 估计；§6.3
tangential difference quotient	切向差商；§6.3
boundary regularity	边界正则性；§6.3
Hopf boundary lemma	Hopf 边界引理；§6.4
strong maximum principle	强最大值原理；§6.4
Harnack inequality	Harnack 不等式；§6.4
Rayleigh quotient	Rayleigh 商；§6.5
principal eigenvalue	主特征值；§6.5
Hadamard variation formula	Hadamard 变分公式；§6.6
Sommerfeld radiation condition	Sommerfeld 辐射条件；§6.6

Linear evolution / 线性发展方程

English	中文及正文位置
Gelfand triple	Gelfand 三重空间；§7.1
initial trace	初值迹；§7.1
Galerkin approximation	Galerkin 逼近；§7.1
energy estimate	能量估计；§7.1
parabolic boundary	抛物边界；§7.1
corner compatibility	角点相容条件；§7.1
parabolic Harnack inequality	抛物型 Harnack 不等式；§7.1
domain of dependence	依赖区域；§7.2
finite propagation speed	有限传播速度；§7.2
optical distance	光学距离；§7.2
symmetric hyperbolic system	对称双曲系统；§7.3
principal symbol	主符号；§7.3
derivative loss	导数损失；§7.3
strongly continuous semigroup	强连续半群；§7.4
infinitesimal generator	无穷小生成元；§7.4
graph norm	图范数；§7.5
resolvent identity	预解恒等式；§7.4
Yosida approximation	Yosida 逼近；§7.4
mild solution	温和解；§7.4
variation of constants	常数变易公式；§7.4

Variational theory / 变分理论

English	中文及正文位置
first variation	一阶变分；§8.1
second variation	二阶变分；§8.1
null Lagrangian	零 Lagrange 函数；§8.1
Piola identity	Piola 恒等式；§8.1
direct method	直接法；§8.2
weak lower semicontinuity	弱下半连续性；§8.2
polyconvexity	多凸性；§8.2
field of extremals	极值场；§8.2
constraint qualification	约束资格条件；§8.4
Lagrange multiplier	Lagrange 乘子；§8.4
variational inequality	变分不等式；§8.4
contact set	接触集；§8.4
free boundary	自由边界；§8.4
harmonic map	调和映射；§8.4
Palais–Smale condition	Palais–Smale 条件；§8.5
mountain pass	山路；§8.5
Noether current	Noether 流；§8.6
stress tensor	应力张量；§8.6
conformal energy	共形能量；§8.6
Almgren frequency	Almgren 频率；§8.7

Nonvariational techniques / 非变分技术

English	中文及正文位置
monotone vector field	单调向量场；§9.1
strong monotonicity	强单调性；§9.1
coercivity	强制性；§9.1
Minty argument	Minty 论证；§9.1
strict contraction	严格压缩映射；§9.2
compact mapping	紧映射；§9.2
a priori estimate	先验估计；§9.2
Schaefer fixed point theorem	Schaefer 不动点定理；§9.2
subsolution and supersolution	下解与上解；§9.3
monotone iteration	单调迭代；§9.3
blow-up	爆破；§9.4
critical exponent	临界指数；§9.4
Derrick–Pohozaev identity	Derrick–Pohozaev 恒等式；§9.4
star-shaped level set	星形水平集；§9.5
moving plane method	移动平面法；§9.5
subdifferential	次微分；§9.6
nonlinear resolvent	非线性预解算子；§9.6
gradient flow	梯度流；§9.6
normal cone	法锥；§9.7
penalty method	罚方法；§9.7

Hamilton–Jacobi / 黏性解与控制

English	中文及正文位置
viscosity solution	黏性解；§10.1
viscosity subsolution	黏性次解；§10.1
viscosity supersolution	黏性上解；§10.1
touching test function	接触测试函数；§10.1
vanishing viscosity	消黏法；§10.1
locally uniform convergence	局部一致收敛；§10.1
comparison principle	比较原理；§10.2
doubling of variables	双变量法；§10.2
modulus of continuity	连续模；§10.2
sup-norm contraction	上确界范数收缩；§10.2
admissible control	容许控制；§10.3
value function	价值函数；§10.3
running cost	运行代价；§10.3
terminal cost	终端代价；§10.3
dynamic programming principle	动态规划原理；§10.3
feedback control	反馈控制；§10.3
verification theorem	验证定理；§10.3
eikonal equation	程函方程；§10.4
discounted control	折扣控制；§10.4
degenerate ellipticity	退化椭圆性；§10.4

Conservation laws / 守恒律组

English	中文及正文位置
conserved variable	守恒变量；§11.1
integral solution	积分解；§11.1
Rankine–Hugoniot condition	跳跃条件；§11.1
strict hyperbolicity	严格双曲性；§11.1
left eigenvector	左特征向量；§11.1
state space	状态空间；§11.2
Riemann problem	Riemann 问题；§11.2
simple wave	简单波；§11.2
rarefaction wave	稀疏波；§11.2
Hugoniot locus	Hugoniot 轨迹；§11.2
genuine nonlinearity	真正非线性；§11.2
linear degeneracy	线性退化；§11.2
contact discontinuity	接触间断；§11.2
Lax entropy condition	Lax 熵条件；§11.2
Riemann invariant	Riemann 不变量；§11.3
gradient blow-up	梯度爆破；§11.3
viscous profile	黏性行波剖面；§11.4
Liu entropy criterion	Liu 熵准则；§11.4
entropy/entropy-flux pair	熵与熵通量对；§11.4
sonic shock	声速激波；§11.5

Nonlinear waves / 非线性波动方程

English	中文及正文位置
backward wave cone	后向波锥；§12.1
energy flux	能量通量；§12.1
finite propagation speed	有限传播速度；§12.1
domain of dependence	依赖域；§12.1
energy solution	能量解；§12.2
continuation criterion	延拓准则；§12.3
uniform integrability	一致可积性；§12.3
retarded potential	推迟势；§12.3
defocusing nonlinearity	退聚焦非线性；§12.3
energy-critical power	能量临界幂；§12.4
Strichartz estimate	Strichartz 估计；§12.4
Morawetz multiplier	Morawetz 乘子；§12.4
nonconcentration	不集中性；§12.4
concavity method	凹性方法；§12.5
Huygens' principle	Huygens 原理；§12.5
Strauss exponent	Strauss 指数；§12.5
hyperbolic Kelvin transform	双曲 Kelvin 变换；§12.6
Bäcklund transformation	Bäcklund 变换；§12.6
wave map	波映射；§12.6
virial identity	virial 恒等式；§12.6